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Inventor(s)

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Methods of treating HIV-1 infection utilizing broadly neutralizing human immunodeficiency virus type 1 (HIV-1) GP120-specific monoclonal antibodies

Abstract

The invention provides a method for obtaining a broadly neutralizing antibody (bNab), including screening memory B cell cultures from a donor PBMC sample for neutralization activity against a plurality of HIV-1 species, cloning a memory B cell that exhibits broad neutralization activity; and rescuing a monoclonal antibody from that memory B cell culture. The resultant monoclonal antibodies may be characterized by their ability to selectively bind epitopes from the Env proteins in native or monomeric form, as well as to inhibit infection of HIV-1 species from a plurality of clades. Compositions containing human monoclonal anti-HIV antibodies used for prophylaxis, diagnosis and treatment of HIV infection are provided. Methods for generating such antibodies by immunization using epitopes from conserved regions within the variable loops of gp120 are provided. Immunogens for generating anti-HIV1 bNAbs are also provided. Furthermore, methods for vaccination using suitable epitopes are provided.

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Background/Summary

RELATED APPLICATIONS AND INCORPORATION BY REFERENCE (1) This application is a continuation of U.S. application Ser. No. 17/376,276 filed Jul. 15, 2021, now allowed, which is a continuation of U.S. application Ser. No. 16/591,175 filed Oct. 2, 2019, now U.S. Pat. No. 11,319,362, which issued on May 3, 2022, which is a continuation of U.S. application Ser. No. 15/701,679 filed Sep. 12, 2017, now U.S. Pat. No. 10,836,811, which issued on Nov. 17, 2020, which is a continuation of U.S. application Ser. No. 15/152,630 filed May 12, 2016, now U.S. Pat. No. 10,087,239, which issued on Oct. 2, 2018, which is a continuation of U.S. application Ser. No. 13/780,776 filed Feb. 28, 2013, now U.S. Pat. No. 9,464,131, which issued on Oct. 11, 2016, which is a continuation-in-part application of international patent application Serial No. PCT/US2011/049880 filed Aug. 31, 2011, which published as PCT Publication No. WO 2012/030904 on Mar. 8, 2012, which claims priority to U.S. Provisional Patent Application Serial Nos. 61/378,604 filed Aug. 31, 2010, 61/386,940 filed Sep. 27, 2010, 61/476,978 filed Apr. 19, 2011 and 61/515,548 filed Aug. 5, 2011. (2) The foregoing applications, and all documents cited therein or during their prosecution (“appln cited documents”) and all documents cited or referenced in the appln cited documents, and all documents cited or referenced herein (“herein cited documents”), and all documents cited or referenced in herein cited documents, together with any manufacturer's instructions, descriptions, product specifications, and product sheets for any products mentioned herein or in any document incorporated by reference herein, are hereby incorporated herein by reference, and may be employed in the practice of the invention. More specifically, all referenced documents are incorporated by reference to the same extent as if each individual document was specifically and individually indicated to be incorporated by reference.

SEQUENCE STATEMENT

(1) The instant application contains a Sequence Listing which has been submitted electronically in XML format and is hereby incorporated by reference in its entirety. Said XML copy, was amended on Jan. 9, 2023, is named Y7991_10001.xml and is 11,128,824 bytes in size.

FIELD OF THE INVENTION

(2) The present invention relates generally to therapy, diagnosis and monitoring of human immunodeficiency virus (HIV) infection. The invention is more specifically related to human neutralizing monoclonal antibodies specific for HIV-1, such as broad and potent neutralizing monoclonal antibodies specific for HIV-1 and their manufacture and use. Broad neutralization suggests that the antibodies can neutralize HIV-1 isolates from different individuals. Such antibodies are useful in pharmaceutical compositions for the prevention and treatment of HIV, and for the diagnosis and monitoring of HIV infection and for design of HIV vaccine immunogens.

BACKGROUND OF THE INVENTION

(3) AIDS was first reported in the United States in 1981 and has since become a major worldwide epidemic. AIDS is caused by the human immunodeficiency virus, or HIV. By killing or damaging cells of the body's immune system, HIV progressively destroys the body's ability to fight infections and certain cancers. People diagnosed with AIDS may get life-threatening diseases called opportunistic infections. These infections are caused by microbes such as viruses or bacteria that usually do not make healthy people sick. HIV is spread most often through unprotected sex with an infected partner. HIV also is spread through contact with infected blood. The human immunodeficiency virus (HIV) is the cause of acquired immune deficiency syndrome (AIDS) (Barre-Sinoussi, F., et al., 1983, Science 220:868-870; Gallo, R., et al., 1984, Science 224:500-503). There are currently 1.25 million people in the US infected with HIV-induced acquired immunodeficiency syndrome according to a Center for Disease Control report. The epidemic is growing most rapidly among minority populations and is a leading killer of African-American males ages 25 to 44. According, AIDS affects nearly seven times more African Americans and three times more Hispanics than whites. In recent years, an increasing number of African-American women and children are being affected by HIV/AIDS. With over 40 million people infected worldwide, the current global HIV pandemic ranks among the greatest infectious disease scourges in human history.

(4) There is therefore a need for the efficient identification and production of neutralizing antibodies effective against multiple clades and strains of HIV as well as the elucidation of the target and antigenic determinants to which such antibodies bind.

(5) Citation or identification of any document in this application is not an admission that such document is available as prior art to the present invention.

SUMMARY OF THE INVENTION

(6) The present invention provides a novel method for isolating potent, broadly neutralizing monoclonal antibodies against HIV. Peripheral Blood Mononuclear Cells (PBMCs) are obtained from an HIV-infected donor selected for HIV-1 neutralizing activity in the plasma, and memory B cells are isolated for culture in vitro. The B cell culture supernatants may then be screened by a primary neutralization assay in a high throughput format, and B cell cultures exhibiting neutralizing activity may be selected for rescue of monoclonal antibodies. It is surprisingly observed that neutralizing antibodies obtained by this method do not always exhibit gp120 or gp41 binding at levels that correlate with neutralization activity. The method of the invention therefore allows identification of novel antibodies with cross-clade neutralization properties.

(7) The present invention provides human monoclonal antibodies specifically directed against HIV. In certain embodiments, the invention provides human anti-HIV monoclonal antibodies including, but not limited to, 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_101 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158) and sister clones thereof. For instance, an exemplary sister clone of the 1443_C16 (PG16) (TCN-116) antibody is the 1503_H05 (PG16) (TCN-119) antibody, the 1456_A12 (PG16) (TCN-117) antibody, the 1469_M23 (PG16) (TCN-118) antibody, the 1489_I13 (PG16) (TCN-120) antibody, or the 1480_I08 (PG16) antibody.

(8) Specifically, the invention provides an isolated anti-HIV antibody, wherein said antibody may have heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(9) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GFTFHK (SEQ ID NO: 266), LISDDGMRKY (SEQ ID NO: 267), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(10) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTRSDVGGFDSVS (SEQ ID NO: 92), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(11) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GFTFHK (SEQ ID NO: 266), LISDDGMRKY (SEQ ID NO: 267), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTRSDVGGFDSVS (SEQ ID NO: 92), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(12) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSRDVGGFDSVS (SEQ ID NO: 93), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(13) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GFTFHK (SEQ ID NO: 266), LISDDGMRKY (SEQ ID NO: 267), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSRDVGGFDSVS (SEQ ID NO: 93), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(14) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of KYGMH (SEQ ID NO: 88), LISDDGMRKYHSNSMWG (SEQ ID NO: 98), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), and a light chain with three

CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(15) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of SYAFT (SEQ ID NO: 104), MVTPIFGGEAKYSQRFEG (SEQ ID NO: 105), and DRRVPIATDNWLDP (SEQ ID NO: 9), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQTINNYLN (SEQ ID NO: 107), GASNLQN (SEQ ID NO: 108), and QQSFSPTPT (SEQ ID NO: 42).

(16) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GGTFSS (SEQ ID NO: 268), MVTPIFGGEAK (SEQ ID NO: 269), and DRRVPIATDNWLDP (SEQ ID NO: 9), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQTINNYLN (SEQ ID NO: 107), GASNLQN (SEQ ID NO: 108), and QQSFSPTPT (SEQ ID NO: 42).

(17) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of SYAFS (SEQ ID NO: 110), MITPVFGGETKYAPRFQG (SEQ ID NO: 111), and DRRVVPMTDNWLDP (SEQ ID NO: 8), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQTIHTYLN (SEQ ID NO: 113), GASTLQS (SEQ ID NO: 114), and QQSYSTPT (SEQ ID NO: 43).

(18) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GGAFSS (SEQ ID NO: 270), MITPVFGGETK (SEQ ID NO: 271), DRRVVPMTDNWLDP (SEQ ID NO: 8), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQTIHTYLN (SEQ ID NO: 113), GASTLQS (SEQ ID NO: 114), and QQSYSTPT (SEQ ID NO: 43).

(19) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of DYVLH (SEQ ID NO: 116), LIDPENGGEARYAEKFQG (SEQ ID NO: 117), GAVGADSGSWFDP (SEQ ID NO: 10), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of SGSKLGDKYVS (SEQ ID NO: 120), ENDRRPS (SEQ ID NO: 121), QAWETTTTTFVF (SEQ ID NO: 44).

(20) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GYSFID (SEQ ID NO: 102), LIDPENGGEAR (SEQ ID NO: 103), GAVGADSGSWFDP (SEQ ID NO: 10), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of SGSKLGDKYVS (SEQ ID NO: 120), ENDRRPS (SEQ ID NO: 121), QAWETTTTTFVF (SEQ ID NO: 44).

(21) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of RQGM1-1 (SEQ ID NO: 123), FIKYDGSEKYHADSVWG (SEQ ID NO: 124), and EAGGPDYRNGYNYDFYDGYNYHYMDV (SEQ ID NO: 7), and a light chain with three

CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSNDVGGYESVS (SEQ ID NO: 126), DVSKRPS (SEQ ID NO: 127), and KSLTSTRRRV (SEQ ID NO: 45).

(22) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GFDFSR (SEQ ID NO: 118), FIKYDGSEKY (SEQ ID NO: 272), and EAGGPDYRNGYNYDFYDGYNYHYMDV (SEQ ID NO: 7), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSNDVGGYESVS (SEQ ID NO: 126), DVSKRPS (SEQ ID NO: 127), and KSLTSTRRRV (SEQ ID NO: 45).

(23) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of DSYWS (SEQ ID NO: 90), YVHKSGDTNYSPLKS (SEQ ID NO: 265), TLHGRRYIGIVAFNEWFTYFYMDV (SEQ ID NO: 143), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of GEKSLGSRAVQ (SEQ ID NO: 150), NNQDRPS (SEQ ID NO: 151), HIWDSRVPTKWV (SEQ ID NO: 152).

(24) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GASISD (SEQ ID NO: 144), YVHKSGDTN (SEQ ID NO: 145), TLHGRRYIGIVAFNEWFTYFYMDV (SEQ ID NO: 143), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of GEKSLGSRAVQ (SEQ ID NO: 150), NNQDRPS (SEQ ID NO: 151), HIWDSRVPTKWV (SEQ ID NO: 152).

(25) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of DNYWS (SEQ ID NO: 261), YVHDSGDTNYPNLSLKS (SEQ ID NO: 157), and TKHGRRYIGVVAFAKEWFTYFYMDV (SEQ ID NO: 262), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of GEESLGSRSVI (SEQ ID NO: 162), NNNDRPS (SEQ ID NO: 163), and HIWDSRRPTNWV (SEQ ID NO: 164).

(26) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GTLVRD (SEQ ID NO: 263), YVHDSGDTN (SEQ ID NO: 264), and TKHGRRYIGVVAFAKEWFTYFYMDV (SEQ ID NO: 262), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of GEESLGSRSVI (SEQ ID NO: 162), NNNDRPS (SEQ ID NO: 163), and HIWDSRRPTNWV (SEQ ID NO: 164).

(27) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of DAYWS (SEQ ID NO: 169), YVHHSGDTNYPNLSLKR (SEQ ID NO: 170), and ALHGKRIYGI VALGELFTYFYMDV (SEQ ID NO: 171), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of GKESIGSRAVQ (SEQ ID NO: 178), NNQDRPA (SEQ ID NO: 179), and HIYDARGGTNWV (SEQ ID NO: 180).

(28) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GASIND (SEQ ID NO: 172), YVHHSGDTN (SEQ ID NO: 173), and ALHGKRIYGI VALGELFTYFYMDV (SEQ ID NO: 171), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid

sequences of GKESIGSRAVQ (SEQ ID NO: 178), NNQDRPA (SEQ ID NO: 179), and HIYDARGGTNWV (SEQ ID NO: 180).

(29) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of ACTYFWG (SEQ ID NO: 185), SLSHCQSFWSGSGWTFHNPSLKS (SEQ ID NO: 186), and FDGEVLVYNHWPKPAWVDL (SEQ ID NO: 187), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTATNFVS (SEQ ID NO: 194), GVDKRPP (SEQ ID NO: 195), and GSLVGNWDVI (SEQ ID NO: 196).

(30) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GESTGACT (SEQ ID NO: 188), SLSHCQSFWSGSGWTF (SEQ ID NO: 189), and FDGEVLVYNHWPKPAWVDL (SEQ ID NO: 187), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTATNFVS (SEQ ID NO: 194), GVDKRPP (SEQ ID NO: 195), and GSLVGNWDVI (SEQ ID NO: 196).

(31) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of ACDYFWG (SEQ ID NO: 201), GLSHCAGYYNTGWTYHNPSLKS (SEQ ID NO: 202), and FDGEVLVYHDWPKPAWVDL (SEQ ID NO: 203), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TGTSNRFVS (SEQ ID NO: 210), GVNKRPS (SEQ ID NO: 211), and SSLVGNWDVI (SEQ ID NO: 212).

(32) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GDSTAACD (SEQ ID NO: 204), GLSHCAGYYNTGWTY (SEQ ID NO: 205), and FDGEVLVYHDWPKPAWVDL (SEQ ID NO: 203), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TGTSNRFVS (SEQ ID NO: 210), GVNKRPS (SEQ ID NO: 211), and SSLVGNWDVI (SEQ ID NO: 212).

(33) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of TGHYYWG (SEQ ID NO: 217), HIHYTTAVLHNPSLKS (SEQ ID NO: 218), and SGGDILYYYEWQKPHWFSP (SEQ ID NO: 219), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSSDIGGWNFVS (SEQ ID NO: 226), EVNKRPS (SEQ ID NO: 227), and SSLFGRWDVV (SEQ ID NO: 228).

(34) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GESINTGH (SEQ ID NO: 220), HIHYTTAVL (SEQ ID NO: 221), and SGGDILYYYEWQKPHWFSP (SEQ ID NO: 219), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSSDIGGWNFVS (SEQ ID NO: 226), EVNKRPS (SEQ ID NO: 227), and SSLFGRWDVV (SEQ ID NO: 228).

(35) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GGEWGD KDYHWG (SEQ ID NO: 233), SIHWRGTTHYKESLRR (SEQ ID NO: 234), and HRHHDVFM LVPIAGWFDV (SEQ ID NO: 235), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQNINKNLA (SEQ ID NO: 243), ETYSKIA

(SEQ ID NO: 244), and QQYEEWPRT (SEQ ID NO: 245).

(36) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GDSIRGGEWGDKD (SEQ ID NO: 236), SIHWRTGTH (SEQ ID NO: 237), and HRHHDVFMLVPIAGWFDV (SEQ ID NO: 235), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQNINKNLA (SEQ ID NO: 243), ETYSKIA (SEQ ID NO: 244), and QQYEEWPRT (SEQ ID NO: 245).

(37) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GTDWGENDFHYG (SEQ ID NO: 250), SIHWRTGTHYKTSFRS (SEQ ID NO: 251), HKYHDIFRVVPVAGWFDP (SEQ ID NO: 252), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), QQYEEWPRT (SEQ ID NO: 245).

(38) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of GGSMRGTDWGEND (SEQ ID NO: 253), SIHWRTGTH (SEQ ID NO: 254), HKYHDIFRVVPVAGWFDP (SEQ ID NO: 252), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), QQYEEWPRT (SEQ ID NO: 245).

(39) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of KYDVH (SEQ ID NO: 277), WMSHEGDKTESAQRFGK (SEQ ID NO: 278), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(40) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GNTFSK (SEQ ID NO: 280), WMSHEGDKTE (SEQ ID NO: 281), GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(41) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of KYDVH (SEQ ID NO: 277), WISHERDKTESAQRFGK (SEQ ID NO: 293), GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(42) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GNTFSK (SEQ ID NO: 280), WISHERDKTE (SEQ ID NO: 294), GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of

the amino acid sequences of TSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(43) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of KYDVH (SEQ ID NO: 277), WMSHEGDKTESAQRFKG (SEQ ID NO: 278), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TSTQSLRHSNGANYLA (SEQ ID NO: 303), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(44) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GNTFSK (SEQ ID NO: 280), WMSHEGDKTE (SEQ ID NO: 281), GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TSTQSLRHSNGANYLA (SEQ ID NO: 303), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(45) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of KYDVH (SEQ ID NO: 277), WMSHEGDKTESAQRFKG (SEQ ID NO: 278), GSKHRLRDYVLYDDYGLINQQEWNDYLEFLDV (SEQ ID NO: 308), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TSTQSLRHSNGANYLA (SEQ ID NO: 303), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(46) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GNTFRK (SEQ ID NO: 309), WMSHEGDKTE (SEQ ID NO: 281), and GSKHRLRDYVLYDDYGLINQQEWNDYLEFLDV (SEQ ID NO: 308), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TSTQSLRHSNGANYLA (SEQ ID NO: 303), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(47) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of RCNYFWG (SEQ ID NO: 320), SLSHCRSYYNTDWTYHNPSLKS (SEQ ID NO: 321), and FGGEVLVYRDWPKPAWVDL (SEQ ID NO: 322), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TGTSNNFVS (SEQ ID NO: 325), EVNKRPS (SEQ ID NO: 227), and SSLVGNWDVI (SEQ ID NO: 212).

(48) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GDSTGRCN (SEQ ID NO: 323), SLSHCRSYYNTDWTY (SEQ ID NO: 324), and FGGEVLVYRDWPKPAWVDL (SEQ ID NO: 322), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TGTSNNFVS (SEQ ID NO: 325), EVNKRPS (SEQ ID NO: 227), and SSLVGNWDVI (SEQ ID NO: 212).

(49) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of ACNSFWG (SEQ ID NO: 326), SLSHCASYWNRGWTYHNPSLKS (SEQ ID NO: 335), and FGGEVLRYTDWPKPAWVDL

(SEQ ID NO: 336), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TGTSNNFVS (SEQ ID NO: 325), (SEQ ID NO: 343), and (SEQ ID NO: 196).

(50) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GDSTAACN (SEQ ID NO: 337), SLSHCASYWNRGWTY (SEQ ID NO: 338), and FGGEVLRDTPKPAWVDL (SEQ ID NO: 336), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of TGTSNNFVS (SEQ ID NO: 325), DVNKRPS (SEQ ID NO: 343), and GSLVGNWDVI (SEQ ID NO: 196).

(51) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of TGHHYWG (SEQ ID NO: 348), HIHYNTAVLHNPALKS (SEQ ID NO: 349), and SGGDILYYIEWQKPHWFYP (SEQ ID NO: 350), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of SGTGSDIGSWNFVS (SEQ ID NO: 357), EVNRRRS (SEQ ID NO: 358), and SSLSGRWDIV (SEQ ID NO: 359).

(52) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GDSINTGH (SEQ ID NO: 351), HIHYNTAVL (SEQ ID NO: 352), and SGGDILYYIEWQKPHWFYP (SEQ ID NO: 350), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of SGTGSDIGSWNFVS (SEQ ID NO: 357), EVNRRRS (SEQ ID NO: 358), and SSLSGRWDIV (SEQ ID NO: 359).

(53) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GGEWGDSYHWG (SEQ ID NO: 364), SIHWRGTTTHYNAPFRG (SEQ ID NO: 365), and HKYHDIVMVVPIAGWFDP (SEQ ID NO: 366), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQSVKNNLA (SEQ ID NO: 372), DTSSRAS (SEQ ID NO: 373), and QQYEEWPRT (SEQ ID NO: 245).

(54) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GGSIRGGEWGDS (SEQ ID NO: 367), SIHWRGTTTH (SEQ ID NO: 237), and HKYHDIVMVVPIAGWFDP (SEQ ID NO: 366), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of RASQSVKNNLA (SEQ ID NO: 372), DTSSRAS (SEQ ID NO: 373), and QQYEEWPRT (SEQ ID NO: 245).

(55) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of NHDVH (SEQ ID NO: 378), WMSHEGDKTGLAQKFQG (SEQ ID NO: 379), and GSKHRLRDYFLYNEYGPNYEEWGDYLATLDV (SEQ ID NO: 380), and a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of KCSHSLQHSTGANYLA (SEQ ID NO: 387), LATHRAS (SEQ ID NO: 388), and MQGLHSPWT (SEQ ID NO: 389).

(56) The invention provides an isolated anti-HIV antibody, wherein said antibody may have a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GNSFSN (SEQ ID NO: 381), WMSHEGDKTG (SEQ ID NO: 382), and GSKHRLRDYFLYNEYGPNYEEWGDYLATLDV (SEQ ID NO: 380), and a

light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of KCSHSLQHSTGANYLA (SEQ ID NO: 387), LATHRAS (SEQ ID NO: 388), and MQGLHSPWT (SEQ ID NO: 389).

(57) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of NYJWT (SEQ ID NO: 406); a V.sub.H_CDR2 region which may comprise the amino acid sequence of YISDRETTTYNPSLNS (SEQ ID NO: 407); a V.sub.H_CDR3 region which may comprise the amino acid sequence of ARRQRIYGVVSFGEFFYYYYMDV (SEQ ID NO: 408); a V.sub.L_CDR1 region which may comprise the amino acid sequence of GRQALGSRAVQ (SEQ ID NO: 415); a V.sub.L_CDR2 region which may comprise the amino acid sequence of NNQDRPS (SEQ ID NO: 151); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of HMWDSRSGFSWS (SEQ ID NO: 416).

(58) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of GRFWS (SEQ ID NO: 421); a V.sub.H_CDR2 region which may comprise the amino acid sequence of YFSDTDRSEYNPSLRS (SEQ ID NO: 422); a V.sub.H_CDR3 region which may comprise the amino acid sequence of AQQGKRIYGIVSFGEFFYYYYMDA (SEQ ID NO: 423); a V.sub.L_CDR1 region which may comprise the amino acid sequence of GERSRGSRAVQ (SEQ ID NO: 430); a V.sub.L_CDR2 region which may comprise the amino acid sequence of NNQDRPA (SEQ ID NO: 179); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of HYWDSRSPISWI (SEQ ID NO: 431).

(59) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of GRFWS (SEQ ID NO: 421); a V.sub.H_CDR2 region which may comprise the amino acid sequence of YFSDTDRSEYNPSLRS (SEQ ID NO: 422); a V.sub.H_CDR3 region which may comprise the amino acid sequence of AQQGKRIYGIVSFGELFYYYYYMDA (SEQ ID NO: 436); a V.sub.L_CDR1 region which may comprise the amino acid sequence of GERSRGSRAVQ (SEQ ID NO: 430); a V.sub.L_CDR2 region which may comprise the amino acid sequence of NNQDRPA (SEQ ID NO: 179); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of HYWDSRSPISWI (SEQ ID NO: 431).

(60) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of TGHHYWG (SEQ ID NO: 348); a V.sub.H_CDR2 region which may comprise the amino acid sequence of HIHYNTAVLHNPALKS (SEQ ID NO: 349); a V.sub.H_CDR3 region which may comprise the amino acid sequence of SGGDILYYNEWQKPHWFYP (SEQ ID NO: 445); a V.sub.L_CDR1 region which may comprise the amino acid sequence of SGTASDIGSWNFVS (SEQ ID NO: 450); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EVNRRRS (SEQ ID NO: 358); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of SSLSGRWDIV (SEQ ID NO: 359).

(61) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of ACDYFWG (SEQ ID NO: 201); a V.sub.H_CDR2 region which may comprise the amino acid sequence of SLSHCAGYYNSGWTYHNPSLKS (SEQ ID NO: 455); a V.sub.H_CDR3 region which may comprise the amino acid sequence of FGGDVLVYHDWPKPAWVDL (SEQ ID NO: 456); a V.sub.L_CDR1 region which may comprise the amino acid sequence of TGNINNVS (SEQ ID NO: 458); a V.sub.L_CDR2 region which may comprise the amino acid sequence of GVNKRPS (SEQ ID NO: 211); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of GSLAGNWDVV (SEQ ID NO: 459).

(62) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of GCDYFWG (SEQ ID NO:

464); a V.sub.H_CDR2 region which may comprise the amino acid sequence of GLSHCAGYYNTGWTYHNPSLKS (SEQ ID NO: 202); a V.sub.H_CDR3 region which may comprise the amino acid sequence of FDGEVLVYNDWPKPAWVDL (SEQ ID NO: 465); a V.sub.L_CDR1 region which may comprise the amino acid sequence of TGTSNNFVS (SEQ ID NO: 325); a V.sub.L_CDR2 region which may comprise the amino acid sequence of GVNKRPS (SEQ ID NO: 211); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of GSLVGNWDVI (SEQ ID NO: 196).

(63) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of KYPMY (SEQ ID NO: 475); a V.sub.H_CDR2 region which may comprise the amino acid sequence of AISGDAWHVVYSNSVQG (SEQ ID NO: 476); a V.sub.H_CDR3 region which may comprise the amino acid sequence of MFQESGPPRLDRWSGRNYYYYSGMDV (SEQ ID NO: 477); a V.sub.L_CDR1 region which may comprise the amino acid sequence of KSSESLRQSNGKTSLY (SEQ ID NO: 484); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EVSNRFS (SEQ ID NO: 485); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of MQSKDFPLT (SEQ ID NO: 486).

(64) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of KYPMY (SEQ ID NO: 475); a V.sub.H_CDR2 region which may comprise the amino acid sequence of AISADAWHVVSYSVQG (SEQ ID NO: 491); a V.sub.H_CDR3 region which may comprise the amino acid sequence of MFQESGPPRFDSWSGRNYYYYSGMDV (SEQ ID NO: 492); a V.sub.L_CDR1 region which may comprise the amino acid sequence of KSSQSLRQSNGKTSLY (SEQ ID NO: 498); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EVSNRFS (SEQ ID NO: 485); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of (MQSKDFPLT (SEQ ID NO: 486).

(65) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of KRHIVIH (SEQ ID NO: 503); a V.sub.H_CDR2 region which may comprise the amino acid sequence of VISSDAIHVDYASSVRG (SEQ ID NO: 504); a V.sub.H_CDR3 region which may comprise the amino acid sequence of DRDGYGPPQIQTWSGRYLHLYSGIDA (SEQ ID NO: 505); a V.sub.L_CDR1 region which may comprise the amino acid sequence of KSSQSLRQSNGKTYLY (SEQ ID NO: 512); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EVSIRFS (SEQ ID NO: 513); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of MQSKDFPLT (SEQ ID NO: 486).

(66) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of KYPMY (SEQ ID NO: 475); a V.sub.H_CDR2 region which may comprise the amino acid sequence of AISADAWHVDYAASVKD (SEQ ID NO: 518); a V.sub.H_CDR3 region which may comprise the amino acid sequence of NIEEFSVPQFDSWSGRSYHYFGMDV (SEQ ID NO: 519); a V.sub.L_CDR1 region which may comprise the amino acid sequence of SSSESLGRGDGRTYLH (SEQ ID NO: 526); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EVSTRFS (SEQ ID NO: 527); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of MQSRDFPIT (SEQ ID NO: 528).

(67) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of EYPMY (SEQ ID NO: 533); a V.sub.H_CDR2 region which may comprise the amino acid sequence of AISADAWHVDYAGSVRG (SEQ ID NO: 534); a V.sub.H_CDR3 region which may comprise the amino acid sequence of DGEEHKVPQLHSWSGRNLYHYTGFDV (SEQ ID NO: 535); a V.sub.L_CDR1 region which may comprise the amino acid sequence of KSSQSVRQSDGKTFLY (SEQ ID NO: 541); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EGSSRFS

(SEQ ID NO: 542); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of LQTKDFPLT (SEQ ID NO: 543).

(68) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of QYPMY (SEQ ID NO: 548); a V.sub.H_CDR2 region which may comprise the amino acid sequence of AISADAWHVDYPGSVRG (SEQ ID NO: 549); a V.sub.H_CDR3 region which may comprise the amino acid sequence of DGEEHKVPQLHSWSGRNLYHYTGFDV (SEQ ID NO: 535); a V.sub.L_CDR1 region which may comprise the amino acid sequence of KSSQTVRQSDGKTFLY (SEQ ID NO: 555); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EGSNRFS (SEQ ID NO: 556); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of LQTKDFPLT (SEQ ID NO: 543).

(69) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of QYPMY (SEQ ID NO: 548); a V.sub.H_CDR2 region which may comprise the amino acid sequence of AISADAWHVDYAGSVRG (SEQ ID NO: 534); a V.sub.H_CDR3 region which may comprise the amino acid sequence of DGEEHEVPQLHSWSGRNLYHYTGVDI (SEQ ID NO: 561); a V.sub.L_CDR1 region which may comprise the amino acid sequence of KSSQSLRQSDGKTFLY (SEQ ID NO: 567); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EASNRFS (SEQ ID NO: 568); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of MQTKDFPLT (SEQ ID NO: 569).

(70) The invention provides an isolated anti-HIV antibody, wherein said antibody may comprise a V.sub.H_CDR1 region which may comprise the amino acid sequence of KYPMY (SEQ ID NO: 475); a V.sub.H_CDR2 region which may comprise the amino acid sequence of AISADAWHVDYPGSVRG (SEQ ID NO: 549); a V.sub.H_CDR3 region which may comprise the amino acid sequence of DGEEHEVPQLHSWSGRNLYHYTGVDV (SEQ ID NO: 574); a V.sub.L_CDR1 region which may comprise the amino acid sequence of KSSQSVRQSDGKTFLY (SEQ ID NO: 541); a V.sub.L_CDR2 region which may comprise the amino acid sequence of EASKRFS (SEQ ID NO: 580); and a V.sub.L_CDR3 region which may comprise the amino acid sequence of MQTKDFPLT (SEQ ID NO: 569).

(71) The invention also provides an isolated anti-HIV antibody, wherein said antibody has a heavy chain with three CDRs including an amino acid sequence selected from the group consisting of the amino acid sequences of KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), EAGGPiWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), SYAFT (SEQ ID NO: 104), MVTPIFGEAKYSQRFEG (SEQ ID NO: 105), DRRAPPIATDNWLDP (SEQ ID NO: 9), SYAFS (SEQ ID NO: 110), MITPVFGETKYAPRFQG (SEQ ID NO: 111), DRRVVPMATDNWLDP (SEQ ID NO: 8), DYLLH (SEQ ID NO: 116), LIDPENGARYAEKFQG (SEQ ID NO: 117), GAVGADSGSWFDP (SEQ ID NO: 10), RQGMH (SEQ ID NO: 123), FIKYDGSEKYHADSVWG (SEQ ID NO: 124), EAGGPDYRNGYNYDFYDGYNYHYMDV (SEQ ID NO: 7), LISDDGMRKYHSNSMWG (SEQ ID NO: 98), DSYWS (SEQ ID NO: 90), YVHKSGDTNYSPLKS (SEQ ID NO: 265), TLHGRRIGIVAFNEWFTYFYMDV (SEQ ID NO: 143), DNYWS (SEQ ID NO: 261), YVHDSGDTNYPNLSLKS (SEQ ID NO: 157), TKHGRRIGVVAFKWFTYFYMDV (SEQ ID NO: 262), DAYWS (SEQ ID NO: 169), YVHHS GDTNYPNLSLKR (SEQ ID NO: 170), ALHGKRIYGIVALGELFTYFYMDV (SEQ ID NO: 171), ACTYFWG (SEQ ID NO: 185), SLSHCQSFWSGSGWTFHNPSLKS (SEQ ID NO: 186), FDGEVLVYNHWPKPAWVDL (SEQ ID NO: 187), ACDYFWG (SEQ ID NO: 201), GLSHCAGYYNTGWTYHNPSLKS (SEQ ID NO: 202), FDGEVLVYHDWPKPAWVDL (SEQ ID NO: 203), TGHYYWG (SEQ ID NO: 217), HIHYTTAVLHNPSLKS (SEQ ID NO: 218), SGGDILYYEWQKPHWFSP (SEQ ID NO: 219), GGEWGD KDYHWG (SEQ ID NO: 233), SIHWRGTTTHYKESLRR (SEQ ID NO: 234), HRHHDVFMLVPIAGWFDV (SEQ ID NO: 235), GTDWGENDFHYG (SEQ ID NO: 250),

SIHWRGRTHYKTSFRS (SEQ ID NO: 251), HKYHDI FRVVPVAGWFD (SEQ ID NO: 252), KYDVH (SEQ ID NO: 277), WMSHEGDKTESAQRFGK (SEQ ID NO: 278), GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), WISHERDKTESAQRFGK (SEQ ID NO: 293), and GSKHRLRDYVLYDDYGLINQQEWNDYLEFLDV (SEQ ID NO: 308), wherein said antibody binds to and neutralizes HIV-1. Optionally, this antibody has a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of (72) TABLE-US-00001 (SEQ ID NO: 97) NGTSSDVGGFDSVS, (SEQ ID NO: 95) DVSHRPS, (SEQ ID NO: 41) SSLTDRSHRI, (SEQ ID NO: 107) RASQTINNYLN, (SEQ ID NO: 108) GASNLQN, (SEQ ID NO: 42) QQSFSPTPT, (SEQ ID NO: 113) RASQTIHTYLN, (SEQ ID NO: 114) GASTLQS, (SEQ ID NO: 43) QQSYSTPT, (SEQ ID NO: 120) SGSKLGDKYVS, (SEQ ID NO: 121) ENDRPS, (SEQ ID NO: 44) QAWETTTTTTFVF, (SEQ ID NO: 126) NGTSNDVGGYESVS, (SEQ ID NO: 127) DVSKRPS, (SEQ ID NO: 45) KSLTSTRRRV, (SEQ ID NO: 92) NGTRSDVGGFDSVS, (SEQ ID NO: 93) NGTSRDVGGFDSVS, (SEQ ID NO: 150) GEKSLGSRAVQ, (SEQ ID NO: 151) NNQDRPS, (SEQ ID NO: 152) HIWDSRVPTKWV, (SEQ ID NO: 162) GEESLGSRSVI, (SEQ ID NO: 163) NNNDRPS, (SEQ ID NO: 164) HIWDSRRPTNWV, (SEQ ID NO: 178) GKESIGSRAVQ, (SEQ ID NO: 179) NNQDRPA, (SEQ ID NO: 180) HIYDARGGTNWV, (SEQ ID NO: 194) NGTATNFVS, (SEQ ID NO: 195) GVDKRPP, (SEQ ID NO: 196) GSLVGNWDVI, (SEQ ID NO: 210) TGTSNRFVS, (SEQ ID NO: 211) GVNKRPS, (SEQ ID NO: 212) SSLVGNWDVI, (SEQ ID NO: 226) NGTSSDIGGWNFVS, (SEQ ID NO: 227) EVNKRPS, (SEQ ID NO: 228) SSLFGRWDVV, (SEQ ID NO: 243) RASQNINKNLA, (SEQ ID NO: 244) ETYSKIA, (SEQ ID NO: 245) QQYEEWPRT, (SEQ ID NO: 259) RASQNVKNNLA, (SEQ ID NO: 260) DASSRAG, (SEQ ID NO: 286) SSTQSLRHSNGANYLA, (SEQ ID NO: 287) LGSQRAS, and (SEQ ID NO: 288) MQGLNRPWT, and (SEQ ID NO: 303) TSTQSLRHSNGANYLA.

(73) The invention provides an isolated anti-HIV antibody, wherein said antibody has a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), SSLTDRSHRI (SEQ ID NO: 41), RASQTINNYLN (SEQ ID NO: 107), GASNLQN (SEQ ID NO: 108), QQSFSPTPT (SEQ ID NO: 42), RASQTIHTYLN (SEQ ID NO: 113), GASTLQS (SEQ ID NO: 114), QQSYSTPT (SEQ ID NO: 43), SGSKLGDKYVS (SEQ ID NO: 120), ENDRPS (SEQ ID NO: 121), QAWETTTTTTFVF (SEQ ID NO: 44), NGTSNDVGGYESVS (SEQ ID NO: 126), DVSKRPS (SEQ ID NO: 127), KSLTSTRRRV (SEQ ID NO: 45), NGTRSDVGGFDSVS (SEQ ID NO: 92), NGTSRDVGGFDSVS (SEQ ID NO: 93), GEKSLGSRAVQ (SEQ ID NO: 150), NNQDRPS (SEQ ID NO: 151), HIWDSRVPTKWV (SEQ ID NO: 152), GEESLGSRSVI (SEQ ID NO: 162), NNNDRPS (SEQ ID NO: 163), HIWDSRRPTNWV (SEQ ID NO: 164), GKESIGSRAVQ (SEQ ID NO: 178), NNQDRPA (SEQ ID NO: 179), HIYDARGGTNWV (SEQ ID NO: 180), NGTATNFVS (SEQ ID NO: 194), GVDKRPP (SEQ ID NO: 195), GSLVGNWDVI (SEQ ID NO: 196), TGTSNRFVS (SEQ ID NO: 210), GVNKRPS (SEQ ID NO: 211), SSLVGNWDVI (SEQ ID NO: 212), NGTSSDIGGWNFVS (SEQ ID NO: 226), EVNKRPS (SEQ ID NO: 227), SSLFGRWDVV (SEQ ID NO: 228), RASQNINKNLA (SEQ ID NO: 243), ETYSKIA (SEQ ID NO: 244), QQYEEWPRT (SEQ ID NO: 245), RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288), and TSTQSLRHSNGANYLA (SEQ ID NO: 303), wherein said antibody binds to and neutralizes HIV-1.

(74) The invention provides an isolated anti-HIV antibody, wherein said antibody has a heavy chain with three CDRs which may comprise an amino acid sequence selected from the group consisting of the amino acid sequences of GFTFHK (SEQ ID NO: 266), LISDDGMRKY (SEQ ID NO: 267), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6), GGTFSS (SEQ ID NO: 268), MVTPIFGEAK (SEQ ID NO: 269), and DRRAPPIATDNWLDP (SEQ ID NO: 9), GGAFSS (SEQ ID NO: 270), MITPVFGETK (SEQ ID NO: 271), DRRVVPMATDNWLDP (SEQ ID NO: 8), GYSFID (SEQ ID NO: 102), LIDPENGAR (SEQ ID NO: 103), GAVGADSGSWFDP (SEQ ID NO: 10), GFDFSR (SEQ ID NO: 118), FIKYDGSEKY (SEQ ID NO: 272), and EAGGPDYRNGYNYDFYDGYNYHYMDV (SEQ ID NO: 7), GASISD (SEQ ID NO: 144), YVHKSGDTN (SEQ ID NO: 145), TLHGRRIRYGIVAFNEWFTYFYMDV (SEQ ID NO: 143), GTLVRD (SEQ ID NO: 263), YVHDSGDTN (SEQ ID NO: 264), TKHGRRIRYGIVAFKEWFTYFYMDV (SEQ ID NO: 262), GASIND (SEQ ID NO: 172), YVHHS GDTN (SEQ ID NO: 173), ALHGKRIYGIVALGELFTYFYMDV (SEQ ID NO: 171), GESTGACT (SEQ ID NO: 188), SLSHCQSFWSGSGWTF (SEQ ID NO: 189), FDGEVLVYNHWPKPAWVDL (SEQ ID NO: 187), GDSTAACD (SEQ ID NO: 204), GLSHCAGYYNTGWTY (SEQ ID NO: 205), FDGEVLVYHDWPKPAWVDL (SEQ ID NO: 203), GESINTGH (SEQ ID NO: 220), HIHYTTAVL (SEQ ID NO: 221), SGGDILYYYEWQKPHWFSP (SEQ ID NO: 219), GDSIRGGEWGDKD (SEQ ID NO: 236), SIHWRGTTTH (SEQ ID NO: 237), HRHHDVFMLVPIAGWFDV (SEQ ID NO: 235), GGSMRGTDWGEND (SEQ ID NO: 253), SIHWRGRTTH (SEQ ID NO: 254), HKYHDIRVVPVAGWFDP (SEQ ID NO: 252), GNTFSK (SEQ ID NO: 280), WMSHEGDKTE (SEQ ID NO: 281), GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), WISHERDKTE (SEQ ID NO: 294), GNTFRK (SEQ ID NO: 309), and GSKHRLRDYVLYDDYGLINQQEWNDYLEFLDV (SEQ ID NO: 308), wherein said antibody binds to and neutralizes HIV-1. Optionally, this antibody has a light chain with three CDRs that include an amino acid sequence selected from the group consisting of the amino acid sequences of (75) TABLE-US-00002 (SEQ ID NO: 97) NGTSSDVGGFDSVS, (SEQ ID NO: 95) DVSHRPS, (SEQ ID NO: 41) SSLTDRSHRI, (SEQ ID NO: 107) RASQTINNYLN, (SEQ ID NO: 108) GASNLQN, (SEQ ID NO: 42) QQSFSPTPT, (SEQ ID NO: 113) RASQTIHTYLN, (SEQ ID NO: 114) GASTLQS, (SEQ ID NO: 43) QQSYSTPT, (SEQ ID NO: 120) SGSKLGDKYVS, (SEQ ID NO: 121) ENDRRPS, (SEQ ID NO: 44) QAWETTTTTTFVE, (SEQ ID NO: 126) NGTSNDVGGYESVS, (SEQ ID NO: 127) DVSKRPS, (SEQ ID NO: 45) KSLTSTRRRV, (SEQ ID NO: 92) NGTRSDVGGFDSVS, (SEQ ID NO: 93) NGTSRDVGGFDSVS, (SEQ ID NO: 150) GEKSLGSRAVQ, (SEQ ID NO: 151) NNQDRPS, (SEQ ID NO: 152) HIWDSRVPTKWV, (SEQ ID NO: 162) GEESLGSRSVI, (SEQ ID NO: 163) NNNDRPS, (SEQ ID NO: 164) HIWDSRRPTNWV, (SEQ ID NO: 178) GKESIGSRAVQ, (SEQ ID NO: 179) NNQDRPA, (SEQ ID NO: 180) HIYDARGGTNWV, (SEQ ID NO: 194) NGTATNFVS, (SEQ ID NO: 195) GVDKRPP, (SEQ ID NO: 196) GSLVGNWDVI, (SEQ ID NO: 210) TGTSNRFVS, (SEQ ID NO: 211) GVNKRPS, (SEQ ID NO: 212) SSLVGNWDVI, (SEQ ID NO: 226) NGTSSDIGGWNFVS, (SEQ ID NO: 227) EVNKRPS, (SEQ ID NO: 228) SSLFGRWDVV, (SEQ ID NO: 243) RASQNINKNLA, (SEQ ID NO: 244) ETYSKIA, (SEQ ID NO: 245) QQYEEWPRT, (SEQ ID NO: 259) RASQNVKNNLA, (SEQ ID NO: 260) DASSRAG, (SEQ ID NO: 286) SSTQSLRHSNGANYLA, (SEQ ID NO: 287) LGSQRAS, and (SEQ ID NO: 288) MQGLNRPWT, and (SEQ ID NO: 303) TSTQSLRHSNGANYLA.

(76) Moreover, the invention provides an isolated anti-HIV antibody or fragment thereof, wherein said antibody includes: (a) a V.sub.H_CDR1 region including the amino acid sequence of SEQ ID

NO: 88, 104, 110, 116, 123, 90, 261, 169, 185, 201, 217, 233, 250, or 277; (b) a V.sub.H_CDR2 region including the amino acid sequence of SEQ ID NO: 98, 89, 105, 111, 117, 124, 265, 157, 170, 186, 202, 218, 234, 251, 278, or 293; and (c) a V.sub.H_CDR3 region including the amino acid sequence of SEQ ID NO: 6, 9, 8, 10, 7, 143, 262, 171, 187, 203, 219, 235, 252, 279, or 308; wherein said antibody binds to and neutralizes HIV-1. This antibody may further include: (a) a V.sub.L_CDR1 region including the amino acid sequence of SEQ ID NO: 93, 92, 97, 94, 107, 113, 120, 126, 150, 162, 178, 194, 210, 226, 243, 259, 286 or 303; (b) a V.sub.L_CDR2 region including the amino acid sequence of SEQ ID NO: 95, 108, 114, 121, 127, 151, 163, 179, 195, 211, 227, 244, 260, or 287; and (c) a V.sub.L_CDR3 region including the amino acid sequence of SEQ ID NO: 41, 42, 43, 44, 45, 152, 164, 180, 196, 212, 228, 245, or 288.

(77) Alternatively, the invention provides an isolated anti-HIV antibody or fragment thereof, wherein said antibody includes: (a) a V.sub.H_CDR1 region including the amino acid sequence of SEQ ID NO: 266, 268, 270, 201, 118, 144, 263, 172, 188, 204, 220, 236, 253, 280 or 309; (b) a V.sub.H_CDR2 region including the amino acid sequence of SEQ ID NO: 267, 269, 271, 103, 272, 145, 264, 173, 189, 205, 221, 237, 254, 281, or 294; and (c) a V.sub.H_CDR3 region including the amino acid sequence of SEQ ID NO: 6, 9, 8, 10, 7, 143, 262, 171, 187, 203, 219, 235, 252, 279, or 308; wherein said antibody binds to and neutralizes HIV-1. This antibody may further include: (a) a V.sub.L_CDR1 region including the amino acid sequence of SEQ ID NO: 93, 92, 97, 94, 107, 113, 120, 126, 150, 162, 178, 194, 210, 226, 243, 259, 286 or 303; (b) a V.sub.L_CDR2 region including the amino acid sequence of SEQ ID NO: 95, 108, 114, 121, 127, 151, 163, 179, 195, 211, 227, 244, 260, or 287; and (c) a V.sub.L_CDR3 region including the amino acid sequence of SEQ ID NO: 41, 42, 43, 44, 45, 152, 164, 180, 196, 212, 228, 245, or 288.

(78) The invention provides an isolated fully human monoclonal anti-HIV antibody including: a) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 31 and a light chain sequence including amino acid sequence SEQ ID NO: 32, or b) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 33 and a light chain sequence including amino acid sequence SEQ ID NO: 34, or c) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 35 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 36, or d) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 37 and a light chain sequence including amino acid sequence SEQ ID NO: 38, or e) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 39 and a light chain sequence including amino acid sequence SEQ ID NO: 40, or f) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 140 and a light chain sequence including amino acid sequence SEQ ID NO: 96, or g) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 48 and a light chain sequence including amino acid sequence SEQ ID NO: 51, or h) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 54 and a light chain sequence including amino acid sequence SEQ ID NO: 57, or i) a heavy chain sequence including the amino acid sequence of SEQ ID NO: 60 and a light chain sequence including amino acid sequence SEQ ID NO: 32, or j) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 79 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 149, or k) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 156 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 161, or l) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 168 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 177, or m) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 184 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 193, or n) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 200 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 209, or o) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 216 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 225, or p) a heavy chain

sequence which may comprise the amino acid sequence of SEQ ID NO: 232 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 242 or q) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 249 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 258 or r) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 276 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 285 or s) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 292 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 285 or t) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 298 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 302 or u) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 307 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 313 or v) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 319 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 330 or w) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 334 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 393 or x) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 347 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 356 or y) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 363 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 397 or z) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 401 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 386, or aa) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 405 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 414, or ab) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 420 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 429, or ac) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 435 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 440, or ad) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 444 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 449, or ae) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 454 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 584, or af) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 463 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 470, or ag) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 474 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 483, or ah) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 490 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 497, or ai) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 502 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 511, or aj) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 517 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 525, or ak) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 532 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 540, or al) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 547 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 554, or am) a heavy chain sequence which may comprise the amino acid sequence of SEQ ID NO: 560 and a light chain sequence which may comprise amino acid sequence SEQ ID NO: 566.

(79) The invention provides a composition including any one of the isolated anti-HIV antibodies described herein.

(80) Optionally, an anti-HIV human monoclonal antibody of the invention is isolated from a B-cell

from an HIV-1-infected human donor. In some embodiments, the antibody is effective in neutralizing a plurality of different clades of HIV. In some embodiments, the antibody is effective in neutralizing a plurality of different strain within the same Glade of HIV-1. In some embodiments, the neutralizing antibody binds to the HIV envelope proteins gp120, or gp41 or envelope protein on HIV-1 pseudovirions or expressed on transfected or infected cell surfaces. In some embodiments, the neutralizing antibody does not bind to recombinant or monomeric envelope proteins gp120, or gp41 or envelope protein on HIV-1 pseudovirions or expressed on transfected or infected cell surfaces but binds to natural trimeric forms of the HIV-1 Env proteins.

(81) The present invention provides human monoclonal antibodies wherein the antibodies are potent, broadly neutralizing antibody (bNAb). In some embodiments, a broadly neutralizing antibody is defined as a bNAb that neutralizes HIV-1 species belonging to two or more different clades. In some embodiments the different clades are selected from the group consisting of clades A, B, C, D, E, AE, AG, G or F. In some embodiments the HIV-1 strains from two or more clades comprise virus from non-B clades.

(82) In some embodiments, a broadly neutralizing antibody is defined as a bNAb that neutralizes at least 60% of the HIV-1 strains listed in Tables 18A-18F. In some embodiments, at least 70%, or at least 80%, or at least 90% of the HIV-1 strains listed in Tables 18A-18F are neutralized.

(83) In some embodiments, a potent, broadly neutralizing antibody is defined as a bNAb that displays a potency of neutralization of at least a plurality of HIV-1 species with an IC₅₀ value of less than 0.2 µg/mL. In some embodiments the potency of neutralization of the HIV-1 species has an IC₅₀ value of less than 0.15 µg/mL, or less than 0.10 µg/mL, or less than 0.05 µg/mL. A potent, broadly neutralizing antibody is also defined as a bNAb that displays a potency of neutralization of at least a plurality of HIV-1 species with an IC₉₀ value of less than 2.0 µg/mL. In some embodiments the potency of neutralization of the HIV-1 species has an IC₉₀ value of less than 1.0 µg/mL, or less than 0.5 µg/mL.

(84) Exemplary monoclonal antibodies that neutralize HIV-1 include 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123 A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993 K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158) described herein. Alternatively, the monoclonal antibody is an antibody that binds to the same epitope as 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123 A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), or 6881_N05 (PGT-158). Specifically, monoclonal antibodies PG9 and

PG16 are broad and potent neutralizing antibodies. The antibodies are respectively referred to herein as HIV antibodies.

(85) The invention provides a number of isolated human monoclonal antibodies, wherein each said monoclonal antibody binds to HIV-1 infected or transfected cells; and binds to HIV-1 virus. A neutralizing antibody having potency in neutralizing HIV-1, or a fragment thereof is provided. In some embodiments a neutralizing antibody of the invention exhibits higher neutralization index and/or a higher affinity for binding to the envelope proteins gp120, or gp41 than anti-HIV mAbs known in the art, such as the mAb b12. (Burton D R et al., Science Vol. 266. no. 5187, pp. 1024-1027). Exemplary monoclonal antibodies 1496_C09 (PG9), 1443_C16 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), and 1495_C14 (PGC14) exhibit binding to the envelope glycoprotein gp120, but not gp41, in an ELISA assay, however gp120 binding does not always correlate with neutralization activity against specific strains of HIV-1. In some embodiments, monoclonal antibodies, for example 1443_C16 (PG16) and 1496_C09 (PG9), display none or weak gp120 binding activity against a particular strain but bind to HIV-1 trimer on transfected or infected cell surface and/or virion and exhibit broad and potent neutralization activity against that strain of HIV-1.

(86) In one aspect the antibody is a monoclonal antibody which may comprise one or more polypeptides selected from the group consisting of 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123 A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158); which may comprise a heavy chain selected from the group consisting of the heavy chain of 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995 E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158); which may comprise a heavy chain which may comprise a CDR selected from the group consisting of the CDRs of the heavy chain of 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995 E20 (PGT-142),

4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158); which may comprise a light chain selected from the group consisting of the light chain of 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158); which may comprise a light chain which may comprise a CDR selected from the group consisting of the CDRs of the light chain of 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158).

(87) The invention relates to an antibody or a fragment thereof, such as Fab, Fab', F(ab')₂ and Fv fragments that binds to an epitope or immunogenic polypeptide capable of binding to an antibody selected from 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158).

(88) The invention also relates to immunogenic polypeptides encoding such epitopes.

(89) Nucleic acid molecules encoding such antibodies, and vectors and cells carrying such nucleic acids are also provided.

(90) The invention relates to a pharmaceutical composition which may comprise at least one antibody or fragment as recited herein, together with a pharmaceutically acceptable carrier.

(91) The invention relates to a method of immunizing, preventing or inhibiting HIV infection or an

HIV-related disease which may comprise the steps of identifying a patient in need of such treatment and administering to said patient a therapeutically effective amount of at least one monoclonal antibody as recited herein.

(92) In a further aspect the HIV antibodies according to the invention are linked to a therapeutic agent or a detectable label.

(93) Additionally, the invention provides methods for stimulating an immune response, treating, preventing or alleviating a symptom of an HIV viral infection by administering an HIV antibody to a subject

(94) In another aspect, the invention provides methods of administering the HIV antibody of the invention to a subject prior to, and/or after exposure to an HIV virus. For example, the HIV antibody of the invention is used to treat or prevent HIV infection. The HIV antibody is administered at a dose sufficient to promote viral clearance or eliminate HIV infected cells.

(95) Also included in the invention is a method for determining the presence of an HIV virus infection in a patient, by contacting a biological sample obtained from the patient with an HIV antibody; detecting an amount of the antibody that binds to the biological sample; and comparing the amount of antibody that binds to the biological sample to a control value.

(96) The invention further provides a diagnostic kit which may comprise an HIV monoclonal antibody.

(97) The invention relates to a broadly neutralizing antibody (bNAb) wherein the antibody neutralizes at least one member of each Glade with a potency greater than that of the bNAbs b12, 2G12, 2F5 and 4E10 respectively.

(98) The invention relates to a broadly neutralizing antibody (bNAb) wherein the antibody binds or does not bind monomeric gp120 or gp41 proteins of the HIV-1 env gene. The antibody binds with higher affinity to trimeric forms of the HIV-1 Env expressed on a cell surface than to the monomeric gp120 or artificially trimerized gp140. In some aspects, the antibody binds with high affinity to uncleaved HIV-1 gp160 trimers on a cell surface.

(99) The invention relates to a broadly neutralizing antibody (bNAb) wherein the antibody binds an epitope within the variable loop of gp120, wherein the epitope may comprise the conserved regions of V2 and V3 loops of gp120, wherein the epitope may comprise N-glycosylation site at residue Asn-160 within the V2 loop of gp120, wherein the antibody binds an epitope presented by a trimeric spike of gp120 on a cell surface, wherein the epitope is not presented when gp120 is artificially trimerized. In some embodiments, the antibody does not neutralize the HIV-1 in the absence of N-glycosylation site at residue Asn-160 within the V2 loop of gp120.

(100) The invention relates to a broadly neutralizing antibody (bNAb) selected from the group consisting of PG16 and PG9. Moreover, the invention relates to a broadly neutralizing antibody (bNAb) selected from the group consisting of 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158).

(101) The invention relates to an antigen or an immunogenic polypeptide, or a vaccine which may comprise such antigen or immunogenic polypeptide, for producing a broadly neutralizing antibody (bNAb) by an immune response, the antigen which may comprise an epitope within the variable

loop of gp120 according to the invention.

(102) The invention relates to method for passive or active immunization of an individual against a plurality of HIV-1 species across one or more clades, the method which may comprise: providing a broadly neutralizing antibody (bNAb) wherein the bNAb neutralizes HIV-1 species belonging to two or more clades, and further wherein the potency of neutralization of at least one member of each Glade is determined by an IC50 value of less than 0.005 µg/mL. In some embodiments, the antibody is selected from the group consisting of PG9 and PG16. Alternatively, or in addition, the antibody is selected from the group consisting of 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_101 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158).

(103) In some embodiments, the antibody is produced by active immunization with an antigen which may comprise an epitope within the variable loop of gp120, wherein the epitope may comprise the conserved regions of V2 and V3 loops of gp120 or, wherein the epitope may comprise an N-glycosylation site at residue Asn-160 within the V2 loop of gp120. In some aspects, the epitope is presented by a trimeric spike of gp120 on a cell surface, and the epitope is not presented when gp120 is monomeric or artificially trimerized.

(104) The invention provides a method for obtaining a broadly neutralizing human monoclonal antibody, the method including: (a) screening memory B cell cultures from a donor PBMC sample for a broad neutralization activity against a plurality of HIV-1 species; (b) cloning a memory B cell that exhibits broad neutralization activity; and then (c) rescuing the monoclonal antibody from the clonal memory B cell culture that exhibits broad neutralization activity. In one embodiment the method, the screening step includes screening polyclonal transfectants for neutralization activity prior to the cloning step of monoclonal transfection. In this embodiment, the screening step is optionally repeated following monoclonal transfection. Finally, in this embodiment, the DNA sequence of the monoclonal antibody is determined as part of the rescue step. Exemplary antibodies that are generated using this embodiment include, but are not limited to, 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_101 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158).

(105) Alternatively, or in addition, the screening step includes determining variable gene sequences from selected B cell wells by deep sequencing, which is optionally followed by sequence alignment to cluster related antibodies. In this alternative embodiment, following the screening step, a

monoclonal transfection is performed as part of the cloning step. Subsequently, in this alternative embodiment, monoclonal transfectants are screened for neutralization activity against an HIV virus from one or more clades. Exemplary antibodies that are generated using this embodiment include, but are not limited to, 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158).

(106) Accordingly, it is an object of the invention to not encompass within the invention any previously known product, process of making the product, or method of using the product such that Applicants reserve the right and hereby disclose a disclaimer of any previously known product, process, or method. It is further noted that the invention does not intend to encompass within the scope of the invention any product, process, or making of the product or method of using the product, which does not meet the written description and enablement requirements of the USPTO (35 U.S.C. § 112, first paragraph) or the EPO (Article 83 of the EPC), such that Applicants reserve the right and hereby disclose a disclaimer of any previously described product, process of making the product, or method of using the product.

(107) It is noted that in this disclosure and particularly in the claims and/or paragraphs, terms such as “comprises”, “comprised”, “comprising” and the like can have the meaning attributed to it in U.S. Patent law; e.g., they can mean “includes”, “included”, “including”, and the like; and that terms such as “consisting essentially of” and “consists essentially of” have the meaning ascribed to them in U.S. Patent law, e.g., they allow for elements not explicitly recited, but exclude elements that are found in the prior art or that affect a basic or novel characteristic of the invention.

(108) These and other embodiments are disclosed or are obvious from and encompassed by, the following Detailed Description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a schematic tree diagram of Clustal W-aligned variable region sequences of heavy chains of the monoclonal antibodies.

(2) FIG. 1B is a schematic tree diagram of Clustal W-aligned variable region sequences of light chains of the monoclonal antibodies.

(3) FIG. 2 is a flow chart of the process for isolation of monoclonal antibodies according to the invention.

(4) FIG. 3A is a schematic diagram that summarizes the screening results for neutralization and HIV-env protein (gp120 and gp41) binding assays from which B cell cultures were selected for antibody rescue and the monoclonal antibodies 1496_C09 (PG9), 1443_C16 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), and 1495_C14 (PGC14) were derived. A neutralization index value Of 1.5 was used as a cut-off.

(5) FIG. 3B is a schematic diagram that summaries the neutralizing activity and HIV-env protein (gp120 and gp41) binding activities of the monoclonal antibodies 1496_C09 (PG9), 1443_C16

(PG16), 1456_P20 (PG20), 1460_G14 (PGG14), and 1495_C14 (PGC14) as determined by ELISA assays among the B cell supernatants using a neutralization index cut-off value of 2.0. The neutralization index was expressed as the ratio of normalized relative luminescence units (RLU) of SIVmac239 to that of test viral strain derived from the same test B cell culture supernatant. The cut-off values used to distinguish neutralizing hits were determined by the neutralization index of a large number of negative control wells containing B cell culture supernatants derived from healthy donors.

(6) FIG. 4 is a series of graphs depicting the neutralization activity of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to additional pseudoviruses not included in Tables 17A and 17B.

(7) FIG. 5 is a graph depicting the dose response curves of 1456_P20 (PG20), 1495_C14 (PGC14) and 1460_G14 (PGG14) binding to recombinant gp120 in ELISA as compared to control anti-gp120 (b12). Data is presented as average OD values of triplicate ELISA wells obtained on the same plate.

(8) FIG. 6 is a series of graphs depicting the results from ELISA binding assays of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to HIV-1 YU2 gp140, JR-CSFgp120, membrane-proximal external regions (MPER) peptide of gp41 and V3 polypeptide.

(9) FIG. 7 is a graph depicting the results of a binding assay using monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to HIV-1 YU2 gp160 expressed on the cell surface in the presence and absence of soluble CD4 (sCD4).

(10) FIG. 8 is a graph depicting the results of a binding assay using monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to HIV-1 gp160 transfected cells.

(11) FIG. 9 is a series of graphs depicting the results of a capture assay. The data describe capturing of entry-competent JRCSF pseudovirus by neutralizing monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) in a dose-dependent manner.

(12) FIG. 10A is a graph depicting the results of a competitive binding assay using monoclonal antibodies sCD4, PG16 and PG9, wherein the claimed antibodies compete for the binding of monoclonal antibody 1443_C16 (PG16) to pseudovirus but control antibodies b12, 2G12, 2F5 and 4E10 do not competitively bind to the pseudovirus.

(13) FIG. 10B is a graph depicting the results of a competitive binding assay using monoclonal antibodies sCD4, PG16 and PG9, wherein the claimed antibodies compete for the binding of monoclonal antibody 1496_C09 (PG9) to pseudovirus but control antibodies b12, 2G12, 2F5 and 4E10 do not competitively bind to the pseudovirus.

(14) FIG. 11A is a series of graphs depicting the results of a binding assay using PG9 and PG16. The data show that PG9 and PG16 bind to monomeric gp120 and artificially trimerized gp140 constructs as determined by ELISA. IgG b12 was used as a control for ELISA assays.

(15) FIG. 11B is a series of graphs depicting the results of a binding assay using PG9 and PG16. The data show that PG9 and PG16 bind to Env expressed on the surface of 293T cells as determined by flow cytometry. The bNAbs b12 and the non-neutralizing antibody b6 are included in the cell surface binding assays to show the expected percentages of cleaved and uncleaved Env expressed on the cell surface.

(16) FIG. 12 is a series of graphs depicting the results of a binding assay using PG9 and PG16 and cleavage-defective HIV-1YU2 trimers. PG9 and PG16 bind with high affinity to cleavage-defective HIV-1YU2 trimers as determined by flow cytometry. Binding curves were generated by plotting the MFI of antigen binding as a function of antibody concentration.

(17) FIG. 13A-13E is a series of graphs depicting the mapping the PG9 and PG16 epitopes. Competitor antibody is indicated at the top of each graph. 2G12 is included to control for cell surface Env expression. A: PG9 and PG16 compete with each other for cell surface Env binding and neither antibody competes with the CD4bs antibody b12 for Env binding. B: Ligation of cell surface Env with sCD4 diminishes binding of PG9 and PG16. 2G12 is included to control for CD4-

induced shedding of gp120. C: sCD4 inhibits binding of PG9 to artificially trimerized gp140YU-2 as determined by ELISA. D: PG9 competes with 10/76b (anti-V2), F425/b4e8 (anti-V3) and X5 (CD4i) for gp120 binding in competition ELISA assays. E: PG9 and PG16 fail to bind variable loop deleted HIV-1JR-CSF variants expressed on the surface of 293T cells.

(18) FIG. **14** is a series of graphs depicting the results of competition ELISA assays using the monoclonal antibody PG9.

(19) FIG. **15** is a graph depicting monoclonal antibody binding, PG9 or PG16, to HIV-1JR-FLACT E168K Env expressed on the surface of 293T cells as determined by flow cytometry.

(20) FIG. **16** is a graph depicting monoclonal antibody PG9 binding to deglycosylated gp120.

(21) FIG. **17** is a series of graphs depicting the neutralization activity of PG9 and PG16 against HIV-1SF162 and HIV-1SF162 K160N, which was determined using a single-round replication luciferase reporter assay of pseudotyped virus.

(22) FIG. **18** is a series of graphs depicting the binding of PG9 and PG16 to mixed trimers. Alanine substitutions at positions 160 and 299 were introduced into HIV-1YU2 Env to abolish binding of PG9 and PG16. An alanine substitution at position 295 was also introduced into the same construct to abrogate binding of 2G12. Co-transfection of 293T cells with WT and mutant plasmids in a 1:2 ratio resulted in the expression of 29% mutant homotrimers, 44% heterotrimers with two mutant subunits, 23% heterotrimers with one mutant subunit, and 4% wild-type homotrimers.

(23) FIG. **19** is a series of graphical depictions of the number of nucleotide or amino acid differences in the heavy chain sequences of sister clones of 1443_C16 (PG16) among each other. Note that the single nucleotide difference of 1408_I08 translates into an identical protein sequence of 1443_C16. The nucleotide sequence of the 1408_I08 light chain is identical to the nucleotide sequence of the light chain of 1443_C16.

(24) FIG. **20A** is a tree diagram illustrating the correlation of the heavy chain of 1443_C16 sister clones to the heavy chain of 1496_C09 at the nucleotide level.

(25) FIG. **20B** is a tree diagram illustrating the correlation of the light chain of 1443_C16 sister clones to the light chain of 1496_C09 at the nucleotide level.

(26) FIG. **21A** is a tree diagram illustrating the correlation of the heavy chain of 1443_C16 sister clones to the heavy chain of 1496_C09 at the protein level.

(27) FIG. **21B** is a tree diagram illustrating the correlation of the light chain of 1443_C16 sister clones to the light chain of 1496_C09 at the protein level.

(28) FIG. **22** is a Venn diagram depicting the viruses used in primary HIV-neutralization screening (JR-CSF, MGRM-C-26, 92BR020, 94UG103, 93IN905, 92TH021) and the number of neutralizing antibodies identified using these viruses alone, or in the demonstrated combinations. The results of screening antibodies isolated from B-cell cultures established from four human donors (#517, 039, 196, and 584) are shown.

(29) FIG. **23** is a tree diagram illustrating the relationships between the heavy chain variable gene sequences of antibodies PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, and PGT-136. Scale bar=0.03. A value of zero demonstrates that an identical antibody was produced by two separate B-cell clones. Antibodies are less closely-related as the provided values increase.

(30) FIG. **24** is a tree diagram illustrating the relationships between the light chain variable gene sequences of antibodies PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, and PGT-136. Scale bar=0.04. A value of zero demonstrates that an identical antibody was produced by two separate B-cell clones. Antibodies are less closely-related as the provided values increase.

(31) FIG. **25** is a tree diagram illustrating the relationships between the heavy chain variable gene sequences of antibodies PGT-141, PGT-142, and PGT-143. Scale bar=0.04. A value of zero demonstrates that an identical antibody was produced by two separate B-cell clones. Antibodies are less closely-related as the provided values increase.

(32) FIG. **26** is a tree diagram illustrating the relationships between the light chain variable gene sequences of antibodies PGT-141, PGT-142, and PGT-143. Scale bar=0.04. A value of zero demonstrates that an identical antibody was produced by two separate B-cell clones. Antibodies are less closely-related as the provided values increase.

(33) FIG. **27** is a pie chart showing that a limited number of antibody specificities mediate broad and potent serum neutralization in elite neutralizers (Walker L M, et al. PLOS Pathogen, 2010).

(34) FIG. **28** is a schematic diagram depicting the I-Star™ Human bNAb (broadly Neutralizing Antibody) Discovery Platform developed by Theraclone Sciences.

(35) FIG. **29** is a schematic diagram depicting the method of bNMab (broadly Neutralizing Monoclonal Antibody) isolation from IgG-positive (IgG.sup.+) Memory B Cells developed by Theraclone Sciences.

(36) FIG. **30** is a computer-generated three-dimensional depiction of trimer-specific PG9 and PG16 antibodies in close proximity of conserved regions of V2 and V3, where they bind.

(37) FIG. **31** is a computer-generated three-dimensional depiction of highly conserved epitopes on the HIV spike, including the V1/V2 and V3 loops to which PG9 and PG16 bind and the epitopes to which PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, PGT-136, PGT-141, PGT-142, PGT-143, and PGT-144.

(38) FIG. **32** is a graph depicting the potency of monoclonal anti-HIV antibodies PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-127, PGT-128, PGT-130, PGT-131, PGT-135, PGT-136, PGT-137, PGT-141, PGT-142, PGT-143, PGT-144, PGT-145, and PG9, expressed as the half-maximal inhibitory concentration, or IC.sub.50 (μg/ml). The bar for each antibody represents the median IC.sub.50 value.

(39) FIG. **33A-33D** is a series of graphs depicting the newly identified PGT antibodies redefine broad and potent neutralisation of HIV. A, Key MAbs fully recapitulate serum neutralization by the corresponding donor serum. Serum breadth was correlated with the breadth of the broadest MAb for each donor (% viruses neutralized at NT.sub.50>100 or IC.sub.50<50 μg/ml, respectively). Of note, MAbs isolated from donor 39 could not completely recapitulate the serum neutralization breadth. B-D. Certain antibodies or antibody combinations are able to cover a broad range of HIV isolates at low, vaccine achievable, concentrations. B, Cumulative frequency distribution of IC.sub.50 values of broadly neutralizing MAbs tested against a 162-virus panel. The y-axis shows the cumulative frequency of IC.sub.50 values up to the concentration shown on the x-axis and can therefore also be interpreted as the breadth at a specific IC.sub.50 cut-off. C-D, Percent viruses covered by single MAbs (solid lines) or by at least one of the MAbs in dual combinations (dashed black lines) dependent on individual concentrations. The grey area in both panels is the coverage of 26 MAbs tested on the 162-virus panel (PGT121-123, PGT125-128, PGT130-131, PGT135-137, PGT141-145, PG9, PG16, PGC14, VRC01, PGV04, b12, 2G12, 4E10, 2F5) and depicts the theoretical maximal achievable coverage known to date.

(40) FIG. **34A** is a table depicting the competition of PGT MAbs with sCD4 (soluble CD4), b12 (anti-CD4bs), 2G12 (anti-glycan), F425/b4e8 (anti-V3), X5 (CD4i), PG9 (anti-V1/V2 and V3, quaternary) and each other. Competition assays were performed by ELISA using gp120.sub.Bal or gp120.sub.JR-FL, except for the PG9 competition assay, which was performed on the surface of JR-FL.sub.E168K or JR-CSF transfected cells. Boxes are color coded as follows: red, 75-100% competition; orange, 50-75% competition; yellow, 25-50% competition; gray, <25% competition. Experiments were performed in duplicate, and data represent an average of at least two independent experiments.

(41) FIG. **34B-34D** is a series of graphs depicting the epitope mapping of PGT antibodies. b, Glycan microarray analysis (Consortium for Functional Glycomics, CFG, v 5.0) reveals that PGT MAbs 125, 126, 127, 128, and 130 contact Man.sub.8 (313), Man.sub.8GlcNAc.sub.2 (193), Man.sub.9 (314) and Man.sub.9GlcNAc.sub.2 (194) glycans directly. Only glycans structures with RFU (relative fluorescent units)>3000 are shown. PGT-131 showed no detectable binding to the

CFG glycan array but bound to Man.sub.9-oligodendrons.sup.30 (data not shown). Error bars represent standard deviation. c, d, Binding of PGT MAb 125, 126, 127, 128 and 130 to gp120 is competed by Man.sub.9 oligodendrons but not Man.sub.4 oligodendrons. Binding of 131 to immobilized gp120 was too low to measure any competition.

(42) FIG. 35A-35D is a series of graphs depicting the lack of polyreactivity of PGT monoclonal antibodies (mAbs) in ELISA assay. PGT mAbs were tested for ELISA reactivity against a panel of antigens. The bNAbs b12 and 4E10 were also included for comparison. d.s, double-stranded; s.s, single-stranded.

(43) FIG. 36A-36D is a series of graphs depicting the results of an analysis of neutralization activity by virus clades. Cumulative frequency distribution of IC₅₀ values of broadly neutralizing Mabs tested against a 162 virus panel separated by clades A, B, C, D, F, G, AE and AG. VRC01 was tested on a different virus panel (n=190, ref 6).

(44) FIG. 37A-37D is a series of graphs showing that MAb neutralization correlates strongly with serum neutralization. Correlation of IC₅₀s of the MAb and serum NT₅₀s of the corresponding donors 17 (a), 36 (b), 39 (c) and 84 (d) is shown. Spearman correlation was used for statistical analyses. Only viruses neutralized by either the MAb (IC₅₀<50 µg/ml) or the serum (NT₅₀>100) were included.

(45) FIG. 38A-38C is a series of graphs showing that PGT 141-145 bind preferentially to cell-surface expressed trimers. A) Binding of PGTs 141-145 to monomeric gp120 and artificially trimerized gp140 constructs as determined by ELISA. The bNAbs b12 and PG9 are included for comparison. OD, optical density (absorbance at 450 nm). B) Binding of PGTs 141-145 to Env expressed on the surface of 293T cells as determined by flow cytometry. The bNAbs 2G12 and PG9 are included for comparison.

(46) FIG. 39A-39B is a series of graphs showing that PGT mAbs 141-145 bind to epitopes overlapping those of PG9 and PG16. A) PGTs 141-145 are sensitive to the N160K mutation and PGTs 141-144 fail to neutralize pseudoviruses produced in the presence of kifunensine. The bNAb 2G12 was also included for comparison. B) PG9 competes with PGTs 141-145 for binding to cell-surface trimers. The bNAb 2G12 was included as a negative control.

(47) FIG. 40A-40B is a series of graphs showing PGTs 121, 122 and 123 in competition with oligodendrons. Unlike PGTs 125, 126, 127, 128 and 130, the binding of PGTs 121, 122 and 123 to gp120 could not be competed by A) Man.sub.4 or B) Man.sub.9 dendrons.

(48) FIG. 41 is a series of graphs showing the neutralization activity of Fab fragments. Fab fragments of PGTs-125, 126, 127, 128, 130 and 131 were generated by Lys-C digestion and the neutralizing activity tested against HIV-1 JR-CSF using a single round of replication pseudovirus assay.

(49) FIG. 42A-42B is a series of graphs showing that the combination of two or three antibody specificities is sufficient to cover a broad range of HIV isolates at vaccine achievable concentration. Cumulative frequency distribution of IC₅₀ values of double (a) and triple (b) combinations of neutralization activities (overall lowest IC₅₀ against each isolate). The grey area depicts the theoretical maximal achievable neutralization activity known to date.

(50) FIG. 43A-43C is a series of graphs showing that combinations of two or three antibody specificities are sufficient to cover a broad range of HIV isolates at vaccine achievable concentrations. A-C Cumulative frequency distribution of IC₅₀ values of single MAb (solid lines) and combined neutralisation activity (overall lowest IC₅₀ against each isolate) of two or three MABs (dashed lines). The grey area is the combined neutralisation activity of 25 MABs tested on the 162-virus panel (b12, 2G12, 4E10, 2F5, PG9, PG16, PGC14, PGV04, PGTs 121-123, PGTs 125-128, PGTs 130-131, PGTs 135-137, PGTs 141-145) and depicts the theoretical maximal achievable neutralisation activity known to date. VRC01 and PGV04 in panel c are measured on a different virus panel (n=97).

(51) FIG. 44A-44M is a series of graphs depicting the percent of viruses covered by single MABs

(solid lines) or by at least one of the MABs in dual combinations (dashed black lines) dependent on individual concentrations. The grey area in all panels is the coverage of 26 MABs tested on the 162-virus panel (PGT121-123, PGT125-128, PGT130-131, PGT135-137, PGT141-145, PG9, PG16, PGC14, VRC01, PGV04, b12, 2G12, 4E10, 2F5) and depicts the theoretical maximal achievable coverage known to date.

(52) FIG. **45** is an alignment of heavy chain coding sequences of the variable domain of 1443_C16 sister clones to 1443_C16 and 1496_C09 (SEQ ID NO: 652-653).

(53) FIG. **46** is an alignment of light chain coding sequences of the variable domain of 1443_C16 sister clones to 1443_C16 and 1496_C09 (SEQ ID NO: 654-655).

(54) FIG. **47** is an alignment of heavy chain protein sequences of the variable domain of 1443_C16 sister clones to 1443_C16 and 1496_C09 (SEQ ID NO: 656-658).

(55) FIG. **48** is an alignment of light chain protein sequences of the variable domain of 1443_C16 sister clones to 1443_C16 and 1496_C09 (SEQ ID NO: 659-660).

(56) FIG. **49** is a table showing antibody sequence characteristics of MABs PGT121-123, PGT125-128, PGT130-131, PGT135-137, PGT141-145 (SEQ ID NO: 143, 262, 171, 187, 203, 661, 336, 219, 350, 235, 252, 366, 279, 308, 380, 152, 164, 180, 196, 212, 228, 359, 245, 288, 389).

(57) FIG. **50** is a table showing heavy chain variable gene alignment for PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT 135, and PGT-136 (SEQ ID NO: 662-673 from top to bottom).

(58) FIG. **51** is a table showing heavy chain variable protein alignment for PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT 135, and PGT-136 (SEQ ID NO: 674-684 from top to bottom).

(59) FIG. **52** is a table showing light chain variable gene alignment for PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, and PGT-136 (SEQ ID NO: 685-697 from top to bottom).

(60) FIG. **53** is a table showing light chain variable protein alignment for PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, and PGT-136 (SEQ ID NO: 698-706 from top to bottom).

(61) FIG. **54** is a table showing heavy chain variable gene alignment for PGT-141, PGT-142, PGT-143, and PGT-144 (SEQ ID NO: 707-712 from top to bottom).

(62) FIG. **55** is a table showing heavy chain variable protein alignment for PGT-141, PGT-142, PGT-143, and PGT-144 (SEQ ID NO: 717-720 from top to bottom).

(63) FIG. **56** is a table showing light chain variable gene alignment for PGT-141, PGT 142, PGT-143, and PGT-144 (SEQ ID NO: 713-716 from top to bottom).

(64) FIG. **57** is a table showing light chain variable protein alignment for PGT-141, PGT 142, PGT-143, and PGT-144 (SEQ ID NO: 721-724 from top to bottom).

DETAILED DESCRIPTION OF THE INVENTION

(65) In the sera of human immunodeficiency virus type 1 (HIV-1) infected patients, anti-virus antibodies can be detected over a certain period after infection without any clinical manifestations of the acquired immunodeficiency syndrome (AIDS). At this state of active immune response, high numbers of antigen-specific B-cells are expected in the circulation. These B-cells are used as fusion partners for the generation of human monoclonal anti-HIV antibodies. One major drawback to finding a vaccine composition suitable for more reliable prevention of human individuals from HIV-1 infection and/or for more successful therapeutic treatment of infected patients is the ability of the HIV-1 virus to escape antibody capture by genetic variation, which very often renders the remarkable efforts of the researchers almost useless. Such escape mutants may be characterized by a change of only one or several of the amino acids within one of the targeted antigenic determinants and may occur, for example, as a result of spontaneous or induced mutation. In addition to genetic variation, certain other properties of the HIV-1 envelope glycoprotein makes it difficult to elicit neutralizing antibodies making generation of undesirable non-neutralizing

antibodies a major concern (see, Phogat S K and Wyatt R T, *Curr Pharm Design* 2007; 13(2):213-227).

(66) HIV-1 is among the most genetically diverse viral pathogens. Of the three main branches of the HIV-1 phylogenetic tree, the M (main), N (new), and O (outlier) groups, group M viruses are the most widespread, accounting for over 99% of global infections. This group is presently divided into nine distinct genetic subtypes, or clades (A through K), based on full-length sequences. Env is the most variable HIV-1 gene, with up to 35% sequence diversity between clades, 20% sequence diversity within clades, and up to 10% sequence diversity in a single infected person (Shankarappa, R. et al. 1999. *J. Virol.* 73:10489-10502). Clade B is dominant in Europe, the Americas, and Australia. Clade C is common in southern Africa, China, and India and presently infects more people worldwide than any other Glade (McCutchan, F E. 2000. Understanding the genetic diversity of HIV-1. *AIDS* 14(Suppl. 3): S31-S44). Clades A and D are prominent in central and eastern Africa.

(67) Neutralizing antibodies (NAbs) against viral envelope proteins (Env) provide adaptive immune defense against human immunodeficiency virus type 1 (HIV-1) exposure by blocking the infection of susceptible cells (Kwong P D et al., 2002. *Nature* 420: 678-682). The efficacy of vaccines against several viruses has been attributed to their ability to elicit NAbs. However, despite enormous efforts, there has been limited progress toward an effective immunogen for HIV-1. (Burton, D. R. 2002. *Nat. Rev. Immunol.* 2:706-713).

(68) HIV-1 has evolved with an extensive array of strategies to evade antibody-mediated neutralization. (Barouch, D. H. *Nature* 455, 613-619 (2008); Kwong, P. D. & Wilson, I. A. *Nat Immunol* 10, 573-578 (2009); Karlsson Hedestam, G. B., et al. *Nat Rev Microbiol* 6, 143-155 (2008)). However, broadly neutralizing antibodies (bNAbs) develop over time in a proportion of HIV-1 infected individuals. (Leonidas Stamatatos, L. M., Dennis R Burton, and John Mascola. *Nature Medicine* (E-Pub: Jun. 14, 2009); PMID: 19525964.) A handful of broadly neutralizing monoclonal antibodies have been isolated from Glade B infected donors. (Burton, D. R., et al. *Science* 266, 1024-1027 (1994); Trkola, A., et al. *J Virol* 69, 6609-6617 (1995); Stiegler, G., et al. *AIDS Res Hum Retroviruses* 17, 1757-1765 (2001)). These antibodies tend to display less breadth and potency against non-Glade B viruses, and they recognize epitopes on the virus that have so far failed to elicit broadly neutralizing responses when incorporated into a diverse range of immunogens. (Phogat, S. & Wyatt, R. *Curr Pharm Design* 13, 213-227 (2007); Montero, M., van Houten, N. E., Wang, X. & Scott, J. K. *Microbiol Mol Biol Rev* 72, 54-84, table of contents (2008); Scanlan, C. N., Offer, J., Zitzmann, N. & Dwek, R. A. *Nature* 446, 1038-1045 (2007)). Despite the enormous diversity of the human immunodeficiency virus (HIV), all HIV viruses known to date interact with the same cellular receptors (CD4 and/or a co-receptor, CCR5 or CXCR4). Most neutralizing antibodies bind to functional regions involved in receptor interactions and cell membrane fusion. However, the vast majority of neutralizing antibodies isolated to date do not recognize more than one Glade, therefore exhibiting limited protective efficacy in vitro or in vivo. (See Binley J M et al., 2004. *J. Virol.* 78(23):13232-13252). The rare broadly neutralizing human monoclonal antibodies (mAbs) that have been isolated from HIV+Glade B-infected human donors bind to products of the env gene of HIV-1, gp120 and the transmembrane protein gp41. (Parren, P W et al. 1999. *AIDS* 13:S137-S162). However, a well-known characteristic of the HIV-1 envelope glycoprotein is its extreme variability. It has been recognized that even relatively conserved epitopes on HIV-1, such as the CD4 binding site, show some variability between different isolates (Poignard, P., et al., *Ann. Rev. Immunol.* (2001) 19:253-274). Even an antibody targeted to one of these conserved sites can be expected to suffer from a reduced breadth of reactivity across multiple different isolates.

(69) The few cross-clade reactive monoclonal antibodies known to date have been isolated by processes involving generation of panels of specific viral antibodies from peripheral blood lymphocytes (PBLs) of HIV-infected individuals, either via phage display, or via conventional

immortalization techniques such as hybridoma or Epstein Barr virus transformation, electrofusion and the like. These are selected based on reactivity in vitro to HIV-1 proteins, followed by testing for HIV neutralization activity.

(70) An antibody phage surface expression system was used to isolate the cross-clade neutralizing Fab (fragment, antigen binding) b12 occurring in a combinatorial library. The Fab b12 was screened by panning for envelope glycoprotein gp120 binding activity and neutralizing activity against the HIV-1 (HXBc2) isolate was observed. (Roben P et al., J. Virol. 68(8): 4821-4828(1994); Barbas C F et al., Proc. Natl. Acad. Sci. USA Vol. 89, pp. 9339-9343, (1992); Burton D P et al., Proc. Natl. Acad. Sci. USA Vol. 88, pp. 10134-10137 (1991)).

(71) Human B cell immortalization was used to isolate the cross-clade neutralizing monoclonal antibodies 2G12, 2F5, and 4E10 from HIV-infected individuals. The monoclonal antibody 2G12 binds to a glycotope on the gp120 surface glycoprotein of HIV-1 and had been shown to display broad neutralizing patterns. (Trkola A., et al., J. Virol. 70(2):1100-1108 (1996), Buchacher, A., et al., 1994. AIDS Res. Hum. Retroviruses 10:359-369). The monoclonal antibody 2F5 which had been shown to bind a sequence within the external domain of the gp41 envelope glycoprotein of HIV-1 was found to have broad neutralization properties. (Conley A J Proc. Natl. Acad. Sci. USA Vol. 91, pp. 3348-3352 (1994); Muster T et al., J. Virol. 67(11):6642-6647 (1993); Buchacher A et al., 1992, Vaccines 92:191-195). The monoclonal antibody 4E10, which binds to a novel epitope C terminal of the ELDKWA sequence in gp41 recognized by 2F5, has also been found to have potent cross-clade neutralization activity. (Buchacher, A., et al., 1994. AIDS Res. Hum. Retroviruses 10:359-369; Stiegler, G., et al., 2001. AIDS Res. Hum. Retroviruses 17(18): 1757-1765)).

(72) Other studies on antibody neutralization of HIV-1 (Nara, P. L., et al. (1991) FASEB J. 5:2437-2455.) focused on a single linear epitope in the third hypervariable region of the viral envelope glycoprotein gp120 known as the V3 loop. Antibodies to this loop are suggested to neutralize by inhibiting fusion of viral and cell membranes. However there is sequence variability within the loop and neutralizing antibodies are sensitive to sequence variations outside the loop (Albert J. et al., (1990) AIDS 4, 107-112). Hence anti-V3 loop antibodies are often strain-specific and mutations in the loop in vivo may provide a mechanism for viral escape from antibody neutralization. There is some indication that not all neutralizing antibodies act by blocking the attachment of virus, since a number of mouse monoclonal antibodies inhibiting CD4 binding to gp120 are either non-neutralizing (Lasky L A, et al., (1987) Cell 50:975-985.) or only weakly neutralizing (Sun N., et al., (1989) J. Virol. 63, 3579-3585).

(73) It is widely accepted that such a vaccine will require both T-cell mediated immunity as well as the elicitation of a broadly neutralizing antibody (bNAb) response. (Barouch, D. H. Nature 455, 613-619 (2008); Walker, B. D. & Burton, D. R. Science 320, 760-764 (2008); Johnston, M. I. & Fauci, A. S. N Engl J Med 356, 2073-2081 (2007)). All of the known bNAbs provide protection in the best available primate models (Veazey, R. S., et al. Nat Med 9, 343-346 (2003); Hessel, A. J., et al. PLoS Pathog 5, e1000433 (2009); Parren, P. W., et al. J Virol 75, 8340-8347 (2001); Mascola, J. R. Vaccine 20, 1922-1925 (2002); Mascola, J. R., et al. Nat Med 6, 207-210 (2000); Mascola, J. R., et al. J Virol 73, 4009-4018 (1999)). Therefore, broadly neutralizing antibodies (bNAbs) are considered to be the types of antibodies that should be elicited by a vaccine. Unfortunately, existing immunogens, often designed based on these bNAbs, have failed to elicit NAb responses of the required breadth and potency. Therefore, it is of high priority to identify new bNAbs that bind to epitopes that may be more amenable to incorporation into immunogens for elicitation of NAb responses.

(74) The present invention provides a novel method for isolating novel broad and potent neutralizing monoclonal antibodies against HIV. The method involves selection of a PBMC donor with high neutralization titer of antibodies in the plasma. B cells are screened for neutralization activity prior to rescue of antibodies. Novel broadly neutralizing antibodies are obtained by emphasizing neutralization as the initial screen.

(75) The invention relates to potent, broadly neutralizing antibody (bNAb) wherein the antibody neutralizes HIV-1 species belonging to two or more clades, and further wherein the potency of neutralization of at least one member of each Glade is determined by an IC50 value of less than 0.2 µg/mL. In some aspects, the clades are selected from Clade A, Clade B, Clade C, Clade D and Clade A E. In some aspects, the HIV-1 belonging two or more clades are non-Clade B viruses. In some aspects, the broadly neutralizing antibody neutralizes at least 60% of the HIV-1 strains listed in Tables 18A-18F. In some embodiments, at least 70%, or at least 80%, or at least 90% of the HIV-1 strains listed in Tables 18A-18F are neutralized.

(76) The invention relates to potent, broadly neutralizing antibody (bNAb) wherein the antibody neutralizes HIV-1 species with a potency of neutralization of at least a plurality of HIV-1 species with an IC50 value of less than 0.2 µg/mL. In some embodiments the potency of neutralization of the HIV-1 species has an IC50 value of less than 0.15 µg/mL, or less than 0.10 µg/mL, or less than 0.05 µg/mL. In some aspects, a potent, broadly neutralizing antibody is defined as a bNAb that displays a potency of neutralization of at least a plurality of HIV-1 species with an IC90 value of less than 2.0 µg/mL. In some embodiments the potency of neutralization of the HIV-1 species has an IC90 value of less than 1.0 µg/mL, or less than 0.5 µg/mL.

(77) An exemplary method is illustrated in the schematic shown in FIG. 4. Peripheral Blood Mononuclear Cells (PBMCs) were obtained from an HIV-infected donor selected for HIV-1 neutralizing activity in the plasma. Memory B cells were isolated and B cell culture supernatants were subjected to a primary screen of neutralization assay in a high throughput format. Optionally, HIV antigen binding assays using ELISA or like methods were also used as a screen. B cell lysates corresponding to supernatants exhibiting neutralizing activity were selected for rescue of monoclonal antibodies by standard recombinant methods.

(78) In one embodiment, the recombinant rescue of the monoclonal antibodies involves use of a B cell culture system as described in Weitzcamp J-H et al., J. Immunol. 171:4680-4688 (2003). Any other method for rescue of single B cells clones known in the art also may be employed such as EBV immortalization of B cells (Traggiai E., et al., Nat. Med. 10(8):871-875 (2004)), electrofusion (Buchacher, A., et al., 1994. AIDS Res. Hum. Retroviruses 10:359-369), and B cell hybridoma (Karpas A. et al., Proc. Natl. Acad. Sci. USA 98:1799-1804 (2001)).

(79) In some embodiments, monoclonal antibodies were rescued from the B cell cultures using variable chain gene-specific RT-PCR, and transfectant with combinations of H and L chain clones were screened again for neutralization and HIV antigen binding activities. mAbs with neutralization properties were selected for further characterization.

(80) A novel high-throughput strategy was used to screen IgG-containing culture screening supernatants from approximately 30,000 activated memory B cells from a Glade A infected donor for recombinant, monomeric gp120JR-CSF and gp41HxB2 (Env) binding as well as neutralization activity against HIV-1JR-CSF and HIV-1SF162 (See Table 1).

(81) TABLE-US-00003 TABLE 1 Memory B cell Screening. Total number of wells screened 23,328 Number of sIgG.sup.+ memory B cells screened 30,300 gp120 ELISA hits 411 (1.36%) gp41 ELISA hits 167 (0.55%) SF162 neutralization hits 401 (1.32%) JR-CSF neutralization hits 401 (1.32%)

(82) Unexpectedly, a large proportion of the B cell supernatants that neutralized HIV-1JR-CSF did not bind monomeric gp120JR-CSF or gp41HxB2, and there were only a limited number of cultures that neutralized both viruses (FIG. 3B). Antibody genes were rescued from five B cell cultures selected for differing functional profiles; one bound to gp120 and only neutralized HIV-1SF162, two bound to gp120 and weakly neutralized both viruses, and two potentially neutralized HIV-1JR-CSF, failed to neutralize HIV-1SF162, and did not bind to monomeric gp120 or gp41. Five antibodies identified according to these methods are disclosed herein. The antibodies were isolated from a human sample obtained through International AIDS Vaccine Initiative's (IAVI's) Protocol G, and are produced by the B cell cultures referred to as 1443_C16 (PG16) (TCN-116), 1503_H05

(PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456 P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123 A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_101 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158). Antibodies referred to as 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123 A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_101 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158), were isolated from the corresponding B cell cultures. These antibodies have been shown to neutralize HIV in vitro.

(83) Analysis of the antibody variable genes revealed that two antibody pairs were related by somatic hypermutation and that two of the somatic variants contained unusually long CDRH3 loops (Table 2). Long CDRH3 loops have previously been associated with polyreactivity. (Ichiyoshi, Y. & Casali, P. J Exp Med 180, 885-895 (1994)). The antibodies were tested against a panel of antigens and the antibodies were confirmed to be not polyreactive.

(84) TABLE-US-00004 TABLE 2 Sequence Analysis of mAb Variable Genes Germline Germline SEQ ID SEQ ID Clone IGV.L.sup.a IGV.H.sup.a CDRL3.sup.b NO: CDRH3.sup.b NO: PG16 VL2-14*01 VH3-33*05 **SSLTDRSHRIF 1 EAGGPIWHDDV 6 KYYDFNDGYYN YHYMDV PG9 VL2-14*01 VH3-33*05 KSLTSTRRRVF 2 EAGGPDYRNGY 7 NYYDFYDGYYN YHYMDV PGG14 VK1-39*01 VH1-69*12 SYSTPRTF 3 DRRVVP MATDN 8 WLDP PG20 VK2-14*01 VH1-69*12 SFSTPRTF 4 DRRAVPIATDN 9 WLDP PGC14 VL3-1*01 VH1-24*01 AWETTTTTTFV 5 GAVGADSGSWF 10 FF DP .sup.a** Germ line gene sequences were determined using the IMGT database, which is publicly available at imgt.cines.fr. “L” and “K” refer to lambda and kappa chains, respectively, .sup.b Bolded amino acids denote differences between somatic variants.

(85) TABLE-US-00005 TABLE 3A Heavy Chain Gene Usage Summary mAb mAb V-Gene & V-Gene J-Gene & J-Gene ID Specificity allele identity allele identity CDR3 1443_C16 ELISA- IGHV3- 85.07% IGHJ6* 85.48% AREAGGPIWHDDVKY negative 33*05 (245/ 03 (53/ YDFNDGYNYHYMDV 288 nt) 62 nt) (SEQ ID NO: 46) 1456_P20 gp120 IGHV1- 85.07% IGHJ5* 88.24% ARDRRAVPIATDNWL 69*11 (245/ 02 (45/ DP or 288 nt) 51 nt) (SEQ ID NO: 47) IGHV1- 69*12 1460_G14 gp120 IGHV1- 86.11% IGHJ5* 86.27% TRDRRVVP MATDNWL 69*11 (248/ 02 (44/ DP or 288 nt) 51 nt) (SEQ ID NO: 48) IGHV1- 69*12 1495_C14 gp120 IGHV1- 88.89% IGHJ5* 84.31% AAGAVGADSGSWFDP f*01 (256/ 02 (43/ (SEQ ID NO: 49) 288 nt) 51 nt) 1496_C09 ELISA- IGHV3- 85.07% IGHJ6* 83.87% VREAGGPDYRNGYNY negative 33*05 (245/ 03 (52/ YDFYDGYNYHYMDV 288 nt) 62 nt) (SEQ ID NO: 50)

(86) TABLE-US-00006 TABLE 3B Light Chain Summary V-Gene J-GENE mAb mAb and V-gene and J-Gene SEQ ID ID Specificity allele identity allele identity CDR3 NO: 1443_C16 ELISA- IGLV2- 88.19% IGLJ2*01. 83.33% SSLTDRSHRI 41 negative 14*01 (254/ or (30/ 288 nt) IGLJ3*01 36 nt) or IGLJ3*02 1456_P20 gp120 IGKV1- 92.11% IGKJ5*01 92.11% QQSFSPTPT 42 39*01, (257/ (35/ or 279 nt) 38 nt) IGKV1D- 39*01 1460_G14 gp120 IGKV1- 92.11% IGKJ5*01 89.47% QQSYSTPTPT 43 39*01, (257/ (34/ or 279 nt) 38 nt) IGKV1D- 39*01 1495_C14 gp120 IGLV3- 88.89% IGLJ2*01. 86.84% QAWETTTTT 44 1*01 (248/ or (33/ FVF 279 nt) IGLJ3*01 38 nt) 1496_C09 ELISA- IGLV2- 91.32% IGLJ3*02 86.11% KSLTSTRRR 45 negative 14*01 (263/ (31/ V 288 nt) 36 nt)

(87) The broadly neutralizing antibodies from 1443_C16 (PG16) and 1496_C09 (PG9) clones obtained by this method did not exhibit soluble gp120 or gp41 binding at levels that correlate with neutralization activity. The method of the invention therefore allows identification of novel antibodies with broad cross-clade neutralization properties regardless of binding activities in an ELISA screen. Further characterization of PG16 and PG9 is disclosed herein.

(88) All five antibodies were first tested for neutralization activity against a multi-clade 16-pseudovirus panel (Table 4). Two of the antibodies that bound to monomeric gp120 in the initial screen (PGG14 and PG20) did not show substantial neutralization breadth or potency against any of the viruses tested, and the third antibody that bound to gp120 (PGC14) neutralized 4/16 viruses with varying degrees of potency. In contrast, the two antibodies that failed to bind recombinant Env in the initial screen (PG9 and PG16) neutralized a large proportion of the viruses at sub-microgram per ml concentrations. PG9 and PG16 neutralized non-clade B viruses with greater breadth than three out of the four existing bNAbs. This is significant considering that the majority of HIV-1 infected individuals worldwide are infected with non-clade B viruses.

(89) TABLE-US-00007 TABLE 4 Neutralization Profiles of Rescued mAbs IC50 (μg/mL) Isolate PGC14 PG9 PG16 PGG14 P20 Clade A 94UG103 >50 0.17 0.008 >50 >50 92RW020 28.60 0.06 0.004.sup.a >50 50 93UG077 >50 >50 >50 >50 >50 Clade B 92BR020 0.64 >50 >50 >50 >50 APV-13 >50 >50 >50 >50 >50 JRCSF >50 <0.0025 <0.0025 >50 >50 APV-17 >50 26.45 >50 >50 >50 APV-6 7.41 0.09 0.08.sup.a >50 25.770 Clade C 93IN905 >50 N/A 0.10.sup.a >50 >50 IAVI-C18 >50 0.05 0.007 >50 >50 IAVI-C22 >50 N/A 0.069.sup.a >50 >50 IAVI-C3 9.50 12.91 14.80 >50 >50 Clade D 92UG024 >50 10.96 >50 >50 >50 92UG005 >50 >50 >50 >50 CRF01_92TH021 >50 0.11 0.13.sup.a >50 >50 AE CMU02 >50 >50 >50 >50 >50 negative aMLV >50 >50 >50 >50 >50 control .sup.aPlateau observed in curve.

(90) Table 17A shows neutralization profiles (IC50 values) of monoclonal antibodies 1443_C16 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14) and 1496_C09 (PG9) and the known cross-clade neutralizing antibodies b12, 2G12, 2F5 and 4E10 on a diverse panel of 16 HIV pseudoviruses from different clades. 1443_C16 (PG16) and 1496_C09 (PG9) neutralize HIV-1 species from Clades A, B, C, D and CRF01_AE with better potency for most viral strains tested than known and generally accepted broad and potent neutralizing antibodies. However, neutralization profiles of individual species of HIV-1 belonging to these clades vary between 1443_C16 (PG16) and 1496_C09 (PG9) and the known cross-clade neutralizing antibodies b12, 2G12, 2F5 and 4E10. 1495_C14 (PGC14) neutralizes fewer HIV-1 species from Clades A, B and C comparable to other neutralizing antibodies. Table 17B shows IC90 values of the monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) and the known cross-clade neutralizing antibodies b12, 2G12, 2F5 and 4E10 on the same panel of pseudoviruses. FIG. 4 shows neutralization activities of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to six other HIV pseudoviruses (YU2, Bal, ADA, DU172, DU422, and ZM197) for clades B and C not included in Tables 17A and 17B.

(91) PG9, PG16, and PGC14 were next evaluated on a large multi-clade pseudovirus panel consisting of 162 viruses to further assess the neutralization breadth and potency of these three antibodies (Tables 5A-5B, Tables 18A-18F and Tables 19A-19B). The bNAbs b12, 2G12, 2F5, and

4E10, as well as the donor's serum, were also included in the panel for comparison. Overall, PG9 neutralized 127 out of 162 and PG16 neutralized 119 out of 162 viruses with a potency that frequently considerably exceeded that noted for the four control bNAbs.

(92) The median IC50 and IC90 values for neutralized viruses across all clades were an order of magnitude lower for PG9 and PG16 than any of the four existing bNAbs (Table 5A, Tables 18A-18F and Tables 19A-19B). Both mAbs showed overall greater neutralization breadth than b12, 2G12, and 2F5 (Table 5B, Tables 18A-18F and Tables 19A-19B). At low antibody concentrations, PG9 and PG16 also demonstrated greater neutralization breadth than 4E10 (Table 5B).

Furthermore, both mAbs potently neutralized one virus (IAVI-C18) that exhibits resistance to all four existing bNAbs (Tables 18A-18F). The mAb neutralization curves reveal that, whereas the PG9 neutralization curves usually exhibit sharp slopes, the neutralization curves for PG16 sometimes exhibit gradual slopes or plateaus at less than 100% neutralization. Although neutralization curves with similar profiles have been reported previously (W. J. Honnen et al., J Virol 81, 1424 (February, 2007), A. Pinter et al., J Virol 79, 6909 (June, 2005)), the mechanism for this is not well understood.

(93) Comparison of the neutralization profile of the serum with the neutralization profile of PG9, PG16 and PGC14 revealed that these three antibodies could recapitulate the breadth of the serum neutralization in most cases (Tables 18A-18F). For example, almost all of the viruses that were neutralized by the serum with an IC50>1:500 were neutralized by PG9 and/or PG16 at <0.05 µg/mL. The one case where this did not occur was against HIV-1SF162, but this virus was potently neutralized by PGC14. Despite the fact that PG9 and PG16 are somatic variants, they exhibited different degrees of potency against a number of the viruses tested. For instance, PG9 neutralized HIV-16535.30 approximately 185 times more potently than PG16, and PG16 neutralized HIV-1MGRM-AG-001 approximately 440 times more potently than PG9. In some cases, the two antibodies also differed in neutralization breadth; PG9 neutralized nine viruses that were not affected by PG16, and PG16 neutralized two viruses that were not affected by PG9. Based on these results, it is postulated that broad serum neutralization might be mediated by somatic antibody variants that recognize slightly different epitopes and display varying degrees of neutralization breadth and potency against any given virus. In the face of an evolving viral response, it seems reasonable that the immune system might select for these types of antibodies.

(94) Comparison of the neutralization profile of the serum with the neutralization profile of PG9, PG16 and PGC14 revealed that these three antibodies could recapitulate the breadth of the serum neutralization in most cases. For example, almost all of the viruses that were neutralized by the serum with an IC50>1:1000 were neutralized by PG9 and/or PG16 at <0.005 µg/mL. The one case where this did not occur was against HIV-1SF162, but this virus was potently neutralized by PGC14. Tables 5(a) and 5(b) show the neutralization activities—breadth and potency, respectively—of PG9, PG16, and PGC14 as well as four control bNAbs as measured by IC50 values. Tables 19A-19B show results of the same analysis using IC90 values.

(95) TABLE-US-00008 TABLE 5(A) Neutralization Potency of mAbs Median IC.sub.50 (µg/mL) against viruses neutralized with an IC.sub.50 <50 µg/mL Clade.sup.a # viruses b12 2G12 2F5 4E10 PG9 PG16 PGC14 A 27 6.98 17.10 5.70 6.20 0.16 0.11 41.59 B 31 0.80 0.82 2.41 5.22 0.43 0.70 21.88 C 27 6.46 2.93 31.51 2.97 0.22 0.25 11.97 D 25 1.47 7.71 3.17 4.60 0.10 0.02 38.57 CRF01_AE 10 21.53 >50 0.26 0.51 0.08 0.03 >50 CRF_AG 10 10.40 0.95 0.64 1.42 0.80 0.03 45.10 G 15 3.07 31.03 1.24 1.44 0.29 1.21 >50 F 15 >50 9.23 1.78 2.30 0.09 0.08 25.71 Total 162 2.82 2.43 2.30 3.24 0.22 0.15 25.99 Boxes are color coded as follows: white, median potency >50 µg/mL; light grey, median potency between 2 and 20 µg/mL; medium grey, median potency between 0.2 and 2 µg/mL; dark grey, median potency <0.2 µg/mL. .sup.aCRF_07BC and CRF_08BC viruses are not included in the clade analysis because there was only one virus tested from each of these clades.

(96) TABLE-US-00009 TABLE 5(B) Neutralization Breadth of mAbs Clade.sup.a # viruses b12

2G12 2F5 4E10 PG9 PG16 PGC14 % viruses neutralized with an IC_{sub.50} <50 µg/mL A 27 30 37 74 96 85 85 11 B 31 58 71 68 97 74 74 29 C 27 33 11 7 96 78 78 19 D 25 48 24 56 96 76 60 8 CRF01_AE 10 30 0 89 100 100 100 0 CRF_AG 10 30 50 80 100 80 60 10 G 15 13 20 80 100 87 73 7 F 15 0 21 87 100 67 64 13 Total 162 35 32 60 98 79 73 15 % viruses neutralized with an IC_{sub.50} <1.0 µg/mL A 27 0 4 4 0 70 63 0 B 31 32 39 23 0 45 42 3 C 27 7 0 0 11 56 48 0 D 25 12 8 12 8 48 44 0 CRF01_AE 10 11 0 88 80 70 70 0 CRF_AG 10 10 30 60 30 40 50 0 G 15 0 0 27 0 60 33 0 F 15 0 14 13 28 80 79 0 Total 162 11 12 19 12 57 51 1 Boxes are color coded as follows: white, no viruses neutralized; black, 1 to 30% of viruses neutralized; light grey, 30 to 60% of viruses neutralized; medium grey, 60 to 90% of viruses neutralized; dark grey, 90 to 100% of viruses neutralized. .sup.aCRF_07BC and CRF_08BC viruses are not included in the clade analysis because there was only one virus tested from each of these clades.

(97) Despite the fact that PG9 and PG16 are somatic variants, they exhibited different degrees of potency against a number of the viruses tested. For instance, PG9 neutralized the virus 6535.30 about 100 times more potently than PG16, and PG16 neutralized the virus MGRM-AG-001 about 3000 times more potently than PG9. In some cases, the two antibodies also differed in neutralization breadth; PG9 neutralized seven viruses that were not neutralized by PG16, and PG16 neutralized three viruses that were not neutralized by PG9. Without being bound by theory, it appears that broad serum neutralization might be mediated by somatic variants that recognize slightly different epitopes and display varying degrees of neutralization breadth and potency against any given virus. In the face of an evolving viral response, the immune system likely selects for these types of antibodies.

(98) The antibodies were also tested for ability to bind soluble recombinant HIV envelope proteins. FIG. 5 shows dose response curves of 1456_P20 (PG20), 1495_C14 (PGC14) and 1460_G14 (PGG14) binding to recombinant gp120 in ELISA as compared to control anti-gp120 (b12). FIG. 6 shows ELISA binding assays of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to HIV-1 strain YU2 gp140 and JR-CSF gp120, the membrane proximal region (MPER) of HIV-1 envelope glycoprotein gp41, and the V3 polypeptide. PG-9 binds to YU2 gp140 (IC₅₀–20–40 nM), YU2 gp120 and weakly binds to JR-CSF gp120. However, PG16 weakly binds Yu2 gp120, but not the soluble form of HIV-1 envelope glycoprotein, gp120 JR-CSF. Neither mAb binds to JR-FL gp120, JR-FL gp140, MPER peptide of gp41 or V3 peptide.

(99) FIG. 7 shows binding of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to HIV-1 YU2 gp160 expressed on the cell surface in the presence and absence of sCD4. Competitive inhibition of the binding by sCD4 indicates that the binding of monoclonal antibody 1496_C09 to HIV-1 envelope protein gp160 expressed on the cell surface is presumably affected due to the conformational changes induced by sCD4. The data further suggest that 1443_C16 (PG16) and 1496_C09 (PG9) exhibit relatively stronger binding to trimeric forms of the HIV-1 Env (gp160 and gp140) than to the monomeric gp120.

(100) FIG. 8 shows binding of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to HIV-1 transfected cells. PG9 and PG16 do not bind untransfected cells. PG9 and PG16 bind JR-CSF, ADA, and YU2 gp160 transfected cells. PG9 and PG16 do not bind JR-FL gp160 transfected cells (cleaved or uncleaved). PG9 and PG16 do not bind ADA AV1/AV2 transfected cells. PG9 and PG16 binding to JR-CSF gp160 transfected cells is inhibited by sCD4.

(101) FIG. 9 shows the capturing of entry-competent JR-CSF pseudovirus by neutralizing monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) in a dose-dependent manner. The ability of both antibodies to capture JR-CSF pseudovirus is higher than IgG b12 but comparable to IgG 2G12. It is postulated that the capture may be mediated by the binding of the mAbs to the HIV-1 Env on the virions.

(102) FIG. 10A shows that sCD4, PG16 and PG9 compete for the binding of monoclonal antibody 1443_C16 (PG16) to JR-CSF pseudovirus but b12, 2G12, 2F5 and 4E10 do not. FIG. 10B shows sCD4, PG16 and PG9 compete for the binding of monoclonal antibody 1496_C09 (PG9) to JR-

CSF pseudovirus but b12, 2G12, 2F5 and 4E10 do not. This suggests that the PG16 and PG9 mAbs bind gp120 at a site different from those bound by b12 and 2G12. PG9 and PG16 binding to HIV-1 envelope protein is competitively inhibited by sCD4. Given that the MAbs are not inhibited by the CD4 binding site MAb b12, this suggests that PG9 and PG16 are binding to an epitope that is unavailable for sCD4 binding to gp120 as a result of conformational changes. The inability of PG9 and PG16 to bind monomeric gp120JR-CSF or gp41HxB2 in the initial screen while potentially neutralizing HIV-1JR-CSF suggests that the epitope targeted by these antibodies is preferentially expressed on trimeric HIV envelope protein. The ability of PG9 and PG16 to bind monomeric gp120 from several different strains, artificially trimerized gp140 constructs, and trimeric Env expressed on the surface of transfected cells respectively, was compared. Although both antibodies bound with high affinity to cell surface Env, PG16 did not bind to any of the soluble gp120 or gp140 constructs and PG9 bound only weakly to monomeric gp120 and trimerized gp140 from certain strains (FIG. 11). It has been previously shown that a substantial fraction of cell surface Env is comprised of uncleaved gp160 molecules. (Pancera, M. & Wyatt, R. *Virology* 332, 145-156 (2005)). That PG9 and PG16 do not exhibit exclusive specificity for native HIV-1 trimers was confirmed by the fact that both antibodies bound with high affinity to cleavage-defective HIV-1YU2 trimers expressed on the surface of transfected cells (FIG. 12).

(103) The epitopes recognized by PG9 and PG16 were investigated. Since the PG9 and PG16 antibodies are somatic variants, they recognize the same or overlapping epitopes. Both antibodies cross-competed for binding to HIV-1JR-CSF transfected cells (FIG. 13A). Ligation of monomeric gp120 or cell surface Env with soluble CD4 diminished binding of both PG9 and PG16, although neither antibody competed with CD4-binding site antibodies for trimer binding (FIG. 13A-13C). This result suggests that CD4-induced conformational changes cause a loss of the epitope targeted by the antibodies.

(104) Since PG9 bound well enough to gp120 from certain isolates to generate ELISA binding curves, competition ELISAs were performed with PG9 using a panel of neutralizing and non-neutralizing antibodies. These data revealed that PG9 cross-competed with anti-V2, anti-V3, and to a lesser extent, CD4i antibodies for gp120. (FIGS. 13D and 14).

(105) Neither PG9 nor PG16 bound to V1/V2 or V3 deleted HIV-1JR-CSF variants expressed on the surface of transfected cells, further suggesting contributions of variable loops in forming their epitopes (FIG. 13E).

(106) To dissect the fine specificity of PG9 and PG16, alanine scanning was performed using a large panel of HIV-1JR-CSF Env alanine mutants that have been described previously (Pantophlet, R., et al. *J Virol* 77, 642-658 (2003); Pantophlet, R., et al. *J Virol* 83, 1649-1659 (2009); Darbha, R., et al. *Biochemistry* 43, 1410-1417 (2004); Scanlan, C. N., et al. *J Virol* 76, 7306-7321 (2002)) as well as several new alanine mutants. Pseudoviruses incorporating single Env alanine mutations were generated, and PG9 and PG16 were tested for neutralization activity against each mutant pseudovirus. Mutations that resulted in viral escape from PG9 and PG16 neutralization were considered important for formation of the PG9 and PG16 epitopes (Tables 12 and 13).

(107) Based on these criteria, and consistent with the competition experiments, residues that form the epitopes recognized by PG9 and PG16 appear to be located in conserved regions of the V2 and V3 loops of gp120. Certain co-receptor binding site mutations also had an effect on PG9 and PG16 neutralization, albeit to a lesser extent. Generally, PG9 and PG16 were dependent on the same residues, although PG16 was more sensitive to mutations located in the tip of the V3 loop than PG9. Interestingly, although neither antibody bound to wild-type HIV-1JR-FL transfected cells, a D to K mutation at position 168 in the V2 loop of HIV-1JR-FL generated high-affinity PG9 and PG16 recognition (Tables 18A-18F). N156 and N160, sites of V2 N-glycosylation, also appear to be critical in forming the epitope since substitutions at these positions resulted in escape from PG9 and PG16 neutralization. Deglycosylation of gp120 abolished binding of PG9 (FIG. 16), confirming that certain glycans may be important in forming the epitope.

(108) HIV-1 SF162 contains a rare N to K polymorphism at position 160, and mutation of this residue to an Asn renders this isolate sensitive to PG9 and PG16 (FIG. 17).

(109) The preferential binding of PG9 and PG16 to native trimers could either be a consequence of gp120 subunit cross-linking or recognition of a preferred oligomeric gp120 conformation. To address this question, the binding profiles of PG9 and PG16 to mixed HIV-1YU2 trimers were examined, in which two gp120 subunits containing point mutations abolished binding of the two antibodies. A third substitution that abrogates binding of 2G12, which binds with high affinity to both monomeric gp120 and trimeric Env, was also introduced into the same construct as an internal control. Cell surface binding analysis revealed that all three antibodies bound to the mixed trimers with similar apparent affinity as to wild-type trimers and all saturated at a similar lower level (FIG. 18). This result suggests that the preference of PG9 and PG16 for trimeric Env is due to gp120 subunit presentation in the context of the trimeric spike rather than gp120 cross-linking.

(110) It has been shown that NAbS that bind to epitopes encompassing parts of the V2 or both the V2 and V3 domains can exhibit potency comparable to that of PG9 and PG16, although these antibodies have thus far displayed strong strain-specificity. (Honnen, W. J., et al. *J Virol* 81, 1424-1432 (2007); Gorny, M. K., et al. *J Virol* 79, 5232-5237 (2005)). Importantly, the epitopes recognized by these antibodies have been shown to differ from that of the Glade B consensus sequence only by single amino acid substitutions, which suggested the existence of a relatively conserved structure within the V2 domain. (Honnen, W. J., et al. *J Virol* 81, 1424-1432 (2007)). The results observed with PG9 and PG16 confirm that this region serves as a potent neutralization target and demonstrates that antibodies that recognize conserved parts of V2 and V3 can possess broad reactivity.

(111) The invention is based on novel monoclonal antibodies and antibody fragments that broadly and potently neutralize HIV infection. In some embodiments, these monoclonal antibodies and antibody fragments have a particularly high potency in neutralizing HIV infection in vitro across multiple clades or across a large number of different HIV species. Such antibodies are desirable, as only low concentrations are required to neutralize a given amount of virus. This facilitates higher levels of protection while administering lower amounts of antibody. Human monoclonal antibodies and the immortalized B cell clones that secrete such antibodies are included within the scope of the invention.

(112) The invention provides methods for using high throughput functional screening to select neutralizing antibodies with unprecedented breadth and potency. The invention relates to other potent and broadly neutralizing antibodies that can be developed using the same methods. In particular, the invention relates to potent, broadly neutralizing antibodies against different strains of HIV, wherein the bNAbs bind poorly to recombinant forms of Env. The invention provides two neutralizing antibodies, PG9 and PG16, with broad neutralizing activities particularly against non-clade B isolates. The invention provides vaccine-induced antibodies of high specificity that provide protection against a diverse range of the most prevalent isolates of HIV circulating worldwide. The invention provides antibodies with very high and broad neutralization potency, such as that exhibited by PG9 and PG16 in vitro, which provides protection at relatively modest serum concentrations, and are generated by vaccination unlike the broad NAbS known in the art. The invention provides immunogens that can be designed that focus the immune response on conserved regions of variable loops in the context of the trimeric spike of the gp120 subunit of the Env protein.

(113) The invention also relates to the characterization of the epitope to which the antibodies bind and the use of that epitope in raising an immune response.

(114) The invention also relates to various methods and uses involving the antibodies of the invention and the epitopes to which they bind. For example, monoclonal antibodies according to the invention can be used as therapeutics. In some aspects, the monoclonal antibodies are used for adjuvant therapy. Adjuvant therapy refers to treatment with the therapeutic monoclonal antibodies,

wherein the adjuvant therapy is administered after the primary treatment to increase the chances of a cure or reduce the statistical risk of relapse.

(115) The invention provides novel monoclonal or recombinant antibodies having particularly high potency in neutralizing HIV. The invention also provides fragments of these recombinant or monoclonal antibodies, particularly fragments that retain the antigen-binding activity of the antibodies, for example which retain at least one complementarity determining region (CDR) specific for HIV proteins. In this specification, by “high potency in neutralizing HIV” is meant that an antibody molecule of the invention neutralizes HIV in a standard assay at a concentration lower than antibodies known in the art.

(116) Preferably, the antibody molecule of the present invention can neutralize at a concentration of 0.16 µg/ml or lower (i.e. 0.15, 0.125, 0.1, 0.075, 0.05, 0.025, 0.02, 0.016, 0.015, 0.0125, 0.01, 0.0075, 0.005, 0.004 or lower), preferably 0.016 µg/ml or lower (an antibody concentration of 10.sup.-8 or lower, preferably 10.sup.-9 M or lower, preferably 10-10 M or lower, i.e. 10-11 M, 10-12 M, 10-13 M or lower). This means that only very low concentrations of antibody are required for 50% neutralization of a clinical isolate of HIV in vitro. Potency can be measured using a standard neutralization assay as described in the art.

(117) The antibodies of the invention are able to neutralize HIV. Monoclonal antibodies can be produced by known procedures, e.g., as described by R. Kennet et al. in “Monoclonal Antibodies and Functional Cell Lines; Progress and Applications”. Plenum Press (New York), 1984. Further materials and methods applied are based on known procedures, e.g., such as described in J. Virol. 67:6642-6647, 1993.

(118) These antibodies can be used as prophylactic or therapeutic agents upon appropriate formulation, or as a diagnostic tool.

(119) A “neutralizing antibody” is one that can neutralize the ability of that pathogen to initiate and/or perpetuate an infection in a host and/or in target cells in vitro. The invention provides a neutralizing monoclonal human antibody, wherein the antibody recognizes an antigen from HIV.

(120) Preferably an antibody according to the invention is a novel monoclonal antibody referred to herein as 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158). These antibodies were initially isolated from human samples and are produced by the B cell cultures referred to as 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869_K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-

154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158). These antibodies have been shown to neutralize HIV in vitro. 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_I01 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and 6881_N05 (PGT-158) have been shown to have broad, potent HIV neutralizing activity.

(121) The CDRs of the antibody heavy chains are referred to as CDRH1, CDRH2 and CDRH3, respectively. Similarly, the CDRs of the antibody light chains are referred to as CDRL1, CDRL2 and CDRL3, respectively. The position of the CDR amino acids is defined according to the IMGT numbering system as: CDR1—IMGT positions 27 to 38, CDR2—IMGT positions 56 to 65 and CDR3—IMGT positions 105 to 117. (Lefranc, M P. et al. 2003 IMGT unique numbering for immunoglobulin and T cell receptor variable regions and Ig superfamily V-like domains. *Dev Comp Immunol.* 27(1):55-77; Lefranc, M P. 1997. Unique database numbering system for immunogenetic analysis. *Immunology Today*, 18:509; Lefranc, M P. 1999. The IMGT unique numbering for Immunoglobulins, T cell receptors and Ig-like domains. *The Immunologist*, 7:132-136.)

(122) The amino acid sequences of the CDR3 regions of the light and heavy chains of the antibodies are shown in Tables 3A and 3B.

(123) A phylogram is a branching diagram (tree) assumed to be an estimate of phylogeny, branch lengths are proportional to the amount of inferred evolutionary change. Tree diagrams of the five heavy chains and the five light chains were prepared using ClustalW (Larkin M. A., Blackshields G., Brown N. P., Chenna R., McGettigan P. A., McWilliam H., Valentin F., Wallace I. M., Wilm A., Lopez R., Thompson J. D., Gibson T. J. and Higgins D. G. *Bioinformatics* 23(21): 2947-2948 (2007); Higgins D G et al. *Nucleic Acids Research* 22: 4673-4680. (1994)) and are shown in FIGS. 1A and 1B respectively.

(124) The sequences of the antibodies were determined, including the sequences of the variable regions of the Gamma heavy and Kappa or Lambda light chains of the antibodies designated 1496_C09 (PG9), 1443_C16 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), and 1495_C14 (PGC14). In addition, the sequence of each of the polynucleotides encoding the antibody sequences was determined. Shown below are the polypeptide and polynucleotide sequences of the gamma heavy chains and kappa light chains, with the signal peptides at the N-terminus (or 5' end) and the constant regions at the C-terminus (or 3' end) of the variable regions, which are shown in bolded text.

(125) TABLE-US-00010 1443_C16 (PG16) (TCN-116) gamma heavy chain
nucleotide sequence: 1443 C16 γ 3 coding sequence (variable region in bold)
(SEQ ID NO: 11)

ATGGAGTTTGGGCTGAGCTGGGTTTTCTCGCAACTCTGTTAAGAGT
TGTGAAGTGTCAGGAACAACCTGGTGGAGTCTGGGGGAGGCGTGGTCC
AGCCGGGGGGGTCCCTGAGACTCTCCTGTTTAGCGTCTGGATTCACG
TTTCACAAATATGGCATGCACTGGGTCCGCCAGGCTCCAGGCAAGGG
CCTGGAGTGGGTGGCACTCATCTCAGATGACGGAATGAGGAAATATC
ATTCAGACTCCATGTGGGGCCGAGTCACCATCTCCAGAGACAATTCC

AAAGAACTCTTCTTATGATAGCAGCTTGAAAGTCGAAGACAC
GGCTATGTTCTTCTGTGCGAGAGAGGCTGGTGGGCCAATCTGGCATG
ACGACGTCAAATATTACGATTTTAATGACGGCTACTACAACCTACCAC
TACATGGACGTCTGGGGCAAGGGGACCACGGTCACCGTCTCGAGCGC
CTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGA
GCACCTCTGGGGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTAC
TTCCCCGAACCGGTGACGGTGTCTGTGGAACCTCAGGCGCCCTGACCAG
CGGCGTGCACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACT
CCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGGCACCCAG
ACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGA
CAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACACATGCCCAC
CGTGCCCAGCACCTGAACTCCTGGGGGGGACCGTCAGTCTTCCTCTTC
CCCCCAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGT
CACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGT
TCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAG
CCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCT
CACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCA
AGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCC
AAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCC
ATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGG
TCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAAT
GGGCAGCCGGAGAACAACCTACAAGACCACGCCTCCCGTGCTGGACTC
CGACGGCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAAGAGCA
GGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCT
CTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAA

ATGA

1443_C16 (PG16) (TCN-116) gamma heavy chain variable region nucleotide
sequence: (SEQ ID NO: 99)

CAGGAACAACCTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCGGGGGG
GTCCCTGAGACTCTCCTGTTTAGCGTCTGGATTACGTTTCACAAAT
ATGGCATGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCTGGAGTGG
GTGGCACTCATCTCAGATGACGGAATGAGGAAATATCATTGAGACTC
CATGTGGGGCCGAGTCACCATCTCCAGAGACAATTCCAAGAACAACCTC
TTTATCTGCAATTCAGCAGCCTGAAAGTCGAAGACACGGCTATGTTC
TTCTGTGCGAGAGAGGCTGGTGGGCCAATCTGGCATGACGACGTCAA
ATATTACGATTTTAATGACGGCTACTACAACCTACCCTACATGGACG

TCTGGGGCAAGGGGACCACGGTCACCGTCTCGAGC 1443_C16 (PG16) (TCN-
116) gamma heavy chain amino acid sequence: expressed protein with variable
region in bold. (SEQ ID NO: 12)

QEQLVESGGGVVQPGGSLRLSCLASGFTFHKYGMHWVRQAPGKGLEW
VALISDDGMRKYHSDSMWGRVTISRDN SKNTLYLQFSSLKVEDTAMF
FCAREAGGP**IWHDDV****KYYDFNDGY****NYHYMDVW****GKGTTVT****VSSASTK**
GPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVH
TFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRV
EPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVL
HQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRE
EMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGS

FFLYSKLTVDKSRWQQGNV FSCSV MHEALHNHYTQKSLSLSPGK 1443_C16 (PG16)
(TCN-116) gamma heavy chain variable region amino acid sequence: (Kabat
CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 31)

QEQLVESGGGVVQLGQGLRLSLASGLFTFH custom character WVRQAPGKLEWVA

custom character custom character RVTISRDN SKNTLYLQFSSLKVEDTAMFFCAR

custom character custom character custom character WGKGTTVTVSS 1443_C16

(PG16) (TCN-116) gamma heavy chain Kabat CDRs: CDR 1: (SEQ ID NO: 88) KYGMH CDR 2: (SEQ ID NO: 89) LISDDGMRKYHSDSMWG CDR 3: (SEQ ID NO: 6) EAGGPIWHDDVKYYDENDGYNYHYMDV 1443_C16 (PG16) (TCN-116) gamma heavy chain Chothia CDRs: CDR 1: (SEQ ID NO: 266) GFTFHK CDR 2: (SEQ ID NO: 267) LISDDGMRKY CDR 3: (SEQ ID NO: 6) EAGGPIWHDDVKYYDFNDGYNYHYMDV 1443_C16 (PG16) (TCN-116) lambda light chain nucleotide sequence: 1443_C16 22 coding sequence (variable region in bold) (SEQ ID NO: 13)

ATGGCCTGGGCTCTGCTATTCCTCACCCCTCTTCACTCAGGGCACAGGGTC
CTGGGGCCAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTG
GACAGACGATCACCATCTCCTGCAATGGAACCAGCAGTGACGTTGGTGGA
TTTGACTCTGTCTCCTGGTACCAACAATCCCCAGGGAAAGCCCCCAAAGT
CATGGTTTTTGATGTCAGTCATCGGCCCTCAGGTATCTCTAATCGCTTCT
CTGGCTCCAAGTCCGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCAC
ATTGAGGACGAGGGCGATTATTTCTGCTCTTCACTGACAGACAGAAGCCA
TCGCATATTCGGCGGGCGGGACCAAGGTGACCGTTCTAGGTCAGCCCAAGG
CTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAGCC
AACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGCCGT
GACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGAGGA
CCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGCAGCTAC
CTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACAGCTGCCA
GGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCCCTACAGAAT
GTTCATAG 1443_C16 (PG16) (TCN-116) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 100)

CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTGGACAGAC
GATCACCATCTCCTGCAATGGAACCAGCAGTGACGTTGGTGGATTGACT
CTGTCTCCTGGTACCAACAATCCCCAGGGAAAGCCCCCAAAGTCATGGTT
TTTGATGTCAGTCATCGGCCCTCAGGTATCTCTAATCGCTTCTCTGGCTC
CAAGTCCGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCACATTGAGG
ACGAGGGCGATTATTTCTGCTCTTCACTGACAGACAGAAGCCATCGCATA
TTCGGCGGGCGGGACCAAGGTGACCGTTCTA 1443_C16 (PG16) (TCN-116)
lambda light chain amino acid sequence: expressed protein with variable region
in bold. (SEQ ID NO: 14)

QSALTQPASVSGSPGQTITISCNGTSSDVGGFDSVSWYQQSPGKAPKVMV

FDVSHRPSGISNRFSGSKSGNTASLTISGLHIEDEGDYFCSSLTDRSHRI

FGGGTKVTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTV

AWKADSSPVKAGVETTPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVT

HEGSTVEKTVAPTECS 1443_C16 (PG16) (TCN-116) lambda light chain

variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs

in bold italics) (SEQ ID NO: 32) QSALTQPASVSGSPGQTITISC custom character

WYQQSPGKAPKVMVF custom character custom character

GISNRFSGSKSGNTASLTISGLHIEDEGDYFC custom character FGGGTKVTVL

1443_C16 (PG16) (TCN-116) lambda light chain Kabat CDRs: CDR 1: (SEQ

ID NO: 97) NGTSSDVGGFDSVS CDR 2: (SEQ ID NO: 95) DVSHRPS

CDR 3: (SEQ ID NO: 41) SSLTDRSHRI 1443_C16 (PG16) (TCN-116)

lambda light chain Chothia CDRs: CDR 1: (SEQ ID NO: 97)

NGTSSDVGGFDSVS CDR 2: (SEQ ID NO: 95) DVSHRPS CDR 3: (SEQ

ID NO: 41) SSLTDRSHRI 1456_P20 (PG20) gamma heavy chain nucleotide sequence: 1456_P20 y1 coding sequence (variable region in bold)(SEQ ID NO: 15) ATGGACTGGATTGAGGTTCTCTTTGTGGTGGCAGCAGCTACAGGTGTCCAGTCCCAGGTCCGCCTGGTACAGTCTGGGCCTGAGGTGAAGAAGCCTGGGTCCTCGGTGACGGTCTCCTGCCAGGCTTCTGGAGGCACCTTCAGCAGTTATGCTTTCACCTGGGTGCGCCAGGCCCGGACAAGGTCTTGAGTGGTTGGGCATGGTCACCCCAATCTTTGGTGAGGCCAAGTACTCACAAAGATTCGAGGGCAGAGTCACCATCACCGCGGACGAATCCACGAGCACAACTCCATGAATTGAGAGGCCTGACATCCGAAGACACGGCCATTTATTACTGTGCGCGAGATCGGCGCGCGGTTCCAATTGCCACGGACAACCTGGTTAGACCCCTGGGGCCAGGGGACCCTGGTCAACGTCTCGAGCGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGGCACAGCGGCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCGTGGAATCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACACATGCCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCCTCTTCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGCGAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 1456_P20 (PG20) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 101) CAGGTCCGCCTGGTACAGTCTGGGCCTGAGGTGAAGAAGCCTGGGTCTCCTGGTGACGGTCTCCTGCCAGGCTTCTGGAGGCACCTTCAGCAGTTATGCTTTCACCTGGGTGCGCCAGGCCCGGACAAGGTCTTGAGTGGTTGGGCATGTACCCCAATCTTTGGTGAGGCCAAGTACTCACAAAGATTCGAGGGCAGAGTCACCATCACCGCGGACGAATCCACGAGCACAACTCCATAGAATTGAGAGCCTGACATCCGAAGACACGGCCATTTATTACTGTGCGCGAGATCGGCGCGCGGTTCCAATTGCCACGGACAACCTGGTTAGACCCCTGGGGCCAGGGGACCCTGGTCACCGTCTCGAGC 1456_P20 (PG20) gamma heavy chain amino acid sequence: expressed protein with variable region in bold. (SEQ ID NO: 16) **QVRLVQSGPEVKKPGSSVTVSCQASGGTFSSYAFTWVRQAPGQGLEWLGMVTPIFGEAKYSQRFEGRVTITADESTSTTSIELRGLTSEDTAIYYCARDRR****RAVPIATDNWLDPWGQGT****LVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK**

1456_P20 (PG20) gamma heavy chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)(SEQ ID NO: 33) QVRLVQSGPEVKKPGSSVTVSCQASGGTFS custom character WVRQAPGQGLEWLG custom character custom character RVTITADESTSTTSIELRGLTSEDTAIYYC AR custom character custom character WGQGTTLVTVSS 1456_P20 (PG20) gamma heavy chain Kabat CDRs: CDR 1: (SEQ ID NO: 104) SYAFT CDR 2: (SEQ ID NO: 105) MVTPIFGAEAKYSQRFEG CDR 3: (SEQ ID NO: 9) DRRAPVIATDNWLDP 1456_P20 (PG20) gamma heavy chain Chothia CDRs: CDR 1: (SEQ ID NO: 268) GGTFSS CDR 2: (SEQ ID NO: 269) MVTPIFGAEAK CDR 3: (SEQ ID NO: 9) DRRAPVIATDNWLDP 1456_P20 (PG20) kappa light chain nucleotide sequence: 1456_P20 κ1 coding sequence (variable region in bold)(SEQ ID NO: 17)

ATGGACATGAGGGTCCCCGCTCAGCTCCTGGGGCTCCTGCTACTCTGGCT
CCGAGGTGCCAGATGTGACATCCAGTTGACCCAGTCTCCATCCTCCCTGT
CTGCATCTGTTGGCGACAGAGTCTCCATCACTTGCCGGGCGAGTCAGACC
ATTAACAACTACTTAAATTGGTATCAACAGACACCCGGGAAAGCCCCCTAA
ACTCCTGATCTATGGTGCCTCCAATTTGCAAAATGGGGTCCCATCAAGGT
TCAGCGGCAGTGGCTCTGGGACAGACTTCACTCTCACCATCAGCAGTCTG
CAACCTGAGGATTTTGCAACTTACTACTGTCAACAGAGTTTCAGTACTCC
GAGGACCTTCGGCCAAGGGACACGACTGGATATTAAACGTACGGTGGCTG
CACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGTTGAAATCTGGA
ACTGCCTCTGTTGTGTGCCTGCTGAATAACTTCTATCCCAGAGAGGCCAA
AGTACAGTGGAAGGTGGATAACGCCCTCCAATCGGGTAACTCCCAGGAGA
GTGTCACAGAGCAGGACAGCAAGGACAGCACCTACAGCCTCAGCAGCACC
CTGACGCTGAGCAAAGCAGACTACGAGAAACACAAAGTCTACGCCTGCGA
AGTCACCCATCAGGGCCTGAGCTCGCCCGTCACAAAGAGCTTCAACAGGG
GAGAGTGTTAG 1456_P20 (PG20) kappa light chain variable region
nucleotide sequence: (SEQ ID NO: 106)

GACATCCAGTTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTTGGCGA
CAGAGTCTCCATCACTTGCCGGGCGAGTCAGACCATTAACAACACTACTTAA
ATTGGTATCAACAGACACCCGGGAAAGCCCCCTAAACTCCTGATCTATGGT
GCCTCCAATTTGCAAAATGGGGTCCCATCAAGGTTTCAGCGGCAGTGGCTC
TGGGACAGACTTCACTCTCACCATCAGCAGTCTGCAACCTGAGGATTTTG
CAACTTACTACTGTCAACAGAGTTTCAGTACTCCGAGGACCTTCGGCCAA
GGGACACGACTGGATATTAAA 1456_P20 (PG20) kappa light chain amino
acid sequence: expressed protein with variable region in bold.(SEQ ID NO:
18) **DIQLTQSPSSLSASVGDRVSITCRASQTINNYLNWYQQTPGKAPKLLIYGA**
SNLQNGVPSRFSGSGSGTDFTLTISSLQPEDFATYYCQQSFSTPRTFGQGT
RLDIKRVAAPS VFIFPPSDEQLKSGTASVVCLLN NFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSP

VTKSFNRGEC 1456_P20 (PG20) kappa light chain variable region amino
acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)(SEQ
ID NO: 34) DIQLTQSPSSLSASVGDRVSITC custom character WYQQTPGKAPKLLIY
custom character custom character GVPSRFSGSGSGTDFTLTISSLQPEDFATYYC
custom character FGQGTRLDIK 1456_P20 (PG20) kappa light chain Kabat CDRs:
CDR 1: (SEQ ID NO: 107) RASQTINNYLN CDR 2: (SEQ ID NO: 108)
GASNLQN CDR 3: (SEQ ID NO: 42) QQSFSPTPT 1456_P20 (PG20) kappa
light chain Chothia CDRs: CDR 1: (SEQ ID NO: 107) RASQTINNYLN CDR
2: (SEQ ID NO: 108) GASNLQN CDR 3: (SEQ ID NO: 42) QQSFSPTPT
1460_G14 (PGG14) gamma heavy chain nucleotide sequence: 1460_G14 γ1

coding sequence (variable region in bold) (SEQ ID NO: 19)
ATGGACTGGATTTGGAGGTTCTCTTGGTGGTGGCAGCAGCTACAGGTGT
CCAGTCCCAGGTCCTGCTGGTGCAGTCTGGGACTGAGGTGAAGAAGCCTG
GGTCCTCGGTGAAGGTCTCCTGTCAGGCTTCTGGAGGCGCCTTCAGTAGT
TATGCTTTCAGCTGGGTGCGACAGGCCCTGGACAGGGGCTTGAATGGAT
GGGCATGATCACCCCTGTCTTTGGTGAGACTAAATATGCACCGAGGTTCC
AGGGCAGACTCACACTTACCGCGGAAGAATCCTTGAGCACCACTACATG
GAATTGAGAAGCCTGACATCTGATGACACGGCCTTTTATTATTGTACGAG
AGATCGGCGCGTAGTTCCAATGGCCACAGACAACCTGGTTAGACCCCTGGG
GCCAGGGGACGCTGGTCACCGTCTCGAGCGCCTCCACCAAGGGGCCATCG
GTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGC
CCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCGT
GGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTA
CAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAG
CAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCA
ACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCAC
ACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTT
CCTCTTCCCCCCTAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTG
AGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAG
TTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCC
GCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCG
TCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCC
AACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGG
GCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGA
TGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCC
AGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAATA
CAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCTATA
GCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCA
TGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCT
CTCCCTGTCTCCGGGTAAATGA

1460_G14 (PGG14) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 109)

CAGGTCCTGCTGGTGCAGTCTGGGACTGAGGTGAAGAAGCCTGGGTCTC
GGTGAAGGTCTCCTGTCAGGCTTCTGGAGGCGCCTTCAGTAGTTATGCTT
TCAGCTGGGTGCGACAGGCCCTGGACAGGGGCTTGAATGGATGGGCATG
ATCACCCCTGTCTTTGGTGAGACTAAATATGCACCGAGGTTCCAGGGCAG
ACTCACACTTACCGCGGAAGAATCCTTGAGCACCACTACATGGAATTGA
GAAGCCTGACATCTGATGACACGGCCTTTTATTATTGTACGAGAGATCGG
CGCGTAGTTCCAATGGCCACAGACAACCTGGTTAGACCCCTGGGGCCAGGG
GACGCTGGTCACCGTCTCGAGC

1460_G14 gamma heavy chain amino acid sequence: expressed protein with variable region in bold. (SEQ ID NO: 20)

QVLLVQSGTEVKKPGSSVKVSCQASGGAFFSWVRQAPGQGLEWMGM
ITPVFGETKYAPRFQGRLTLAEESLSTTYMELRSLTSDDTAFYYCTRDR
RVVPMATDNWLDPWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALG
CLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSL
GTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLF
PPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQP
REPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKT
TPPVLDSDGSFFLYSKLTVDKSRWQQGNVFSVMSHEALHNHYTQKSLSLSPG

1460_G14 gamma heavy chain variable region amino acid sequence: (Kabat

CDRs underlined, Chothia CDRs in bold italics)(SEQ ID NO: 35)
QVLLVQSGTEVKKPGSSVKVSCQAS***GGAFS***custom character WVRQAPGQGLEWMG
custom character custom character RLTLTAESLSTTYMELRSLTSDDTAFYYC TR
custom character custom character WGQGTLLTVSS 1460_G14 gamma heavy
chain Kabat CDRs: CDR 1: (SEQ ID NO: 110)SYAFS CDR 2: (SEQ ID
NO: 111)MITPVFGETKYAPRFQG CDR 3: (SEQ ID NO: 8)
DRRVVPMATDNWLDP 1460_G14 gamma heavy chain Chothia CDRs: CDR 1:
(SEQ ID NO: 270)GGAFSS CDR 2: (SEQ ID NO: 271)MITPVFGETK
CDR 3: (SEQ ID NO: 8)DRRVVPMATDNWLDP 1460_G14 (PGG14) kappa
light chain nucleotide sequence: 1460_G14 κl coding sequence (variable
region in bold)(SEQ ID NO: 21)
ATGGACATGAGGGTCCCCGCTCAGCTCCTGGGGCTCCTGCTCCTCTGGCT
CCGAGGTGCCACATGTGACATCCAGTTGACCCAGTCTCCATCCTCCCTGT
CTGCATCTGTAGGAGACAGGGTCACCGTCACTTGCCGGGCGAGTCAGACC
ATACACACCTATTTAAATTGGTATCAGCAAATTCCAGGAAAAGCCCCTAA
GCTCCTGATCTATGGTGCCTCCACCTTGCAAAGTGGGGTCCCGTCAAGGT
TCAGTGGCAGTGGATCTGGGACAGATTTCACTCTCACCATCAACAGTCTC
CAACCTGAGGACTTTGCAACTTACTACTGTCAACAGAGTTACAGTACCCC
AAGGACCTTCGGCCAAGGGACACGACTGGATATTAAACGTACGGTGGCTG
CACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGTTGAAATCTGGA
ACTGCCTCTGTTGTGTGCCTGCTGAATAACTTCTATCCCAGAGAGGCCAA
AGTACAGTGGAAGGTGGATAACGCCCTCCAATCGGGTAACTCCCAGGAGA
GTGTCACAGAGCAGGACAGCAAGGACAGCACCTACAGCCTCAGCAGCACC
CTGACGCTGAGCAAAGCAGACTACGAGAAACACAAAGTCTACGCCTGCGA
AGTCACCCATCAGGGCCTGAGCTCGCCCGTCACAAAGAGCTTCAACAGGG
GAGAGTGTTAG 1460_G14 (PGG14) kappa light chain variable region
nucleotide sequence: (SEQ ID NO: 112)
GACATCCAGTTGACCCAGTCTCCATCCTCCCTGTCTGCATCTGTAGGAGA
CAGGGTCACCGTCACTTGCCGGGCGAGTCAGACCATACACACCTATTTAA
ATTGGTATCAGCAAATTCCAGGAAAAGCCCCTAAGCTCCTGATCTATGGT
GCCTCCACCTTGCAAAGTGGGGTCCCGTCAAGGTTCAGTGGCAGTGGATC
TGGGACAGATTTCACTCTCACCATCAACAGTCTCCAACCTGAGGACTTTG
CAACTTACTACTGTCAACAGAGTTACAGTACCCCAAGGACCTTCGGCCAA
GGGACACGACTGGATATTAAA 1460_G14 kappa light chain amino acid
sequence: expressed protein with variable region in bold.(SEQ ID NO: 22)
DIQLTQSPSSLSASVGDRVTVTCRASQTIHTYLNWYQQIPGKAPKLLIYG
ASTLQSGVPSRFSGSGSGTDFTLTINSLQPEDFATYYCQQSYSTPRTFGQ
GTRLDIKRTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKV
DNALQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQG
LSSPVTKSFNRGEC 1460_G14 kappa light chain variable region amino acid
sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)(SEQ ID
NO: 36)DIQLTQSPSSLSASVGDRVTVTCcustom character WYQQIPGKAPKLLIY
custom character custom character GVPSRFSGSGSGTDFTLTINSLQPEDFATYYC
custom character FGQGTRLDIK 1460_G14 kappa light chain Kabat CDRs: CDR
1: (SEQ ID NO: 113)RASQTIHTYLN CDR 2: (SEQ ID NO: 114)
GASTLQS CDR 3: (SEQ ID NO: 43)QQSYSTPRT 1460_G14 kappa light
chain Chothia CDRs: CDR 1: (SEQ ID NO: 113)RASQTIHTYLN CDR 2:
(SEQ ID NO: 114)GASTLQS CDR 3: (SEQ ID NO: 43)QQSYSTPRT
1495_C14 (PGC14) gamma heavy chain nucleotide sequence: 1495_C14 γl
coding sequence (variable region in bold)(SEQ ID NO: 23)

ATGGACTTGATTGGATTTGGATCTTGGTGGCAGCTCTTGGCCAGAC
CCTCGCCGACGGCCACCTGGTTCAGTCTGGGGTTGAGGTGAAGAAGACTG
GGGCTACAGTCAAATCTCCTGCAAGGTTTCTGGATACAGCTTCATCGAC
TACTACCTTCATTGGGTGCAACGGGCCCCCTGGAAAAGGCCTTGAGTGGGT
GGGACTTATTGATCCTGAAAATGGTGAGGCTCGATATGCAGAGAAGTTCC
AGGGCAGAGTCACCATAATCGCGGACACGTCTATAGATACAGGCTACATG
GAAATGAGGAGCCTGAAATCTGAGGACACGGCCGTGTATTTCTGTGCAGC
AGGTGCCGTGGGGGCTGATTCCGGGAGCTGGTTCGACCCCTGGGGGCCAGG
GAACTCTGGTCACCGTCTCGAGCGCCTCCACCAAGGGCCCATCGGTCTTC
CCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGGCACAGCGGCCCTGGG
CTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGTGAACT
CAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTACAGTCC
TCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTT
GGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCA
AGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACACATGC
CCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCCTCTT
CCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCA
CATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAAC
TGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCGCGGGA
GGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGC
ACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAA
GCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCC
CCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGATGACCA
AGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGAC
ATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAACTACAAGAC
CACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCTATAGCAAGC
TCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCC
GTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCT
GTCTCCGGGTAAATGA

1495_C14 (PGC14) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 115)

GACGGCCACCTGGTTCAGTCTGGGGTTGAGGTGAAGAAGACTGGGGCTAC
AGTCAAATCTCCTGCAAGGTTTCTGGATACAGCTTCATCGACTACTACC
TTCATTGGGTGCAACGGGCCCCCTGGAAAAGGCCTTGAGTGGGTGGGACTT
ATTGATCCTGAAAATGGTGAGGCTCGATATGCAGAGAAGTTCCAGGGCAG
AGTCACCATAATCGCGGACACGTCTATAGATACAGGCTACATGGAAATGA
GGAGCCTGAAATCTGAGGACACGGCCGTGTATTTCTGTGCAGCAGGTGCC
GTGGGGGCTGATTCCGGGAGCTGGTTCGACCCCTGGGGCCAGGGAACTCT
GGTCACCGTCTCGAGC

1495_C14 (PGC14) gamma heavy chain amino acid sequence: expressed protein with variable region in bold. (SEQ ID NO: 24)

DGHLVQSGVEVKKTGATVKISCKVSGYSFIDYYLHWVQRAPGKGLEWVGL
IDPENGEARYAEKFQGRVTIIADTSIDTGYMEMRSLKSEDTAVYFCAAGA
VGADSGSWFDPWGQGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCL
VKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGT
QTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPP
KPKDTLMISRTPEVTCVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQ
YNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPRE
PQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTP
PVLDSDSGSFFLYSKLTVDKSRWQQGNVFCSSVMHEALHNHYTQKSLSLSP GK

1495_C14 (PGC14) gamma heavy chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 37)

DGHLVQSGVEVKKTGATVKISKSGYSFI custom character WVQRAPGKGLEWVG
custom character custom character RVTIIADTSIDTGYMEMRSLKSEDTAVYFCA A
custom character custom character WGQGTTLTVSS 1495_C14 gamma heavy
chain Kabat CDRs: CDR 1: (SEQ ID NO: 116) DYYLH CDR 2: (SEQ ID
NO: 117) LIDPENGARYAEKFQG CDR 3: (SEQ ID NO: 10)
GAVGADSGSWFDP 1495_C14 gamma heavy chain Chothia CDRs: CDR 1:
(SEQ ID NO: 102) GYSFID CDR 2: (SEQ ID NO: 103) LIDPENGAR
CDR 3: (SEQ ID NO: 10) GAVGADSGSWFDP 1495_C14 (PGC14) lambda
light chain nucleotide sequence: 1495_C14 λ 3 coding sequence (variable region
in bold) (SEQ ID NO: 25)
ATGGCCTGGATCCCTCTCTTCCTCGGCGTCCTTGCTTACTGCACAGATTC
CGTAGTCTCCTATGAACTGACTCAGCCACCCTCAGTGTCCGTGTCCCCAG
GACAGACAGCCAGCATCACCTGTTCTGGATCTAAATTGGGGGATAAATAT
GTTTCCTGGTATCAACTGAGGCCAGGCCAGTCCCCCATACTGGTCATGTA
TGAAAATGACAGGCGGCCCTCCGGGATCCCTGAGCGATTCTCCGGTTCCA
ATTCTGGCGACACTGCCACTCTGACCATCAGCGGGACCCAGGCTTTGGAT
GAGGCTGACTTCTACTGTCAGGCGTGGGAGACCACCACCACCTTTTGT
TTTCTTCGGCGGAGGGACCCAGCTGACCGTTCTAGGTCAGCCCAAGGCTG
CCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAGCCAAC
AAGGCCCACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGCCGTGAC
AGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGGAGACCA
CCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGCAGCTACCTG
AGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACAGCTGCCAGGT
CACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCCCTACAGAATGTT CATAG
1495_C14 (PGC14) lambda light chain variable region nucleotide sequence:
(SEQ ID NO: 119)
TCCTATGAACTGACTCAGCCACCCTCAGTGTCCGTGTCCCCAGGACAGAC
AGCCAGCATCACCTGTTCTGGATCTAAATTGGGGGATAAATATGTTTCCT
GGTATCAACTGAGGCCAGGCCAGTCCCCCATACTGGTCATGTATGAAAAT
GACAGGCGGCCCTCCGGGATCCCTGAGCGATTCTCCGGTTCCAATTCTGG
CGACACTGCCACTCTGACCATCAGCGGGACCCAGGCTTTGGATGAGGCTG
ACTTCTACTGTCAGGCGTGGGAGACCACCACCACCTTTTGTTCCTTC
GGCGGAGGGACCCAGCTGACCGTTCTA 1495_C14 (PGC14) lambda light chain
amino acid sequence: expressed protein with variable region in bold. (SEQ
ID NO: 26)
SYELTQPPSVSVSPGQTASITCSGSKLGDKEYVSWYQLRPGQSPILVMYEN
DRRPSGIPERFSGSNSGDTATLTISGTQALDEADFYCQAWETTTTTTFVFF
GGGTQLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVA
WKADSSPVKAGVETTTTPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVTH
EGSTVEKTVAPTECS 1495_C14 (PGC14) lambda light chain variable region
amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 38) SYELTQPPSVSVSPGQTASITC custom character
WYQLRPGQSPILVMY custom character custom character
GIPERFSGSNSGDTATLTISGTQALDEADFYCF custom character GGGTQLTVL
1495_C14 (PGC14) lambda light chain Kabat CDRs: CDR 1: (SEQ ID NO:
120) SGSKLGDKEYVS CDR 2: (SEQ ID NO: 121) ENDRRPS CDR 3: (SEQ
ID NO: 44) QAWETTTTTTFVF 1495_C14 (PGC14) lambda light chain Chothia
CDRs: CDR 1: (SEQ ID NO: 120) SGSKLGDKEYVS CDR 2: (SEQ ID NO:
121) ENDRRPS CDR 3: (SEQ ID NO: 44) QAWETTTTTTFVF 1496_C09
(PG9) (TCN-109) gamma heavy chain nucleotide sequence: 1496_C09 γ 3

coding sequence (variable region in bold) (SEQ ID NO: 27)
ATGGAGTTTGGGCTGAGCTGGGTTTTCTCGTTGCTTTCTTAAGAGGTGT
CCAGTGT**CAGCGATTAGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGT**
CGTCCCTGAGACTCTCCTGTGCAGCGTCCGGATTTCGACTTCAGTAGACAA
GGCATGCACTGGGTCCGCCAGGCTCCAGGCCAGGGGGCTGGAGTGGGTGGC
ATTTATTAAATATGATGGAAGTGAGAAATATCATGCTGACTCCGTATGGG
GCCGACTCAGCATCTCCAGAGACAATTCCAAGGATACGCTTTATCTCCAA
ATGAATAGCCTGAGAGTCGAGGACACGGCTACATATTTTTGTGTGAGAGA
GGCTGGTGGGCCCCGACTACCGTAATGGGTACAACCTATTACGATTTCTATG
ATGGTTATTATAACTACCACTATATGGACGTCTGGGGCAAAGGGACCACG
GTCACCGTCTCGAGCGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGC
ACCCTCCTCCAAGAGCACCTCTGGGGGACAGCGGCCCTGGGCTGCCTGG
TCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGTGGAACCTCAGGCGCC
CTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTACAGTCCTCAGGACT
CTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCC
AGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGAC
AAGAGAGTTGAGCCCCAAATCTTGTGACAAAACCTCACACATGCCACCGTG
CCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCTCTTCCCCCCAA
AACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTG
GTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGT
GGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGT
ACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGAC
TGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCC
AGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGAGAAC
CACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAG
GTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGT
GGAGTGGGAGAGCAATGGGCAGCCGGAGAACAACTACAAGACCACGCCTC
CCGTGCTGGACTCCGACGGCTCCTTCTTCTCTATAGCAAGCTCACCGTG
GACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCA
TGAGGCTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGG
GTAAATGA 1496_C09 (PG9) (TCN-109) gamma heavy chain variable region
nucleotide sequence: (SEQ ID NO: 122)

CAGCGATTAGTGGAGTCTGGGGGAGGCGTGGTCCAGCCTGGGTCTGCCCT
GAGACTCTCCTGTGCAGCGTCCGGATTTCGACTTCAGTAGACAAGGCATGC
ACTGGGTCCGCCAGGCTCCAGGCCAGGGGGCTGGAGTGGGTGGCATTAT
AAATATGATGGAAGTGAGAAATATCATGCTGACTCCGTATGGGGCCGACT
CAGCATCTCCAGAGACAATTCCAAGGATACGCTTTATCTCCAAATGAATA
GCCTGAGAGTCGAGGACACGGCTACATATTTTTGTGTGAGAGAGGCTGGT
GGGCCCCGACTACCGTAATGGGTACAACCTATTACGATTTCTATGATGGTTA
TTATAACTACCACTATATGGACGTCTGGGGCAAAGGGACCACGGTCACCG
TCTCGAGC 1496_C09 (PG9) (TCN-109) gamma heavy chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 28)
QRLVESGGGVVQPGSSLRLSCAASGFDFSRQGMHWVRQAPGQGLEWVAFI
KYDGSEKYHADSVWGRLSISRDNSKDTLYLQMNSLR VEDTATYFCVREA
GGPDYRNGYNYYDFYDGYNYHYMDVWVGKTTVTVSSASTKGPSVFPLAP
SSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLY
SLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCP
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVD
GVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPA
PIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVE

WESNGQPNENYKTPPVLDSGFFLYSKLTVDKSRWQGNVFCSCVMHE
ALHNHYTQKSLSLSPGK 1496_C09 (PG9) (TCN-109) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 39) QRLVESGGGVVQPGSSRLSCAAS**GFDGS**
WVRQAPGQGLEWVA 
RLSISRDNSTLTLYLQMNSLRVEDTATYFC VR
WGKGTTVTSS 1496_C09 (PG9) (TCN-109) gamma heavy
chain Kabat CDRs: CDR 1: (SEQ ID NO: 123) RQGMH CDR 2: (SEQ
ID NO: 124) FIKYDGSEKYHADSVWG CDR 3: (SEQ ID NO: 7)
EAGGPDYRNGYNYYDFYDGYNYHYMDV 1496_C09 (PG9) (TCN-109) gamma
heavy chain Chothia CDRs: CDR 1: (SEQ ID NO: 118) GFDFSR CDR 2:
(SEQ ID NO: 272) FIKYDGSEKY CDR 3: (SEQ ID NO: 7)
EAGGPDYRNGYNYYDFYDGYNYHYMDV 1496_C09 (PG9) (TCN-109) lambda
light chain nucleotide sequence: 1496_C09 λ 2 coding sequence (variable region
in bold) (SEQ ID NO: 29)
ATGGCCTGGGCTCTGCTTTTCCTCACCTCCTCACTCAGGGCACAGGGTC
CTGGGCCCAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTG
GACAGTCGATCACCATCTCCTGCAATGGAACCAGCAATGATGTTGGTGGC
TATGAATCTGTCTCCTGGTACCAACAACATCCCGGCAAAGCCCCCAAAGT
CGTGATTTATGATGTCAGTAAACGGGCCCTCAGGGGTTTCTAATCGCTTCT
CTGGCTCCAAGTCCGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCAG
GCTGAGGACGAGGGTGACTATTACTGCAAGTCTCGACAAGCACGAGACTG
TCGGGTTTTTCGGCACTGGGACCAAGCTGACCGTTCTAGGTCAGCCCAAGG
CTGCCCCCTCGGTCCTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAGCC
AACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGCCGT
GACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGGAGA
CCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGCAGCTAC
CTGAGCCTGACGCCTGAGCAGTGGAAGTCCCAACAAAAGCTACAGCTGCCA
GGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCCCTACAGAAT
GTTTCATAG 1496_C09 (PG9) (TCN-109) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 125)
CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTGGACAGTC
GATCACCATCTCCTGCAATGGAACCAGCAATGATGTTGGTGGCTATGAAT
CTGTCTCCTGGTACCAACAACATCCCGGCAAAGCCCCCAAAGTCGTGATT
TATGATGTCAGTAAACGGGCCCTCAGGGGTTTCTAATCGCTTCTCTGGCTC
CAAGTCCGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCAGGCTGAGG
ACGAGGGTGACTATTACTGCAAGTCTCTGACAAGCACGAGACGTCGGGTT
TTCGGCACTGGGACCAAGCTGACCGTTCTA 1496_C09 (PG9) (TCN-109) lambda
light chain amino acid sequence: expressed protein with variable region in bold.
(SEQ ID NO: 30)

QSALTQPASVSGSPGQSITISCNGTSDNDVGGYESVSWYQQHPGKAPKVVI
YDVSKRPSGVSNRFSGSKSGNTASLTISGLQAEDEGDYYCKSLTSTRRRV
FGTGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTV
AWKADSSPVKAGVETTTSPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVT
HEGSTVEKTVAPTECS 1496_C09 (PG9) (TCN-109) lambda light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 40) QSALTQPASVSGSPGQSITISC
WYQQHPGKAPKV VIY
GVSNRFSGSKSGNTASLTISGLQAEDEGDYYC FGTGTKLTVL
1496_C09 (PG9) (TCN-109) lambda light chain Kabat CDRs: CDR 1: (SEQ

ID NO: 126) NGTSNDVGGYESVS CDR 2: (SEQ ID NO: 127) DVSKRPS
CDR 3: (SEQ ID NO: 45) KSLTSTRRRV 1496_C09 (PG9) (TCN-109)
lambda light chain Chothia CDRs: CDR 1: (SEQ ID NO: 126)
NGTSNDVGGYESVS CDR 2: (SEQ ID NO: 127) DVSKRPS CDR 3:
(SEQ ID NO: 45) KSLTSTRRRV

(126) The 1443_C16 (PG16) antibody includes a heavy chain variable region (SEQ ID NO: 31), encoded by the nucleic acid sequence shown in SEQ ID NO: 99, and a light chain variable region (SEQ ID NO: 32) encoded by the nucleic acid sequence shown in SEQ ID NO: 100.

(127) The heavy chain CDRs of the 1443_C16 (PG16) antibody have the following sequences per Kabat definition: KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6). The light chain CDRs of the 1443_C16 (PG16) antibody have the following sequences per Kabat definition: NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(128) The heavy chain CDRs of the 1443_C16 (PG16) antibody have the following sequences per Chothia definition: GFTFHK (SEQ ID NO: 266), LISDDGMRKY (SEQ ID NO: 267), and EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6). The light chain CDRs of the 1443_C16 (PG16) antibody have the following sequences per Chothia definition: NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(129) The 1456 P20 (PG20) antibody includes a heavy chain variable region (SEQ ID NO: 33), encoded by the nucleic acid sequence shown in SEQ ID NO: 101, and a light chain variable region (SEQ ID NO: 34) encoded by the nucleic acid sequence shown in SEQ ID NO: 106.

(130) The heavy chain CDRs of the 1456 P20 (PG20) antibody have the following sequences per Kabat definition: SYAFT (SEQ ID NO: 104), MVTPIFGAEAKYSQRFEG (SEQ ID NO: 105), and DRRAPVIATDNWLDP (SEQ ID NO: 9). The light chain CDRs of the 1456 P20 (PG20) antibody have the following sequences per Kabat definition: RASQTINNYLN (SEQ ID NO: 107), GASNLQN (SEQ ID NO: 108), and QQSFSPTPT (SEQ ID NO: 42).

(131) The heavy chain CDRs of the 1456 P20 (PG20) antibody have the following sequences per Chothia definition: GGTFSS (SEQ ID NO: 268), MVTPIFGAEAK (SEQ ID NO: 269), and DRRAPVIATDNWLDP (SEQ ID NO: 9). The light chain CDRs of the 1456 P20 (PG20) antibody have the following sequences per Chothia definition: RASQTINNYLN (SEQ ID NO: 107), GASNLQN (SEQ ID NO: 108), and QQSFSPTPT (SEQ ID NO: 42).

(132) The 1460_G14 (PGG14) antibody includes a heavy chain variable region (SEQ ID NO: 35), encoded by the nucleic acid sequence shown in SEQ ID NO: 109, and a light chain variable region (SEQ ID NO: 36) encoded by the nucleic acid sequence shown in SEQ ID NO: 112.

(133) The heavy chain CDRs of the 1460_G14 (PGG14) antibody have the following sequences per Kabat definition: SYAFS (SEQ ID NO: 110), MITPVFGETKYAPRFQG (SEQ ID NO: 111), and DRRVVPMTDNWLDP (SEQ ID NO: 8). The light chain CDRs of the 1460_G14 (PGG14) antibody have the following sequences per Kabat definition: RASQTIHTYLN (SEQ ID NO: 113), GASTLQS (SEQ ID NO: 114), QQSYSTPTPT (SEQ ID NO: 43).

(134) The heavy chain CDRs of the 1460_G14 (PGG14) antibody have the following sequences per Chothia definition: GGAFFSS (SEQ ID NO: 270), MITPVFGETK (SEQ ID NO: 271), DRRVVPMTDNWLDP (SEQ ID NO: 8). The light chain CDRs of the 1460_G14 (PGG14) antibody have the following sequences per Chothia definition: RASQTIHTYLN (SEQ ID NO: 113), GASTLQS (SEQ ID NO: 114), QQSYSTPTPT (SEQ ID NO: 43).

(135) The 1495_C14 (PGC14) antibody includes a heavy chain variable region (SEQ ID NO: 37), encoded by the nucleic acid sequence shown in SEQ ID NO: 115, and a light chain variable region (SEQ ID NO: 38) encoded by the nucleic acid sequence shown in SEQ ID NO: 119.

(136) The heavy chain CDRs of the 1495_C14 (PGC14) antibody have the following sequences per

10 20 1495_C14_L3 ref S Y E L T Q — P P S V S V S P G Q T A S I T C 1460_G14_K1 ref D
 I Q L T Q S P S S L S A S V G D R V T V T C 1456_P20_K1 ref D I Q L T Q S P S S L S A S V
 G D R V S I T C 1443_C16_L2 ref Q S A L T Q — P A S V S G S P G Q T I T I S C
 1496_C09_L2 ref Q S A L T Q — P A S V S G S P G Q S I T I S C 30 40 1495_C14_L3 ref S
 G S K — — — L G D K Y V S W Y Q L R P G Q S 1460_G14_K1 ref R A S Q T — — — I H T
 Y L N W Y Q Q I P G K A 1456_P20_K1 ref R A S Q T — — — I N N Y L N W Y Q Q T P G K
 A 1443_C16_L2 ref N G T S S D V G G F D S V S W Y Q Q S P G K A 1496_C09_L2 ref N G
 T S N D V G G Y E S V S W Y Q Q H P G K A 50 60 1495_C14_L3 ref P I L V M Y E N D R R
 P S G I P E R F S G S N 1460_G14_K1 ref P K L L I Y G A S T L Q S G V P S R F S G S G
 1456_P20_K1 ref P K L L I Y G A S N L Q N G V P S R F S G S G 1443_C16_L2 ref P K V M
 V F D V S H R P S G I S N R F S G S K 1496_C09_L2 ref P K V V I Y D V S K R P S G V S N
 R F S G S K 70 80 90 1495_C14_L3 ref S G D T A T L T I S G T Q A L D E A D F Y C Q
 1460_G14_K1 ref S G T D F T L T I N S L Q P E D F A T Y Y C Q 1456_P20_K1 ref S G T D
 F T L T I S S L Q P E D F A T Y Y C Q 1443_C16_L2 ref S G N T A S L T I S G L H I E D E G
 D Y F C S 1496_C09_L2 ref S G N T A S L T I S G L Q A E D E G D Y Y C K 100 110
 1495_C14_L3 ref A W E T T T T T F V F F G G G T Q L T V L G 1460_G14_K1 ref Q — —
 — S Y S T P R T F G Q G T R L D I K — 1456_P20_K1 ref Q — — — S F S T P R T F G Q G
 T R L D I K — 1443_C16_L2 ref S — — L T D R S H R I F G G G T K V T V L G
 1496_C09_L2 ref S — — L T S T R R R V F G T G T K L T V L G

(143) The sequences of sister clones to human monoclonal antibody 1443_C16 (PG16) were determined, including the sequences of the variable regions of the Gamma heavy and Kappa or Lambda light chains. In addition, the sequence of each of the polynucleotides encoding the antibody sequences was determined. Shown below are the polypeptide and polynucleotide sequences of the gamma heavy chains and kappa light chains, with the signal peptides at the N-terminus (or 5' end) and the constant regions at the C-terminus (or 3' end) of the variable regions, which are shown in bolded text.

(144) TABLE-US-00013 1469_M23 (PG16)(TCN-118) gamma heavy chain nucleotide sequence: 1469_M23 y3 coding sequence (variable region in bold) (SEQ ID NO: 138)

ATGGAGTTTGGGCTGAGCTGGGTTTTTCCTCGCAACTCTGTTAAGAGTTGTGAAGTG
 TCAGGAAAAACTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCC
GGGGGGGGTCCCTGAGACTCTCCTGTTTAGCGTCTGGATTACCTTTC
ACAAATATGGCATGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCTG
GAGTGGGTGGCACTCATCTCAGATGACGGAATGAGGAAATATCATTC
AGACTCCATGTGGGGCCGAGTCACCATCTCCAGAGACAATTCCAAGA
ACACTCTATATCTGCAATTCaGCAGCCTGAAAGTCGAAGACACGGCTA
TGTTCTTCTGTGCGAGAGAGGCTGGTGGGCCAATCTGGCATGACGAC
GTCAAATATTACGATTTTAATGACGGCTACTACAACCTACCACTACATG
GACGTCTGGGGCAAGGGGACCACGGTCACCGtCTCCTCAGCGTCGACC
 AAGGGCCCATCGGTCTTCCCTCTGGCACCATCATCCAAGTCGACCTCTGG
 GGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGG
 TGACGGTGTCTGTGAACTCAGGCGCCCTGACCAGCGGCGTGCACACCTTC
 CCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGAC
 CGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATC
 ACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTG
 TGACAAAACCTCACACATGCCACCGTGCCACGACCTGAACTCCTGGGGG
 GACCGTCAGTCTTCTCTTCCCCCAAAACCCAAGGACACCCTCATGATCT
 CCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGA
 CCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATG
 CCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGT

CAGTCTCTGCTGCACTGAGCTGAGTGGCAAGGAGGTACAC
AGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATC
TCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCC
CATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTC
AAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGC
AGCCGGAGAACAATAACAAGACCACGCCTCCCGTGCTGGACTCCGACGG
CTCCTTCTTCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGC
AGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCAC
TACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 1469_M23 (PG16) (TCN-
118) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 128)
CAGGAAAACTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCGGGGGGGT
CCCTGAGACTCTCCTGTTTAGCGTCTGGATTACCTTTACAAATATGGCA
TGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCTGGAGTGGGTGGCACTC
ATCTCAGATGACGGAATGAGGAAATATCATTACAGACTCCATGTGGGGCCG
AGTCACCATCTCCAGAGACAATTCCAAGAACACTCTATATCTGCAATTCaG
CAGCCTGAAAGTCGAAGACACGGCTATGTTCTTCTGTGCGAGAGAGGCTG
GTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACGGC
TACTACAATACTACTACATGGACGTCTGGGGCAAGGGGACCACGGTCAC
CGtCTCCTCA 1469_M23 (PG16)(TCN-118) gamma heavy chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 139)

**QEKLVESGGGVVQPGGSLRLSCLASGFTFHKYGMHWVRQAPGKGLEW
VALISDDGMRKYHSDSMWGRVTISRDN SKNTLYLQFSSLKVEDTAMFFC
AREAGGPIWHDDVKYYDFNDGYNYHYMDVWVGKGT TVTVSSASTKGPS
VFPLAPSSKSTSGGTAALGCLVKDYFPEPVT VSWNSGALTSGVHTFPAVLQSS
GLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDRVEPKSCDKTHTCPPCP
APELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGV
EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHY
TQKSLSLSPGK 1469_M23 (PG16) (TCN-118) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 140) QEKLVESGGGVVQPGGSLRLSCLAScustom character
YGMHWVRQAPGKGLEWVAcustom character custom character
HSDSMWGRVTISRDN SKNTLYLQFSSLKVEDTAMFFCARcustom character
custom character custom character WVGKGT TVTVSS 1469_M23 (PG16)(TCN-118)
gamma heavy chain Kabat CDRs: (SEQ ID NO: 88) CDR 1: KYGMH (SEQ
ID NO: 89) CDR 2: LISDDGMRKYHSDSMWG (SEQ ID NO: 6) CDR 3:
EAGGPIWHDDVKYYDFNDGYNYHYMDV 1469_M23 (PG16)(TCN-118) gamma
heavy chain Chothia CDRs: (SEQ ID NO: 266) CDR 1: GFTFHK (SEQ ID
NO: 267) CDR 2: LISDDGMRKY (SEQ ID NO: 6) CDR 3:
EAGGPIWHDDVKYYDENNDGYNYHYMDV 1469_M23(PG16)(TCN-118) lambda light
chain nucleotide sequence: 1469_M23 λ2 coding sequence (variable region in
bold) (SEQ ID NO: 141)**

ATGGCCTGGGCTCTGCTATTCTCACCCTCTTCACTCAGGGCACAGGGTCC
TGGGGCCAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCT
GGACAGACGATCACCATCTCCTGCAATGGAACCAGAAGTGACGTTGG
TGGATTTGACTCTGTCTCCTGGTACCAACAATCCCCAGGGAGAGCCC
CAAAGTCATGGTTTTTGTATGTCAGTCATCGGCCCTCAGGTATCTCTA
ATCGCTTCTCTGGCTCCAAGTCCGGCAACACGGCCTCCCTGACCATC
TCTGGGCTCCACATTGAGGACGAGGGCGATTATTTCTGCTCTTCACT

GACAGACAGCAATCGCATCGCGGCGGATCCCAAGCTGACC
GTTCTAGGTCAGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTC
CTCTGAGGAGCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTG
ACTTCTACCCGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCC
GTCAAGGCGGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACA
AGTACGCGGCCAGCAGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCC
CACAAAAGCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGA
AGACAGTGGCCCCTACAGAATGTTCATAG 1469_M23 (PG16)(TCN-118) lambda
light chain variable region nucleotide sequence: (SEQ ID NO: 129)
CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTGGACAGACG
ATCACCATCTCCTGCAATGGAACCAGAAGTGACGTTGGTGGATTGACTC
TGTCTCCTGGTACCAACAATCCCCAGGGAGAGCCCCCAAAGTCATGGTTT
TTGATGTCAGTCATCGGCCCTCAGGTATCTCTAATCGCTTCTCTGGCTCCA
AGTCCGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCACATTGAGGAC
GAGGGCGATTATTTCTGCTCTTCACTGACAGACAGAAGCCATCGCATATT
CGGCGGCGGGACCAAGCTGACCGTTCTA 1469_M23 (PG16)(TCN-118) lambda
light chain amino acid sequence: expressed protein with variable region in bold.
(SEQ ID NO: 142)

QSALTQPASVSGSPGQTITISCNGTRSDVGGFDSVSWYQQSPGRAPKVMV
FDVSHRPSGISNRFSGSKSGNTASLTISGLHIEDEGDYFCSSLTDRSHRIFG
GGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKA
DSSPVKAGVETTPSKQSNKYAASSYLSLTPEQWKSHKSYSCQVTHEGSTV
EKTVPTECS 1469_M23 (PG16) (TCN-118) lambda light chain variable region
amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)
(SEQ ID NO: 96) **QSALTQPASVSGSPGQTITISC** custom character
WYQQSPGRAPKVMVF custom character custom character
GISNRFSGSKSGNTASLTISGLHIEDEGDYFC custom character **FGGGTKL TVL**
1469_M23 (PG16)(TCN-118) lambda light chain Kabat CDRs: (SEQ ID NO: 92)
CDR 1: NGTRSDVGGFDSVS (SEQ ID NO: 95) CDR 2: DVSHRPS (SEQ ID
NO: 41) CDR 3: SSLTDRSHRI 1469_M23(PG16)(TCN-118) lambda light chain
Chothia CDRs: (SEQ ID NO: 92) CDR 1: NGTRSDVGGFDSVS (SEQ ID NO:
95) CDR 2: DVSHRPS (SEQ ID NO: 41) CDR 3: SSLTDRSHRI 1456_A12 (PG16)
(TCN-117) gamma heavy chain nucleotide sequence: 1456_A12 γ3 coding
sequence (variable region in bold) (SEQ ID NO: 46)
ATGGAGTTTGGGCTGAGCTGGGTTTTCTCGCAACTCTGTTAAGAGTTGTG
AAGTGTACGAACAACACTGGTGGAGGCCGGGGGAGGCGTGGTCCAGC
CGGGGGGGTCCCTGAGACTCTCCTGTTTAGCGTCTGGATTACGTTT
CACAAATATGGCATGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCT
GGAGTGGGTGGCACTCATCTCAGATGACGGAATGAGGAAATATCATT
CAGACTCCATGTGGGGCCGAGTCACCATCTCCAGAGACAATTCCAAG
AACACTCTTTATCTGCAATTCAGCAGCCTGAGAGTCGAAGACACGGC
TATGTTCTTCTGTGCGAGAGAGGCCGGTGGGCCAATCTGGCATGACG
ACGTCAAATATTACGATTTTAATGACGGCTACTACAACCTATCACTACA
TGGACGTCTGGGGCAAGGGGACCAAGGTCACCGTCTCCTCAGCGTCG
ACCAAGGGCCCATCGGTCTTCCCTCTGGCACCATCATCCAAGTCGACCTCT
GGGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACC
GGTGACGGTGTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGACACCT
TCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGA
CCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAAT
CACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTT

GTGACAAACCAACTACATGACCATGCCCACCGTCCAGCCACCTGCAACCTGCTGGGG
GGACCGTCAGTCTTCCTCTTCCCCCCTCAAAACCCAAGGACACCCTCATGAT
CTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAG
ACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAAT
GCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGG
TCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTAC
AAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCAT
CTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCC
CCATCCCGGGAGGAGATGACCAAGAACCAGGTGACCTGACCTGCCTGGT
CAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGG
CAGCCGGAGAACAACACTACAAGACCACGCCTCCCGTGCTGGACTCCGACG
GCTCCTTCTTCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAG
CAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCA
CTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 1456_A12 (PG16) (TCN-
117) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 130)
CACGAACAACCTGGTGGAGGCCGGGGGAGGCGTGGTCCAGCCGGGGGGGGT
CCCTGAGACTCTCCTGTTTAGCGTCTGGATTACGTTTCACAAATATGGCA
TGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCTGGAGTGGGTGGCACTC
ATCTCAGATGACGGAATGAGGAAATATCATTGAGACTCCATGTGGGGCCG
AGTCACCATCTCCAGAGACAATTCCAAGAACACTCTTTATCTGCAATTCA
GCAGCCTGAGAGTCGAAGACACGGCTATGTTCTTCTGTGCGAGAGAGGCC
GGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACGG
CTACTACAACCTATCACTACATGGACGTCTGGGGCAAGGGGACCAAGGTCA
CCGTCTCCTCA 1456_A12 (PG16)(TCN-117) gamma heavy chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 47)
HEQLVEAGGGVVQPGGSLRLSCLASGFTFHKYGMHWVRQAPGKGLEW
VALISDDGMRKYHSDSMWGRVTISRDN SKNTLYLQFSSLRVEDTAMFFC
AREAGGPIWHDDVKYYDFNDGYNYHYMDVWGKGTKVTVSSASTKGPS
VFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS
GLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCP
APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGV
EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSVCSVMHEALHNHY
TQKSLSLSPGK 1456_A12 (PG16) (TCN-117) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 48) HEQLVEAGGGVVQPGGSLRLSCLAS
YGMHWVRQAPGKGLEWVA
HSDSMWGRVTISRDN SKNTLYLQFSSLRVEDTAMFFCAR
WGKGTKVTVSS 1456_A12 (PG16)(TCN-117)
gamma heavy chain Kabat CDRs: (SEQ ID NO: 88) CDR 1: KYGMH (SEQ
ID NO: 89) CDR 2: LISDDGMRKYHSDSMWG (SEQ ID NO: 6) CDR 3:
EAGGPIWHDDVKYYDFNDGYNYHYMDV 1456_A12 (PG16)(TCN-117) gamma
heavy chain Chothia CDRs: (SEQ ID NO: 266) CDR 1: GFTFHK (SEQ ID
NO: 267) CDR 2: LISDDGMRKY (SEQ ID NO: 6) CDR 3:
EAGGPIWHDDVKYYDFNDGYNYHYMDV 1456_A12 (PG16)(TCN-117) lambda
light chain nucleotide sequence: 1456_A12 λ2 coding sequence (variable region
in bold) (SEQ ID NO: 49)
ATGGCCTGGGCTTGCTATTCCCTCACCCCTCTTCACTCAGGGCACAGGGTCCT
GGGGCCAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCT

GGACGATCATGCTATCTCTGCAATGGAACTGCGCTGCTGG
TGGATTTGACTCTGTCTCCTGGTATCAACAATCCCCAGGGAAAGCCC
CCAAAGTCATGGTTTTTTGATGTCAGTCATCGGCCCTCAGGTATGTCTA
ATCGCTTCTCTGGCTCCAAGTCCGGCAACACGGCCTCCCTGACCATT
TCTGGGCTCCACATTGAGGACGAGGGCGATTATTTCTGCTCTTCATT
GACAGACAGAAGCCATCGCATATTCGGCGGGCGGGACCAAGCTGACC
GTTCTAGGTCAGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTC
CTCTGAGGAGCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTG
ACTTCTACCCGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCC
GTCAAGGCGGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACA
AGTACGCGGCCAGCAGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCC
CACAAAAGCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGAGAGA
AGACAGTGGCCCCTACAGAATGTTTCATAG 1456_A12 (PG16)(TCN-117) lambda
light chain variable region nucleotide sequence: (SEQ ID NO: 131)
CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTGGACAGACG
ATCACCATCTCCTGCAATGGAACCAGCCGTGACGTTGGTGGATTGACTCT
GTCTCCTGGTATCAACAATCCCCAGGGAAAGCCCCCAAAGTCATGGTTTT
TGATGTCAGTCATCGGCCCTCAGGTATGTCTAATCGCTTCTCTGGCTCCAA
GTCCGGCAACACGGCCTCCCTGACCATTCTGGGCTCCACATTGAGGACG
AGGGCGATTATTTCTGCTCTTCATTGACAGACAGAAGCCATCGCATATTCG
GCGGCGGGACCAAGCTGACCGTTCTA 1456_A12 (PG16)(TCN-117) lambda light
chain amino acid sequence: expressed protein with variable region in bold.
(SEQ ID NO: 50)

QSALTQPASVSGSPGQTITISCNGTSRDVGGFDSVSWYQQSPGKAPKVMV
FDVSHRPSGMSNRFSGSKSGNTASLTISGLHIEDEGDYFCSSLTDRSHRI
GGGTKLTLVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAW
KADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVTHEGS
TVEKTVAPTECS 1456_A12 (PG16) (TCN-117) lambda light chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 51) **QSALTQPASVSGSPGQTITISC** custom character
WYQQSPGKAPKVMV custom character custom character
GMSNRFSGSKSGNTASLTISGLHIEDEGDYFC custom character FGGGTK LTVL
1456_A12 (PG16)(TCN-117) lambda light chain Kabat CDRs: (SEQ ID NO: 93)
CDR 1: NGTSRDVGGFDSVS (SEQ ID NO: 95) CDR 2: DVSHRPS (SEQ ID
NO: 41) CDR 3: SSLTDRSHRI 1456_A12 (PG16)(TCN-117) lambda light chain
Chothia CDRs: (SEQ ID NO: 93) CDR 1: **NGTSRDVGGFDSVS** (SEQ ID NO:
95) CDR 2: **DVSHRPS** (SEQ ID NO: 41) CDR 3: **SSLTDRSHRI** 1503_H05 (PG16)
(TCN-119) gamma heavy chain nucleotide sequence: 1503_H05 y3 coding
sequence (variable region in bold) (SEQ ID NO: 52)

ATGGAGTTTGGCTGAGCTGGGTTTTCTCGCAACTCTGTAAAGAGTTGTGA
AGTGTCAAGGAAAACTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCG
GGGGGGTCCCTGAGACTCTCCTGTTTAGCGTCTGGATTCACCTTTCA
CAAATATGGCATGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCTGG
AGTGGGTGGCACTCATCTCAGATGACGGAATGAGGAAATATCATTCA
GACTCCATGTGGGGCCGAGTCACCATCTCCAGAGACAATTCCAAGAA
CACTTTATATCTGCAATTCAGCAGCCTGAAAGTCGAAGACACGGCTA
TGTTCTTCTGTGCGAGAGAGGCTGGTGGGCCAATCTGGCATGACGAC
GTCAAATATTACGATTTTAATGACGGCTACTACAATTACCACTACATG
GACGTCTGGGGCAAGGGGACCATTGTCACCGTCTCCTCAGCGTCGAC
CAAGGGCCCATCGGTCTTCCCTCTGGCACCATCATCCAAGTCGACCTCTG

GGGGCGGACGCGGACGCGGCTGAGGCAAGGCACTGCTTCCCGGACCGG
GTGACGGTGTTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTT
CCCGGCTGTCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGA
CCGTGCCCTCCAGCAGCTTGGGACCCAGACCTACATCTGCAACGTGAAT
CACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTT
GTGACAAAACCTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGG
GGACCGTCAGTCTTCTCTTCCCCCCTTCCCAAGGACACCCTCATGAT
CTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAG
ACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAAT
GCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGG
TCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTAC
AAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCAT
CTCCAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCC
CCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGT
CAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGG
CAGCCGGAGAACAACTACAAGACCACGCCTCCCGTGCTGGACTCCGACG
GCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAG
CAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCA
CTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 1503_H05 (PG16) (TCN-
119) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 132)
CAGGAAAACTGGTGGAGTCTGGGGGAGGCGTGGTCCAGCCGGGGGGGT
CCCTGAGACTCTCTGTGTTAGCGTCTGGATTCACCTTTCACAAATATGGCA
TGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCTGGAGTGGGTGGCACTC
ATCTCAGATGACGGAATGAGGAAATATCATTCAGACTCCATGTGGGGCCG
AGTCACCATCTCCAGAGACAATTCCAAGAACACTTTATATCTGCAATTCA
GCAGCCTGAAAGTCGAAGACACGGCTATGTTCTTCTGTGCGAGAGAGGCT
GGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACGG
CTACTACAATTACCACTACATGGACGTCTGGGGCAAGGGGACCATTGTCA
CCGTCTCCTCA 1503_H05 (PG16)(TCN-119) gamma heavy chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 53)
QEKLVESGGGVVQPGGSLRLSCLASGFTFHKYGMHWVRQAPGKGLEW
VALISDDGMRKYHSDSMWGRVTISRDN SKNTLYLQFSSLKVEDTAMFFC
AREAGGPIWHDDVKYYDENDGYNYHYMDVWGKGTIVTVSSASTKGPS
VFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS
GLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVKDKRVEPKSCDKTHTCPPCP
APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGV
EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHY
TQKSLSLSPGK. 1503_H05 (PG16) (TCN-119) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 54) QEKLVESGGGVVQPGGSLRLSCLAS
YGMHWVRQAPGKGLEWVA
HSDSMWGRVTISRDN SKNTLYLQFSSLKVEDTAMFFCAR
WGKGTIVTVSS 1503_H05 (PG16)(TCN-119)
gamma heavy chain Kabat CDRs: (SEQ ID NO: 88) CDR 1: KYGMH (SEQ
ID NO: 89) CDR 2: LISDDGMRKYHSDSMWG (SEQ ID NO: 6) CDR 3:
EAGGPIWHDDVKYYDFNDGYNYHYMDV 1503_H05 (PG16)(TCN-119) gamma
heavy chain Chothia CDRs: (SEQ ID NO: 266) CDR 1: GFTFHK (SEQ ID
NO: 267) CDR 2: LISDDGMRKY (SEQ ID NO: 6) CDR 3:

1503_H05 (PG16)(TCN-119) lambda
light chain nucleotide sequence: 1503_H05 λ2 coding sequence (variable region
in bold)(SEQ ID NO: 55)

ATGGCCTGGGCTTGCTATTCTCCTCACCCTCTTCACTCAGGGCACAGGGTCCT
GGGGCCAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCT
GGACAGACGATCACCATCTCCTGCAATGGAACCAGAAGTGACGTTGG
TGGATTTGACTCTGTCTCCTGGTACCAACAATCCCCAGGGAAAGCCC
CCAAAGTCATGGTTTTTGATGTCAGTCATCGGCCCTCAGGTATCTCTA
ATCGCTTCTCTGGCTCCAAGTCCGGCAACACGGCCTCCCTGACCATC
TCTGGGCTCCACATTGAGGACGAGGGCGATTATTTCTGCTCTTCACT
GACAGACAGAAGCCATCGCATATTCGGCGGGCGGGACCAAGGTGACC
GTTCTAGGTCAGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTC
CTCTGAGGAGCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTG
ACTTCTACCCGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCC
GTCAAGGCGGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACA
AGTACGCGGCCAGCAGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCC
CACAAAAGCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGA
AGACAGTGGCCCCTACAGAATGTTTCATAG 1503_H05 (PG16)(TCN-119) lambda

light chain variable region nucleotide sequence: (SEQ ID NO: 133)
CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTGGACAGACG
ATCACCATCTCCTGCAATGGAACCAGAAGTGACGTTGGTGGATTTGACTC
TGTCTCCTGGTACCAACAATCCCCAGGGAAAGCCCCCAAAGTCATGGTTT
TTGATGTCAGTCATCGGCCCTCAGGTATCTCTAATCGCTTCTCTGGCTCCA
AGTCCGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCACATTGAGGAC
GAGGGCGATTATTTCTGCTCTTCACTGACAGACAGAAGCCATCGCATATT
CGGCGGGCGGGACCAAGGTGACCGTTCTA 1503_H05 (PG16) (TCN-119) lambda
light chain amino acid sequence: expressed protein with variable region in bold.
(SEQ ID NO: 56)

QSALTQPASVSGSPGQTITISCNGTRSDVGGFDSVSWYQQSPGKAPKVMV
FDVSHRPSGISNRFSGSKSGNTASLTISGLHIEDEGDYFCSSLTDRSHRIFG
GGTKVTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK
ADSSPVKAGVETTPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVTHEGST
VEKTVAPTECS 1503_H05 (PG16)(TCN-119) lambda light chain variable region
amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)
(SEQ ID NO: 57) QSALTQPASVSGSPGQTITISC

WYQQSPGKAPKVMVF custom character custom character
GISNRFSGSKSGNTASLTISGLHIEDEGDYFC custom character FGGGTK VTVL
1503_H05 (PG16)(TCN-119) lambda light chain Kabat CDRs: (SEQ ID NO: 92)
CDR 1: NGTRSDVGGFDSVS (SEQ ID NO: 95) CDR 2: DVSHRPS (SEQ ID
NO: 41) CDR 3: SSLTDRSHRI 1503_H05 (PG16)(TCN-119) lambda light chain
Chothia CDRs: (SEQ ID NO: 92) CDR 1: NGTRSDVGGFDSVS (SEQ ID NO:
95) CDR 2: DVSHRPS (SEQ ID NO: 41) CDR 3: SSLTDRSHRI 1489_I13 (PG16)
(TCN-120) gamma heavy chain nucleotide sequence: 1489_113 γ3 coding
sequence (variable region in bold)(SEQ ID NO: 58)

ATGGAGTTTGGGCTGAGCTGGGTTTTCTCGCAACTCTGTTAAGAGTTGTG
AAGTGTCAGGAACAACCTGTTGGAGTCTGGGGGAGGCGTGGTCCAGCC
GGGGGGGTCCCTGAGACTCTCCTGTTTAGCGTCTGGATTCACGTTTC
ACAAATATGGCATGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCTG
GAGTGGGTGGCACTCATCTCAGATGACGGAATGAGGAAATATCATTC
AAACTCCATGTGGGGCCGAGTCACCATCTCCAGAGACAATTCCAAGA

ACACTTCTTATCTGAAGTCAAGACACGGCT
ATGTTCTTCTGTGCGAGAGAGGCTGGTGGGCCAATCTGGCATGACGA
CGTCAAATATTACGATTTTAATGACGGCTACTACAACCTACCACTACAT
GGACGTCTGGGGCAAGGGGACCACGGTCACCGTCTCCTCAGCGTCTGA
CCAAGGGCCCATCGGTCTTCCCTCTGGCACCATCATCCAAGTCGACCTCTG
GGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCG
GTGACGGTGTCTGGAAGTCAAGGCGCCCTGACCAGCGGCGTGCACACCTT
CCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGA
CCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAAT
CACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTT
GTGACAAAACCTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGG
GGACCGTCAGTCTTCTCTTCCCCCAAAACCCAAGGACACCCTCATGAT
CTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAG
ACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAAT
GCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGG
TCAGCGTCCTCACCGTCTTGCACCAGGACTGGCTGAATGGCAAGGAGTAC
AAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCAT
CTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCC
CCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGT
CAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGG
CAGCCGGAGAACAACCTACAAGACCACGCCTCCCGTGCTGGACTCCGACG
GCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAG
CAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCA
CTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 1489_I13 (PG16)(TCN-
120) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO:
134) CAGGAACAACCTGTTGGAGTCTGGGGGAGGCGTGGTCCAGCCGGGGGGGT
CCCTGAGACTCTCCTGTTTAGCGTCTGGATTCACGTTTCACAAATATGGCA
TGCACTGGGTCCGCCAGGCTCCAGGCAAGGGCCTGGAGTGGGTGGCACTC
ATCTCAGATGACGGAATGAGGAAATATCATTCAAACTCCATGTGGGGCCG
AGTCACCATCTCCAGAGACAATTCCAAGAACTCTTTATCTGCAATTCA
GCAGCCTGAAAGTCGAAGACACGGCTATGTTCTTCTGTGCGAGAGAGGCT
GGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACGG
CTACTACAACCTACCACTACATGGACGTCTGGGGCAAGGGGACCACGGTCA
CCGTCTCCTCA 1489_I13 (PG16) (TCN-120) gamma heavy chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 59)

QEQLLESGGGVVQPGGSLRLSCLASGFTFHKYGMHWVRQAPGKGLEW
VALISDDGMRKYHSNSMWGRVTISRDN SKNTLYLQFSSLKVEDTAMFFC
AREAGGPIWHDDVKYYDFNDGYNYHYMDVWVGKGT TVTVSSASTKGPS
VFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSS
GLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCP
APPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGV
EVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV FSCSVMHEALHNHY
TQKSLSLSPGK 1489_I13 (PG16) (TCN-120) gamma heavy chain variable region
amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)
(SEQ ID NO: 60) QEQLLESGGGVVQPGGSLRLSCLAScustom character
YGMHWVRQAPGKGLEWVAcustom character custom character
HSNSMWGRVTISRDN SKNTLYLQFSSLKVEDTAMFFCARcustom character
custom character custom character WGKGT TVTVSS 1489_I13 (PG16)(TCN-120)

gamma heavy chain Kabat CDRs: (SEQ ID NO: 88) CDR 1: KYGMH (SEQ ID NO: 98) CDR 2: LISDDGMRKYHSNSMWG (SEQ ID NO: 6) CDR 3: EAGGPIWHDDVKYYDFNDGYNYHYMDV 1489_I13 (PG16)(TCN-120) gamma heavy chain Chothia CDRs: (SEQ ID NO: 266) CDR 1: GFTFHK (SEQ ID NO: 267) CDR 2: LISDDGMRKY (SEQ ID NO: 6) CDR 3: EAGGPIWHDDVKYYDFNDGYNYHYMDV 1489_I13 (PG16)(TCN-120) lambda light chain nucleotide sequence: 1489_113 λ 2 coding sequence (variable region in bold) (SEQ ID NO: 61)

ATGGCCTGGGCTCTGCTATTCCTCACCTCTTCACTCAGGGCACAGGGTCC
CGGGGCCAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCT
GGACAGACGATCACCATCTCCTGCAATGGAACCAGCAGTGACGTTGG
TGGATTTGACTCTGTCTCCTGGTATCAACAATCCCCAGGGAAAGCCC
CCAAAGTCATGGTTTTTGATGTCAGTCATCGGCCCTCAGGTATCTCTA
ATCGCTTCTCTGGCTCCAAGTCCGGCAACACGGCCTCCCTGACCATC
TCTGGGCTCCACATTGAGGACGAGGGCGATTATTTCTGCTCTTCACT
GACAGACAGAAGCCATCGCATATTCGGCGGGCGGGACCAAGGTGACC
GTTCTAGGTCAGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTC
CTCTGAGGAGCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTG
ACTTCTACCCGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCC
GTCAAGGCGGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACA
AGTACGCGGCCAGCAGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCC
CACAAAAGCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGAGAGA
AGACAGTGGCCCCTACAGAATGTTTCATAG 1489_I13 (PG16)(TCN-120) lambda light chain variable region nucleotide sequence: (SEQ ID NO: 135)

CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTGGACAGACG
ATCACCATCTCCTGCAATGGAACCAGCAGTGACGTTGGTGGATTTGACTC
TGTCTCCTGGTATCAACAATCCCCAGGGAAAGCCCCCAAAGTCATGGTTT
TTGATGTCAGTCATCGGCCCTCAGGTATCTCTAATCGCTTCTCTGGCTCCA
AGTCCGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCACATTGAGGAC
GAGGGCGATTATTTCTGCTCTTCACTGACAGACAGAAGCCATCGCATATT
CGGCGGGCGGGACCAAGGTGACCGTTCTA 1489_I13 (PG16) (TCN-120) lambda light chain amino acid sequence: expressed protein with variable region in bold. (SEQ ID NO: 14)

QSALTQPASVSGSPGQTITISCNGTSSDVGGFDSVSWYQQSPGKAPKVMV
FDVSHRPSGISNRFSGSKSGNTASLTISGLHIEDEGDYFCSSLTDRSHRIFG
GGTKVTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWK
ADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVTHEGST
VEKTVAPTECS 1489_I13 (PG16)(TCN-120) lambda light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics).

(SEQ ID NO: 32) QSALTQPASVSGSPGQTITISCcustom character

WYQQSPGKAPKVMVFcustom character custom character

GISNRFSGSKSGNTASLTISGLHIEDEGDYFCcustom character FGGGTK VTVL 1489_I13 (PG16)(TCN-120) lambda light chain Kabat CDRs: (SEQ ID NO: 97) CDR 1: NGTSSDVGGFDSVS (SEQ ID NO: 95) CDR 2: DVSHRPS (SEQ ID NO: 41) CDR 3: SSLTDRSHRI 1489_I13 (PG16)(TCN-120) lambda light chain Chothia CDRs: (SEQ ID NO: 97) CDR 1: NGTSSDVGGFDSVS (SEQ ID NO: 95) CDR 2: DVSHRPS (SEQ ID NO: 41) CDR 3: SSLTDRSHRI 1480_I08 gamma heavy chain nucleotide sequence: 1480_108 γ 3 coding sequence (variable region in bold) (SEQ ID NO: 64)

ATGGAGTTTGGCTGAGCTGGGTTTTCTCGCAACTCTGTAAAGAGTTGTGA

sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 31) QEQLVESGGGVVQPGGSLRLSCLAS YGMHWVRQAPGKGLEWVA  HSDSMWGRVTISRDNSKNTLYLQFSSLKVEDTAMFFCAR   WGKGTTVTVSS 1480_I08 gamma heavy chain Kabat CDRs: (SEQ ID NO: 88) CDR 1: KYGMH (SEQ ID NO: 89) CDR 2: LISDDGMRKYHSDSMWG (SEQ ID NO: 6) CDR 3: EAGGPIWHDDVKYYDFNDGYNYHYMDV 1480_I08 gamma heavy chain Chothia CDRs: (SEQ ID NO: 266) CDR 1: GFTFHK (SEQ ID NO: 267) CDR 2: LISDDGMRKY (SEQ ID NO: 6) CDR 3: EAGGPIWHDDVKYYDFNDGYNYHYMDV 1480_I08 lambda light chain nucleotide sequence: 1480_I08 λ 2 coding sequence (variable region in bold) (SEQ ID NO: 67)

ATGGCCTGGGCTCTGCTATTCGTCACCCTCCTCACTCAGGGCACAGGGTCC
TGGGGCCAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCT
GGACAGACGATCACCATCTCCTGCAATGGAACCAGCAGTGACGTTGG
TGGATTTGACTCTGTCTCCTGGTATCAACAATCCCCAGGGAAAGCCC
CCAAAGTCATGGTTTTTGATGTCAGTCATCGGCCCTCAGGTATCTCTA
ATCGCTTCTCTGGCTCCAAGTCCGGCAACACGGCCTCCCTGACCATC
TCTGGGCTCCACATTGAGGACGAGGGCGATTATTTCTGCTCTTCACT
GACAGACAGAAGCCATCGCATATTCGGCGGGCGGGACCAAGGTGACC
GTTCTAGGTCAGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTC
CTCTGAGGAGCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTG
ACTTCTACCCGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCC
GTCAAGGCGGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACA
AGTACGCGGCCAGCAGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCC
CACAAAAGCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGA
AGACAGTGGCCCCTACAGAATGTTTCATAG 1480_I08 lambda light chain variable
region nucleotide sequence: (SEQ ID NO: 137)

CAGTCTGCCCTGACTCAGCCTGCCTCCGTGTCTGGGTCTCCTGGACAGACG
ATCACCATCTCCTGCAATGGAACCAGCAGTGACGTTGGTGGATTTGACTC
TGTCTCCTGGTATCAACAATCCCCAGGGAAAGCCCCCAAAGTCATGGTTT
TTGATGTCAGTCATCGGCCCTCAGGTATCTCTAATCGCTTCTCTGGCTCCA
AGTCCGGCAACACGGCCTCCCTGACCATCTCTGGGCTCCACATTGAGGAC
GAGGGCGATTATTTCTGCTCTTCACTGACAGACAGAAGCCATCGCATATT
CGGCGGCGGGACCAAGGTGACCGTTCTA 1480_I08 lambda light chain amino
acid sequence: expressed protein with variable region in bold. (SEQ ID NO: 14)

QSALTQPASVSGSPGQTITISC **WYQQSPGKAPKVMV**
FDVSHRPSGISNRFSGSKSGNTASLTISGLHIEDEGDYFC**SSLTDRSHRIFG**
GGTKVTVLGQPKAAPSVTFLFPSSEELQANKATLVCLISDFYPGAVTVAWK
ADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVTHEGST
VEKTVAPTECS 1480_I08 lambda light chain variable region amino acid
sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 32) QSALTQPASVSGSPGQTITISC WYQQSPGKAPKVMV
  GISNRFSGSKSGNTASLTISGLHIEDEGDYFC
 FGGGTK VTVL 1480_I08 lambda light chain Kabat CDRs: (SEQ ID NO: 97) CDR 1: NGTSSDVGGFDSVS (SEQ ID NO: 95) CDR 2: DVSHRPS (SEQ ID NO: 41) CDR 3: SSLTDRSHRI 1480_I08 lambda light chain Chothia CDRs: (SEQ ID NO: 97) CDR 1: NGTSSDVGGFDSVS (SEQ ID NO: 95) CDR 2: DVSHRPS (SEQ ID NO: 41) CDR 3: SSLTDRSHRI

(145) The 1469_M23 (PG16) antibody includes a heavy chain variable region (SEQ ID NO: 139), encoded by the nucleic acid sequence shown in SEQ ID NO: 128, and a light chain variable region (SEQ ID NO: 142) encoded by the nucleic acid sequence shown in SEQ ID NO: 129.

(146) The heavy chain CDRs of the 1469_M23 (PG16) antibody have the following sequences per Kabat and Chothia definitions: KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), and EAGGPWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6). The light chain CDRs of the 1469 M23 (PG16) antibody have the following sequences per Kabat and Chothia definitions: NGTRSDVGGFDSVS (SEQ ID NO: 92), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(147) The 1456_A12 (PG16) antibody includes a heavy chain variable region (SEQ ID NO: 47), encoded by the nucleic acid sequence shown in SEQ ID NO: 130, and a light chain variable region (SEQ ID NO: 50) encoded by the nucleic acid sequence shown in SEQ ID NO: 131.

(148) The heavy chain CDRs of the 1456_A12 (PG16) antibody have the following sequences per Kabat and Chothia definitions: KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), and EAGGPWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6). The light chain CDRs of the 1456 A12 (PG16) antibody have the following sequences per Kabat and Chothia definitions: NGTSRDVGGFDSVS (SEQ ID NO: 93), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(149) The 1503_H05 (PG16) antibody includes a heavy chain variable region (SEQ ID NO: 53), encoded by the nucleic acid sequence shown in SEQ ID NO: 132, and a light chain variable region (SEQ ID NO: 56) encoded by the nucleic acid sequence shown in SEQ ID NO: 133.

(150) The heavy chain CDRs of the 1503_H05 (PG16) antibody have the following sequences per Kabat and Chothia definitions: KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), and EAGGPWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6). The light chain CDRs of the 1503_H05 (PG16) antibody have the following sequences per Kabat and Chothia definitions: NGTRSDVGGFDSVS (SEQ ID NO: 92), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(151) The 1489_I13 (PG16) antibody includes a heavy chain variable region (SEQ ID NO: 59), encoded by the nucleic acid sequence shown in SEQ ID NO: 134, and a light chain variable region (SEQ ID NO: 14) encoded by the nucleic acid sequence shown in SEQ ID NO: 135.

(152) The heavy chain CDRs of the 1489_I13 (PG16) antibody have the following sequences per Kabat and Chothia definitions: KYGMH (SEQ ID NO: 88), LISDDGMRKYHSNSMWG (SEQ ID NO: 98), and EAGGPWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6). The light chain CDRs of the 1489_I13 (PG16) antibody have the following sequences per Kabat and Chothia definitions: NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(153) The 1480_I08 (PG16) antibody includes a heavy chain variable region (SEQ ID NO: 65), encoded by the nucleic acid sequence shown in SEQ ID NO: 136, and a light chain variable region (SEQ ID NO: 14) encoded by the nucleic acid sequence shown in SEQ ID NO: 137.

(154) The heavy chain CDRs of the 1480_I08 (PG16) antibody have the following sequences per Kabat and Chothia definitions: KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89), and EAGGPWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6). The light chain CDRs of the 1480_I08 (PG16) antibody have the following sequences per Kabat and Chothia definitions: NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), and SSLTDRSHRI (SEQ ID NO: 41).

(155) The sequences of additional human monoclonal antibodies were determined, including the sequences of the variable regions of the Gamma heavy and Kappa or Lambda light chains. In addition, the sequence of each of the polynucleotides encoding the antibody sequences was determined. Shown below are the polypeptide and polynucleotide sequences of the gamma heavy chains and kappa light chains, with the signal peptides at the N-terminus (or 5' end) and the

constant regions at the C-terminus (or 3' end) of the variable regions, which are shown in bolded text.

(156) TABLE-US-00014 4838_L06 (PGT-121) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 62)

ATGAAACACCTGTGGTTCTTCCTTCTCCTGGTGGCAGCTCCCAGATGGGTC
CTGTACAGATGCAGTTACAGGAGTCGGGCCCCGGACTGGTGAAGCC
TTCGGAACCCCTGTCCCTCACGTGCAGTGTGTCTGGTGCCTCCATAA
GTGACAGTTACTGGAGCTGGATCCGGCGGTCCCCAGGGAAGGGACTT
GAGTGGATTGGGTATGTCCACAAAAGCGGCGACACAAATTACAGCCC
CTCCCTCAAGAGTCGAGTCAACTTGTCGTTAGACACGTCCAAAAATC
AGGTGTCCCTGAGCCTTGTGGCCGCGACCGCTGCGGACTCGGGCAA
TATTATTGCGCGAGAACACTGCACGGGAGGAGAATTTATGGAATCGT
TGCCTTCAATGAGTGGTTCACCTACTTCTACATGGACGTCTGGGGCA
ATGGGACTCAGGTCACCGTCTCCTCAGCCTCCACCAAGGGCCCATCGG
TCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCC
CTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGTG
GAACTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTAC
AGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGC
AGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAA
CACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCAC
ACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTT
CCTCTTCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTG
AGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAA
GTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAG
CCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCAC
CGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCT
CCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAA
GGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGG
AGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTAT
CCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACA
ACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCT
ATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTT
CTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGA
GCCTCTCCCTGTCTCCGGGTAAATGA

4838_L06 (PGT-121) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 63)

CAGATGCAGTTACAGGAGTCGGGCCCCGGACTGGTGAAGCCTTCGGAAA
CCCTGTCCCTCACGTGCAGTGTGTCTGGTGCCTCCATAAGTGACAGTTACT
GGAGCTGGATCCGGCGGTCCCCAGGGAAGGGACTTGAGTGGATTGGGTAT
GTCCACAAAAGCGGCGACACAAATTACAGCCCCTCCCTCAAGAGTCGAGT
CAACTTGTCGTTAGACACGTCCAAAAATCAGGTGTCCCTGAGCCTTGTGG
CCGCGACCGCTGCGGACTCGGGCAAATATTATTGCGCGAGAACACTGCAC
GGGAGGAGAATTTATGGAATCGTTGCCTTCAATGAGTGGTTCACCTACTT
CTACATGGACGTCTGGGGCAATGGGACTCAGGTCACCGTCTCCTCA

4838_L06 (PGT-121) gamma heavy chain amino acid sequence: expressed protein with variable region in bold. (SEQ ID NO: 66)

QMQLQESGPLVKPSETLSLTCSVSGASISDSYWSWIRRSPGKGLEWIGY
VHKSGDTNYSPLKSRVNLSLDTSKNQVSLSLVAATAADSGKYCARTLH
GRRYIGIVAFNEWFTYFYMDVWGNGTQVTVSSASTKGPSVFPLAPSSKSTS
GGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTV
PSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVF

LFPPKDTLTKMPTPEVTVDFVKNVYVDGVVEVHNAKTKPR
EEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPP
VLDSGDGSSFFLYSKLTVDKSRWQQGNVFCSSVMHEALHNHYTQKSLSLSPGK 4838_L06
(PGT-121) gamma heavy chain variable region amino acid sequence: (Kabat
CDRs underlined, Chothia CDRs in bold italics)(SEQ ID NO: 79)
QMQLQESGPGLVKPSETLSLTCSVScustom character SYWSWIRRSPPGKGLEWIG
custom character custom character

NYSPSLKSRVNLSLDTSKNQVSLSLVAATAADSGKYCAR
custom character WGNGTQVTVSS 4838_L06 (PGT-121) gamma heavy chain
Kabat CDRs:(SEQ ID NO: 90) CDR 1: DSYWS (SEQ ID NO: 265) CDR 2:
YVHKSGDTNYSPSLKS (SEQ ID NO: 143) CDR 3:
TLHGRRRIYGIVAFNEWFTYFYMDV 4838_L06 (PGT-121) gamma heavy chain
Chothia CDRs:(SEQ ID NO: 144) CDR 1: GASISD (SEQ ID NO: 145) CDR
2: YVHKSGDTN (SEQ ID NO: 143) CDR 3: TLHGRRRIYGIVAFNEWFTYFYMDV
4838_L06 (PGT-121) lambda light chain nucleotide sequence: coding sequence
(variable region in bold)(SEQ ID NO: 146)

ATGGCCTGGACCTTTCTCCTCCTCGGCCTCCTCTCTCACTGCACAGCCTCT
GTGACCTCCGATATATCTGTGGCCCCAGGAGAGACGGCCAGGATTTC
CTGTGGGGAAAAGAGCCTTGGAAGTAGAGCTGTACAATGGTATCAAC
ACAGGGCCGGCCAGGCCCCCTCTTTAATCATATATAATAATCAGGAC
CGGCCCTCAGGGATCCCTGAGCGATTCTCTGGCTCCCCTGACTCCCC
TTTTGGGACCACGGCCACCCTGACCATCACCAGTGTCGAAGCCGGGG
ATGAGGCCGACTATTACTGTCATATATGGGATAGTAGAGTTCCCACC
AAATGGGTCTTCGGCGGAGGGACCACGCTGACCGTGTTAGGTCAGCC
CAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCA
AGCCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAG
CCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGT
GGAGACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGC
AGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACA
GCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCC
TACAGAATGTTTCATAG 4838_L06 (PGT-121) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 147)

TCCGATATATCTGTGGCCCCAGGAGAGACGGCCAGGATTTCTGTGGGGA
AAAGAGCCTTGGAAGTAGAGCTGTACAATGGTATCAACACAGGGCCGGC
CAGGCCCCCTCTTTAATCATATATAATAATCAGGACCGGCCCTCAGGGAT
CCCTGAGCGATTCTCTGGCTCCCCTGACTCCCCTTTTGGGACCACGGCCAC
CCTGACCATCACCAGTGTCGAAGCCGGGGATGAGGCCGACTATTACTGTC
ATATATGGGATAGTAGAGTTCCCACCAAATGGGTCTTCGGCGGAGGGACC
ACGCTGACCGTGTTA 4838_L06 (PGT-121) lambda light chain amino acid
sequence: expressed protein with variable region in bold.(SEQ ID NO: 148)

SDISVAPGETARISCGEKSLSRAVQWYQHRAGQAPSLIIYNNQDRPSGIP
ERFSGSPDSPFGTTATLTITSVEAGDEADYYCHIWDSDRVPTKWVFGGGTT
LTVLGQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSP
VKAGVETTTTPSKQSNNKYAASSYLSLTPEQWKSHKSYSCQVTHEGSTVEKTV APTECS
4838_L06 (PGT-121) lambda light chain variable region amino acid sequence:
(Kabat CDRs underlined, Chothia CDRs in bold italics)(SEQ ID NO: 149)
SDISVAPGETARISCcustom character WYQHRAGQAPSLIIYcustom character GIPERF
SGSPDSPFGTTATLTITSVEAGDEADYYCcustom character FGGGTTTLTVL 4838_L06
(PGT-121) lambda light chain Kabat CDRs:(SEQ ID NO: 150) CDR 1:

GEKSLGSRAVQ (SEQ ID NO: 151) CDR 2: NNQDRPS (SEQ ID NO: 152) CDR 3: HIWDSRVPTKWV 4838_L06 (PGT-121) lambda light chain Chothia CDRs: (SEQ ID NO: 150) CDR 1: GEKSLGSRAVQ (SEQ ID NO: 151) CDR 2: NNQDRPS (SEQ ID NO: 152) CDR 3: HIWDSRVPTKWV 4873_E03 (PGT-121) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 62)

ATGAAACACCTGTGGTTCCTTCCTTCTCCTGGTGGCAGCTCCCAGATGGGTC
CTGTCCAGATGCAGTTACAGGAGTCGGGCCCCGGACTGGTGAAGCCT
TCGGAAACCCTGTCCCTCACGTGCAGTGTGTCTGGTGCCTCCATAAG
TGACAGTTACTGGAGCTGGATCCGGCGGTCCCCAGGGAAGGGACTTG
AGTGGATTGGGTATGTCCACAAAAGCGGCGACACAAATTACATCCCC
TCCCTCAAGAGTCGAGTCAACTTGTCGTTAGACACGTCCAAAAATCA
GGTGTCCCTGAGCCTTGTGGCCGCGACCGCTGCGGACTCGGGCAAAT
ATTATTGCGCGAGAACACTGCACGGGAGGAGAATTTATGGAATCGTT
GCCTTCAATGAGTGGTTCACCTACTTCTACATGGACGTCTGGGGCAA
TGGGACTCAGGTCACCGTCTCCTCAGCCTCCACCAAGGGCCCATCGGTC
TTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCCCT
GGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGGA
ACTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTACAG
TCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAG
CTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACA
CCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACAC
ATGCCACACGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCC
TCTTCCCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAG
GTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTT
CAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCG
CGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGT
CCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCC
AACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAG
GGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGA
GATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATC
CCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAA
CTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCTA
TAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTC
TCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAG
CCTCTCCCTGTCTCCGGGTAAATGA 4873_E03 (PGT-121) gamma heavy chain
variable region nucleotide sequence: (SEQ ID NO: 63)

CAGATGCAGTTACAGGAGTCGGGCCCCGGACTGGTGAAGCCTTCGGAAA
CCCTGTCCCTCACGTGCAGTGTGTCTGGTGCCTCCATAAGTGACAGTTACT
GGAGCTGGATCCGGCGGTCCCCAGGGAAGGGACTTGAGTGGATTGGGTAT
GTCCACAAAAGCGGCGACACAAATTACAGCCCCTCCCTCAAGAGTCGAGT
CAACTTGTCGTTAGACACGTCCAAAAATCAGGTGTCCCTGAGCCTTGTGG
CCGCGACCGCTGCGGACTCGGGCAAATATTATTGCGCGAGAACACTGCAC
GGGAGGAGAAATTTATGGAATCGTTGCCTTCAATGAGTGGTTCACCTACTT
CTACATGGACGTCTGGGGCAATGGGACTCAGGTCACCGTCTCCTCA 4873_E03 (PGT-
121) gamma heavy chain amino acid sequence: expressed protein with variable
region in bold. (SEQ ID NO: 66)

**MKHLWFFLLLVAPRWVLSQMQLQESGPGLVKPSETLSLTCSVSGASIS
DSYWSWIRRS PGKGLEWIGYVHKSGDTNYIPSLKSRVNL SLDTSKNQVSL
SLVAATAADSGKYYCARTLHGRRRIYGIVAFNEWFTYFYMDVWGNGTQV**

TVSSKSTKPSVFLAPSLVGLKDYLPNSPVTVSWNSGALTS
VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKS
CDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKC
KVSNNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSS
VMHEALHNHYTQKSLSLSPGK 4873_E03 (PGT-121) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 79) QMQLQESGPGLVKPSETLSLTCSVS custom character
SYWSWIRRSPGKGLEWIG custom character custom character
NYSPSLKSRVNLSDTSKNQVSLSLVAATAADSGKYCAR custom character
custom character WGNGTQVTVSS 4873_E03 (PGT-121) gamma heavy chain
Kabat CDRs: (SEQ ID NO: 90) CDR 1: DSYWS (SEQ ID NO: 265) CDR 2:
YVHKSGDTNYSPLKS (SEQ ID NO: 143) CDR 3:
TLHGRRIGIVAFNEWFTYFYMDV 4873_E03 (PGT-121) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 144) CDR 1: GASISD (SEQ ID NO: 145) CDR
2: YVHKSGDTN (SEQ ID NO: 143) CDR 3: TLHGRRIGIVAFNEWFTYFYMDV
4873_E03 (PGT-121) lambda light chain nucleotide sequence: coding sequence
(variable region in bold) (SEQ ID NO: 146)
ATGGCCTGGACCTTTCTCCTCCTCGGCCTCCTCTCTCACTGCACAGCCTCT
GTGACCTCCGATATATCTGTGGCCCCAGGAGAGACGGCCAGGATTTC
CTGTGGGGAAAAGAGCCTTGGAAGTAGAGCTGTACAATGGTATCAAC
ACAGGGCCCGGCCAGGCCCCCTCTTTAATCATATATAATAATCAGGAC
CGGCCCTCAGGGATCCCTGAGCGATTCTCTGGCTCCCCTGACTCCCC
TTTTGGGACCACGGCCACCCTGACCATCACCAGTGTCGAAGCCGGGG
ATGAGGCCGACTATTACTGTCATATATGGGATAGTAGAGTTCCCACC
AAATGGGTCTTCGGCGGAGGGACCACGCTGACCGTGTTAGGTCAGCC
CAAGGCTGCCCCCTCGGTCCTCTGTTCCCGCCCTCCTCTGAGGAGCTTCA
AGCCAACAAGGCCACACTGGTGTGTCTCATAAGTGA CTCTACCCGGGAG
CCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTC AAGGCGGGAGT
GGAGACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGC
AGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACA
GCTGCCAGGTCACGCATGAAGGGAGCACCGTG GAGAAGACAGTGGCCCC
TACAGAATGTT CATAG 4873_E03 (PGT-121) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 147)
TCGATATATCTGTGGCCCCAGGAGAGACGGCCAGGATTTCCTGTGGGGA
AAAGAGCCTTGGAAGTAGAGCTGTACAATGGTATCAACACAGGGCCGGC
CAGGCCCCCTCTTTAATCATATATAATAATCAGGACCGGCCCTCAGGGAT
CCCTGAGCGATTCTCTGGCTCCCCTGACTCCCCTTTTGGGACCACGGCCAC
CCTGACCATCACCAGTGTCGAAGCCGGGGATGAGGCCGACTATTACTGTC
ATATATGGGATAGTAGAGTTCCCACCAAATGGGTCTTCGGCGGAGGGACC
ACGCTGACCGTGTTA 4873_E03 (PGT-121) lambda light chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 148)
MAWTFLLGLLSHCTASVTS**DISVAPGETARISCGEKS****LGSR****AVQWYQH**
RAGQAPSLIIYNNQDRPSGIPERFSGSPDSPFGTTATLTITSVEAGDEADYY
CHIWD**SRVPTKWVFGGGTTTLTVLGQPKAAPS****VTLFPPSSEELQANKATLVC**
LISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWK
SHKSYSCQVTHEGSTVEKTVAPTECS 4873_E03 (PGT-121) lambda light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 149) SDISVAPGETARISC custom character

WYQHRAQQAPSLIIF custom character GIPERF SGSPDSPFGTTATLTITSVEAGDEADYYC

custom character FGGGTTLTVL 4873_E03 (PGT-121) lambda light chain Kabat CDRs: (SEQ ID NO: 150) CDR 1: GEKSLGSRVQ (SEQ ID NO: 151) CDR 2: NNQDRPS (SEQ ID NO: 152) CDR 3: HIWDSRVPTKWV 4873_E03 (PGT-121) lambda light chain Chothia CDRs: (SEQ ID NO: 150) CDR 1: GEKSLGSRVQ (SEQ ID NO: 151) CDR 2: NNQDRPS (SEQ ID NO: 152) CDR 3: HIWDSRVPTKWV 4877_D15 (PGT-122) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 153)

ATGAAACACCTGTGGTTCCTTCTCCTGGTGGCAGCTCCCAGATGGGTC
CTGTCCCAGGTTTCATCTGCAGGAGTCGGGCCCCGGACTGGTGAAGCC
TTCGGAGACCCTGTCCCTCACGTGCAATGTGTCTGGGACCCTCGTGC
GTGATAACTACTGGAGCTGGATCAGACAACCCCTCGGGAAGCAACCT
GAGTGGATTGGCTATGTCCATGACAGCGGGGACACGAATTACAACCC
CTCCCTGAAGAGTCGAGTCCACTTATCGTTGGACAAGTCCAAAAACC
TGGTGTCCCTGAGGCTGACCGGCGTGACCGCCGCGGACTCGGCCATA
TATTATTGCGCGACAACAAAACACGGGAGGAGGATTTATGGCGTCGT
TGCCTTCAAAGAGTGGTTCACCTATTTCTACATGGACGTCTGGGGCA
AAGGGACTTCGGTCACCGTCTCCTCAGCCTCCACCAAGGGCCCATCGG
TCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCC
CTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGTG
GAACTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTAC
AGTCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGC
AGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAA
CACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCAC
ACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTT
CCTCTTCCCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTG
AGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAA
GTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAG
CCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCAC
CGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCT
CCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAA
GGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGG
AGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTAT
CCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACA
ACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCT
ATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTT
CTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGA
GCCTCTCCCTGTCTCCGGGTAAATGA 4877_D15 (PGT-122) gamma heavy chain
variable region nucleotide sequence: (SEQ ID NO: 154)

CAGGTTCATCTGCAGGAGTCGGGCCCCGGACTGGTGAAGCCTTCGGAGAC
CCTGTCCCTCACGTGCAATGTGTCTGGGACCCTCGTGCGTGATAACTACTG
GAGCTGGATCAGACAACCCCTCGGGAAGCAACCTGAGTGGATTGGCTATG
TCCATGACAGCGGGGACACGAATTACAACCCCTCCCTGAAGAGTCGAGTC
CACTTATCGTTGGACAAGTCCAAAAACCTGGTGTCCCTGAGGCTGACCGG
CGTGACCGCCGCGGACTCGGCCATATATTATTGCGCGACAACAAAACACG
GGAGGAGGATTTATGGCGTCGTTGCCTTCAAAGAGTGGTTCACCTATTTCT
ACATGGACGTCTGGGGCAAAGGGACTTCGGTCACCGTCTCCTCA 4877_D15 (PGT-
122) gamma heavy chain amino acid sequence: expressed protein with
region in bold. (SEQ ID NO: 155)

MKHLWFFLLLVAAAPRWVLSQVHLQESGPGLVKPSETLSLTCNVSGTLVR

DNYWSQWPLNGKQPEWIGSVHLSLNPSLKSRSVHLSLDKSKNLVS
LRLTGVTAAADSAIYYCATTKHGRRRIYGVVAFKEWFTYFYMDVWGKGTS
VTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPK
SCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKC
KVSNAKALPAIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSS
VMHEALHNHYTQKSLSLSPGK 4877_D15 (PGT-122) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 156) QVHLQESGPGLVKPSETLSLTCNVS custom character
NYWSWIRQPLGKQPEWIG custom character custom character
YNPSLKSRSVHLSLDKSKNLVSLRLTGVTAAADSAIYYCAT custom character
 custom character WGKGTSVTVSS 4877_D15 (PGT-122) gamma heavy chain
Kabat CDRs: (SEQ ID NO: 261) CDR 1: DNYWS (SEQ ID NO: 157) CDR
2: YVHDSGDTNYNPSLKS (SEQ ID NO: 262) CDR 3:
TKHGRRRIYGVVAFKEWFTYFYMDV 4877_D15 (PGT-122) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 263) CDR 1: GTLVRD (SEQ ID NO: 264) CDR
2: YVHDSGDTN (SEQ ID NO: 262) CDR 3: TKHGRRRIYGVVAFKEWFTYFYMDV
4877_D15 (PGT-122) lambda light chain nucleotide sequence: coding sequence
(variable region in bold) (SEQ ID NO: 158)
ATGGCCTGGACCGTTCTCCTCCTCGGCCTCCTCTCTCACTGCACAGGCGCG
GTGTCTACCTTTGTGTCAGTGGCCCCAGGACAGACGGCCAGGATTACT
TGTGGGGAAGAGAGCCTTGGAAGTAGATCTGTTATTTGGTATCAACA
GAGGCCAGGCCAGGCCCTTCATTAATCATCTATAATAATAATGACC
GGCCCTCAGGGATTCTGACCGATTTTCTGGGTCCCCTGGCTCCACT
TTTGGGACCACGGCCACCCTGACCATCACCAGTGTCGAAGCCGGGGA
TGAGGCCGACTATTATTGTCATATCTGGGATAGTAGACGACCAACCA
ATTGGGTCTTCGGCGAAGGGACCACACTGATCGTGTTAGGTCAGCCCA
AGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAG
CCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGCC
GTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGG
AGACCACCACACCCTCCAAACAAGCAACAACAAGTACGCGGCCAGCAG
CTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACAGCT
GCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCCTAC
AGAATGTTCATAG 4877_D15 (PGT-122) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 159)
ACCTTTGTGTCAGTGGCCCCAGGACAGACGGCCAGGATTACTTGTGGGGA
AGAGAGCCTTGGAAGTAGATCTGTTATTTGGTATCAACAGAGGCCAGGCC
AGGCCCTTCATTAATCATCTATAATAATAATGACCGGCCCTCAGGGATTC
CTGACCGATTTTCTGGGTCCCCTGGCTCCACTTTTGGGACCACGGCCACCC
TGACCATCACCAGTGTCGAAGCCGGGGATGAGGCCGACTATTATTGTCAT
ATCTGGGATAGTAGACGACCAACCAATTGGGTCTTCGGCGAAGGGACCAC
ACTGATCGTGTTA 4877_D15 (PGT-122) lambda light chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 160)
MAWTVLLLGLLSHCTGAVSTFVSVAPGQTARITCGEESLGSRSVIWYQQ
RPGQAPSLIYNNNDRPSGIPDRFSGSPGSTFGTTATLTITSVEAGDEADYY
CHIWDSRRPTNWVFGEGTTLIVLGQPKAAPSVTLFPPSSEELQANKATLVC
LISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWK
SHKSYSCQVTHEGSTVEKTVAPTECS 4877_D15 (PGT-122) lambda light chain

variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 161) TFVSVAPGQTARITC custom character WYQQRPGQAPSLIY custom character GIPDRF SGSPGSTFGTTATLTITSVEAGDEADYYC custom character FGEGTTLIVL 4877_D15 (PGT-122) lambda light chain Kabat CDRs: (SEQ ID NO: 162) CDR 1: GEESLGSRSVI (SEQ ID NO: 163) CDR 2: NNNDRPS (SEQ ID NO: 164) CDR 3: HIWDSRRPTNWV 4877_D15 (PGT-122) lambda light chain Chothia CDRs: (SEQ ID NO: 162) CDR 1: GEESLGSRSVI (SEQ ID NO: 163) CDR 2: NNNDRPS (SEQ ID NO: 164) CDR 3: HIWDSRRPTNWV 4858_P08 (PGT-123) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 165)

ATGAAACACCTGTGGATCTTCCTTCTCCTGGTGGCAACTCCCAGATGGGTC
GAGTCCCAGCTGCACCTGCAGGAGTCGGGCCCAGGGCTGGTGAAGCC
TCCGGAGACCCTGTCCCTCACGTGTAGTGTGTCTGGCGCCTCCATCA
ATGATGCCTATTGGAGTTGGATTCGGCAGTCCCCAGGGAAGCGGCCT
GAGTGGGTTGGATATGTCCATCACAGCGGTGACACAAATTATAATCC
CTCACTCAAGAGGCGCGTCACGTTTTTCATTAGACACGGCCAAGAATG
AAGTGTCCCTGAAATTAGTAGACCTGACCGCTGCGGACTCGGCCACA
TATTTTTGTGCGCGAGCACTTCACGGGAAGAGGATTTATGGGATAGT
TGCCCTCGGAGAGTTGTTACCTACTTCTACATGGACGTCTGGGGCA
AGGGGACTGCGGTACACGTCTCCTCAGCCTCCACCAAGGGCCCATCGG
TCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCC
CTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGTG
GAACTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTAC
AGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGC
AGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAA
CACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCAC
ACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTT
CCTCTTCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTG
AGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAA
GTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAG
CCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCAC
CGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCT
CCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAA
GGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGG
AGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTAT
CCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACA
ACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCT
ATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTT
CTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGA
GCCTCTCCCTGTCTCCGGGTAAATGA 4858_P08 (PGT-123) gamma heavy chain

variable region nucleotide sequence: (SEQ ID NO: 166)
CAGCTGCACCTGCAGGAGTCGGGCCCAGGGCTGGTGAAGCCTCCGGAGA
CCCTGTCCCTCACGTGTAGTGTGTCTGGCGCCTCCATCAATGATGCCTATT
GGAGTTGGATTCGGCAGTCCCCAGGGAAGCGGCCTGAGTGGGTTGGATAT
GTCCATCACAGCGGTGACACAAATTATAATCCCTCACTCAAGAGGCGCGT
CACGTTTTTCATTAGACACGGCCAAGAATGAAGTGTCCCTGAAATTAGTAG
ACCTGACCGCTGCGGACTCGGCCACATATTTTTGTGCGCGAGCACTTCAC
GGGAAGAGGATTTATGGGATAGTTGCCCTCGGAGAGTTGTTACCTACTT
CTACATGGACGTCTGGGGCAAGGGGACTGCGGTACACGTCTCCTCA 4858_P08 (PGT-123) gamma heavy chain amino acid sequence: expressed protein with variable

region in bold. (SEQ ID NO: 167)
MKHLWIFLLLVATPRWVESQLHLQESGPGLVKPPETLSLTCSVSGASIND
AYWSWIRQSPGKRPEWVGYVHHSGDTNYNPSLKRRVTFSLDTAKNEVS
LKLVDLTAADSATYFCARALHGKRIYGIVALGELFTYFYMDVWGKGTA
VTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPK
SCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKC
KVSNAKALPAIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGNVFCSS
VMHEALHNHYTQKSLSLSPGK 4858_P08 (PGT-123) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 168) QLHLQESGPGLVKPPETLSLTCSVS custom character
AYWSWIRQSPGKRPEWVG custom character custom character
YNPSLKRRVTFSLDTAKNEVSLKLVDLTAADSATYFCAR custom character
custom character WGKGTA VTVSS 4858_P08 (PGT-123) gamma heavy chain Kabat
CDRs: (SEQ ID NO: 169) CDR 1: DAYWS (SEQ ID NO: 170) CDR 2:
YVHHSGDTNYNPSLKRR (SEQ ID NO: 171) CDR 3:
ALHGKRIYGIVALGELFTYFYMDV 4858_P08 (PGT-123) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 172) CDR 1: GASIND (SEQ ID NO: 173) CDR
2: YVHHSGDTN (SEQ ID NO: 171) CDR 3: ALHGKRIYGIVALGELFTYFYMDV
4858_P08 (PGT-123) lambda light chain nucleotide sequence: coding sequence
(variable region in bold) (SEQ ID NO: 174)
ATGGCCTGGACCGTTCTCCTCCTCGGCCTCCTCTCTCACTGCACAGGCTCT
CTGGCCTCCTCTATGTCCGTGTCCCCGGGGGAGACGGCCAAGATCTC
CTGTGGAAGAGAGCATTGGTAGCAGAGCTGTGCAATGGTATCAGC
AGAAGCCAGGCCAGCCCCCTCATTGATTATCTATAATAATCAGGAC
CGCCCCGCAGGGGTCCCTGAGCGATTCTCTGCCTCCCCTGACTTCCG
TCCTGGGACCACGGCCACCCTGACCATCACCAATGTCGACGCCGAGG
ATGAGGCCGACTATTACTGTCATATATATGATGCTAGAGGTGGCACC
AATTGGGTCTTCGACAGAGGGACCACACTGACCGTCTTAGGTCAGCCC
AAGGCTGCCCCCTCGGTCACCTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAA
GCCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGC
CGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTG
GAGACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGCA
GCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACAGC
TGCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCCTA
CAGAAATGTTCATAG 4858_P08 (PGT-123) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 175)
TCCTCTATGTCCGTGTCCCCGGGGGAGACGGCCAAGATCTCCTGTGGAAG
AGAGAGCATTGGTAGCAGAGCTGTGCAATGGTATCAGCAGAAGCCAGGC
CAGCCCCCTCATTGATTATCTATAATAATCAGGACCGCCCCGCAGGGGT
CCCTGAGCGATTCTCTGCCTCCCCTGACTTCCGTCCTGGGACCACGGCCAC
CCTGACCATCACCAATGTCGACGCCGAGGATGAGGCCGACTATTACTGTC
ATATATATGATGCTAGAGGTGGCACCAATTGGGTCTTCGACAGAGGGACC
ACACTGACCGTCTTA 4858_P08 (PGT-123) lambda light chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 176)
MAWTVLLLGLLSHCTGSLASSMSVSPGETAKISCGKESIGSRAVQWYQQ
KPGQPPSLIYNNQDRPAGVPERFSASPDFRPGTTATLTITNVDAEDEADY
YCHIIDARGGTNWVFDRTTLTVLGQPKAAPSVTLFPPSSEELQANKATLV

CLISDFYPKADSVFVWAKDVSPTVQSNKQSVNNKYAASSYLSLTPEQW
KSHKSYSCQVTHEGSTVEKTVAPTECS 4858_P08 (PGT-123) lambda light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 177) SSMSVSPGETAKISCcustom character
WYQQKPGQPPSLIYcustom character GVPER
FSASPDFRPGTTATLTITNVDAEDEADYYCcustom character FDRGTTTLTVL 4858_P08
(PGT-123) lambda light chain Kabat CDRs: (SEQ ID NO: 178) CDR 1:
GKESIGSRAVQ (SEQ ID NO: 179) CDR 2: NNQDRPA (SEQ ID NO: 180)
CDR 3: HIYDARGGTNWV 4858_P08 (PGT-123) lambda light chain Chothia
CDRs: (SEQ ID NO: 178) CDR 1: GKESIGSRAVQ (SEQ ID NO: 179) CDR
2: NNQDRPA (SEQ ID NO: 180) CDR 3: HIYDARGGTNWV 5123_A06 (PGT-
125) gamma heavy chain nucleotide sequence: coding sequence (variable region
in bold) (SEQ ID NO: 181)
ATGAAACACCTGTGGTTCTTCTTCCTGCTGGTGGCGGCTCCCAGATGCGTC
CTGTCCCAGTCGCAGCTGCAGGAGTCGGGCCCACGACTGGTGGAGGC
CTCGGAGACCCTGTCACTCACGTGCAATGTGTCCGGCGAGTCCACTG
GTGCCTGTACTTATTTCTGGGGCTGGGTCCGGCAGGCCCCAGGGAAG
GGGCTGGAGTGGATCGGGAGTTTGTCCCATTTGTCAGAGTTTCTGGGG
TTCCGGTTGGACCTTCCACAACCCGTCTCTCAAGAGTCGACTCACGA
TTTCACTCGACACGCCCAAGAATCAGGTCTTCCTCAAGCTCACTTCTC
TGACTGCCGCGGACACGGCCACTTACTACTGTGCGCGATTTCGACGGC
GAAGTCTTGGTCTATAATCATTGGCCAAAGCCGGCCTGGGTGGACCT
CTGGGGCCCGCGGAATACCGGTCAACGTCTCCTCAGCCTCCACCAAGGG
CCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCA
CAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACG
GTGTCTGTGAACTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGC
TGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCC
CTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGC
CCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAA
AACTCACACATGCCACCGTGCCAGCACCTGAACTCCTGGGGGGACCGT
CAGTCTTCCTCTTCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGA
CCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAG
GTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGAC
AAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTC
CTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5123_A06 (PGT-125) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 182)
CAGTCGCAGCTGCAGGAGTCGGGCCCACGACTGGTGGAGGCCTCGGAGA
CCCTGTCACTCACGTGCAATGTGTCCGGCGAGTCCACTGGTGCCTGTACTT
ATTTCTGGGGCTGGGTCCGGCAGGCCCCAGGGAAGGGGCTGGAGTGGATC
GGGAGTTTGTCCCATTTGTCAGAGTTTCTGGGGTTCCGGTTGGACCTTCCAC
AACCCGTCTCTCAAGAGTCGACTCACGATTTCACTCGACACGCCCAAGAA
TCAGGTCTTCCTCAAGCTCACTTCTCTGACTGCCGCGGACACGGCCACTTA

CTACTGTCGGTGGTCTGATGCTTGGTCTATAATCATTTGGCCAA
AGCCGGCCTGGGTGGACCTCTGGGGCCGCGGAATACCGGTCACCGTCTCC TCA
5123_A06 (PGT-125) gamma heavy chain amino acid sequence: expressed protein
with variable region in bold.(SEQ ID NO: 183)

**MKHLWFFFLVAAPRCVLSQSQLQESGPRLVEASETLSLTCNVSGESTG
ACTYFWGWVRQAPGKGLEWIGSLSHCQSFWGSWTFHNPSLKSRLTIS
LDTPKNQVFLKLTSLTAADTATYYCARFDGEVLVYNHWPKPAWVDLW
GRGIPVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDK
RVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV
FSCSVMHEALHNHYTQKSLSLSPGK** 5123_A06 (PGT-125) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 184) QSQLQESGPRLVEASETLSLTCNVS

 YFWGWVRQAPGKGLEWIG   HNPSLKSRLTISLDTPKNQVFLKLTSLTAADTATYYCA  WGRGIPVTVSS
5123_A06 (PGT-125) gamma heavy chain Kabat CDRs: (SEQ ID NO: 185)
CDR 1: ACTYFWG (SEQ ID NO: 186) CDR 2: SLSHCQSFWGSWTFHNPSLKS
(SEQ ID NO: 187) CDR 3: FDGEVLVYNHWPKPAWVDL 5123_A06 (PGT-125)
gamma heavy chain Chothia CDRs: (SEQ ID NO: 188) CDR 1: **GESTGACT**
(SEQ ID NO: 189) CDR 2: **SLSHCQSFWGSWTF** (SEQ ID NO: 187) CDR
3: **FDGEVLVYNHWPKPAWVDL** 5123_A06 (PGT-125) lambda light chain
nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 190)

ATGGCCTGGGCTCTGCTCCTCCTCACCTCCTCACTCAGGGCACAGGGGC
CTGGGCCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCC
**TGGACAGTCAATCACCATCTCCTGCAATGGAACCGCCACTAACTTTGT
CTCCTGGTACCAACAATTCCCAGACAAGGCCCCCAAACATCATCTTT
TGGGGTCGATAAGCGCCCCCCCCGGGTGTCCCGATCGTTTCTCTGGCT
CCCGGTCTGGCACGACGGCCTCCCTTACCGTCTCCCGACTCCAGACT
GACGATGAGGCTGTCTATTATTGCGGTTCACTTGTCGGCAACTGGGA
TGTGATTTTCGGCGGAGGGACCACCTTGACCGTCCTAGGTCAGCCCCAA
GGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAGC
CAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGCCG
TGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGGA
GACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGCAGC
TACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACAGCTG
CCAGGTCACGCATGAAGGGAGCACCGTGGAAGAAGACAGTGGCCCCTACA
GAATGTTCATAG** 5123_A06 (PGT-125) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 191)

**CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCCTGGACAGTCA
ATCACCATCTCCTGCAATGGAACCGCCACTAACTTTGTCTCCTGGTACCAA
CAATTCCCAGACAAGGCCCCCAAACATCATCTTTTGGGGTCGATAAGCG
CCCCCCCCGGTGTCCCGATCGTTTCTCTGGCTCCCGGTCTGGCACGACGGC
CTCCCTTACCGTCTCCCGACTCCAGACTGACGATGAGGCTGTCTATTATTG
CGGTTCACTTGTCGGCAACTGGGATGTGATTTTCGGCGGAGGGACCACCT
TGACCGTCCTA** 5123_A06 (PGT-125) lambda light chain amino acid sequence:
expressed protein with variable region in bold. (SEQ ID NO: 192)

MAWALLLLTLLTQGTGAWAQSALTQPPSASGSPGQSITISCNGTATNFVS

WYQQFDPDKAPKLIIFVPDRFSGSRSGTTASLTVSRLQTDDEAVYYC
VYYCGSLVGNWDVIFGGGTTLTVLGQPKAAPSVTLFPPSSEELQANKATLV
CLISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQW
KSHKSYSCQVTHEGSTVEKTVAPTECS 5123_A06 (PGT-125) lambda light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 193) QSALTQPPSASGSPGQSITISCcustom character
WYQQFDPDKAPKLIIFcustom character VPDRFSGSRSGTTASLTVSRLQTDDEAVYYC
custom character FGGGTTLTVL 5123_A06 (PGT-125) lambda light chain Kabat
CDRs: (SEQ ID NO: 194) CDR 1: NGTATNFVS (SEQ ID NO: 195) CDR 2:
GVDKRPP (SEQ ID NO: 196) CDR 3: GSLVGNWDVI 5123_A06 (PGT-125)
lambda light chain Chothia CDRs: (SEQ ID NO: 194) CDR 1: NGTATNFVS
(SEQ ID NO: 195) CDR 2: GVDKRPP (SEQ ID NO: 196) CDR 3:
GSLVGNWDVI 5141_B17 (PGT-126) gamma heavy chain nucleotide sequence:
coding sequence (variable region in bold) (SEQ ID NO: 197)
ATGAAACACCTGTGGTTCCTCCTGCTGGTGGCGGCTCCCAGATGGGTC
CTGTCCCAGCCGCAGCTGCAGGAGTCGGGGGCCAGGACTGGTGGAGGC
TTCGGAGACCCTGTCCCTCACCTGCACTGTGTCCGGCGACTCCACTG
CTGCTTGTGACTATTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAG
GGCCTGGAGTGGATTGGGGGTTTGTACATTGTGCAGGTTACTACAA
TACTGGCTGGACCTACCACAACCCGTCTCTCAAGAGTCGGCTCACGA
TTTCACTCGACACCCCCAAGAATCAGGTCTTCCTGAAGTTAAATTCTG
TGACCGCCGCGGACACGGCCATTTACTACTGTGCGCGATTTCGACGGC
GAAGTTTTGGTGTACCACGATTGGCCAAAGCCGGCCTGGGTTCGACCT
CTGGGGCCGGGGAACTTTGGTCACCGTCTCCTCAGCCTCCACCAAGGG
CCCATCGGTCTTCCCCCTGGCACCTCCTCCAAGAGCACCTCTGGGGGCA
CAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACG
GTGTCGTGGAATCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGC
TGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCC
CTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGC
CCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAA
AACTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGT
CAGTCTTCTCTTCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGA
CCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAG
GTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGAC
AAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTC
CTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAATAACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5141_B17 (PGT-126) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 198)
CAGCCGCAGCTGCAGGAGTCGGGGGCCAGGACTGGTGGAGGCTTCGGAGA
CCCTGTCCCTCACCTGCACTGTGTCCGGCGACTCCACTGCTGCTTGTGACT
ATTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAGGGCCTGGAGTGGATT
GGGGGTTTGTACATTGTGCAGGTTACTACAATACTGGCTGGACCTACCA
CAACCCGTCTCTCAAGAGTCGGCTCACGATTTCACTCGACACCCCCAAGA

ATCAGGCTCTTCTTCAATCTGTGACCGCCGCGGACACGCGATT
ACTACTGTGCGCGATTTCGACGGCGAAGTTTTTGGTGTACCACGATTGGCCA
AAGCCGGCCTGGGTCGACCTCTGGGGCCGGGGAACCTTTGGTCACCGTCTC CTCA
5141_B17 (PGT-126) gamma heavy chain amino acid sequence: expressed protein
with variable region in bold.(SEQ ID NO: 199)

**MKHLWFFLLVAAPRWVLSQPQLQESGPGLVEASETLSLTCTVSGDSTA
ACDYFWGWVRQPPGKGLEWIGGLSHCAGYYNTGWTYHNPSLKSRLTIS
LDTPKNQVFLKLNSVTAADTAIYYCARFDGEVLVYHDWPKPAWVDLWG
RGT**LTVTSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDK
RVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV
FSCSVMHEALHNHYTQKSLSLSPGK 5141_B17 (PGT-126) gamma heavy chain

variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 200) QPQLQESGPGLVEASETLSLTCTVS

 YFWGWVRQPPGKGLEWIG   HNPSLKSRLTISLDTPKNQVFLKLNSVTAADTAIYYCAR 

 WGRGTLVTSS 5141_B17 (PGT-126) gamma heavy chain Kabat
CDRs: (SEQ ID NO: 201) CDR 1: ACDYFWG (SEQ ID NO: 202) CDR 2:
GLSHCAGYYNTGWTYHNPSLKS (SEQ ID NO: 203) CDR 3:

FDGEVLVYHDWPKPAWVDL 5141_B17 (PGT-126) gamma heavy chain Chothia
CDRs: (SEQ ID NO: 204) CDR 1: GDSTAACD (SEQ ID NO: 205) CDR 2:
GLSHCAGYYNTGWTY (SEQ ID NO: 203) CDR 3: FDGEVLVYHDWPKPAWVDL

5141_B17 (PGT-126) lambda light chain nucleotide sequence: coding sequence
(variable region in bold) (SEQ ID NO: 206)

ATGGCCTGGGCTCTGCTCCTCCTCACCCTCCTCACTCAGGGCACAGGGGC
CTGGGCCCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCC
**TGGACAGTCAATCTCCATCTCCTGCACTGGAACCAGCAATAGGTTTG
TCTCCTGGTACCAGCAACACCCAGGCAAGGCCCCCAAAC**TCGTCATT
**TATGGGGTCAATAAGCGCCCCCTCAGGTGTCCCTGATCGTTTTTCTGG
CTCCAAGTCTGGCAACACGGCCTCCCTGACCGTCTCTGGGCTCCAGA
CTGACGATGAGGCTGTCTATTACTGCAGCTCACTTGTAGGCAACTGG
GATGTGATTTTCGGCGGAGGGACCAAGTTGACCGTCCTGGGTCAGCC
CAAGGCTGCCCCCTCGGTC**ACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCA
AGCCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAG
CCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGT
GGAGACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGC
AGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACA
GCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCC
TACAGAATGTTCATAG

5141_B17 (PGT-126) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 207)

CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCCTGGACAGTCA
ATCTCCATCTCCTGCACTGGAACCAGCAATAGGTTTGTCTCCTGGTACCAG
CAACACCCAGGCAAGGCCCCCAAACTCGTCATTTATGGGGTCAATAAGCG
CCCCTCAGGTGTCCCTGATCGTTTTTCTGGCTCCAAGTCTGGCAACACGGC
CTCCCTGACCGTCTCTGGGCTCCAGACTGACGATGAGGCTGTCTATTACTG
CAGCTCACTTGTAGGCAACTGGGATGTGATTTTCGGCGGAGGGACCAAGT
TGACCGTCCTG

5141_B17 (PGT-126) lambda light chain amino acid sequence:

expressed protein with variable region in bold. (SEQ ID NO: 208)
MAWALLLLTLLTQGTGAWAQSALTQPPSASGSPGQSSISISCTGTSNRFVS
WYQQHPGKAPKLVIYGVNKRPSGVPDRFSGSKSGNTASLTVSGLQTDDE
AVYYCSSLVGNWDVIFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATL
VCLISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQ
WKSHKSYSCQVTHEGSTVEKTVAPTECS 5141_B17 (PGT-126) lambda light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 209) Q**SALTQPPSASGSPGQSSISIS**Ccustom character
WYQQHPGKAPKLVIYcustom character GVPDRFSGSKSGNTASLTVSGLQTDDEAVYYC
custom character FGGGTKLTVL 5141_B17 (PGT-126) lambda light chain Kabat
CDRs: (SEQ ID NO: 210) CDR 1: TGTSNRFVS (SEQ ID NO: 211) CDR 2:
GVNKRPS (SEQ ID NO: 212) CDR 3: SSLVGNWDVI 5141_B17 (PGT-126)
lambda light chain Chothia CDRs: (SEQ ID NO: 210) CDR 1: TGTSNRFVS
(SEQ ID NO: 211) CDR 2: GVNKRPS (SEQ ID NO: 212) CDR 3:
SSLVGNWDVI 5147_N06 (PGT-130) gamma heavy chain nucleotide sequence:
coding sequence (variable region in bold) (SEQ ID NO: 213)
ATGAAACACCTGTGGTTCCTCCTGCTGGTGGCGGCTCCCAGATGGGTC
CTGTCCCAGGTGCAGCTGCAGGAGTCGGGCCCAGGACTGGTGAAGCC
TGCGGAGACCCTGTCCCTCACCTGCAGTGTCTCTGGAGAATCTATCA
ATACTGGTCATTACTACTGGGGCTGGGTCCGTCAGGTCCCAGGGAAG
GGACTTGAGTGGATAGGTCATATCCATTATACGACGGCTGTCCTGCA
CAACCCGTCCCTCAAGAGTCGACTCACCATCAAATTTACACGTTGA
GAAACCAGATTACCCTGAGGCTCAGTAATGTGACGGCCGCGGACACG
GCCGTCTATCACTGCGTACGATCCGGCGGGCGACATCTTATATTATTAT
GAGTGGCAAAAGCCGCACTGGTTCCTCCTGGGGCCCCGGGAATCCA
CGTCACCGTCTCGAGCGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTG
GCACCCTCCTCCAAGAGCACCTCTGGGGGACAGCGGCCCTGGGCTGCCT
GGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGGAAGTCAAGGCG
CCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTACAGTCCTCAGGA
CTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCAC
CCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTG
GACAAGAGAGTTGAGCCCAAATCTTGAGCAAACTCACACATGCCACC
GTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCCTCTTCCCCC
AAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCG
TGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTAC
GTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGC
AGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAG
GACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCC
TCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCG
AGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAG
AACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACAT
CGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAACTACAAGACC
ACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCTATAGCAAGCTC
ACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGT
GATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGT
CTCCGGGTAAATGA 5147_N06 (PGT-130) gamma heavy chain variable region
nucleotide sequence: (SEQ ID NO: 214)
CAGGTGCAGCTGCAGGAGTCGGGCCCAGGACTGGTGAAGCCTGCGGAGA
CCCTGTCCCTCACCTGCAGTGTCTCTGGAGAATCTATCAATACTGGTCATT
ACTACTGGGGCTGGGTCCGTCAGGTCCCAGGGAAGGGACTTGAGTGGATA

GGTCATCATTCGATCAACCAACCCGTCCTCCCTCAAGAG
TCGACTCACCATCAAAATTTACACGTTGAGAAACCAGATTACCCTGAGGC
TCAGTAATGTGACGGCCGCGGACACGGCCGTCTATCACTGCGTACGATCC
GGCGGCGACATCTTATATTATTATGAGTGGCAAAGCCGCACTGGTTCTC
TCCCTGGGGCCCCGGAATCCACGTCACCGTCTCGAGC 5147 N06 (PGT-130)
gamma heavy chain amino acid sequence: expressed protein with variable
region in bold. (SEQ ID NO: 215)

MKHLWFFLLVAAPRWVLSQVQLQESGPGLVKPAETLSLTCSVSGESIN
TGHYYWGWVRQVPGKGLEWIGHIHYTTAVLHNPSLKSRLTIKIYTLRN
QITLRLSNVTAADTAVYHCVRSGGDILYYYEWQKPHWFSPWGP**GIHVTV**
SSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVH
TFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDRVEPKSCD
KTHHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKF
NWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKV
NKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDI
AVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFC
SVMHEALHNHYTQKSLSLSPGK 5147_N06 (PGT-130) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 216) QVQLQESGPGLVKPAETLSLTCSVS custom character
YYWGWVRQVPGKGLEWIG  custom character

HNPSLKSRLTIKIYTLRNQITLRLSNVTAADTAVYHCVR custom character
 custom character WGPGIHVTVSS 5147_N06 (PGT-130) gamma heavy chain Kabat
CDRs: (SEQ ID NO: 217) CDR 1: TGHYYWG (SEQ ID NO: 218) CDR 2:
HIHYTTAVLHNPSLKS (SEQ ID NO: 219) CDR 3: SGGDILYYYEWQKPHWFSP
5147_N06 (PGT-130) gamma heavy chain Chothia CDRs: (SEQ ID NO: 220)
CDR 1: GESINTGH (SEQ ID NO: 221) CDR 2: HIHYTTAVL (SEQ ID NO:
219) CDR 3: SGGDILYYYEWQKPHWFSP 5147_N06 (PGT-130) lambda light
chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID
NO: 222) ATGGCCTGGGCTCTGCTCCTCCTCACCTCCTCACTCAGGGCACAGGGTCC
TGGGCCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCTT
GGACAGTCAGTCACCATCTCCTGCAATGGAACCAGCAGTGACATTGG
CGGTTGGAATTTTGTCTCCTGGTATCAACAGTTCCCGGGCAGAGCCC
CCAGACTCATTATTTTGTAGGTCAATAAGCGGCCCTCAGGGGTCCCT
GGTCGCTTCTCTGGCTCCAAGTCGGGCAATTCGGCCTCCCTGACCGT
CTCTGGGCTCCAGTCTGACGATGAGGGTCAATATTTCTGCAGTTCAC
TTTTCGGCAGGTGGGATGTTGTTTTTGGCGGGGGGACCAAGCTGACC
GTCTAGGTCAGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTC
CTCTGAGGAGCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTG
ACTTCTACCCGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCC
GTCAAGGCGGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACA
AGTACGCGGCCAGCAGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCC
CACAAAAGCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGAGGA
AGACAGTGGCCCCTACAGAATGTTTCATAG 5147_N06 (PGT-130) lambda light
chain variable region nucleotide sequence: (SEQ ID NO: 223)

CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCTTGGACAGTCA
GTCACCATCTCCTGCAATGGAACCAGCAGTGACATTGGCGGTTGGAATTT
TGTCTCCTGGTATCAACAGTTCCCGGGCAGAGCCCCCAGACTCATTATTTT
TGAGGTCAATAAGCGGCCCTCAGGGGTCCCTGGTCGCTTCTCTGGCTCCA
AGTCGGGCAATTCGGCCTCCCTGACCGTCTCTGGGCTCCAGTCTGACGAT
GAGGGTCAATATTTCTGCAGTTCACTTTTTCGGCAGGTGGGATGTTGTTTTT

GGCGGGGGACCAAGCTGACCGTCCTA 5147_N06 (PGT-130) lambda light chain amino acid sequence: expressed protein with variable region in bold. (SEQ ID NO: 224) **MAWALLLLTLLTQGTGSWAQSAITQPPSASGSLGQSVTISCNGTSSDIGGWNFVSWYQQFPGRAPRLIIFEVNKRPSGVPGRFSGSKSGNSASLTVSGLQSDDEGQYFCSSLFGRWDVVFGGGTKLTVLGQPKAAPSVTLFPPSSEELQAN**KATLVCLISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNKYAASSYLSL

TPEQWKSHKSYSCQVTHEGSTVEKTVAPTECS 5147_N06 (PGT-130) lambda light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 225) QSALTQPPSASGSLGQSVTISC
WYQQFPGRAPRLIIFGVPGRFSGSKSGNSASLTVSGLQSDDEGQYFCFGGGTK LTVL

5147_N06 (PGT-130) lambda light chain Kabat CDRs: (SEQ ID NO: 226) CDR 1: NGTSSDIGGWNFVS (SEQ ID NO: 227) CDR 2: EVNKRPS (SEQ ID NO: 228) CDR 3: SSLFGRWDVV 5147_N06 (PGT-130) lambda light chain Chothia CDRs: (SEQ ID NO: 226) CDR 1: NGTSSDIGGWNFVS (SEQ ID NO: 227) CDR 2: EVNKRPS (SEQ ID NO: 228) CDR 3: SSLFGRWDVV 5343_B08 (PGT-135) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 229)

ATGAAACACCTGTGGTTCCTCTTGCTGGTGGCGGCTCCCAGATGGGTCCTGTCCCAGTTGCAGATGCAGGAGTCGGGCCCAGGACTGGTGAAGCCTTCGGAGACCCTGTCTCTGAGTTGCACTGTCTCTGGTGACTCCATAAGGGGTGGCGAGTGGGGCGATAAAGATTATCATTGGGGCTGGGTCCGCCACTCAGCAGGAAAGGGCCTGGAGTGGATTGGGAGTATCCATTGGAAGGGGACCACCCACTACAAAGAGTCCCTCAGGAGAAGAGTGAGTATGTCGATCGACACGTCCAGGAATTGGTTCTCCCTGAGGCTGGCCTCTGTGACCGCCGCGGACACGGCCGTCTACTTTTGTGCGAGACACCGACATCATGATGTTTTCATGTTGGTCCCTATTGCGGGCTGGTTCGACGTCTGGGCCCGGGAGTCCAGGTCACCGTCTCGAGCGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGTGAACTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACACATGCCCACCGTGCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCTCTTCCCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGACAACACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5343_B08 (PGT-135) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 230)

CAGTTGCAGATGCAGGAGTCGGGCCCAGGACTGGTGAAGCCTTCGGAGA

CCCTGTCTCTGAGTTGCACTGTCTCTGTTGACTCCATAAGGGGTGGCGAGT
GGGGCGATAAAGATTATCATTGGGGCTGGGTCCGCCACTCAGCAGGAAA
GGGCTGGAGTGGATTGGGAGTATCCATTGGAGGGGGACCACCCACTACA
AAGAGTCCCTCAGGAGAAGAGTGAGTATGTCGATCGACACGTCCAGGAA
TTGGTTCTCCCTGAGGCTGGCCTCTGTGACCGCCGCGGACACGGCCGTCT
ACTTTTGTGCGAGACACCGACATCATGATGTTTTCATGTTGGTCCCTATTG
CGGGCTGGTTCGACGTCTGGGGCCCCGGGAGTCCAGGTACCGTCTCGAGC 5343_B08
(PGT-135) gamma heavy chain amino acid sequence: expressed protein with
variable region in bold. (SEQ ID NO: 231)

FSCSVMHEALHNHYTQKSLSLSPGK 5343_B08 (PGT-135) gamma heavy chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 232) QLQMQUESGPGLVKPSSETLSLSCTVS

YKESLRRRVMSIDTSRNWFLRLASVTAADTAVYFC AR custom character

5343_B08 (PGT-135) gamma heavy chain Chothia CDRs: (SEQ ID NO: 236)
CDR 1: GDSIRGGEWGDKD (SEQ ID NO: 237) CDR 2: SIHWRTGTH (SEQ
ID NO: 235) CDR 3: HRHHDVFMLVPIAGWFDV 5343_B08 (PGT-135) kappa
light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ
ID NO: 238)

TACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGTT
GAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCTGAATAACTTCTATCCCAG
AGAGGCCAAAGTACAGTGGAAGGTGGATAACGCCCTCCAATCGGGTAAC
TCCCAGGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACCTACAGCC
TCAGCAGCACCTGACGCTGAGCAAAGCAGACTACGAGAAACACAAAGT
CTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTCGCCCGTCACAAAGA
GCTTCAACAGGGGAGAGTGTTAG 5343_B08 (PGT-135) kappa light chain
variable region nucleotide sequence: (SEQ ID NO: 239)

ACATATAGCAAGTTCCTGCTGCGGTTTCGTTGCCAGTGGTTC
GGGACAGAGTTCACCTCTCACCATCAACAACATGCAGTCTGAAGATGTTGC
AGTTTATTACTGTCAACAATATGAAGAGTGGCCTCGGACGTTCCGGGCAAG
GGACCAAGGTGGATATCAAA 5343_B08 (PGT-135) kappa light chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 240)

METPAQLLFLLLWL**PDTTGEIVMTQSPDTLSVSPGETVTLS****CRASQNIN**
KNLAWYQYKPGQSPRLVIFETYSKIAAFPARFVASGSGTEFTLTIN**MQS**
EDVAVYYCQQYEEWPRTFGQGTKVDIKRTVAAPSVFIFPPSDEQLKSGTAS
VVCLLN**FYPREAKVQWKVDNALQSGNSQESVTEQDSK****DSTYSL****SSTLTLSK**
ADYEKHKVYACEVTHQGLSSPVTKSFNRGEC 5343_B08 (PGT-135) kappa light
chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia
CDRs in bold italics) (SEQ ID NO: 242) **EIVMTQSPDTLSVSPGETVTLS****C**

WYQYKPGQSPRLVIF
AFPARFVASGSGTEFTLTINMQSE**EDVAVYYC**FGQGTKVD IK
5343_B08 (PGT-135) kappa light chain Kabat CDRs: (SEQ ID NO: 243) CDR
1: **RASQNIN****KNLA** (SEQ ID NO: 244) CDR 2: **ETYSKIA** (SEQ ID NO: 245)
CDR 3: **QQYEEWPRT** 5343_B08 (PGT-135) kappa light chain Chothia CDRs:
(SEQ ID NO: 243) CDR 1: **RASQNIN****KNLA** (SEQ ID NO: 244) CDR 2:
ETYSKIA (SEQ ID NO: 245) CDR 3: **QQYEEWPRT** 5344_E16 (PGT-135)

gamma heavy chain nucleotide sequence: coding sequence (variable region in
bold) (SEQ ID NO: 229)

ATGAAACACCTGTGGTTCCTCCTTGCTGGTGGCGGCTCCCAGATGGGTC
CTGTCCCAGTTGCAGATGCAGGAGTCGGGCCCAGGACTGGTGAAGCC
TTCGGAGACCCTGTCTCTGAGTTGCACTGTCTCTGGTGACTCCATAA
GGGGTGGCGAGTGGGGCGATAAAGATTATCATTGGGGCTGGGTCCG
CCACTCAGCAGGAAAGGGCCTGGAGTGGATTGGGAGTATCCATTGGA
GGGGGACCACCCACTACAAAGAGTCCCTCAGGAGAAGAGTGAGTATG
TCGATCGACACGTCCAGGAATTGGTTCTCCCTGAGGCTGGCCTCTGT
GACCGCCGCGGACACGGCCGTCTACTTTTGTGCGAGACACCGACATC
ATGATGTTTTTCATGTTGGTCCCTATTGCGGGCTGGTTTCGACGTCTGG
GGCCCGGGAGTCCAGGTCACCGTCTCGAGCGCCTCCACCAAGGGCCCA
TCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACAGC
GGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGT
CGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTC
CTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCC
AGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAG
CAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCT
CACACATGCCACCGTGCCAGCACCTGAACTCCTGGGGGGACCGTCAGT
CTTCTCTTCCCCCCTCCCAAGGACACCCCTCATGATCTCCCGGACCCC
TGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCA
AGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAA
GCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTC
ACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGG
TCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCC
AAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGG
AGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTC
TATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGA
ACAACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTC
CTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACG
TCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAG

AAGAGCCTTCCCTGTCTCCGGGTAAATGA 5344_E16 (PGT-135) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 230)
CAGTTGCAGATGCAGGAGTCGGGCCCAGGACTGGTGAAGCCTTCGGAGA
CCCTGTCTCTGAGTTGCACTGTCTCTGGTGACTCCATAAGGGGTGGCGAGT
GGGGCGATAAAGATTATCATTGGGGCTGGGTCCGCCACTCAGCAGGAAA
GGGCCTGGAGTGGATTGGGAGTATCCATTGGAGGGGGGACCACCCACTACA
AAGAGTCCCTCAGGAGAAGAGTGAGTATGTCGATCGACACGTCCAGGAA
TTGGTTCTCCCTGAGGCTGGCCTCTGTGACCGCCGCGGACACGGCCGTCT
ACTTTTGTGCGAGACACCGACATCATGATGTTTTTCATGTTGGTCCCTATTG
CGGGCTGGTTCGACGTCTGGGGCCCCGGGAGTCCAGGTCACCGTCTCGAGC 5344_E16 (PGT-135) gamma heavy chain amino acid sequence: expressed protein with variable region in bold. (SEQ ID NO: 231)

**MKHLWFFLLVAAPRWVLSQLQMQUESGPGLVKPSETLSLSCTVSGDSIR
GGEWGD KDYHWGWVRHSAGKGLEWIGSIHWRGTTTHYKESLRRRVSM
SIDTSRNWFSLRLASVTAADTAVYFCARHRHHDVFMLVPIAGWFDVWGP
GVQVTVSSASTKGPSVFLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSG
ALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKR
VEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNV
FSCSVMHEALHNHYTQKSLSLSPGK** 5344_E16 (PGT-135) gamma heavy chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 232) QLQMQUESGPGLVKPSETLSLSCTVS

 YHWGWVRHSAGK GLEWIG  YKESLRRRVSM  SIDTSRNWFSLRLASVTAADTAVYFC AR  WGPGVQVTVSS 5344_E16 (PGT-135) gamma heavy chain Kabat CDRs: (SEQ ID NO: 233) CDR 1: GGEWGD KDYHWG (SEQ ID NO: 234) CDR 2: SIHWRGTTTHYKESLRR (SEQ ID NO: 235) CDR 3: HRHHDVFMLVPIAGWFDV 5344_E16 (PGT-135) gamma heavy chain Chothia CDRs: (SEQ ID NO: 236) CDR 1: GDSIRGGEWGD KD (SEQ ID NO: 237) CDR 2: SIHWRGTTTH (SEQ ID NO: 235) CDR 3: HRHHDVFMLVPIAGWFDV 5344_E16 (PGT-135) kappa light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 238)

ATGGAACCCAGCTCAGCTTCTCTTCCTCCTGCTACTCTGGCTCCCAGAT
ACCACTGGAGAAATTGTGATGACGCAGTCTCCCGACACCCTGTCTGTC
**TCTCCAGGGGAGACAGTCACACTCTCCTGCAGGGCCAGTCAGAATAT
TAACAAGAATTTAGCCTGGTACCAATACAAACCTGGCCAGTCTCCCA
GGCTCGTAATTTTTTGAACATATAGCAAGATCGCTGCTTTCCCTGCCA
GGTTCGTTGCCAGTGGTTCTGGGACAGAGTTCACTCTCACCATCAAC
AACATGCAGTCTGAAGATGTTGCAGTTTATTACTGTCAACAATATGAA
GAGTGGCCTCGGACGTTCTGGGCAAGGGACCAAGGTGGATATCAAACG
TACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGTT
GAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCTGAATAACTTCTATCCCAG
AGAGGCCAAAGTACAGTGGAAGGTGGATAACGCCCTCCAATCGGGTAAC
TCCCAGGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACCTACAGCC
TCAGCAGCACCCCTGACGCTGAGCAAAGCAGACTACGAGAAACACAAAGT
CTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTCGCCCGTCACAAAGA
GCTTCAACAGGGGAGAGTGTTAG** 5344_E16 (PGT-135) kappa light chain variable region nucleotide sequence: (SEQ ID NO: 239)

GAAATGTAGTGCAGTCTCCGACACACCTGCTCCTCCTCCAGGGA
GACAGTCACACTCTCCTGCAGGGCCAGTCAGAATATTAACAAGAATTTAG
CCTGGTACCAATAACAAACCTGGCCAGTCTCCCAGGCTCGTAATTTTTGAA
ACATATAGCAAGATCGCTGCTTTCCCTGCCAGGTTTCGTTGCCAGTGGTTCT
GGGACAGAGTTCACTCTCACCATCAACAACATGCAGTCTGAAGATGTTGC
AGTTTATTACTGTCAACAATATGAAGAGTGGCCTCGGACGTTTCGGGCAAG
GGACCAAGGTGGATATCAAA 5344_E16 (PGT-135) kappa light chain amino acid
sequence: expressed protein with variable region in bold. (SEQ ID NO: 240)

METPAQLLFLLLWL**PDTTGEIVMTQSPDTLSVSPGETVTLS****CRASQNIN**
KNLAWYQYKPGQSPRLVIFETYSKIAAFPARFVASGSGTEFTLTIN**MQS**
EDVAVYYCQQYEEWPRTFGQGTKVDIKRTVAAPSVFIFPPSDEQLKSGTAS
VVCLLNNFY**PREAKVQWKVDNALQSGNSQESVTEQDSK****DSTYLS****SSTLTLSK**

ADYEKHKVYACEVTHQGLSSPVTKSFNRGEC 5344_E16 (PGT-135) kappa light
chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia
CDRs in bold italics) (SEQ ID NO: 242) *EIVMTQSPDTLSVSPGETVTLS**C*

 *WYQYKPGQSPRLVIF*  *AFPARFVASGSGTEFTLTIN**MQS*  *EDVAVYYC*  *FGQGTKVD* *IK*

5344_E16 (PGT-135) kappa light chain Kabat CDRs: (SEQ ID NO: 243) CDR
1: *RASQNINKNLA* (SEQ ID NO: 244) CDR 2: *ETYSKIA* (SEQ ID NO: 245)
CDR 3: *QQYEEWPRT* 5344_E16 (PGT-135) kappa light chain Chothia CDRs:
(SEQ ID NO: 243) CDR 1: *RASQNINKNLA* (SEQ ID NO: 244) CDR 2:
ETYSKIA (SEQ ID NO: 245) CDR 3: *QQYEEWPRT* 5329_C19 (PGT-136)

gamma heavy chain nucleotide sequence: coding sequence (variable region in
bold) (SEQ ID NO: 246)

ATGAAACACCTGTGGTTCTTCCTCCTGCTAGTGGCGGCTCCCAGATGGGTC
CTGTTCGCAGCTGCAGTTGCAGGAATCGGGCCCAGGACTGGTGAAGCC
TTCGGAGACCCTGTCCCTGACTTGCACAGTTTCTGGTGGCTCCATGA
GGGGCACCGACTGGGGCGAGAATGACTTCCACTACGGCTGGATCCG
CCAGTCCTCCGCAAAGGGGCTGGAGTGGATTGGGAGCATCCATTGGA
GGGGGAGGACCACCCACTACAAGACGTCCTTCAGGAGTCGGGCCAC
CTTGTCGATAGACACGTCCAATAATCGCTTCTCCCTGACGTTTAGTTT
TGTGACCGCCGCGGACACGGCCGTCTACTATTGTGCGAGACATAAAT
ATCATGATATTTTCAGGGTGGTCCCTGTTGCGGGGCTGGTTCGACCCC
TGGGGCCAGGGATTACTGGTCACCGTCTCGAGCGCCTCCACCAAGGGC
CCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCAC
AGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGG
TGTCGTGGA**ACTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCT**
GTCTTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCC
CTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGC
CCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAA
AACTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGT
CAGTCTTCTCTTCCCCC**AAAACCCAAGGACACCCTCATGATCTCCCGGA**
CCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAG
GTCAAGTTCA**ACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGAC**
AAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTC
CTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA

GAACAACAGACACCGCTGGACTCCGACGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5329_C19 (PGT-136) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 247)
CAGCTGCAGTTGCAGGAATCGGGCCCAGGACTGGTGAAGCCTTCGGAGA
CCCTGTCCCTGACTTGCACAGTTTCTGGTGGCTCCATGAGGGGACCGACT
GGGGCGAGAATGACTTCCACTACGGCTGGATCCGCCAGTCCTCCGCAAAG
GGGCTGGAGTGGATTGGGAGCATCCATTGGAGGGGGAGGACCACCCACT
ACAAGACGTCCTTCAGGAGTCGGGGCCACCTTGTCGATAGACACGTCCAAT
AATCGCTTCTCCCTGACGTTTAGTTTTGTGACCGCCGCGGACACGGCCGTC
TACTATTGTGCGAGACATAAATATCATGATATTTTCAGGGTGGTCCCTGTT
GCGGGCTGGTTCGACCCCTGGGGCCAGGGATTACTGGTCACCGTCTCGAG C
5329_C19 (PGT-136) gamma heavy chain amino acid sequence: expressed protein
with variable region in bold. (SEQ ID NO: 248)

MKHLWFFLLVAAPRWVLSQLQLQESGPGLVKPSETLSLTCTVSGGSM
RGTDWGENDFHYGWIRQSSAKGLEWIGSIHWRGRTHYKTSFRSRATLS
IDTSNNRFSLTFSFVTAADTAVYYCARHKYHDI**FRVVPVAGWFDPWGQG**
LLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGAL
TSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKRVE
PKSCDKHTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHE
DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK 5329_C19 (PGT-136) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 249) QLQLQESGPGLVKPSETLSLTCTVS

 FHYGWIRQSSAKG LEWIG 
YKTSFRSRATLSIDTSNNRFSLTFSFVTAADTAVYYCA R 
WGQGLLVTVSS 5329_C19 (PGT-136) gamma heavy chain Kabat CDRs: (SEQ
ID NO: 250) CDR 1: GTDWGENDFHYG (SEQ ID NO: 251) CDR 2:
SIHWRGRTHYKTSFRS (SEQ ID NO: 252) CDR 3: HKYHDI FRVVPVAGWFDP
5329_C19 (PGT-136) gamma heavy chain Chothia CDRs: (SEQ ID NO: 253)
CDR 1: GGS MRGTDWGEND (SEQ ID NO: 254) CDR 2: SIHWRGRTH (SEQ
ID NO: 252) CDR 3: HKYHDI FRVVPVAGWFDP 5329_C19 (PGT-136) kappa
light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ
ID NO: 255)

ATGGAACCCCAGCTCAGCTTCTCTTCCTCCTGCTACTCTGGCTCCCAGAT
AGCACTGGAGAAATAGTGATGACGCAGTCTCCACCCACCCTGTCTGTG
TCTCCAGGGGAAACAGCCACACTCTCCTGTAGGGCCAGTCAGAATGT
TAAGAATAATTTAGCCTGGTACCAGCTGAAACCTGGCCAGGCTCCCA
GGCTCCTCATCTTTGATGCGTCCAGCAGGGCCGGTGGTATTCCTGAC
AGGTTCAGTGGCAGCGGTTATGGGACAGACTTCACTCTCACCGTCAA
CAGTGTGCAGTCCGAAGATTTTGGAGATTATTTTGTGTCAGCAATATGA
AGAGTGGCCTCGGACGTTCCGGCCAAGGGACCAAGGTGGATATCAAAC
GTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGT
TGAAATCTGGAACTGCCTCTGTTGTGTGCCTGCTGAATAACTTCTATCCCA
GAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCCCTCCAATCGGGTAA
CTCCAGGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACCTACAGC
CTCAGCAGCACCTGACGCTGAGCAAAGCAGACTACGAGAAACACAAAG

TCTACCTGCTGCGCTGAGCTCGAGCTCGCCCGTCACAAAG
AGCTTCAACAGGGGAGAGTGTTAG 5329_C19 (PGT-136) kappa light chain
variable region nucleotide sequence: (SEQ ID NO: 256)
GAAATAGTGATGACGCAGTCTCCACCCACCCTGTCTGTGTCTCCAGGGGA
AACAGCCACACTCTCCTGTAGGGCCAGTCAGAATGTTAAGAATAATTTAG
CCTGGTACCAGCTGAAACCTGGCCAGGCTCCCAGGCTCCTCATCTTTGAT
GCGTCCAGCAGGGCCGGTGGTATTCCTGACAGGTTCAAGTGGCAGCGGTTA
TGGGACAGACTTCACTCTCACCGTCAACAGTGTGCAGTCCGAAGATTTTG
GAGATTATTTTTGTGAGCAATATGAAGAGTGGCCTCGGACGTTCCGCCAA
GGGACCAAGGTGGATATCAAA 5329_C19 (PGT-136) kappa light chain amino
acid sequence: expressed protein with variable region in bold. (SEQ ID NO:
257) **METPAQLLFLLLLWLPDSTGEIVMTQSPPTLSVSPGETATLSCRASQNVK**
NNLAWYQLKPGQAPRLIFDASSRAGGIPDRFSGSGYGTDFTLTVNSVQS
EDFGDYFCQQYEEWPRTFGQGTKVDIKRTVAAPSVFIFPPSDEQLKSGTAS
VVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSYSTLSSTLTLSK
ADYEKHKVYACEVTHQGLSPVTKSFNRGEC 5329_C19 (PGT-136) kappa light
chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia
CDRs in bold italics) (SEQ ID NO: 258) EIVMTQSPPTLSVSPGETATLSC
custom character WYQLKPGQAPRLIF custom character custom character
GIPDRFSGSGYGTDFTLTVNSVQSEDFGDYFC custom character FGQGTKVD IK
5329_C19 (PGT-136) kappa light chain Kabat CDRs: (SEQ ID NO: 259) CDR
1: RASQNVKNNLA (SEQ ID NO: 260) CDR 2: DASSRAG (SEQ ID NO: 245)
CDR 3: QQYEEWPRT 5329_C19 (PGT-136) kappa light chain Chothia CDRs:
(SEQ ID NO: 259) CDR 1: RASQNVKNNLA (SEQ ID NO: 260) CDR 2:
DASSRAG (SEQ ID NO: 245) CDR 3: QQYEEWPRT 5366_P21 (PGT-136)
gamma heavy chain nucleotide sequence: coding sequence (variable region in
bold) (SEQ ID NO: 246)
ATGAAACACCTGTGGTTCTTCCTCCTGCTAGTGGCGGCTCCCAGATGGGTC
CTGTGCGCAGCTGCAGTTGCAGGAATCGGGCCCAGGACTGGTGAAGCC
TTCGGAGACCCTGTCCCTGACTTGCACAGTTTCTGGTGGCTCCATGA
GGGGCACCGACTGGGGCGAGAATGACTTCCACTACGGCTGGATCCG
CCAGTCCTCCGCAAAGGGGCTGGAGTGGATTGGGAGCATCCATTGGA
GGGGGAGGACCACCCACTACAAGACGTCCTTCAGGAGTCGGGCCAC
CTTGTCGATAGACACGTCCAATAATCGCTTCTCCCTGACGTTTAGTTT
TGTGACCGCCGCGGACACGGCCGTCTACTATTGTGCGAGACATAAAT
ATCATGATATTTTCAGGGTGGTCCCTGTTGCGGGGCTGGTTCGACCCC
TGGGGCCAGGGATTACTGGTCACCGTCTCGAGCGCCTCCACCAAGGGC
CCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCAC
AGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGG
TGTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCT
GTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCC
CTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGC
CCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAA
AACTCACACATGCCCACCGTGCCAGCACCTGAACTCCTGGGGGGACCGT
CAGTCTTCTCTTCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGA
CCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAG
GTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGAC
AAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTC
CTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG

CCCAAGGAGCGAGCCCGAGCCAGCCCTGACACCCCTGCCCCCTGCCG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACCTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5366_P21 (PGT-136) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 247)
CAGCTGCAGTTGCAGGAATCGGGCCCAGGACTGGTGAAGCCTTCGGAGA
CCCTGTCCCTGACTTGCACAGTTTCTGGTGGCTCCATGAGGGGGCACCGACT
GGGGCGAGAATGACTTCCACTACGGCTGGATCCGCCAGTCCTCCGCAAAG
GGGCTGGAGTGGATTGGGAGCATCCATTGGAGGGGGAGGACCACCCACT
ACAAGACGTCCTTCAGGAGTCGGGGCCACCTTGTGATAGACACGTCCAAT
AATCGCTTCTCCCTGACGTTTAGTTTTGTGACCGCCGCGGACACGGCCGTC
TACTATTGTGCGAGACATAAATATCATGATATTTTCAGGGTGGTCCCTGTT
GCGGGCTGGTTCGACCCCTGGGGCCAGGGATTACTGGTCACCGTCTCGAG C
5366_P21 (PGT-136) gamma heavy chain amino acid sequence: expressed
protein with variable region in bold. (SEQ ID NO: 248)

MKHLWFFLLVAAPRWVLSQLQLQESGPGLVKPSETLSLTCTVSGGSM
RGTDWGENDFHYGWIRQSSAKGLEWIGSIHWRGRTHYKTSFRSRATLS
IDTSNNRFSLTFSFVTAADTAVYYCARHKYHDI**FRVVPVAGWFDPWGQG**
LLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGAL
TSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVE
PKSCDKHTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDV
SHE DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK 5366_P21 (PGT-136) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 249) QLQLQESGPGLVKPSETLSLTCTVS

 FHYGWIRQSSAKG LEWIG 
YKTSFRSRATLSIDTSNNRFSLTFSFVTAADTAVYYCA R 
WGQGLLVTVSS 5366_P21 (PGT-136) gamma heavy chain Kabat CDRs: (SEQ
ID NO: 250) CDR 1: GTDWGENDFHYG (SEQ ID NO: 251) CDR 2:
SIHWRGRTHYKTSFRS (SEQ ID NO: 252) CDR 3: HKYHDI FRVVPVAGWFDP
5366_P21 (PGT-136) gamma heavy chain Chothia CDRs: (SEQ ID NO: 253)
CDR 1: GGSMRGTDWGEND (SEQ ID NO: 254) CDR 2: SIHWRGRTH (SEQ
ID NO: 252) CDR 3: HKYHDI FRVVPVAGWFDP 5366_P21 (PGT-136) kappa
light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ
ID NO: 255)

ATGGAACCCCAGCTCAGCTTCTCTTCCTCCTGCTACTCTGGCTCCCAGAT
AGCACTGGAGAAATAGTGATGACGCAGTCTCCACCCACCCTGTCTGTG
TCTCCAGGGGAAACAGCCACACTCTCCTGTAGGGCCAGTCAGAATGT
TAAGAATAATTTAGCCTGGTACCAGCTGAAACCTGGCCAGGCTCCCA
GGCTCCTCATCTTTGATGCGTCCAGCAGGGCCGGTGGTATTCCTGAC
AGGTTCAGTGGCAGCGGTTATGGGACAGACTTCACTCTCACCGTCAA
CAGTGTGCAGTCCGAAGATTTTGGAGATTATTTTGTGTCAGCAATATGA
AGAGTGGCCTCGGACGTTTCGGCCAAGGGACCAAGGTGGATATCAAAC
GTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGT
TGAAATCTGGAACCTGCCTCTGTTGTGTGCCTGCTGAATAACTTCTATCCCA

AGAGCCAAAGTACGATACGACCCCTGCGAATCGGTTAA
CTCCCAGGAGAGTGTACACAGAGCAGGACAGCAAGGACAGCACCTACAGC
CTCAGCAGCACCCCTGACGCTGAGCAAAGCAGACTACGAGAAACACAAAG
TCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTCGCCCGTCACAAAG
AGCTTCAACAGGGGAGAGTGTTAG 5366_P21 (PGT-136) kappa light chain
variable region nucleotide sequence: (SEQ ID NO: 256)

GAAATAGTGATGACGCAGTCTCCACCCACCCTGTCTGTGTCTCCAGGGGA
AACAGCCACACTCTCCTGTAGGGCCAGTCAGAATGTTAAGAATAATTTAG
CCTGGTACCAGCTGAAACCTGGCCAGGCTCCCAGGCTCCTCATCTTTGAT
GCGTCCAGCAGGGCCGGTGGTATTCCTGACAGGTTCAAGTGGCAGCGGTTA
TGGGACAGACTTCACTCTCACCGTCAACAGTGTGCAGTCCGAAGATTTTG
GAGATTATTTTTGTCAGCAATATGAAGAGTGGCCTCGGACGTTCCGGCCAA
GGGACCAAGGTGGATATCAAA 5366_P21 (PGT-136) kappa light chain amino
acid sequence: expressed protein with variable region in bold. (SEQ ID NO:

257) **METPAQLLFLLLLWLPDSTGEIVMTQSPPTLSVSPGETATLSCRASQNVK**
NNLAWYQLKPGQAPRLLIFDASSRAGGIPDRFSGSGYGTDFTLTVNSVQS
EDFGDYFCQQYEEWPRTFGQGTKVDIKRTVAAPSVFIFPPSDEQLKSGTAS
VVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSLSTLTSK

ADYEKHKVYACEVTHQGLSPVTKSFNRGEC 5366_P21 (PGT-136) kappa light
chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia
CDRs in bold italics) (SEQ ID NO: 258) EIVMTQSPPTLSVSPGETATLSC

WYQLKPGQAPRLLIF
GIPDRFSGSGYGTDFTLTVNSVQSEDFGDYFCFGQGTKVD IK
5366_P21 (PGT-136) kappa light chain Kabat CDRs: (SEQ ID NO: 259) CDR
1: RASQNVKNNLA (SEQ ID NO: 260) CDR 2: DASSRAG (SEQ ID NO: 245)
CDR 3: QQYEEWPRT 5366_P21 (PGT-136) kappa light chain Chothia CDRs:
(SEQ ID NO: 259) CDR 1: RASQNVKNNLA (SEQ ID NO: 260) CDR 2:
DASSRAG (SEQ ID NO: 245) CDR 3: QQYEEWPRT 4964_G22 (PGT-141)
gamma heavy chain nucleotide sequence: coding sequence (variable region in
bold) (SEQ ID NO: 273)

ATGGACTGGATTTGGAGGATCCTCTTCTTGGTGGCAGCAGTTGCAAGTGC
CCACTCGCAGGTGCAGCTGGTGCAGTCTGGGGCCGGAGGTGAAGAAGC
CTGGGTCTCTCAGTGAAGGTCTCCTGCAAGGCCTCTGGAAACACCTTC
AGTAAATATGATGTCCACTGGGTACGACAGGCCACTGGACAGGGGGCT
TGAATGGGTGGGATGGATGAGTCATGAGGGTGATAAGACAGAATCTG
CACAGAGATTAAAGGGCCGAGTCACCTTCACGAGGGGACACTTCCGCA
AGCACAGCCTACATGGAAGTGC GCGGCCTGACATCTGACGACACGGC
CATCTATTATTGTACGAGAGGCTCAAACATCGTTTGCGAGACTACGT
TCTCTACGATGACTACGGCTTAATTAATTATCAAGAGTGGAATGACTA
CCTTGAATTTTTGGACGTCTGGGGCCATGGAACCGCGGTCACCGTCT
CCTCAGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
AAGAGCACCTCTGGGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACT
ACTTCCCCGAACCGGTGACGGTGTCTGTGGAAGTCAAGGCGCCCTGACCAGC
GGCGTGACACCTTCCCGGCTGTCTTACAGTCCTCAGGACTCTACTCCCTC
AGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACAT
CTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTT
GAGCCCAAATCTTGTGACAAAACCTCACACATGCCCACCGTGCCCAGCACC
TGAAGTCTTGGGGGGACCGTCAGTCTTCTTCTTCCCCCAAACCCCAAGG
ACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTGGTGGTGGAC
GTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGT

TTCGGGTCCTCCCTGAGGCTGGATCAGGCACCTT
TACACTGAAAATCAGTAGAGTGGAGGCTGAAGATGCTGCAATTTATT
ATTGCATGCAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACC
AAGTTGGAAATCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCC
GCCATCTGATGAGCAGTTGAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCT
GAATAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAAC
GCCCTCCAATCGGGTAACTCCCAGGAGAGTGTACACAGAGCAGGACAGCA
AGGACAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGA
CTACGAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGA
GCTCGCCCGTCAACAAAGAGCTTCAACAGGGGAGAGTGTTAG 4964_G22 (PGT-141)

kappa light chain variable region nucleotide sequence: (SEQ ID NO: 283)

GATACTGTCGTGACTCAGTCTCCACTCTCCCTGCCCCGTCACCCCTGG
AGAGGCGGGCCTCCATGTCCTGTTTCGTCGACTCAGAGCCTCCGGGCATA
GTAATGGAGCCAACCTATTTGGCTTGGTATCAGCACAAACCGGGGCAG
TCTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGGCCTCCGGGGT
CCCTGACAGATTCAAGTGGCAGTGGATCAGGCACTCATTTTACACTGA
AAATCAGTAGAGTGGAGGCTGAAGATGCTGCAATTTATTATTGCATG
CAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACCAAGTTGGA AATCAAA

4964_G22 (PGT-141) kappa light chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 284)

*MRLPAQLLGLLMLWVSGSSADTVVTQSPLSLPVT***PGEAASMSCSSTQSLRHS**
NGANYLAWYQH**KPGQSPRL***LIRLGSQRASGV***PDRFSGSGSGTHFTLKISR**
VEAEDA**AIYYCMQGLNRPWT****FGKGTKLEIK***RTVAAPSVFIFPPSDEQLKSG*
*TASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSK***STYSLSSTLT**

LSKADYEKHKVYACEVTHQGLSPVTKSFNRGEC 4964_G22 (PGT-141) kappa light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 285) *DTVVTQSPLSLPVT***PGEAASMSC**

 *WYQHKPGQSPRL***LIR** 

*GV***PDRFSGSGSGTHFTLKISR****VEAEDA****AIYYC**  *FGK* **GTKLEIK**

4964_G22 (PGT-141) kappa light chain Kabat CDRs: (SEQ ID NO: 286) CDR 1: **SSTQSLRHS****NGANYLA** (SEQ ID NO: 287) CDR 2: **LGSQRAS** (SEQ ID NO: 288) CDR 3: **MQGLNRPWT** 4964_G22 (PGT-141) kappa light chain Chothia CDRs: (SEQ ID NO: 286) CDR 1: **SSTQSLRHS****NGANYLA** (SEQ ID NO: 287) CDR 2: **LGSQRAS** (SEQ ID NO: 288) CDR 3: **MQGLNRPWT**

4993_K13 (PGT-141) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 289)

ATGGACTGGATTTGGAGGATCCTCTTCTTGGTGGCAGCAGTTGCAAGTGC
CCACTCGCAGGTGCAGCTGGTGCAGTCTGGGGCCTGAGGTGAAGAAGC
CTGGGTCCTCAGTGAAGGTCTCCTGCAAGGCCTCTGGAAACACCTTC
AGTAAATATGATGTCCACTGGGTACGGCAGGCCACTGGACAGGGGCT
TGAATGGGTGGGATGGATGAGTCATGAGGGTGATAAGACAGAATCTG
CACAGAGATTTAAGGGCCGAGTCACCTTCACGAGGGGACACTTCCGCA
AGCACAGCCTACATGGAAGTGC**CGCGGCCTGACATCTGACGACACGGC**
CATTTATTATTGTACGAGAGGCTCAA**AACATCGCTTGCGAGACTATGT**
TCTCTACGATGACTACGGCTTAATTAATTATCAAGAGTGGAATGACTA
CCTTGAATTTTTTGGACGTCTGGGGCCATGGAACCGCGGTCACCGTCT
CCTCAGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCTCCTCC
AAGAGCACCTCTGGGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACT
ACTTCCCCGAACCGGTGACGGTGT**CGTGGAAGTCAAGGCGCCCTGACCAGC**
GGCGTGACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTC

AGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACAT
CTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTT
GAGCCCAAATCTTGTGACAAAACCTCACACATGCCACCGTGCCCAGCACC
TGAATCCTGGGGGGACCGTCAGTCTTCCTCTTCCCCC AAAACCCAAGG
ACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTGGTGGTGGAC
GTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGT
GGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGC
ACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAA
TGGAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCA
TCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGT
GTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTGAGC
CTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTG
GGAGAGCAATGGGCAGCCGGAGAACAACTACAAGACCACGCCTCCCGTG
CTGGACTCCGACGGCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAA
GAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGG
CTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAA TGA
4993_K13 (PGT-141) gamma heavy chain variable region nucleotide sequence:
(SEQ ID NO: 290)

MDWTWRILFLVAVASAHSQVQLVQSGPEVKKPGSSVKVSCKASGNTFSKY
DVHWVRQATGQGLEWVGWMSHEGDKTESAQRFKGRVTFTRDTSASTA
YMELRGLTSDDTAIYYCTRGSKHRLRDYVLYDDYGLINYQEWNDYLEFL
DVWGHGTAVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTV
SWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNT
KVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVS
LTCLVKGFYPSDIAVEWESNGQPENNYKTTTPPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK 4993_K13 (PGT-141) gamma heavy
chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia
CDRs in bold italics) (SEQ ID NO: 276) QVQLVQSGPEVKKPGSSVKVSCKAS

chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 282) ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGGTCTCTGGA TCCAGTGCGGATACTGTCGTGACTCAGTCTCCACTCTCCCTGCCCCGTC ACCCCTGGAGAGGGCGGCCTCCATGTCCTGTTTCGTCGACTCAGAGCCT CCGGCATAGTAATGGAGCCAACTATTTGGCTTGGTATCAGCACAAAC CGGGGCAGTCTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGGCC TCCGGGGTCCCTGACAGATTCAGTGGCAGTGGATCAGGCACTCATTT TAACTGAAAATCAGTAGAGTGGAGGCTGAAGATGCTGCAATTTATT ATTGCATGCAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACC AAGTTGGAAATCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCC GCCATCTGATGAGCAGTTGAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCT GAATAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAAC GCCCTCCAATCGGGTAACTCCCAGGAGAGTGTACAGAGCAGGACAGCA AGGACAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGA CTACGAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGA GCTCGCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 4993_K13 (PGT-141)

kappa light chain variable region nucleotide sequence: (SEQ ID NO: 283) **GATACTGTCGTGACTCAGTCTCCACTCTCCCTGCCCCGTCACCCCTGG AGAGGCGGCCTCCATGTCCTGTTTCGTCGACTCAGAGCCTCCGGGCATA GTAATGGAGCCAACTATTTGGCTTGGTATCAGCACAAACCGGGGCGAG TCTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGGCCTCCGGGGT CCCTGACAGATTCAGTGGCAGTGGATCAGGCACTCATTTTAACTGA AAATCAGTAGAGTGGAGGCTGAAGATGCTGCAATTTATTATTGCATG CAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACCAAGTTGGA AATCAAA**

4993_K13 (PGT-141) kappa light chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 284)

*MRLPAQLLGLLMLWVSGSSADTVVTQSPLSLPVT***PGEAASMSCSSTQSLRHS NGANYLAWYQHKGPGQSPRL***LIRLGSQRASGV***PDRFSGSGSGTHFTLKISR VEAEDA***AIYYCMQGLNRPWT***FGKGTKLEIKRTVAAPSVFIFPPSDEQLKSG TASVVCLLN***NFYPREAKVQWKVDNALQSGNSQESVTEQDSK***DSTYSLSSTLT**

LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC 4993_K13 (PGT-141) kappa light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 285) *DTVVTQSPLSLPVT***PGEAASMSC**

custom character **WYQHKGPGQSPRL***LI* custom character

GVPDRFSGSGSGTHFTLKISRVEAEDA*AIYYC*custom character **FGK GTKLEIK** 4993

K13 (PGT-141) kappa light chain Kabat CDRs: (SEQ ID NO: 286) CDR 1:

SSTQSLRHSNGANYLA (SEQ ID NO: 287) CDR 2: **LGSQRAS** (SEQ ID NO:

288) CDR 3: **MQGLNRPWT** 4993_K13 (PGT-141) kappa light chain Chothia

CDRs: (SEQ ID NO: 286) CDR 1: **SSTQSLRHS**NGANYLA (SEQ ID NO: 287)

CDR 2: **LGSQRAS** (SEQ ID NO: 288) CDR 3: **MQGLNRPWT** 4995_E20 (PGT-

142) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 314)

ATGGACTGGATTTGGAGGATCCTCTTCTTGGTGGCAGCAGTTGCAAGTGC CCACTCGCAGGTGCAGCTGGTGCAGTCTGGGCCTGAGGTGAAGAAGC CTGGGTCCTCAGTGAAGGTCTCCTGCAAGGCCTCTGGAAACACCTTC AGTAAATATGATGTCCACTGGGTACGACAGGCCACTGGACAGGGGGCT TGAATGGGTGGGATGGATTAGTCATGAGCGTGATAAGACAGAATCTG CACAGAGATTTAAGGGCCGAGTCACCTTCACGAGGGGACACTTCCGCA ACCACAGCCTACATGGAAGTGCAGCGGCCTGACATCTGACGACACGGC CATTATATTATTGTACGAGAGGCTCAAAACATCGCTTGCGAGACTACGT

heavy chain Kabat CDRs: (SEQ ID NO: 277) CDR 1: KYDVH (SEQ ID NO: 293) CDR 2: WISHERDKTESAQRFKG (SEQ ID NO: 279) CDR 3: GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV 4995_E20 (PGT-142) gamma heavy chain Chothia CDRs: (SEQ ID NO: 280) CDR 1: GNTFSK (SEQ ID NO: 294) CDR 2: WISHERDKTE (SEQ ID NO: 279) CDR 3: GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV 4995_E20 (PGT-142) kappa light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 282) ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGGTCTCTGGA TCCAGTGCGGATACTGTCGTGACTCAGTCTCCACTCTCCCTGCCCCGTC **ACCCCTGGAGAGGGCGGCCTCCATGTCCTGTTTCGTGACTCAGAGCCT CCGGCATAGTAATGGAGCCAACTATTTGGCTTGGTATCAGCACAAAC CGGGGCAGTCTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGGC TCCGGGGTCCCTGACAGATTCAGTGGCAGTGGATCAGGCACTCATTT TACACTGAAAATCAGTAGAGTGGAGGCTGAAGATGCTGCAATTTATT ATTGCATGCAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACC AAGTTGGAAATCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCC GCCATCTGATGAGCAGTTGAAATCTGGAACCTGCCTCTGTTGTGTGCCTGCT GAATAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAAC GCCCTCCAATCGGGTAACTCCCAGGAGAGTGTACACAGAGCAGGACAGCA AGGACAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGA CTACGAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGA GCTCGCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 4995_E20 (PGT-142) kappa light chain variable region nucleotide sequence: (SEQ ID NO: 283) **GATACTGTCGTGACTCAGTCTCCACTCTCCCTGCCCCGTCACCCCTGG AGAGGCGGCCTCCATGTCCTGTTTCGTGACTCAGAGCCTCCGGGCATA GTAATGGAGCCAACTATTTGGCTTGGTATCAGCACAAACCGGGGCGAG TCTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGGCCTCCGGGGT CCCTGACAGATTCAGTGGCAGTGGATCAGGCACTCATTTTACACTGA AAATCAGTAGAGTGGAGGCTGAAGATGCTGCAATTTATTATTGCATG CAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACCAAGTTGGA AATCAAA** 4995_E20 (PGT-142) kappa light chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 284) *MRLPAQLLGLLMLWVSGSSADTVVTQSPLSLPVT*PGEAASMSCSSTQSLRHS **NGANYLAWYQHKPGQSPRL**LIRLGSQRASGVPDRFSGSGSGTHFTLKISR **VEAEDA**AIYYCMQGLNRPWTFGKGTKLEIKRTVAAPSVFIFPPSDEQLKSG TASVVCLLNFPYPREAKVQWKVDNALQSGNSQESVTEQDSKDSSTLSSTLT LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC 4995_E20 (PGT-142) kappa light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 285) DTVVTQSPLSLPVT**PGEAASMSC** WYQHKPGQSPRL**L** R GVPDRFSGSGSGTHFTLKISR**VEAEDA**AIYYC FGK GTKLEIK 4995_E20 (PGT-142) kappa light chain Kabat CDRs: (SEQ ID NO: 286) CDR 1: SSTQSLRHSNGANYLA (SEQ ID NO: 287) CDR 2: LGSQRAS (SEQ ID NO: 288) CDR 3: MQGLNRPWT 4995_E20 (PGT-142) kappa light chain Chothia CDRs: (SEQ ID NO: 286) CDR 1: SSTQSLRHSNGANYLA (SEQ ID NO: 287) CDR 2: LGSQRAS (SEQ ID NO: 288) CDR 3: MQGLNRPWT 4980_N08 (PGT-143) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 295) ATGGACTGGATTTGGAGGATCCTCTTCTTGGTGGCAGCAGTTGCAAGTGC CCACGCGCAGGTGCAGCTGGAGCAGTCTGGGGCTGAGGTGAAGAAGC**

CTGGGTCCTGCTAAGGCTCTCTGGAAACAGCTTC
AGTAAATATGATGTCCACTGGGTACGACAGGCCACTGGACAGGGGGCT
TGAATGGGTGGGATGGATGAGTCATGAGGGTGATAAGACAGAATCTG
CACAGAGATTTAAGGGGGCGAGTCACCTTCACGAGGGGACACTTCCGCA
AGCACAGCCTACATGGAAGTGC GCGGCCTGACATCTGACGACACGGC
CATTTATTATTGTACGAGAGGTTCAA AACATCGCTTGCGAGACTACGT
TCTCTACGATGACTACGGCTTAATTAATTATCAAGAGTGGAATGACTA
CCTTGAATTTTTGGACGTCTGGGGCCATGGAACCGCGGTCACCGTCT
CCTCAGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCC
AAGAGCACCTCTGGGGGACAGCGGCCCTGGGCTGCCTGGTCAAGGACT
ACTTCCCCGAACCGGTGACGGTGTCTGTGGAAGTCAAGGCGCCCTGACCAGC
GGCGTGACACCTTCCCGGCTGTCTTACAGTCCTCAGGACTCTACTCCCTC
AGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACAT
CTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTT
GAGCCCAAATCTTGTGACAAAACCTCACACATGCCCACCGTGCCCAGCACC
TGAAGTCTTGGGGGGACCGTCAGTCTTCTCTTCCCCCAAACCCCAAGG
ACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTGGTGGTGGAC
GTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGT
GGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGC
ACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAA
TGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCA
TCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGT
GTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGC
CTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTG
GGAGAGCAATGGGCAGCCGGAGAACAACCTACAAGACCACGCCTCCCGTG
CTGGACTCCGACGGCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAA
GAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGG
CTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAA TGA

4980_N08 (PGT-143) gamma heavy chain variable region nucleotide sequence:
(SEQ ID NO: 296)

CAGGTGCAGCTGGAGCAGTCTGGGGCTGAGGTGAAGAAGCCTGGGT
CCTCAGTGAAGGTCTCCTGCAAGGCCTCTGGAAACACCTTCAGTAAA
TATGATGTCCACTGGGTACGACAGGCCACTGGACAGGGGGCTTGAATG
GGTGGGATGGATGAGTCATGAGGGTGATAAGACAGAATCTGCACAGA
GATTTAAGGGGGCGAGTCACCTTCACGAGGGGACACTTCCGCAAGCACA
GCCTACATGGAAGTGC GCGGCCTGACATCTGACGACACGGCCATTTA
TTATTGTACGAGAGGTTCAA AACATCGCTTGCGAGACTACGTTCTCTA
CGATGACTACGGCTTAATTAATTATCAAGAGTGGAATGACTACCTTGA
ATTTTTGGACGTCTGGGGCCATGGAACCGCGGTCACCGTCTCCTCA 4980_N08

(PGT-143) gamma heavy chain amino acid sequence: expressed protein with
leader sequence in italics and variable region in bold. (SEQ ID NO: 297)

MDWIWRILFLVA AVASAHQVQLEQSGAEVKKPGSSVKV SCKASGNTFSKY
DVHWVRQATGQGLEWVGWMSHEGDKTESAQRFKGRVTFTRDTSASTA
Y MELRGLTSDDTAIYYCTRGSKHRLRDYVLYDDYGLINYQE WNDYLEFL
DVWGHGTA*VT***VSSAST***KGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTV*
SWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNT
KVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVS
LTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR

WQQGNVFSVMHEALNHYTQKSLSPGK 4980_N08 (PGT-143) gamma heavy chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 298) QVQLEQSGAEVKKPGSSSVKVSCKAS  custom character YDVHWVRQATGQGLEWVG  custom character SAQRFKGRVTFTTRDTSASTAYMELRGLTSDDTAIYYCTR  custom character  custom character  custom character WGHGTAVTVSS 4980_N08 (PGT-143) gamma heavy chain Kabat CDRs: (SEQ ID NO: 277) CDR 1: KYDVH (SEQ ID NO: 278) CDR 2: WMSHEGDKTESAQRFKG (SEQ ID NO: 279) CDR 3: GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV 4980_N08 (PGT-143) gamma heavy chain Chothia CDRs: (SEQ ID NO: 280) CDR 1: GNTFSK (SEQ ID NO: 281) CDR 2: WMSHEGDKTE (SEQ ID NO: 279) CDR 3: GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV 4980_N08 (PGT-143) kappa light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 299) ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGGTCTCTGGA TCCAGTGCGGATACTGTCGTGACTCAGTCTCCACTCTCCCTGCCCCGTC ACCCCTGGAGAGGCGGCCTCCATGTCCTGTACGTCGACTCAGAGCCT CCGTCATAGTAATGGAGCCAACTATTTGGCTTGGTACCAGCACAAAC CAGGGCAGTCTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGGCC TCCGGGGTCCCTGACAGATTCAGTGGCAGTGGATCAGGCACTCATT TACTACTGAAAATCAGTCGAGTGGAGCCTGAAGATGCTGCAATTTATT ATTGCATGCAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACC AAGTTGGAAATCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCC GCCATCTGATGAGCAGTTGAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCT GAATAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAAC GCCCTCCAATCGGGTAACTCCCAGGAGAGTGTACACAGAGCAGGACAGCA AGGACAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGA CTACGAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGA GCTCGCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 4980_N08 (PGT-143) kappa light chain variable region nucleotide sequence: (SEQ ID NO: 300) GATACTGTCGTGACTCAGTCTCCACTCTCCCTGCCCCGTCACCCCTGG AGAGGCGGCCTCCATGTCCTGTACGTCGACTCAGAGCCTCCGTCATA GTAATGGAGCCAACTATTTGGCTTGGTACCAGCACAAACCAGGGCAG TCTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGCCTCCGGGGT CCCTGACAGATTCAGTGGCAGTGGATCAGGCACTCATTTTACTACTGA AAATCAGTCGAGTGGAGCCTGAAGATGCTGCAATTTATTATTGCATG CAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACCAAGTTGGA AATCAA 4980_N08 (PGT-143) kappa light chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 301) *MRLPAQLLGLLMLWVSGSSADTVVTQSPLSLPVT* **PGEAASMSCTSTQSLRHS** **NGANYLAWYQHKPGQSPRL** **LIRLGSQRASGV** **PDRFSGSGSGTHFTLKISR** **VEPEDAAIYYCMQGLNRPWTFGKG** **GTKLEIKRTVAAPSVFIFPPSDEQLKSG** **TASVVCLLN** **NFY** **PREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLT** **LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC** 4980_N08 (PGT-143) kappa light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 302) *DTVVTQSPLSLPVT* **PGEAASMSC**  custom character WYQHKPGQSPRL **LIR**  custom character **FGK** **GTKLEIK** 4980_N08 (PGT-143) kappa light chain Kabat CDRs: (SEQ ID NO: 303) CDR 1: **TSTQSLRHS** **NGANYLA** (SEQ ID NO: 287) CDR 2: **LGSQRAS** (SEQ ID NO: 288) CDR 3: **MQGLNRPWT** 4980_N08 (PGT-143) kappa light chain

Chothia CDRs: (SEQ ID NO: 303) CDR 1: TSTQSLRHSNGANYLA (SEQ ID NO: 287) CDR 2: LGSQRAS (SEQ ID NO: 288) CDR 3: MQGLNRPWT 4970_K22 (PGT-144) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 304)

ATGGACTGGATTTGGAGGATCCTCTTCTTGGTGGCAGCAGTTGCAAGTGC
CCACTCGCAGGTGCAGCTGGTGCAGTCTGGGGCTGAGGTGAAGAAGC
CTGGGTCTCTCAGTGAAGGTCTCCTGCAAGGCCTCTGGAAACACCTTC
AGGAAATATGATGTCCACTGGGTACGACAGGCCACTGGACAGGGGCT
TGAATGGGTGGGATGGATGAGTCATGAGGGTGATAAGACAGAATCTG
CACAGAGATTTAAGGGCCGAGTCTCTTTTCACGAGGGACAATTCCGCA
AGCACAGCCTACATTGAACTGCGCGGCCTGACATCTGACGACACGGC
CATTIATTATTGTACCGGAGGCTCAAAACATCGCTTGCGAGACTACGT
TCTCTACGATGATTACGGCCTAATAAATCAGCAAGAGTGGAATGACT
ACCTTGAATTTTTGGACGTCTGGGGCCATGGAACCGCGGTACACCGTC
TCCTCAGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCTCCTC
CAAGAGCACCTCTGGGGGCACAGCGGCCCTGGGCTGCCTGGTCAAGGACT
ACTTCCCCGAACCGGTGACGGTGTCTGGAAGTCAAGGCGCCCTGACCAGC
GGCGTGACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTC
AGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACAT
CTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTT
GAGCCCAAATCTTGTGACAAAACCTCACACATGCCCACCGTGCCCAGCACC
TGAAGTCTTGGGGGGACCGTCAGTCTTCTCTTCCCCCAAACCCAAGG
ACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTGGTGGTGGAC
GTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGT
GGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGC
ACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAA
TGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCA
TCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGT
GTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGC
CTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTG
GGAGAGCAATGGGCAGCCGGAGAACAACACTACAAGACCACGCCTCCCGTG
CTGGACTCCGACGGCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAA
GAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGG
CTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAA TGA

4970_K22 (PGT-144) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 305)

CAGGTGCAGCTGGTGCAGTCTGGGGCTGAGGTGAAGAAGCCTGGGT
CCTCAGTGAAGGTCTCCTGCAAGGCCTCTGGAAACACCTTCAGGAAA
TATGATGTCCACTGGGTACGACAGGCCACTGGACAGGGGCTTGAATG
GGTGGGATGGATGAGTCATGAGGGTGATAAGACAGAATCTGCACAGA
GATTTAAGGGCCGAGTCTCTTTTCACGAGGGACAATTCCGCAAGCACA
GCCTACATTGAACTGCGCGGCCTGACATCTGACGACACGGCCATTTA
TTATTGTACCGGAGGCTCAAAACATCGCTTGCGAGACTACGTTCTCTA
CGATGATTACGGCCTAATAAATCAGCAAGAGTGGAATGACTACCTTG
AATTTTTGGACGTCTGGGGCCATGGAACCGCGGTACACCGTCTCCTCA 4970_K22

(PGT-144) gamma heavy chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 306)

*MDWIWRILFLVA*AVASAHSQVQLVQSGAEVKKPGSSVKVSCKASGNTFRKY
DVHWVRQATGQGLEWVGWMSHEGDKTESAQRFKGRVSFTRDNSASTA
YIELRGLTSDDTAIYYCTGGSKHRLRDYVLYDDYGLINQQEWNDYLEFL

DVWVSGHTAVTVSSAATGSLVSTSGGTAALFGCLVVKDYFPEPVTY
SWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNT
KVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV
VVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQD
WLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVS
LTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK 4970_K22 (PGT-144) gamma
chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia
CDRs in bold italics) (SEQ ID NO: 307) QVQLVQSGAEVKKPGSSVKVSCKAS
custom character YDVHWVRQATGQGLEWVG custom character
SAQRFKGRVSFTRDNSASTAYIELRGLTSDDTAIYYCTG custom character
custom character custom character WGHGTA VTVSS 4970_K22 (PGT-144) gamma
heavy chain Kabat CDRs: (SEQ ID NO: 277) CDR 1: KYDVH (SEQ ID NO:
278) CDR 2: WMSHEGDKTESAQRFKG (SEQ ID NO: 308) CDR 3:
GSKHRLRDYVLYDDYGLINQQEWNDYLEFLDV 4970_K22 (PGT-144) gamma heavy
chain Chothia CDRs: (SEQ ID NO: 309) CDR 1: GNTFRK (SEQ ID NO: 281)
CDR 2: WMSHEGDKTE (SEQ ID NO: 308) CDR 3:
GSKHRLRDYVLYDDYGLINQQEWNDYLEFLDV 4970_K22 (PGT-144) kappa light
chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID
NO: 310) ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGGTCTCTGGA
TCCAGTGCGGATACTGTCGTGACTCAGTCTCCACTCTCCCTGTCCGTC
ACCCCTGGAGAGGCGGCCTCCATGTCCTGTACGTCGACTCAGAGCCT
CCGGCATAGTAATGGAGCCAACTATTTGGCTTGGTACCAGCACAAAC
CAGGGCAGTCTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGCC
TCCGGGGTCCCTGACAGATTCAGTGGCAGTGGATCAGGCACTCATTT
TACACTGAAAATCAGTAGAGTGGAGGCTGACGATGCTGCAATTTATT
ATTGCATGCAAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACC
AAGTTGGAGATCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCC
GCCATCTGATGAGCAGTTGAAATCTGGA ACTGCCTCTGTTGTGTGCCTGCT
GAATAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAAC
GCCCTCCAATCGGGTAACTCCCAGGAGAGTGTACAGAGCAGGACAGCA
AGGACAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGA
CTACGAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGA
GCTCGCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 4970_K22 (PGT-144)
kappa light chain variable region nucleotide sequence: (SEQ ID NO: 311)
GATACTGTCGTGACTCAGTCTCCACTCTCCCTGTCCGTCACCCCTGGA
GAGGCGGCCTCCATGTCCTGTACGTCGACTCAGAGCCTCCGGCATAG
TAATGGAGCCAACTATTTGGCTTGGTACCAGCACAAACCAGGGCAGT
CTCCACGACTCCTAATCCGTTTAGGTTCTCAACGGGCCTCCGGGGTC
CCTGACAGATTCAGTGGCAGTGGATCAGGCACTCATTTTACACTGAA
AATCAGTAGAGTGGAGGCTGACGATGCTGCAATTTATTATTGCATGC
AAGGTCTGAACCGTCCCTGGACGTTTCGGCAAGGGGACCAAGTTGGAG ATCAAA
4970_K22 (PGT-144) kappa light chain amino acid sequence: expressed protein
with leader sequence in italics and variable region in bold. (SEQ ID NO: 312)
MRLPAQLLGLLMLWVSGSSADTVVTQSPLSLSVTPGEAASMSCTSTQSLRHS
NGANYLAWYQHKPGQSPRL*LIRLGSQRASGV***PDRFSGSGSGTHFTLKISR**
VEADDAAIYYCMQGLNRPWTFGKG*TKLEIKRTVAAPSVFIFPPSDEQLKSG*
*TASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSK***DSTYSL***SSTLT*
*LSKADY***EKH***KVYACEVTHQGLSSPVTKSFNR***GEC** 4970_K22 (PGT-144) kappa light
chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia

CDRs in **bold** (italics) (SEQ ID NO: 313) DTVVTQSPLSVTPGEAASMSC

 custom character WYQHKPGQSPRLLI R  custom character

GVPDRFSGSGSGTHFTLKISRVEADDAAIYYC  custom character FGK GTKLEIK

4970_K22 (PGT-144) kappa light chain Kabat CDRs: (SEQ ID NO: 303) CDR

1: TSTQSLRHSNGANYLA (SEQ ID NO: 287) CDR 2: LGSQRAS (SEQ ID

NO: 288) CDR 3: MQGLNRPWT 4970_K22 (PGT-144) kappa light chain

Chothia CDRs: (SEQ ID NO: 303) CDR 1: TSTQSLRHSNGANYLA (SEQ ID

NO: 287) CDR 2: LGSQRAS (SEQ ID NO: 288) CDR 3: MQGLNRPWT

(157) The 4838 L06 (PGT-121) antibody includes a heavy chain variable region (SEQ ID NO: 79),

encoded by the nucleic acid sequence shown in SEQ ID NO: 63, and a light chain variable region

(SEQ ID NO: 149) encoded by the nucleic acid sequence shown in SEQ ID NO: 147.

(158) The heavy chain CDRs of the 4838_L06 (PGT-121) antibody have the following sequences

per Kabat definition: DSYWS (SEQ ID NO: 90), YVHKSGDTNYSPLKS (SEQ ID NO: 265),

and TLHGRRIGIVAFNEWFTYFYMDV (SEQ ID NO: 143). The light chain CDRs of the 4838

L06 (PGT-121) antibody have the following sequences per Kabat definition: GEKSLGSRAVQ

(SEQ ID NO: 150), NNQDRPS (SEQ ID NO: 151), and HIWDSRVPTKWV (SEQ ID NO: 152).

(159) The heavy chain CDRs of the 4838_L06 (PGT-121) antibody have the following sequences

per Chothia definition: GASISD (SEQ ID NO: 144), YVHKSGDTN (SEQ ID NO: 145), and

TLHGRRIGIVAFNEWFTYFYMDV (SEQ ID NO: 143). The light chain CDRs of the 4838_L06

(PGT-121) antibody have the following sequences per Chothia definition: GEKSLGSRAVQ (SEQ

ID NO: 150), NNQDRPS (SEQ ID NO: 151), and HIWDSRVPTKWV (SEQ ID NO: 152).

(160) The 4873 E03 (PGT-121) antibody includes a heavy chain variable region (SEQ ID NO: 79),

encoded by the nucleic acid sequence shown in SEQ ID NO: 63, and a light chain variable region

(SEQ ID NO: 149) encoded by the nucleic acid sequence shown in SEQ ID NO: 147.

(161) The heavy chain CDRs of the 4873 E03 (PGT-121) antibody have the following sequences

per Kabat definition: DSYWS (SEQ ID NO: 90), YVHKSGDTNYSPLKS (SEQ ID NO: 265),

and TLHGRRIGIVAFNEWFTYFYMDV (SEQ ID NO: 143). The light chain CDRs of the 4873

E03 (PGT-121) antibody have the following sequences per Kabat definition: GEKSLGSRAVQ

(SEQ ID NO: 150), NNQDRPS (SEQ ID NO: 151), and HIWDSRVPTKWV (SEQ ID NO: 152).

(162) The heavy chain CDRs of the 4873 E03 (PGT-121) antibody have the following sequences

per Chothia definition: GASISD (SEQ ID NO: 144), YVHKSGDTN (SEQ ID NO: 145), and

TLHGRRIGIVAFNEWFTYFYMDV (SEQ ID NO: 143). The light chain CDRs of the 4873 E03

(PGT-121) antibody have the following sequences per Chothia definition: GEKSLGSRAVQ (SEQ

ID NO: 150), NNQDRPS (SEQ ID NO: 151), and HIWDSRVPTKWV (SEQ ID NO: 152).

(163) The 4877_D15 (PGT-122) antibody includes a heavy chain variable region (SEQ ID NO:

156), encoded by the nucleic acid sequence shown in SEQ ID NO: 154, and a light chain variable

region (SEQ ID NO: 161) encoded by the nucleic acid sequence shown in SEQ ID NO: 159.

(164) The heavy chain CDRs of the 4877_D15 (PGT-122) antibody have the following sequences

per Kabat definition: DNYWS (SEQ ID NO: 261), YVHDSGDTNYPNPSLKS (SEQ ID NO: 157),

and TKHGRRIGVVAFKEWFTYFYMDV (SEQ ID NO: 262). The light chain CDRs of the 4877

D15 (PGT-122) antibody have the following sequences per Kabat definition: GEESLGSRSVI

(SEQ ID NO: 162), NNNDRPS (SEQ ID NO: 163), and HIWDSRRPTNWW (SEQ ID NO: 164).

(165) The heavy chain CDRs of the 4877_D15 (PGT-122) antibody have the following sequences

per Chothia definition: GTLVRD (SEQ ID NO: 263), YVHDSGDTN (SEQ ID NO: 264), and

TKHGRRIGVVAFKEWFTYFYMDV (SEQ ID NO: 262). The light chain CDRs of the

4877_D15 (PGT-122) antibody have the following sequences per Chothia definition:

GEESLGSRSVI (SEQ ID NO: 162), NNNDRPS (SEQ ID NO: 163), and HIWDSRRPTNWW

(SEQ ID NO: 164).

(166) The 4858 P08 (PGT-123) antibody includes a heavy chain variable region (SEQ ID NO:

168), encoded by the nucleic acid sequence shown in SEQ ID NO: 166, and a light chain variable

region (SEQ ID NO: 177) encoded by the nucleic acid sequence shown in SEQ ID NO: 175.

(167) The heavy chain CDRs of the 4858 P08 (PGT-123) antibody have the following sequences per Kabat definition: DAYWS (SEQ ID NO: 169), YVHHSGDTNYPNPSLKR (SEQ ID NO: 170), ALHGKRIYGIVALGELFTYFYMDV (SEQ ID NO: 171). The light chain CDRs of the 4858 P08 (PGT-123) antibody have the following sequences per Kabat definition: GKESIGSRAVQ (SEQ ID NO: 178), NNQDRPA (SEQ ID NO: 179), and HIYDARGGTNWV (SEQ ID NO: 180).

(168) The heavy chain CDRs of the 4858_P08 (PGT-123) antibody have the following sequences per Chothia definition: GASIND (SEQ ID NO: 172), YVHHSGDTN (SEQ ID NO: 173), ALHGKRIYGIVALGELFTYFYMDV (SEQ ID NO: 171). The light chain CDRs of the 4858 P08 (PGT-123) antibody have the following sequences per Chothia definition: GKESIGSRAVQ (SEQ ID NO: 178), NNQDRPA (SEQ ID NO: 179), HIYDARGGTNWV (SEQ ID NO: 180).

(169) The 5123_A06 (PGT-125) antibody includes a heavy chain variable region (SEQ ID NO: 164), encoded by the nucleic acid sequence shown in SEQ ID NO: 182, and a light chain variable region (SEQ ID NO: 193) encoded by the nucleic acid sequence shown in SEQ ID NO: 191.

(170) The heavy chain CDRs of the 5123_A06 (PGT-125) antibody have the following sequences per Kabat definition: ACTYFWG (SEQ ID NO: 185), SLSHCQSFWSGWTGHNPSLKS (SEQ ID NO: 186), and FDGEVLVYNHWPKPAWVDL (SEQ ID NO: 187). The light chain CDRs of the 5123_A06 (PGT-125) antibody have the following sequences per Kabat definition: NGTATNFVS (SEQ ID NO: 194), GVDKRPP (SEQ ID NO: 195), and GSLVGNWDVI (SEQ ID NO: 196).

(171) The heavy chain CDRs of the 5123_A06 (PGT-125) antibody have the following sequences per Chothia definition: GESTGACT (SEQ ID NO: 188), SLSHCQSFWSGWTG (SEQ ID NO: 189), and FDGEVLVYNHWPKPAWVDL (SEQ ID NO: 187). The light chain CDRs of the 5123_A06 (PGT-125) antibody have the following sequences per Chothia definition: NGTATNFVS (SEQ ID NO: 194), GVDKRPP (SEQ ID NO: 195), and GSLVGNWDVI (SEQ ID NO: 196).

(172) The 5141_B17 (PGT-126) antibody includes a heavy chain variable region (SEQ ID NO: 200), encoded by the nucleic acid sequence shown in SEQ ID NO: 198, and a light chain variable region (SEQ ID NO: 209) encoded by the nucleic acid sequence shown in SEQ ID NO: 207.

(173) The heavy chain CDRs of the 5141_B17 (PGT-126) antibody have the following sequences per Kabat definition: ACDYFWG (SEQ ID NO: 201), GLSHCAGYYNTGWTYHNPSLKS (SEQ ID NO: 202), and FDGEVLVYHDWPKPAWVDL (SEQ ID NO: 203). The light chain CDRs of the 5141_B17 (PGT-126) antibody have the following sequences per Kabat definition: TGTSNRFVS (SEQ ID NO: 210), GVNKRPS (SEQ ID NO: 211), and SSLVGNWDVI (SEQ ID NO: 212).

(174) The heavy chain CDRs of the 5141_B17 (PGT-126) antibody have the following sequences per Chothia definition: GDSTAACD (SEQ ID NO: 204), GLSHCAGYYNTGWTY (SEQ ID NO: 205), and FDGEVLVYHDWPKPAWVDL (SEQ ID NO: 203). The light chain CDRs of the 5141_B17 (PGT-126) antibody have the following sequences per Chothia definition: TGTSNRFVS (SEQ ID NO: 210), GVNKRPS (SEQ ID NO: 211), and SSLVGNWDVI (SEQ ID NO: 212).

(175) The 5147_N06 (PGT-130) antibody includes a heavy chain variable region (SEQ ID NO: 216), encoded by the nucleic acid sequence shown in SEQ ID NO: 214, and a light chain variable region (SEQ ID NO: 225) encoded by the nucleic acid sequence shown in SEQ ID NO: 223.

(176) The heavy chain CDRs of the 5147_N06 (PGT-130) antibody have the following sequences per Kabat definition: TGHYYWG (SEQ ID NO: 217), HIHYTTAVLHNPSLKS (SEQ ID NO: 218), and SGGDILYYYEWQKPHWFSP (SEQ ID NO: 219). The light chain CDRs of the 5147_N06 (PGT-130) antibody have the following sequences per Kabat definition: NGTSSDIGGWNFVS (SEQ ID NO: 226), EVNKRPS (SEQ ID NO: 227), and SSLFGRWDVV (SEQ ID NO: 228).

(177) The heavy chain CDRs of the 5147_N06 (PGT-130) antibody have the following sequences per Chothia definition: GESINTGH (SEQ ID NO: 220), HIHYTTAVL (SEQ ID NO: 221), and

SGGDILYYEYWQKPHWFSP (SEQ ID NO: 219). The light chain CDRs of the 5147_N06 (PGT-130) antibody have the following sequences per Chothia definition: NGTSSDIGGWNFVS (SEQ ID NO: 226), EVNKRPS (SEQ ID NO: 227), and SSLFGRWDVV (SEQ ID NO: 228).

(178) The 5343_B08 (PGT-135) antibody includes a heavy chain variable region (SEQ ID NO: 232), encoded by the nucleic acid sequence shown in SEQ ID NO: 230, and a light chain variable region (SEQ ID NO: 242) encoded by the nucleic acid sequence shown in SEQ ID NO: 239.

(179) The heavy chain CDRs of the 5343_B08 (PGT-135) antibody have the following sequences per Kabat definition: GGEWGD KDYHWG (SEQ ID NO: 233), SIHWRGTTHYKESLRR (SEQ ID NO: 234), and HRHHDVFMLVPIAGWFDV (SEQ ID NO: 235). The light chain CDRs of the 5343_B08 (PGT-135) antibody have the following sequences per Kabat definition: RASQNINKNLA (SEQ ID NO: 243), ETYSKIA (SEQ ID NO: 244), and QQYEEWPRT (SEQ ID NO: 245).

(180) The heavy chain CDRs of the 5343_B08 (PGT-135) antibody have the following sequences per Chothia definition: GDSIRGG EWGDKD (SEQ ID NO: 236), SIHWRGTTH (SEQ ID NO: 237), and HRHEIDVFMLVPIAGWFDV (SEQ ID NO: 235). The light chain CDRs of the 5343_B08 (PGT-135) antibody have the following sequences per Chothia definition: RASQNINKNLA (SEQ ID NO: 243), ETYSKIA (SEQ ID NO: 244), and QQYEEWPRT (SEQ ID NO: 245).

(181) The 5344_E16 (PGT-135) antibody includes a heavy chain variable region (SEQ ID NO: 232), encoded by the nucleic acid sequence shown in SEQ ID NO: 230, and a light chain variable region (SEQ ID NO: 242) encoded by the nucleic acid sequence shown in SEQ ID NO: 239.

(182) The heavy chain CDRs of the 5344_E16 (PGT-135) antibody have the following sequences per Kabat definition: GGEWGD KDYHWG (SEQ ID NO: 233), SIHWRGTTHYKESLRR (SEQ ID NO: 234), and HRHHDVFMLVPIAGWFDV (SEQ ID NO: 235). The light chain CDRs of the 5344_E16 (PGT-135) antibody have the following sequences per Kabat definition: RASQNINKNLA (SEQ ID NO: 243), ETYSKIA (SEQ ID NO: 244), and QQYEEWPRT (SEQ ID NO: 245).

(183) The heavy chain CDRs of the 5344_E16 (PGT-135) antibody have the following sequences per Chothia definition: GDSIRGG EWGDKD (SEQ ID NO: 236), SIHWRGTTH (SEQ ID NO: 237), and HRHEIDVFMLVPIAGWFDV (SEQ ID NO: 235). The light chain CDRs of the 5344_E16 (PGT-135) antibody have the following sequences per Chothia definition: RASQNINKNLA (SEQ ID NO: 243), ETYSKIA (SEQ ID NO: 244), and QQYEEWPRT (SEQ ID NO: 245).

(184) The 5329_C19 (PGT-136) antibody includes a heavy chain variable region (SEQ ID NO: 249), encoded by the nucleic acid sequence shown in SEQ ID NO: 247, and a light chain variable region (SEQ ID NO: 258) encoded by the nucleic acid sequence shown in SEQ ID NO: 256.

(185) The heavy chain CDRs of the 5329_C19 (PGT-136) antibody have the following sequences per Kabat definition: GTDWGENDFHYG (SEQ ID NO: 250), SIHWRGRTTHYKTSFRS (SEQ ID NO: 251), and HKYHDIFRVVPVAGWFDP (SEQ ID NO: 252). The light chain CDRs of the 5329_C19 (PGT-136) antibody have the following sequences per Kabat definition: RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), and QQYEEWPRT (SEQ ID NO: 245).

(186) The heavy chain CDRs of the 5329_C19 (PGT-136) antibody have the following sequences per Chothia definition: GGSMRGTDWGEND (SEQ ID NO: 253), SIHWRGRTTH (SEQ ID NO: 254), and HKYHDIFRVVPVAGWFDP (SEQ ID NO: 252). The light chain CDRs of the 5329_C19 (PGT-136) antibody have the following sequences per Chothia definition: RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), and QQYEEWPRT (SEQ ID NO: 245).

(187) The 5366_P21 (PGT-136) antibody includes a heavy chain variable region (SEQ ID NO: 249), encoded by the nucleic acid sequence shown in SEQ ID NO: 247, and a light chain variable region (SEQ ID NO: 258) encoded by the nucleic acid sequence shown in SEQ ID NO: 256.

(188) The heavy chain CDRs of the 5366_P21 (PGT-136) antibody have the following sequences

per Kabat definition: GTDWGENDFHYG (SEQ ID NO: 250), SIHWRGRTHYKTSFRS (SEQ ID NO: 251), and HKYHDIFRVVPVAGWFDP (SEQ ID NO: 252). The light chain CDRs of the 5366_P21 (PGT-136) antibody have the following sequences per Kabat definition: RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), and QQYEEWPRT (SEQ ID NO: 245).

(189) The heavy chain CDRs of the 5366_P21 (PGT-136) antibody have the following sequences per Chothia definition: GGSMRGTDWGEND (SEQ ID NO: 253), SIHWRGRTH (SEQ ID NO: 254), and HKYHDIFRVVPVAGWFDP (SEQ ID NO: 252). The light chain CDRs of the 5366 P21 (PGT-136) antibody have the following sequences per Chothia definition: RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), and QQYEEWPRT (SEQ ID NO: 245).

(190) The 5964_G22 (PGT-141) antibody includes a heavy chain variable region (SEQ ID NO: 276), encoded by the nucleic acid sequence shown in SEQ ID NO: 274, and a light chain variable region (SEQ ID NO: 285) encoded by the nucleic acid sequence shown in SEQ ID NO: 283.

(191) The heavy chain CDRs of the 5964_G22 (PGT-141) antibody have the following sequences per Kabat definition: KYDVH (SEQ ID NO: 277), WMSHEGDKTESAQRFKG (SEQ ID NO: 278), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279). The light chain CDRs of the 5964_G22 (PGT-141) antibody have the following sequences per Kabat definition: SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(192) The heavy chain CDRs of the 5964_G22 (PGT-141) antibody have the following sequences per Chothia definition: GNTFSK (SEQ ID NO: 280), WMSHEGDKTE (SEQ ID NO: 281), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279). The light chain CDRs of the 5964_G22 (PGT-141) antibody have the following sequences per Chothia definition: SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(193) The 4993_K13 (PGT-141) antibody includes a heavy chain variable region (SEQ ID NO: 276), encoded by the nucleic acid sequence shown in SEQ ID NO: 290, and a light chain variable region (SEQ ID NO: 285) encoded by the nucleic acid sequence shown in SEQ ID NO: 283.

(194) The heavy chain CDRs of the 4993_K13 (PGT-141) antibody have the following sequences per Kabat definition: KYDVH (SEQ ID NO: 277), WMSHEGDKTESAQRFKG (SEQ ID NO: 278), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279). The light chain CDRs of the 4993 K13 (PGT-141) antibody have the following sequences per Kabat definition: SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(195) The heavy chain CDRs of the 4993_K13 (PGT-141) antibody have the following sequences per Chothia definition: GNTFSK (SEQ ID NO: 280), WMSHEGDKTE (SEQ ID NO: 281), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279). The light chain CDRs of the 4993 K13 (PGT-141) antibody have the following sequences per Chothia definition: SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(196) The 4995 E20 (PGT-142) antibody includes a heavy chain variable region (SEQ ID NO: 292), encoded by the nucleic acid sequence shown in SEQ ID NO: 315, and a light chain variable region (SEQ ID NO: 285) encoded by the nucleic acid sequence shown in SEQ ID NO: 283.

(197) The heavy chain CDRs of the 4995 E20 (PGT-142) antibody have the following sequences per Kabat definition: KYDVH (SEQ ID NO: 277), WISHERDKTESAQRFKG (SEQ ID NO: 293), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279). The light chain CDRs of the 4995 E20 (PGT-142) antibody have the following sequences per Kabat definition: SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(198) The heavy chain CDRs of the 4995 E20 (PGT-142) antibody have the following sequences

per Chothia definition: GNTFSK (SEQ ID NO: 280), WISHERDKTE (SEQ ID NO: 294), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279). The light chain CDRs of the 4995 E20 (PGT-142) antibody have the following sequences per Chothia definition: SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(199) The 4980_N08 (PGT-143) antibody includes a heavy chain variable region (SEQ ID NO: 298), encoded by the nucleic acid sequence shown in SEQ ID NO: 296, and a light chain variable region (SEQ ID NO: 302) encoded by the nucleic acid sequence shown in SEQ ID NO: 300.

(200) The heavy chain CDRs of the 4980_N08 (PGT-143) antibody have the following sequences per Kabat definition: KYDVH (SEQ ID NO: 277), WMSHEGDKTESAQRFKG (SEQ ID NO: 278), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279). The light chain CDRs of the 4980_N08 (PGT-143) antibody have the following sequences per Kabat definition: TSTQSLRHSNGANYLA (SEQ ID NO: 303), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(201) The heavy chain CDRs of the 4980_N08 (PGT-143) antibody have the following sequences per Chothia definition: GNTFSK (SEQ ID NO: 280), WMSHEGDKTE (SEQ ID NO: 281), and GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279). The light chain CDRs of the 4980_N08 (PGT-143) antibody have the following sequences per Chothia definition: TSTQSLRHSNGANYLA (SEQ ID NO: 303), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(202) The 4970_K22 (PGT-144) antibody includes a heavy chain variable region (SEQ ID NO: 307), encoded by the nucleic acid sequence shown in SEQ ID NO: 305, and a light chain variable region (SEQ ID NO: 313) encoded by the nucleic acid sequence shown in SEQ ID NO: 311.

(203) The heavy chain CDRs of the 4970 K22 (PGT-144) antibody have the following sequences per Kabat definition: KYDVH (SEQ ID NO: 277), WMSHEGDKTESAQRFKG (SEQ ID NO: 278), and GSKHRLRDYVLYDDYGLINQEWNDYLEFLDV (SEQ ID NO: 308). The light chain CDRs of the 4970_K22 (PGT-144) antibody have the following sequences per Kabat definition: TSTQSLRHSNGANYLA (SEQ ID NO: 303), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(204) The heavy chain CDRs of the 4970_K22 (PGT-144) antibody have the following sequences per Chothia definition: GNTFRK (SEQ ID NO: 309), WMSHEGDKTE (SEQ ID NO: 281), and GSKHRLRDYVLYDDYGLINQEWNDYLEFLDV (SEQ ID NO: 308). The light chain CDRs of the 4970 K22 (PGT-144) antibody have the following sequences per Chothia definition: TSTQSLRHSNGANYLA (SEQ ID NO: 303), LGSQRAS (SEQ ID NO: 287), and MQGLNRPWT (SEQ ID NO: 288).

(205) The sequences of additional human monoclonal antibodies were determined, including the sequences of the variable regions of the Gamma heavy and Kappa or Lambda light chains. In addition, the sequence of each of the polynucleotides encoding the antibody sequences was determined. Shown below are the polypeptide and polynucleotide sequences of the gamma heavy chains and kappa light chains, with the signal peptides at the N-terminus (or 5' end) and the constant regions at the C-terminus (or 3' end) of the variable regions, which are shown in bolded text.

(206) TABLE-US-00015 5145_B14 (PGT-127) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 316)
ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCGGCTCCCAGATGGGTC
CTGTCCCAGCCGCAGCTGCAGGAGTCGGGCCCAGGACTGGTGGAGGC
TTCGGAGACCCTGTCCCTCACGTGCACTGTGTCCGGCGACTCCACTG
GTCGTTGTAATTATTTCTGGGGCTGGGTCCGGCAGCCCCAGGGAAG
GGGCTGGAGTGGATTGGGAGTTTGTCCCACTGTAGAAGTTACTACAA
TACTGACTGGACCTACCACAACCCGTCTCTCAAGAGTCGACTCACTAT

TTCACTCGACACGACGAAGAACTTCCTGAGATTGACCTGT
GACCGCCGCGGACACGGCCACTTATTACTGTGCGCGATTTCGGCGGGCG
AAGTTCTAGTGTACAGAGATTGGCCAAAGCCGGCCTGGGTTCGACCTC
TGGGGCCCGGGGAACGCTGGTTCGTACCGTCTCGAGCGCCTCCACCAA
GGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGG
GCACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTG
ACGGTGTCTGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCC
GGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCG
TGCCCTCCAGCAGCTTGGGACCCAGACCTACATCTGCAACGTGAATCAC
AAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTG
ACAAAACCTCACACATGCCACCGTGCCCAGCACCTGAACTCCTGGGGGGA
CCGTCAGTCTTCTCTTCCCCCAAAACCCAAGGACACCCTCATGATCTCC
CGGACCCCTGAGGTACATGCGTGGTGGTGGACGTGAGCCACGAAGACC
CTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCC
AAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCA
GCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAG
TGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTC
CAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCA
TCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCA
AAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCA
GCCGGAGAACAACACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCT
CCTTCTTCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAG
GGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTA
CACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5145_B14 (PGT-127) gamma
heavy chain variable region nucleotide sequence: (SEQ ID NO: 317)
CAGCCGCAGCTGCAGGAGTCGGGCCCAGGACTGGTGGAGGCTTCGGAGA
CCCTGTCCCTCACGTGCACTGTGTCCGGCGACTCCACTGGTCGTTGTAATT
ATTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAGGGGCTGGAGTGGATT
GGGAGTTTGTCCCCTGTAGAAAGTTACTACAATACTGACTGGACCTACCA
CAACCCGTCTCTCAAGAGTCGACTCACTATTTCACTCGACACGCCCAAGA
ATCAGGTCTTCCTGAGATTGACCTCTGTGACCGCCGCGGACACGGCCACT
TATTACTGTGCGCGATTTCGGCGGCGAAGTTCTAGTGTACAGAGATTGGCC
AAAGCCGGCCTGGGTTCGACCTCTGGGGCCGGGGAACGCTGGTCGTACCG
TCTCGAGC 5145_B14 (PGT-127) gamma heavy chain amino acid sequence:
expressed protein with leader sequence in italics and variable region in bold.
(SEQ ID NO: 318)

*MKHLWFFLLLVAAPRWVLSQPQLQESGPGLVEASET***LSLTCTVSGDSTGRC**
NYFWGWVRQPPGKGLEWIGSLSHCRSYNTDWTYHNPSLKSRLTISLDT
PKNQVFLRLTSVTAADTATYYCARFGGEVLVYRDWPKPAWVDLWGRG
TLVVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSG
ALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKR
VEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV
FSCSVMHEALHNHYTQKSLSLSPGK 5145_B14 (PGT-127) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 319)
*QPQLQESGPGLVEASET***LSLTCTVSGDSTGRC***NYFWGWVRQPPGKGLEWIGS*
LSHCRSYNTDWTYHNPSLKSRLTISLDT*PKNQVFLRLTSVTAADTATYYCAR*

FGGEVLVYRDWPKPAWVDL WGRGTLVTVSS 5145_B14 (PGT-127) gamma heavy chain Kabat CDRs: (SEQ ID NO: 320) CDR 1: RCNYFWG (SEQ ID NO: 321) CDR 2: SLSHCRSYNTDWTYHNPSLKS (SEQ ID NO: 322) CDR 3: **FGGEVLVYRDWPKPAWVDL** 5145_B14 (PGT-127) gamma heavy chain Chothia CDRs: (SEQ ID NO: 323) CDR 1: GDSTGRCN (SEQ ID NO: 324) CDR 2: SLSHCRSYNTDWTY (SEQ ID NO: 322) CDR 3: **FGGEVLVYRDWPKPAWVDL** 5145_B14 (PGT-127) lambda light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 327)

ATGGCCTGGGCTCTGCTCCTCCTCACCTCCTCACTCAGGGCACAGGGGC
CTGGGCCCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCC
TGGACAGTCAATCACCATCTCCTGCACTGGAACCAGCAATAACTTTGT
CTCCTGGTACCAACAATACCCAGGCAAGGCCCCCAAACCTCGTCATTT
ATGAGGTCAATAAGCGCCCCCTCAGGTGTCCCTGATCGTTTCTCTGGC
TCCAAGTCTGGCAGCACGGCCTCCCTGACCGTCTCTGGACTCCAGGC
TGACGATGAGGGTGTCTATTATTGTAGTTCACTTGTAGGCAACTGGG
ATGTGATTTTCGGCGGAGGGACCAAGTTGACCGTCCTAGGTCAGCCCA
AGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAG
CCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGCC
GTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGG
AGACCACCACACCCTCCAAACAAGCAACAACAAGTACGCGGCCAGCAG
CTATCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAGAAGCTACAGCT
GCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCCCTAC
AGAATGTTTCATAG 5145_B14 (PGT-127) lambda light chain variable region
nucleotide sequence: (SEQ ID NO: 328)

CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCCTGGACAGTCA
ATCACCATCTCCTGCACTGGAACCAGCAATAACTTTGTCTCCTGGTACCAA
CAATACCCAGGCAAGGCCCCCAAACCTCGTCATTTATGAGGTCAATAAGCG
CCCCTCAGGTGTCCCTGATCGTTTCTCTGGCTCCAAGTCTGGCAGCACGGC
CTCCCTGACCGTCTCTGGACTCCAGGCTGACGATGAGGGTGTCTATTATTG
TAGTTCACTTGTAGGCAACTGGGATGTGATTTTCGGCGGAGGGACCAAGT
TGACCGTCCTA 5145_B14 (PGT-127) lambda light chain amino acid sequence:
expressed protein with leader sequence in italics and variable region in bold.
(SEQ ID NO: 329)

*MAWALLLLTLLTQGTGAWAQ***SALTQPPSASGSPGQSITISCTGTSNNFVSWY**
QQYPGKAPKLVIIYEVNKRPSGVPDRFSGSKSGSTASLTVSGLQADDEGVY
YCSSLVGNWDVIFGGG**TKLTVLGQPKAAPSVTLFPPSSEELQANKATLVCLI**
*SDFYPGAVTVAWKADSSPVKAGVETTT***PSKQSNNKYAASSYLSLTPEQWKS**
HRSYSCQVTHEGSTVEKTVAPTECS 5145_B14 (PGT-127) lambda light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 330)

*QSALTQPPSASGSPGQSITISCT***TGTSNNFVSWYQQYPGKAPKLVIIYEVNKRPS**
*GVPDRFSGSKSGSTASLTVSGLQADDEGVYYC***SSLVGNWDVIFGGG****TKLTVL** 5145_B14
(PGT-127) lambda light chain Kabat CDRs: (SEQ ID NO: 325) CDR 1:
TGTSNNFVS (SEQ ID NO: 227) CDR 2: EVNKRPS (SEQ ID NO: 212) CDR
3: SSLVGNWDVI 5145_B14 (PGT-127) lambda light chain Chothia CDRs: (SEQ
ID NO: 325) CDR 1: TGTSNNFVS (SEQ ID NO: 227) CDR 2: EVNKRPS
(SEQ ID NO: 212) CDR 3: SSLVGNWDVI 5114_A19 (PGT-128) gamma heavy
chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID
NO: 331) ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCGGCTCCCAGATGGGTC
CTGTCCCAGCCGCAGCTGCAGGAGTCGGGGCCCAACACTGGTGGAGGC

TTCCGGAGACTCTGTCCTCCCTCACTTCCGCTGTGTCTCCGGCGACTCCCACTG
CTGCATGTAATTCTTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAG
GGGCTGGAGTGGGTGAGTTTGTCCCATTGTGCAAGCTATTGGAA
TCGTGGGTGGACCTACCACAACCCGTCTCTCAAGAGTCGGCTCACGC
TTGCTCTCGACACACCCAAGAATCTGGTCTTCCTCAAATTAAATTCTG
TGACTGCCGCGGACACGGCCACTTACTACTGTGCGCGATTTCGGCGGC
GAAGTTTTACGCTACACGGATTGGCCAAAGCCGGCCTGGGTTCGACCT
CTGGGGGCCGGGGAACGCTGGTCACCGTCTCGAGCGCCTCCACCAAGG
GCCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGC
ACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGAC
GGTGTCTGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGG
CTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGC
CCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAG
CCCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACA
AAACTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCG
TCAGTCTTCCTCTTCCCCCCTTCCCAAGGACACCCCTCATGATCTCCCGG
ACCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGA
GGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGA
CAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGT
CCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCA
AGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAA
GCCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCG
GGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGC
TTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGG
AGAACAACACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTC
TTCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGA
ACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACG
CAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5114_A19 (PGT-128) gamma
heavy chain variable region nucleotide sequence: (SEQ ID NO: 332)
CAGCCGCAGCTGCAGGAGTCGGGCCCCAACACTGGTGGAGGCTTCGGGAGA
CTCTGTCCCTCACCTGCGCTGTGTCCGGCGACTCCACTGCTGCATGTAATT
CTTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAGGGGCTGGAGTGGGT
GGGAGTTTGTCCCATTGTGCAAGCTATTGGAATCGTGGGTGGACCTACCA
CAACCCGTCTCTCAAGAGTCGGCTCACGCTTGCTCTCGACACACCCAAGA
ATCTGGTCTTCCTCAAATTAAATTCTGTGACTGCCGCGGACACGGCCACTT
ACTACTGTGCGCGATTTCGGCGGCGAAGTTTTACGCTACACGGATTGGCCA
AAGCCGGCCTGGGTTCGACCTCTGGGGCCGGGGAACGCTGGTCACCGTCTC GAGC
5114_A19 (PGT-128) gamma heavy chain amino acid sequence: expressed
protein with leader sequence in italics and variable region in bold. (SEQ ID
NO: 333) *MKHLWFFLLLVAPRWVLSQPQLQESGPTLVEASETLSLTCAVSGDSTAAC*
NSFWGWVRQPPGKGLEWVGSLSHCASYWNRGWTYHNPSLKSRLTLAL
DTPKNLVFLKLNSVTAADTATYYCARFGGEVLRYTDWPKPAWVDLWG
RGTLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDK
RVEPKSCDKHTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV
***FSCSVMHEALHNHYTQKSLSLSPGK* 5114_A19 (PGT-128) gamma heavy chain**
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs

in **bold** italics) (SEQ ID NO: 334)

QPQLQESGPTLVEASETSLTCAVSGDSTAACNSFWGWVRQPPGKGLEWVG
SLSHCASYWNRGWTYHNPSLKSRLTLALDTPKNLVFLKLNSVTAADTATYYC

ARFGGEVLRYTDWPKPAWVDLWGRGTLVTVSS 5114_A19 (PGT-128) gamma heavy chain Kabat CDRs: (SEQ ID NO: 326) CDR 1: ACNSFWG (SEQ ID NO: 335) CDR 2: SLSHCASYWNRGWTYHNPSLKS (SEQ ID NO: 336) CDR 3: FGGEVLRYTDWPKPAWVDL 5114_A19 (PGT-128) gamma heavy chain Chothia CDRs: (SEQ ID NO: 337) CDR 1: GDSTAACN (SEQ ID NO: 338) CDR 2: SLSHCASYWNRGWTY (SEQ ID NO: 336) CDR 3: FGGEVLRYTDWPKPAWVDL 5114_A19 (PGT-128) lambda light chain nucleotide sequence: coding sequence (variable region in **bold**) (SEQ ID NO: 390)

ATGGCCTGGGCTCTGCTCCTCCTCACCTCCTCACTCAGGGCACAGGGGC
CTGGGCCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCC
TGGACAGTCAATCACCATCTCCTGCACTGGAACCAGCAATAACTTTGT
CTCCTGGTACCAGCAACACGCAGGCAAGGCCCCCAAGCTCGTCATTT
ATGACGTCAATAAGCGCCCCCTCAGGTGTCCCTGATCGTTTCTCTGGC
TCCAAGTCTGGCAACACGGCCTCCCTGACCGTCTCTGGACTCCAGAC
TGACGATGAGGCTGTCTATTACTGCGGCTCACTTGTAGGCAACTGGG
ATGTGATTTTCGGCGGAGGGACCAAGTTGACCGTCCTAGGTCAGCCCA
AGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAG
CCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGCC
GTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGG
AGACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGCAG
CTATCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAGAAGCTACAGCT
GCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCCTAC
AGAATGTTCATAG 5114_A19 (PGT-128) lambda light chain variable region nucleotide sequence: (SEQ ID NO: 391)

CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCCTGGACAGTCA
ATCACCATCTCCTGCACTGGAACCAGCAATAACTTTGTCTCCTGGTACCAG
CAACACGCAGGCAAGGCCCCCAAGCTCGTCATTTATGACGTCAATAAGCG
CCCCTCAGGTGTCCCTGATCGTTTCTCTGGCTCCAAGTCTGGCAACACGGC
CTCCCTGACCGTCTCTGGACTCCAGACTGACGATGAGGCTGTCTATTACTG
CGGCTCACTTGTAGGCAACTGGGATGTGATTTTCGGCGGAGGGACCAAGT
TGACCGTCCTA 5114_A19 (PGT-128) lambda light chain amino acid sequence:

expressed protein with leader sequence in italics and variable region in **bold**. (SEQ ID NO: 392)

MAWALLLLTLLTQGTGAWAQ***SALTQPPSASGSPGQSITISCTGTSNNFVSWY***
QQHAGKAPKLVIYDVNKRPSGVPDRFSGSKSGNTASLTVSGLQTDDEAV
YYCGSLVGNWDVIFGGGTKLTVLGQPKAAPSVTLFPPSSEELQANKATLVC
LISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWK
SHRSYSCQVTHEGSTVEKTVAPTECS (SEQ ID NO: 393) 5114_A19 (PGT-128) lambda light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in **bold** italics)

QSALTQPPSASGSPGQSITISCT***TGTSNNFVSWYQQHAGKAPKLVIY******DVNKRPS***
GVPDRFSGSKSGNTASLTVSGLQTDDEAVYYC***GSLVGNWDVIFGGGTKLTVL*** 5114_A19 (PGT-128) lambda light chain Kabat CDRs: (SEQ ID NO: 325) CDR 1: TGTSNNFVS (SEQ ID NO: 343) CDR 2: DVNKRPS (SEQ ID NO: 196) CDR 3: GSLVGNWDVI 5114_A19 (PGT-128) lambda light chain Chothia CDRs: (SEQ ID NO: 325) CDR 1: TGTSNNFVS (SEQ ID NO: 343) CDR 2: DVNKRPS (SEQ ID NO: 196) CDR 3: GSLVGNWDVI 5136_H01 (PGT-131) gamma heavy

chain nucleotide sequence: (variable region in bold) (SEQ ID NO: 344) ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCGGCTCCCAGATGGGTCCTTTCCCAGGTGCAACTACAGGAGTCGGGGCCCAGGACTGGTGAAGCCTTCGGAGACCCTTTCCCTCACCTGCACTGTCTCTGGTGACTCCATCAACACTGGTCATCACTACTGGGGCTGGGTCCGTCAGGTCCCAGGGAAGG GACCGGAATGGATTGCTCACATCCACTATAATACGGCTGTCTTACAC AATCCGGGCCCTCAAGAGTCGAGTCACCATTTCGATTTTACCCTGAA GAATCTGATTACCCTGAGCCTCAGTAATGTGACCGCCGCGGACACGG CCGTCTATTTCTGCGTTCGATCCGGCGGGCGACATTTTATACTATATTG AGTGGCAAAAACCCCACTGGTTCTATCCCTGGGGCCCCGGGAATTTTG GTCACCGTCTCGAGCGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGG CACCCTCCTCCAAGAGCACCTCTGGGGGCACAGCGGCCCTGGGCTGCCTG GTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCGTGGAACCTCAGGCGC CCTGACCAGCGGCGTGACACCTTCCCGGCTGTCCTACAGTCCTCAGGAC TCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACC CAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGG ACAAGAGAGTTGAGCCCAAATCTTGTGACAAACTCACACATGCCACCG TGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCCTCTTCCCCCA AAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGT GGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACG TGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCA GTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGG ACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCT CCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGA GAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGA ACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATC GCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAACACTACAAGACCA CGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCTATAGCAAGCTCA CCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTG ATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTC TCCGGGTAAATGA 5136_H01 (PGT-131) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 345)

CAGGTGCAACTACAGGAGTCGGGGCCCAGGACTGGTGAAGCCTTTCGGAGACCCTTTCCCTCACCTGCACTGTCTCTGGTGACTCCATCAACACTGGTCATCACTACTGGGGCTGGGTCCGTCAGGTCCCAGGGAAGGGACCGGAATGGATTGCTCACATCCACTATAATACGGCTGTCTTACACAATCCGGGCCCTCAAGAGTCGAGTCACCATTTCGATTTTACCCTGAAGAATCTGATTACCCTGAGCCTCAGTAATGTGACCGCCGCGGACACGGCCGTCTATTTCTGCGTTTCGATCCGGCGGCGACATTTTATACTATATTGAGTGGCAAAAACCCCACTGGTTCTATCCCTGGGGCCCCGGGAATTTTGGTCAACCGTCTCGAGC 5136_H01 (PGT-131) gamma heavy chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 346)

MKHLWFFLLVAAPRWVLSQVQLQESGPGLVKPSSETLSLTCTVSGDSINTGHHYWG WVRQVPGKGPEWIAHIHYNTAVLENPALKSRVTISIFTLKNLITLSLSNVTAADTAVYFCVRSGGDILYYIEWQKPHWFYPWGP GILVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHTCP PCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWES

NGQPNKYKTPPVLDSPKSFSLYDQSLTVDKSRWQQGNVFSQVMSVMHEALHN
HYTQKSLSLSPGK 5136_H01 (PGT-131) gamma heavy chain variable region
amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)
(SEQ ID NO: 347)

QVQLQESGPGLVKPSETLSLTCTVSGDSINTGHHYWG WVRQVPGKGPEWIA
HIHYNTAVLHNPALKSRVTISIFTLKNLITLSLSNVTAADTAVYFCVR**SGGDIL**
YYIEWQKPHWFYPWGPGILVTVSS 5136_H01 (PGT-131) gamma heavy chain
Kabat CDRs: (SEQ ID NO: 348) CDR 1: TGHYWG (SEQ ID NO: 349) CDR
2: HIHYNTAVLHNPALKS (SEQ ID NO: 350) CDR 3: SGGDILYYIEWQKPHWFYP
5136_H01 (PGT-131) gamma heavy chain Chothia CDRs: (SEQ ID NO: 351)
CDR 1: GDSINTGH (SEQ ID NO: 352) CDR 2: HIHYNTAVL (SEQ ID NO:
350) CDR 3: SGGDILYYIEWQKPHWFYP 5136_H01 (PGT-131) lambda light
chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID
NO: 353) ATGGCCTGGGCTCTGCTCCTCCTCACCTCCTCACTCAGGGCACAGGGTCC

TGGGCCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCTT
GGACAGTCACTCACCATCTCCTGCAGTGAACCGGCAGTGACATTGG
CAGTTGGAATTTTGTCTCCTGGTATCAACAATTCCCAGGCAGAGCCC
CCAACCTCATTATTTTTGAGGTCAATAGGCGGCGATCAGGGGTCCCT
GATCGCTTCTCTGGTTCCAAGTCGGGCAATACGGCCTCCCTGACCGT
CTCTGGGCTCCGGTCTGAGGATGAGGCTGAATATTTTTGCAGTTCCC
TTTCAGGCAGGTGGGACATTGTTTTTGGCGGAGGGACCAAGGTGACC
GTCCTAGGTCAGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTC
CTCTGAGGAGCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTG
ACTTCTACCCGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCC
GTCAAGGCGGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACA
AGTACGCGGCCAGCAGCTATCTGAGCCTGACGCCTGAGCAGTGGAAGTCC
CACAGAAGCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGA
AGACAGTGGCCCCTACAGAATGTTTCATAG 5136_H01 (PGT-131) lambda light
chain variable region nucleotide sequence: (SEQ ID NO: 354)

CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCTTGGACAGTCA
CTCACCATCTCCTGCAGTGGAACCGGCAGTGACATTGGCAGTTGGAATTT
TGTCTCCTGGTATCAACAATTCCCAGGCAGAGCCCCCAACCTCATTATTTT
TGAGGTCAATAGGCGGCGATCAGGGGTCCCTGATCGCTTCTCTGGTTCCA
AGTCGGGCAATACGGCCTCCCTGACCGTCTCTGGGCTCCGGTCTGAGGAT
GAGGCTGAATATTTTTGCAGTTCCCTTTCAGGCAGGTGGGACATTGTTTTT
GGCGGAGGGACCAAGGTGACCGTCCTA 5136_H01 (PGT-131) lambda light chain
amino acid sequence: expressed protein with leader sequence in italics and
variable region in bold. (SEQ ID NO: 355)

MAWALLLLTLLTQGTGSWAQ**SALTQPPSASGSLGQSLTISC**SGTGS**DIGSWNF**
VSWYQQFPGRAPNLIIFEVNRRRSGVPDRFSGSKSGNTASLTVSGLRSEDE
AEYFCSSLSGRWDIVFGGGTKVTVLGQPKAAPSVTLFPPSSEELQANKATL
VCLISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQ
WKSHRSYSCQVTHEGSTVEKTVAPTECS 5136_H01 (PGT-131) lambda light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 356)

QSALTQPPSASGSLGQSLTISC**SGTGSDIGSWNFV**S**WYQQFPGRAPNLIIFE**VN
RRRSGVPDRFSGSKSGNTASLTVSGLRSEDEAEYFC**SSLSGRWDIV**FGGGTKV TVL
5136_H01 (PGT-131) lambda light chain Kabat CDRs: (SEQ ID NO: 357)
CDR 1: SGTGSDIGSWNFVS (SEQ ID NO: 358) CDR 2: EVNRRRS (SEQ ID
NO: 359) CDR 3: SSLSGRWDIV 5136_H01 (PGT-131) lambda light chain

Chothia CDRs: (SEQ ID NO: 357) CDR 1: SGTGSDIGSWNW (SEQ ID NO: 358) CDR 2: EVNRRRS (SEQ ID NO: 359) CDR 3: SSLSGRWDIV 5345_I01 (PGT-137) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 360)

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTTGCGGCTCCCAGATGTGTC
CTGTCTGAGGTGCATCTGGAGGAGTCGGGCCCAGGACTGGTGAGGCC
CTCGGAGACCTTGTCCCTGACTTGCACGGCCTCTGGTGGCTCCATAA
GGGGGGGGAGTGGGGCGATAGTGA**CTACCACTGGGGCTGGGTCCG**
CCACTCTCCCGAAAAGGGACTGGAATGGATTGGAAGTATTCATTGGC
GGGGGACCACCCACTACAACGCGCCCTTCCGGGGGGCGAGGCAGATT
GTCGATAGACCTCTCCCGGAATCAATTCTCCCTGCGCCTGACGTCTG
TGACCGCCGAAGACACTGCCGTCTATTATTGTGTGAAGCACAAATAT
CATGACATTGTCATGGTGGTCCCCATTGCGGGGCTGGTTCGACCCCTG
GGGCCAGGGACTCCAGGTCACCGTCTCGAGCGCCTCCACCAAGGGCCC
ATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGGCACAG
CGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTG
TCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGT
CCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTC
CAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCA
GCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAAC
TCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAG
TCTTCCTCTTCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCC
CTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTC
AAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAA
AGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCT
CACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAG
GTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGC
CAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACCTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC

AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5345_I01 (PGT-137) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 361)

GAGGTGCATCTGGAGGAGTCGGGCCCAGGACTGGTGAGGCCCTCGGAGA
CCTTGTCCCTGACTTGCACGGCCTCTGGTGGCTCCATAAGGGGGGGCGAG
TGGGGCGATAGTGA**CTACCACTGGGGCTGGGTCCGCCACTCTCCCGAAA**
GGGACTGGAATGGATTGGAAGTATTCATTGGCGGGGGACCACCCACTACA
ACGCGCCCTTCCGGGGGGCGAGGCAGATTGTCGATAGACCTCTCCCGGAAT
CAATTCTCCCTGCGCCTGACGTCTGTGACCGCCGAAGACACTGCCGTCTAT
TATTGTGTGAAGCACAAATATCATGACATTGTCATGGTGGTCCCCATTGCG
GGCTGGTTCGACCCCTGGGGCCAGGGACTCCAGGTCACCGTCTCGAGC 5345_I01

(PGT-137) gamma heavy chain amino acid sequence: expressed protein with leader sequence in *italics* and variable region in **bold**. (SEQ ID NO: 362)

*MKHLWFFLLLVAAPRCVLSEVHLEESGPGLVRPSETLSLTCTASGG***SIRGGE**
WGDS**DYHWGWVRHSPEKGLEWIGSIHWRGTTHYNAPFRGRGR****LSIDLS**
RNQFSLRLTSVTAEDTAVYYCVKHKYHDIVMVVPIAGW**FDPWGQGLQV**
TVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSG
VHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKR VEPKS

CDKTHCPCPPCLPPELPGFPPKMTLSRTPFVTVVVDVSHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKC
KVSNAKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYP SDIA
VEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
VMHEALHNHYTQKSLSLSPGK 5345_I01 (PGT-137) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 363)
EVHLEESGPGLVRPSETLSLTCTAS***GG******SIRGG******EWG******DS******DY******HW******GW******VR******HS******PE******K******G******L***
EWIG***SIHWRGTTH***YNAPFRGRRLSIDLSRNQFSLRLTSVTAEDTAVYYCVK
HKYHDIVMVVPIAGWEDPWGQGLQVTVSS 5345_I01 (PGT-137) gamma heavy
chain Kabat CDRs: (SEQ ID NO: 364) CDR 1: GGEWGDSDYHWG (SEQ ID
NO: 365) CDR 2: SIHWRGTTHYNAPFRG (SEQ ID NO: 366) CDR 3:
HKYHDIVMVVPIAGWFDP 5345_I01 (PGT-137) gamma heavy chain Chothia CDRs:
(SEQ ID NO: 367) CDR 1: GGSIRGGEWGDSD (SEQ ID NO: 237) CDR 2:
SIHWRGTTH (SEQ ID NO: 366) CDR 3: HKYHDIVMVVPIAGWFDP 5345_I01
(PGT-137) kappa light chain nucleotide sequence: coding sequence (variable
region in bold) (SEQ ID NO: 394)
ATGGAAACCCCAGCTCAGCTTCTCTTCCTCCTGCTACTCTGGCTCCCAGAT
ACTACTGGAGAAATAATGATGACGCAGTCTCCAGCCATCCTGTCTGTG
TCTCCAGGAGACAGAGCCACACTCTCCTGCAGGGCCAGTCAGAGTGT
GAAGAATAATTTAGCCTGGTACCAGAAGAGACCTGGCCAGGCTCCCA
GACTCCTCATCTTTGATACATCCAGCAGGGCCTCTGGTATCCCTGCCA
GGTTCAGTGGCGGTGGTTCTGGGACAGAGTTCACCTCACCAGTCAAC
AGCATGCAGTCTGAAGACTTTGCGACTTATTACTGTCAGCAATATGAA
GAGTGGCCTCGGACGTTCCGGCCAGGGGACCAAGGTGGAAATCAAAC
GTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCCATCTGATGAGCAGT
TGAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCTGAATAACTTCTATCCCA
GAGAGGCCAAAGTACAGTGGAAAGGTGGATAACGCCCTCCAATCGGGTAA
CTCCCAGGAGAGTGTACAGAGCAGGACAGCAAGGACAGCACCTACAGC
CTCAGCAGCACCTGACGCTGAGCAAAGCAGACTACGAGAAACACAAAG
TCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTCGCCCGTCACAAAG
AGCTTCAACAGGGGAGAGTGTTAG 5345_I01 (PGT-137) kappa light chain
variable region nucleotide sequence: (SEQ ID NO: 395)
GAAATAATGATGACGCAGTCTCCAGCCATCCTGTCTGTGTCTCCAGGAGA
CAGAGCCACACTCTCCTGCAGGGCCAGTCAGAGTGTGAAGAATAATTTAG
CCTGGTACCAGAAGAGACCTGGCCAGGCTCCCAGACTCCTCATCTTTGAT
ACATCCAGCAGGGCCTCTGGTATCCCTGCCAGGTTCAAGTGGCGGTGGTTC
TGGGACAGAGTTCACCTCACCAGTCAACAGCATGCAGTCTGAAGACTTTG
CGACTTATTACTGTCAGCAATATGAAGAGTGGCCTCGGACGTTCCGGCCAG
GGGACCAAGGTGGAAATCAAA 5345_I01 (PGT-137) kappa light chain amino
acid sequence: expressed protein with leader sequence in italics and variable
region in bold. (SEQ ID NO: 396)
METPAQLLFLLLWLPDTTGEIMMTQSPAILS***SV******SP******GDRATL******SCRASQSVKNN***
LAWYQKRPGQAPRLLIFDTSSRASGIPARFSGGGSGTEFTLT***VNSMQSED***
FATYYCQQYEEWPRTFGQGTKVEIKRTVAAPSVFIFPPSDEQLKSGTASVV
CLLNNFY***P******RE******AKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLTLSKAD***
YEKHKVYACEVTHQGLSSPVTKSFNRGEC 5345_I01 (PGT-137) kappa light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 397)
EIMMTQSPAILS***SV******SP******GDRATL******SC******RASQSVKNN******LAWYQKRPGQAPRLLIF******DTSS***

RASGIPARFSGTLGTEFTLTVMQSEDFATYYC**QQYEEWPRT**FGQGTKVEI K

5345_I01 (PGT-137) kappa light chain Kabat CDRs: (SEQ ID NO: 372) CDR 1: RASQSVKNNLA (SEQ ID NO: 373) CDR 2: DTSSRAS (SEQ ID NO: 245) CDR 3: QQYEEWPRT 5345_I01 (PGT-137) kappa light chain Chothia CDRs: (SEQ ID NO: 372) CDR 1: RASQSVKNNLA (SEQ ID NO: 373) CDR 2: DTSSRAS (SEQ ID NO: 245) CDR 3: QQYEEWPRT 4995_P16 (PGT-145) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 398)

ATGGACTGGATTTGGAGGATCCTCTTCTTGGTGGCAGCAGCTACAAGTGC
CCACTCCCAGGTGCAGTTGGTGCAGTCTGGGGCTGAAGTGAAGAAGC
CTGGGTCTCTCAGTGAAGGTCTCCTGCAAGGCCTCTGGAAACAGTTTC
AGTAATCATGATGTCCACTGGGTACGACAGGCCACTGGACAGGGGCT
TGAATGGATGGGATGGATGAGTCATGAGGGTGATAAGACAGGCTTGG
CACAAAAGTTTCAGGGCAGAGTCACCATCACGAGGGACAGTGGCGCA
AGTACAGTCTACATGGAGTTGCGCGGCCTGACAGCTGACGACACGGC
CATTTATTATTGTTTGACCGGCTCAAAACATCGCCTGCGAGATTATTT
TCTGTACAATGAATATGGCCCCAATTATGAAGAGTGGGGTGACTACC
TTGCGACTTTGGACGTCTGGGGCCATGGGACCGCGGTCACCGTCTCG
AGCGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCCTCCTCCAA
GAGCACCTCTGGGGGACACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACT
TCCCCGAACCGGTGACGGTGTCGTGGAACCTCAGGCGCCCTGACCAGCGGC
GTGCACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGC
AGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTG
CAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAGAGAGTTGAG
CCCAAATCTTGTGACAAAACCTCACACATGCCACCCGTGCCAGCACCTGA
ACTCCTGGGGGGACCGTCAGTCTTCTCTTCCCCCCTCAAAACCCAAGGACA
CCCTCATGATCTCCCGGACCCCTGAGGTCACATGCGTGGTGGTGGACGTG
AGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGG
AGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCAC
GTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATG
GCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATC
GAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGAGAACCACAGGTGT
ACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCT
GACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGG
AGAGCAATGGGCAGCCGGAGAACAACCTACAAGACCACGCCTCCCGTGCT
GGACTCCGACGGCTCCTTCTTCTCTATAGCAAGCTCACCGTGGACAAGA
GCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCT
CTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATG A

4995_P16 (PGT-145) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 399)

CAGGTGCAGTTGGTGCAGTCTGGGGCTGAAGTGAAGAAGCCTGGGTCTCT
AGTGAAGGTCTCCTGCAAGGCCTCTGGAAACAGTTTCAGTAATCATGATG
TCCACTGGGTACGACAGGCCACTGGACAGGGGCTTGAATGGATGGGATG
GATGAGTCATGAGGGTGATAAGACAGGCTTGGCACAAAAGTTTCAGGGC
AGAGTCACCATCACGAGGGACAGTGGCGCAAGTACAGTCTACATGGAGTT
GCGCGGCCTGACAGCTGACGACACGGCCATTTATTATTGTTTGACCGGCT
CAAAACATCGCCTGCGAGATTATTTTCTGTACAATGAATATGGCCCCAATT
ATGAAGAGTGGGGTGACTACCTTGCGACTTTGGACGTCTGGGGCCATGGG

ACCGCGGTCACCGTCTCGAGC 4995_P16 (PGT-145) gamma heavy chain amino acid sequence: expressed protein with leader sequence in italics and variable

region in bold. (SEQ ID NO: 400)

MDWIWRILFLVAAATSAHSQVQLVQSGAEVKKPGSSVKVSCKASGNSFSNH
DVHWVRQATGQGLEWMGWMSHEGDKTGLAQKFQGRVTITRDSGAST
VYMELRGLTADDTAIYYCLTGSKHRLRDYFLYNEYGPNYEEWGDYLAT
LDVWGHGTAVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT
VSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSN
TKVDKRVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTC
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQ
DWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQV
SLTCLVKGFYPSDIA VEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSR
WQQGNVFSCSVMHEALHNHYTQKSLSLSPGK 4995_P16 (PGT-145) gamma
chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia
CDRs in bold italics) (SEQ ID NO: 401)

QVQLVQSGAEVKKPGSSVKVSCKASGNSFSNHDVHWVRQATGQGLEWMG
WMSHEGDKTGLAQKFQGRVTITRDSGASTVYMELRGLTADDTAIYYCLTGS
KHRLRDYFLYNEYGPNYEEWGDYLATLDVWGHGTAVTVSS 4995_P16 (PGT-145)

gamma heavy chain Kabat CDRs: (SEQ ID NO: 378) CDR 1: NHDVH (SEQ
ID NO: 379) CDR 2: WMSHEGDKTGLAQKFQG (SEQ ID NO: 380) CDR 3:
GSKHRLRDYFLYNEYGPNYEEWGDYLATLDV 4995_P16 (PGT-145) gamma heavy
chain Chothia CDRs: (SEQ ID NO: 381) CDR 1: GNSFSN (SEQ ID NO: 382)
CDR 2: WMSHEGDKTG (SEQ ID NO: 380) CDR 3:

GSKHRLRDYFLYNEYGPNYEEWGDYLATLDV 4995_P16 (PGT-145) kappa light
chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID
NO: 383) ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGGTCTCTGGA
TCCGGTGCGGAGGTTGTCATAACTCAGTCTCCACTCTTCCTGCCCCGTC
ACCCCTGGAGAGGGCGGCCTCCTTGTCTTGCAAGTGCAGCCACAGCCT
CCAACATTCAACTGGAGCCAACCTATTTGGCTTGGTACCTGCAGAGAC
CAGGGCAAACCTCCACGCCTGTTGATCCATTTGGCCACTCATCGGGCC
TCCGGGGTCCCTGACAGATTCAGTGGCAGTGGATCAGGCACAGATTT
TACACTTAAAATCAGTCGAGTGGAGTCTGACGATGTTGGAACCTTATTA
TTGCATGCAGGGTCTGCACAGTCCCTGGACGTTTCGGCCAAGGGACCA
AGGTGGAGATCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCCG
CCATCTGATGAGCAGTTGAAATCTGGAACCTGCCTCTGTTGTGTGCCTGCTG
AATAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACG
CCCTCCAATCGGGTAACTCCCAGGAGAGTGTACAGAGCAGGACAGCAA
GGACAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGAC
TACGAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGA
GCTCGCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 4995_P16 (PGT-145)

kappa light chain variable region nucleotide sequence: (SEQ ID NO: 384)

GAGGTTGTCATAACTCAGTCTCCACTCTTCCTGCCCCGTCACCCCTGGAGAG
GCGGCCTCCTTGTCTTGCAAGTGCAGCCACAGCCTCCAACATTCAACTGG
AGCCAACCTATTTGGCTTGGTACCTGCAGAGACCAGGGCAAACCTCCACGCC
TGTTGATCCATTTGGCCACTCATCGGGCCTCCGGGGTCCCTGACAGATTCA
GTGGCAGTGGATCAGGCACAGATTTTACACTTAAAATCAGTCGAGTGGAG
TCTGACGATGTTGGAACCTTATTATTGCATGCAGGGTCTGCACAGTCCCTGG
ACGTTTCGGCCAAGGGACCAAGGTGGAGATCAAA 4995_P16 (PGT-145) kappa light
chain amino acid sequence: expressed protein with leader sequence in italics
and variable region in bold. (SEQ ID NO: 385)

MRLPAQLLGLLMLWVSGSGAEVVITQSPLFLPVTGPGEAASLSCKCSHSLQHS
TGANYLAWYLQRPQTPLLHLATHRASGVPDREFSGSGSGTDFTLKISR

VESDDVGTYYCQGLHSPWTFGQGTKVEIKRTVAAPSVFIFPPSDEQLKS

GTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKSTYSLSST

LTLISKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC 4995_P16 (PGT-145) kappa light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 386)

EVVITQSPLFLPVTPGEAASLSC**KCSHSLQHSTGANYLA**WYLQRPQGTPRLLI

HLATHRASGVPDRFSGSGSGTDFTLKISR VESDDVGTYYC**MQGLHSPWTFGQ**

GTKVEIK 4995_P16 (PGT-145) kappa lambda light chain Kabat CDRs: (SEQ ID NO: 387) CDR 1: KCSHSLQHSTGANYLA (SEQ ID NO: 388) CDR 2:

LATHRAS (SEQ ID NO: 389) CDR 3: MQGLHSPWT 4995_P16 (PGT-145)

kappa light chain Chothia CDRs: (SEQ ID NO: 387) CDR 1:

KCSHSLQHSTGANYLA (SEQ ID NO: 388) CDR 2: LATHRAS (SEQ ID NO: 389) CDR 3: MQGLHSPWT

(207) The 5145_B14 (PGT-127) antibody includes a heavy chain variable region (SEQ ID NO: 319), encoded by the nucleic acid sequence shown in SEQ ID NO: 317, and a light chain variable region (SEQ ID NO: 330) encoded by the nucleic acid sequence shown in SEQ ID NO: 328.

(208) The heavy chain CDRs of the 5145_B14 (PGT-127) antibody have the following sequences per Kabat definition: RCNYFWG (SEQ ID NO: 320), SLSHCRSYYNTDWTYHNPSLKS (SEQ ID NO: 321), and FGGEVLVYRDWPKPAWVDL (SEQ ID NO: 322). The light chain CDRs of the 5145_B14 (PGT-127) antibody have the following sequences per Kabat definition: TGTSNNFVS (SEQ ID NO: 325), EVNKRPS (SEQ ID NO: 227), and SSLVGNWDVI (SEQ ID NO: 212).

(209) The heavy chain CDRs of the 5145_B14 (PGT-127) antibody have the following sequences per Chothia definition: GDSTGRCN (SEQ ID NO: 323), SLSHCRSYYNTDWTY (SEQ ID NO: 324), and FGGEVLVYRDWPKPAWVDL (SEQ ID NO: 322). The light chain CDRs of the 5145_B14 (PGT-127) antibody have the following sequences per Chothia definition: TGTSNNFVS (SEQ ID NO: 325), EVNKRPS (SEQ ID NO: 227), and SSLVGNWDVI (SEQ ID NO: 212).

(210) The 5114_A19 (PGT-128) antibody includes a heavy chain variable region (SEQ ID NO: 334), encoded by the nucleic acid sequence shown in SEQ ID NO: 332, and a light chain variable region (SEQ ID NO: 393) encoded by the nucleic acid sequence shown in SEQ ID NO: 391.

(211) The heavy chain CDRs of the 5114_A19 (PGT-128) antibody have the following sequences per Kabat definition: ACNSFWG (SEQ ID NO: 326), SLSHCASYWNRGWTYHNPSLKS (SEQ ID NO: 335), and FGGEVLRYTDWPKPAWVDL (SEQ ID NO: 336). The light chain CDRs of the 5114_A19 (PGT-128) antibody have the following sequences per Kabat definition: TGTSNNFVS (SEQ ID NO: 325), DVNKRPS (SEQ ID NO: 343), GSLVGNWDVI (SEQ ID NO: 196).

(212) The heavy chain CDRs of the 5114_A19 (PGT-128) antibody have the following sequences per Chothia definition: GDSTAACN (SEQ ID NO: 337), SLSHCASYWNRGWTY (SEQ ID NO: 338), FGGEVLRYTDWPKPAWVDL (SEQ ID NO: 336)

(213) The light chain CDRs of the 5114_A19 (PGT-128) antibody have the following sequences per Chothia definition: TGTSNNFVS (SEQ ID NO: 325), DVNKRPS (SEQ ID NO: 343), GSLVGNWDVI (SEQ ID NO: 196).

(214) The 5136_H01 (PGT-131) antibody includes a heavy chain variable region (SEQ ID NO: 347), encoded by the nucleic acid sequence shown in SEQ ID NO: 345, and a light chain variable region (SEQ ID NO: 356) encoded by the nucleic acid sequence shown in SEQ ID NO: 354.

(215) The heavy chain CDRs of the 5136_H01 (PGT-131) antibody have the following sequences per Kabat definition: TGHHYWG (SEQ ID NO: 348), HIHYNTAVLHNPALKS (SEQ ID NO: 349), and SGGDILYYIEWQKPHWFYP (SEQ ID NO: 350). The light chain CDRs of the 5136_H01 (PGT-131) antibody have the following sequences per Kabat definition:

SGTGSDIGSWNFVS (SEQ ID NO: 357), EVNRRRS (SEQ ID NO: 358), and SSLSGRWDIV (SEQ ID NO: 359).

(216) The heavy chain CDRs of the 5136_H01 (PGT-131) antibody have the following sequences

per Chothia definition: GDSINTGH (SEQ ID NO: 351), HIHYNTAVL (SEQ ID NO: 352), and SGGDILYYIEWQKPHWFYP (SEQ ID NO: 350). The light chain CDRs of the 5136 HO1 (PGT-131) antibody have the following sequences per Chothia definition: SGTGSDIGSWNFVS (SEQ ID NO: 357), EVNRRRS (SEQ ID NO: 358), and SSLSGRWDIV (SEQ ID NO: 359).

(217) The 5345 I01 (PGT-137) antibody includes a heavy chain variable region (SEQ ID NO: 363), encoded by the nucleic acid sequence shown in SEQ ID NO: 361, and a light chain variable region (SEQ ID NO: 397) encoded by the nucleic acid sequence shown in SEQ ID NO: 395.

(218) The heavy chain CDRs of the 5345 I01 (PGT-137) antibody have the following sequences per Kabat definition: GGEWGDSDYHWG (SEQ ID NO: 364), SIHWRGTTHYNAPFRG (SEQ ID NO: 365), and HKYHDIVMVVPIAGWFDP (SEQ ID NO: 366). The light chain CDRs of the 5345_101 (PGT-137) antibody have the following sequences per Kabat definition: RASQSVKNNLA (SEQ ID NO: 372), DTSSRAS (SEQ ID NO: 373), and QQYEEWPRT (SEQ ID NO: 245).

(219) The heavy chain CDRs of the 5345_101 (PGT-137) antibody have the following sequences per Chothia definition: GGSIRGGEWGDSD (SEQ ID NO: 367), SIHWRGTTH (SEQ ID NO: 237), and HKYHDIVMVVPIAGWFDP (SEQ ID NO: 366). The light chain CDRs of the 5345 I01 (PGT-137) antibody have the following sequences per Chothia definition: RASQSVKNNLA (SEQ ID NO: 372), DTSSRAS (SEQ ID NO: 373), and QQYEEWPRT (SEQ ID NO: 245).

(220) The 4995_P16 (PGT-145) antibody includes a heavy chain variable region (SEQ ID NO: 401), encoded by the nucleic acid sequence shown in SEQ ID NO: 399, and a light chain variable region (SEQ ID NO: 386) encoded by the nucleic acid sequence shown in SEQ ID NO: 384.

(221) The heavy chain CDRs of the 4995_P16 (PGT-145) antibody have the following sequences per Kabat definition: NHDVH (SEQ ID NO: 378), WMSHEGDKTGLAQKFQG (SEQ ID NO: 379), and GSKHRLRDYFLYNEYGPNYEEWGDYLATLDV (SEQ ID NO: 380). The light chain CDRs of the 4995_P16 (PGT-145) antibody have the following sequences per Kabat definition: KCSHSLQHSTGANYLA (SEQ ID NO: 387), LATHRAS (SEQ ID NO: 388), and MQGLHSPWT (SEQ ID NO: 389).

(222) The heavy chain CDRs of the 4995 P16 (PGT-145) antibody have the following sequences per Chothia definition: GNSFSN (SEQ ID NO: 381), WMSHEGDKTG (SEQ ID NO: 382), and GSKHRLRDYFLYNEYGPNYEEWGDYLATLDV (SEQ ID NO: 380). The light chain CDRs of the 4995 P16 (PGT-145) antibody have the following sequences per Chothia definition: KCSHSLQHSTGANYLA (SEQ ID NO: 387), LATHRAS (SEQ ID NO: 388), and MQGLHSPWT (SEQ ID NO: 389).

(223) The sequences of additional human monoclonal antibodies were determined, including the sequences of the variable regions of the Gamma heavy and Kappa or Lambda light chains. In addition, the sequence of each of the polynucleotides encoding the antibody sequences was determined. Shown below are the polypeptide and polynucleotide sequences of the gamma heavy chains and kappa light chains, with the signal peptides at the N-terminus (or 5' end) and the constant regions at the C-terminus (or 3' end) of the variable regions, which are shown in bolded text.

(224) TABLE-US-00016 4835_F12 (PGT-124) gamma heavy chain nucleotide sequence: coding sequence (leader sequence in italics, variable region in bold) (SEQ ID NO: 402)

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTCCT
ATCCCAGGTGCAGCTGCAGGAGTCGGGCCCAGGACTGGTGAGACCTT
CGGAGACCCTGTCCGTCACCTGCATCGTCTCTGGGGGCTCCATCAGC
AATTACTACTGGACTTGGATCCGACAGTCCCAGGAAAGGGACTGGA
GTGGATAGGCTATATTTCTGACAGAGAAACAACGACTTACAATCCCT
CCCTCAACAGTCGAGCCGTCATATCACGAGACACGTCGAAAAACCAA
TTGTCCCTACAATTACGTTCCGTCACCACTGCGGACACGGCCATCTAT

TTCTGTGCGGACGAGGAGGATTATGGAGTTTC
ATTTGGAGAGTTCTTCTACTACTACTACATGGACGTCTGGGGCAAAG
GGACTGCGGTCACCGTCTCCTCAGCGTCGACCAAGGGCCCATCGGTCTT
CCCTCTGGCACCATCATCCAAGTCGACCTCTGGGGGCACAGCGGCCCTGG
GCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCTGTGGAAC
TCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTGTCTACAGTC
CTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCT
TGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACC
AAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACACAT
GCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCTCCTC
TTCCCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGT
CACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCA
ACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCGCG
GGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGTCC
TGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAA
CAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGG
CAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGAT
GACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCA
GCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAATA
CAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCTCTATAG
CAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCA
TGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCT
CTCCCTGTCTCCGGGTAAATGA 4835_F12 (PGT-124) gamma heavy chain variable
region nucleotide sequence: (SEQ ID NO: 403)

CAGGTGCAGCTGCAGGAGTCGGGCCCAGGACTGGTGAGACCTTCGG
AGACCCTGTCCGTCACCTGCATCGTCTCTGGGGGCTCCATCAGCAAT
TACTACTGGACTTGGATCCGACAGTCCCCAGGAAAGGGACTGGAGTG
GATAGGCTATATTTCTGACAGAGAAACAACGACTTACAATCCCTCCCT
CAACAGTCGAGCCGTCATATCACGAGACACGTCGAAAAACCAATTGT
CCCTACAATTACGTTCCGTCACCACTGCGGACACGGCCATCTATTTCT
GTGCGACAGCGCGCCGAGGACAGAGGATTTATGGAGTGGTTTCATTT
GGAGAGTTCTTCTACTACTACTACATGGACGTCTGGGGCAAAGGGAC
TGCGGTACCGTCTCCTCA 4835_F12 (PGT-124) gamma heavy chain amino
acid sequence: expressed protein with leader sequence in italics and variable
region in bold. (SEQ ID NO: 404)

*MKHLWFFLLLVAAPRWVLSQVQLQESGPG***LVRPSETLSVTCIVSGGSISNYY**
WTWIRQSPGKGLEWIGYISDRETTTYNPSLNSRAVISRDTSKNQLSLQLR
SVTTADTAIYFCATARRGQRIYGVVSFGEFFYYYYMDVWGKGTAVTVSS
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF
PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKCDKT
HTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFN
WYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSN
KALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIA
VEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSV
MH EALHNHYTQKSLSLSPGK 4835_F12 (PGT-124) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 405) QVQLQESGPG**LVRPSETLSVTCIVS**
NY^WWTWIRQSPGKGLEWIG^{custom character}
YNPSLNSRAVISRDTSKNQLSLQLRSVTTADTAIYFCAT^{custom character}
^{custom character}WGKGTAVTVSS 4835_F12 (PGT-124) gamma heavy chain Kabat

CDRs: (SEQ ID NO: 406) CDR 1: NYWT (SEQ ID NO: 407) CDR 2: YISDRETTTYPNSLNS (SEQ ID NO: 408) CDR 3: ARRQRIYGVVSFGEFFYYYYMDV 4835_F12 (PGT-124) gamma heavy chain Chothia CDRs: (SEQ ID NO: 409) CDR 1: GGSISN (SEQ ID NO: 410) CDR 2: YISDRETTT (SEQ ID NO: 408) CDR 3: ARRQRIYGVVSFGEFFYYYYMDV 4835_F12 (PGT-124) light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 411)

*ATGGCCTGGATCCCTCTCCTCCTCGGCCTCCTCTCTCACTGCACAGGGTCTGT
GACGTCCTATGTGAGCCCACTGTCAGTGGCCCTGGGGGAGACGGCCA
GGATTTCTGTGGACGACAGGCCCTTGGAAAGTAGAGCTGTGCAGTGG
TATCAACATAAGCCAGGCCAGGCCCTATTTTGCTCATCTATAATAAT
CAAGACCGGCCCTCAGGGATCCCTGAGCGGTTCTCTGGCACCCCTGA
TATTAATTTTGGGACCACGGCCACCCTGACTATCAGCGGGGGTCTGAAG
TCGGGGGATGAAGCCGACTATTACTGTCACATGTGGGACTCTAGAAGT
GGTTTCAGTTGGTCTTTTCGGCGGGGGCGACCAGGCTGACCGTCCTAGG
TCAGCCCAAGGCTGCCCCCTCGGTCCTCTGTTCCCGCCCTCCTCTGAGGA
GCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACC
CGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGC
GGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCG
GCCAGCAGCTATCTGAGCCTGACGCCTGAGCAGTGGAAAGTCCCACAAAA
GCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGT
GGCCCCTACAGAATGTTTCATAG 4835_F12 (PGT-124) light chain variable region
nucleotide sequence: (SEQ ID NO: 412)*

***TCCTATGTGAGCCCACTGTCAGTGGCCCTGGGGGAGACGGCCAGGAT
TTCCTGTGGACGACAGGCCCTTGGAAAGTAGAGCTGTGCAGTGGTATC
AACATAAGCCAGGCCAGGCCCTATTTTGCTCATCTATAATAATCAAG
ACCGGCCCTCAGGGATCCCTGAGCGGTTCTCTGGCACCCCTGATATT
AATTTTGGGACCACGGCCACCCTGACTATCAGCGGGGGTCTGAAGTCGG
GGATGAAGCCGACTATTACTGTCACATGTGGGACTCTAGAAGTGGTT
TCAGTTGGTCTTTTCGGCGGGGGCGACCAGGCTGACCGTCCTA*** 4835_F12 (PGT-124) light chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 413)

*MAWIPLLLGLLSHCTGSVTSYVSPLSVALGETARISCGRQALGSRAVQWYQ
HKPGQAPILLIYNQDRPSGIPERFSGTPDINFGTTATLTISGVEVGDEAD
YYCHMWDSRSGFSWSFGGATRLTVLGQPKAAPSVTLFPPSSEELQANKAT
LVCLISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTP
QWKSHKSYSCQVTHEGSTVEKTVAPTECS 4835_F12 (PGT-124) light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 414) SYVSPLSVALGETARISC custom character
WYQHKPGQAPILLIY custom character GI PERFSGTPDINFGTTATLTISGVEVGDEADYYC
 custom character FGGATRL TVL 4835_F12 (PGT-124) light chain Kabat CDRs:
(SEQ ID NO: 415) CDR 1: GRQALGSRAVQ (SEQ ID NO: 151) CDR 2:
NNQDRPS (SEQ ID NO: 416) CDR 3: HMWDSRSGFSWS 4835_F12 (PGT-124)
light chain Chothia CDRs: (SEQ ID NO: 415) CDR 1: GRQALGSRAVQ (SEQ
ID NO: 151) CDR 2: NNQDRPS (SEQ ID NO: 416) CDR 3: HMWDSRSGFSWS
4869_K15 (PGT-133) gamma heavy chain nucleotide sequence: coding sequence
(variable region in bold) (SEQ ID NO: 417)*

*ATGAAACACCTGTGGTTCTTCCTTCTCCTGGTGGCAGCTCCCAGATGGGTCGT
GTCCCAGGTGCATCTGCAAGAGTCGGGGGCCAGGACTGGTGACGCCTT
CGGAAACCCTGTCCCTCACTTGCACTGTGTCTGAATGGCTCCGTCAGT*

italics) (SEQ ID NO: 420) QVHLQESGPGSLVTPSETLSLTCTVS
 RFWSWIRQSPGRGLEWIG
 YNPSLRSRLTSLVDRSKNQLSLRLKSVTAADSATYYCAR
WGKGTPVTVSS 4869_K15 (PGT-133) gamma heavy chain
 Kabat CDRs: (SEQ ID NO: 421) CDR 1: GRFWS (SEQ ID NO: 422) CDR 2:
 YFSDTDRSEYNPSLRS (SEQ ID NO: 423) CDR 3:
 AQQGKRIYGIVSFGEFFYYYYYMDA 4869_K15 (PGT-133) gamma heavy chain
 Chothia CDRs: (SEQ ID NO: 424) CDR 1: NGSVSG (SEQ ID NO: 425) CDR
 2: YFSDTDRSE (SEQ ID NO: 423) CDR 3: AQQGKRIYGIVSFGEFFYYYYYMDA
 4869_K15 (PGT-133) light chain nucleotide sequence: coding sequence (variable
 region in bold) (SEQ ID NO: 426)
 ATGGCCTGGATCCCTCTCCTCCTCGGCCTCCTCTCTCACTGCACAGGTTCTGA
 CACTTCGTAAACCCACTGTCTGCTGGCCCCAGGAGCGACGGCCAAAA
 TTCCCTGCGGAGAAAGGAGCCGTGGAAGTAGGGCTGTCCAGTGGTAT
 CAGCAGAAGCCAGGCCAGGCCCCCCACATTGATCATTTATAATAATCA
 AGACCGGCCCCGCAGGGGTCTCTGAACGATTTTCTGGCAATCCTGACG
 TCGCTATTGGGGTGACGGCCACCCTGACCATCAGTCGGGTCTGAAGTC
 GGGGATGAGGCCGACTATTATTGTCACTATTGGGACAGTAGAAGTCC
 CATCAGCTGGATTTTCGGCGGAGGGACCCAGCTGACCGTCCTGGGTC
 AGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGC
 TTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCG
 GGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGG
 GAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGC
 CAGCAGCTATCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCT
 ACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGC
 CCCTACAGAATGTTTCATAG 4869_K15 (PGT-133) light chain variable
 nucleotide sequence: (SEQ ID NO: 427)
 TCGTTAAACCCACTGTCTGCTGGCCCCAGGAGCGACGGCCAAAATTCC
 CTGCGGAGAAAGGAGCCGTGGAAGTAGGGCTGTCCAGTGGTATCAG
 CAGAAGCCAGGCCAGGCCCCCCACATTGATCATTTATAATAATCAAGA
 CCGGCCCGCAGGGGTCTCTGAACGATTTTCTGGCAATCCTGACGTCG
 CTATTGGGGTGACGGCCACCCTGACCATCAGTCGGGTCTGAAGTCGGG
 GATGAGGCCGACTATTATTGTCACTATTGGGACAGTAGAAGTCCCAT
 CAGCTGGATTTTCGGCGGAGGGACCCAGCTGACCGTCCTG 4869_K15 (PGT-133)
 light chain amino acid sequence: expressed protein with leader sequence in
 italics and variable region in bold. (SEQ ID NO: 428)
 MAWIPLLGLLSHCTGSDTSLNPLSLAPGATAKIPCGERSRGSRAVQWYQQ
 KPGQAPTLIIYNNQDRPAGVSERFSGNPDVAIGVTATLTISRVEVGDEADY
 YCHYWDSRSPISWIFGGGTQLTVLGQPKAAPSVTLFPPSSEELQANKATLV
 CLISDFYPGAVTVAWKADSSPVKAGVETTTSPSKQSNNKYAASSYLSLTPEQW
 KSHKSYSCQVTHEGSTVEKTVAPTECS 4869_K15 (PGT-133) light chain variable
 region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
 italics) (SEQ ID NO: 429) SLNPLSLAPGATAKIPC
 WYQQKPGQAPTLIIYGVS
 ERFSGNPDVAIGVTATLTISRVEVGDEADYYCFGGGTQLTV L
 4869_K15 (PGT-133) light chain Kabat CDRs: (SEQ ID NO: 430) CDR 1:
 GERSRGSRAVQ (SEQ ID NO: 179) CDR 2: NNQDRPA (SEQ ID NO: 431)
 CDR 3: HYWDSRSPISWI 4869_K15 (PGT-133) light chain Chothia CDRs: (SEQ
 ID NO: 430) CDR 1: GERSRGSRAVQ (SEQ ID NO: 179) CDR 2: NNQDRPA
 (SEQ ID NO: 431) CDR 3: HYWDSRSPISWI 4876_M06 (PGT-134) gamma

heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 432)

**ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTTCGT
GTCCCAGGTGCATCTGCAAGAGTCGGGGGCCAGGACTGGTGACGCCTT
CGGAAACCCTGTCCCTCACTTGCACTGTGTCTGAATGGCTCCGTCAGT
GGTCGCTTCTGGAGCTGGATCCGGCAGTCCCCAGGGAGAGGACTGG
AATGGATCGGTTATTTTTCTGACACTGACAGGTCTGAATATAATCCTT
CTCTCAGGAGTCGACTCACCTTATCAGTCGATAGATCCAAGAACCAG
TTGTCCCTAAAATTGAAGTCCGTGACCGCTGCGGATTCTGGGCCACTTA
TACTGTGCGAGAGCACAAACAGGGGAAGAGGATCTATGGAATAGTGT
CTTTCGGAGAGTTGTTCTATTATTATTACATGGACGCCTGGGGGCAA
GGGACTCCGGTCACCGTCTCCTCAGCGTCGACCAAGGGCCCATCGGTC
TTCCCTCTGGCACCATCATCCAAGTCGACCTCTGGGGGGCACAGCGGCCCT
GGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCGTGGA
ACTCAGGCGCCCTGACCAGCGGCGTGACACCTTCCCGGCTGTCCTACAG
TCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAG
CTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACA
CCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACAC
ATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCC
TCTTCCCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACCCCTGAG
GTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCAAGTT
CAACTGGTACGTGGACGGCGTGAGGTGCATAATGCCAAGACAAAGCCG
CGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGT
CCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCC
AACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAG
GGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGGAGGA
GATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATC
CCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAA
CTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCTA
TAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTC
TCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAG
CCTCTCCCTGTCTCCGGGTAAATGA** 4876_M06 (PGT-134) gamma heavy chain

variable region nucleotide sequence: (SEQ ID NO: 433)

**CAGGTGCATCTGCAAGAGTCGGGGGCCAGGACTGGTGACGCCTTCGG
AAACCCTGTCCCTCACTTGCACTGTGTCTGAATGGCTCCGTCAGTGGT
CGCTTCTGGAGCTGGATCCGGCAGTCCCCAGGGAGAGGACTGGAAT
GGATCGGTTATTTTTCTGACACTGACAGGTCTGAATATAATCCTTCTC
TCAGGAGTCGACTCACCTTATCAGTCGATAGATCCAAGAACCAGTTG
TCCCTAAAATTGAAGTCCGTGACCGCTGCGGATTCTGGGCCACTTATTA
CTGTGCGAGAGCACAAACAGGGGAAGAGGATCTATGGAATAGTGTCTT
TCGGAGAGTTGTTCTATTATTATTACATGGACGCCTGGGGCAAAGGG**

ACTCCGGTCACCGTCTCCTCA 4876_M06 (PGT-134) gamma heavy chain

amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 434)

*MKHLWFFLLVAAPRWVVSQVHLQESGPGLVTPSETLSLTCTVSNGSVSGRF
WSWIRQSPGRGLEWIGYFSDTDRSEYNPSLRSRLTLSVDRSKNQLSLKLK
SVTAADSATYYCARAQQGKRIYGIVSFGELFYYYMDAWGKGTPVTVSS
ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTF
PAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSDKT
HTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSHEDPEVKFN*

WYVVDGVEVHNEQYKSTYRVNQLTVLHQDWLNGKEYKCKVSN
KALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIA
VEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSSVMH
EALHNHYTQKSLSLSPGK 4876_M06 (PGT-134) gamma heavy chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 435) QVHLQESGPGLVTPSETLSLTCTVS custom character
RFWSWIRQSPGRGLEWIG custom character custom character
YNPSLRSRLTSLVDRSKNQLSLKLKSVTAADSATYYCAR custom character
 custom character WGKGTPVTVSS 4876_M06 (PGT-134) gamma heavy chain
Kabat CDRs: (SEQ ID NO: 421) CDR 1: GRFWS (SEQ ID NO: 422) CDR 2:
YFSDTDRSEYNPSLRS (SEQ ID NO: 436) CDR 3:
AQQGKRIYGIVSFGELFYYYMDA 4876_M06 (PGT-134) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 424) CDR 1: NGSVSG (SEQ ID NO: 425) CDR
2: YFSDTDRSE (SEQ ID NO: 436) CDR 3: AQQGKRIYGIVSFGELFYYYMDA
4876_M06 (PGT-134) light chain nucleotide sequence: coding sequence (variable
region in bold) (SEQ ID NO: 437)
ATGGCCTGGATCCCTCTCCTCCTCGGCCTCCTCTCTCACTGCACAGGTTCTGA
CACTTCGTTAAACCCACTGTGCTGGCCCCGGGAGCGACGGCCAAAA
TTCCCTGCGGAGAAAGGAGCCGTGGAAGTAGGGCTGTCCAGTGGTAT
CAGCAGAAGCCAGGCCAGGCCCCCCACATTGATCATTTATAATAATCA
AGACCGGCCCCGCAGGGGTCTCTGAACGATTTTCTGGCAATCCTGACG
TCGCTATTGGGGTGACGGCCACCCTGACCATCAGTCGGGTCTGAAGTC
GGGGATGAGGGCGACTATTATTGTCACTATTGGGACAGTAGAAGTCC
CATCAGCTGGATTTTCGCCGGAGGGACCCAGTTGACCGTCCTGGGTC
AGCCCAAGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGC
TTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCG
GGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGG
GAGTGGAGACCACACACCCTCCAAACAAAGCAACAACAAGTACGCGGC
CAGCAGCTATCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCT
ACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGC
CCCTACAGAATGTTTCATAG 4876_M06 (PGT-134) light chain variable region
nucleotide sequence: (SEQ ID NO: 438)
TCGTTAAACCCACTGTGCTGGCCCCGGGAGCGACGGCCAAAATTCC
CTGCGGAGAAAGGAGCCGTGGAAGTAGGGCTGTCCAGTGGTATCAG
CAGAAGCCAGGCCAGGCCCCCCACATTGATCATTTATAATAATCAAGA
CCGGCCCCGCAGGGGTCTCTGAACGATTTTCTGGCAATCCTGACGTCG
CTATTGGGGTGACGGCCACCCTGACCATCAGTCGGGTCTGAAGTCGGG
GATGAGGGCGACTATTATTGTCACTATTGGGACAGTAGAAGTCCCAT
CAGCTGGATTTTCGCCGGAGGGACCCAGTTGACCGTCCTG 4876_M06 (PGT-
134) light chain amino acid sequence: expressed protein with leader sequence
in italics and variable region in bold. (SEQ ID NO: 439)
MAWIPLLGLLSHCTGSDTSLNPLSLAPGATAKIPCGERSRGSRAVQWYQQ
KPGQAPTLIIYNNQDRPAGVSERFSGNPDVAIGVTATLTISRVEVGDEGDY
YCHYWDSRSPISWIFAGGTQLTVLGQPKAAPSVTLFPPSSEELQANKATLV
CLISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNKYAASSYLSLTPEQW
KSHKSYSCQVTHEGSTVEKTVAPTECS 4876_M06 (PGT-134) light chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 440) SLNPLSLAPGATAKIPC custom character
WYQQKPGQAPTLIIY custom character GVS
ERFSGNPDVAIGVTATLTISRVEVGDEGDYYC custom character FAGGTQLTV L

4876_M06 (PGT-134) light chain Kabat CDRs: (SEQ ID NO: 430) CDR 1: GERSRGSRAVQ (SEQ ID NO: 179) CDR 2: NNQDRPA (SEQ ID NO: 431) CDR 3: HYWDSRSPISWI 4876_M06 (PGT-134) light chain Chothia CDRs: (SEQ ID NO: 430) CDR 1: GERSRGSRAVQ (SEQ ID NO: 179) CDR 2: NNQDRPA (SEQ ID NO: 431) CDR 3: HYWDSRSPISWI 5131_A17 (PGT-132) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 441)

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTCTCTTCCCAGGTGCAACTACAGGAGTCGGGGCCCAGGACTGGTGAAGCCTTCGGAGACCCTTTCCCTCACCTGCACTGTCTCTGGTGACTCCATCAACACTGGTCATCACTACTGGGGCTGGGTCCGTCAGGTCCCAGGGAAGGGACCGGAATGGATTGCTCACATCCACTATAATACGGCTGTCTTGCACAATCCGGCCCTCAAGAGTCGAGTCACCATTTTCGATTTTTCACCCTGAAGAACTCTGATTACCCTGAGGCTCAGTAATATGACCGCCGCGGACACGGCCGTCTATTTCTGCGTTCGATCCGGCGGGCGACATTTTATACTATAATGAGTGGCAAAAACCCCACTGGTTCTATCCCTGGGGGCCCGGGAATTTTGGTCAACCGTCTCGAGCGCCTCCACCAAGGGCCCATCGGTCTTCCCCCTGGCACCCTCCTCCAAGAGCACCTCTGGGGGACACAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGTCGTGGAACCTCAGGCGCCCTGACCAGCGCGTGCACACCTTCCCGGCTGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAGCAACACCAAGGTGGACAAAGAGAGTTGAGCCCAAATCTTGTGACAAAACCTCACACATGCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGTCTTCCTCTTCCCCCAAAA
CCCAAGGACACCCTCATGATCTCCCGGACCCCTGAGGTACATGCGTGGTGGTGACGTGAGCCACGAAGACCCTGAGGTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCCAAAGGGCAGCCCCGAGAACACAGGTGTACACCCTGCCCCCATCCCGGGAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGAACAACACTACAAGACCACGCTCCCGTGCTGGACTCCGACGGCTCCTTCTTCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5131_A17 (PGT-132) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 442)

CAGGTGCAACTACAGGAGTCGGGGCCCAGGACTGGTGAAGCCTTCGGAGACCCTTTCCCTCACCTGCACTGTCTCTGGTGACTCCATCAAACTGTGCATCACTACTGGGGCTGGGTCCGTCAGGTCCCAGGGAAGGGACCGGAATGGATTGCTCACATCCACTATAATACGGCTGTCTTGCACAATCCGGCCCTCAAGAGTCGAGTCACCATTTTCGATTTTTCACCCTGAAGAATCTGATTACCCTGAGGCTCAGTAATATGACCGCCGCGGACACGGCCGTCTATTTCTGCGTTCGATCCGGCGGGCGACATTTTATACTATAATGAGTGGCAAAAACCCCACTGGTTCTATCCCTGGGGGCCCGGGAATTTTGGTCACCGTCTCGAGC 5131_A17 (PGT-132) gamma heavy chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 443)

MKHLWFFLLLVAAPRWVLSQVQLQESGPGLVKPSSETLSLTCTVSGDSINTGH

HYWGWVRQVPGKGPGEWIAHNPALKSRVTISIFTLRLSNMTAADTAVYFCVRS
RLSNMTAADTAVYFCVRSGGDILYYNEWQKPHWFYWPWGPILVTVSSAS
TKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPA
VLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPKSCDKTHT
CPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWY
VDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKAL
PAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEW
ESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEAL
HNHYTQKSLSLSPGK 5131_A17 (PGT-132) gamma heavy chain variable
amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)
(SEQ ID NO: 444) QVQLQESGPGLVKPKSETLSLTCTVS custom character
HYWGWVRQVPGKGPGEWIA custom character
HNPALKSRVTISIFTLKNLITLRLSNMTAADTAVYFCVR custom character
 custom character WGPILVTVSS 5131_A17 (PGT-132) gamma heavy chain Kabat
CDRs: (SEQ ID NO: 348) CDR 1: TGHHYWG (SEQ ID NO: 349) CDR 2:
HIHYNTAVLHNPALKS (SEQ ID NO: 445) CDR 3: SGGDILYYNEWQKPHWFY
5131_A17 (PGT-132) gamma heavy chain Chothia CDRs: (SEQ ID NO: 351)
CDR 1: GDSINTGH (SEQ ID NO: 352) CDR 2: HIHYNTAVL (SEQ ID NO:
445) CDR 3: SGGDILYYNEWQKPHWFY 5131_A17 (PGT-132) light chain
nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 446)
ATGGCCTGGGCTCTGCTCCTCCTCACCTCCTCACTCAGGGCACAGGGTCCTG
GGCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCTTG
GACAGTCACTCACCATCTCCTGCAGTGGAACCGCCAGTGACATTGGC
AGTTGGAATTTTGTCTCCTGGTATCAACAATTCCCAGGCAGAGCCCC
CAACCTCATTATTTTGGAGGTCAATAGGCGGCGATCAGGGGTCCCTG
ATCGCTTCTCTGGTTCCAAGTCGGGCAATACGGCCTCCCTGACCGTC
TCTGGGCTCCGGTCTGAGGATGAGGCTGAATATTTTTGCAGTTCCCT
TTCAGGCAGGTGGGACATTGTTTTTGGCGGAGGGACCAAGGTGACCG
TCCTAGGTCAGCCCAAGGCTGCCCCCTCGGTCCTCTGTTCCCGCCCTCCT
CTGAGGAGCTTCAAGCCAACAAGGCCACACTGGTGTGTCTCATAAGTGAC
TTCTACCCGGGAGCCGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGT
CAAGGCGGGAGTGGAGACCACCACACCCTCCAAACAAAGCAACAACAAG
TACGCGGCCAGCAGCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCA
CAAAAGCTACAGCTGCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAG
ACAGTGGCCCCTACAGAATGTTCATAG 5131_A17 (PGT-132) light chain variable
region nucleotide sequence: (SEQ ID NO: 447)
CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCTTGGACA
GTCCTCACCATCTCCTGCAGTGGAACCGCCAGTGACATTGGCAGTT
GGAATTTTGTCTCCTGGTATCAACAATTCCCAGGCAGAGCCCCCAAC
CTCATTATTTTGGAGGTCAATAGGCGGCGATCAGGGGTCCCTGATCG
CTTCTCTGGTTCCAAGTCGGGCAATACGGCCTCCCTGACCGTCTCTG
GGCTCCGGTCTGAGGATGAGGCTGAATATTTTTGCAGTTCCCTTTCA
GGCAGGTGGGACATTGTTTTTGGCGGAGGGACCAAGGTGACCGTCCT A
5131_A17 (PGT-132) light chain amino acid sequence: expressed protein with
leader sequence in italics and variable region in bold. (SEQ ID NO: 448)
MAWALLLLTLLTQGTGSWAQSAITQPPSASGSLGQSLTISCSGTASDIGSWNF
VSWYQQFPGRAPNLIIFEVNRRRSGVPDRFSGSKSGNTASLTVSGLRSEDE
AEYFCSSLSGRWDIVFGGGTKVTVLGQPKAAPSVTLFPPSSEELQANKATL
VCLISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNKYYASSYLSLTPEQ
WKSHKSYSCQVTHEGSTVEKTVAPTECS 5131_A17 (PGT-132) light chain variable

region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 449) QSALTQPPSASGSLGQSLTISC custom character WYQQFPGRAPNLIIF custom character custom character

GVPDRFSGSKSGNTASLTVSGLRSEDEAEYFC custom character FGGGTKV TVL

5131_A17 (PGT-132) light chain Kabat CDRs: (SEQ ID NO: 450) CDR 1: SGTASDIGSWNFVS (SEQ ID NO: 358) CDR 2: EVNRRRS (SEQ ID NO: 359)

CDR 3: SSLSGRWDIV 5131_A17 (PGT-132) light chain Chothia CDRs: (SEQ ID NO: 450) CDR 1: SGTASDIGSWNFVS (SEQ ID NO: 358) CDR 2:

EVNRRRS (SEQ ID NO: 359) CDR 3: SSLSGRWDIV 5138_G07 (PGT-138)

gamma heavy chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 451)

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTCCT
GTCCCAGCCGCAGCTGCAGGAGTCGGGGCCAGGACTGGTGGAGGCTT
CGGAGACCCTGTCCCTCACCTGCACTGTGTCCGGCGACTCCACTGCT
GCTTGTGACTATTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAGGG
GCTGGAGTGGATTGGAAGTTTGTCACATTGTGCAGGTTACTACAATA
GTGGCTGGACCTACCACAACCCGTCTCTCAAGAGTCGACTCACGATT
TCACTCGACACGCCCAAGAATCAGGTCTTCCTGAAGTTAAATTCTGTG
ACCGCCGCGGACACGGCCATTTACTACTGTGCGCGATTCCGGTGGCGA
CGTTTTTGGTGTACCACGATTGGCCAAAGCCGGCCTGGGTTCGACCTCT
GGGGCCGGGGAGTTTTTGGTCAACCGTCTCGAGCGCCTCCACCAAGGGC
CCATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCGAC
AGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGG
TGTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCT
GTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCC
CTCCAGCAGCTTGGGACCCAGACCTACATCTGCAACGTGAATCACAAGC
CCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAA
AACTCACACATGCCCACCGTGCCAGCACCTGAACTCCTGGGGGGACCGT
CAGTCTTCCTCTTCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGA
CCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAG
GTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGAC
AAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTC
CTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAATAACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5138_G07 (PGT-138) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 452)

CAGCCGCAGCTGCAGGAGTCGGGGCCAGGACTGGTGGAGGCTTCGG
AGACCCTGTCCCTCACCTGCACTGTGTCCGGCGACTCCACTGCTGCT
TGTGACTATTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAGGGGGCT
GGAGTGGATTGGAAGTTTGTCACATTGTGCAGGTTACTACAATAGTG
GCTGGACCTACCACAACCCGTCTCTCAAGAGTCGACTCACGATTTCA
CTCGACACGCCCAAGAATCAGGTCTTCCTGAAGTTAAATTCTGTGAC
CGCCGCGGACACGGCCATTTACTACTGTGCGCGATTCCGGTGGCGACG
TTTTGGTGTACCACGATTGGCCAAAGCCGGCCTGGGTTCGACCTCTGG

GGCCGGGAGTTTGTGGTACCGTCTCGAGC 5138_G07 (PGT-138) gamma

heavy chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold.(SEQ ID NO: 453)

***MKHLWFFLLLVAAPRWVLSQPQLQESGPGLVEASETLSLTCTVSGDSTAAC
DYFWGWVRQPPGKGLEWIGSLSHCAGYYNSGWTYHNPSLKSRLTISLD
TPKNQVFLKLNSVTAADTAIYYCARFGGDVLVYHDWPKPAWVDLWGR
GVLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSG
ALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKR
VEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGNV
FSCSVMHEALHNHYTQKSLSLSPGK*** 5138_G07 (PGT-138) gamma heavy chain

variable region amino acid sequence:(Kabat CDRs underlined, Chothia CDRs in bold italics)(SEQ ID NO: 454) ***QPQLQESGPGLVEASETLSLTCTVS***

 ***YFWGWVRQPPGKGLEWIG***  ***HNPSLKSRLTISLDTPKNQVFLKLNSVTAADTAIYYCAR*** 

WGRGVLVTVSS 5138_G07 (PGT-138) gamma heavy chain Kabat CDRs:(SEQ ID NO: 201) CDR 1: ***ACDYFWG***(SEQ ID NO: 455) CDR 2:

SLSHCAGYYNSGWTYHNPSLKS(SEQ ID NO: 456) CDR 3:

FGGDVLVYHDWPKPAWVDL 5138_G07 (PGT-138) gamma heavy chain Chothia CDRs:(SEQ ID NO: 204) CDR 1: ***GDSTAACD***(SEQ ID NO: 457) CDR 2: ***SLSHCAGYYNSGWTY***(SEQ ID NO: 456) CDR 3: ***FGGDVLVYHDWPKPAWVDL*** 5138_G07 (PGT-138) light chain nucleotide sequence: coding sequence (variable region in bold)(SEQ ID NO: 581)

***ATGGCCTGGGCTCTGCTCCTCCTCACCTCCTCACTCAGGGCACAGGGGCCT
GGGCCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCCT
GGACAGTCAATCACCATCTCCTGCACTGGAAATATCAATAACTTTGTC
TCCTGGTACCAACAACACCCTGGCAAGGCCCCCAAACCTCGTCATTTA
TGGGGTCAATAAGCGCCCCCTCAGGTGTCCCTGATCGTTTTTCTGGCT
CCAAGTCTGGCAACGCGGCCTCCCTGACCGTCTCTGGACTCCAGACT
GACGATGAGGCTGTCTATTACTGCGGCTCACTTGCAGGCAACTGGGA
TGTGGTTTTTCGGCGGAGGGACCAAGTTGACTGTCCTGGGTCAGCCCAT
GGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAAGC
CAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGCCG
TGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTGGA
GACCACCACACCCTCCAAACAAAGCAACAACAAGTACGCGGCCAGCAGC
TACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACAGCTG
CCAGGTCACGCATGAAGGGAGCACCGTGGAAGAAGACAGTGGCCCCTACA
GAATGTTCATAG*** 5138_G07 (PGT-138) light chain variable region nucleotide

sequence:(SEQ ID NO: 582)

***CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCCTGGACA
GTCAATCACCATCTCCTGCACTGGAAATATCAATAACTTTGTCTCCTG
GTACCAACAACACCCTGGCAAGGCCCCCAAACCTCGTCATTTATGGGG
TCAATAAGCGCCCCCTCAGGTGTCCCTGATCGTTTTTCTGGCTCCAAGT
CTGGCAACGCGGCCTCCCTGACCGTCTCTGGACTCCAGACTGACGAT
GAGGCTGTCTATTACTGCGGCTCACTTGCAGGCAACTGGGATGTGGT
TTTCGGCGGAGGGACCAAGTTGACTGTCCTG*** 5138_G07 (PGT-138) light chain

amino acid sequence: expressed protein with leader sequence in italics and variable region in bold.(SEQ ID NO: 583)

MAWLLLLTQTGATGAQSAALTPPPSASGSPQSGSITISCTGNINNFVSWY
QQHPGKAPKLVIYGVNKRPSGVPDRFSGSGSKSGNAASLTVSGLQTDDEAV
YYCGSLAGNWDVVFGGGTKLTVLGQPMAAPSVTLFPPSSEELQANKATLV
CLISDFYPGAVTVAWKADSSPVKAGVETTTSPSKQSNNKYAASSYLSLTPEQW
KSHKSYSCQVTHEGSTVEKTVAPTECS 5138_G07 (PGT-138) light chain variable
region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold
italics) (SEQ ID NO: 584) QSALTQPPSASGSPGQSITISCcustom character
WYQQHPGKAPKLVIYcustom character GVPDRFSGSGSKSGNAASLTVSGLQTDDEAVYYC
custom character FGGGTKLTV L 5138_G07 (PGT-138) light chain Kabat CDRs:
(SEQ ID NO: 458) CDR 1: TGNINNFVS (SEQ ID NO: 211) CDR 2:
GVNKRPS (SEQ ID NO: 459) CDR 3: GSLAGNWDVV 5138_G07 (PGT-138)
light chain Chothia CDRs: (SEQ ID NO: 458) CDR 1: TGNINNFVS (SEQ ID
NO: 211) CDR 2: GVNKRPS (SEQ ID NO: 459) CDR 3: GSLAGNWDVV
5120_N10 (PGT-139) gamma heavy chain nucleotide sequence: coding sequence
(variable region in bold) (SEQ ID NO: 460)

ATGAAACACCTGTGGTTCTTCCTCCTGCTGGTGGCAGCTCCCAGATGGGTCCT
GTCCCAGCCGCAGCTGCAGGAGTCGGGGCCAGGACTGGTGGAGGCTT
CGGAGACCCTGTCCCTCACCTGCACTGTGTCCGGCGACTCCACTGCT
GGTTGTGACTATTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAGG
GGCTGGAGTGGATTGGGGGTTTGTACATTGTGCAGGTTACTACAAT
ACTGGCTGGACCTACCACAACCCGTCTCTCAAGAGTCGACTCACGAT
TTCACTCGACACGCCAAGAATCAGGTCTTCCTGAAGTTAAATTCTGT
GACCGCCGCGGACACGGCCATTTACTACTGTGCGCGATTTCGACGGCG
AAGTTTTGGTGTACAACGATTGGCCAAAGCCGGCCTGGGTTCGACCTC
TGGGGCCCGGGGAACCTTGGTCCACCGTCTCGAGCGCCTCCACCAAGGG
CCCATCGGTCTTCCCCCTGGCACCTCCTCCAAGAGCACCTCTGGGGGCA
CAGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACG
GTGTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGC
TGTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCC
CTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGC
CCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAA
AACTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGT
CAGTCTTCCTCTTCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGA
CCCCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAG
GTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGAC
AAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTC
CTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 5120_N10 (PGT-139) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 461)
CAGCCGCAGCTGCAGGAGTCGGGGCCAGGACTGGTGGAGGCTTCGG
AGACCCTGTCCCTCACCTGCACTGTGTCCGGCGACTCCACTGCTGGT
TGTGACTATTTCTGGGGCTGGGTCCGGCAGCCCCCAGGGAAGGGGCT
GGAGTGGATTGGGGGTTTGTACATTGTGCAGGTTACTACAATACTG

GCTGACACGCCCCAAGAATCAGGTCTTCCTGAAGTTAAATTCTGTGAC
CTCGACACGCCCCAAGAATCAGGTCTTCCTGAAGTTAAATTCTGTGAC
CGCCGCGGACACGGCCATTTACTACTGTGCGCGATTTCGACGGCGAAG
TTTTGGTGTACAACGATTGGCCAAAGCCGGCCTGGGTCGACCTCTGG
GGCCGGGGAACTTTGGTCACCGTCTCGAGC 5120_N10 (PGT-139) gamma
heavy chain amino acid sequence: expressed protein with leader sequence in
italics and variable region in bold.(SEQ ID NO: 462)

*MKHLWFFLLLVAAPRWVLSQPQLQESGPGLVEASET***LSLTCTVSGDSTAGC**
DYFWGWVRQPPGKGLEWIGGLSHCAGYYNTGWTYHNPSLKSRLTISLD
TPKNQVFLKLNSVTAADTAIYYCARFDGEVLVYNDWPKPAWVDLWGRG
TLVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGAL
TSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVE
PKSCDKHTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSHE
DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK 5120_N10 (PGT-139) gamma heavy chain
variable region amino acid sequence:(Kabat CDRs underlined, Chothia CDRs
in bold italics)(SEQ ID NO: 463) QPQLQESGPGLVEASET**LSLTCTVS**

 YFWGWVRQPPGKGLEWIG 
HNPSLKSRLTISLD**TPKNQVFLKLNSVTAADTAIYYC** 
WGRGTLVTVSS 5120_N10 (PGT-139) gamma heavy chain Kabat CDRs:(SEQ
ID NO: 464) CDR 1: GCDYFWG (SEQ ID NO: 202) CDR 2:
GLSHCAGYYNTGWTYHNPSLKS (SEQ ID NO: 465) CDR 3:
FDGEVLVYNDWPKPAWVDL 5120_N10 (PGT-139) gamma heavy chain Chothia
CDRs:(SEQ ID NO: 466) CDR 1: GDSTAGCD (SEQ ID NO: 205) CDR 2:
GLSHCAGYYNTGWTY (SEQ ID NO: 465) CDR 3: FDGEVLVYNDWPKPAWVDL
5120_N10 (PGT-139) light chain nucleotide sequence: coding sequence (variable
region in bold)(SEQ ID NO: 467)

ATGGCCTGGGCTCTGCTCCTCCTCACCTCCTCACTCAGGGCACAGGGGCCT
GGGCCCAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCCT
GGACAGTCAATCACCATCTCCTGCACTGGAACCAGCAATAACTTTGT
CTCCTGGTACCAGCAACACCCAGCCAAGGCCCCCAAACCTCGTCATTT
ATGGGGTCAATAAGCGCCCCTCAGGTGTCCCTGATCGTTTTTCTGGC
TCCAAGTCTGGCAACACGGCCTCCCTGACCGTCTCTGGACTCCAGAC
TGACGATGAGGCTGTCTATTACTGCGGCTCACTTGTAGGCAACTGGG
ATGTGATTTTCGGCGGAGGGACCAAGTTGACCGTCCTGGGTCAGCCC
ATGGCTGCCCCCTCGGTCACTCTGTTCCCGCCCTCCTCTGAGGAGCTTCAA
GCCAACAAGGCCACACTGGTGTGTCTCATAAGTGACTTCTACCCGGGAGC
CGTGACAGTGGCCTGGAAGGCAGATAGCAGCCCCGTCAAGGCGGGAGTG
GAGACCACACACCTCCAAACAAAGCAACAACAAGTACGCGGCCAGCA
GCTACCTGAGCCTGACGCCTGAGCAGTGGAAGTCCCACAAAAGCTACAGC
TGCCAGGTCACGCATGAAGGGAGCACCGTGGAGAAGACAGTGGCCCCCTA
CAGATGTTTCATAG 5120_N10 (PGT-139) light chain variable region nucleotide
sequence:(SEQ ID NO: 468)

CAGTCTGCCCTGACTCAGCCTCCCTCCGCGTCCGGGTCTCCTGGACA
GTCAATCACCATCTCCTGCACTGGAACCAGCAATAACTTTGTCTCCTG
GTACCAGCAACACCCAGCCAAGGCCCCCAAACCTCGTCATTTATGGGG
TCAATAAGCGCCCCTCAGGTGTCCCTGATCGTTTTTCTGGCTCCAAGT
CTGGCAACACGGCCTCCCTGACCGTCTCTGGACTCCAGACTGACGAT

GAGGTGTGCTGACTGTCGGGTAGGCACTGGGATGTGAT

TTTCGGCGGAGGGACCAAGTTGACCGTCCTG 5120_N10 (PGT-139) light chain amino acid sequence: expressed protein with leader sequence in *italics* and variable region in **bold**. (SEQ ID NO: 469)

MAWALLLLTLLTQGTGAWAQ**SALTQPPSASGSPGQSITISCTGTSNNFVSWY**
QQHPAKAPKLVIYGVNKRPSGVPDRFSGSKSGNTASLTVSGLQTDDEAV
YYCGSLVGNWDVIFGGG**TKLTVLGQPMAPS****VTLPFPSSEELQANKATLV**
CLISDFYPGAVTVAWKADSSPVKAGVETTT**PSKQSN****NKYAASSYLSLTPEQW**

KSHKSYSCQVTHEGSTVEKTVAPTECS 5120_N10 (PGT-139) light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in *bold italics*) (SEQ ID NO: 470) **QSALTQPPSASGSPGQSITIS****custom character**
WYQQHPAKAPKLVIY**custom character** **GVPDRFSGSKSGNTASLTVSGLQTDDEAVYYC**
**custom character** **FGGGTKLTVL** 5120_N10 (PGT-139) light chain Kabat CDRs:

(SEQ ID NO: 325) CDR 1: **TGTSNNFVS** (SEQ ID NO: 211) CDR 2: **GVNKRPS** (SEQ ID NO: 196) CDR 3: **GSLVGNWDVI** 5120_N10 (PGT-139) light chain Chothia CDRs: (SEQ ID NO: 325) CDR 1: **TGTSNNFVS** (SEQ ID NO: 211) CDR 2: **GVNKRPS** (SEQ ID NO: 196) CDR 3: **GSLVGNWDVI**

(225) The 4835_F12 (PGT-124) antibody includes a heavy chain variable region (SEQ ID NO: 405), encoded by the nucleic acid sequence shown in SEQ ID NO: 403, and a light chain variable region (SEQ ID NO: 414) encoded by the nucleic acid sequence shown in SEQ ID NO: 412.

(226) The heavy chain CDRs of the 4835 F12 (PGT-124) antibody have the following sequences per Kabat definition: **NYYWT** (SEQ ID NO: 406), **YISDRETTTYNPSLNS** (SEQ ID NO: 407), and **ARRGQRIYGVVSFGEFFYYYYMDV** (SEQ ID NO: 408). The light chain CDRs of the 4835 F12 (PGT-124) antibody have the following sequences per Kabat definition: **GRQALGSRAVQ** (SEQ ID NO: 415), **NNQDRPS** (SEQ ID NO: 151), and **HMWDSRSGFSWS** (SEQ ID NO: 416).

(227) The heavy chain CDRs of the 4835_F12 (PGT-124) antibody have the following sequences per Chothia definition: **GGSISN** (SEQ ID NO: 409), **YISDRETTT** (SEQ ID NO: 410), and **ARRGQRIYGVVSFGEFFYYYYMDV** (SEQ ID NO: 408). The light chain CDRs of the 4835 F12 (PGT-124) antibody have the following sequences per Chothia definition: **GRQALGSRAVQ** (SEQ ID NO: 415), **NNQDRPS** (SEQ ID NO: 151), and **HMWDSRSGFSWS** (SEQ ID NO: 416).

(228) The 4869_K15 (PGT-133) antibody includes a heavy chain variable region (SEQ ID NO: 420), encoded by the nucleic acid sequence shown in SEQ ID NO: 418, and a light chain variable region (SEQ ID NO: 429) encoded by the nucleic acid sequence shown in SEQ ID NO: 427.

(229) The heavy chain CDRs of the 4869_K15 (PGT-133) antibody have the following sequences per Kabat definition: **GRFWS** (SEQ ID NO: 421), **YFSDTDRSEYNPSLRS** (SEQ ID NO: 422), and **AQQGKRIYGIVSFGEFFYYYYMDA** (SEQ ID NO: 423). The light chain CDRs of the 4869 K15 (PGT-133) antibody have the following sequences per Kabat definition: **GERSRGSRAVQ** (SEQ ID NO: 430), **NNQDRPA** (SEQ ID NO: 179), and **HYWDSRSPISWI** (SEQ ID NO: 431).

(230) The heavy chain CDRs of the 4869_K15 (PGT-133) antibody have the following sequences per Chothia definition: **NGSVSG** (SEQ ID NO: 424), **YFSDTDRSE** (SEQ ID NO: 425), and **AQQGKRIYGIVSFGEFFYYYYMDA** (SEQ ID NO: 423). The light chain CDRs of the 4869 K15 (PGT-133) antibody have the following sequences per Chothia definition: **GERSRGSRAVQ** (SEQ ID NO: 430), **NNQDRPA** (SEQ ID NO: 179), and **HYWDSRSPISWI** (SEQ ID NO: 431).

(231) The 4876 M06 (PGT-134) antibody includes a heavy chain variable region (SEQ ID NO: 435), encoded by the nucleic acid sequence shown in SEQ ID NO: 433, and a light chain variable region (SEQ ID NO: 440) encoded by the nucleic acid sequence shown in SEQ ID NO: 438.

(232) The heavy chain CDRs of the 4876_M06 (PGT-134) antibody have the following sequences per Kabat definition: **GRFWS** (SEQ ID NO: 421), **YFSDTDRSEYNPSLRS** (SEQ ID NO: 422), and **AQQGKRIYGIVSFGEFFYYYYMDA** (SEQ ID NO: 436). The light chain CDRs of the 4876 M06 (PGT-134) antibody have the following sequences per Kabat definition: **GERSRGSRAVQ**

(SEQ ID NO: 430), NNQDRPA (SEQ ID NO: 179), and HYWDSRSPISWI (SEQ ID NO: 431).

(233) The heavy chain CDRs of the 4876 M06 (PGT-134) antibody have the following sequences per Chothia definition: NGSVSG (SEQ ID NO: 424), YFSDTDRSE (SEQ ID NO: 425), and AQQGKRIYGIVSFGELFYYYMDA (SEQ ID NO: 436). The light chain CDRs of the 4876 M06 (PGT-134) antibody have the following sequences per Chothia definition: GERSRGSRAVQ (SEQ ID NO: 430), NNQDRPA (SEQ ID NO: 179), and HYWDSRSPISWI (SEQ ID NO: 431).

(234) The 5131_A17 (PGT-132) antibody includes a heavy chain variable region (SEQ ID NO: 444), encoded by the nucleic acid sequence shown in SEQ ID NO: 442, and a light chain variable region (SEQ ID NO: 449) encoded by the nucleic acid sequence shown in SEQ ID NO: 447.

(235) The heavy chain CDRs of the 5131_A17 (PGT-132) antibody have the following sequences per Kabat definition: TGHHYWG (SEQ ID NO: 348), HIHYNTAVLHNPALKS (SEQ ID NO: 349), and SGGDILYYNEWQKPHWFYP (SEQ ID NO: 445). The light chain CDRs of the 5131 A17 (PGT-132) antibody have the following sequences per Kabat definition: SGTASDIGSWNFVS (SEQ ID NO: 450), EVNRRRS (SEQ ID NO: 358), and SSLSGRWDIV (SEQ ID NO: 359).

(236) The heavy chain CDRs of the 5131 A17 (PGT-132) antibody have the following sequences per Chothia definition: GDSINTGH (SEQ ID NO: 351), HIHYNTAVL (SEQ ID NO: 352), and SGGDILYYNEWQKPHWFYP (SEQ ID NO: 445). The light chain CDRs of the 5131 A17 (PGT-132) antibody have the following sequences per Chothia definition: SGTASDIGSWNFVS (SEQ ID NO: 450), EVNRRRS (SEQ ID NO: 358), and SSLSGRWDIV (SEQ ID NO: 359).

(237) The 5138_G07 (PGT-138) antibody includes a heavy chain variable region (SEQ ID NO: 454), encoded by the nucleic acid sequence shown in SEQ ID NO: 452, and a light chain variable region (SEQ ID NO: 461) encoded by the nucleic acid sequence shown in SEQ ID NO: 459.

(238) The heavy chain CDRs of the 5138_G07 (PGT-138) antibody have the following sequences per Kabat definition: ACDYFWG (SEQ ID NO: 201), SLSHCAGYYNSGWTYHNPSLKS (SEQ ID NO: 455), and FGGDVLVYHDWPKPAWVDL (SEQ ID NO: 456). The light chain CDRs of the 5138_G07 (PGT-138) antibody have the following sequences per Kabat definition: TGNINNFVS (SEQ ID NO: 458), GVNKRPS (SEQ ID NO: 211), and GSLAGNWDVV (SEQ ID NO: 459).

(239) The heavy chain CDRs of the 5138_G07 (PGT-138) antibody have the following sequences per Chothia definition: GDSTAACD (SEQ ID NO: 204), SLSHCAGYYNSGWTY (SEQ ID NO: 457), and FGGDVLVYHDWPKPAWVDL (SEQ ID NO: 456). The light chain CDRs of the 5138_G07 (PGT-138) antibody have the following sequences per Chothia definition: TGNINNFVS (SEQ ID NO: 458), GVNKRPS (SEQ ID NO: 211), and GSLAGNWDVV (SEQ ID NO: 459).

(240) The 5120_N10 (PGT-139) antibody includes a heavy chain variable region (SEQ ID NO: 463), encoded by the nucleic acid sequence shown in SEQ ID NO: 461, and a light chain variable region (SEQ ID NO: 470) encoded by the nucleic acid sequence shown in SEQ ID NO: 468.

(241) The heavy chain CDRs of the 5120_N10 (PGT-139) antibody have the following sequences per Kabat definition: GCDYFWG (SEQ ID NO: 464), GLSHCAGYYNTGWTYHNPSLKS (SEQ ID NO: 202), and FDGEVLVYNDWPKPAWVDL (SEQ ID NO: 465). The light chain CDRs of the 5120_N10 (PGT-139) antibody have the following sequences per Kabat definition: TGTSNNFVS (SEQ ID NO: 325), GVNKRPS (SEQ ID NO: 211), and GSLVGNWDVI (SEQ ID NO: 196).

(242) The heavy chain CDRs of the 5120_N10 (PGT-139) antibody have the following sequences per Chothia definition: GDSTAGCD (SEQ ID NO: 466), GLSHCAGYYNTGWTY (SEQ ID NO: 205), and FDGEVLVYNDWPKPAWVDL (SEQ ID NO: 465). The light chain CDRs of the 5120_N10 (PGT-139) antibody have the following sequences per Chothia definition: TGTSNNFVS (SEQ ID NO: 325), GVNKRPS (SEQ ID NO: 211), and GSLVGNWDVI (SEQ ID NO: 196).

(243) The sequences of additional human monoclonal antibodies were determined, including the sequences of the variable regions of the Gamma heavy and Kappa or Lambda light chains. In

addition, the sequence of each of the polynucleotides encoding the antibody sequences was determined. Shown below are the polypeptide and polynucleotide sequences of the gamma heavy chains and kappa light chains, with the signal peptides at the N-terminus (or 5' end) and the constant regions at the C-terminus (or 3' end) of the variable regions, which are shown in bolded text.

(244) TABLE-US-00017 6831_A21 (PGT-151) gamma heavy chain nucleotide sequence: coding sequence (leader sequence in *italics*, variable region in **bold**) (SEQ ID NO: 471)

ATGGAATTGGGGCTGAGCTGGGTTTTCTCGTTGGTCTCTTAAGAGGTGTCCA
GTGTCTGGGTGCAGTTGGTGGAGTCGGGGGGAGGCGTGGTCCAGCCTG
GGAAGTCCGTGAGACTTTCCTGTGTAGTCTCCGATTTCCCCTTCAGCA
AGTATCCTATGTATTGGGTTCGCCAGGCTCCAGGCAAGGGGCTGGAG
TGGGTGGCAGCCATCTCCGGTGATGCCTGGCATGTGGTCTACTCAA
TTCCGTGCAGGGCCGATTTCTCGTCTCCAGGGACAATGTCAAGAACA
CTCTATATTTAGAAATGAACAGCCTGAAAATTGAGGATACGGCCGTA
TATCGCTGCGCGAGAATGTTCCAGGAGTCTGGTCCACCACGTTTGG
TCGTTGGAGCGGTTCGAAATTATTACTATTATTCTGGTATGGACGTCTG
GGGCCAAGGGACCACGGTCACCGTCTCGAGCGCCTCCACCAAGGGCC
CATCGGTCTTCCCCCTGGCACCCTCCTCCAAGAGCACCTCTGGGGGCACA
GCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGT
GTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTG
TCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCT
CCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCC
AGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAA
CTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCA
GTCTTCCTCTTCCCCCCTCAAAACCCAAGGACACCCTCATGATCTCCCGGACC
CCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGT
CAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACA
AAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCC
TCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACCTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA

6831_A21 (PGT-151) gamma heavy chain variable region nucleotide sequence: (SEQ ID NO: 472)

CGGGTGCAGTTGGTGGAGTCGGGGGGAGGCGTGGTCCAGCCTGGGA
AGTCCGTGAGACTTTCCTGTGTAGTCTCCGATTTCCCCTTCAGCAAGT
ATCCTATGTATTGGGTTCGCCAGGCTCCAGGCAAGGGGCTGGAGTGG
GTGGCAGCCATCTCCGGTGATGCCTGGCATGTGGTCTACTCAAATTC
CGTGCAGGGCCGATTTCTCGTCTCCAGGGACAATGTCAAGAACA
CTCTATATTTAGAAATGAACAGCCTGAAAATTGAGGATACGGCCGTATAT
CGCTGCGCGAGAATGTTCCAGGAGTCTGGTCCACCACGTTTGGATCG
TTGGAGCGGTTCGAAATTATTACTATTATTCTGGTATGGACGTCTGGG
GCCAAGGGACCACGGTCACCGTCTCGAGC

6831_A21 (PGT-151) gamma heavy chain amino acid sequence: expressed protein with leader sequence in *italics* and variable region in **bold**. (SEQ ID NO: 473)

MELGLSVFLVQCRVQLVQPGKSVRLSCVVSDFPFISKY
PMYWVRQAPGKGLEWVA AISGDAWHVVYSNSVQGRFLVSRDNVKNTL
YLEMNSLKIEDTAVYRCARMFQESGPPRLDRWSGRNYYYYSGMDVWG
QGTTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNS
GALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDK
RVEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNV
FSCSVMHEALHNHYTQKSLSLSPGK 6831_A21 (PGT-151) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 474) RVQLVESGGGVVQPGKSVRLSCVVS
custom character YPMYWVRQAPGKGLEWVAcustom character custom character
YSNSVQGRFLVSRDNVKNTLYLEMNSLKIEDTAVYRCAR
custom character WGQGTTVTVSS 6831_A21 (PGT-151) gamma heavy chain
Kabat CDRs: (SEQ ID NO: 475) CDR 1: KYPMY (SEQ ID NO: 476) CDR
2: AISGDAWHVVYSNSVQG (SEQ ID NO: 477) CDR 3:
MFQESGPPRLDRWSGRNYYYYSGMDV 6831_A21 (PGT-151) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 478) CDR 1: DFPFSK (SEQ ID NO: 479) CDR
2: AISGDAWHVV (SEQ ID NO: 477) CDR 3:
MFQESGPPRLDRWSGRNYYYYSGMDV 6831_A21 (PGT-151) light chain nucleotide
sequence: coding sequence (variable region in bold) (SEQ ID NO: 480)
ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGATACCTGAATT
CACTGCAGACATTGTGATGACCCAGACTCCTCTCTCTTTGTCCGTCAC
CCCTGGACAGCCGGCCTCCATCTCCTGCAAGTCCAGTGAGAGCCTCC
GACAAAGTAATGGAAAGACCTCTTTGTATTGGTATCGGCAGAAGCCA
GGCCAGTCTCCACAACCTCCTAGTGTTTGAAGTTTCTAATCGATTCTCT
GGCGTGTCGGATAGGTTTGTGTTGGCAGCGGGTCAGGGACAGACTTCAC
ACTGAGAATCAGCCGGGTAGAGGCTGAGGATGTTGGATTTTATTACT
GCATGCAAAGTAAAGACTTCCCACTTACATTTGGCGGCGGGACCAAG
GTGGATCTCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCC
ATCTGATGAGCAGTTGAAATCTGGAACCTGCCTCTGTTGTGTGCCTGCTGAA
TAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCC
TCCAATCGGGTAACTCCCAGGAGAGTGTCACAGAGCAGGACAGCAAGGA
CAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGACTAC
GAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTC
GCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 6831_A21 (PGT-151) light
chain variable region nucleotide sequence: (SEQ ID NO: 481)
GACATTGTGATGACCCAGACTCCTCTCTCTTTGTCCGTCACCCCTGGA
CAGCCGGCCTCCATCTCCTGCAAGTCCAGTGAGAGCCTCCGACAAAG
TAATGGAAAGACCTCTTTGTATTGGTATCGGCAGAAGCCAGGCCAGT
CTCCACAACCTCCTAGTGTTTGAAGTTTCTAATCGATTCTCTGGCGTGT
CGGATAGGTTTGTGTTGGCAGCGGGTCAGGGACAGACTTCACACTGAGA
ATCAGCCGGGTAGAGGCTGAGGATGTTGGATTTTATTACTGCATGCA
AAGTAAAGACTTCCCACTTACATTTGGCGGCGGGACCAAGGTGGATC TCAA
6831_A21 (PGT-151) light chain amino acid sequence: expressed protein with
leader sequence in italics and variable region in bold. (SEQ ID NO: 482)
MRLPAQLLGLLMLWIPEFTADIVMTQTPLSLSVTPGQPASISCKSSES LRQSN
GKTS*LYWYRQKPGQSPQLLVFEVSNRFS***GVSDRFVSGSGTDFTLRISRV**
EAEDVGFYYCMQSKDFPLTFGGG**TKVDLKRTVAAPSVFIFPPSDEQLKSGT**

ASVLCNLYNREAKVQWKLQSQSVTEQDSKDSSTLTL
SKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC 6831_A21 (PGT-151) light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 483) DIVMTQTPLSLSVTPGQPASISCcustom character
WYRQKPGQSPQLLVFcustom character GVSDRFVGS GSGTDFTLRISRVEAEDVGFYYC
custom character FGGG TKVDLK 6831_A21 (PGT-151) light chain Kabat CDRs:
(SEQ ID NO: 484) CDR 1: KSSESLRQSNGKTSLY (SEQ ID NO: 485) CDR
2: EVSNRFS (SEQ ID NO: 486) CDR 3: MQSKDFPLT 6831_A21 (PGT-151)
light chain Chothia CDRs: (SEQ ID NO: 484) CDR 1: KSSESLRQSNGKTSLY
(SEQ ID NO: 485) CDR 2: EVSNRFS (SEQ ID NO: 486) CDR 3:
MQSKDFPLT 6889_117 (PGT-152) gamma heavy chain nucleotide sequence:
coding sequence (variable region in bold) (SEQ ID NO: 487)

ATGGAATTGGGGCTGAGCTGGGTTTTCTCGTTGGTCTCTTAAGAGGTGTCCA
CTGTCTGGGTGCAGTTGGTGGAGTCGGGGGGAGGCGTGGTCCAGCCTG
GGAAGTCCGTGAGACTTTCCTGTGTAGTCTCTGATTTCCCCTTCAGCA
AGTATCCTATGTATTGGGTTTCGCCAGGCTCCAGGCAAGGGGCTGGAG
TGGGTGGCAGCCATCTCCGCTGATGCCTGGCATGTGGTCTACTCAGG
CTCCGTGCAGGGCCGATTTCTCGTCTCCAGGGACAACCTCCAAGAACA
TTCTGTATTTGGAAATGAACACCCTGAAAATTGAGGACACGGCCGTA
TATCGCTGCGCGAGAATGTTCCAGGAGTCTGGTCCACCACGTTTCGA
TTCTTGGAGCGGTTCGAAATTACTACTATTACTCTGGTATGGACGTCTG
GGGCAAGGGACCACGGTCACCGTCTCGAGCGCCTCCACCAAGGGCC
CATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACA
GCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGT
GTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTG
TCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCT
CCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCC
AGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAA
CTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCA
GTCTTCCTCTTCCCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACC
CCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGT
CAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACA
AAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCC
TCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACCTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 6889_117 (PGT-152) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 488)

CGGGTGCAAGTTGGTGGAGTCGGGGGGAGGCGTGGTCCAGCCTGGGA
AGTCCGTGAGACTTTCCTGTGTAGTCTCTGATTTCCCCTTCAGCAAGT
ATCCTATGTATTGGGTTTCGCCAGGCTCCAGGCAAGGGGCTGGAGTGG
GTGGCAGCCATCTCCGCTGATGCCTGGCATGTGGTCTACTCAGGCTC
CGTGCAGGGCCGATTTCTCGTCTCCAGGGACAACCTCCAAGAACATTC
TGTATTTGGAAATGAACACCCTGAAAATTGAGGACACGGCCGTATAT
CGCTGCGCGAGAATGTTCCAGGAGTCTGGTCCACCACGTTTCGATTC

TTGGAGCGGTACGTAATCTGCTATGACCTCTGGG
GCCAAGGGACCACGGTCACCGTCTCGAGC 6889_117 (PGT-152) gamma heavy chain amino acid sequence: expressed protein with leader sequence in *italics* and variable region in **bold**. (SEQ ID NO: 489)

MELGLSWVFLVGLLRGVHCRVQLVESGGGVVQPGKSVRLSCVVSDFPFSKY
PMYWVRQAPGKGLEWVA*AI***SADAWHV***V***YSGSVQGRFLVSRDNSKNILY**
LEMNTLKIEDTAVYRCAR*MFQESGPPRFDSWSGRNYYYYSGMDVWGQ*
GTTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSG
ALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKR
VEPKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVS
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGK
EYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYP*SDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNV*

FSCSVMHEALHNHYTQKSLSLSPGK 6889_I17 (PGT-152) gamma heavy chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in **bold italics**) (SEQ ID NO: 490) *RVQLVESGGGVVQPGKSVRLSCVVS*

custom character *YPMYWVRQAPGKGLEWVA**custom character* *custom character*
*YSGSVQGRFLVSRDNSKNILYLEMNTLKIEDTAVYRCAR**custom character*

custom character **WGQGTTVTVSS** 6889_117 (PGT-152) gamma heavy chain Kabat CDRs: (SEQ ID NO: 475) CDR 1: **KYPMY** (SEQ ID NO: 491) CDR 2: **AI****SADAWHV***V***YSGSVQG** (SEQ ID NO: 492) CDR 3: **MFQESGPPRFDSWSGRNYYYYSGMDV** 6889_117 (PGT-152) gamma heavy chain Chothia CDRs: (SEQ ID NO: 478) CDR 1: **DFPFSK** (SEQ ID NO: 493) CDR 2: **AI****SADAWHV***V* (SEQ ID NO: 492) CDR 3:

MFQESGPPRFDSWSGRNYYYYSGMDV 6889_117 (PGT-152) light chain nucleotide sequence: coding sequence (variable region in **bold**) (SEQ ID NO: 494)

ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGATACCTGAATT
TATTGCCGACATTGTGATGACCCAGACTCCTCTCTCTTTGTCCGTCGAC
CCTGGACAGCCGGCCTCCATCTCCTGCAAGTCCAGTCAGAGCCTCCG
ACAAAGTAATGGAAAGACCTCTTTGTATTGGTATCAGCAGAAGCCAG
GCCAGTCTCCACA**ACTCCTAATATTTGAAGTTTCTAATCGATTCTCTG**
GCGTGTCGGATAGGTTTGTGTTGGCAGCGGGTCAGGGACAGACTTCACA
CTGAGAATCAGCCGGGTAGAGGCTGAGGATGTTGGATTTTATTACTG
CATGCAAAGTAAAGACTTCCC**ACTCACCTTTGGCGGGCGGGACCAAGG**
TGGATCTCAACCGTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCCA
TCTGATGAGCAGTTGAAATCTGGA**ACTGCCTCTGTTGTGTGCCTGCTGAAT**
AACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCCCT
CCAATCGGGTAACTCCCAGGAGAGTGTACAGAGCAGGACAGCAAGGAC
AGCACCTACAGCCTCAGCAGCACCCCTGACGCTGAGCAAAGCAGACTACG
AGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTCG
CCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 6889_117 (PGT-152) light chain variable region nucleotide sequence: (SEQ ID NO: 495)

GACATTGTGATGACCCAGACTCCTCTCTCTTTGTCCGTCGACCCTGGA
CAGCCGGCCTCCATCTCCTGCAAGTCCAGTCAGAGCCTCCGACAAAG
TAATGGAAAGACCTCTTTGTATTGGTATCAGCAGAAGCCAGGCCAGT
CTCCACA**ACTCCTAATATTTGAAGTTTCTAATCGATTCTCTGGCGTGT**
CGGATAGGTTTGTGTTGGCAGCGGGTCAGGGACAGACTTCACACTGAGA
ATCAGCCGGGTAGAGGCTGAGGATGTTGGATTTTATTACTGCATGCA
AAGTAAAGACTTCCC**ACTCACCTTTGGCGGGCGGGACCAAGGTGGATC** **TCAAC**
6889_I17 (PGT-152) light chain amino acid sequence: expressed protein with

leader sequence in *italics*, variable region in **bold**. (SEQ ID NO: 496)
MRLPAQLLGLLMLWIPEFIADIVMTQTPLSLSVDPGQPASISCKSSQSLRQSN
GKTS*LYWYQQKPGQSPQLLIFEVSNRFS***GVSDRFVGS***SGSGTDFTLRISRV*
EAEDVGFYYCMQSKDFPLTFGGG*TKVDLNRTVAAPSVFIFPPSDEQLKSGT*
*ASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSK**DSTYLS***SSTLTL**
SKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC 6889_117 (PGT-152) light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in **bold italics**) (SEQ ID NO: 497) *DIVMTQTPLSLSVDPGQPASIS**WYQQKPGQSPQLLIF* *GVSDRFVGS***SGSGTDFTLRISRVEAEDVGFYYC**
*FGGG TKVDLN* 6889_117 (PGT-152) light chain Kabat CDRs:
(SEQ ID NO: 498) CDR 1: KSSQSLRQSN**GKTS***LY* (SEQ ID NO: 485) CDR
2: EVS*NRFS* (SEQ ID NO: 486) CDR 3: MQSKDFPLT 6889_117 (PGT-152)
light chain Chothia CDRs: (SEQ ID NO: 498) CDR 1: KSSQSLRQSN**GKTS***LY*
(SEQ ID NO: 485) CDR 2: EVS*NRFS* (SEQ ID NO: 486) CDR 3:
MQSKDFPLT 6891_F06 (PGT-153) gamma heavy chain nucleotide sequence:
coding sequence (leader sequence in *italics*, variable region in **bold**) (SEQ ID
NO: 499) *ATGGAATTGGGGCTGAGCTGGGTTTTCTCGTTGCTCTCTTAAGAGGTGTCCA*
GTGTCAGGTGCAGTTGGTGGAGTCGGGCGGAGGCGTGGTCCAGCCTG
GGAAGTCCCTGAGACTCTCCTGTGTAGTCTCTAATTTTCTCTTCAATA
AACGTCACATGCACTGGGTCCGCCAGGCTCCAGGCAAGGGACTAGAG
TGGATAGCAGTCATTTCTCTGATGCCATTCACGTAGACTACGCAAGT
TCCGTGCGGGGCCGATCCCTCATCTCCAGAGACAATTCCAAAAATAG
TTTATATCTAGACATGAATAACCTGAAAATTGAGGACACGGCCACATA
TTATTGTGCAAGAGATAGAGACGGATATGGTCCACCACAGATCCAGA
CTTGGAGCGGTCGATACCTCCACCTTTATTCTGGAATAGACGCCTGG
GGCCTAGGGACCACGGTCACCGTCTCGAGCGCCTCCACCAAGGGCCCA
TCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACAGC
GGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGTGT
*CGTGGA**ACTCAGGCGCCCTGACCAGCGGCGTGACACACCTTCCCGGCTGTC*
*CTACAGT**CCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCTCC*
AGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCCAG
*CAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAA**ACT*
CACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCAGT
*CTTCCTCTTCCCCC**AAAACCCAAGGACACCCTCATGATCTCCCGGACCCC*
*TGAGGT**CACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGTCA*
*AGTTCA**ACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACAAA*
GCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCCTC
*ACCGT**CCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAAGG*
TCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAGCC
AAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGGG
AGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCTTC
TATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGAGA
ACAACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCTTC
CTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAACG
TCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGCAG
AAGAGCCTCTCCCTGTCTCCGGGTAAATGA 6891_F06 (PGT-153) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 500)
CAGGTGCAGTTGGTGGAGTCGGGCGGAGGCGTGGTCCAGCCTGGGA
AGTCCCTGAGACTCTCCTGTGTAGTCTCTAATTTTCTCTTCAATAAAC
GTCACATGCACTGGGTCCGCCAGGCTCCAGGCAAGGGACTAGAGTG

GATGACCTAGCTACCTTCTGCTACCTAGCTACGCAAGTTC
CGTGCGGGGCGGATCCCTCATCTCCAGAGACAATTCCAAAAATAGTT
TATATCTAGACATGAATAACCTGAAAATTGAGGACACGGCCACATATT
ATTGTGCAAGAGATAGAGACGGATATGGTCCACCACAGATCCAGACT
TGGAGCGGTCGATACCTCCACCTTTATTCTGGAATAGACGCCTGGGG
CCTAGGGACCACGGTCACCGTCTCGAGC 6891_F06 (PGT-153) gamma heavy chain amino acid sequence: expressed protein with leader sequence in and variable region in bold.(SEQ ID NO: 501)

**MELGLSWVFLVALLRGVQCQVQLVESGGGVVQPGKSLRLSCVVSNFLFNKR
HMHWVRQAPGKGLEWIAVISSDAIHVDYASSVRGRSLISRDNSKNSLYLD
MNNLKIEDTATYYCARDRDGYGPPQIQTWSGRYLHLYSGIDAWGLGTT
VTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTS
GVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVEPK
SCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKC
KVSNAKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYP
SDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSS
VMHEALHNHYTQKSLSLSPGK 6891_F06 (PGT-153) gamma heavy chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 502) QVQLVESGGGVVQPGKSLRLSCVVSN custom character
RHMHWVRQAPGKGLEWIA custom character custom character
YASSVRGRSLISRDNSKNSLYLDMNNLKIEDTATYYCAR custom character**

custom character WGLGTTVTVSS 6891_F06 (PGT-153) gamma heavy chain Kabat CDRs: (SEQ ID NO: 503) CDR 1: KRHMH (SEQ ID NO: 504) CDR 2: VISSDAIHVDYASSVRG (SEQ ID NO: 505) CDR 3: DRDGYGPPQIQTWSGRYLHLYSGIDA 6891_F06 (PGT-153) gamma heavy chain Chothia CDRs: (SEQ ID NO: 506) CDR 1: NFLFNK (SEQ ID NO: 507) CDR 2: VISSDAIHVD (SEQ ID NO: 505) CDR 3: DRDGYGPPQIQTWSGRYLHLYSGIDA 6891_F06 (PGT-153) light chain nucleotide sequence: coding sequence (variable region in bold) (SEQ ID NO: 508)

ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGATACCTGAATT
CACTGCGGACATTGTGCTGACCCAGAGCCCCCTCTTTCTGTCCGTCAG
TCCTGGACAGCCGGCCTCCATCTCCTGTAAGTCTAGTCAGAGCCTCC
GACAAAGTAATGGAAAGACATATTTGTATTGGTACGTACAAAAGTCC
GGCCAGTCTCCACAACCCCTGATCCAGGAAGTTTCCATTTCGCTTCTCT
GGAGTGCCAGGTAGATTGCTGGCAGCGGATCAGGGACAGACTTCAC
ACTGAAAATCAGCCGGGTGGAGGCTGAAGATGTTGGAGTTTATTTCT
GCATGCAAAGTAAAGACTTTCCACTCACTTTTGGCGGAGGGACCAAG
GTGGACCTCAATCGTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCC
ATCTGATGAGCAGTTGAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCTGAA
TAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCC
TCCAATCGGGTAACTCCCAGGAGAGTGTACAGAGCAGGACAGCAAGGA
CAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGACTAC
GAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTC
GCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 6891_F06 (PGT-153) light chain variable region nucleotide sequence: (SEQ ID NO: 509)

**GACATTGTGCTGACCCAGAGCCCCCTCTTTCTGTCCGTCAGTCCTGG
ACAGCCGGCCTCCATCTCCTGTAAGTCTAGTCAGAGCCTCCGACAAA
GTAATGGAAAGACATATTTGTATTGGTACGTACAAAAGTCCGGCCAG
TCTCCACAACCCCTGATCCAGGAAGTTTCCATTTCGCTTCTCTGGAGTG**

**CCAGGTAGGCTGCTGCGGAGCGAGCACTTCACACTGAA
AATCAGCCGGGTGGAGGCTGAAGATGTTGGAGTTTATTTCTGCATGC
AAAGTAAAGACTTTCCACTCACTTTTGGCGGAGGGACCAAGGTGGAC CTCAAT**

6891_F06 (PGT-153) light chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold.(SEQ ID NO: 510)

*MRLPAQLLGLLMLWIPEFTADIVLTQSPFLSVSPGQPASISCKSSQSLRQSN
GKTYLYWYVQKSGQSPQPLIQEVSIRFSGVPGRFAGSGSGTDFTLKISRV
EAEDVG VYFCMQSKDFPLTFGGG TKVDLNRTVAAPSVFIFPPSDEQLKSGT
ASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLT
SKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC* 6891_F06 (PGT-153) light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics)(SEQ ID NO: 511) *DIVLTQSPFLSVSPGQPASIS* *WYVQKSGQSPQPLIQ* *GVPGRFAGSGSGTDFTLKISRVEAEDVGVYFC* *FGGGT KVDLN* 6891_F06 (PGT-153) light chain Kabat CDRs:

(SEQ ID NO: 512) CDR 1: KSSQSLRQSNGKTYLY (SEQ ID NO: 513) CDR 2: EVSIRFS (SEQ ID NO: 486) CDR 3: MQSKDFPLT 6891_F06 (PGT-153)

light chain Chothia CDRs: (SEQ ID NO: 512) CDR 1: *KSSQSLRQSNGKTYLY* (SEQ ID NO: 513) CDR 2: *EVSIRFS* (SEQ ID NO: 486) CDR 3: *MQSKDFPLT* 6843_G20 (PGT-154) gamma heavy chain nucleotide sequence: coding sequence (variable region in bold)(SEQ ID NO: 514)

*ATGGAATTGGGGCTGAGCTGGGTTTTCTCGTTGCTCTCTTAAGAGGTGTCCA
GTGTCAGGTGCAGCTGGTGGAAATCGGGAGGAGGCGTGGTCCAGCCTG
GAAAGTCCCTCAGACTCTCATGTGTCGTCTCTAATTTTCATCTTTAATA
AATATCCTATGTATTGGGTCCGCCAGGCTCCAGGCAAGGGGCTGGAG
TGGGTGGCAGCCATCTCCGCTGATGCCTGGCATGTAGACTACGCAGC
CTCCGTGAAGGACCGATTTCTCATCTCCAGAGACAATTCCAAGAATG
CTCTATATTTGGAAATGAACACCCTGAGAGTTGAAGACACGGGTATC
TACTACTGTGCGAGAAATATAGAGGAGTTTAGTGTTCCACAGTTCGA
TTCTTGGAGCGGTCTGAAGCTACTACCACTATTTTGGGATGGACGTCT
GGGGCCAAGGGACCAACGGTCACCGTCTCGAGCGCCTCCACCAAGGGC
CCATCGGTCTTCCCCCTGGCACCTCCTCCAAGAGCACCTCTGGGGGCAC
AGCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGG
TGTCGTGGAATCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCT
GTCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCC
CTCCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGC
CCAGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAA
AACTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGT
CAGTCTTCTCTTCCCCCCTGAGGACACCTCATGATCTCCCGGA
CCCCTGAGGTACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAG
GTCAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGAC
AAAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTC
CTCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACACTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA* 6843_G20 (PGT-154) gamma heavy

chain variable region nucleotide sequence: (SEQ ID NO: 515)
**CAGGTGCAGCTGGTGGAAATCGGGAGGAGGCGTGGTCCAGCCTGGAA
AGTCCCTCAGACTCTCATGTGTCTCTCTAATTTTCATCTTTAATAAAT
ATCCTATGTATTGGGTCCGCCAGGCTCCAGGCAAGGGGCTGGAGTGG
GTGGCAGCCATCTCCGCTGATGCCTGGCATGTAGACTACGCAGCCTC
CGTGAAGGACCGATTTCTCATCTCCAGAGACAATTCCAAGAATGCTC
TATATTTGGAAATGAACACCCTGAGAGTTGAAGACACGGGTATCTAC
TACTGTGCGAGAAATATAGAGGAGTTTAGTGTTCCACAGTTCGATTCT
TGGAGCGGTCGAAGCTACTACCACTATTTTGGGATGGACGTCTGGGG
CCAAGGGACCACGGTCACCGTCTCGAGC** 6843_G20 (PGT-154) gamma heavy
chain amino acid sequence: expressed protein with leader sequence in italics
and variable region in bold. (SEQ ID NO: 516)

*MELGLSWVFLVALLRGVQCQVQLVESGGGVVQPGKSLRLSCVVS***NFIENKY
PMYWVRQAPGKGLEWVA***AI***SADAWHVDYAASVKDRFLISRDNSKNALY
LEMNTLRVEDTGIYYCARNIEEF***SV***PQFDSWSGRSYYHYFGMDVWGQG
TTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVT***VS***WNSGAL
TSGVHTFPAVLQSSGLYSLSSV***TV***PSSSLGTQTYICNVNHKPSNTKVDKRV
PKSCDKHTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCV***V***VDVSHE
DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK** 6843_G20 (PGT-154) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 517) *QVQLVESGGGVVQPGKSLRLSCVVS*

 *YPMYWVRQAPGKGLEWVA*  *YAASVKDRFLISRDNSKNALYLEMNTLRVEDTGIYYCAR* 
 *WGQGTTVTVSS* 6843_G20 (PGT-154) gamma heavy chain
Kabat CDRs: (SEQ ID NO: 475) CDR 1: *KYPMY* (SEQ ID NO: 518) CDR
2: *AI***SADAWHVDYAASVKD** (SEQ ID NO: 519) CDR 3:
NIEEFSVPQFDSWSGRSYYHYFGMDV 6843_G20 (PGT-154) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 520) CDR 1: *NFIFNK* (SEQ ID NO: 521) CDR
2: *AI***SADAWHVD** (SEQ ID NO: 519) CDR 3:
NIEEFSVPQFDSWSGRSYYHYFGMDV 6843_G20 (PGT-154) light chain nucleotide
sequence: coding sequence (variable region in bold) (SEQ ID NO: 522)

**ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGATACCTGAGTT
CGCTGCAGACATTGTGATGACTCAGACTCCTGTCTCTCTGTCCGTCAG
TCTTGGACAGGCGGCCTCCATCTCCTGCAGCTCCAGTGAGAGTCTCG
GACGTGGTGTGGAAGGACCTATTTGCATTGGTACCGACAGAAGCCA
GGCCAGACTCCACAATTACTCATGTATGAAGTTTCTACTCGATTCTCT
GGAGTGTCCGACAGGTTTCGCTGGCAGCGGGTCACGTACACAATTCAC
ATTGAAAATTAGTCGGGTGGAGGCTGAAGATGTTGGCGTTTATTACT
GCATGCAAAGTAGAGACTTCCCAATCACTTTTGGCGGAGGGACCAGG
GTGGATCTCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCC
ATCTGATGAGCAGTTGAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCTGAA
TAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCCC
TCCAATCGGGTAACTCCCAGGAGAGTGTACAGAGCAGGACAGCAAGGA
CAGCACCTACAGCCTCAGCAGCACCCCTGACGCTGAGCAAAGCAGACTAC
GAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTC
GCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG** 6843_G20 (PGT-154) light
chain variable region nucleotide sequence: (SEQ ID NO: 523)

GACATTGATGATGACTGACTGACTGCTCTCTGTCCTGCTCAGTTCAGTGA
CAGGCGGCCTCCATCTCCTGCAGCTCCAGTGAGAGTCTCGGACGTGG
TGATGGAAGGACCTATTTGCATTGGTACCGACAGAAGCCAGGCCAGA
CTCCACAATTACTCATGTATGAAGTTTCTACTCGATTCTCTGGAGTGT
CCGACAGGTTTCGCTGGCAGCGGGTCACGTACACAATTCACATTGAAA
ATTAGTCGGGTGGAGGCTGAAGATGTTGGCGTTTATTACTGCATGCA
AAGTAGAGACTTCCCAATCACTTTTGGCGGAGGGACCAGGGTGGATC TCAA

6843 G20 (PGT-154) light chain amino acid sequence: expressed protein with leader sequence in *italics* and variable region in **bold**. (SEQ ID NO: 524)

*MRLPAQLLGLLMLWIPEFAADIVMTQTPVSLSVSLGQAASISCSSES***LGRGD
GRTYLHWYRQKPGQTPQLLMYE***VSTRFSGVSDRFAGSGSRTQFTLKISR
VEAEDVG***VYYCMQSRDFPITFGGGTRVDLKR***TVAAPSVFIFPPSDEQLKSG
TASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSK***DSTYSL***SSTLT
LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC* 6843_G20 (PGT-154) light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in **bold italics**) (SEQ ID NO: 525) *DIVMTQTPVSLSVSLGQAASISCSSES***LGRGDGRTYLHWYRQKPGQTPQLLMYE***VSTRFSGVSDRFAGSGSRTQFTLKISRVEAEDVG***VYYCMQSRDFPITFGGGTRVDLKR**

*TVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSK***DSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC** 6843_G20 (PGT-154) light chain Kabat CDRs: (SEQ ID NO: 526) CDR 1: *SSSESLGRGDGR***TYLH** (SEQ ID NO: 527) CDR 2: *EVSTRFS* (SEQ ID NO: 528) CDR 3: *MQSRDFPIT* 6843_G20 (PGT-154) light chain Chothia CDRs: (SEQ ID NO: 526) CDR 1: *SSSESLGRGDGR***TYLH** (SEQ ID NO: 527) CDR 2: *EVSTRFS* (SEQ ID NO: 528) CDR 3: *MQSRDFPIT*

6892_D19 (PGT-155) gamma heavy chain nucleotide sequence: coding sequence (leader sequence in *italics*, variable region in **bold**) (SEQ ID NO: 529) *ATGGAATTGGGGCTGAGCTGGGTTTCTCTCGTCGTTCTCCTAAGAGGTGTCCA*

*CTGT***CAGGTGCATCTGGTGGAGTCGGGGGGGAGGCGTGGTCCAACCTG
GGAAGTCCCTAAGACTCTCCTGTGAAACCTCTGGCTTCATCTTCAACG
AATATCCCATGTATTGGATCCGCCAGGCTCCAGGCAAGGGACCGGAG
TGGGTGGCCGCCATCTCCGCTGACGCCTGGCATGTGGACTACGCAGG
CTCCGTGCGGGGGCCGATTTACCGTCTCCAGAGACAATTCTAAGAATT
CTCTATATTTAGACATGAAGAGTCTGAAAGTTGAAGACACGGCTATAT
ATTTCTGTGCGAAAGATGGGGAGGAACACAAGGTACCACAATTGCAT
TCCTGGAGCGGACGAACTTATATCACTACACTGGTTTTTGACGTCTG
GGGCCCAGGGACCACGGTCACCGTCTCGAGCGCCTCCACCAAGGGCC
CATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACA
GCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGT
GTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTG
TCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCT
CCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCC
AGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAA
CTCACACATGCCACCGTGCCAGCACCTGAACTCCTGGGGGGACCGTCA
GTCTTCCTCTTCCCCCCTCAACCAAGGACACCCTCATGATCTCCCGGACC
CCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGT
CAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACA
AAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCC
TCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA**

GAACCAACTCAACAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 6892_D19 (PGT-155) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 530)
CAGGTGCATCTGGTGGAGTCGGGGGGGAGGCGTGGTCCAACCTGGGA
AGTCCCTAAGACTCTCCTGTGAAACCTCTGGCTTCATCTTCAACGAAT
ATCCCATGTATTGGATCCGCCAGGCTCCAGGCAAGGGACCGGAGTGG
GTGGCCGCCATCTCCGCTGACGCCTGGCATGTGGACTACGCAGGCTC
CGTGCGGGGGCCGATTTACCGTCTCCAGAGACAATTCTAAGAATTCTC
TATATTTAGACATGAAGAGTCTGAAAGTTGAAGACACGGCTATATATT
TCTGTGCGAAAGATGGGGAGGAACACAAGGTACCACAATTGCATTCC
TGGAGCGGACGAACTTATATCACTACACTGGTTTTGACGTCTGGGG
CCCAGGGACCACGGTCACCGTCTCGAGC 6892_D19 (PGT-155) gamma heavy
chain amino acid sequence: expressed protein with leader sequence in *italics*
and variable region in **bold**. (SEQ ID NO: 531)

MELGLSWVFLVLLRGVHCQVHLVESGGGVVQPGKSLRLSCETSGFIFNEY
PMYWIRQAPGKGPEWVA*AI***SADAWHVDYAGSVRGRFTVSRDNSKNSLY**
LDMKSLKVEDTAIYFCAKDGE*EHKVPQLHSW***SGRNLYHYTGFDVWGPG**
TTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGAL
TSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKRVE
PKSCDKHTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDV
SHE
DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK 6892_D19 (PGT-155) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in **bold italics**) (SEQ ID NO: 532) *QVHLVESGGGVVQPGKSLRLSCETS*
custom character *YPMYWIRQAPGKGPEWVA* *custom character* *custom character*
YAGSVRGRFTVSRDNSKNSLYLDMKSLKVEDTAIYFCAK *custom character*
custom character *WGPGTTVTVSS* 6892_D19 (PGT-155) gamma heavy chain Kabat
CDRs: (SEQ ID NO: 533) CDR 1: *EYPMY* (SEQ ID NO: 534) CDR 2:
*AI***SADAWHVDYAGSVRG** (SEQ ID NO: 535) CDR 3:
*DGEEHKVPQLHSW***SGRNLYHYTGFDV** 6892_D19 (PGT-155) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 536) CDR 1: *GFIFNE* (SEQ ID NO: 521) CDR
2: *AI***SADAWHVD** (SEQ ID NO: 535) CDR 3:
*DGEEHKVPQLHSW***SGRNLYHYTGFDV** 6892_D19 (PGT-155) light chain nucleotide
sequence: coding sequence (variable region in **bold**) (SEQ ID NO: 537)

ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGATACCTGA
ACT
TGCTGCAGACATTGTGATGACCCAGTCTCCTGTCTCTCTGTCCGTCAC
CCTCGGACAGCCGGCCTCCATGTCCTGCAAGTCCAGTCAGAGTGTCC
GACAGAGTGATGGCAAGACTTTCTTATATTGGTATCGACAGAAGCCA
*GGCCAGTCTCCACA***ACTGTTAATATATGAGGGTTCGAGTCGATTCTCT**
GGAGTGT*CAGATAGGATCTCTGGCAGCGGGTCAGGGACAGACTTCAC*
ACTGAGGATCAGTCGAGTGGAGGCTGAGGATGCTGGCGTTTACTTCT
*GCTTGCAA***ACTAAAGACTTCCCCCTCACTTTTGGCGGAGGGACCAGG**
GTGGATCTCAA*ACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCC*
*ATCTGATGAGCAGTTGAAATCTGGA***ACTGCCTCTGTTGTGTGCCTGCTGAA**
TAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCC
TCCAATCGGGTAACTCCCAGGAGAGTGTCACAGAGCAGGACAGCAAGGA

CAGCACCTACAGTCAAGCTGAGCAAGAGCACTAC
GAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTC
GCCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 6892_D19 (PGT-155) light
chain variable region nucleotide sequence: (SEQ ID NO: 538)

**GACATTGTGATGACCCAGTCTCCTGTCTCTCTGTCCGTCACCCTCGGA
CAGCCGGCCTCCATGTCTTGCAAGTCCAGTCAGAGTGTCCGACAGAG
TGATGGCAAGACTTTTCTTATATTGGTATCGACAGAAGCCAGGCCAGT
CTCCACAACCTGTTAATATATGAGGGTTCGAGTCGATTCTCTGGAGTGT
CAGATAGGATCTCTGGCAGCGGGTCAGGGACAGACTTCACACTGAGG
ATCAGTCGAGTGGAGGCTGAGGATGCTGGCGTTTACTTCTGCTTGCA
AACTAAAGACTTCCCCCTCACTTTTGGCGGAGGGACCAGGGTGGATC TCAA**
6892_D19 (PGT-155) light chain amino acid sequence: expressed protein with
leader sequence in *italics* and variable region in **bold**. (SEQ ID NO: 539)

*MRLPAQLLGLLMLWIPELAADIVMTQSPVLSVTLGQPASMSCKSSQSVRQS
DGKTFLYWYRQKPGQSPQLLIYEGSSRFSGVSDRISGSGSGTDFTLRISRV
EAEDAGVYFCLQTKDFPLTFGGGTRVDLKR**TVAAPSVFIFPPSDEQLKSGT
ASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYLSSTLT
SKADYEEKHKVYACEVTHQGLSSPVTKSFNRGEC* 6892_D19 (PGT-155) light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in **bold italics**) (SEQ ID NO: 540) *DIVMTQSPVLSVTLGQPASMSC*

 *WYRQKPGQSPQLLI* 
GVSDRISGSGSGTDFTLRISRVEAEDAGVYFC  *FGGG TRVDLK*
6892_D19 (PGT-155) light chain Kabat CDRs: (SEQ ID NO: 541) CDR 1:
KSSQSVRQSDGKTFLY (SEQ ID NO: 542) CDR 2: EGSSRFS (SEQ ID NO: 543)
CDR 3: LQTKDFPLT 6892_D19 (PGT-155) light chain Chothia CDRs: (SEQ ID
NO: 541) CDR 1: KSSQSVRQSDGKTFLY (SEQ ID NO: 542) CDR 2: EGSSRFS
(SEQ ID NO: 543) CDR 3: LQTKDFPLT 6808_B09 (PGT-156) gamma heavy
chain nucleotide sequence: coding sequence (variable region in **bold**) (SEQ ID
NO: 544) *ATGGAATTGGGGCTGAGCTGGGTTTTCTCGTCGTTCTCCTAAGAGGTGTCCA*

CTGT**CAGGTGCATCTGGTGGAGTCGGGGGGGAGGCGTTGTCCAACCTG
GAAAGTCCCTAAGACTCTCCTGTGAAACCTCTGGCTTCATCTTCAATC
AATATCCCATGTATTGGGTCCGCCAGGCTCCAGGCAAGGGACCGGAG
TGGGTGGCCGCCATCTCCGCTGATGCCTGGCATGTGGACTACCCAGG
CTCCGTGCGGGGGCCGATTTACCGTCTCCAGAGACAATTCCAAGAGTT
CTCTATATTTAGACATGAAGAGTCTGAAAGTTGAAGACACGGCTATAT
ATTTCTGTGCGAAAGATGGGGAGGAACACAAGGTACCACAATTGCAT
TCCTGGAGCGGACGAAACTTATATCACTACACTGGTTTTTGACGTCTG
GGGCCCAGGGACCACGGTCACCGTCTCGAGCGCCTCCACCAAGGGCC
CATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACA
GCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGT
GTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTG
TCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCT
CCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCC
AGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAA
CTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCA
GTCTTCCTCTTCCCCCCCCAAAACCCAAGGACACCCTCATGATCTCCCGGACC
CCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGT
CAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACA
AAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCC
TCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA**

GGTCTCAACCAACCCACCCCTCCAGCAAAACCCATCTCCAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCCG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACCTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 6808_B09 (PGT-156) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 545)

**CAGGTGCATCTGGTGGAGTCGGGGGGGAGGCGTTGTCCAACCTGGAA
AGTCCCTAAGACTCTCCTGTGAAACCTCTGGCTTCATCTTCAATCAAT
ATCCCATGTATTGGGTCCGCCAGGCTCCAGGCAAGGGACCGGAGTGG
GTGGCCGCCATCTCCGCTGATGCCTGGCATGTGGACTACCCAGGCTC
CGTGCGGGGGCCGATTACCGTCTCCAGAGACAATTCCAAGAGTTCTC
TATATTTAGACATGAAGAGTCTGAAAGTTGAAGACACGGCTATATATT
TCTGTGCGAAAGATGGGGAGGAACACAAGGTACCACAATTGCATTCC
TGGAGCGGACGAAACTTATATCACTACACTGGTTTTTGACGTCTGGGG
CCCAGGGACCACGGTCACCGTCTCGAGC** 6808_B09 (PGT-156) gamma heavy
chain amino acid sequence: expressed protein with leader sequence in italics
and variable region in bold. (SEQ ID NO: 546)

*MELGLSWVFLVVLLRGVHCQVHLVESGGGVVQPGKSLRLSCETSGFIENQY
PMYWVRQAPGKGPEWVA AISADAWHVDYPGSVRGRFTVSRDNSKSSLY
LDMKSLKVEDTAIYFCAKDGEHVKVPQLHSWSGRNLYHYTGFDVWGPG
TTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGAL
TSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVKDRVE
PKSCDKTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVSH
DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK* 6808_B09 (PGT-156) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 547) *QVHLVESGGGVVQPGKSLRLSCETS*

 YPMYWVRQAPGKGPEWVA   
YPGSVRGRFTVSRDNSKSSLYLDMKSLKVEDTAIYFCAK 
 WGPGTTVTVSS 6808_B09 (PGT-156) gamma heavy chain Kabat
CDRs: (SEQ ID NO: 548) CDR 1: QYPMY (SEQ ID NO: 549) CDR 2:
AISADAWHVDYPGSVRG (SEQ ID NO: 535) CDR 3:

DGEEHKVPQLHSWSGRNLYHYTGFDV 6808_B09 (PGT-156) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 550) CDR 1: GFIFNQ (SEQ ID NO: 521) CDR
2: AISADAWHVD (SEQ ID NO: 535) CDR 3:
DGEEHKVPQLHSWSGRNLYHYTGFDV 6808_B09 (PGT-156) light chain nucleotide
sequence: coding sequence (variable region in bold) (SEQ ID NO: 551)

*ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGATACCTGAACT
TGCTGCAGACATTGTGATGACCCAGTCTCCTGTCTCTCTGTCCGTCAC
CCTCGGACAGCCGGCCTCCATGTCCTGCAAGTCCAGTCAGACTGTCC
GACAGAGTGATGGCAAGACTTTCTTATATTGGTATCGACAGAAGGCA
GGCCAGTCTCCACAACCTGTTAATATATGAGGGTTCGAATCGATTCTCT
GGAGTGTGAGATAGGATCTCTGGCAGCGGGTCGGGGACAGATTTCAC
ACTGAGAATCAGTCGAGTGGAGGCTGAGGATGTTGGCGTTTATTTCT
GCCTGCAAACTAAAGACTTCCCCCTCACTTTTGGCGGAGGGACCAGG*

GTGGATATCAACATGACCTACGGTCTGTCCTTCATCTTCCCCTCC
ATCTGATGAGCAGTTGAAATCTGGAAGTGCCTCTGTTGTGTGCCTGCTGAA
TAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCCC
TCCAATCGGGTAAGTCCCAGGAGAGTGTACAGAGCAGGACAGCAAGGA
CAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGACTAC
GAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTC
GCCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 6808_B09 (PGT-156) light
chain variable region nucleotide sequence: (SEQ ID NO: 552)

**GACATTGTGATGACCCAGTCTCCTGTCTCTCTGTCCGTCACCCTCGGA
CAGCCGGCCTCCATGTCCTGCAAGTCCAGTCAGACTGTCCGACAGAG
TGATGGCAAGACTTTCTTATATTGGTATCGACAGAAGGCAGGCCAGT
CTCCACAAGTGTAAATATATGAGGGTTCGAATCGATTCTCTGGAGTGT
CAGATAGGATCTCTGGCAGCGGGTCGGGGACAGATTTACACTGAGA
ATCAGTCGAGTGGAGGCTGAGGATGTTGGCGTTTATTTCTGCCTGCA
AACTAAAGACTTCCCCCTCACTTTTGGCGGAGGGACCAGGGTGGATA TCAA**
6808 B09 (PGT-156) light chain amino acid sequence: expressed protein with
leader sequence in italics and variable region in bold. (SEQ ID NO: 553)

*MRLPAQLLGLLMLWIPELAADIVMTQSPVSLSVTLGQPASMSCKSSQTVRQS
DGKTFLYWYRQKAGQSPQLLIYEGSNRFSGVSDRISGSGSGTDFTLRISR
VEAEDVG VYFCLQTKDFPLTFGGGTRVDIKRTVAAPSVFIFPPSDEQLKSG
TASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSYSTLSSTLT
LSKADYEKHKVYACEVTHQGLSPVTKSFNRGEC* 6808_B09 (PGT-156) light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 554) *DIVMTQSPVSLSVTLGQPASMSC*

 *WYRQKAGQSPQLLIY*  *GVSDRISGSGSGTDFTLRISRVEAEDVG VYFC*  *FGG GTRVDIK*
6808_B09 (PGT-156) light chain Kabat CDRs: (SEQ ID NO: 555) CDR 1:
KSSQTVRQSDGKTFLY (SEQ ID NO: 556) CDR 2: EGSNRFS (SEQ ID NO:
543) CDR 3: LQTKDFPLT 6808_B09 (PGT-156) light chain Chothia CDRs: (SEQ
ID NO: 555) CDR 1: KSSQTVRQSDGKTFLY (SEQ ID NO: 556) CDR 2:
EGSNRFS (SEQ ID NO: 543) CDR 3: LQTKDFPLT 6892_C23 (PGT-157) gamma
heavy chain nucleotide sequence: coding sequence (leader sequence in italics,
variable region in bold) (SEQ ID NO: 557)

*ATGGAATTGGGGCTGAGCTGGGTTTTCTCGTCTCCTAAGAGGTGTCCA
CTGTGAAGTGCATCTGGTGGAGTCGGGGGGAGGCGTGGTCCAACCTG
GAAAGTCCCTCAGACTCTCCTGTGTAAGTCTGGCTTCATCTTCAAAC
AATATCCTATGTATTGGATCCGCCAGGCTCCAGGCAAGGGGCTGGAG
TGGGTGGCCGCCATCTCCGCTGATGCCTGGCATGTGGACTACGCAGG
CTCCGTGCGGGGGCCGATTTACCGTCTCCAGAGACAAGTCCAAGAATT
CTCTATATTTAGACATGAACAGTCTGACAGTTGAAGACACGGCTATAT
ATTTCTGTGCGAAAGATGGGGAAGAACAGAAAGTACCACAGTTGCAC
TCCTGGAGCGGACGAAATTTATATCACTACACTGGTGTGGACATCTG
GGGCCCAGGGACCACGGTCACCGTCTCGAGCGCCTCCACCAAGGGCC
CATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACA
GCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGT
GTCGTGGAAGTCAAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTG
TCCTACAGTCCTCAGGACTCTACTCCCTCAGCAGCGTGGTGACCGTGCCCT
CCAGCAGCTTGGGCACCCAGACCTACATCTGCAACGTGAATCACAAGCCC
AGCAACACCAAGGTGGACAAGAGAGTTGAGCCCAAATCTTGTGACAAAA
CTCACACATGCCCACCGTGCCCAGCACCTGAACTCCTGGGGGGACCGTCA*

GTCTTCTCTTCCCTCCCAACCCCTCCATGATCTCCCGGACC
CCTGAGGTCACATGCGTGGTGGTGGACGTGAGCCACGAAGACCCTGAGGT
CAAGTTCAACTGGTACGTGGACGGCGTGGAGGTGCATAATGCCAAGACA
AAGCCGCGGGAGGAGCAGTACAACAGCACGTACCGTGTGGTCAGCGTCC
TCACCGTCCTGCACCAGGACTGGCTGAATGGCAAGGAGTACAAGTGCAA
GGTCTCCAACAAAGCCCTCCCAGCCCCCATCGAGAAAACCATCTCCAAAG
CCAAAGGGCAGCCCCGAGAACCACAGGTGTACACCCTGCCCCCATCCCGG
GAGGAGATGACCAAGAACCAGGTCAGCCTGACCTGCCTGGTCAAAGGCT
TCTATCCCAGCGACATCGCCGTGGAGTGGGAGAGCAATGGGCAGCCGGA
GAACAACCTACAAGACCACGCCTCCCGTGCTGGACTCCGACGGCTCCTTCT
TCCTCTATAGCAAGCTCACCGTGGACAAGAGCAGGTGGCAGCAGGGGAA
CGTCTTCTCATGCTCCGTGATGCATGAGGCTCTGCACAACCACTACACGC
AGAAGAGCCTCTCCCTGTCTCCGGGTAAATGA 6892_C23 (PGT-157) gamma heavy
chain variable region nucleotide sequence: (SEQ ID NO: 558)

**GAAGTGCATCTGGTGGAGTCGGGGGGGAGGCGTGGTCCAACCTGGAA
AGTCCCTCAGACTCTCCTGTGTAACCTTCTGGCTTCATCTTCAAACAAT
ATCCTATGTATTGGATCCGCCAGGCTCCAGGCAAGGGGCTGGAGTGG
GTGGCCGCCATCTCCGCTGATGCCTGGCATGTGGACTACGCAGGCTC
CGTGCGGGGGCCGATTTACCGTCTCCAGAGACAACCTCCAAGAATTCTC
TATATTTAGACATGAACAGTCTGACAGTTGAAGACACGGCTATATATT
TCTGTGCGAAAGATGGGGAAGAACACGAAGTACCACAGTTGCACTCC
TGGAGCGGACGAAATTTATATCACTACACTGGTGTGGACATCTGGGG
CCCAGGGACCACGGTCACCGTCTCGAGC 6892_C23 (PGT-157) gamma heavy
chain amino acid sequence: expressed protein with leader sequence in italics
and variable region in bold. (SEQ ID NO: 559)**

*MELGLSWVFLVALLRGVHCEVHLVESGGGVVQPGKSLRLSCVTSGFIFKQY
PMYWIRQAPGKGLEWVA AISADAWHVDYAGSVRGRFTVSRDNSKNSLY
LDMNSLTVEDTAIYFCAKDGE EHEVPQLHSWSGRNLYHYTGVDIWGP
TTVTVSSASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGAL
TSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVKDRVE
PKSCDKHTHTCPPCPAPELLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSHE
DPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEY
KCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGF
YPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK 6892_C23 (PGT-157) gamma heavy chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in bold italics) (SEQ ID NO: 560) EVHLVESGGGVVQPGKSLRLSCVTS*

 YPMYWIRQAPGKGLEWVA  YAGSVRGRFTVSRDNSKNSLYLDMNSLTVEDTAIYFCAK 

 WPGTTTVTVSS 6892_C23 (PGT-157) gamma heavy chain Kabat
CDRs: (SEQ ID NO: 548) CDR 1: QYPMY (SEQ ID NO: 534) CDR 2:
AISADAWHVDYAGSVRG (SEQ ID NO: 561) CDR 3:

DGEEHEVPQLHSWSGRNLYHYTGVDI 6892_C23 (PGT-157) gamma heavy chain
Chothia CDRs: (SEQ ID NO: 562) CDR 1: GFIFKQ (SEQ ID NO: 521) CDR
2: AISADAWHVD (SEQ ID NO: 561) CDR 3:

DGEEHEVPQLHSWSGRNLYHYTGVDI 6892_C23 (PGT-157) light chain nucleotide
sequence: coding sequence (variable region in bold) (SEQ ID NO: 563)

**ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGATACCTGA
ACTTACTGCAGACATTGTGATGACCCAGACTCCTGTCTCTCTGTCCGTCAC
CCTCGGACAGCCGGCCTCCATGTCCTGTAAGTCCAGTCAGAGCCTCC**

**GACAAGTGTAGTGCACCACTTTCTTGGTATTCGATCGACAGACAGGCA
GGCCAGTCTCCACAACCTCCTAATATCTGAGGCTTCGAATCGATTCTCT
GGAGTGTGATAGGTTCTCTGGCAGCGGTTTCAGGGACAGACTTCAC
ACTGAAAATCAGTCGGGTGGAGGCTGAGGATGTTGGCATTATTTCT
GCATGCAAACCTAAAGACTTCCCCCTCACTTTTGGCGGAGGGACCAAG
GTGGATCTCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCC
ATCTGATGAGCAGTTGAAATCTGGAACCTGCCTCTGTTGTGTGCCTGCTGAA
TAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCCC
TCCAATCGGGTAACTCCCAGGAGAGTGTACAGAGCAGGACAGCAAGGA
CAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGACTAC
GAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTC
GCCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 6892_C23 (PGT-157) light
chain variable region nucleotide sequence: (SEQ ID NO: 564)**

**GACATTGTGATGACCCAGACTCCTGTCTCTCTGTCCGTCACCCTCGG
ACAGCCGGCCTCCATGTCCTGTAAGTCCAGTCAGAGCCTCCGACAAA
GTGATGGCAAGACTTTCTTGTATTGGTATCGACAGAAGGCAGGCCAG
TCTCCACAACCTCCTAATATCTGAGGCTTCGAATCGATTCTCTGGAGTG
TCAGATAGGTTCTCTGGCAGCGGTTTCAGGGACAGACTTCACACTGAA
AATCAGTCGGGTGGAGGCTGAGGATGTTGGCATTATTTCTGCATGC
AACTAAAGACTTCCCCCTCACTTTTGGCGGAGGGACCAAGGTGGAT CTCAA**
6892_C23 (PGT-157) light chain amino acid sequence: expressed protein with
leader sequence in *italics* and variable region in **bold**. (SEQ ID NO: 565)

*MRLPAQLLGLLMLWIPELTADIVMTQTPVSLSVTLGQPASMSCKSSQSLRQS
DGKTFLYWYRQKAGQSPQLLISEASNRFSGVSDRFSGSGSGTDFTLKISR
VEAEDVGIYFCMQTKDFPLTFGGGTKVDLKR**TVAAPSVFIFPPSDEQLKSG
TASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYSLSSTLT
LSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC* 6892_C23 (PGT-157) light chain
variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs
in **bold italics**) (SEQ ID NO: 566) *DIVMTQTPVSLSVTLGQPASMSC*

 *WYRQKAGQSPQLL*  *GVSDRFSGSGSGTDFTLKISRVEAEDVGIYFC*  *FGG* *GTKVDLK*
6892_C23 (PGT-157) light chain Kabat CDRs: (SEQ ID NO: 567) CDR 1:
KSSQSLRQSDGKTFLY (SEQ ID NO: 568) CDR 2: EASNRFS (SEQ ID NO: 569)
CDR 3: MQTKDFPLT 6892_C23 (PGT-157) light chain Chothia CDRs: (SEQ ID
NO: 567) CDR 1: KSSQSLRQSDGKTFLY (SEQ ID NO: 568) CDR 2: EASNRFS
(SEQ ID NO: 569) CDR 3: MQTKDFPLT 6881_N05 (PGT-158) gamma heavy
chain nucleotide sequence: coding sequence (variable region in **bold**) (SEQ ID
NO: 570) *ATGGAATTGGGGCTGAGCTGGGTTTTCTCGTCGCTCTCCTAAGAGGTGTCCA*

*CTGTGAGGTGCGTCTGATGGAGTCGGGGGGGAGGCGTGGTCCAGCCTG
GGAAGTCCCTCAGACTCTCCTGTGTAACCTCTGGCTTCATCTTCAAAA
AATATCCTATGTACTGGATCCGCCAGGCTCCAGGCAAGGGGCTGGAG
TGGGTGGCCGCCATCTCCGCTGATGCCTGGCATGTGGACTACCCAGG
CTCCGTGCGGGGCGGATTTACCGTCTCAAGAGACAACCTCCAAGAATT
CTCTATATTTAGACATGAATAGTCTGACAGTAGAAGACACGGCTATAT
ATTTTTGTGCGAAAGATGGGGAGGAACACGAAGTCCCACAACCTGCAC
TCCTGGAGCGGACGAAATTTATATCACTACACTGGTGTAGACGTCTG
GGGCCCAGGGACCAACGGTCACCGTCTCGAGCGCCTCCACCAAGGGCC
CATCGGTCTTCCCCCTGGCACCCCTCCTCCAAGAGCACCTCTGGGGGCACA
GCGGCCCTGGGCTGCCTGGTCAAGGACTACTTCCCCGAACCGGTGACGGT
GTCGTGGAACCTCAGGCGCCCTGACCAGCGGCGTGCACACCTTCCCGGCTG*

sequence: coding sequence (variable region in bold) (SEQ ID NO: 576)
ATGAGGCTCCCTGCTCAGCTCCTGGGGCTGCTAATGCTCTGGATACCTGAAGT
GACTGCAGACATTGTGATGACCCAGACTCCTGTCTCTGTGTCCGTCAC
CCTCGGACAGCCGGCCTCCATGTCCTGCAAGTCCAGTCAGAGCGTCC
GACAAAGTGATGGCAAGACTTTTTTATATTGGTATCGACAGAAGGCA
GGCCAGTCTCCACAACCTCTTAATATATGAGGCTTCGAAGCGATTCTCT
GGAGTGTGAGATAGGTTCTCTGGCAGCGGGTCAGGGACAGACTTCAC
ACTGAAAATCAGTCGGGTGGGGGCTGAGGATGTTGGCGTTTATTTCT
GCATGCAAACATAAAGACTTCCCCCTTACTTTTGGCGGAGGGACCAAG
GTGGATCTCAAACGTACGGTGGCTGCACCATCTGTCTTCATCTTCCCGCC
ATCTGATGAGCAGTTGAAATCTGGAACCTGCCTCTGTTGTGTGCCTGCTGAA
TAACTTCTATCCCAGAGAGGCCAAAGTACAGTGGAAGGTGGATAACGCCC
TCCAATCGGGTAACTCCCAGGAGAGTGTACACAGAGCAGGACAGCAAGGA
CAGCACCTACAGCCTCAGCAGCACCTGACGCTGAGCAAAGCAGACTAC
GAGAAACACAAAGTCTACGCCTGCGAAGTCACCCATCAGGGCCTGAGCTC
GCCCGTCACAAAGAGCTTCAACAGGGGAGAGTGTTAG 6881_N05 (PGT-158) light chain variable region nucleotide sequence: (SEQ ID NO: 577)

GACATTGTGATGACCCAGACTCCTGTCTCTGTGTCCGTCACCCTCGG
ACAGCCGGCCTCCATGTCCTGCAAGTCCAGTCAGAGCGTCCGACAAA
GTGATGGCAAGACTTTTTTATATTGGTATCGACAGAAGGCAGGCCAG
TCTCCACAACCTCTTAATATATGAGGCTTCGAAGCGATTCTCTGGAGTG
TCAGATAGGTTCTCTGGCAGCGGGTCAGGGACAGACTTCACACTGAA
AATCAGTCGGGTGGGGGCTGAGGATGTTGGCGTTTATTTCTGCATGC
AACTAAAGACTTCCCCCTTACTTTTGGCGGAGGGACCAAGGTGGAT CTCAA

6881_N05 (PGT-158) light chain amino acid sequence: expressed protein with leader sequence in italics and variable region in bold. (SEQ ID NO: 578)

MRLPAQLLGLLMLWIPEVTADIVMTQTPVSVSVTLGQPASMSCKSSQSVRQS
DGKTFLYWYRQKAGQSPQLLIYEASKRFSGVSDRFSGSGSGTDFTLKISR
VGAEDVG VYFCMQTKDFPLTFGGGTKVDLKR TVAAPS VFIFPPSDEQLKS
GTASVVCLLN NFYPREAKVQWKVDNALQSGNSQESVTEQDSKDYSLSS
LTL SKADY EKHKVYACEVTHQGLSSPVTKSFNRGEC 6881_N05 (PGT-158) light chain variable region amino acid sequence: (Kabat CDRs underlined, Chothia CDRs in bold italics) (SEQ ID NO: 579) DIVMTQTPVSVSVTLGQPASMSC

 WYRQKAGQSPQLL IY 

GVSDRFSGSGSGTDFTLKISR  VGAEDVG VYFC  FGG GTKVDLK

6881_N05 (PGT-158) light chain Kabat CDRs: (SEQ ID NO: 541) CDR 1:

KSSQSVRQSDGKTFLY (SEQ ID NO: 580) CDR 2: EASKRFS (SEQ ID NO:

569) CDR 3: MQTKDFPLT 6881_N05 (PGT-158) light chain Chothia CDRs:

(SEQ ID NO: 541) CDR 1: KSSQSVRQSDGKTFLY (SEQ ID NO: 580) CDR

2: EASKRFS (SEQ ID NO: 569) CDR 3: MQTKDFPLT

(245) The 6831 A21 (PGT-151) antibody includes a heavy chain variable region (SEQ ID NO: 474), encoded by the nucleic acid sequence shown in SEQ ID NO: 472, and a light chain variable region (SEQ ID NO: 483) encoded by the nucleic acid sequence shown in SEQ ID NO: 481.

(246) The heavy chain CDRs of the 6831 A21 (PGT-151) antibody have the following sequences per Kabat definition: KYPMY (SEQ ID NO: 475), AISGDAWHVVYSNSVQG (SEQ ID NO: 476), and MFQESGPPRLDRWSGRNYYYYSGMDV (SEQ ID NO: 477). The light chain CDRs of the 6831 A21 (PGT-151) antibody have the following sequences per Kabat definition: KSSESLRQSNGKTSLY (SEQ ID NO: 484), EVSNRFS (SEQ ID NO: 485), and MQSKDFPLT (SEQ ID NO: 486).

(247) The heavy chain CDRs of the 6831_A21 (PGT-151) antibody have the following sequences

per Chothia definition: DFPFSK (SEQ ID NO: 478), AISGDAWHVV (SEQ ID NO: 479), and MFQESGPPRLDRWSGRNYYYYSGMDV (SEQ ID NO: 477). The light chain CDRs of the 6831_A21 (PGT-151) antibody have the following sequences per Chothia definition: KSSESLRQSNGKTSLY (SEQ ID NO: 484), EVSNRFS (SEQ ID NO: 485), and MQSKDFPLT (SEQ ID NO: 486).

(248) The 6889_I17 (PGT-152) antibody includes a heavy chain variable region (SEQ ID NO: 490), encoded by the nucleic acid sequence shown in SEQ ID NO: 488, and a light chain variable region (SEQ ID NO: 497) encoded by the nucleic acid sequence shown in SEQ ID NO: 495.

(249) The heavy chain CDRs of the 6889_I17 (PGT-152) antibody have the following sequences per Kabat definition: KYPMY (SEQ ID NO: 475), AISADAWHVVYSGSVQG (SEQ ID NO: 491), and MFQESGPPRFDSWSGRNYYYYSGMDV (SEQ ID NO: 492). The light chain CDRs of the 6889_I17 (PGT-152) antibody have the following sequences per Kabat definition: KSSQSLRQSNGKTSLY (SEQ ID NO: 498), EVSNRFS (SEQ ID NO: 485), and MQSKDFPLT (SEQ ID NO: 486).

(250) The heavy chain CDRs of the 6889_I17 (PGT-152) antibody have the following sequences per Chothia definition: DFPFSK (SEQ ID NO: 478), AISADAWHVV (SEQ ID NO: 493), and MFQESGPPRFDSWSGRNYYYYSGMDV (SEQ ID NO: 492). The light chain CDRs of the 6889_I17 (PGT-152) antibody have the following sequences per Chothia definition: KSSQSLRQSNGKTSLY (SEQ ID NO: 498), EVSNRFS (SEQ ID NO: 485), and MQSKDFPLT (SEQ ID NO: 486).

(251) The 6891_F06 (PGT-153) antibody includes a heavy chain variable region (SEQ ID NO: 502), encoded by the nucleic acid sequence shown in SEQ ID NO: 500, and a light chain variable region (SEQ ID NO: 511) encoded by the nucleic acid sequence shown in SEQ ID NO: 509.

(252) The heavy chain CDRs of the 6891_F06 (PGT-153) antibody have the following sequences per Kabat definition: KRHHM (SEQ ID NO: 503), VISSDAIHVDYASSVRG (SEQ ID NO: 504), and DRDGYGPPQIQTWSGRYLHLYSGIDA (SEQ ID NO: 505). The light chain CDRs of the 6891_F06 (PGT-153) antibody have the following sequences per Kabat definition: KSSQSLRQSNGKTYLY (SEQ ID NO: 512), EVSIRFS (SEQ ID NO: 513), and MQSKDFPLT (SEQ ID NO: 486).

(253) The heavy chain CDRs of the 6891_F06 (PGT-153) antibody have the following sequences per Chothia definition: NFLFNK (SEQ ID NO: 506), VISSDAIHVD (SEQ ID NO: 507), and DRDGYGPPQIQTWSGRYLHLYSGIDA (SEQ ID NO: 505). The light chain CDRs of the 6891_F06 (PGT-153) antibody have the following sequences per Chothia definition: KSSQSLRQSNGKTYLY (SEQ ID NO: 512), EVSIRFS (SEQ ID NO: 513), and MQSKDFPLT (SEQ ID NO: 486).

(254) The 6843_G20 (PGT-154) antibody includes a heavy chain variable region (SEQ ID NO: 517), encoded by the nucleic acid sequence shown in SEQ ID NO: 515, and a light chain variable region (SEQ ID NO: 525) encoded by the nucleic acid sequence shown in SEQ ID NO: 523.

(255) The heavy chain CDRs of the 6843_G20 (PGT-154) antibody have the following sequences per Kabat definition: KYPMY (SEQ ID NO: 475), AISADAWHVDYAASVKD (SEQ ID NO: 518), and NIEEFVPPQFDSWSGRSYYHYFGMDV (SEQ ID NO: 519). The light chain CDRs of the 6843_G20 (PGT-154) antibody have the following sequences per Kabat definition: SSSESLGRGDGRTYLH (SEQ ID NO: 526), EVSTRFS (SEQ ID NO: 527), and MQSRDFPIT (SEQ ID NO: 528).

(256) The heavy chain CDRs of the 6843_G20 (PGT-154) antibody have the following sequences per Chothia definition: NFIFNK (SEQ ID NO: 520), AISADAWHVD (SEQ ID NO: 521), and NIEEFVPPQFDSWSGRSYYHYFGMDV (SEQ ID NO: 519). The light chain CDRs of the 6843_G20 (PGT-154) antibody have the following sequences per Chothia definition: SSSESLGRGDGRTYLH (SEQ ID NO: 526), EVSTRFS (SEQ ID NO: 527), and MQSRDFPIT (SEQ ID NO: 528).

(257) The 6892_D19 (PGT-155) antibody includes a heavy chain variable region (SEQ ID NO: 532), encoded by the nucleic acid sequence shown in SEQ ID NO: 530, and a light chain variable region (SEQ ID NO: 540) encoded by the nucleic acid sequence shown in SEQ ID NO: 538.

(258) The heavy chain CDRs of the 6892_D19 (PGT-155) antibody have the following sequences per Kabat definition: EYPMY (SEQ ID NO: 533), AISADAWHVDYAGSVRG (SEQ ID NO: 534), and DGEEHKVPQLHSWSGRNLYHYTGFDV (SEQ ID NO: 535). The light chain CDRs of the 6892_D19 (PGT-155) antibody have the following sequences per Kabat definition: KSSQSVRQSDGKTFLY (SEQ ID NO: 541), EGSSRFS (SEQ ID NO: 542), and LQTKDFPLT (SEQ ID NO: 543).

(259) The heavy chain CDRs of the 6892_D19 (PGT-155) antibody have the following sequences per Chothia definition: GFIFNE (SEQ ID NO: 536), AISADAWHVD (SEQ ID NO: 521), and DGEEHKVPQLHSWSGRNLYHYTGFDV (SEQ ID NO: 535). The light chain CDRs of the 6892_D19 (PGT-155) antibody have the following sequences per Chothia definition: KSSQSVRQSDGKTFLY (SEQ ID NO: 541), EGSSRFS (SEQ ID NO: 542), and LQTKDFPLT (SEQ ID NO: 543).

(260) The 6808_B09 (PGT-156) antibody includes a heavy chain variable region (SEQ ID NO: 547), encoded by the nucleic acid sequence shown in SEQ ID NO: 545, and a light chain variable region (SEQ ID NO: 554) encoded by the nucleic acid sequence shown in SEQ ID NO: 552.

(261) The heavy chain CDRs of the 6808_B09 (PGT-156) antibody have the following sequences per Kabat definition: QYPMY (SEQ ID NO: 548), AISADAWHVDYPGSVRG (SEQ ID NO: 549), and DGEEHKVPQLHSWSGRNLYHYTGFDV (SEQ ID NO: 535). The light chain CDRs of the 6808_B09 (PGT-156) antibody have the following sequences per Kabat definition: KSSQTVRQSDGKTFLY (SEQ ID NO: 555), EGSNRFS (SEQ ID NO: 556), and LQTKDFPLT (SEQ ID NO: 543).

(262) The heavy chain CDRs of the 6808_B09 (PGT-156) antibody have the following sequences per Chothia definition: GFIFNQ (SEQ ID NO: 550), AISADAWHVD (SEQ ID NO: 521), and DGEEHKVPQLHSWSGRNLYHYTGFDV (SEQ ID NO: 535). The light chain CDRs of the 6808_B09 (PGT-156) antibody have the following sequences per Chothia definition: KSSQTVRQSDGKTFLY (SEQ ID NO: 555), EGSNRFS (SEQ ID NO: 556), and LQTKDFPLT (SEQ ID NO: 543).

(263) The 6892_C23 (PGT-157) antibody includes a heavy chain variable region (SEQ ID NO: 560), encoded by the nucleic acid sequence shown in SEQ ID NO: 558, and a light chain variable region (SEQ ID NO: 566) encoded by the nucleic acid sequence shown in SEQ ID NO: 564.

(264) The heavy chain CDRs of the 6892_C23 (PGT-157) antibody have the following sequences per Kabat definition: QYPMY (SEQ ID NO: 548), AISADAWHVDYAGSVRG (SEQ ID NO: 534), and DGEEHEVPQLHSWSGRNLYHYTGVDI (SEQ ID NO: 561). The light chain CDRs of the 6892_C23 (PGT-157) antibody have the following sequences per Kabat definition: KSSQSLRQSDGKTFLY (SEQ ID NO: 567), EASNRFS (SEQ ID NO: 568), and MQTKDFPLT (SEQ ID NO: 569).

(265) The heavy chain CDRs of the 6892_C23 (PGT-157) antibody have the following sequences per Chothia definition: GFIFKQ (SEQ ID NO: 562), AISADAWHVD (SEQ ID NO: 521), and DGEEHEVPQLHSWSGRNLYHYTGVDI (SEQ ID NO: 561). The light chain CDRs of the 6892_C23 (PGT-157) antibody have the following sequences per Chothia definition: KSSQSLRQSDGKTFLY (SEQ ID NO: 567), EASNRFS (SEQ ID NO: 568), and MQTKDFPLT (SEQ ID NO: 569).

(266) The 6881_N05 (PGT-158) antibody includes a heavy chain variable region (SEQ ID NO: 573), encoded by the nucleic acid sequence shown in SEQ ID NO: 571, and a light chain variable region (SEQ ID NO: 579) encoded by the nucleic acid sequence shown in SEQ ID NO: 577.

(267) The heavy chain CDRs of the 6881_N05 (PGT-158) antibody have the following sequences per Kabat definition: KYPMY (SEQ ID NO: 475), AISADAWHVDYPGSVRG (SEQ ID NO:

549), and DGEHEVPQLHSWSGRNLYHYTGVDV (SEQ ID NO: 574). The light chain CDRs of the 6881_N05 (PGT-158) antibody have the following sequences per Kabat definition: KSSQSVRQSDGKTFLY (SEQ ID NO: 541), EASKRFS (SEQ ID NO: 580), and MQTKDFPLT (SEQ ID NO: 569).

(268) The heavy chain CDRs of the 6881_N05 (PGT-158) antibody have the following sequences per Chothia definition: GFIFKK (SEQ ID NO: 575), AISADAWHVD (SEQ ID NO: 521), and DGEHEVPQLHSWSGRNLYHYTGVDV (SEQ ID NO: 574). The light chain CDRs of the 6881_N05 (PGT-158) antibody have the following sequences per Chothia definition: KSSQSVRQSDGKTFLY (SEQ ID NO: 541), EASKRFS (SEQ ID NO: 580), and MQTKDFPLT (SEQ ID NO: 569).

(269) In one aspect, an antibody according to the invention contains a heavy chain having the amino acid sequence of SEQ ID NOs: 12, 16, 20, 24, 28, 139, 47, 53, 59, 65, 62, 153, 165, 181, 197, 213, 229, 246, 275, 291, 297, 306, 318, 333, 346, 362, 400, 404, 419, 434, 443, 453, 462, 473, 489, 501, 516, 531, 546, 559, or 572, and a light chain having the amino acid sequence of SEQ ID NOs: 14, 18, 22, 26, 30, 142, 50, 56, 148, 158, 174, 190, 206, 222, 238, 255, 284, 301, 312, 329, 392, 355, 396, 385, 413, 428, 439, 448, 583, 469, 482, 496, 510, 524, 539, 553, 565, or 578.

Alternatively, an antibody according to the invention contains a heavy chain variable region having the amino acid sequence of SEQ ID NOs: 31, 33, 35, 37, 39, 140, 48, 54, 60, 79, 156, 168, 184, 200, 216, 232, 149, 276, 292, 298, 307, 319, 334, 347, 363, 401, 405, 420, 435, 444, 454, 463, 474, 490, 502, 517, 532, 547, 560, or 573, and a light chain variable region having the amino acid sequence of SEQ ID NOs: 32, 34, 36, 38, 40, 96, 51, 57, 149, 161, 177, 193, 209, 225, 242, 258, 285, 302, 313, 330, 393, 356, 397, 386, 414, 429, 440, 449, 584, 470, 483, 497, 511, 525, 540, 554, 566, or 579.

(270) In another aspect, an antibody according to the invention contains a heavy chain having the amino acid sequence encoded by the nucleic acid sequence of SEQ ID NOs: 11, 15, 19, 23, 27, 138, 46, 52, 58, 64, 66, 166, 167, 183, 199, 215, 231, 248, 273, 289, 295, 304, 314, 316, 331, 344, 360, 398, 402, 417, 432, 441, 451, 460, 471, 487, 499, 514, 529, 544, 557, or 570, and a light chain having the amino acid sequence encoded by the nucleic acid sequence of SEQ ID NOs: 13, 17, 21, 25, 29, 141, 49, 55, 61, 67, 146, 160, 176, 192, 208, 224, 240, 257, 282, 299, 310, 327, 390, 353, 394, 383, 411, 426, 437, 446, 581, 467, 480, 494, 508, 522, 537, 551, 563, or 576. Alternatively, an antibody according to the invention contains a heavy chain variable region having the amino acid sequence encoded by the nucleic acid sequence of SEQ ID NOs: 99, 101, 109, 115, 122, 128, 130, 132, 134, 136, 63, 154, 166, 182, 198, 214, 230, 247, 274, 290, 296, 305, 315, 317, 332, 345, 361, 399, 403, 418, 433, 442, 452, 461, 472, 488, 500, 515, 530, 545, 558, or 571, and a light chain variable region having the amino acid sequence encoded by the nucleic acid sequence of SEQ ID NOs: 100, 106, 112, 119, 125, 129, 131, 133, 135, 137, 147, 159, 175, 191, 207, 223, 239, 256, 283, 300, 311, 328, 391, 354, 395, 384, 412, 427, 438, 447, 582, 468, 481, 495, 509, 523, 538, 552, 564, or 577. Furthermore, an antibody according to the invention contains a heavy chain having the amino acid sequence encoded by a nucleic acid sequence of SEQ ID NOs: 11, 15, 19, 23, 27, 138, 46, 52, 58, 64, 66, 166, 167, 183, 199, 215, 231, 248, 273, 289, 295, 304, 314, 316, 331, 344, 360, 398, 402, 417, 432, 441, 451, 460, 471, 487, 499, 514, 529, 544, 557, or 570, which contains a silent or degenerate mutation, and a light chain having the amino acid sequence encoded by the nucleic acid sequence of SEQ ID NOs: 13, 17, 21, 25, 29, 141, 49, 55, 61, 67, 146, 160, 176, 192, 208, 224, 240, 257, 282, 299, 310, 327, 390, 353, 394, 383, 411, 426, 437, 446, 581, 467, 480, 494, 508, 522, 537, 551, 563, or 576, which contains a silent or degenerate mutation. Silent and degenerate mutations alter the nucleic acid sequence, but do not alter the resultant amino acid sequence.

(271) Preferably the three heavy chain CDRs include an amino acid sequence of at least 90%, 92%, 95%, 97%, 98%, 99%, or more identical to the amino acid sequence of KYGMH (SEQ ID NO: 88), LISDDGMRKYHSDSMWG (SEQ ID NO: 89),

EAGGPIWDDVQYDFNDGYNYHYMDV (SEQ ID NO: 6), SYAFT (SEQ ID NO: 104),
MVTPIFGEAKYSQRFEG (SEQ ID NO: 105), DRRVPIATDNWLDP (SEQ ID NO: 9), SYAFS
(SEQ ID NO: 110), MITPVFGETKYAPRFQG (SEQ ID NO: 111), DRRVVPMATDNWLDP
(SEQ ID NO: 8), DYYLH (SEQ ID NO: 116), LIDPENGEARYAEKFQG (SEQ ID NO: 117),
GAVGADSGSWFDP (SEQ ID NO: 10), RQGMH (SEQ ID NO: 123),
FIKYDGSEKYHADSVWG (SEQ ID NO: 124), EAGGPDYRNGYNYDFYDGYNYHYMDV
(SEQ ID NO: 7), LISDDGMRKYHSNSMWG (SEQ ID NO: 98), DSYWS (SEQ ID NO: 90),
YVHKSGDTNYSPLKS (SEQ ID NO: 265), TLHGRRYIGIVAFNEWFTYFYMDV (SEQ ID
NO: 143), DNYWS (SEQ ID NO: 261), YVHDSGDTNYPNPSLKS (SEQ ID NO: 157),
TKHGRRYIGVAFKEWFTYFYMDV (SEQ ID NO: 262), DAYWS (SEQ ID NO: 169),
YVHHSGDTNYPNPSLKR (SEQ ID NO: 170), ALHGKRIYGIVALGELFTYFYMDV (SEQ ID
NO: 171), ACTYFWG (SEQ ID NO: 185), SLSHCQSFWSGWTGHNPSLKS (SEQ ID NO:
186), FDGEVLVYNHWPKPAPVVDL (SEQ ID NO: 187), ACDYFWG (SEQ ID NO: 201),
GLSHCAGYYNTGWTYHNPSLKS (SEQ ID NO: 202), FDGEVLVYHDWPKPAPVVDL (SEQ
ID NO: 203), TGHYYWG (SEQ ID NO: 217), HIHYTTAVLHNPALKS (SEQ ID NO: 218),
SGGDILYYEWQKPHWFSP (SEQ ID NO: 219), GGEWGDKDYHWG (SEQ ID NO: 233),
SIHWRGTTTHYKESLRR (SEQ ID NO: 234), HRHHDVFMVPIAGWFDV (SEQ ID NO: 235),
GTDWGENDFHYG (SEQ ID NO: 250), SIHWRGRTTHYKTSFRS (SEQ ID NO: 251),
HKYHDIFRVVPVAGWFDP (SEQ ID NO: 252), KYDVH (SEQ ID NO: 277),
WMSHEGDKTESAQRFGK (SEQ ID NO: 278),
GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279),
WISHERDKTESAQRFGK (SEQ ID NO: 293),
GSKHRLRDYVLYDDYGLINQQEWNDYLEFLDV (SEQ ID NO: 308), RCNYFWG (SEQ ID
NO: 320), SLSHCRSYYNTDWTYHNPSLKS (SEQ ID NO: 321),
FGGEVLVYRDWPKPAPVVDL (SEQ ID NO: 322), ACNSFWG (SEQ ID NO: 326),
SLSHCASYWNRGWTYHNPSLKS (SEQ ID NO: 335), FGGEVLRYTDWPKPAPVVDL (SEQ ID
NO: 336), TGHYYWG (SEQ ID NO: 348), HIHYNTAVLHNPALKS (SEQ ID NO: 349),
SGGDILYYIEWQKPHWFYP (SEQ ID NO: 350), GGEWGDSDYHWG (SEQ ID NO: 364),
SIHWRGTTTHYNAPFRG (SEQ ID NO: 365), HKYHDIVMVVPIAGWFDP (SEQ ID NO: 366),
NHDVH (SEQ ID NO: 378), WMSHEGDKTGLAQKFQG (SEQ ID NO: 379),
GSKHRLRDYFLYNEYGPNYEEWGDYLATLDV (SEQ ID NO: 380), NYYWT (SEQ ID NO:
406), YISDRETTTYPNPSLNS (SEQ ID NO: 407), ARRGQRIYGVVSFGEFFYYYYMDV (SEQ
ID NO: 408), GRFWS (SEQ ID NO: 421), YFSDTDRSEYNPSLRS (SEQ ID NO: 422),
AQQGKRIYGIVSFGEFFYYYYMDA (SEQ ID NO: 423),
AQQGKRIYGIVSFGEFFYYYYMDA (SEQ ID NO: 436), SGGDILYYNEWQKPHWFYP (SEQ
ID NO: 445), SLSHCAGYYNSGWTYHNPSLKS (SEQ ID NO: 455),
FGGDVLVYHDWPKPAPVVDL (SEQ ID NO: 456), GCDYFWG (SEQ ID NO: 464),
FDGEVLVYNDWPKPAPVVDL (SEQ ID NO: 465), KYPMY (SEQ ID NO: 475),
AISGDAWHVVYSNSVQG (SEQ ID NO: 476), MFQESGPPRLDRWSGRNYYYYSGMMDV
(SEQ ID NO: 477), AISADAWHVVYSGSVQG (SEQ ID NO: 491),
MFQESGPPRFDSWSGRNYYYYSGMMDV (SEQ ID NO: 492), KRHMH (SEQ ID NO: 503),
VISSDAIHVDYASSVRG (SEQ ID NO: 504), DRDGYGPPQIQTWSGRYLHLYSGIDA (SEQ ID
NO: 505), AISADAWHVDYAASVKD (SEQ ID NO: 518),
NIEEFSPQFDSWSGRSYYHYFGMDV (SEQ ID NO: 519), EYPMY (SEQ ID NO: 533),
AISADAWHVDYAGSVRG (SEQ ID NO: 534), DGEEHKVPQLHSWSGRNLHYHTGFDV
(SEQ ID NO: 535), QYPMY (SEQ ID NO: 548), AISADAWHVDYPGSVRG (SEQ ID NO: 549),
DGEEHEVPQLHSWSGRNLHYHTGVDI (SEQ ID NO: 561),
DGEEHEVPQLHSWSGRNLHYHTGVDDV (SEQ ID NO: 574), (as determined by the Kabat
method) or GFTFHK (SEQ ID NO: 266), LISDDGMRKY (SEQ ID NO: 267),
EAGGPIWDDVKYDFNDGYNYHYMDV (SEQ ID NO: 6), GGTFSS (SEQ ID NO: 268),

MVTPIFGEAK (SEQ ID NO: 269), DRRVPIATDNWLDL (SEQ ID NO: 9), GGAFSS (SEQ ID NO: 270), MITPVFGETK (SEQ ID NO: 271), DRRVVPMATDNWLDL (SEQ ID NO: 8), GYSFID (SEQ ID NO: 102), LIDPENGAR (SEQ ID NO: 103), GAVGADSGSWFDP (SEQ ID NO: 10), GFDFSR (SEQ ID NO: 118), FIKYDGSEKY (SEQ ID NO: 272), EAGGPDYRNGYNYDFYDGYNYHYMDV (SEQ ID NO: 7), GASISD (SEQ ID NO: 144), YVHKSGDTN (SEQ ID NO: 145), TLHGRRYIGIVAFNEWFTYFYMDV (SEQ ID NO: 143), GTLVDR (SEQ ID NO: 263), YVHDSGDTN (SEQ ID NO: 264), TKHGRRYIGVAFKEWFTYFYMDV (SEQ ID NO: 262), GASIND (SEQ ID NO: 172), YVHHSMDTN (SEQ ID NO: 173), ALHGKRYIGIVALGELFTYFYMDV (SEQ ID NO: 171), GESTGACT (SEQ ID NO: 188), SLSHCQSFWSGSGWTF (SEQ ID NO: 189), FDGEVLVYNHWPKPAWVDL (SEQ ID NO: 187), GDSTAACD (SEQ ID NO: 204), GLSHCAGYYNTGWTY (SEQ ID NO: 205), FDGEVLVYHDWPKPAWVDL (SEQ ID NO: 203), GESINTGH (SEQ ID NO: 220), HIHYTTAVL (SEQ ID NO: 221), SGGDILYYYEWQKPHWFSP (SEQ ID NO: 219), GDSIRGGEWGDKD (SEQ ID NO: 236), SIHWRTTH (SEQ ID NO: 237), HRHHDVFMLVPIAGWFDV (SEQ ID NO: 235), GGSMRGTDWGEND (SEQ ID NO: 253), SIHWRTTH (SEQ ID NO: 254), HKYHDIFRVVPVAGWFDL (SEQ ID NO: 252), GNTFSK (SEQ ID NO: 280), WMSHEGDKTE (SEQ ID NO: 281), GSKHRLRDYVLYDDYGLINYQEWNDYLEFLDV (SEQ ID NO: 279), WISHERDKTE (SEQ ID NO: 294), GSKHRLRDYVLYDDYGLINQQEWNDYLEFLDV (SEQ ID NO: 308), or GNTFRK (SEQ ID NO: 309), GDSTGRCN (SEQ ID NO: 323), SLSHCRSYNTDWTY (SEQ ID NO: 324), FGGEVLVYRDWPKPAWVDL (SEQ ID NO: 322), GDSTAACN (SEQ ID NO: 337), SLSHCASYWNRGWTY (SEQ ID NO: 338), FGGEVLRYTDWPKPAWVDL (SEQ ID NO: 336), GDSINTGH (SEQ ID NO: 351), HIHYNTAVL (SEQ ID NO: 352), SGGDILYIEWQKPHWFYP (SEQ ID NO: 350), GGSIRGGEWGDSD (SEQ ID NO: 367), SIHWRTTH (SEQ ID NO: 237), HKYHDIVMVPIAGWFDL (SEQ ID NO: 366), GNSFSN (SEQ ID NO: 381), WMSHEGDKTG (SEQ ID NO: 382), GSKHRLRDYFLYNEYGPNYEEWGDYLATLDV (SEQ ID NO: 380), GGSISN (SEQ ID NO: 409), YISDRETTT (SEQ ID NO: 410), ARRQRIYGVVSFGEFFYYYYMDV (SEQ ID NO: 408), NGSVSG (SEQ ID NO: 424), YFSDTDRSE (SEQ ID NO: 425), AQQGKRYGIVSFGEFFYYYYMDA (SEQ ID NO: 423), AQQGKRYGIVSFGEFFYYYYMDA (SEQ ID NO: 436), SGGDILYNEWQKPHWFYP (SEQ ID NO: 445), SLSHCAGYYNSGWTY (SEQ ID NO: 457), FGGEVLVYHDWPKPAWVDL (SEQ ID NO: 456), GDSTAGCD (SEQ ID NO: 466), FDGEVLVYNDWPKPAWVDL (SEQ ID NO: 465), DFPFSK (SEQ ID NO: 478), AISGDAWHVV (SEQ ID NO: 479), MFQESGPPRLDRWSGRNYYYYSGMDV (SEQ ID NO: 477), AISADAWHVV (SEQ ID NO: 493), MFQESGPPRFDSWSGRNYYYYSGMDV (SEQ ID NO: 492), NFLFNK (SEQ ID NO: 506), VISSDAIHVD (SEQ ID NO: 507), DRDGYGPPQIQTWSGRYLHLYSGIDA (SEQ ID NO: 505), NFIFNK (SEQ ID NO: 520), AISADAWHVD (SEQ ID NO: 521), NIEEFSVPQFDSWSGRSYHYFGMDV (SEQ ID NO: 519), GFIFNE (SEQ ID NO: 536), DGEEHKVPQLHSWSGRNLYHYTGFDV (SEQ ID NO: 535), GFIFNQ (SEQ ID NO: 550), GFIFKQ (SEQ ID NO: 562), DGEEHEVPQLHSWSGRNLYHYTGVDI (SEQ ID NO: 561), GFIFKK (SEQ ID NO: 575), DGEEHEVPQLHSWSGRNLYHYTGVDV (SEQ ID NO: 574), (as determined by the Chothia method), and a light chain with three CDRs that include an amino acid sequence of at least 90%, 92%, 95%, 97%, 98%, 99%, or more identical to the amino acid sequence of NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), SSLTDRSHRI (SEQ ID NO: 41), RASQTINNYLN (SEQ ID NO: 107), GASNLQN (SEQ ID NO: 108), QQSFTPT (SEQ ID NO: 42), RASQTIHTYLN (SEQ ID NO: 113), GASTLQS (SEQ ID NO: 114), QQSYSTPT (SEQ ID NO: 43), SGSKLGDKYVS (SEQ ID NO: 120), ENDRPS (SEQ ID NO: 121), QAWETTTTTFVF (SEQ ID NO: 44), NGTSNDVGGYESVS (SEQ ID NO: 126), DVSKRPS (SEQ ID NO: 127), KSLTSTRRRV (SEQ ID NO: 45), NGTRSDVGGFDSVS (SEQ ID NO: 92), NGTSRDVGGFDSVS (SEQ ID NO: 93), GEKSLGSRAVQ (SEQ ID NO:

150), NNQDRPS (SEQ ID NO: 151), HIWDSRVPTKWV (SEQ ID NO: 152), GEESLGSRSVI (SEQ ID NO: 162), NNNDRPS (SEQ ID NO: 163), HIWDSRRPTNWV (SEQ ID NO: 164), GKESIGSRAVQ (SEQ ID NO: 178), NNQDRPA (SEQ ID NO: 179), HIYDARGGTNWV (SEQ ID NO: 180), NGTATNFVS (SEQ ID NO: 194), GVDKRPP (SEQ ID NO: 195), GSLVGNWDVI (SEQ ID NO: 196), TGTSNRFVS (SEQ ID NO: 210), GVNKRPS (SEQ ID NO: 211), SSLVGNWDVI (SEQ ID NO: 212), NGTSSDIGGWNFVS (SEQ ID NO: 226), EVNKRPS (SEQ ID NO: 227), SSLFGRWDVV (SEQ ID NO: 228), RASQNINKNLA (SEQ ID NO: 243), ETYSKIA (SEQ ID NO: 244), QQYEEWPRT (SEQ ID NO: 245), RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), MQGLNRPWT (SEQ ID NO: 288), or TSTQSLRHSNGANYLA (SEQ ID NO: 303), TGTSNNFVS (SEQ ID NO: 325), DVNKRPS (SEQ ID NO: 343), GSLVGNWDVI (SEQ ID NO: 196), SGTGSDIGSWNFVS (SEQ ID NO: 357), EVNRRRS (SEQ ID NO: 358), SSLSGRWDIV (SEQ ID NO: 359), RASQSVKNNLA (SEQ ID NO: 372), DTSSRAS (SEQ ID NO: 373), KCSHSLQHSTGANYLA (SEQ ID NO: 387), LATHRAS (SEQ ID NO: 388), MQGLHSPWT (SEQ ID NO: 389), GRQALGSRAVQ (SEQ ID NO: 415), HMWDSRSGFSWS (SEQ ID NO: 416), GERSRGSRAVQ (SEQ ID NO: 430), HYWDSRSPISWI (SEQ ID NO: 431), SGTASDIGSWNFVS (SEQ ID NO: 450), TGNINNFVS (SEQ ID NO: 458), GSLAGNWDVV (SEQ ID NO: 459), KSSESLRQSNGKTSLY (SEQ ID NO: 484), EVSNRFS (SEQ ID NO: 485), MQSKDFPLT (SEQ ID NO: 486), KSSQSLRQSNGKTSLY (SEQ ID NO: 498), KSSQSLRQSNGKTYLY (SEQ ID NO: 512), EVSIRFS (SEQ ID NO: 513), SSSESLGRGDGRTYLH (SEQ ID NO: 526), EVSTRFS (SEQ ID NO: 527), MQSRDFPIT (SEQ ID NO: 528), KSSQSVRQSDGKTFLY (SEQ ID NO: 541), EGSSRFS (SEQ ID NO: 542), LQTKDFPLT (SEQ ID NO: 543), KSSQTVRQSDGKTFLY (SEQ ID NO: 555), EGSNRFS (SEQ ID NO: 556), KSSQSLRQSDGKTFLY (SEQ ID NO: 567), EASNRFS (SEQ ID NO: 568), MQTKDFPLT (SEQ ID NO: 569), EASKRFS (SEQ ID NO: 580), (as determined by the Kabat method), or NGTSSDVGGFDSVS (SEQ ID NO: 97), DVSHRPS (SEQ ID NO: 95), SSLTDRSHRI (SEQ ID NO: 41), RASQTINNYLN (SEQ ID NO: 107), GASNLQN (SEQ ID NO: 108), QQSFSPTPT (SEQ ID NO: 42), RASQTIHTYLN (SEQ ID NO: 113), GASTLQS (SEQ ID NO: 114), QQSYSTPRT (SEQ ID NO: 43), SGSKLGDKYVS (SEQ ID NO: 120), ENDRRPS (SEQ ID NO: 121), QAWETTTTTTFVF (SEQ ID NO: 44), NGTSNDVGGYESVS (SEQ ID NO: 126), DVSKRPS (SEQ ID NO: 127), KSLTSTRRRV (SEQ ID NO: 45), NGTRSDVGGFDSVS (SEQ ID NO: 92), NGTSRDVGGFDSVS (SEQ ID NO: 93) GEKSLGSRAVQ (SEQ ID NO: 150), NNQDRPS (SEQ ID NO: 151), HIWDSRVPTKWV (SEQ ID NO: 152), GEESLGSRSVI (SEQ ID NO: 162), NNNDRPS (SEQ ID NO: 163), HIWDSRRPTNWV (SEQ ID NO: 164), GKESIGSRAVQ (SEQ ID NO: 178), NNQDRPA (SEQ ID NO: 179), HIYDARGGTNWV (SEQ ID NO: 180), NGTATNFVS (SEQ ID NO: 194), GVDKRPP (SEQ ID NO: 195), GSLVGNWDVI (SEQ ID NO: 196), TGTSNRFVS (SEQ ID NO: 210), GVNKRPS (SEQ ID NO: 211), SSLVGNWDVI (SEQ ID NO: 212), NGTSSDIGGWNFVS (SEQ ID NO: 226), EVNKRPS (SEQ ID NO: 227), SSLFGRWDVV (SEQ ID NO: 228), RASQNINKNLA (SEQ ID NO: 243), ETYSKIA (SEQ ID NO: 244), QQYEEWPRT (SEQ ID NO: 245), RASQNVKNNLA (SEQ ID NO: 259), DASSRAG (SEQ ID NO: 260), SSTQSLRHSNGANYLA (SEQ ID NO: 286), LGSQRAS (SEQ ID NO: 287), MQGLNRPWT (SEQ ID NO: 288), TSTQSLRHSNGANYLA (SEQ ID NO: 303), TGTSNNFVS (SEQ ID NO: 325), DVNKRPS (SEQ ID NO: 343), GSLVGNWDVI (SEQ ID NO: 196), SGTGSDIGSWNFVS (SEQ ID NO: 357), EVNRRRS (SEQ ID NO: 358), SSLSGRWDIV (SEQ ID NO: 359), RASQSVKNNLA (SEQ ID NO: 372), DTSSRAS (SEQ ID NO: 373), KCSHSLQHSTGANYLA (SEQ ID NO: 387), LATHRAS (SEQ ID NO: 388), MQGLHSPWT (SEQ ID NO: 389), GRQALGSRAVQ (SEQ ID NO: 415), HMWDSRSGFSWS (SEQ ID NO: 416), GERSRGSRAVQ (SEQ ID NO: 430), HYWDSRSPISWI (SEQ ID NO: 431), SGTASDIGSWNFVS (SEQ ID NO: 450), TGNINNFVS (SEQ ID NO: 458), GSLAGNWDVV (SEQ ID NO: 459), KSSESLRQSNGKTSLY (SEQ ID NO: 484), EVSNRFS

(SEQ ID NO: 485), MQSKDFPLT (SEQ ID NO: 486), KS SQSLRQSNKGKTSY (SEQ ID NO: 498), KS SQSLRQSNKGKTYLY (SEQ ID NO: 512), EVSIRFS (SEQ ID NO: 513), SSSESLGRGDGRTYLH (SEQ ID NO: 526), EVSTRFS (SEQ ID NO: 527), MQSRDFPIT (SEQ ID NO: 528), KSSQSVRQSDGKTFLY (SEQ ID NO: 541), EGSSRFS (SEQ ID NO: 542), LQTKDFPLT (SEQ ID NO: 543), KSSQTVRQSDGKTFLY (SEQ ID NO: 555), EGSNRFS (SEQ ID NO: 556), KSSQSLRQSDGKTFLY (SEQ ID NO: 567), EASNRFS (SEQ ID NO: 568), MQTKDFPLT (SEQ ID NO: 569), EASKRFS (SEQ ID NO: 580), (as determined by the Chothia method).

(272) The heavy chain of the anti-HIV monoclonal antibody is derived from a germ line variable (V) gene such as, for example, the IGHV1, IGHV3, or IGHV4 germline gene or an allele thereof.

(273) The anti-HIV antibodies of the invention include a variable heavy chain (V.sub.H) region encoded by a human IGHV1, IGHV3, or IGHV4 germline gene sequence or an allele thereof. Antibodies of the invention are derived from the IGHV1-2, IGHV1-8, or IGHV1-46 genes, or an allele thereof. Exemplary alleles of the IGHV1 germline gene include, but are not limited to, IGHV1-2*02, IGHV1-2*04, IGHV1-8*01, IGHV1-46*01, IGHV1-46*02, or IGHV1-46*03. IGHV1 germline gene sequences are shown, e.g., in Accession numbers L22582, X27506, X92340, M83132, X67905, L22583, Z29978, Z14309, Z14307, Z14300, Z14296, and Z14301. IGHV3 germline gene sequences are shown, e.g., in Accession numbers AB019439, M99665, M77305, M77335, and M77334. Antibodies of the invention are derived from the IGHV4-59, IGHV4-64, IGHV4-b, IGHV4-39, or IGHV4-28 genes, or an allele thereof. Exemplary alleles of the IGHV4 germline gene include, but are not limited to, IGHV4-59*01, IGHV4-59*07, IGHV4-59*02, IGHV4-59*03, IGHV4-59*04, IGHV4-61*08, IGHV4-b*02, IGHV4-b*01, IGHV4-39*07, IGHV4-39*03, IGHV4-39*06, IGHV4-39*01, IGHV4-39*02, or IGHV4-28*05. IGHV4 germline gene sequences are shown, e.g., in Accession numbers AB019439, L10094, X05715, X92259, X92297, M95116, Z14236, AM940222, X54447, X56362, Z14075, Z75352, AB019438, M29812, M95114, M95117, M95118, M95119, X56360, X87091, Z75359, Z14243, L10088, U03896, X56355, X56359, X92248, X92296, Z12371, M29811, L10097, X92230, X92250, X56356, Z75347, Z75348, AB019437, M95111, X92249, X92251, Z12366, Z75346, Z75361, Z12367, X56365, and X92289. The anti-HIV antibodies of the invention include a V.sub.H region that is encoded by a nucleic acid sequence that is at least 80% homologous to the IGHV1, IGHV3, or IGHV4 germline gene sequence or an allele thereof. Preferably, the nucleic acid sequence is at least 90%, 95%, 96%, 97% homologous to the IGHV1, IGHV3, or IGHV4 germline gene sequence, and more preferably, at least 98%, 99% homologous to the IGHV1, IGHV3, or IGHV4 germline gene sequence or an allele thereof. The V.sub.H region of the anti-HIV antibody is at least 80% homologous to the amino acid sequence of the V.sub.H region encoded by the IGHV1, IGHV3, or IGHV4 germline gene sequence or an allele thereof. Preferably, the amino acid sequence of V.sub.H region of the anti-HIV antibody is at least 90%, 95%, 96%, 97% homologous to the amino acid sequence encoded by the IGHV1, IGHV3, or IGHV4 germline gene sequence or an allele thereof, and more preferably, at least 98%, 99% homologous to the sequence encoded by the IGHV1, IGHV3, or IGHV4 germline gene sequence or an allele thereof.

(274) The light chain of the anti-HIV monoclonal antibody is derived from a germ line variable (V) gene such as, for example, the IGLV2, IGLV3, IGKV1, IGKV2, IGKV2D, or IGKV3 germline gene or an allele thereof.

(275) The anti-HIV antibodies of the invention also include a variable light chain (V.sub.L) region encoded by a human IGLV2, IGLV3, IGKV1, IGKV2, IGKV2D, IGKV3, or IGKV3D germline gene or an allele thereof. A human IGLV2 V.sub.L germline gene sequence is shown, e.g., Accession numbers Z73664, L27822, Y12412, and Y12413. A human IGLV3 V.sub.L germline gene sequence is shown, e.g., Accession number X57826. Antibodies of the invention are derived from the IGLV2-8 germline gene, or an allele thereof.

(276) Exemplary alleles of the IGLV2-8 germline gene include, but are not limited to, IGLV2-8*01

and IGLV2-8*02. Antibodies of the invention are derived from the IGLV3-21 germline gene, or an allele thereof. Exemplary alleles of the IGLV3-21 germline gene include, but are not limited to, IGLV3-21*01, IGLV3-21*02, and IGLV3-21*03. Antibodies of the invention are derived from the IGKV2-28 and IGKV2D-28 germline genes, or an allele thereof. Exemplary alleles of the IGKV2-28 and IGKV2D-28 germline genes include, but are not limited to, IGKV2-28*01 and IGKV2D-28*01. Antibodies of the invention are derived from the IGKV3-15 and IGKV3D-15 germline genes, or an allele thereof. Exemplary alleles of the IGKV3-15 and IGKV3D-15 germline genes include, but are not limited to, IGKV3-15*01, IGKV3D-15*01, and IGKV3D-15*02(P).

(277) A human IGLV2 V.sub.L germline gene sequence is shown, e.g., Accession numbers Z73657, Z73664, Z73642, X14616, X97466, Z73643, D87013, Z73641, X97462, D87021, Y12417, L27695, and Z22209. A human IGLV3 V.sub.L germline gene sequence is shown, e.g., Accession numbers X57826, X97464, Z73658, X97463, D87015, X97471, X97472, X56178, X97468, X71966, D87007, M94115, Z73666, X71968, X97474, X97467, D86994, Z73644, Z73646, X97469, Z73645, D87024, X97465, X97470, and X97473. A human IGKV1 V.sub.L germline gene sequence is shown, e.g., Accession numbers AF306358, AF490911, L12062, L12064, L12065, L12066, L12068, L12072, L12075, L12076, L12079, L12080, L12081, L12082, L12083, L12084, L12085, L12086, :12088, L12091, L12093, L12101, L12106, L12108, L12110, L12112, M95721, M95722, M95723, X73855, X73860, X98972, X98973, Z15073, Z15074, Z15075, Z15077, Z15079, Z15081. A human IGKV3 V.sub.L germline gene sequence is shown, e.g., Accession numbers X01668, M23090, X12686, X06583, X71883, X71891, X02725, L37728, L37727, L37730, L19271, L19272, X17264, X72815, X12687, X71886, X71896, X71895, X72820.

(278) Alternatively, the anti-HIV antibodies include a V.sub.L region that is encoded by a nucleic acid sequence that is at least 80% homologous to the IGLV2, IGLV3, IGKV1, IGKV2, IGKV2D, IGKV3, or IGKV3D germline gene or an allele thereof. Preferably, the nucleic acid sequence is at least 90%, 95%, 96%, 97% homologous to the IGLV2, IGLV3, IGKV1, IGKV2, IGKV2D, IGKV3, or IGKV3D germline gene or an allele thereof, and more preferably, at least 98%, 99% homologous to the IGLV2, IGLV3, IGKV1, IGKV2, IGKV2D, IGKV3, or IGKV3D germline gene or an allele thereof. The V.sub.L region of the anti-HIV antibody is at least 80% homologous to the amino acid sequence of the V.sub.L region encoded the IGLV2, IGLV3, IGKV1, IGKV2, IGKV2D, IGKV3, or IGKV3D germline gene or an allele thereof. Preferably, the amino acid sequence of V.sub.L region of the anti-HIV antibody is at least 90%, 95%, 96%, 97% homologous to the amino acid sequence encoded by the IGLV2, IGLV3, IGKV1, IGKV2, IGKV2D, IGKV3, or IGKV3D germline gene or an allele thereof, and more preferably, at least 98%, 99% homologous to the sequence encoded by the IGLV2, IGLV3, IGKV1, IGKV2, IGKV2D, IGKV3, or IGKV3D germline gene or an allele thereof.

(279) TABLE-US-00018 TABLE 11 Consensus nucleotide sequences of Kabat CDRs of heavy chains of 1443 PG16 sister clones. CDR1 (kabat): 1443 C16
AAATATGGCATGCAC (SEQ ID NO: 68) 1469 M23 AAATATGGCATGCAC (SEQ ID NO: 68) 1456 A12 AAATATGGCATGCAC (SEQ ID NO: 68) 1503 H05
AAATATGGCATGCAC (SEQ ID NO: 68) 1489 I13 AAATATGGCATGCAC (SEQ ID NO: 68) 1480 I08 AAATATGGCATGCAC (SEQ ID NO: 68) Consensus
AAATATGGCATGCAC (SEQ ID NO: 68) CDR1 (chothia): 1443 C16
TCTGGATTCACGTTTCACAAA (SEQ ID NO: 69) 1469 M23
TCTGGATTCACCTTTCACAAA (SEQ ID NO: 70) 1456 A12
TCTGGATTCACGTTTCACAAA (SEQ ID NO: 69) 1503 H05
TCTGGATTCACCTTTCACAAA (SEQ ID NO: 70) 1489 I13
TCTGGATTCACGTTTCACAAA (SEQ ID NO: 69) 1480 I08
TCTGGATTCACGTTTCACAAA (SEQ ID NO: 69) Consensus*
TCTGGATTCACXTTTCACAAA (SEQ ID NO: 71) Variation1

TCTGGATTTCACCTTTTACAAA (SEQ ID NO: 69) Variation2
 TCTGGATTTCACCTTTTACAAA (SEQ ID NO: 70) *Wherein X is C or G.
 CDR2: 1443 C16 CTCATCTCAGATGACGGAATGAGGAAATATCATTTCAGACTCCATGT
 GGGGC (SEQ ID NO: 72) 1469 M23
 CTCATCTCAGATGACGGAATGAGGAAATATCATTTCAGACTCCATGT GGGGC (SEQ
 ID NO: 72) 1456 A12
 CTCATCTCAGATGACGGAATGAGGAAATATCATTTCAGACTCCATGT GGGGC (SEQ
 ID NO: 72) 1503 H05
 CTCATCTCAGATGACGGAATGAGGAAATATCATTTCAGACTCCATGT GGGGC (SEQ
 ID NO: 72) 1489 I13
 CTCATCTCAGATGACGGAATGAGGAAATATCATTCAAACCTCCATGT GGGGC (SEQ
 ID NO: 73) 1480 I08
 CTCATCTCAGATGACGGAATGAGGAAATATCATTTCAGACTCCATGT GGGGC (SEQ
 ID NO: 72) Consensus*
 CTCATCTCAGATGACGGAATGAGGAAATATCATTTCAXACTCCATGT GGGGC (SEQ
 ID NO: 74) Variation1
 CTCATCTCAGATGACGGAATGAGGAAATATCATTTCAGACTCCATGT GGGGC (SEQ
 ID NO: 72) Variation2
 CTCATCTCAGATGACGGAATGAGGAAATATCATTCAAACCTCCATGT GGGGC (SEQ
 ID NO: 73) *Wherein X is A or G. CDR3: 1443 C16 (SEQ ID NO: 75)
 GAGGCTGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAACCTACCTACATGGACGTC 1469 M23 (SEQ ID NO: 75)
 GAGGCTGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAACCTACCTACATGGACGTC 1456 A12 (SEQ ID NO: 77)
 GAGGCCGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAACCTATCACTACATGGACGTC 1503 H05 (SEQ ID NO: 79)
 GAGGCTGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAATTACCTACATGGACGTC 1489 I13 (SEQ ID NO: 75)
 GAGGCTGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAACCTACCTACATGGACGTC 1480 I08 (SEQ ID NO: 75)
 GAGGCTGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAACCTACCTACATGGACGTC Consensus (SEQ ID NO: 76)
 GAGGCXGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAACCTATCACTACATGGACGTC Variation1 (SEQ ID NO: 78)
 GAGGCCGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAACCTATCACTACATGGACGTC Variation2 (SEQ ID NO: 77)
 GAGGCCGGTGGGCCAATCTGGCATGACGACGTCAAATATTACGATTTTAATGACG
 GCTACTACAACCTATCACTACATGGACGTC *Wherein X is T, C or G.
 (280) TABLE-US-00019 TABLE 12 Consensus nucleotide sequences of Kabat CDRs
 of light chains of 1443 PG16 sister clones. CDR1: 1443 C16
 AATGGAACCAGCAGTGACGTTGGTGGATTGACTCTGTCTCC (SEQ ID NO: 80)
 1469 M23 AATGGAACCAGAAAGTGACGTTGGTGGATTGACTCTGTCTCC (SEQ ID
 NO: 82) 1456 A12 AATGGAACCAGCCGTGACGTTGGTGGATTGACTCTGTCTCC
 (SEQ ID NO: 83) 1503 H05
 AATGGAACCAGAAAGTGACGTTGGTGGATTGACTCTGTCTCC (SEQ ID NO: 82)
 1489 I13 AATGGAACCAGCAGTGACGTTGGTGGATTGACTCTGTCTCC (SEQ ID
 NO: 80) 1480 I08 AATGGAACCAGCAGTGACGTTGGTGGATTGACTCTGTCTCC
 (SEQ ID NO: 80) Consensus*
 AATGGAACCAGX.sub.1X.sub.2GTGACGTTGGTGGATTGACTCTGTCTCC (SEQ ID
 NO: 81) Variation1 AATGGAACCAGCAGTGACGTTGGTGGATTGACTCTGTCTCC

(SEQ ID NO: 80) Variation2

AATGGAACCAGAAAGTGACGTTGGTGGATTGACTCTGTCTCC (SEQ ID NO: 82)
Variation2 AATGGAACCAGCCGTGACGTTGGTGGATTGACTCTGTCTCC (SEQ ID
NO: 83) *Wherein X.sub.1 is C or A. Wherein X.sub.2 is C or A. CDR2:
1443 C16 GATGTCAGTCATCGGCCCTCA (SEQ ID NO: 84) 1469 M23
GATGTCAGTCATCGGCCCTCA (SEQ ID NO: 84) 1456 A12
GATGTCAGTCATCGGCCCTCA (SEQ ID NO: 84) 1503 H05
GATGTCAGTCATCGGCCCTCA (SEQ ID NO: 84) 1489 I13
GATGTCAGTCATCGGCCCTCA (SEQ ID NO: 84) 1480 I08
GATGTCAGTCATCGGCCCTCA (SEQ ID NO: 84) Consensus
GATGTCAGTCATCGGCCCTCA (SEQ ID NO: 84) CDR3: 1443 C16
TCTTCACTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 85) 1469 M23
TCTTCACTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 85) 1456 A12
TCTTCATTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 86) 1503 H05
TCTTCACTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 85) 1489 I13
TCTTCACTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 85) 1480 I08
TCTTCACTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 85) Consensus*
TCTTCAXTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 87) Variation1
TCTTCACTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 85) Variation2
TCTTCATTGACAGACAGAAGCCATCGCATA (SEQ ID NO: 86) *Wherein X.sub.1
is C or T and wherein X.sub.2 is C or T.

(281) TABLE-US-00020 TABLE 13 Consensus protein sequences of Kabat CDRs
of Heavy chains of 1443 PG16 sister clones. CDR1: 1443 C16 KYGMH (SEQ
ID NO: 88) 1469 M23 KYGMH (SEQ ID NO: 88) 1456 A12 KYGMH (SEQ
ID NO: 88) 1503 H05 KYGMH (SEQ ID NO: 88) 1489 I13 KYGMH (SEQ
ID NO: 88) 1480 I08 KYGMH (SEQ ID NO: 88) Consensus KYGMH (SEQ ID
NO: 88) CDR2: 1443 C16 LISDDGMRKYHSDSMWG (SEQ ID NO: 89) 1469 M23
LISDDGMRKYHSDSMWG (SEQ ID NO: 89) 1456 A12
LISDDGMRKYHSDSMWG (SEQ ID NO: 89) 1503 H05
LISDDGMRKYHSDSMWG (SEQ ID NO: 89) 1489 I13 LISDDGMRKYHSNSMWG
(SEQ ID NO: 98) 1480 I08 LISDDGMRKYHSDSMWG (SEQ ID NO: 89)
Consensus* LISDDGMRKYHSXSMWG (SEQ ID NO: 91) Variation1
LISDDGMRKYHSDSMWG (SEQ ID NO: 89) Variation2 LISDDGMRKYHSNSMWG
(SEQ ID NO: 98) *Wherein X is D or N, or wherein X is an amino
acid with similar physical properties to either D or N. CDR3: 1443 C16
EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6) 1469 M23
EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6) 1456 A12
EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6) 1503 H05
EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6) 1489 I13
EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6) 1480 I08
EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6) Consensus
EAGGPIWHDDVKYYDFNDGYNYHYMDV (SEQ ID NO: 6)

(282) TABLE-US-00021 TABLE 14 Consensus protein sequences of Kabat CDRs
of light chains of 1443 PG16 sister clones. CDR1: 1443 C16 NGTSSDVGGFDSVS
(SEQ ID NO: 97) 1469 M23 NGTRSDVGGFDSVS (SEQ ID NO: 92) 1456 A12
NGTSRDVGGFDSVS (SEQ ID NO: 93) 1503 H05 NGTRSDVGGFDSVS (SEQ
ID NO: 92) 1489 I13 NGTSSDVGGFDSVS (SEQ ID NO: 97) 1480 I08
NGTSSDVGGFDSVS (SEQ ID NO: 97) Consensus*
NGTX.sub.1X.sub.2DVGGFDSVS (SEQ ID NO: 94) Variation1 NGTSSDVGGFDSVS
(SEQ ID NO: 97) Variation2 NGTRSDVGGFDSVS (SEQ ID NO: 92) Variation3

NGTSRDVGFDVS (SEQ ID NO: 93) *Wherein X.sub.1 is S or R, or wherein X.sub.1 is an amino acid with similar physical properties to either S or R. Wherein X.sub.2 is S or R, or wherein X.sub.2 is an amino acid with similar physical properties to either S or R. CDR2: 1443 C16 DVSHRPS (SEQ ID NO: 95) 1469 M23 DVSHRPS (SEQ ID NO: 95) 1456 A12 DVSHRPS (SEQ ID NO: 95) 1503 H05 DVSHRPS (SEQ ID NO: 95) 1489 I13 DVSHRPS (SEQ ID NO: 95) 1408 I08 DVSHRPS (SEQ ID NO: 95) Consensus DVSHRPS (SEQ ID NO: 95) CDR3: 1443 C16 SSLTDRSHRI (SEQ ID NO: 41) 1469 M23 SSLTDRSHRI (SEQ ID NO: 41) 1456 A12 SSLTDRSHRI (SEQ ID NO: 41) 1503 H05 SSLTDRSHRI (SEQ ID NO: 41) 1489 I13 SSLTDRSHRI (SEQ ID NO: 41) 1480 I08 SSLTDRSHRI (SEQ ID NO: 41) Consensus SSLTDRSHRI (SEQ ID NO: 41)

(283) Monoclonal and recombinant antibodies are particularly useful in identification and purification of the individual polypeptides or other antigens against which they are directed. The antibodies of the invention have additional utility in that they may be employed as reagents in immunoassays, radioimmunoassays (MA) or enzyme-linked immunosorbent assays (ELISA). In these applications, the antibodies can be labeled with an analytically-detectable reagent such as a radioisotope, a fluorescent molecule or an enzyme. The antibodies may also be used for the molecular identification and characterization (epitope mapping) of antigens.

(284) As mentioned above, the antibodies of the invention can be used to map the epitopes to which they bind. Applicants have discovered that the antibodies 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456 P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123 A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_101 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and/or 6881_N05 (PGT-158) neutralize HIV. Although the Applicant does not wish to be bound by this theory, it is postulated that the antibodies 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456 A12 (PG16) (TCN-117), 1469 M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873 E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123 A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128), 5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_101 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and/or 6881_N05 (PGT-158) bind to one or more conformational epitopes formed by HIV1-encoded proteins.

(285) Neutralization activity of human monoclonal antibodies was tested against HIV-1 strains SF162 and JR-CSF. HIV-1 strains SF162 and JR-CSF both belong to HIV Glade B. Each clonal monoclonal antibody was screened for neutralization activity and for anti-gp120, anti-gp41 and total IgG in quantitative ELISA. For the monoclonal antibodies 1456_P20, 1495_C14, and

1460_G14 anti-gp120 antigen-specific binding was detected. Neutralizing activity against SF162, but not JR-CSF was detected for 1456 P20 (PG20), 1495_C14 (PGC14), and 1460_G14 (PGG14). For the two monoclonal antibody preparations that did not show binding to gp120 in the ELISA assay, 1443_C16 (PG16) and 1496_C09 (PG9), high quantities of human IgG were determined to be present in the assay. However, 1443_C16 (PG16) and 1496_C09 (PG9) both were found to exhibit neutralizing activity against HIV-1 strain JR-CSF, but not against strain SF162. 1443_C16 (PG16) and 1496_C09 (PG9) also were found to lack gp41 binding activity in the ELISA assay. (286) The epitopes recognized by these antibodies may have a number of uses. The epitopes and mimotopes in purified or synthetic form can be used to raise immune responses (i.e. as a vaccine, or for the production of antibodies for other uses) or for screening patient serum for antibodies that immunoreact with the epitopes or mimotopes. Preferably, such an epitope or mimotope, or antigen comprising such an epitope or mimotope is used as a vaccine for raising an immune response. The antibodies of the invention can also be used in a method to monitor the quality of vaccines in particular to check that the antigen in a vaccine contains the correct immunogenic epitope in the correct conformation.

(287) The epitopes may also be useful in screening for ligands that bind to said epitopes. Such ligands preferably block the epitopes and thus prevent infection. Such ligands are encompassed within the scope of the invention.

(288) Standard techniques of molecular biology may be used to prepare DNA sequences coding for the antibodies or fragments of the antibodies of the present invention. Desired DNA sequences may be synthesized completely or in part using oligonucleotide synthesis techniques. Site-directed mutagenesis and polymerase chain reaction (PCR) techniques may be used as appropriate.

(289) Any suitable host cell/vector system may be used for expression of the DNA sequences encoding the antibody molecules of the present invention or fragments thereof. Bacterial, for example *E. coli*, and other microbial systems may be used, in part, for expression of antibody fragments such as Fab and F(ab')₂ fragments, and especially Fv fragments and single chain antibody fragments, for example, single chain Fvs. Eukaryotic, e.g. mammalian, host cell expression systems may be used for production of larger antibody molecules, including complete antibody molecules. Suitable mammalian host cells include CHO, HEK293T, PER.C6, myeloma or hybridoma cells.

(290) The present invention also provides a process for the production of an antibody molecule according to the present invention comprising culturing a host cell comprising a vector of the present invention under conditions suitable for leading to expression of protein from DNA encoding the antibody molecule of the present invention, and isolating the antibody molecule. The antibody molecule may comprise only a heavy or light chain polypeptide, in which case only a heavy chain or light chain polypeptide coding sequence needs to be used to transfect the host cells. For production of products comprising both heavy and light chains, the cell line may be transfected with two vectors, a first vector encoding a light chain polypeptide and a second vector encoding a heavy chain polypeptide. Alternatively, a single vector may be used, the vector including sequences encoding light chain and heavy chain polypeptides.

(291) Alternatively, antibodies according to the invention may be produced by i) expressing a nucleic acid sequence according to the invention in a cell, and ii) isolating the expressed antibody product. Additionally, the method may include iii) purifying the antibody. Transformed B cells are screened for those producing antibodies of the desired antigen specificity, and individual B cell clones can then be produced from the positive cells. The screening step may be carried out by ELISA, by staining of tissues or cells (including transfected cells), a neutralization assay or one of a number of other methods known in the art for identifying desired antigen specificity. The assay may select on the basis of simple antigen recognition, or may select on the additional basis of a desired function e.g. to select neutralizing antibodies rather than just antigen-binding antibodies, to select antibodies that can change characteristics of targeted cells, such as their signaling cascades,

their shape, their growth rate, their capability of influencing other cells, their response to the influence by other cells or by other reagents or by a change in conditions, their differentiation status, etc.

(292) The cloning step for separating individual clones from the mixture of positive cells may be carried out using limiting dilution, micromanipulation, single cell deposition by cell sorting or another method known in the art. Preferably the cloning is carried out using limiting dilution.

(293) The immortalized B cell clones of the invention can be used in various ways e.g. as a source of monoclonal antibodies, as a source of nucleic acid (DNA or mRNA) encoding a monoclonal antibody of interest, for research, etc.

(294) Unless otherwise defined, scientific and technical terms used in connection with the present invention shall have the meanings that are commonly understood by those of ordinary skill in the art. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. Generally, nomenclatures utilized in connection with, and techniques of, cell and tissue culture, molecular biology, and protein and oligo- or polynucleotide chemistry and hybridization described herein are those well known and commonly used in the art. Standard techniques are used for recombinant DNA, oligonucleotide synthesis, and tissue culture and transformation (e.g., electroporation, lipofection). Enzymatic reactions and purification techniques are performed according to manufacturer's specifications or as commonly accomplished in the art or as described herein. The practice of the present invention will employ, unless indicated specifically to the contrary, conventional methods of virology, immunology, microbiology, molecular biology and recombinant DNA techniques within the skill of the art, many of which are described below for the purpose of illustration. Such techniques are explained fully in the literature. See, e.g., Sambrook, et al. *Molecular Cloning: A Laboratory Manual* (2nd Edition, 1989); Maniatis et al. *Molecular Cloning: A Laboratory Manual* (1982); *DNA Cloning: A Practical Approach*, vol. I & II (D. Glover, ed.); *Oligonucleotide Synthesis* (N. Gait, ed., 1984); *Nucleic Acid Hybridization* (B. Hames & S. Higgins, eds., 1985); *Transcription and Translation* (B. Hames & S. Higgins, eds., 1984); *Animal Cell Culture* (R. Freshney, ed., 1986); Perbal, *A Practical Guide to Molecular Cloning* (1984). The nomenclatures utilized in connection with, and the laboratory procedures and techniques of, analytical chemistry, synthetic organic chemistry, and medicinal and pharmaceutical chemistry described herein are those well known and commonly used in the art. Standard techniques are used for chemical syntheses, chemical analyses, pharmaceutical preparation, formulation, and delivery, and treatment of patients.

(295) The following definitions are useful in understanding the present invention: The term "antibody" (Ab) as used herein includes monoclonal antibodies, polyclonal antibodies, multispecific antibodies (e.g., bispecific antibodies), and antibody fragments, as long as they exhibit the desired biological activity. The term "immunoglobulin" (Ig) is used interchangeably with "antibody" herein.

(296) A "neutralizing antibody" may inhibit the entry of HIV-1 virus for example SF162 and/or JR-CSF with a neutralization index >1.5 or >2.0 . (Kostrikis L G et al. *J Virol.* 1996; 70(1): 445-458.) By "broad and potent neutralizing antibodies" are meant antibodies that neutralize more than one HIV-1 virus species (from diverse clades and different strains within a Glade) in a neutralization assay. A broad neutralizing antibody may neutralize at least 2, 3, 4, 5, 6, 7, 8, 9 or more different strains of HIV-1, the strains belonging to the same or different clades. A broad neutralizing antibody may neutralize multiple HIV-1 species belonging to at least 2, 3, 4, 5, or 6 different clades. The inhibitory concentration of the monoclonal antibody may be less than about 25 mg/ml to neutralize about 50% of the input virus in the neutralization assay.

(297) An "isolated antibody" is one that has been separated and/or recovered from a component of its natural environment. Contaminant components of its natural environment are materials that would interfere with diagnostic or therapeutic uses for the antibody, and may include enzymes, hormones, and other proteinaceous or nonproteinaceous solutes. In preferred embodiments, the

antibody is purified: (1) to greater than 95% by weight of antibody as determined by the Lowry method, and most preferably more than 99% by weight; (2) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence by use of a spinning cup sequenator; or (3) to homogeneity by SD S-PAGE under reducing or non-reducing conditions using Coomassie blue or, preferably, silver stain. Isolated antibody includes the antibody in situ within recombinant cells since at least one component of the antibody's natural environment will not be present.

Ordinarily, however, isolated antibody will be prepared by at least one purification step.

(298) The basic four-chain antibody unit is a heterotetrameric glycoprotein composed of two identical light (L) chains and two identical heavy (H) chains. An IgM antibody consists of 5 basic heterotetramer units along with an additional polypeptide called J chain, and therefore contain 10 antigen binding sites, while secreted IgA antibodies can polymerize to form polyvalent assemblages comprising 2-5 of the basic 4-chain units along with J chain. In the case of IgGs, the 4-chain unit is generally about 150,000 daltons. Each L chain is linked to an H chain by one covalent disulfide bond, while the two H chains are linked to each other by one or more disulfide bonds depending on the H chain isotype. Each H and L chain also has regularly spaced intrachain disulfide bridges. Each H chain has at the N-terminus, a variable region (V.sub.H) followed by three constant domains (C.sub.H) for each of the α and γ chains and four C.sub.H domains for μ and ϵ isotypes. Each L chain has at the N-terminus, a variable region (V.sub.L) followed by a constant domain (C.sub.L) at its other end. The V.sub.L is aligned with the V.sub.H and the C.sub.L is aligned with the first constant domain of the heavy chain (C.sub.H1). Particular amino acid residues are believed to form an interface between the light chain and heavy chain variable regions. The pairing of a V.sub.H and V.sub.L together forms a single antigen-binding site. For the structure and properties of the different classes of antibodies, see, e.g., Basic and Clinical Immunology, 8th edition, Daniel P. Stites, Abba I. Terr and Tristram G. Parslow (eds.), Appleton & Lange, Norwalk, Conn., 1994, page 71, and Chapter 6.

(299) The L chain from any vertebrate species can be assigned to one of two clearly distinct types, called kappa (κ) and lambda (λ), based on the amino acid sequences of their constant domains (C.sub.L). Depending on the amino acid sequence of the constant domain of their heavy chains (C.sub.H), immunoglobulins can be assigned to different classes or isotypes. There are five classes of immunoglobulins: IgA, IgD, IgE, IgG, and IgM, having heavy chains designated alpha (α), delta (δ), epsilon (ϵ), gamma (γ) and mu (μ), respectively. The γ and α classes are further divided into subclasses on the basis of relatively minor differences in C.sub.H sequence and function, e.g., humans express the following subclasses: IgG.sub.1, IgG.sub.2, IgG.sub.3, IgG.sub.4, IgA.sub.1, and IgA.sub.2.

(300) The term "variable" refers to the fact that certain segments of the V domains differ extensively in sequence among antibodies. The V domain mediates antigen binding and defines specificity of a particular antibody for its particular antigen. However, the variability is not evenly distributed across the 110-amino acid span of the variable regions. Instead, the V regions consist of relatively invariant stretches called framework regions (FRs) of 15-30 amino acids separated by shorter regions of extreme variability called "hypervariable regions" that are each 9-12 amino acids long. The variable regions of native heavy and light chains each comprise four FRs, largely adopting a β -sheet configuration, connected by three hypervariable regions, which form loops connecting, and in some cases forming part of, the β -sheet structure. The hypervariable regions in each chain are held together in close proximity by the FRs and, with the hypervariable regions from the other chain, contribute to the formation of the antigen-binding site of antibodies (see Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991)). The constant domains are not involved directly in binding an antibody to an antigen, but exhibit various effector functions, such as participation of the antibody in antibody dependent cellular cytotoxicity (ADCC).

(301) The term "hypervariable region" when used herein refers to the amino acid residues of an

antibody that are responsible for antigen binding. The hypervariable region generally comprises amino acid residues from a “complementarity determining region” or “CDR” (e.g., around about residues 24-34 (L1), 50-56 (L2) and 89-97 (L3) in the V.sub.L, and around about 31-35 (H1), 50-65 (H2) and 95-102 (H3) in the V.sub.H when numbered in accordance with the Kabat numbering system; Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991)); and/or those residues from a “hypervariable loop” (e.g., residues 24-34 (L1), 50-56 (L2) and 89-97 (L3) in the V.sub.L, and 26-32 (H1), 52-56 (H2) and 95-101 (H3) in the V.sub.H when numbered in accordance with the Chothia numbering system; Chothia and Lesk, J. Mol. Biol. 196:901-917 (1987)); and/or those residues from a “hypervariable loop”/CDR (e.g., residues 27-38 (L1), 56-65 (L2) and 105-120 (L3) in the V.sub.L, and 27-38 (H1), 56-65 (H2) and 105-120 (H3) in the V.sub.H when numbered in accordance with the IMGT numbering system; Lefranc, M. P. et al. Nucl. Acids Res. 27:209-212 (1999), Ruiz, M. e al. Nucl. Acids Res. 28:219-221 (2000)). Optionally the antibody has symmetrical insertions at one or more of the following points 28, 36 (L1), 63, 74-75 (L2) and 123 (L3) in the V.sub.L, and 28, 36 (H1), 63, 74-75 (H2) and 123 (H3) in the V.sub.H when numbered in accordance with AHO; Honneger, A. and Plunkthun, A. J. Mol. Biol. 309:657-670 (2001)).

(302) By “germline nucleic acid residue” is meant the nucleic acid residue that naturally occurs in a germline gene encoding a constant or variable region. “Germline gene” is the DNA found in a germ cell (i.e., a cell destined to become an egg or in the sperm). A “germline mutation” refers to a heritable change in a particular DNA that has occurred in a germ cell or the zygote at the single-cell stage, and when transmitted to offspring, such a mutation is incorporated in every cell of the body. A germline mutation is in contrast to a somatic mutation which is acquired in a single body cell. In some cases, nucleotides in a germline DNA sequence encoding for a variable region are mutated (i.e., a somatic mutation) and replaced with a different nucleotide.

(303) The term “monoclonal antibody” as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies, i.e., the individual antibodies comprising the population are identical except for possible naturally occurring mutations that may be present in minor amounts. Monoclonal antibodies are highly specific, being directed against a single antigenic site. Furthermore, in contrast to polyclonal antibody preparations that include different antibodies directed against different determinants (epitopes), each monoclonal antibody is directed against a single determinant on the antigen. In addition to their specificity, the monoclonal antibodies are advantageous in that they may be synthesized uncontaminated by other antibodies. The modifier “monoclonal” is not to be construed as requiring production of the antibody by any particular method. For example, the monoclonal antibodies useful in the present invention may be prepared by the hybridoma methodology first described by Kohler et al., Nature, 256:495 (1975), or may be made using recombinant DNA methods in bacterial, eukaryotic animal or plant cells (see, e.g., U.S. Pat. No. 4,816,567). The “monoclonal antibodies” may also be isolated from phage antibody libraries using the techniques described in Clackson et al., Nature, 352:624-628 (1991) and Marks et al., J. Mol. Biol., 222:581-597 (1991), for example.

(304) In some aspects, the alternative EBV immortalization method described in WO2004/076677 is used. Using this method, B-cells producing the antibody of the invention can be transformed with EBV in the presence of a polyclonal B cell activator. Transformation with EBV is a standard technique and can easily be adapted to include polyclonal B cell activators. Additional stimulants of cellular growth and differentiation may be added during the transformation step to further enhance the efficiency. These stimulants may be cytokines such as IL-2 and IL-15. In a particularly preferred aspect, IL-2 is added during the immortalization step to further improve the efficiency of immortalization, but its use is not essential.

(305) The monoclonal antibodies herein include “chimeric” antibodies in which a portion of the heavy and/or light chain is identical with or homologous to corresponding sequences in antibodies derived from a particular species or belonging to a particular antibody class or subclass, while the

remainder of the chain(s) is identical with or homologous to corresponding sequences in antibodies derived from another species or belonging to another antibody class or subclass, as well as fragments of such antibodies, so long as they exhibit the desired biological activity (see U.S. Pat. No. 4,816,567; and Morrison et al., Proc. Natl. Acad. Sci. USA, 81:6851-6855 (1984)). The present invention provides variable region antigen-binding sequences derived from human antibodies. Accordingly, chimeric antibodies of primary interest herein include antibodies having one or more human antigen binding sequences (e.g., CDRs) and containing one or more sequences derived from a non-human antibody, e.g., an FR or C region sequence. In addition, chimeric antibodies of primary interest herein include those comprising a human variable region antigen binding sequence of one antibody class or subclass and another sequence, e.g., FR or C region sequence, derived from another antibody class or subclass. Chimeric antibodies of interest herein also include those containing variable region antigen-binding sequences related to those described herein or derived from a different species, such as a non-human primate (e.g., Old World Monkey, Ape, etc.). Chimeric antibodies also include primatized and humanized antibodies.

(306) Furthermore, chimeric antibodies may comprise residues that are not found in the recipient antibody or in the donor antibody. These modifications are made to further refine antibody performance. For further details, see Jones et al., Nature 321:522-525 (1986); Riechmann et al., Nature 332:323-329 (1988); and Presta, Curr. Op. Struct. Biol. 2:593-596 (1992).

(307) A “humanized antibody” is generally considered to be a human antibody that has one or more amino acid residues introduced into it from a source that is non-human. These non-human amino acid residues are often referred to as “import” residues, which are typically taken from an “import” variable region. Humanization is traditionally performed following the method of Winter and co-workers (Jones et al., Nature, 321:522-525 (1986); Reichmann et al., Nature, 332:323-327 (1988); Verhoeyen et al., Science, 239:1534-1536 (1988)), by substituting import hypervariable region sequences for the corresponding sequences of a human antibody. Accordingly, such “humanized” antibodies are chimeric antibodies (U.S. Pat. No. 4,816,567), wherein substantially less than an intact human variable region has been substituted by the corresponding sequence from a non-human species.

(308) A “human antibody” is an antibody containing only sequences present in an antibody naturally produced by a human. However, as used herein, human antibodies may comprise residues or modifications not found in a naturally occurring human antibody, including those modifications and variant sequences described herein. These are typically made to further refine or enhance antibody performance.

(309) An “intact” antibody is one that comprises an antigen-binding site as well as a C.sub.L and at least heavy chain constant domains, C.sub.H1, C.sub.H2 and C.sub.H3. The constant domains may be native sequence constant domains (e.g., human native sequence constant domains) or amino acid sequence variant thereof. Preferably, the intact antibody has one or more effector functions.

(310) An “antibody fragment” comprises a portion of an intact antibody, preferably the antigen binding or variable region of the intact antibody. Examples of antibody fragments include Fab, Fab', F(ab')₂, and Fv fragments; diabodies; linear antibodies (see U.S. Pat. No. 5,641,870; Zapata et al., Protein Eng. 8(10): 1057-1062 [1995]); single-chain antibody molecules; and multispecific antibodies formed from antibody fragments.

(311) The phrase “functional fragment or analog” of an antibody is a compound having qualitative biological activity in common with a full-length antibody. For example, a functional fragment or analog of an anti-IgE antibody is one that can bind to an IgE immunoglobulin in such a manner so as to prevent or substantially reduce the ability of such molecule from having the ability to bind to the high affinity receptor, Fc .sub.εRI.

(312) Papain digestion of antibodies produces two identical antigen-binding fragments, called “Fab” fragments, and a residual “Fc” fragment, a designation reflecting the ability to crystallize readily. The Fab fragment consists of an entire L chain along with the variable region domain of the

H chain (V.sub.H), and the first constant domain of one heavy chain (C.sub.H1). Each Fab fragment is monovalent with respect to antigen binding, i.e., it has a single antigen-binding site. Pepsin treatment of an antibody yields a single large F(ab')₂ fragment that roughly corresponds to two disulfide linked Fab fragments having divalent antigen-binding activity and is still capable of cross-linking antigen. Fab' fragments differ from Fab fragments by having additional few residues at the carboxy terminus of the C.sub.H1 domain including one or more cysteines from the antibody hinge region. Fab'-SH is the designation herein for Fab' in which the cysteine residue(s) of the constant domains bear a free thiol group. F(ab')₂ antibody fragments originally were produced as pairs of Fab' fragments that have hinge cysteines between them. Other chemical couplings of antibody fragments are also known.

(313) The "Fc" fragment comprises the carboxy-terminal portions of both H chains held together by disulfides. The effector functions of antibodies are determined by sequences in the Fc region, which region is also the part recognized by Fc receptors (FcR) found on certain types of cells.

(314) "Fv" is the minimum antibody fragment that contains a complete antigen-recognition and -binding site. This fragment consists of a dimer of one heavy- and one light-chain variable region domain in tight, non-covalent association. From the folding of these two domains emanate six hypervariable loops (three loops each from the H and L chain) that contribute the amino acid residues for antigen binding and confer antigen binding specificity to the antibody. However, even a single variable region (or half of an Fv comprising only three CDRs specific for an antigen) has the ability to recognize and bind antigen, although at a lower affinity than the entire binding site.

(315) "Single-chain Fv" also abbreviated as "sFv" or "scFv" are antibody fragments that comprise the V.sub.H and V.sub.L antibody domains connected into a single polypeptide chain. Preferably, the sFv polypeptide further comprises a polypeptide linker between the V.sub.H and V.sub.L domains that enables the sFv to form the desired structure for antigen binding. For a review of sFv, see Pluckthun in *The Pharmacology of Monoclonal Antibodies*, vol. 113, Rosenberg and Moore eds., Springer-Verlag, New York, pp. 269-315 (1994); Borrebaeck 1995, *infra*.

(316) The term "diabodies" refers to small antibody fragments prepared by constructing sFv fragments (see preceding paragraph) with short linkers (about 5-10 residues) between the V.sub.H and V.sub.L domains such that inter-chain but not intra-chain pairing of the V domains is achieved, resulting in a bivalent fragment, i.e., fragment having two antigen-binding sites. Bispecific diabodies are heterodimers of two "crossover" sFv fragments in which the V.sub.H and V.sub.L domains of the two antibodies are present on different polypeptide chains. Diabodies are described more fully in, for example, EP 404,097; WO 93/11161; and Hollinger et al., *Proc. Natl. Acad. Sci. USA*, 90:6444-6448 (1993).

(317) Domain antibodies (dAbs), which can be produced in fully human form, are the smallest known antigen-binding fragments of antibodies, ranging from 11 kDa to 15 kDa. dAbs are the robust variable regions of the heavy and light chains of immunoglobulins (V.sub.H and V.sub.L respectively). They are highly expressed in microbial cell culture, show favourable biophysical properties including solubility and temperature stability, and are well suited to selection and affinity maturation by in vitro selection systems such as phage display. dAbs are bioactive as monomers and, owing to their small size and inherent stability, can be formatted into larger molecules to create drugs with prolonged serum half-lives or other pharmacological activities. Examples of this technology have been described in WO9425591 for antibodies derived from Camelidae heavy chain Ig, as well in US20030130496 describing the isolation of single domain fully human antibodies from phage libraries.

(318) As used herein, an antibody that "internalizes" is one that is taken up by (i.e., enters) the cell upon binding to an antigen on a mammalian cell (e.g., a cell surface polypeptide or receptor). The internalizing antibody will of course include antibody fragments, human or chimeric antibody, and antibody conjugates. For certain therapeutic applications, internalization in vivo is contemplated. The number of antibody molecules internalized will be sufficient or adequate to kill a cell or inhibit

its growth, especially an infected cell. Depending on the potency of the antibody or antibody conjugate, in some instances, the uptake of a single antibody molecule into the cell is sufficient to kill the target cell to which the antibody binds. For example, certain toxins are highly potent in killing such that internalization of one molecule of the toxin conjugated to the antibody is sufficient to kill the infected cell.

(319) As used herein, an antibody is said to be “immunospecific,” “specific for” or to “specifically bind” an antigen if it reacts at a detectable level with the antigen, preferably with an affinity constant, $K_{sub.a}$, of greater than or equal to about $10^{sup.4} M^{sup.-1}$, or greater than or equal to about $10^{sup.5} M^{sup.-1}$, greater than or equal to about $10^{sup.6} M^{sup.-1}$, greater than or equal to about $10^{sup.7} M^{sup.-1}$, or greater than or equal to $10^{sup.8} M^{sup.-1}$. Affinity of an antibody for its cognate antigen is also commonly expressed as a dissociation constant $K_{sub.D}$, and in certain embodiments, HIV1 antibody specifically binds to an HIV1 polypeptide if it binds with a $K_{sub.D}$ of less than or equal to $10^{sup.-4} M$, less than or equal to about $10^{sup.-5} M$, less than or equal to about $10^{sup.-6} M$, less than or equal to $10^{sup.-7} M$, or less than or equal to $10^{sup.-8} M$. Affinities of antibodies can be readily determined using conventional techniques, for example, those described by Scatchard et al. (*Ann. N.Y. Acad. Sci. USA* 51:660 (1949)).

(320) Binding properties of an antibody to antigens, cells or tissues thereof may generally be determined and assessed using immunodetection methods including, for example, immunofluorescence-based assays, such as immuno-histochemistry (IHC) and/or fluorescence-activated cell sorting (FACS).

(321) An antibody having a “biological characteristic” of a designated antibody is one that possesses one or more of the biological characteristics of that antibody which distinguish it from other antibodies. For example, in certain embodiments, an antibody with a biological characteristic of a designated antibody will bind the same epitope as that bound by the designated antibody and/or have a common effector function as the designated antibody.

(322) The term “antagonist” antibody is used in the broadest sense, and includes an antibody that partially or fully blocks, inhibits, or neutralizes a biological activity of an epitope, polypeptide, or cell that it specifically binds. Methods for identifying antagonist antibodies may comprise contacting a polypeptide or cell specifically bound by a candidate antagonist antibody with the candidate antagonist antibody and measuring a detectable change in one or more biological activities normally associated with the polypeptide or cell.

(323) An “antibody that inhibits the growth of infected cells” or a “growth inhibitory” antibody is one that binds to and results in measurable growth inhibition of infected cells expressing or capable of expressing an HIV1 epitope bound by an antibody. Preferred growth inhibitory antibodies inhibit growth of infected cells by greater than 20%, preferably from about 20% to about 50%, and even more preferably, by greater than 50% (e.g., from about 50% to about 100%) as compared to the appropriate control, the control typically being infected cells not treated with the antibody being tested. Growth inhibition can be measured at an antibody concentration of about 0.1 to 30 $\mu\text{g/ml}$ or about 0.5 nM to 200 nM in cell culture, where the growth inhibition is determined 1-10 days after exposure of the infected cells to the antibody. Growth inhibition of infected cells in vivo can be determined in various ways known in the art.

(324) The antibody is growth inhibitory in vivo if administration of the antibody at about 1 $\mu\text{g/kg}$ to about 100 mg/kg body weight results in reduction the percent of infected cells or total number of infected cells within about 5 days to 3 months from the first administration of the antibody, preferably within about 5 to 30 days.

(325) An antibody that “induces apoptosis” is one which induces programmed cell death as determined by binding of annexin V, fragmentation of DNA, cell shrinkage, dilation of endoplasmic reticulum, cell fragmentation, and/or formation of membrane vesicles (called apoptotic bodies). Preferably the cell is an infected cell. Various methods are available for evaluating the cellular events associated with apoptosis. For example, phosphatidyl serine (PS)

translocation can be measured by annexin binding; DNA fragmentation can be evaluated through DNA laddering; and nuclear/chromatin condensation along with DNA fragmentation can be evaluated by any increase in hypodiploid cells. Preferably, the antibody that induces apoptosis is one that results in about 2 to 50 fold, preferably about 5 to 50 fold, and most preferably about 10 to 50 fold, induction of annexin binding relative to untreated cell in an annexin binding assay.

(326) Antibody “effector functions” refer to those biological activities attributable to the Fc region (a native sequence Fc region or amino acid sequence variant Fc region) of an antibody, and vary with the antibody isotype. Examples of antibody effector functions include: C1q binding and complement dependent cytotoxicity; Fc receptor binding; antibody-dependent cell-mediated cytotoxicity (ADCC); phagocytosis; down regulation of cell surface receptors (e.g., B cell receptor); and B cell activation.

(327) “Antibody-dependent cell-mediated cytotoxicity” or “ADCC” refers to a form of cytotoxicity in which secreted Ig bound to Fc receptors (FcRs) present on certain cytotoxic cells (e.g., Natural Killer (NK) cells, neutrophils, and macrophages) enable these cytotoxic effector cells to bind specifically to an antigen-bearing target cell and subsequently kill the target cell with cytotoxins. The antibodies “arm” the cytotoxic cells and are required for such killing. The primary cells for mediating ADCC, NK cells, express FcγRIII only, whereas monocytes express FcγRI, FcγRII and FcγRIII. FcR expression on hematopoietic cells is summarized in Table 4 on page 464 of Ravetch and Kinet, *Annu. Rev. Immunol* 9:457-92 (1991). To assess ADCC activity of a molecule of interest, an in vitro ADCC assay, such as that described in U.S. Pat. No. 5,500,362 or U.S. Pat. No. 5,821,337 may be performed. Useful effector cells for such assays include peripheral blood mononuclear cells (PBMC) and Natural Killer (NK) cells.

(328) Alternatively, or additionally, ADCC activity of the molecule of interest may be assessed in vivo, e.g., in a animal model such as that disclosed in Clynes et al., *Proc. Natl. Acad. Sci. (USA)* 95:652-656 (1998).

(329) “Fc receptor” or “FcR” describes a receptor that binds to the Fc region of an antibody. In certain embodiments, the FcR is a native sequence human FcR. Moreover, a preferred FcR is one that binds an IgG antibody (a gamma receptor) and includes receptors of the FcγRI, FcγRII, and FcγRIII subclasses, including allelic variants and alternatively spliced forms of these receptors. FcγRII receptors include FcγRIIA (an “activating receptor”) and FcγRIIB (an “inhibiting receptor”), which have similar amino acid sequences that differ primarily in the cytoplasmic domains thereof. Activating receptor FcγRIIA contains an immunoreceptor tyrosine-based activation motif (ITAM) in its cytoplasmic domain. Inhibiting receptor FcγRIIB contains an immunoreceptor tyrosine-based inhibition motif (ITIM) in its cytoplasmic domain. (see review M. in Daeron, *Annu. Rev. Immunol.* 15:203-234 (1997)). FcRs are reviewed in Ravetch and Kinet, *Annu. Rev. Immunol* 9:457-92 (1991); Capel et al., *Immunomethods* 4:25-34 (1994); and de Haas et al., *J. Lab. Clin. Med.* 126:330-41 (1995). Other FcRs, including those to be identified in the future, are encompassed by the term “FcR” herein. The term also includes the neonatal receptor, FcRn, which is responsible for the transfer of maternal IgGs to the fetus (Guyer et al., *J. Immunol.* 117:587 (1976) and Kim et al., *J. Immunol.* 24:249 (1994)).

(330) “Human effector cells” are leukocytes that express one or more FcRs and perform effector functions. Preferably, the cells express at least FcγRIII and perform ADCC effector function. Examples of human leukocytes that mediate ADCC include PBMC, NK cells, monocytes, cytotoxic T cells and neutrophils; with PBMCs and NK cells being preferred. The effector cells may be isolated from a native source, e.g., from blood.

(331) “Complement dependent cytotoxicity” or “CDC” refers to the lysis of a target cell in the presence of complement. Activation of the classical complement pathway is initiated by the binding of the first component of the complement system (C1q) to antibodies (of the appropriate subclass) that are bound to their cognate antigen. To assess complement activation, a CDC assay, e.g., as described in Gazzano-Santoro et al., *J. Immunol. Methods* 202:163 (1996), may be performed.

(332) A “mammal” for purposes of treating an infection, refers to any mammal, including humans, domestic and farm animals, and zoo, sports, or pet animals, such as dogs, cats, cattle, horses, sheep, pigs, goats, rabbits, etc. Preferably, the mammal is human.

(333) “Treating” or “treatment” or “alleviation” refers to both therapeutic treatment and prophylactic or preventative measures; wherein the object is to prevent or slow down (lessen) the targeted pathologic condition or disorder. Those in need of treatment include those already with the disorder as well as those prone to have the disorder or those in whom the disorder is to be prevented. A subject or mammal is successfully “treated” for an infection if, after receiving a therapeutic amount of an antibody according to the methods of the present invention, the patient shows observable and/or measurable reduction in or absence of one or more of the following: reduction in the number of infected cells or absence of the infected cells; reduction in the percent of total cells that are infected; and/or relief to some extent, one or more of the symptoms associated with the specific infection; reduced morbidity and mortality, and improvement in quality of life issues. The above parameters for assessing successful treatment and improvement in the disease are readily measurable by routine procedures familiar to a physician.

(334) The term “therapeutically effective amount” refers to an amount of an antibody or a drug effective to “treat” a disease or disorder in a subject or mammal. See preceding definition of “treating.”

(335) “Chronic” administration refers to administration of the agent(s) in a continuous mode as opposed to an acute mode, so as to maintain the initial therapeutic effect (activity) for an extended period of time. “Intermittent” administration is treatment that is not consecutively done without interruption, but rather is cyclic in nature.

(336) Administration “in combination with” one or more further therapeutic agents includes simultaneous (concurrent) and consecutive administration in any order.

(337) “Carriers” as used herein include pharmaceutically acceptable carriers, excipients, or stabilizers that are nontoxic to the cell or mammal being exposed thereto at the dosages and concentrations employed. Often the physiologically acceptable carrier is an aqueous pH buffered solution. Examples of physiologically acceptable carriers include buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid; low molecular weight (less than about 10 residues) polypeptide; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, arginine or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugar alcohols such as mannitol or sorbitol; salt-forming counterions such as sodium; and/or nonionic surfactants such as TWEEN® (polysorbate), polyethylene glycol (PEG), and PLURONICS® (poloxamer).

(338) The term “cytotoxic agent” as used herein refers to a substance that inhibits or prevents the function of cells and/or causes destruction of cells. The term is intended to include radioactive isotopes (e.g., At.sup.211, I.sup.131, I.sup.125, y.sup.90, Re.sup.186, Re.sup.188, Sm.sup.153, Bi.sup.212, P.sup.32 and radioactive isotopes of Lu), chemotherapeutic agents e.g., methotrexate, adriamycin, vinca alkaloids (vincristine, vinblastine, etoposide), doxorubicin, melphalan, mitomycin C, chlorambucil, daunorubicin or other intercalating agents, enzymes and fragments thereof such as nucleolytic enzymes, antibiotics, and toxins such as small molecule toxins or enzymatically active toxins of bacterial, fungal, plant or animal origin, including fragments and/or variants thereof, and the various antitumor or anticancer agents disclosed below. Other cytotoxic agents are described below.

(339) A “growth inhibitory agent” when used herein refers to a compound or composition which inhibits growth of a cell, either in vitro or in vivo. Examples of growth inhibitory agents include agents that block cell cycle progression, such as agents that induce G1 arrest and M-phase arrest. Classical M-phase blockers include the *vinca* alkaloids (vincristine, vinorelbine and vinblastine), taxanes, and topoisomerase II inhibitors such as doxorubicin, epirubicin, daunorubicin, etoposide,

and bleomycin. Those agents that arrest G1 also spill over into S-phase arrest, for example, DNA alkylating agents such as tamoxifen, prednisone, dacarbazine, mechlorethamine, cisplatin, methotrexate, 5-fluorouracil, and ara-C. Further information can be found in *The Molecular Basis of Cancer*, Mendelsohn and Israel, eds., Chapter 1, entitled “Cell cycle regulation, oncogenes, and antineoplastic drugs” by Murakami et al. (W B Saunders: Philadelphia, 1995), especially p. 13. The taxanes (paclitaxel and docetaxel) are anticancer drugs both derived from the yew tree. Docetaxel (TAXOTERE™, Rhone-Poulenc Rorer), derived from the European yew, is a semisynthetic analogue of paclitaxel (TAXOL®, Bristol-Myers Squibb). Paclitaxel and docetaxel promote the assembly of microtubules from tubulin dimers and stabilize microtubules by preventing depolymerization, which results in the inhibition of mitosis in cells.

(340) “Label” as used herein refers to a detectable compound or composition that is conjugated directly or indirectly to the antibody so as to generate a “labeled” antibody. The label may be detectable by itself (e.g., radioisotope labels or fluorescent labels) or, in the case of an enzymatic label, may catalyze chemical alteration of a substrate compound or composition that is detectable.

(341) The term “epitope tagged” as used herein refers to a chimeric polypeptide comprising a polypeptide fused to a “tag polypeptide.” The tag polypeptide has enough residues to provide an epitope against which an antibody can be made, yet is short enough such that it does not interfere with activity of the polypeptide to which it is fused. The tag polypeptide is also preferably fairly unique so that the antibody does not substantially cross-react with other epitopes. Suitable tag polypeptides generally have at least six amino acid residues and usually between about 8 and 50 amino acid residues (preferably, between about 10 and 20 amino acid residues).

(342) A “small molecule” is defined herein to have a molecular weight below about 500 Daltons.

(343) The terms “nucleic acid” and “polynucleotide” are used interchangeably herein to refer to single- or double-stranded RNA, DNA, or mixed polymers. Polynucleotides may include genomic sequences, extra-genomic and plasmid sequences, and smaller engineered gene segments that express, or may be adapted to express polypeptides.

(344) An “isolated nucleic acid” is a nucleic acid that is substantially separated from other genome DNA sequences as well as proteins or complexes such as ribosomes and polymerases, which naturally accompany a native sequence. The term embraces a nucleic acid sequence that has been removed from its naturally occurring environment, and includes recombinant or cloned DNA isolates and chemically synthesized analogues or analogues biologically synthesized by heterologous systems. A substantially pure nucleic acid includes isolated forms of the nucleic acid. Of course, this refers to the nucleic acid as originally isolated and does not exclude genes or sequences later added to the isolated nucleic acid by the hand of man.

(345) The term “polypeptide” is used in its conventional meaning, i.e., as a sequence of amino acids. The polypeptides are not limited to a specific length of the product. Peptides, oligopeptides, and proteins are included within the definition of polypeptide, and such terms may be used interchangeably herein unless specifically indicated otherwise. This term also does not refer to or exclude post-expression modifications of the polypeptide, for example, glycosylations, acetylations, phosphorylations and the like, as well as other modifications known in the art, both naturally occurring and non-naturally occurring. A polypeptide may be an entire protein, or a subsequence thereof. Particular polypeptides of interest in the context of this invention are amino acid subsequences comprising CDRs and being capable of binding an antigen or HIV-infected cell.

(346) An “isolated polypeptide” is one that has been identified and separated and/or recovered from a component of its natural environment. In preferred embodiments, the isolated polypeptide will be purified (1) to greater than 95% by weight of polypeptide as determined by the Lowry method, and most preferably more than 99% by weight, (2) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence by use of a spinning cup sequenator, or (3) to homogeneity by SDS-PAGE under reducing or non-reducing conditions using Coomassie blue or, preferably, silver stain. Isolated polypeptide includes the polypeptide in situ within recombinant

cells since at least one component of the polypeptide's natural environment will not be present.

Ordinarily, however, isolated polypeptide will be prepared by at least one purification step.

(347) A “native sequence” polynucleotide is one that has the same nucleotide sequence as a polynucleotide derived from nature. A “native sequence” polypeptide is one that has the same amino acid sequence as a polypeptide (e.g., antibody) derived from nature (e.g., from any species). Such native sequence polynucleotides and polypeptides can be isolated from nature or can be produced by recombinant or synthetic means.

(348) A polynucleotide “variant,” as the term is used herein, is a polynucleotide that typically differs from a polynucleotide specifically disclosed herein in one or more substitutions, deletions, additions and/or insertions. Such variants may be naturally occurring or may be synthetically generated, for example, by modifying one or more of the polynucleotide sequences of the invention and evaluating one or more biological activities of the encoded polypeptide as described herein and/or using any of a number of techniques well known in the art.

(349) A polypeptide “variant,” as the term is used herein, is a polypeptide that typically differs from a polypeptide specifically disclosed herein in one or more substitutions, deletions, additions and/or insertions. Such variants may be naturally occurring or may be synthetically generated, for example, by modifying one or more of the above polypeptide sequences of the invention and evaluating one or more biological activities of the polypeptide as described herein and/or using any of a number of techniques well known in the art.

(350) Modifications may be made in the structure of the polynucleotides and polypeptides of the present invention and still obtain a functional molecule that encodes a variant or derivative polypeptide with desirable characteristics. When it is desired to alter the amino acid sequence of a polypeptide to create an equivalent, or even an improved, variant or portion of a polypeptide of the invention, one skilled in the art will typically change one or more of the codons of the encoding DNA sequence.

(351) For example, certain amino acids may be substituted for other amino acids in a protein structure without appreciable loss of its ability to bind other polypeptides (e.g., antigens) or cells. Since it is the binding capacity and nature of a protein that defines that protein's biological functional activity, certain amino acid sequence substitutions can be made in a protein sequence, and, of course, it's underlying DNA coding sequence, and nevertheless obtain a protein with like properties. It is thus contemplated that various changes may be made in the peptide sequences of the disclosed compositions, or corresponding DNA sequences that encode said peptides without appreciable loss of their biological utility or activity.

(352) In many instances, a polypeptide variant will contain one or more conservative substitutions. A “conservative substitution” is one in which an amino acid is substituted for another amino acid that has similar properties, such that one skilled in the art of peptide chemistry would expect the secondary structure and hydropathic nature of the polypeptide to be substantially unchanged.

(353) In making such changes, the hydropathic index of amino acids may be considered. The importance of the hydropathic amino acid index in conferring interactive biologic function on a protein is generally understood in the art (Kyte and Doolittle, 1982). It is accepted that the relative hydropathic character of the amino acid contributes to the secondary structure of the resultant protein, which in turn defines the interaction of the protein with other molecules, for example, enzymes, substrates, receptors, DNA, antibodies, antigens, and the like. Each amino acid has been assigned a hydropathic index on the basis of its hydrophobicity and charge characteristics (Kyte and Doolittle, 1982). These values are: isoleucine (+4.5); valine (+4.2); leucine (+3.8); phenylalanine (+2.8); cysteine/cystine (+2.5); methionine (+1.9); alanine (+1.8); glycine (−0.4); threonine (−0.7); serine (−0.8); tryptophan (−0.9); tyrosine (−1.3); proline (−1.6); histidine (−3.2); glutamate (−3.5); glutamine (−3.5); aspartate (−3.5); asparagine (−3.5); lysine (−3.9); and arginine (−4.5).

(354) It is known in the art that certain amino acids may be substituted by other amino acids having

a similar hydropathic index or score and still result in a protein with similar biological activity, i.e. still obtain a biological functionally equivalent protein. In making such changes, the substitution of amino acids whose hydropathic indices are within ± 2 is preferred, those within ± 1 are particularly preferred, and those within ± 0.5 are even more particularly preferred. It is also understood in the art that the substitution of like amino acids can be made effectively on the basis of hydrophilicity. U.S. Pat. No. 4,554,101 states that the greatest local average hydrophilicity of a protein, as governed by the hydrophilicity of its adjacent amino acids, correlates with a biological property of the protein.

(355) As detailed in U.S. Pat. No. 4,554,101, the following hydrophilicity values have been assigned to amino acid residues: arginine (+3.0); lysine (+3.0); aspartate (+3.0 $\pm$ 1); glutamate (+3.0 $\pm$ 1); serine (+0.3); asparagine (+0.2); glutamine (+0.2); glycine (0); threonine (−0.4); proline (−0.5 $\pm$ 1); alanine (−0.5); histidine (−0.5); cysteine (−1.0); methionine (−1.3); valine (−1.5); leucine (−1.8); isoleucine (−1.8); tyrosine (−2.3); phenylalanine (−2.5); tryptophan (−3.4). It is understood that an amino acid can be substituted for another having a similar hydrophilicity value and still obtain a biologically equivalent, and in particular, an immunologically equivalent protein. In such changes, the substitution of amino acids whose hydrophilicity values are within ± 2 is preferred, those within ± 1 are particularly preferred, and those within ± 0.5 are even more particularly preferred.

(356) As outlined above, amino acid substitutions are generally therefore based on the relative similarity of the amino acid side-chain substituents, for example, their hydrophobicity, hydrophilicity, charge, size, and the like. Exemplary substitutions that take various of the foregoing characteristics into consideration are well known to those of skill in the art and include: arginine and lysine; glutamate and aspartate; serine and threonine; glutamine and asparagine; and valine, leucine and isoleucine.

(357) Amino acid substitutions may further be made on the basis of similarity in polarity, charge, solubility, hydrophobicity, hydrophilicity and/or the amphipathic nature of the residues. For example, negatively charged amino acids include aspartic acid and glutamic acid; positively charged amino acids include lysine and arginine; and amino acids with uncharged polar head groups having similar hydrophilicity values include leucine, isoleucine and valine; glycine and alanine; asparagine and glutamine; and serine, threonine, phenylalanine and tyrosine. Other groups of amino acids that may represent conservative changes include: (1) ala, pro, gly, glu, asp, gln, asn, ser, thr; (2) cys, ser, tyr, thr; (3) val, ile, leu, met, ala, phe; (4) lys, arg, his; and (5) phe, tyr, trp, his. A variant may also, or alternatively, contain nonconservative changes. In a preferred embodiment, variant polypeptides differ from a native sequence by substitution, deletion or addition of five amino acids or fewer. Variants may also (or alternatively) be modified by, for example, the deletion or addition of amino acids that have minimal influence on the immunogenicity, secondary structure and hydropathic nature of the polypeptide.

(358) Polypeptides may comprise a signal (or leader) sequence at the N-terminal end of the protein, which co-translationally or post-translationally directs transfer of the protein. The polypeptide may also be conjugated to a linker or other sequence for ease of synthesis, purification or identification of the polypeptide (e.g., poly-His), or to enhance binding of the polypeptide to a solid support. For example, a polypeptide may be conjugated to an immunoglobulin Fc region.

(359) When comparing polynucleotide and polypeptide sequences, two sequences are said to be “identical” if the sequence of nucleotides or amino acids in the two sequences is the same when aligned for maximum correspondence, as described below. Comparisons between two sequences are typically performed by comparing the sequences over a comparison window to identify and compare local regions of sequence similarity. A “comparison window” as used herein, refers to a segment of at least about 20 contiguous positions, usually 30 to about 75, 40 to about 50, in which a sequence may be compared to a reference sequence of the same number of contiguous positions after the two sequences are optimally aligned.

(360) Optimal alignment of sequences for comparison may be conducted using the Megalign

program in the Lasergene suite of bioinformatics software (DNASTAR, Inc., Madison, WI), using default parameters. This program embodies several alignment schemes described in the following references: Dayhoff, M. O. (1978) A model of evolutionary change in proteins—Matrices for detecting distant relationships. In Dayhoff, M. O. (ed.) *Atlas of Protein Sequence and Structure*, National Biomedical Research Foundation, Washington DC Vol. 5, Suppl. 3, pp. 345-358; Hein J. (1990) Unified Approach to Alignment and Phylogenies pp. 626-645 *Methods in Enzymology* vol. 183, Academic Press, Inc., San Diego, CA; Higgins, D. G. and Sharp, P. M. (1989) *CABIOS* 5:151-153; Myers, E. W. and Muller W. (1988) *CABIOS* 4:11-17; Robinson, E. D. (1971) *Comb. Theor* 11:105; Santou, N. Nes, M. (1987) *Mol. Biol. Evol.* 4:406-425; Sneath, P. H. A. and Sokal, R. R. (1973) *Numerical Taxonomy—the Principles and Practice of Numerical Taxonomy*, Freeman Press, San Francisco, CA; Wilbur, W. J. and Lipman, D. J. (1983) *Proc. Natl. Acad., Sci. USA* 80:726-730. (361) Alternatively, optimal alignment of sequences for comparison may be conducted by the local identity algorithm of Smith and Waterman (1981) *Add. APL. Math* 2:482, by the identity alignment algorithm of Needleman and Wunsch (1970) *J. Mol. Biol.* 48:443, by the search for similarity methods of Pearson and Lipman (1988) *Proc. Natl. Acad. Sci. USA* 85: 2444, by computerized implementations of these algorithms (GAP, BESTFIT, BLAST, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group (GCG), 575 Science Dr., Madison, WI), or by inspection.

(362) One preferred example of algorithms that are suitable for determining percent sequence identity and sequence similarity are the BLAST and BLAST 2.0 algorithms, which are described in Altschul et al. (1977) *Nucl. Acids Res.* 25:3389-3402 and Altschul et al. (1990) *J. Mol. Biol.* 215:403-410, respectively. BLAST and BLAST 2.0 can be used, for example with the parameters described herein, to determine percent sequence identity for the polynucleotides and polypeptides of the invention. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information.

(363) In one illustrative example, cumulative scores can be calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a wordlength (W) of 11, and expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff and Henikoff (1989) *Proc. Natl. Acad. Sci. USA* 89:10915) alignments, (B) of 50, expectation (E) of 10, M=5, N=-4 and a comparison of both strands.

(364) For amino acid sequences, a scoring matrix can be used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T and X determine the sensitivity and speed of the alignment.

(365) In one approach, the “percentage of sequence identity” is determined by comparing two optimally aligned sequences over a window of comparison of at least 20 positions, wherein the portion of the polynucleotide or polypeptide sequence in the comparison window may comprise additions or deletions (i.e., gaps) of 20 percent or less, usually 5 to 15 percent, or 10 to 12 percent, as compared to the reference sequences (which does not comprise additions or deletions) for optimal alignment of the two sequences. The percentage is calculated by determining the number of positions at which the identical nucleic acid bases or amino acid residues occur in both sequences to yield the number of matched positions, dividing the number of matched positions by

the total number of positions in the reference sequence (i.e., the window size) and multiplying the results by 100 to yield the percentage of sequence identity.

(366) “Homology” refers to the percentage of residues in the polynucleotide or polypeptide sequence variant that are identical to the non-variant sequence after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent homology. In particular embodiments, polynucleotide and polypeptide variants have at least 70%, at least 75%, at least 80%, at least 90%, at least 95%, at least 98%, or at least 99% polynucleotide or polypeptide homology with a polynucleotide or polypeptide described herein.

(367) “Vector” includes shuttle and expression vectors. Typically, the plasmid construct will also include an origin of replication (e.g., the ColE1 origin of replication) and a selectable marker (e.g., ampicillin or tetracycline resistance), for replication and selection, respectively, of the plasmids in bacteria. An “expression vector” refers to a vector that contains the necessary control sequences or regulatory elements for expression of the antibodies including antibody fragment of the invention, in bacterial or eukaryotic cells. Suitable vectors are disclosed below. As used in this specification and the appended claims, the singular forms “a,” “an” and “the” include plural references unless the content clearly dictates otherwise.

(368) The invention also includes nucleic acid sequences encoding part or all of the light and heavy chains and CDRs of the present invention. Due to redundancy of the genetic code, variants of these sequences will exist that encode the same amino acid sequences.

(369) Variant antibodies are also included within the scope of the invention. Thus, variants of the sequences recited in the application are also included within the scope of the invention. Further variants of the antibody sequences having improved affinity may be obtained using methods known in the art and are included within the scope of the invention. For example, amino acid substitutions may be used to obtain antibodies with further improved affinity. Alternatively, codon optimization of the nucleotide sequence may be used to improve the efficiency of translation in expression systems for the production of the antibody.

(370) Preferably, such variant antibody sequences will share 70% or more (i.e. 80, 85, 90, 95, 97, 98, 99% or more) sequence identity with the sequences recited in the application. Preferably such sequence identity is calculated with regard to the full length of the reference sequence (i.e. the sequence recited in the application). Preferably, percentage identity, as referred to herein, is as determined using BLAST version 2.1.3 using the default parameters specified by the NCBI (the National Center for Biotechnology Information; <http://www.ncbi.nlm.nih.gov/>) [Blosum 62 matrix; gap open penalty=11 and gap extension penalty=1].

(371) Further included within the scope of the invention are vectors such as expression vectors, comprising a nucleic acid sequence according to the invention. Cells transformed with such vectors are also included within the scope of the invention.

(372) As will be understood by the skilled artisan, general description of antibodies herein and methods of preparing and using the same also apply to individual antibody polypeptide constituents and antibody fragments.

(373) The antibodies of the present invention may be polyclonal or monoclonal antibodies. However, in preferred embodiments, they are monoclonal. In particular embodiments, antibodies of the present invention are human antibodies. Methods of producing polyclonal and monoclonal antibodies are known in the art and described generally, e.g., in U.S. Pat. No. 6,824,780.

(374) Typically, the antibodies of the present invention are produced recombinantly, using vectors and methods available in the art, as described further below. Human antibodies may also be generated by in vitro activated B cells (see U.S. Pat. Nos. 5,567,610 and 5,229,275).

(375) Human antibodies may also be produced in transgenic animals (e.g., mice) that are capable of producing a full repertoire of human antibodies in the absence of endogenous immunoglobulin production. For example, it has been described that the homozygous deletion of the antibody heavy-chain joining region (J.sub.H) gene in chimeric and germ-line mutant mice results in

complete inhibition of endogenous antibody production. Transfer of the human germ-line immunoglobulin gene array into such germ-line mutant mice results in the production of human antibodies upon antigen challenge. See, e.g., Jakobovits et al., Proc. Natl. Acad. Sci. USA, 90:2551 (1993); Jakobovits et al., Nature, 362:255-258 (1993); Bruggemann et al., Year in Immuno., 7:33 (1993); U.S. Pat. Nos. 5,545,806, 5,569,825, 5,591,669 (all of GenPharm); U.S. Pat. No. 5,545,807; and WO 97/17852. Such animals may be genetically engineered to produce human antibodies comprising a polypeptide of the present invention.

(376) In certain embodiments, antibodies of the present invention are chimeric antibodies that comprise sequences derived from both human and non-human sources. In particular embodiments, these chimeric antibodies are humanized or Primatized™. In practice, humanized antibodies are typically human antibodies in which some hypervariable region residues and possibly some FR residues are substituted by residues from analogous sites in rodent antibodies.

(377) In the context of the present invention, chimeric antibodies also include human antibodies wherein the human hypervariable region or one or more CDRs are retained, but one or more other regions of sequence have been replaced by corresponding sequences from a non-human animal.

(378) The choice of non-human sequences, both light and heavy, to be used in making the chimeric antibodies is important to reduce antigenicity and human anti-non-human antibody responses when the antibody is intended for human therapeutic use. It is further important that chimeric antibodies retain high binding affinity for the antigen and other favorable biological properties. To achieve this goal, according to a preferred method, chimeric antibodies are prepared by a process of analysis of the parental sequences and various conceptual chimeric products using three-dimensional models of the parental human and non-human sequences. Three-dimensional immunoglobulin models are commonly available and are familiar to those skilled in the art. Computer programs are available which illustrate and display probable three-dimensional conformational structures of selected candidate immunoglobulin sequences.

(379) Inspection of these displays permits analysis of the likely role of the residues in the functioning of the candidate immunoglobulin sequence, i.e., the analysis of residues that influence the ability of the candidate immunoglobulin to bind its antigen. In this way, FR residues can be selected and combined from the recipient and import sequences so that the desired antibody characteristic, such as increased affinity for the target antigen(s), is achieved. In general, the hypervariable region residues are directly and most substantially involved in influencing antigen binding.

(380) As noted above, antibodies (or immunoglobulins) can be divided into five different classes, based on differences in the amino acid sequences in the constant region of the heavy chains. All immunoglobulins within a given class have very similar heavy chain constant regions. These differences can be detected by sequence studies or more commonly by serological means (i.e. by the use of antibodies directed to these differences). Antibodies, or fragments thereof, of the present invention may be any class, and may, therefore, have a gamma, mu, alpha, delta, or epsilon heavy chain. A gamma chain may be gamma 1, gamma 2, gamma 3, or gamma 4; and an alpha chain may be alpha 1 or alpha 2.

(381) In a preferred embodiment, an antibody of the present invention, or fragment thereof, is an IgG. IgG is considered the most versatile immunoglobulin, because it is capable of carrying out all of the functions of immunoglobulin molecules. IgG is the major Ig in serum, and the only class of Ig that crosses the placenta. IgG also fixes complement, although the IgG.sub.4 subclass does not. Macrophages, monocytes, PMN's and some lymphocytes have Fc receptors for the Fc region of IgG. Not all subclasses bind equally well: IgG.sub.2 and IgG.sub.4 do not bind to Fc receptors. A consequence of binding to the Fc receptors on PMN's, monocytes and macrophages is that the cell can now internalize the antigen better. IgG is an opsonin that enhances phagocytosis. Binding of IgG to Fc receptors on other types of cells results in the activation of other functions. Antibodies of the present invention may be of any IgG subclass.

(382) In another preferred embodiment, an antibody, or fragment thereof, of the present invention is an IgE. IgE is the least common serum Ig since it binds very tightly to Fc receptors on basophils and mast cells even before interacting with antigen. As a consequence of its binding to basophils and mast cells, IgE is involved in allergic reactions. Binding of the allergen to the IgE on the cells results in the release of various pharmacological mediators that result in allergic symptoms. IgE also plays a role in parasitic helminth diseases. Eosinophils have Fc receptors for IgE and binding of eosinophils to IgE-coated helminths results in killing of the parasite. IgE does not fix complement.

(383) In various embodiments, antibodies of the present invention, and fragments thereof, comprise a variable light chain that is either kappa or lambda. The lambda chain may be any of subtype, including, e.g., lambda 1, lambda 2, lambda 3, and lambda 4.

(384) As noted above, the present invention further provides antibody fragments comprising a polypeptide of the present invention. In certain circumstances there are advantages of using antibody fragments, rather than whole antibodies. For example, the smaller size of the fragments allows for rapid clearance, and may lead to improved access to certain tissues, such as solid tumors. Examples of antibody fragments include: Fab, Fab', F(ab')₂ and Fv fragments; diabodies; linear antibodies; single-chain antibodies; and multispecific antibodies formed from antibody fragments.

(385) Various techniques have been developed for the production of antibody fragments. Traditionally, these fragments were derived via proteolytic digestion of intact antibodies (see, e.g., Morimoto et al., *Journal of Biochemical and Biophysical Methods* 24:107-117 (1992); and Brennan et al., *Science*, 229:81 (1985)). However, these fragments can now be produced directly by recombinant host cells. Fab, Fv and ScFv antibody fragments can all be expressed in and secreted from *E. coli*, thus allowing the facile production of large amounts of these fragments. Fab'-SH fragments can be directly recovered from *E. coli* and chemically coupled to form F(ab')₂ fragments (Carter et al., *Bio/Technology* 10:163-167 (1992)). According to another approach, F(ab')₂ fragments can be isolated directly from recombinant host cell culture. Fab and F(ab')₂ fragment with increased in vivo half-life comprising a salvage receptor binding epitope residues are described in U.S. Pat. No. 5,869,046. Other techniques for the production of antibody fragments will be apparent to the skilled practitioner.

(386) In other embodiments, the antibody of choice is a single chain Fv fragment (scFv). See WO 93/16185; U.S. Pat. Nos. 5,571,894; and 5,587,458. Fv and sFv are the only species with intact combining sites that are devoid of constant regions. Thus, they are suitable for reduced nonspecific binding during in vivo use. sFv fusion proteins may be constructed to yield fusion of an effector protein at either the amino or the carboxy terminus of an sFv. See *Antibody Engineering*, ed. Borrebaeck, supra. The antibody fragment may also be a "linear antibody", e.g., as described in U.S. Pat. No. 5,641,870 for example. Such linear antibody fragments may be monospecific or bispecific.

(387) In certain embodiments, antibodies of the present invention are bispecific or multispecific. Bispecific antibodies are antibodies that have binding specificities for at least two different epitopes. Exemplary bispecific antibodies may bind to two different epitopes of a single antigen. Other such antibodies may combine a first antigen binding site with a binding site for a second antigen. Alternatively, an anti-HIV1 arm may be combined with an arm that binds to a triggering molecule on a leukocyte, such as a T-cell receptor molecule (e.g., CD3), or Fc receptors for IgG (FcγR), such as FcγRI (CD64), FcγRII (CD32) and FcγRIII (CD16), so as to focus and localize cellular defense mechanisms to the infected cell. Bispecific antibodies may also be used to localize cytotoxic agents to infected cells. These antibodies possess an HIV1-binding arm and an arm that binds the cytotoxic agent (e.g., saporin, anti-interferon-α, *vinca* alkaloid, ricin A chain, methotrexate or radioactive isotope hapten). Bispecific antibodies can be prepared as full length antibodies or antibody fragments (e.g., F(ab')₂ bispecific antibodies). WO 96/16673 describes a bispecific anti-ErbB2/anti-FcγRIII antibody and U.S. Pat. No. 5,837,234 discloses a bispecific anti-

ErbB2/anti-FcγRI antibody. A bispecific anti-ErbB2/Fcα antibody is shown in WO98/02463. U.S. Pat. No. 5,821,337 teaches a bispecific anti-ErbB2/anti-CD3 antibody.

(388) Methods for making bispecific antibodies are known in the art. Traditional production of full length bispecific antibodies is based on the co-expression of two immunoglobulin heavy chain-light chain pairs, where the two chains have different specificities (Millstein et al., *Nature*, 305:537-539 (1983)). Because of the random assortment of immunoglobulin heavy and light chains, these hybridomas (quadromas) produce a potential mixture of ten different antibody molecules, of which only one has the correct bispecific structure. Purification of the correct molecule, which is usually done by affinity chromatography steps, is rather cumbersome, and the product yields are low. Similar procedures are disclosed in WO 93/08829, and in Traunecker et al., *EMBO J.*, 10:3655-3659 (1991).

(389) According to a different approach, antibody variable regions with the desired binding specificities (antibody-antigen combining sites) are fused to immunoglobulin constant domain sequences. Preferably, the fusion is with an Ig heavy chain constant domain, comprising at least part of the hinge, C.sub.H2, and C.sub.H3 regions. It is preferred to have the first heavy-chain constant region (C.sub.H1) containing the site necessary for light chain bonding, present in at least one of the fusions. DNAs encoding the immunoglobulin heavy chain fusions and, if desired, the immunoglobulin light chain, are inserted into separate expression vectors, and are co-transfected into a suitable host cell. This provides for greater flexibility in adjusting the mutual proportions of the three polypeptide fragments in embodiments when unequal ratios of the three polypeptide chains used in the construction provide the optimum yield of the desired bispecific antibody. It is, however, possible to insert the coding sequences for two or all three polypeptide chains into a single expression vector when the expression of at least two polypeptide chains in equal ratios results in high yields or when the ratios have no significant effect on the yield of the desired chain combination.

(390) In a preferred embodiment of this approach, the bispecific antibodies are composed of a hybrid immunoglobulin heavy chain with a first binding specificity in one arm, and a hybrid immunoglobulin heavy chain-light chain pair (providing a second binding specificity) in the other arm. It was found that this asymmetric structure facilitates the separation of the desired bispecific compound from unwanted immunoglobulin chain combinations, as the presence of an immunoglobulin light chain in only one half of the bispecific molecule provides for a facile way of separation. This approach is disclosed in WO 94/04690. For further details of generating bispecific antibodies see, for example, Suresh et al., *Methods in Enzymology*, 121:210 (1986).

(391) According to another approach described in U.S. Pat. No. 5,731,168, the interface between a pair of antibody molecules can be engineered to maximize the percentage of heterodimers that are recovered from recombinant cell culture. The preferred interface comprises at least a part of the C.sub.H3 domain. In this method, one or more small amino acid side chains from the interface of the first antibody molecule are replaced with larger side chains (e.g., tyrosine or tryptophan). Compensatory "cavities" of identical or similar size to the large side chain(s) are created on the interface of the second antibody molecule by replacing large amino acid side chains with smaller ones (e.g., alanine or threonine). This provides a mechanism for increasing the yield of the heterodimer over other unwanted end-products such as homodimers.

(392) Bispecific antibodies include cross-linked or "heteroconjugate" antibodies. For example, one of the antibodies in the heteroconjugate can be coupled to avidin, the other to biotin. Such antibodies have, for example, been proposed to target immune system cells to unwanted cells (U.S. Pat. No. 4,676,980), and for treatment of HIV infection (WO 91/00360, WO 92/200373, and EP 03089). Heteroconjugate antibodies may be made using any convenient cross-linking methods. Suitable cross-linking agents are well known in the art, and are disclosed in U.S. Pat. No. 4,676,980, along with a number of cross-linking techniques.

(393) Techniques for generating bispecific antibodies from antibody fragments have also been

described in the literature. For example, bispecific antibodies can be prepared using chemical linkage. Brennan et al., Science, 229: 81 (1985) describe a procedure wherein intact antibodies are proteolytically cleaved to generate F(ab')₂ fragments. These fragments are reduced in the presence of the dithiol complexing agent, sodium arsenite, to stabilize vicinal dithiols and prevent intermolecular disulfide formation. The Fab' fragments generated are then converted to thionitrobenzoate (TNB) derivatives. One of the Fab'-TNB derivatives is then reconverted to the Fab'-thiol by reduction with mercaptoethylamine and is mixed with an equimolar amount of the other Fab'-TNB derivative to form the bispecific antibody. The bispecific antibodies produced can be used as agents for the selective immobilization of enzymes.

(394) Recent progress has facilitated the direct recovery of Fab'-SH fragments from *E. coli*, which can be chemically coupled to form bispecific antibodies. Shalaby et al., J. Exp. Med., 175: 217-225 (1992) describe the production of a humanized bispecific antibody F(ab')₂ molecule. Each Fab' fragment was separately secreted from *E. coli* and subjected to directed chemical coupling in vitro to form the bispecific antibody. The bispecific antibody thus formed was able to bind to cells overexpressing the ErbB2 receptor and normal human T cells, as well as trigger the lytic activity of human cytotoxic lymphocytes against human breast tumor targets.

(395) Various techniques for making and isolating bispecific antibody fragments directly from recombinant cell culture have also been described. For example, bispecific antibodies have been produced using leucine zippers. Kostelny et al., J. Immunol., 148(5):1547-1553 (1992). The leucine zipper peptides from the Fos and Jun proteins were linked to the Fab' portions of two different antibodies by gene fusion. The antibody homodimers were reduced at the hinge region to form monomers and then re-oxidized to form the antibody heterodimers. This method can also be utilized for the production of antibody homodimers. The "diabody" technology described by Hollinger et al., Proc. Natl. Acad. Sci. USA, 90:6444-6448 (1993) has provided an alternative mechanism for making bispecific antibody fragments. The fragments comprise a V_{sub}.H connected to a V_{sub}.L by a linker that is too short to allow pairing between the two domains on the same chain. Accordingly, the V_{sub}.H and V_{sub}.L domains of one fragment are forced to pair with the complementary V_{sub}.L and V_{sub}.H domains of another fragment, thereby forming two antigen-binding sites. Another strategy for making bispecific antibody fragments by the use of single-chain Fv (sFv) dimers has also been reported. See Gruber et al., J. Immunol., 152:5368 (1994).

(396) Antibodies with more than two valencies are contemplated. For example, trispecific antibodies can be prepared. Tutt et al., J. Immunol. 147: 60 (1991). A multivalent antibody may be internalized (and/or catabolized) faster than a bivalent antibody by a cell expressing an antigen to which the antibodies bind. The antibodies of the present invention can be multivalent antibodies with three or more antigen binding sites (e.g., tetravalent antibodies), which can be readily produced by recombinant expression of nucleic acid encoding the polypeptide chains of the antibody. The multivalent antibody can comprise a dimerization domain and three or more antigen binding sites. The preferred dimerization domain comprises (or consists of) an Fc region or a hinge region. In this scenario, the antibody will comprise an Fc region and three or more antigen binding sites amino-terminal to the Fc region. The preferred multivalent antibody herein comprises (or consists of) three to about eight, but preferably four, antigen binding sites. The multivalent antibody comprises at least one polypeptide chain (and preferably two polypeptide chains), wherein the polypeptide chain(s) comprise two or more variable regions. For instance, the polypeptide chain(s) may comprise VD1-(X1)_{sub}.n-VD2-(X2)_{sub}.n-Fc, wherein VD1 is a first variable region, VD2 is a second variable region, Fc is one polypeptide chain of an Fc region, X1 and X2 represent an amino acid or polypeptide, and n is 0 or 1. For instance, the polypeptide chain(s) may comprise: VH-CH1-flexible linker-VH-CH1-Fc region chain; or VH-CH1-VH-CH1-Fc region chain. The multivalent antibody herein preferably further comprises at least two (and preferably four) light chain variable region polypeptides. The multivalent antibody herein may, for instance,

comprise from about two to about eight light chain variable region polypeptides. The light chain variable region polypeptides contemplated here comprise a light chain variable region and, optionally, further comprise a C.sub.L domain.

(397) Antibodies of the invention further include single chain antibodies. In particular embodiments, antibodies of the invention are internalizing antibodies.

(398) Amino acid sequence modification(s) of the antibodies described herein are contemplated. For example, it may be desirable to improve the binding affinity and/or other biological properties of the antibody. Amino acid sequence variants of the antibody may be prepared by introducing appropriate nucleotide changes into a polynucleotide that encodes the antibody, or a chain thereof, or by peptide synthesis. Such modifications include, for example, deletions from, and/or insertions into and/or substitutions of, residues within the amino acid sequences of the antibody. Any combination of deletion, insertion, and substitution may be made to arrive at the final antibody, provided that the final construct possesses the desired characteristics. The amino acid changes also may alter post-translational processes of the antibody, such as changing the number or position of glycosylation sites. Any of the variations and modifications described above for polypeptides of the present invention may be included in antibodies of the present invention.

(399) A useful method for identification of certain residues or regions of an antibody that are preferred locations for mutagenesis is called "alanine scanning mutagenesis" as described by Cunningham and Wells in *Science*, 244:1081-1085 (1989). Here, a residue or group of target residues are identified (e.g., charged residues such as arg, asp, his, lys, and glu) and replaced by a neutral or negatively charged amino acid (most preferably alanine or polyalanine) to affect the interaction of the amino acids with PSCA antigen. Those amino acid locations demonstrating functional sensitivity to the substitutions then are refined by introducing further or other variants at, or for, the sites of substitution. Thus, while the site for introducing an amino acid sequence variation is predetermined, the nature of the mutation per se need not be predetermined. For example, to analyze the performance of a mutation at a given site, ala scanning or random mutagenesis is conducted at the target codon or region and the expressed anti-antibody variants are screened for the desired activity.

(400) Amino acid sequence insertions include amino- and/or carboxyl-terminal fusions ranging in length from one residue to polypeptides containing a hundred or more residues, as well as intrasequence insertions of single or multiple amino acid residues. Examples of terminal insertions include an antibody with an N-terminal methionyl residue or the antibody fused to a cytotoxic polypeptide. Other insertional variants of an antibody include the fusion to the N- or C-terminus of the antibody to an enzyme (e.g., for ADEPT) or a polypeptide that increases the serum half-life of the antibody.

(401) Another type of variant is an amino acid substitution variant. These variants have at least one amino acid residue in the antibody molecule replaced by a different residue. The sites of greatest interest for substitutional mutagenesis include the hypervariable regions, but FR alterations are also contemplated. Conservative and non-conservative substitutions are contemplated.

(402) Substantial modifications in the biological properties of the antibody are accomplished by selecting substitutions that differ significantly in their effect on maintaining (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a sheet or helical conformation, (b) the charge or hydrophobicity of the molecule at the target site, or (c) the bulk of the side chain.

(403) Any cysteine residue not involved in maintaining the proper conformation of the antibody also may be substituted, generally with serine, to improve the oxidative stability of the molecule and prevent aberrant crosslinking. Conversely, cysteine bond(s) may be added to the antibody to improve its stability (particularly where the antibody is an antibody fragment such as an Fv fragment).

(404) One type of substitutional variant involves substituting one or more hypervariable region

residues of a parent antibody. Generally, the resulting variant(s) selected for further development will have improved biological properties relative to the parent antibody from which they are generated. A convenient way for generating such substitutional variants involves affinity maturation using phage display. Briefly, several hypervariable region sites (e.g., 6-7 sites) are mutated to generate all possible amino substitutions at each site. The antibody variants thus generated are displayed in a monovalent fashion from filamentous phage particles as fusions to the gene III product of M13 packaged within each particle. The phage-displayed variants are then screened for their biological activity (e.g., binding affinity) as herein disclosed. In order to identify candidate hypervariable region sites for modification, alanine scanning mutagenesis can be performed to identify hypervariable region residues contributing significantly to antigen binding. Alternatively, or additionally, it may be beneficial to analyze a crystal structure of the antigen-antibody complex to identify contact points between the antibody and an antigen or infected cell. Such contact residues and neighboring residues are candidates for substitution according to the techniques elaborated herein. Once such variants are generated, the panel of variants is subjected to screening as described herein and antibodies with superior properties in one or more relevant assays may be selected for further development.

(405) Another type of amino acid variant of the antibody alters the original glycosylation pattern of the antibody. By altering is meant deleting one or more carbohydrate moieties found in the antibody, and/or adding one or more glycosylation sites that are not present in the antibody. Glycosylation of antibodies is typically either N-linked or O-linked. N-linked refers to the attachment of the carbohydrate moiety to the side chain of an asparagine residue. The tripeptide sequences asparagine-X-serine and asparagine-X-threonine, where X is any amino acid except proline, are the recognition sequences for enzymatic attachment of the carbohydrate moiety to the asparagine side chain. Thus, the presence of either of these tripeptide sequences in a polypeptide creates a potential glycosylation site. O-linked glycosylation refers to the attachment of one of the sugars N-aceylgalactosamine, galactose, or xylose to a hydroxyamino acid, most commonly serine or threonine, although 5-hydroxyproline or 5-hydroxylysine may also be used. Addition of glycosylation sites to the antibody is conveniently accomplished by altering the amino acid sequence such that it contains one or more of the above-described tripeptide sequences (for N-linked glycosylation sites). The alteration may also be made by the addition of, or substitution by, one or more serine or threonine residues to the sequence of the original antibody (for O-linked glycosylation sites).

(406) The antibody of the invention is modified with respect to effector function, e.g., so as to enhance antigen-dependent cell-mediated cytotoxicity (ADCC) and/or complement dependent cytotoxicity (CDC) of the antibody. This may be achieved by introducing one or more amino acid substitutions in an Fc region of the antibody. Alternatively or additionally, cysteine residue(s) may be introduced in the Fc region, thereby allowing interchain disulfide bond formation in this region. The homodimeric antibody thus generated may have improved internalization capability and/or increased complement-mediated cell killing and antibody-dependent cellular cytotoxicity (ADCC). See Caron et al., *J. Exp Med.* 176:1191-1195 (1992) and Shopes, B. J. *Immunol.* 148:2918-2922 (1992). Homodimeric antibodies with enhanced anti-infection activity may also be prepared using heterobifunctional cross-linkers as described in Wolff et al., *Cancer Research* 53:2560-2565 (1993). Alternatively, an antibody can be engineered which has dual Fc regions and may thereby have enhanced complement lysis and ADCC capabilities. See Stevenson et al., *Anti-Cancer Drug Design* 3:219-230 (1989). To increase the serum half-life of the antibody, one may incorporate a salvage receptor binding epitope into the antibody (especially an antibody fragment) as described in U.S. Pat. No. 5,739,277, for example. As used herein, the term "salvage receptor binding epitope" refers to an epitope of the Fc region of an IgG molecule (e.g., IgG.sub.1, IgG.sub.2, IgG.sub.3, or IgG.sub.4) that is responsible for increasing the in vivo serum half-life of the IgG molecule.

(407) Antibodies of the present invention may also be modified to include an epitope tag or label,

e.g., for use in purification or diagnostic applications. The invention also pertains to therapy with immunoconjugates comprising an antibody conjugated to an anti-cancer agent such as a cytotoxic agent or a growth inhibitory agent. Chemotherapeutic agents useful in the generation of such immunoconjugates have been described above.

(408) Conjugates of an antibody and one or more small molecule toxins, such as a calicheamicin, maytansinoids, a trichothene, and CC1065, and the derivatives of these toxins that have toxin activity, are also contemplated herein.

(409) In one preferred embodiment, an antibody (full length or fragments) of the invention is conjugated to one or more maytansinoid molecules. Maytansinoids are mitototic inhibitors that act by inhibiting tubulin polymerization. Maytansine was first isolated from the east African shrub *Maytenus serrata* (U.S. Pat. No. 3,896,111). Subsequently, it was discovered that certain microbes also produce maytansinoids, such as maytansinol and C-3 maytansinol esters (U.S. Pat. No. 4,151,042). Synthetic maytansinol and derivatives and analogues thereof are disclosed, for example, in U.S. Pat. Nos. 4,137,230; 4,248,870; 4,256,746; 4,260,608; 4,265,814; 4,294,757; 4,307,016; 4,308,268; 4,308,269; 4,309,428; 4,313,946; 4,315,929; 4,317,821; 4,322,348; 4,331,598; 4,361,650; 4,364,866; 4,424,219; 4,450,254; 4,362,663; and 4,371,533.

(410) In an attempt to improve their therapeutic index, maytansine and maytansinoids have been conjugated to antibodies specifically binding to tumor cell antigens. Immunoconjugates containing maytansinoids and their therapeutic use are disclosed, for example, in U.S. Pat. Nos. 5,208,020, 5,416,064 and European Patent EP 0 425 235_B1. Liu et al., Proc. Natl. Acad. Sci. USA 93:8618-8623 (1996) described immunoconjugates comprising a maytansinoid designated DM1 linked to the monoclonal antibody C242 directed against human colorectal cancer. The conjugate was found to be highly cytotoxic towards cultured colon cancer cells, and showed antitumor activity in an in vivo tumor growth assay.

(411) Antibody-maytansinoid conjugates are prepared by chemically linking an antibody to a maytansinoid molecule without significantly diminishing the biological activity of either the antibody or the maytansinoid molecule. An average of 3-4 maytansinoid molecules conjugated per antibody molecule has shown efficacy in enhancing cytotoxicity of target cells without negatively affecting the function or solubility of the antibody, although even one molecule of toxin/antibody would be expected to enhance cytotoxicity over the use of naked antibody. Maytansinoids are well known in the art and can be synthesized by known techniques or isolated from natural sources. Suitable maytansinoids are disclosed, for example, in U.S. Pat. No. 5,208,020 and in the other patents and nonpatent publications referred to hereinabove. Preferred maytansinoids are maytansinol and maytansinol analogues modified in the aromatic ring or at other positions of the maytansinol molecule, such as various maytansinol esters.

(412) There are many linking groups known in the art for making antibody conjugates, including, for example, those disclosed in U.S. Pat. No. 5,208,020 or EP Patent 0 425 235_B1, and Chari et al., Cancer Research 52: 127-131 (1992). The linking groups include disulfide groups, thioether groups, acid labile groups, photolabile groups, peptidase labile groups, or esterase labile groups, as disclosed in the above-identified patents, disulfide and thioether groups being preferred.

(413) Immunoconjugates may be made using a variety of bifunctional protein coupling agents such as N-succinimidyl-3-(2-pyridyldithio)propionate (SPDP), succinimidyl-4-(N-maleimidomethyl)cyclohexane-1-carboxylate, iminothiolane (IT), bifunctional derivatives of imidoesters (such as dimethyl adipimidate HCL), active esters (such as disuccinimidyl suberate), aldehydes (such as glutaraldehyde), bis-azido compounds (such as bis (p-azidobenzoyl)hexanediamine), bis-diazonium derivatives (such as bis-(p-diazoniumbenzoyl)-ethylenediamine), diisocyanates (such as toluene 2,6-diisocyanate), and bis-active fluorine compounds (such as 1,5-difluoro-2,4-dinitrobenzene). Particularly preferred coupling agents include N-succinimidyl-3-(2-pyridyldithio) propionate (SPDP) (Carlsson et al., Biochem. J. 173:723-737 [1978]) and N-succinimidyl-4-(2-pyridylthio) pentanoate (SPP) to provide for a

disulfide linkage. For example, a ricin immunotoxin can be prepared as described in Vitetta et al., Science 238: 1098 (1987). Carbon-14-labeled 1-isothiocyanatobenzyl-3-methyldiethylene triaminepentaacetic acid (MX-DTPA) is an exemplary chelating agent for conjugation of radionucleotide to the antibody. See WO94/11026. The linker may be a "cleavable linker" facilitating release of the cytotoxic drug in the cell. For example, an acid-labile linker, Cancer Research 52: 127-131 (1992); U.S. Pat. No. 5,208,020) may be used.

(414) Another immunoconjugate of interest comprises an antibody conjugated to one or more calicheamicin molecules. The calicheamicin family of antibiotics is capable of producing double-stranded DNA breaks at sub-picomolar concentrations. For the preparation of conjugates of the calicheamicin family, see U.S. Pat. Nos. 5,712,374, 5,714,586, 5,739,116, 5,767,285, 5,770,701, 5,770,710, 5,773,001, 5,877,296 (all to American Cyanamid Company). Another drug that the antibody can be conjugated is QFA which is an antifolate. Both calicheamicin and QFA have intracellular sites of action and do not readily cross the plasma membrane. Therefore, cellular uptake of these agents through antibody mediated internalization greatly enhances their cytotoxic effects.

(415) Examples of other agents that can be conjugated to the antibodies of the invention include BCNU, streptozocin, vincristine and 5-fluorouracil, the family of agents known collectively LL-E33288 complex described in U.S. Pat. Nos. 5,053,394, 5,770,710, as well as esperamicins (U.S. Pat. No. 5,877,296).

(416) Enzymatically active toxins and fragments thereof that can be used include, e.g., diphtheria A chain, nonbinding active fragments of diphtheria toxin, exotoxin A chain (from *Pseudomonas aeruginosa*), ricin A chain, abrin A chain, modeccin A chain, alpha-sarcin, *Aleurites fordii* proteins, dianthin proteins, *Phytolaca americana* proteins (PAPI, PAPII, and PAP-S), *Momordica charantia* inhibitor, curcin, crotin, *Sapaonaria officinalis* inhibitor, gelonin, mitogellin, restrictocin, phenomycin, enomycin and the tricothecenes. See, for example, WO 93/21232.

(417) The present invention further includes an immunoconjugate formed between an antibody and a compound with nucleolytic activity (e.g., a ribonuclease or a DNA endonuclease such as a deoxyribonuclease; DNase).

(418) For selective destruction of infected cells, the antibody includes a highly radioactive atom. A variety of radioactive isotopes are available for the production of radioconjugated anti-PSCA antibodies. Examples include At.sup.211, I.sup.131, I.sup.125, y.sup.90, Re.sup.186, Re.sup.188, Sm.sup.153, Bi.sup.212, P.sup.32, Pb.sup.212 and radioactive isotopes of Lu. When the conjugate is used for diagnosis, it may comprise a radioactive atom for scintigraphic studies, for example tc.sup.99m or I.sup.123, or a spin label for nuclear magnetic resonance (NMR) imaging (also known as magnetic resonance imaging, mri), such as iodine-123, iodine-131, indium-111, fluorine-19, carbon-13, nitrogen-15, oxygen-17, gadolinium, manganese or iron.

(419) The radio- or other label is incorporated in the conjugate in known ways. For example, the peptide may be biosynthesized or may be synthesized by chemical amino acid synthesis using suitable amino acid precursors involving, for example, fluorine-19 in place of hydrogen. Labels such as tc.sup.99m or I.sup.123, Re.sup.186, Re.sup.188 and In.sup.111 can be attached via a cysteine residue in the peptide. Yttrium-90 can be attached via a lysine residue. The IODOGEN method (Fraker et al. (1978) Biochem. Biophys. Res. Commun. 80: 49-57 can be used to incorporate iodine-123. "Monoclonal Antibodies in Immunoscintigraphy" (Chatal, CRC Press 1989) describes other methods in detail.

(420) Alternatively, a fusion protein comprising the antibody and cytotoxic agent is made, e.g., by recombinant techniques or peptide synthesis. The length of DNA may comprise respective regions encoding the two portions of the conjugate either adjacent one another or separated by a region encoding a linker peptide which does not destroy the desired properties of the conjugate. The antibodies of the present invention are also used in antibody dependent enzyme mediated prodrug therapy (ADET) by conjugating the antibody to a prodrug-activating enzyme which converts a

prodrug (e.g., a peptidyl chemotherapeutic agent, see WO81/01145) to an active anti-cancer drug (see, e.g., WO 88/07378 and U.S. Pat. No. 4,975,278).

(421) The enzyme component of the immunoconjugate useful for ADEPT includes any enzyme capable of acting on a prodrug in such a way so as to convert it into its more active, cytotoxic form. Enzymes that are useful in the method of this invention include, but are not limited to, alkaline phosphatase useful for converting phosphate-containing prodrugs into free drugs; arylsulfatase useful for converting sulfate-containing prodrugs into free drugs; cytosine deaminase useful for converting non-toxic 5-fluorocytosine into the anti-cancer drug, 5-fluorouracil; proteases, such as serratiopeptidase, thermolysin, subtilisin, carboxypeptidases and cathepsins (such as cathepsins B and L), that are useful for converting peptide-containing prodrugs into free drugs; D-alanylcarboxypeptidases, useful for converting prodrugs that contain D-amino acid substituents; carbohydrate-cleaving enzymes such as β -galactosidase and neuraminidase useful for converting glycosylated prodrugs into free drugs; β -lactamase useful for converting drugs derivatized with β -lactams into free drugs; and penicillin amidases, such as penicillin V amidase or penicillin G amidase, useful for converting drugs derivatized at their amine nitrogens with phenoxyacetyl or phenylacetyl groups, respectively, into free drugs. Alternatively, antibodies with enzymatic activity, also known in the art as "abzymes", can be used to convert the prodrugs of the invention into free active drugs (see, e.g., Massey, *Nature* 328: 457-458 (1987)). Antibody-abzyme conjugates can be prepared as described herein for delivery of the abzyme to a infected cell population.

(422) The enzymes of this invention can be covalently bound to the antibodies by techniques well known in the art such as the use of the heterobifunctional crosslinking reagents discussed above. Alternatively, fusion proteins comprising at least the antigen binding region of an antibody of the invention linked to at least a functionally active portion of an enzyme of the invention can be constructed using recombinant DNA techniques well known in the art (see, e.g., Neuberger et al., *Nature*, 312: 604-608 (1984)).

(423) Other modifications of the antibody are contemplated herein. For example, the antibody may be linked to one of a variety of nonproteinaceous polymers, e.g., polyethylene glycol, polypropylene glycol, polyoxyalkylenes, or copolymers of polyethylene glycol and polypropylene glycol. The antibody also may be entrapped in microcapsules prepared, for example, by coacervation techniques or by interfacial polymerization (for example, hydroxymethylcellulose or gelatin-microcapsules and poly-(methylmethacrylate) microcapsules, respectively), in colloidal drug delivery systems (for example, liposomes, albumin microspheres, microemulsions, nano-particles and nanocapsules), or in macroemulsions. Such techniques are disclosed in Remington's *Pharmaceutical Sciences*, 16th edition, Oslo, A., Ed., (1980).

(424) The antibodies disclosed herein are also formulated as immunoliposomes. A "liposome" is a small vesicle composed of various types of lipids, phospholipids and/or surfactant that is useful for delivery of a drug to a mammal. The components of the liposome are commonly arranged in a bilayer formation, similar to the lipid arrangement of biological membranes. Liposomes containing the antibody are prepared by methods known in the art, such as described in Epstein et al., *Proc. Natl. Acad. Sci. USA*, 82:3688 (1985); Hwang et al., *Proc. Natl. Acad. Sci. USA*, 77:4030 (1980); U.S. Pat. Nos. 4,485,045 and 4,544,545; and WO97/38731 published Oct. 23, 1997. Liposomes with enhanced circulation time are disclosed in U.S. Pat. No. 5,013,556.

(425) Particularly useful liposomes can be generated by the reverse phase evaporation method with a lipid composition comprising phosphatidylcholine, cholesterol and PEG-derivatized phosphatidylethanolamine (PEG-PE). Liposomes are extruded through filters of defined pore size to yield liposomes with the desired diameter. Fab' fragments of the antibody of the present invention can be conjugated to the liposomes as described in Martin et al., *J. Biol. Chem.* 257: 286-288 (1982) via a disulfide interchange reaction. A chemotherapeutic agent is optionally contained within the liposome. See Gabizon et al., *J. National Cancer Inst.* 81(19):1484 (1989). Antibodies of the present invention, or fragments thereof, may possess any of a variety of biological or functional

characteristics. In certain embodiments, these antibodies are HIV1 protein specific antibodies, indicating that they specifically bind to or preferentially bind to HIV1 as compared to a normal control cell.

(426) In particular embodiments, an antibody of the present invention is an antagonist antibody, which partially or fully blocks or inhibits a biological activity of a polypeptide or cell to which it specifically or preferentially binds. In other embodiments, an antibody of the present invention is a growth inhibitory antibody, which partially or fully blocks or inhibits the growth of an infected cell to which it binds. In another embodiment, an antibody of the present invention induces apoptosis. In yet another embodiment, an antibody of the present invention induces or promotes antibody-dependent cell-mediated cytotoxicity or complement dependent cytotoxicity.

(427) HIV1-expressing cells or virus described above are used to screen the biological sample obtained from a patient infected with HIV1 for the presence of antibodies that preferentially bind to the cell expressing HIV1 polypeptides using standard biological techniques. For example, in certain embodiments, the antibodies may be labeled, and the presence of label associated with the cell detected, e.g., using FMAT or FACs analysis. In particular embodiments, the biological sample is blood, serum, plasma, bronchial lavage, or saliva. Methods of the present invention may be practiced using high throughput techniques.

(428) Identified human antibodies may then be characterized further. For example the particular conformational epitopes with in the HIV1 polypeptides that are necessary or sufficient for binding of the antibody may be determined, e.g., using site-directed mutagenesis of expressed HIV1 polypeptides. These methods may be readily adapted to identify human antibodies that bind any protein expressed on a cell surface. Furthermore, these methods may be adapted to determine binding of the antibody to the virus itself, as opposed to a cell expressing recombinant HIV1 or infected with the virus.

(429) Polynucleotide sequences encoding the antibodies, variable regions thereof, or antigen-binding fragments thereof may be subcloned into expression vectors for the recombinant production of human anti-HIV1 antibodies. In one embodiment, this is accomplished by obtaining mononuclear cells from the patient from the serum containing the identified HIV1 antibody was obtained; producing B cell clones from the mononuclear cells; inducing the B cells to become antibody-producing plasma cells; and screening the supernatants produced by the plasma cells to determine if it contains the HIV1 antibody. Once a B cell clone that produces an HIV1 antibody is identified, reverse-transcription polymerase chain reaction (RT-PCR) is performed to clone the DNAs encoding the variable regions or portions thereof of the HIV1 antibody. These sequences are then subcloned into expression vectors suitable for the recombinant production of human HIV1 antibodies. The binding specificity may be confirmed by determining the recombinant antibody's ability to bind cells expressing HIV1 polypeptide.

(430) In particular embodiments of the methods described herein, B cells isolated from peripheral blood or lymph nodes are sorted, e.g., based on their being CD19 positive, and plated, e.g., as low as a single cell specificity per well, e.g., in 96, 384, or 1536 well configurations. The cells are induced to differentiate into antibody-producing cells, e.g., plasma cells, and the culture supernatants are harvested and tested for binding to cells expressing the infectious agent polypeptide on their surface using, e.g., FMAT or FACS analysis. Positive wells are then subjected to whole well RT-PCR to amplify heavy and light chain variable regions of the IgG molecule expressed by the clonal daughter plasma cells. The resulting PCR products encoding the heavy and light chain variable regions, or portions thereof, are subcloned into human antibody expression vectors for recombinant expression. The resulting recombinant antibodies are then tested to confirm their original binding specificity and may be further tested for pan-specificity across various strains of isolates of the infectious agent.

(431) Thus, in one embodiment, a method of identifying HIV1 antibodies is practiced as follows. First, full length or approximately full length HIV1 cDNAs are transfected into a cell line for

expression of HIV1 polypeptides. Secondly, individual human plasma or sera samples are tested for antibodies that bind the cell-expressed HIV1 polypeptides. And lastly, MAbs derived from plasma- or serum-positive individuals are characterized for binding to the same cell-expressed HIV1 polypeptides. Further definition of the fine specificities of the MAbs can be performed at this point.

(432) Polynucleotides that encode the HIV1 antibodies or portions thereof of the present invention may be isolated from cells expressing HIV1 antibodies, according to methods available in the art and described herein, including amplification by polymerase chain reaction using primers specific for conserved regions of human antibody polypeptides. For example, light chain and heavy chain variable regions may be cloned from the B cell according to molecular biology techniques described in WO 92/02551; U.S. Pat. No. 5,627,052; or Babcook et al., Proc. Natl. Acad. Sci. USA 93:7843-48 (1996). In certain embodiments, polynucleotides encoding all or a region of both the heavy and light chain variable regions of the IgG molecule expressed by the clonal daughter plasma cells expressing the HIV1 antibody are subcloned and sequenced. The sequence of the encoded polypeptide may be readily determined from the polynucleotide sequence.

(433) Isolated polynucleotides encoding a polypeptide of the present invention may be subcloned into an expression vector to recombinantly produce antibodies and polypeptides of the present invention, using procedures known in the art and described herein.

(434) Binding properties of an antibody (or fragment thereof) to HIV1 polypeptides or HIV1-infected cells or tissues may generally be determined and assessed using immunodetection methods including, for example, immunofluorescence-based assays, such as immunohistochemistry (IHC) and/or fluorescence-activated cell sorting (FACS). Immunoassay methods may include controls and procedures to determine whether antibodies bind specifically to HIV1 polypeptides from one or more specific clades or strains of HIV, and do not recognize or cross-react with normal control cells.

(435) Following pre-screening of serum to identify patients that produce antibodies to an infectious agent or polypeptide thereof, e.g., HIV1, the methods of the present invention typically include the isolation or purification of B cells from a biological sample previously obtained from a patient or subject. The patient or subject may be currently or previously diagnosed with or suspect or having a particular disease or infection, or the patient or subject may be considered free of a particular disease or infection. Typically, the patient or subject is a mammal and, in particular embodiments, a human. The biological sample may be any sample that contains B cells, including but not limited to, lymph node or lymph node tissue, pleural effusions, peripheral blood, ascites, tumor tissue, or cerebrospinal fluid (CSF). In various embodiments, B cells are isolated from different types of biological samples, such as a biological sample affected by a particular disease or infection. However, it is understood that any biological sample comprising B cells may be used for any of the embodiments of the present invention.

(436) Once isolated, the B cells are induced to produce antibodies, e.g., by culturing the B cells under conditions that support B cell proliferation or development into a plasmacyte, plasmablast, or plasma cell. The antibodies are then screened, typically using high throughput techniques, to identify an antibody that specifically binds to a target antigen, e.g., a particular tissue, cell, infectious agent, or polypeptide. In certain embodiments, the specific antigen, e.g., cell surface polypeptide bound by the antibody is not known, while in other embodiments, the antigen specifically bound by the antibody is known.

(437) According to the present invention, B cells may be isolated from a biological sample, e.g., a tumor, tissue, peripheral blood or lymph node sample, by any means known and available in the art. B cells are typically sorted by FACS based on the presence on their surface of a B cell-specific marker, e.g., CD19, CD138, and/or surface IgG. However, other methods known in the art may be employed, such as, e.g., column purification using CD19 magnetic beads or IgG-specific magnetic beads, followed by elution from the column. However, magnetic isolation of B cells utilizing any marker may result in loss of certain B cells. Therefore, in certain embodiments, the isolated cells

are not sorted but, instead, phicol-purified mononuclear cells isolated from tumor are directly plated to the appropriate or desired number of specificities per well.

(438) In order to identify B cells that produce an infectious agent-specific antibody, the B cells are typically plated at low density (e.g., a single cell specificity per well, 1-10 cells per well, 10-100 cells per well, 1-100 cells per well, less than 10 cells per well, or less than 100 cells per well) in multi-well or microtiter plates, e.g., in 96, 384, or 1536 well configurations. When the B cells are initially plated at a density greater than one cell per well, then the methods of the present invention may include the step of subsequently diluting cells in a well identified as producing an antigen-specific antibody, until a single cell specificity per well is achieved, thereby facilitating the identification of the B cell that produces the antigen-specific antibody. Cell supernatants or a portion thereof and/or cells may be frozen and stored for future testing and later recovery of antibody polynucleotides.

(439) In certain embodiments, the B cells are cultured under conditions that favor the production of antibodies by the B cells. For example, the B cells may be cultured under conditions favorable for B cell proliferation and differentiation to yield antibody-producing plasmablast, plasmacytes, or plasma cells. In particular embodiments, the B cells are cultured in the presence of a B cell mitogen, such as lipopolysaccharide (LPS) or CD40 ligand. In one specific embodiment, B cells are differentiated to antibody-producing cells by culturing them with feed cells and/or other B cell activators, such as CD40 ligand.

(440) Cell culture supernatants or antibodies obtained therefrom may be tested for their ability to bind to a target antigen, using routine methods available in the art, including those described herein. In particular embodiments, culture supernatants are tested for the presence of antibodies that bind to a target antigen using high-throughput methods. For example, B cells may be cultured in multi-well microtiter dishes, such that robotic plate handlers may be used to simultaneously sample multiple cell supernatants and test for the presence of antibodies that bind to a target antigen. In particular embodiments, antigens are bound to beads, e.g., paramagnetic or latex beads) to facilitate the capture of antibody/antigen complexes. In other embodiments, antigens and antibodies are fluorescently labeled (with different labels) and FACS analysis is performed to identify the presence of antibodies that bind to target antigen. In one embodiment, antibody binding is determined using FMAT™ analysis and instrumentation (Applied Biosystems, Foster City, CA). FMAT™ is a fluorescence macro-confocal platform for high-throughput screening, which mix-and-read, non-radioactive assays using live cells or beads.

(441) In the context of comparing the binding of an antibody to a particular target antigen (e.g., a biological sample such as infected tissue or cells, or infectious agents) as compared to a control sample (e.g., a biological sample such as uninfected cells, or a different infectious agent), in various embodiments, the antibody is considered to preferentially bind a particular target antigen if at least two-fold, at least three-fold, at least five-fold, or at least ten-fold more antibody binds to the particular target antigen as compared to the amount that binds a control sample.

(442) Polynucleotides encoding antibody chains, variable regions thereof, or fragments thereof, may be isolated from cells utilizing any means available in the art. In one embodiment, polynucleotides are isolated using polymerase chain reaction (PCR), e.g., reverse transcription-PCR (RT-PCR) using oligonucleotide primers that specifically bind to heavy or light chain encoding polynucleotide sequences or complements thereof using routine procedures available in the art. In one embodiment, positive wells are subjected to whole well RT-PCR to amplify the heavy and light chain variable regions of the IgG molecule expressed by the clonal daughter plasma cells. These PCR products may be sequenced.

(443) The resulting PCR products encoding the heavy and light chain variable regions or portions thereof are then subcloned into human antibody expression vectors and recombinantly expressed according to routine procedures in the art (see, e.g., U.S. Pat. No. 7,112,439). The nucleic acid molecules encoding a tumor-specific antibody or fragment thereof, as described herein, may be

propagated and expressed according to any of a variety of well-known procedures for nucleic acid excision, ligation, transformation, and transfection. Thus, in certain embodiments expression of an antibody fragment may be preferred in a prokaryotic host cell, such as *Escherichia coli* (see, e.g., Pluckthun et al., *Methods Enzymol.* 178:497-515 (1989)). In certain other embodiments, expression of the antibody or an antigen-binding fragment thereof may be preferred in a eukaryotic host cell, including yeast (e.g., *Saccharomyces cerevisiae*, *Schizosaccharomyces pombe*, and *Pichia pastoris*); animal cells (including mammalian cells); or plant cells. Examples of suitable animal cells include, but are not limited to, myeloma, COS, CHO, or hybridoma cells. Examples of plant cells include tobacco, corn, soybean, and rice cells. By methods known to those having ordinary skill in the art and based on the present disclosure, a nucleic acid vector may be designed for expressing foreign sequences in a particular host system, and then polynucleotide sequences encoding the tumor-specific antibody (or fragment thereof) may be inserted. The regulatory elements will vary according to the particular host.

(444) One or more replicable expression vectors containing a polynucleotide encoding a variable and/or constant region may be prepared and used to transform an appropriate cell line, for example, a non-producing myeloma cell line, such as a mouse NSO line or a bacterium, such as *E. coli*, in which production of the antibody will occur. In order to obtain efficient transcription and translation, the polynucleotide sequence in each vector should include appropriate regulatory sequences, particularly a promoter and leader sequence operatively linked to the variable region sequence. Particular methods for producing antibodies in this way are generally well known and routinely used. For example, molecular biology procedures are described by Sambrook et al. (*Molecular Cloning, A Laboratory Manual*, 2nd ed., Cold Spring Harbor Laboratory, New York, 1989; see also Sambrook et al., 3rd ed., Cold Spring Harbor Laboratory, New York, (2001)). While not required, in certain embodiments, regions of polynucleotides encoding the recombinant antibodies may be sequenced. DNA sequencing can be performed as described in Sanger et al. (*Proc. Natl. Acad. Sci. USA* 74:5463 (1977)) and the Amersham International plc sequencing handbook and including improvements thereto.

(445) In particular embodiments, the resulting recombinant antibodies or fragments thereof are then tested to confirm their original specificity and may be further tested for pan-specificity, e.g., with related infectious agents. In particular embodiments, an antibody identified or produced according to methods described herein is tested for cell killing via antibody dependent cellular cytotoxicity (ADCC) or apoptosis, and/or well as its ability to internalize.

(446) The present invention, in other aspects, provides polynucleotide compositions. In preferred embodiments, these polynucleotides encode a polypeptide of the invention, e.g., a region of a variable chain of an antibody that binds to HIV1. Polynucleotides of the invention are single-stranded (coding or antisense) or double-stranded DNA (genomic, cDNA or synthetic) or RNA molecules. RNA molecules include, but are not limited to, HnRNA molecules, which contain introns and correspond to a DNA molecule in a one-to-one manner, and mRNA molecules, which do not contain introns. Alternatively, or in addition, coding or non-coding sequences are present within a polynucleotide of the present invention. Also alternatively, or in addition, a polynucleotide is linked to other molecules and/or support materials of the invention. Polynucleotides of the invention are used, e.g., in hybridization assays to detect the presence of an HIV1 antibody in a biological sample, and in the recombinant production of polypeptides of the invention. Further, the invention includes all polynucleotides that encode any polypeptide of the present invention.

(447) In other related embodiments, the invention provides polynucleotide variants having substantial identity to the sequences of 1443_C16 (PG16) (TCN-116), 1503_H05 (PG16) (TCN-119), 1456_A12 (PG16) (TCN-117), 1469_M23 (PG16) (TCN-118), 1489_I13 (PG16) (TCN-120), 1480_I08 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), 1495_C14 (PGC14), 1496_C09 (PG9) (TCN-109), 4838_L06 (PGT-121), 4873_E03 (PGT-121), 4877_D15 (PGT-122), 4858_P08 (PGT-123), 6123_A06 (PGT-125), 5141_B17 (PGT-126), 5145_B14 (PGT-127), 5114_A19 (PGT-128),

5147_N06 (PGT-130), 5136_H01 (PGT-131), 5343_B08 (PGT-135), 5344_E16 (PGT-135), 5329_C19 (PGT-136), 5366_P21 (PGT-136), 4964_G22 (PGT-141), 5345_101 (PGT-137), 4993_K13 (PGT-141), 4995_E20 (PGT-142), 4980_N08 (PGT-143), 4970_K22 (PGT-144), 4995_P16 (PGT-145), 4835_F12 (PGT-124), 4869-K15 (PGT-133), 4876_M06 (PGT-134), 5131_A17 (PGT-132), 5138_G07 (PGT-138), 5120_N10 (PGT-139), 6831_A21 (PGT-151), 6889_I17 (PGT-152), 6891_F06 (PGT-153), 6843_G20 (PGT-154), 6892_D19 (PGT-155), 6808_B09 (PGT-156), 6892_C23 (PGT-157), and/or 6881_N05 (PGT-158), for example those comprising at least 70% sequence identity, preferably at least 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% or higher, sequence identity compared to a polynucleotide sequence of this invention, as determined using the methods described herein, (e.g., BLAST analysis using standard parameters). One skilled in this art will recognize that these values can be appropriately adjusted to determine corresponding identity of proteins encoded by two nucleotide sequences by taking into account codon degeneracy, amino acid similarity, reading frame positioning, and the like.

(448) Typically, polynucleotide variants contain one or more substitutions, additions, deletions and/or insertions, preferably such that the immunogenic binding properties of the polypeptide encoded by the variant polynucleotide is not substantially diminished relative to a polypeptide encoded by a polynucleotide sequence specifically set forth herein.

(449) In additional embodiments, the present invention provides polynucleotide fragments comprising various lengths of contiguous stretches of sequence identical to or complementary to one or more of the sequences disclosed herein. For example, polynucleotides are provided by this invention that comprise at least about 10, 15, 20, 30, 40, 50, 75, 100, 150, 200, 300, 400, 500 or 1000 or more contiguous nucleotides of one or more of the sequences disclosed herein as well as all intermediate lengths there between. As used herein, the term “intermediate lengths” is meant to describe any length between the quoted values, such as 16, 17, 18, 19, etc.; 21, 22, 23, etc.; 30, 31, 32, etc.; 50, 51, 52, 53, etc.; 100, 101, 102, 103, etc.; 150, 151, 152, 153, etc.; including all integers through 200-500; 500-1,000, and the like.

(450) In another embodiment of the invention, polynucleotide compositions are provided that are capable of hybridizing under moderate to high stringency conditions to a polynucleotide sequence provided herein, or a fragment thereof, or a complementary sequence thereof. Hybridization techniques are well known in the art of molecular biology. For purposes of illustration, suitable moderately stringent conditions for testing the hybridization of a polynucleotide of this invention with other polynucleotides include prewashing in a solution of 5×SSC, 0.5% SDS, 1.0 mM EDTA (pH 8.0); hybridizing at 50° C.-60° C., 5×SSC, overnight; followed by washing twice at 65° C. for 20 minutes with each of 2×, 0.5× and 0.2×SSC containing 0.1% SDS. One skilled in the art will understand that the stringency of hybridization can be readily manipulated, such as by altering the salt content of the hybridization solution and/or the temperature at which the hybridization is performed. For example, in another embodiment, suitable highly stringent hybridization conditions include those described above, with the exception that the temperature of hybridization is increased, e.g., to 60-65° C. or 65-70° C.

(451) In preferred embodiments, the polypeptide encoded by the polynucleotide variant or fragment has the same binding specificity (i.e., specifically or preferentially binds to the same epitope or HIV strain) as the polypeptide encoded by the native polynucleotide. In certain preferred embodiments, the polynucleotides described above, e.g., polynucleotide variants, fragments and hybridizing sequences, encode polypeptides that have a level of binding activity of at least about 50%, preferably at least about 70%, and more preferably at least about 90% of that for a polypeptide sequence specifically set forth herein.

(452) The polynucleotides of the present invention, or fragments thereof, regardless of the length of the coding sequence itself, may be combined with other DNA sequences, such as promoters, polyadenylation signals, additional restriction enzyme sites, multiple cloning sites, other coding segments, and the like, such that their overall length may vary considerably. A nucleic acid

fragment of almost any length is employed, with the total length preferably being limited by the ease of preparation and use in the intended recombinant DNA protocol. For example, illustrative polynucleotide segments with total lengths of about 10,000, about 5000, about 3000, about 2,000, about 1,000, about 500, about 200, about 100, about 50 base pairs in length, and the like, (including all intermediate lengths) are included in many implementations of this invention.

(453) It will be appreciated by those of ordinary skill in the art that, as a result of the degeneracy of the genetic code, there are multiple nucleotide sequences that encode a polypeptide as described herein. Some of these polynucleotides bear minimal homology to the nucleotide sequence of any native gene. Nonetheless, polynucleotides that encode a polypeptide of the present invention but which vary due to differences in codon usage are specifically contemplated by the invention.

Further, alleles of the genes including the polynucleotide sequences provided herein are within the scope of the invention. Alleles are endogenous genes that are altered as a result of one or more mutations, such as deletions, additions and/or substitutions of nucleotides. The resulting mRNA and protein may, but need not, have an altered structure or function. Alleles may be identified using standard techniques (such as hybridization, amplification and/or database sequence comparison).

(454) In certain embodiments of the present invention, mutagenesis of the disclosed polynucleotide sequences is performed in order to alter one or more properties of the encoded polypeptide, such as its binding specificity or binding strength. Techniques for mutagenesis are well-known in the art, and are widely used to create variants of both polypeptides and polynucleotides. A mutagenesis approach, such as site-specific mutagenesis, is employed for the preparation of variants and/or derivatives of the polypeptides described herein. By this approach, specific modifications in a polypeptide sequence are made through mutagenesis of the underlying polynucleotides that encode them. These techniques provides a straightforward approach to prepare and test sequence variants, for example, incorporating one or more of the foregoing considerations, by introducing one or more nucleotide sequence changes into the polynucleotide.

(455) Site-specific mutagenesis allows the production of mutants through the use of specific oligonucleotide sequences include the nucleotide sequence of the desired mutation, as well as a sufficient number of adjacent nucleotides, to provide a primer sequence of sufficient size and sequence complexity to form a stable duplex on both sides of the deletion junction being traversed. Mutations are employed in a selected polynucleotide sequence to improve, alter, decrease, modify, or otherwise change the properties of the polynucleotide itself, and/or alter the properties, activity, composition, stability, or primary sequence of the encoded polypeptide.

(456) In other embodiments of the present invention, the polynucleotide sequences provided herein are used as probes or primers for nucleic acid hybridization, e.g., as PCR primers. The ability of such nucleic acid probes to specifically hybridize to a sequence of interest enables them to detect the presence of complementary sequences in a given sample. However, other uses are also encompassed by the invention, such as the use of the sequence information for the preparation of mutant species primers, or primers for use in preparing other genetic constructions. As such, nucleic acid segments of the invention that include a sequence region of at least about a 15-nucleotide long contiguous sequence that has the same sequence as, or is complementary to, a 15 nucleotide long contiguous sequence disclosed herein is particularly useful. Longer contiguous identical or complementary sequences, e.g., those of about 20, 30, 40, 50, 100, 200, 500, 1000 (including all intermediate lengths) including full length sequences, and all lengths in between, are also used in certain embodiments.

(457) Polynucleotide molecules having sequence regions consisting of contiguous nucleotide stretches of 10-14, 15-20, 30, 50, or even of 100-200 nucleotides or so (including intermediate lengths as well), identical or complementary to a polynucleotide sequence disclosed herein, are particularly contemplated as hybridization probes for use in, e.g., Southern and Northern blotting, and/or primers for use in, e.g., polymerase chain reaction (PCR). The total size of fragment, as well as the size of the complementary stretch (es), ultimately depends on the intended use or application

of the particular nucleic acid segment. Smaller fragments are generally used in hybridization embodiments, wherein the length of the contiguous complementary region may be varied, such as between about 15 and about 100 nucleotides, but larger contiguous complementarity stretches may be used, according to the length complementary sequences one wishes to detect.

(458) The use of a hybridization probe of about 15-25 nucleotides in length allows the formation of a duplex molecule that is both stable and selective. Molecules having contiguous complementary sequences over stretches greater than 12 bases in length are generally preferred, though, in order to increase stability and selectivity of the hybrid, and thereby improve the quality and degree of specific hybrid molecules obtained. Nucleic acid molecules having gene-complementary stretches of 15 to 25 contiguous nucleotides, or even longer where desired, are generally preferred.

(459) Hybridization probes are selected from any portion of any of the sequences disclosed herein. All that is required is to review the sequences set forth herein, or to any continuous portion of the sequences, from about 15-25 nucleotides in length up to and including the full length sequence, that one wishes to utilize as a probe or primer. The choice of probe and primer sequences is governed by various factors. For example, one may wish to employ primers from towards the termini of the total sequence.

(460) Polynucleotide of the present invention, or fragments or variants thereof, are readily prepared by, for example, directly synthesizing the fragment by chemical means, as is commonly practiced using an automated oligonucleotide synthesizer. Also, fragments are obtained by application of nucleic acid reproduction technology, such as the PCRTM technology of U.S. Pat. No. 4,683,202, by introducing selected sequences into recombinant vectors for recombinant production, and by other recombinant DNA techniques generally known to those of skill in the art of molecular biology.

(461) The invention provides vectors and host cells comprising a nucleic acid of the present invention, as well as recombinant techniques for the production of a polypeptide of the present invention. Vectors of the invention include those capable of replication in any type of cell or organism, including, e.g., plasmids, phage, cosmids, and mini chromosomes. In various embodiments, vectors comprising a polynucleotide of the present invention are vectors suitable for propagation or replication of the polynucleotide, or vectors suitable for expressing a polypeptide of the present invention. Such vectors are known in the art and commercially available.

(462) Polynucleotides of the present invention are synthesized, whole or in parts that are then combined, and inserted into a vector using routine molecular and cell biology techniques, including, e.g., subcloning the polynucleotide into a linearized vector using appropriate restriction sites and restriction enzymes. Polynucleotides of the present invention are amplified by polymerase chain reaction using oligonucleotide primers complementary to each strand of the polynucleotide. These primers also include restriction enzyme cleavage sites to facilitate subcloning into a vector. The replicable vector components generally include, but are not limited to, one or more of the following: a signal sequence, an origin of replication, and one or more marker or selectable genes.

(463) In order to express a polypeptide of the present invention, the nucleotide sequences encoding the polypeptide, or functional equivalents, are inserted into an appropriate expression vector, i.e., a vector that contains the necessary elements for the transcription and translation of the inserted coding sequence. Methods well known to those skilled in the art are used to construct expression vectors containing sequences encoding a polypeptide of interest and appropriate transcriptional and translational control elements. These methods include in vitro recombinant DNA techniques, synthetic techniques, and in vivo genetic recombination. Such techniques are described, for example, in Sambrook, J., et al. (1989) *Molecular Cloning, A Laboratory Manual*, Cold Spring Harbor Press, Plainview, N.Y., and Ausubel, F. M. et al. (1989) *Current Protocols in Molecular Biology*, John Wiley & Sons, New York, N.Y.

(464) A variety of expression vector/host systems are utilized to contain and express polynucleotide sequences. These include, but are not limited to, microorganisms such as bacteria transformed with recombinant bacteriophage, plasmid, or cosmid DNA expression vectors; yeast transformed with

yeast expression vectors; insect cell systems infected with virus expression vectors (e.g., baculovirus); plant cell systems transformed with virus expression vectors (e.g., cauliflower mosaic virus, CaMV; tobacco mosaic virus, TMV) or with bacterial expression vectors (e.g., Ti or pBR322 plasmids); or animal cell systems.

(465) Within one embodiment, the variable regions of a gene expressing a monoclonal antibody of interest are amplified from a hybridoma cell using nucleotide primers. These primers are synthesized by one of ordinary skill in the art, or may be purchased from commercially available sources (see, e.g., Stratagene (La Jolla, California), which sells primers for amplifying mouse and human variable regions. The primers are used to amplify heavy or light chain variable regions, which are then inserted into vectors such as ImmunoZAP™ H or ImmunoZAP™ L (Stratagene), respectively. These vectors are then introduced into *E. coli*, yeast, or mammalian-based systems for expression. Large amounts of a single-chain protein containing a fusion of the V H and V L domains are produced using these methods (see Bird et al., Science 242:423-426 (1988)).

(466) The “control elements” or “regulatory sequences” present in an expression vector are those non-translated regions of the vector, e.g., enhancers, promoters, 5' and 3' untranslated regions, that interact with host cellular proteins to carry out transcription and translation. Such elements may vary in their strength and specificity. Depending on the vector system and host utilized, any number of suitable transcription and translation elements, including constitutive and inducible promoters, are used.

(467) Examples of promoters suitable for use with prokaryotic hosts include the phoA promoter, β -lactamase and lactose promoter systems, alkaline phosphatase promoter, a tryptophan (trp) promoter system, and hybrid promoters such as the tac promoter. However, other known bacterial promoters are suitable. Promoters for use in bacterial systems also usually contain a Shine-Dalgarno sequence operably linked to the DNA encoding the polypeptide. Inducible promoters such as the hybrid lacZ promoter of the PBLUESCRIPT phagemid (Stratagene, La Jolla, Calif.) or PSPORT1 plasmid (Gibco BRL, Gaithersburg, MD) and the like are used.

(468) A variety of promoter sequences are known for eukaryotes and any are used according to the present invention. Virtually all eukaryotic genes have an AT-rich region located approximately 25 to 30 bases upstream from the site where transcription is initiated. Another sequence found 70 to 80 bases upstream from the start of transcription of many genes is a CNCAAT region where N may be any nucleotide. At the 3' end of most eukaryotic genes is an AATAAA sequence that may be the signal for addition of the poly A tail to the 3' end of the coding sequence. All of these sequences are suitably inserted into eukaryotic expression vectors.

(469) In mammalian cell systems, promoters from mammalian genes or from mammalian viruses are generally preferred. Polypeptide expression from vectors in mammalian host cells are controlled, for example, by promoters obtained from the genomes of viruses such as polyoma virus, fowlpox virus, adenovirus (e.g., Adenovirus 2), bovine papilloma virus, avian sarcoma virus, cytomegalovirus (CMV), a retrovirus, hepatitis-B virus and most preferably Simian Virus 40 (SV40), from heterologous mammalian promoters, e.g., the actin promoter or an immunoglobulin promoter, and from heat-shock promoters, provided such promoters are compatible with the host cell systems. If it is necessary to generate a cell line that contains multiple copies of the sequence encoding a polypeptide, vectors based on SV40 or EBV may be advantageously used with an appropriate selectable marker. One example of a suitable expression vector is pcDNA-3.1 (Invitrogen, Carlsbad, CA), which includes a CMV promoter.

(470) A number of viral-based expression systems are available for mammalian expression of polypeptides. For example, in cases where an adenovirus is used as an expression vector, sequences encoding a polypeptide of interest may be ligated into an adenovirus transcription/translation complex consisting of the late promoter and tripartite leader sequence. Insertion in a non-essential E1 or E3 region of the viral genome may be used to obtain a viable virus that is capable of expressing the polypeptide in infected host cells (Logan, J. and Shenk, T. (1984) Proc. Natl. Acad.

Sci. 81:3655-3659). In addition, transcription enhancers, such as the Rous sarcoma virus (RSV) enhancer, may be used to increase expression in mammalian host cells.

(471) In bacterial systems, any of a number of expression vectors are selected depending upon the use intended for the expressed polypeptide. For example, when large quantities are desired, vectors that direct high level expression of fusion proteins that are readily purified are used. Such vectors include, but are not limited to, the multifunctional *E. coli* cloning and expression vectors such as BLUESCRIPT (Stratagene), in which the sequence encoding the polypeptide of interest may be ligated into the vector in frame with sequences for the amino-terminal Met and the subsequent 7 residues of β -galactosidase, so that a hybrid protein is produced; pIN vectors (Van Heeke, G. and S. M. Schuster (1989) J. Biol. Chem. 264:5503-5509); and the like. pGEX Vectors (Promega, Madison, WI) are also used to express foreign polypeptides as fusion proteins with glutathione S-transferase (GST). In general, such fusion proteins are soluble and can easily be purified from lysed cells by adsorption to glutathione-agarose beads followed by elution in the presence of free glutathione. Proteins made in such systems are designed to include heparin, thrombin, or factor XA protease cleavage sites so that the cloned polypeptide of interest can be released from the GST moiety at will.

(472) In the yeast, *Saccharomyces cerevisiae*, a number of vectors containing constitutive or inducible promoters such as alpha factor, alcohol oxidase, and PGH are used. Examples of other suitable promoter sequences for use with yeast hosts include the promoters for 3-phosphoglycerate kinase or other glycolytic enzymes, such as enolase, glyceraldehyde-3-phosphate dehydrogenase, hexokinase, pyruvate decarboxylase, phosphofructokinase, glucose-6-phosphate isomerase, 3-phosphoglycerate mutase, pyruvate kinase, triosephosphate isomerase, phosphoglucose isomerase, and glucokinase. For reviews, see Ausubel et al. (supra) and Grant et al. (1987) Methods Enzymol. 153:516-544. Other yeast promoters that are inducible promoters having the additional advantage of transcription controlled by growth conditions include the promoter regions for alcohol dehydrogenase 2, isocytochrome C, acid phosphatase, degradative enzymes associated with nitrogen metabolism, metallothionein, glyceraldehyde-3-phosphate dehydrogenase, and enzymes responsible for maltose and galactose utilization. Suitable vectors and promoters for use in yeast expression are further described in EP 73,657. Yeast enhancers also are advantageously used with yeast promoters.

(473) In cases where plant expression vectors are used, the expression of sequences encoding polypeptides are driven by any of a number of promoters. For example, viral promoters such as the 35S and 19S promoters of CaMV are used alone or in combination with the omega leader sequence from TMV (Takamatsu, N. (1987) EMBO J. 6:307-311. Alternatively, plant promoters such as the small subunit of RUBISCO or heat shock promoters are used (Coruzzi, G. et al. (1984) EMBO J. 3:1671-1680; Broglie, R. et al. (1984) Science 224:838-843; and Winter, J., et al. (1991) Results Probl. Cell Differ. 17:85-105). These constructs can be introduced into plant cells by direct DNA transformation or pathogen-mediated transfection. Such techniques are described in a number of generally available reviews (see, e.g., Hobbs, S. or Murry, L. E. in McGraw Hill Yearbook of Science and Technology (1992) McGraw Hill, New York, N.Y.; pp. 191-196).

(474) An insect system is also used to express a polypeptide of interest. For example, in one such system, *Autographa californica* nuclear polyhedrosis virus (AcNPV) is used as a vector to express foreign genes in *Spodoptera frugiperda* cells or in *Trichoplusia* larvae. The sequences encoding the polypeptide are cloned into a non-essential region of the virus, such as the polyhedrin gene, and placed under control of the polyhedrin promoter. Successful insertion of the polypeptide-encoding sequence renders the polyhedrin gene inactive and produce recombinant virus lacking coat protein. The recombinant viruses are then used to infect, for example, *S. frugiperda* cells or *Trichoplusia* larvae, in which the polypeptide of interest is expressed (Engelhard, E. K. et al. (1994) Proc. Natl. Acad. Sci. 91:3224-3227).

(475) Specific initiation signals are also used to achieve more efficient translation of sequences

encoding a polypeptide of interest. Such signals include the ATG initiation codon and adjacent sequences. In cases where sequences encoding the polypeptide, its initiation codon, and upstream sequences are inserted into the appropriate expression vector, no additional transcriptional or translational control signals may be needed. However, in cases where only coding sequence, or a portion thereof, is inserted, exogenous translational control signals including the ATG initiation codon are provided. Furthermore, the initiation codon is in the correct reading frame to ensure correct translation of the inserted polynucleotide. Exogenous translational elements and initiation codons are of various origins, both natural and synthetic.

(476) Transcription of a DNA encoding a polypeptide of the invention is often increased by inserting an enhancer sequence into the vector. Many enhancer sequences are known, including, e.g., those identified in genes encoding globin, elastase, albumin, a-fetoprotein, and insulin. Typically, however, an enhancer from a eukaryotic cell virus is used. Examples include the SV40 enhancer on the late side of the replication origin (bp 100-270), the cytomegalovirus early promoter enhancer, the polyoma enhancer on the late side of the replication origin, and adenovirus enhancers. See also Yaniv, Nature 297:17-18 (1982) on enhancing elements for activation of eukaryotic promoters. The enhancer is spliced into the vector at a position 5' or 3' to the polypeptide-encoding sequence, but is preferably located at a site 5' from the promoter.

(477) Expression vectors used in eukaryotic host cells (yeast, fungi, insect, plant, animal, human, or nucleated cells from other multicellular organisms) typically also contain sequences necessary for the termination of transcription and for stabilizing the mRNA. Such sequences are commonly available from the 5' and, occasionally 3', untranslated regions of eukaryotic or viral DNAs or cDNAs. These regions contain nucleotide segments transcribed as polyadenylated fragments in the untranslated portion of the mRNA encoding anti-PSCA antibody. One useful transcription termination component is the bovine growth hormone polyadenylation region. See WO94/11026 and the expression vector disclosed therein.

(478) Suitable host cells for cloning or expressing the DNA in the vectors herein are the prokaryote, yeast, plant or higher eukaryote cells described above. Examples of suitable prokaryotes for this purpose include eubacteria, such as Gram-negative or Gram-positive organisms, for example, Enterobacteriaceae such as *Escherichia*, e.g., *E. coli*, *Enterobacter*, *Envinia*, *Klebsiella*, *Proteus*, *Salmonella*, e.g., *Salmonella typhimurium*, *Serratia*, e.g., *Serratia marcescans*, and *Shigella*, as well as *Bacilli* such as *B. subtilis* and *B. licheniformis* (e.g., *B. licheniformis* 41P disclosed in DD 266,710 published 12 Apr. 1989), *Pseudomonas* such as *P. aeruginosa*, and *Streptomyces*. One preferred *E. coli* cloning host is *E. coli* 294 (ATCC 31,446), although other strains such as *E. coli*_B, *E. coli* X1776 (ATCC 31,537), and *E. coli* W3110 (ATCC 27,325) are suitable. These examples are illustrative rather than limiting.

(479) *Saccharomyces cerevisiae*, or common baker's yeast, is the most commonly used among lower eukaryotic host microorganisms. However, a number of other genera, species, and strains are commonly available and used herein, such as *Schizosaccharomyces pombe*; *Kluyveromyces* hosts such as, e.g., *K. lactis*, *K. fragilis* (ATCC 12,424), *K. bulgaricus* (ATCC 16,045), *K. wickerhamii* (ATCC 24,178), *K. waitii* (ATCC 56,500), *K. drosophilum* (ATCC 36,906), *K. thermotolerans*, and *K. marxianus*; *yarrowia* (EP 402,226); *Pichia pastoris*. (EP 183,070); *Candida*; *Trichoderma reesia* (EP 244,234); *Neurospora crassa*; *Schwanniomyces* such as *Schwanniomyces occidentalis*; and filamentous fungi such as, e.g., *Neurospora*, *Penicillium*, *Tolypocladium*, and *Aspergillus* hosts such as *A. nidulans* and *A. niger*.

(480) In certain embodiments, a host cell strain is chosen for its ability to modulate the expression of the inserted sequences or to process the expressed protein in the desired fashion. Such modifications of the polypeptide include, but are not limited to, acetylation, carboxylation, glycosylation, phosphorylation, lipidation, and acylation. Post-translational processing that cleaves a "prepro" form of the protein is also used to facilitate correct insertion, folding and/or function. Different host cells such as CHO, COS, HeLa, MDCK, HEK293, and WI38, which have specific

cellular machinery and characteristic mechanisms for such post-translational activities, are chosen to ensure the correct modification and processing of the foreign protein.

(481) Methods and reagents specifically adapted for the expression of antibodies or fragments thereof are also known and available in the art, including those described, e.g., in U.S. Pat. Nos. 4,816,567 and 6,331,415. In various embodiments, antibody heavy and light chains, or fragments thereof, are expressed from the same or separate expression vectors. In one embodiment, both chains are expressed in the same cell, thereby facilitating the formation of a functional antibody or fragment thereof.

(482) Full length antibody, antibody fragments, and antibody fusion proteins are produced in bacteria, in particular when glycosylation and Fc effector function are not needed, such as when the therapeutic antibody is conjugated to a cytotoxic agent (e.g., a toxin) and the immunoconjugate by itself shows effectiveness in infected cell destruction. For expression of antibody fragments and polypeptides in bacteria, see, e.g., U.S. Pat. Nos. 5,648,237, 5,789,199, and 5,840,523, which describes translation initiation region (TIR) and signal sequences for optimizing expression and secretion. After expression, the antibody is isolated from the *E. coli* cell paste in a soluble fraction and can be purified through, e.g., a protein A or G column depending on the isotype. Final purification can be carried out using a process similar to that used for purifying antibody expressed e.g., in CHO cells.

(483) Suitable host cells for the expression of glycosylated polypeptides and antibodies are derived from multicellular organisms. Examples of invertebrate cells include plant and insect cells. Numerous baculoviral strains and variants and corresponding permissive insect host cells from hosts such as *Spodoptera frugiperda* (caterpillar), *Aedes aegypti* (mosquito), *Aedes albopictus* (mosquito), *Drosophila melanogaster* (fruitfly), and *Bombyx mori* have been identified. A variety of viral strains for transfection are publicly available, e.g., the L-1 variant of *Autographa californica* NPV and the Bm-5 strain of *Bombyx mori* NPV, and such viruses are used as the virus herein according to the present invention, particularly for transfection of *Spodoptera frugiperda* cells. Plant cell cultures of cotton, corn, potato, soybean, *petunia*, tomato, and tobacco are also utilized as hosts.

(484) Methods of propagation of antibody polypeptides and fragments thereof in vertebrate cells in culture (tissue culture) are encompassed by the invention. Examples of mammalian host cell lines used in the methods of the invention are monkey kidney CV1 line transformed by SV40 (COS-7, ATCC CRL 1651); human embryonic kidney line (293 or 293 cells subcloned for growth in suspension culture, Graham et al., J. Gen Virol. 36:59 (1977)); baby hamster kidney cells (BHK, ATCC CCL 10); Chinese hamster ovary cells/-DHFR (CHO, Urlaub et al., Proc. Natl. Acad. Sci. USA 77:4216 (1980)); mouse sertoli cells (TM4, Mather, Biol. Reprod. 23:243-251 (1980)); monkey kidney cells (CV1 ATCC CCL 70); African green monkey kidney cells (VERO-76, ATCC CRL-1587); human cervical carcinoma cells (HELA, ATCC CCL 2); canine kidney cells (MDCK, ATCC CCL 34); buffalo rat liver cells (BRL 3A, ATCC CRL 1442); human lung cells (W138, ATCC CCL 75); human liver cells (Hep G2, HB 8065); mouse mammary tumor (MMT 060562, ATCC CCL51); TR1 cells (Mather et al., Annals N.Y. Acad. Sci. 383:44-68 (1982)); MRC 5 cells; FS4 cells; and a human hepatoma line (Hep G2).

(485) Host cells are transformed with the above-described expression or cloning vectors for polypeptide production and cultured in conventional nutrient media modified as appropriate for inducing promoters, selecting transformants, or amplifying the genes encoding the desired sequences.

(486) For long-term, high-yield production of recombinant proteins, stable expression is generally preferred. For example, cell lines that stably express a polynucleotide of interest are transformed using expression vectors that contain viral origins of replication and/or endogenous expression elements and a selectable marker gene on the same or on a separate vector. Following the introduction of the vector, cells are allowed to grow for 1-2 days in an enriched media before they

are switched to selective media. The purpose of the selectable marker is to confer resistance to selection, and its presence allows growth and recovery of cells that successfully express the introduced sequences. Resistant clones of stably transformed cells are proliferated using tissue culture techniques appropriate to the cell type.

(487) A plurality of selection systems are used to recover transformed cell lines. These include, but are not limited to, the herpes simplex virus thymidine kinase (Wigler, M. et al. (1977) *Cell* 11:223-32) and adenine phosphoribosyltransferase (Lowy, I. et al. (1990) *Cell* 22:817-23) genes that are employed in tk.sup.- or aprt.sup.- cells, respectively. Also, antimetabolite, antibiotic or herbicide resistance is used as the basis for selection; for example, dhfr, which confers resistance to methotrexate (Wigler, M. et al. (1980) *Proc. Natl. Acad. Sci.* 77:3567-70); npt, which confers resistance to the aminoglycosides, neomycin and G-418 (Colbere-Garapin, F. et al. (1981) *J Mol. Biol.* 150:1-14); and als or pat, which confer resistance to chlorsulfuron and phosphinotricin acetyltransferase, respectively (Murry, supra). Additional selectable genes have been described. For example, trpB allows cells to utilize indole in place of tryptophan, and hisD allows cells to utilize histinol in place of histidine (Hartman, S. C. and R. C. Mulligan (1988) *Proc. Natl. Acad. Sci.* 85:8047-51). The use of visible markers has gained popularity with such markers as anthocyanins, beta-glucuronidase and its substrate GUS, and luciferase and its substrate luciferin, being widely used not only to identify transformants, but also to quantify the amount of transient or stable protein expression attributable to a specific vector system (Rhodes, C. A. et al. (1995) *Methods Mol. Biol.* 55:121-131).

(488) Although the presence/absence of marker gene expression suggests that the gene of interest is also present, its presence and expression is confirmed. For example, if the sequence encoding a polypeptide is inserted within a marker gene sequence, recombinant cells containing sequences are identified by the absence of marker gene function. Alternatively, a marker gene is placed in tandem with a polypeptide-encoding sequence under the control of a single promoter. Expression of the marker gene in response to induction or selection usually indicates expression of the tandem gene as well.

(489) Alternatively, host cells that contain and express a desired polynucleotide sequence are identified by a variety of procedures known to those of skill in the art. These procedures include, but are not limited to, DNA-DNA or DNA-RNA hybridizations and protein bioassay or immunoassay techniques which include, for example, membrane, solution, or chip based technologies for the detection and/or quantification of nucleic acid or protein.

(490) A variety of protocols for detecting and measuring the expression of polynucleotide-encoded products, using either polyclonal or monoclonal antibodies specific for the product are known in the art. Nonlimiting examples include enzyme-linked immunosorbent assay (ELISA), radioimmunoassay (MA), and fluorescence activated cell sorting (FACS). A two-site, monoclonal-based immunoassay utilizing monoclonal antibodies reactive to two non-interfering epitopes on a given polypeptide is preferred for some applications, but a competitive binding assay may also be employed. These and other assays are described, among other places, in Hampton, R. et al. (1990; *Serological Methods, a Laboratory Manual*, APS Press, St Paul, Minn.) and Maddox, D. E. et al. (1983; *J. Exp. Med.* 158:1211-1216).

(491) Various labels and conjugation techniques are known by those skilled in the art and are used in various nucleic acid and amino acid assays. Means for producing labeled hybridization or PCR probes for detecting sequences related to polynucleotides include oligolabeling, nick translation, end-labeling or PCR amplification using a labeled nucleotide. Alternatively, the sequences, or any portions thereof are cloned into a vector for the production of an mRNA probe. Such vectors are known in the art, are commercially available, and are used to synthesize RNA probes in vitro by addition of an appropriate RNA polymerase such as T7, T3, or SP6 and labeled nucleotides. These procedures are conducted using a variety of commercially available kits. Suitable reporter molecules or labels, which are used include, but are not limited to, radionucleotides, enzymes,

fluorescent, chemiluminescent, or chromogenic agents as well as substrates, cofactors, inhibitors, magnetic particles, and the like.

(492) The polypeptide produced by a recombinant cell is secreted or contained intracellularly depending on the sequence and/or the vector used. Expression vectors containing polynucleotides of the invention are designed to contain signal sequences that direct secretion of the encoded polypeptide through a prokaryotic or eukaryotic cell membrane.

(493) In certain embodiments, a polypeptide of the invention is produced as a fusion polypeptide further including a polypeptide domain that facilitates purification of soluble proteins. Such purification-facilitating domains include, but are not limited to, metal chelating peptides such as histidine-tryptophan modules that allow purification on immobilized metals, protein A domains that allow purification on immobilized immunoglobulin, and the domain utilized in the FLAGS extension/affinity purification system (Amgen, Seattle, WA). The inclusion of cleavable linker sequences such as those specific for Factor XA or enterokinase (Invitrogen, San Diego, CA) between the purification domain and the encoded polypeptide are used to facilitate purification. An exemplary expression vector provides for expression of a fusion protein containing a polypeptide of interest and a nucleic acid encoding 6 histidine residues preceding a thioredoxin or an enterokinase cleavage site. The histidine residues facilitate purification on IMIAC (immobilized metal ion affinity chromatography) as described in Porath, J. et al. (1992, *Prot. Exp. Purif.* 3:263-281) while the enterokinase cleavage site provides a means for purifying the desired polypeptide from the fusion protein. A discussion of vectors used for producing fusion proteins is provided in Kroll, D. J. et al. (1993; *DNA Cell Biol.* 12:441-453).

(494) In certain embodiments, a polypeptide of the present invention is fused with a heterologous polypeptide, which may be a signal sequence or other polypeptide having a specific cleavage site at the N-terminus of the mature protein or polypeptide. The heterologous signal sequence selected preferably is one that is recognized and processed (i.e., cleaved by a signal peptidase) by the host cell. For prokaryotic host cells, the signal sequence is selected, for example, from the group of the alkaline phosphatase, penicillinase, 1pp, or heat-stable enterotoxin II leaders. For yeast secretion, the signal sequence is selected from, e.g., the yeast invertase leader, a factor leader (including *Saccharomyces* and *Kluyveromyces* factor leaders), or acid phosphatase leader, the *C. albicans* glucoamylase leader, or the signal described in WO 90/13646. In mammalian cell expression, mammalian signal sequences as well as viral secretory leaders, for example, the herpes simplex gD signal, are available.

(495) When using recombinant techniques, the polypeptide or antibody is produced intracellularly, in the periplasmic space, or directly secreted into the medium. If the polypeptide or antibody is produced intracellularly, as a first step, the particulate debris, either host cells or lysed fragments, are removed, for example, by centrifugation or ultrafiltration. Carter et al., *Bio/Technology* 10:163-167 (1992) describe a procedure for isolating antibodies that are secreted to the periplasmic space of *E. coli*. Briefly, cell paste is thawed in the presence of sodium acetate (pH 3.5), EDTA, and phenylmethylsulfonylfluoride (PMSF) over about 30 min. Cell debris is removed by centrifugation. Where the polypeptide or antibody is secreted into the medium, supernatants from such expression systems are generally first concentrated using a commercially available protein concentration filter, for example, an Amicon or Millipore Pellicon ultrafiltration unit. Optionally, a protease inhibitor such as PMSF is included in any of the foregoing steps to inhibit proteolysis and antibiotics are included to prevent the growth of adventitious contaminants.

(496) The polypeptide or antibody composition prepared from the cells are purified using, for example, hydroxylapatite chromatography, gel electrophoresis, dialysis, and affinity chromatography, with affinity chromatography being the preferred purification technique. The suitability of protein A as an affinity ligand depends on the species and isotype of any immunoglobulin Fc domain that is present in the polypeptide or antibody. Protein A is used to purify antibodies or fragments thereof that are based on human γ .sub.1, γ .sub.2, or γ .sub.4 heavy

chains (Lindmark et al., J. Immunol. Meth. 62:1-13 (1983)). Protein G is recommended for all mouse isotypes and for human γ .sub.3 (Guss et al., EMBO J. 5:15671575 (1986)). The matrix to which the affinity ligand is attached is most often agarose, but other matrices are available. Mechanically stable matrices such as controlled pore glass or poly(styrenedivinyl)benzene allow for faster flow rates and shorter processing times than can be achieved with agarose. Where the polypeptide or antibody comprises a C.sub.H3 domain, the Bakerbond ABX™ resin (J. T. Baker, Phillipsburg, N.J.) is useful for purification. Other techniques for protein purification such as fractionation on an ion-exchange column, ethanol precipitation, Reverse Phase HPLC, chromatography on silica, chromatography on heparin SEPHAROSE™ chromatography on an anion or cation exchange resin (such as a polyaspartic acid column), chromatofocusing, SDS-PAGE, and ammonium sulfate precipitation are also available depending on the polypeptide or antibody to be recovered.

(497) Following any preliminary purification step(s), the mixture comprising the polypeptide or antibody of interest and contaminants are subjected to low pH hydrophobic interaction chromatography using an elution buffer at a pH between about 2.5-4.5, preferably performed at low salt concentrations (e.g., from about 0-0.25M salt).

(498) The invention further includes pharmaceutical formulations including a polypeptide, antibody, or modulator of the present invention, at a desired degree of purity, and a pharmaceutically acceptable carrier, excipient, or stabilizer (Remington's Pharmaceutical Sciences 16th edition, Osol, A. Ed. (1980)). In certain embodiments, pharmaceutical formulations are prepared to enhance the stability of the polypeptide or antibody during storage, e.g., in the form of lyophilized formulations or aqueous solutions.

(499) Acceptable carriers, excipients, or stabilizers are nontoxic to recipients at the dosages and concentrations employed, and include, e.g., buffers such as acetate, Tris, phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives (such as octadecyldimethylbenzyl ammonium chloride; hexamethonium chloride; benzalkonium chloride, benzethonium chloride; phenol, butyl or benzyl alcohol; alkyl parabens such as methyl or propyl paraben; catechol; resorcinol; cyclohexanol; 3-pentanol; and m-cresol); low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, histidine, arginine, or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; tonicifiers such as trehalose and sodium chloride; sugars such as sucrose, mannitol, trehalose or sorbitol; surfactant such as polysorbate; salt-forming counter-ions such as sodium; metal complexes (e.g. Zn-protein complexes); and/or non-ionic surfactants such as TWEEN® (polysorbate), PLURONICS® (poloxamer), or polyethylene glycol (PEG). In certain embodiments, the therapeutic formulation preferably comprises the polypeptide or antibody at a concentration of between 5-200 mg/ml, preferably between 10-100 mg/ml.

(500) The formulations herein also contain one or more additional therapeutic agents suitable for the treatment of the particular indication, e.g., infection being treated, or to prevent undesired side-effects. Preferably, the additional therapeutic agent has an activity complementary to the polypeptide or antibody of the present invention, and the two do not adversely affect each other. For example, in addition to the polypeptide or antibody of the invention, an additional or second antibody, anti-viral agent, anti-infective agent and/or cardioprotectant is added to the formulation. Such molecules are suitably present in the pharmaceutical formulation in amounts that are effective for the purpose intended.

(501) The active ingredients, e.g., polypeptides and antibodies of the invention and other therapeutic agents, are also entrapped in microcapsules prepared, for example, by coacervation techniques or by interfacial polymerization, for example, hydroxymethylcellulose or gelatin-microcapsules and polymethylmethacrylate) microcapsules, respectively, in colloidal drug delivery

systems (for example, liposomes, albumin microspheres, microemulsions, nano-particles and nanocapsules) or in macroemulsions. Such techniques are disclosed in Remington's Pharmaceutical Sciences 16th edition, Osol, A. Ed. (1980).

(502) Sustained-release preparations are prepared. Suitable examples of sustained-release preparations include, but are not limited to, semi-permeable matrices of solid hydrophobic polymers containing the antibody, which matrices are in the form of shaped articles, e.g., films, or microcapsules. Nonlimiting examples of sustained-release matrices include polyesters, hydrogels (for example, poly(2-hydroxyethyl-methacrylate), or poly(vinylalcohol)), polylactides (U.S. Pat. No. 3,773,919), copolymers of L-glutamic acid and γ ethyl-L-glutamate, non-degradable ethylene-vinyl acetate, degradable lactic acid-glycolic acid copolymers such as the LUPRON DEPOTTM (injectable microspheres composed of lactic acid-glycolic acid copolymer and leuprolide acetate), and poly-D-(-)-3-hydroxybutyric acid.

(503) Formulations to be used for in vivo administration are preferably sterile. This is readily accomplished by filtration through sterile filtration membranes.

(504) Antibodies of the invention can be coupled to a drug for delivery to a treatment site or coupled to a detectable label to facilitate imaging of a site comprising cells of interest, such as cells infected with HIV. Methods for coupling antibodies to drugs and detectable labels are well known in the art, as are methods for imaging using detectable labels. Labeled antibodies may be employed in a wide variety of assays, employing a wide variety of labels. Detection of the formation of an antibody-antigen complex between an antibody of the invention and an epitope of interest (an HIV epitope) can be facilitated by attaching a detectable substance to the antibody. Suitable detection means include the use of labels such as radionucleotides, enzymes, coenzymes, fluorescers, chemiluminescers, chromogens, enzyme substrates or co-factors, enzyme inhibitors, prosthetic group complexes, free radicals, particles, dyes, and the like. Examples of suitable enzymes include horseradish peroxidase, alkaline phosphatase, B-galactosidase, or acetylcholinesterase; examples of suitable prosthetic group complexes include streptavidin/biotin and avidin/biotin; examples of suitable fluorescent materials include umbelliferone, fluorescein, fluorescein isothiocyanate, rhodamine, dichlorotriazinylamine fluorescein, dansyl chloride or phycoerythrin; an example of a luminescent material is luminol; examples of bioluminescent materials include luciferase, luciferin, and aequorin; and examples of suitable radioactive material include .sup.125I, .sup.131I, .sup.35S, or .sup.3H. Such labeled reagents may be used in a variety of well-known assays, such as radioimmunoassays, enzyme immunoassays, e.g., ELISA, fluorescent immunoassays, and the like.

(505) The antibodies are tagged with such labels by known methods. For instance, coupling agents such as aldehydes, carbodiimides, dimaleimide, imidates, succinimides, bid-diazotized benzadine and the like are used to tag the antibodies with the above-described fluorescent, chemiluminescent, and enzyme labels. An enzyme is typically combined with an antibody using bridging molecules such as carbodiimides, periodate, diisocyanates, glutaraldehyde and the like. Various labeling techniques are described in Morrison, *Methods in Enzymology* 32b, 103 (1974), Syvanen et al., *J. Biol. Chem.* 284, 3762 (1973) and Bolton and Hunter, *Biochem J.* 133, 529(1973).

(506) An antibody according to the invention may be conjugated to a therapeutic moiety such as a cytotoxin, a therapeutic agent, or a radioactive metal ion or radioisotope. Examples of radioisotopes include, but are not limited to, 1-131, 1-123, 1-125, Y-90, Re-188, Re-186, At-211, Cu-67, Bi-212, Bi-213, Pd-109, Tc-99, In-111, and the like. Such antibody conjugates can be used for modifying a given biological response; the drug moiety is not to be construed as limited to classical chemical therapeutic agents. For example, the drug moiety may be a protein or polypeptide possessing a desired biological activity. Such proteins may include, for example, a toxin such as abrin, ricin A, pseudomonas exotoxin, or diphtheria toxin.

(507) Techniques for conjugating such therapeutic moiety to antibodies are well known. See, for example, Arnon et al. (1985) "Monoclonal Antibodies for Immunotargeting of Drugs in Cancer Therapy," in *Monoclonal Antibodies and Cancer Therapy*, ed. Reisfeld et al. (Alan R. Liss, Inc.),

pp. 243-256; ed. Hellstrom et al. (1987) "Antibodies for Drug Delivery," in *Controlled Drug Delivery*, ed. Robinson et al. (2d ed; Marcel Dekker, Inc.), pp. 623-653; Thorpe (1985) "Antibody Carriers of Cytotoxic Agents in Cancer Therapy: A Review," in *Monoclonal Antibodies '84: Biological and Clinical Applications*, ed. Pinchera et al. pp. 475-506 (Editrice Kurtis, Milano, Italy, 1985); "Analysis, Results, and Future Prospective of the Therapeutic Use of Radiolabeled Antibody in Cancer Therapy," in *Monoclonal Antibodies for Cancer Detection and Therapy*, ed. Baldwin et al. (Academic Press, New York, 1985), pp. 303-316; and Thorpe et al. (1982) *Immunol. Rev.* 62:119-158.

(508) Diagnostic methods generally involve contacting a biological sample obtained from a patient, such as, e.g., blood, serum, saliva, urine, sputum, a cell swab sample, or a tissue biopsy, with an HIV1 antibody and determining whether the antibody preferentially binds to the sample as compared to a control sample or predetermined cut-off value, thereby indicating the presence of infected cells. In particular embodiments, at least two-fold, three-fold, or five-fold more HIV1 antibody binds to an infected cell as compared to an appropriate control normal cell or tissue sample. A pre-determined cut-off value is determined, e.g., by averaging the amount of HIV1 antibody that binds to several different appropriate control samples under the same conditions used to perform the diagnostic assay of the biological sample being tested.

(509) Bound antibody is detected using procedures described herein and known in the art. In certain embodiments, diagnostic methods of the invention are practiced using HIV1 antibodies that are conjugated to a detectable label, e.g., a fluorophore, to facilitate detection of bound antibody. However, they are also practiced using methods of secondary detection of the HIV1 antibody. These include, for example, MA, ELISA, precipitation, agglutination, complement fixation and immunofluorescence.

(510) HIV1 antibodies of the present invention are capable of differentiating between patients with and patients without an HIV infection, and determining whether or not a patient has an infection, using the representative assays provided herein. According to one method, a biological sample is obtained from a patient suspected of having or known to have HIV1 infection. In preferred embodiments, the biological sample includes cells from the patient. The sample is contacted with an HIV1 antibody, e.g., for a time and under conditions sufficient to allow the HIV1 antibody to bind to infected cells present in the sample. For instance, the sample is contacted with an HIV1 antibody for 10 seconds, 30 seconds, 1 minute, 5 minutes, 10 minutes, 30 minutes, 1 hour, 6 hours, 12 hours, 24 hours, 3 days or any point in between. The amount of bound HIV1 antibody is determined and compared to a control value, which may be, e.g., a pre-determined value or a value determined from normal tissue sample. An increased amount of antibody bound to the patient sample as compared to the control sample is indicative of the presence of infected cells in the patient sample.

(511) In a related method, a biological sample obtained from a patient is contacted with an HIV1 antibody for a time and under conditions sufficient to allow the antibody to bind to infected cells. Bound antibody is then detected, and the presence of bound antibody indicates that the sample contains infected cells. This embodiment is particularly useful when the HIV1 antibody does not bind normal cells at a detectable level.

(512) Different HIV1 antibodies possess different binding and specificity characteristics. Depending upon these characteristics, particular HIV1 antibodies are used to detect the presence of one or more strains of HIV1. For example, certain antibodies bind specifically to only one or several strains of HIV1, whereas others bind to all or a majority of different strains of HIV1. Antibodies specific for only one strain of HIV1 are used to identify the strain of an infection.

(513) In certain embodiments, antibodies that bind to an infected cell preferably generate a signal indicating the presence of an infection in at least about 20% of patients with the infection being detected, more preferably at least about 30% of patients. Alternatively, or in addition, the antibody generates a negative signal indicating the absence of the infection in at least about 90% of

individuals without the infection being detected. Each antibody satisfies the above criteria; however, antibodies of the present invention are used in combination to improve sensitivity.

(514) The present invention also includes kits useful in performing diagnostic and prognostic assays using the antibodies of the present invention. Kits of the invention include a suitable container comprising an HIV1 antibody of the invention in either labeled or unlabeled form. In addition, when the antibody is supplied in a labeled form suitable for an indirect binding assay, the kit further includes reagents for performing the appropriate indirect assay. For example, the kit includes one or more suitable containers including enzyme substrates or derivatizing agents, depending on the nature of the label. Control samples and/or instructions are also included.

(515) Passive immunization has proven to be an effective and safe strategy for the prevention and treatment of viral diseases. (See Keller et al., Clin. Microbiol. Rev. 13:602-14 (2000); Casadevall, Nat. Biotechnol. 20:114 (2002); Shibata et al., Nat. Med. 5:204-10 (1999); and Igarashi et al., Nat. Med. 5:211-16 (1999), each of which are incorporated herein by reference)). Passive immunization using human monoclonal antibodies, provide an immediate treatment strategy for emergency prophylaxis and treatment of HIV1.

(516) HIV1 antibodies and fragments thereof, and therapeutic compositions, of the invention specifically bind or preferentially bind to infected cells, as compared to normal control uninfected cells and tissue. Thus, these HIV1 antibodies are used to selectively target infected cells or tissues in a patient, biological sample, or cell population. In light of the infection-specific binding properties of these antibodies, the present invention provides methods of regulating (e.g., inhibiting) the growth of infected cells, methods of killing infected cells, and methods of inducing apoptosis of infected cells. These methods include contacting an infected cell with an HIV1 antibody of the invention. These methods are practiced in vitro, ex vivo, and in vivo.

(517) In various embodiments, antibodies of the invention are intrinsically therapeutically active. Alternatively, or in addition, antibodies of the invention are conjugated to a cytotoxic agent or growth inhibitory agent, e.g., a radioisotope or toxin that is used in treating infected cells bound or contacted by the antibody.

(518) Subjects at risk for HIV1-related diseases or disorders include patients who have come into contact with an infected person or who have been exposed to HIV1 in some other way. Administration of a prophylactic agent can occur prior to the manifestation of symptoms characteristic of HIV1-related disease or disorder, such that a disease or disorder is prevented or, alternatively, delayed in its progression.

(519) Methods for preventing an increase in HIV1 virus titer, virus replication, virus proliferation or an amount of an HIV1 viral protein in a subject are further provided. In one embodiment, a method includes administering to the subject an amount of an HIV1 antibody effective to prevent an increase in HIV1 titer, virus replication or an amount of an HIV1 protein of one or more HIV strains or isolates in the subject.

(520) For in vivo treatment of human and non-human patients, the patient is usually administered or provided a pharmaceutical formulation including an HIV1 antibody of the invention. When used for in vivo therapy, the antibodies of the invention are administered to the patient in therapeutically effective amounts (i.e., amounts that eliminate or reduce the patient's viral burden). The antibodies are administered to a human patient, in accord with known methods, such as intravenous administration, e.g., as a bolus or by continuous infusion over a period of time, by intramuscular, intraperitoneal, intracerebrospinal, subcutaneous, intra-articular, intrasynovial, intrathecal, oral, topical, or inhalation routes. The antibodies may be administered parenterally, when possible, at the target cell site, or intravenously. Intravenous or subcutaneous administration of the antibody is preferred in certain embodiments. Therapeutic compositions of the invention are administered to a patient or subject systemically, parenterally, or locally.

(521) For parenteral administration, the antibodies are formulated in a unit dosage injectable form (solution, suspension, emulsion) in association with a pharmaceutically acceptable, parenteral

vehicle. Examples of such vehicles are water, saline, Ringer's solution, dextrose solution, and 5% human serum albumin. Nonaqueous vehicles such as fixed oils and ethyl oleate are also used. Liposomes are used as carriers. The vehicle contains minor amounts of additives such as substances that enhance isotonicity and chemical stability, e.g., buffers and preservatives. The antibodies are typically formulated in such vehicles at concentrations of about 1 mg/ml to 10 mg/ml.

(522) The dose and dosage regimen depends upon a variety of factors readily determined by a physician, such as the nature of the infection and the characteristics of the particular cytotoxic agent or growth inhibitory agent conjugated to the antibody (when used), e.g., its therapeutic index, the patient, and the patient's history. Generally, a therapeutically effective amount of an antibody is administered to a patient. In particular embodiments, the amount of antibody administered is in the range of about 0.1 mg/kg to about 50 mg/kg of patient body weight. Depending on the type and severity of the infection, about 0.1 mg/kg to about 50 mg/kg body weight (e.g., about 0.1-15 mg/kg/dose) of antibody is an initial candidate dosage for administration to the patient, whether, for example, by one or more separate administrations, or by continuous infusion. The progress of this therapy is readily monitored by conventional methods and assays and based on criteria known to the physician or other persons of skill in the art.

(523) In one particular embodiment, an immunoconjugate including the antibody conjugated with a cytotoxic agent is administered to the patient. Preferably, the immunoconjugate is internalized by the cell, resulting in increased therapeutic efficacy of the immunoconjugate in killing the cell to which it binds. In one embodiment, the cytotoxic agent targets or interferes with the nucleic acid in the infected cell. Examples of such cytotoxic agents are described above and include, but are not limited to, maytansinoids, calicheamicins, ribonucleases and DNA endonucleases.

(524) Other therapeutic regimens are combined with the administration of the HIV1 antibody of the present invention. The combined administration includes co-administration, using separate formulations or a single pharmaceutical formulation, and consecutive administration in either order, wherein preferably there is a time period while both (or all) active agents simultaneously exert their biological activities. Preferably such combined therapy results in a synergistic therapeutic effect.

(525) In certain embodiments, it is desirable to combine administration of an antibody of the invention with another antibody directed against another antigen associated with the infectious agent.

(526) Aside from administration of the antibody protein to the patient, the invention provides methods of administration of the antibody by gene therapy. Such administration of nucleic acid encoding the antibody is encompassed by the expression "administering a therapeutically effective amount of an antibody". See, for example, PCT Patent Application Publication WO96/07321 concerning the use of gene therapy to generate intracellular antibodies.

(527) In another embodiment, anti-HIV1 antibodies of the invention are used to determine the structure of bound antigen, e.g., conformational epitopes, the structure of which is then used to develop a vaccine having or mimicking this structure, e.g., through chemical modeling and SAR methods. Such a vaccine could then be used to prevent HIV1 infection.

(528) Although the present invention and its advantages have been described in detail, it should be understood that various changes, substitutions and alterations can be made herein without departing from the spirit and scope of the invention as defined in the appended claims.

(529) The present invention will be further illustrated in the following Examples which are given for illustration purposes only and are not intended to limit the invention in any way.

EXAMPLES

Example 1: Selection of Patient Sample

(530) Serum from approximately 1,800 HIV-1 infected donors from Asia, Australia, Europe, North America and sub-Saharan African countries were screened for neutralization activity and donors who exhibit among the broadest and most potent neutralizing serum activity observed to date were

identified. (Simek, M. D., *J Virol* (2009)). Monoclonal antibodies were generated from these donors using different approaches.

(531) A patient was selected based upon the patient's eligibility for enrollment, which was defined as: male or female at least 18 years of age with documented HIV infection for at least three years, clinically asymptomatic at the time of enrollment, and not currently receiving antiretroviral therapy. (Simek, M. D., *J Virol* (2009 July) 83(14):7337-48). Selection of individuals for monoclonal antibody generation was based on a rank-order high throughput analytical screening algorithm. The volunteer was identified as an individual with broad neutralizing serum based on broad and potent neutralizing activity against a cross-Glade pseudovirus panel.

(532) A novel high-throughput strategy was used to screen IgG-containing culture supernatants from approximately 30,000 activated memory B cells from a Glade A infected donor for recombinant, monomeric gp120.sub.JR-CSF and gp41.sub.HxB2 (Env) binding as well as neutralization activity against HIV-1.sub.JR-CSF and HIV-1.sub.SF162 as shown in Table 1. The memory B cells were cultured at near clonal density such that the authentic antibody heavy and light chain pair could be reconstituted from each culture well.

Example 2: Generation of Monoclonal Antibodies

(533) The human monoclonal antibody discovery platform utilized a short term B cell culture system to interrogate the memory B cell repertoire. 30,300 CD19.sup.+ and surface IgG-expressing memory B cells were isolated from ten million peripheral blood mononuclear cells (PBMC) of the HIV-1 infected donor. CD19.sup.+ /sIgG.sup.+ B cells were then seeded in 384-well microtiter plates at an average of 1.3 cells/well under conditions that promoted B cell activation, proliferation, terminal differentiation and antibody secretion. Culture supernatants were screened in a high throughput format for binding reactivity to recombinant gp120 and gp41 indirectly and directly immobilized on ELISA plates, respectively. In parallel, the culture supernatants were also screened for neutralization activity in a high throughput micro-neutralization assay.

(534) Heavy and light variable regions were isolated from lysates of selected neutralizing hits by RT-PCR amplification using family-specific primer sets. From positive family-specific PCR reactions, pools of the V.sub.H or V.sub.L-region clones were cloned into an expression vector upstream to human IgG.sub.1 constant domain sequence. Minipreps (QIAGEN, Valencia, CA) of these DNA pools, derived from suspension bacterial cultures, were combined in all possible heavy and light chain family-specific pairs and used to transiently transfect 293 cells. All transfectant supernatants containing secreted recombinant antibodies were screened in ELISA and neutralization assays. For B-cell wells that contained more than one B cell clone per culture well, multiple V.sub.H and V.sub.L domain sequences were isolated. ELISA (for B-cell wells positive for ELISA) and neutralization screens identified the heavy and light chain combination pools that reconstituted the binding and neutralizing activity as observed for the B-cell well. DNA sequences of the heavy and light chain variable regions for all neutralizing mAbs were confirmed by multiple sequencing reactions using purified DNA from maxipreps (QIAGEN).

Example 3: Screening of Monoclonal Antibodies for Binding to Recombinant Gp120 and Gp41 by ELISA Assay

(535) Recombinant gp120 with sequence derived from gp120 of primary HIV-1 isolate JR-CSF and expressed in insect cells was obtained from IAVI NAC repository. Recombinant gp41 generated with sequences derived from HxB2 clone of HIV-1 and expressed in *Pichia pastoris* was manufactured by Vybion, Inc., obtained from IAVI NAC repository. Sheep anti-gp120 antibodies used as capturing agent to indirectly immobilize gp120 on ELISA plates was purchased from Aalto Bio Reagents (Dublin, Ireland). All ELISA assays were conducted at 25 µL/well on Maxi Sorp plates from Nunc.

(536) In anti-gp120 ELISA, recombinant gp120 (0.5 µg/ml) was captured on 384 well ELISA plates pre-coated (at 4° C. overnight) with goat anti-gp120 (5 µg/ml) in BSA-containing assay buffer (PBS with 0.05% TWEEN® 20 (polysorbate 20)) for 1 hr at room temperature. After excess

gp120 was removed and plates were washed thrice with assay buffer, B cell culture supernatants diluted 5-fold was added to incubate for 1 hr at room temperature. Following three washes in assay buffer, secondary HRP-conjugated goat anti-human Ig Fc in BSA-containing assay buffer was added and incubated for about 1 hr at room temperature. 3,3',5,5'-tetramethylbenzidine (TMB) substrate was used to develop the colorimetric readouts after washing the ELISA plates 3 times. (537) For anti-gp41 ELISA, recombinant gp41 was directly immobilized on 384 well ELISA plates by adding 1 µg/ml and incubating at 4° C. overnight, followed by blocking with BSA-containing assay buffer. The rest of the assay protocol was similar to that for anti-gp120 ELISA.

(538) Hits from the ELISA assay were identified in a singlet screen based on optical density (OD) values above 3x assay background. A serial titration standard curve of control antibody was included on each plate.

Example 4: Neutralization Assay for Screening Antibodies against Pseudotyped HIV Viruses

(539) The neutralization assay approach has been described previously (Binley J M, et al., (2004). Comprehensive Cross-Clade Neutralization Analysis of a Panel of Anti-Human Immunodeficiency Virus Type 1 Monoclonal Antibodies. *J. Virol.* 78: 13232-13252) and was modified and standardized for implementation in 384-well format.

(540) Neutralization by monoclonal antibodies and patient sera was performed using a single round of replication pseudovirus assay. (Richman, D. D., et al. *Proc Natl Acad Sci USA* 100, 4144-4149 (2003)). Pseudovirus neutralization assays were performed using HIV-1JR-CSF alanine mutants as described in Pantophlet, R., et al. *J Virol* 77, 642-658 (2003). Neutralization activity was measured as a reduction in viral infectivity compared to an antibody-free control using a TZM-BL assay. (Li, M., et al., *J Virol* 79, 10108-10125 (2005)). Monoclonal antibody neutralization assays using phytohaemagglutinin-activated peripheral blood mononuclear cells (PBMC) isolated from three healthy human donors as target cells were performed as described in Scarlatti, G. et al, (1993) *J. Infect. Dis.* 168:207-210; Polonis, V. et al, (2001) *AIDS Res. Hum. Retroviruses* 17:69-79. Memory B cell supernatants were screened in a micro-neutralization assay against HIV-ISF162, HIV-1.sub.JR-CSF, and SIV.sub.mac239 (negative control). This assay was based on the 96-well pseudotyped HIV-1 neutralization assay (Monogram Biosciences) and was modified for screening 15 µl B cell culture supernatants in a 384-well format.

(541) Pseudotyped virus from SF162 and JR-CSF isolates of HIV-1 and SIV mac239 (control virus) were generated by co-transfecting Human Embryonic Kidney 293 cells (293 cells) with 2 plasmids encoding the Envelope cDNA sequence and the rest of the HIV genome separately. In the HIV genome encoding vector, the Env gene was replaced by the firefly luciferase gene.

Transfectant supernatants containing pseudotyped virus were co-incubated overnight (18 hours) with B cell supernatants derived from activation of an infected donor's primary peripheral blood mononuclear cells (PBMCs). U87 cells stably transfected with and expressing CD4 plus the CCR5 and CXCR4 coreceptors were added to the mixture and incubated for 3 days at 37° C. Infected cells were quantified by luminometry. SIVmac239 was used as the negative control virus.

(542) The neutralization index was expressed as the ratio of normalized relative luminescence units (RLU) of the test viral strain to that of the control virus SIVmac239 derived from the same test B cell culture supernatant. The cut-off values used to distinguish neutralizing hits were determined by the neutralization index of a large number of “negative control wells” containing B cell culture supernatants derived from healthy donors. The false positive rate using the cut-off value of 1.5 was very low (1-3%; FIG. 5A), and it was reduced to zero if the cut-off value of 2.0 was used (FIG. 5B).

(543) FIG. 5 summarizes the screening results from which B cell cultures were selected for antibody rescue and the monoclonal antibodies 1496_C09 (PG9), 1443_C16 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), and 1495_C14 (PGC14) were derived. The results reveal that the majority of neutralizing B cell culture supernatants did not have binding reactivity to soluble recombinant gp120 or gp41 proteins.

[illegible]

1503 C14 λ5 020 0.00 0.99 0.87 1460 G14 λ1 023 1460 G14 λ1 017 17.37 12.44 1.64 39.43
 ND ND ND ND ND ND Lightest grey: suggested H&L pair for monoclonal antibody per priority
 well. Medium grey with black lettering: Denotes clones derived from same recombinant H or L
 chain pool of the priority well with identical sequences. Bolded: 1496 C09 λ3 clone 024 is likely a
 cross-contaminant in the recombinant DNA pool as it is identical to 1443 C16 λ2 019 in sequence.
 1496 C09 λ2 017 sequence represents 21/22 clones in the pool. *Anti-gp120 and anti-gp41
 concentrations were extrapolated from b12 and 2F5 standard curves in quantitative ELISA,
 respectively. N/A = not applicable because these hits were neither gp-120 nor gp-41 positive in B
 cell culture. ND = not done.

(546) The purified monoclonal antibodies 1496_C09 (PG9), 1443_C16 (PG16), 1456_P20 (PG20),
 1460_G14 (PGG14), and 1495_C14 (PGC14) were tested for neutralization of 6 additional HIV
 strains from clades A (94UG103), B (92BR020, JR-CSF), C (93IN905, IAVI C22), and CRF01_AE
 (92TH021) (Table 16). The antibodies 1496_C09 (PG9), 1443_C16 (PG16) and 1495_C14
 (PGC14) showed neutralization profile similar to that obtained with the donor sera neutralization
 profile. The pseudoviruses were preincubated with each monoclonal antibody for 1 hour or 18
 hours prior to the infection of target cells. IC.sub.50 values derived from 1 or 18 hours
 preincubation were similar. Therefore, in further neutralization assays testing purified monoclonal
 antibodies, 1 hour of preincubation was used.

(547) Table 17A shows the neutralization profiles for the 5 monoclonal antibodies 1496_C09
 (PG9), 1443_C16 (PG16), 1456_P20 (PG20), 1460_G14 (PGG14), and 1495_C14 (PGC14) in
 IC.sub.50 values on an extended panel of 16 pseudoviruses, together with known cross-clade
 neutralizing antibodies b12, 2G12, 2F5 and 4E10.

(548) Table 17B shows the IC90 of two monoclonal antibodies, 1443_C16 (PG16) and 1496_C09
 (PG9) on the same expanded diverse panel of 16 HIV pseudoviruses from different clades, together
 with known cross-clade neutralizing antibodies b12, 2G12, 2F5 and 4E10. FIG. 4 shows
 neutralization activity of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to 3 other
 pseudoviruses not included in Table 16.

(549) TABLE-US-00023 TABLE 16 Neutralizing Antibody Assay: IC50 Summary IC50 (ug/mL)
 Except Where Noted Virus/Ab Incubation SF162 94UG103 92BR020 93IN905 IAVI_C22
 92TH021 JRCSF NL43 aMLV 1 hour 1443C16 >50 0.0211 >50 0.3302**** 0.1143* 0.1362***
 <0.0025 <0.0025** >50 18 hour 1443C16 >50 0.0085 >50 0.2553* 0.1064*** 0.0435 <0.0025
 4.9874** >50 1 hour 1456P20 0.1946 >50 >50 >50 >50 >50 >50 0.20 >50 18 hour 1456P20
 0.0661 >50 >50 3.8384* >50 >50 >50 0.05 >50 1 hour 1460G14 0.1789 >50 >50 >50 >50 >50
 0.17 >50 18 hour 1460G14 0.0573 >50 >50 3.1738* >50 >50 >50 0.05 >50 1 hour 1495C14
 0.0069 >50 1.1697 >50 >50 >50 >50 0.35 >50 18 hour 1495C14 <0.0025 >50 0.2442 0.1456*
 13.3798 >50 >50 0.15 >50 1 hour 1496C09 >50 0.3336 >50 0.1444 24.8611 0.0612 <0.0025
 0.2944* >50 18 hour 1496C09 >50 0.0942 >50 0.0619 2.1073 0.0571 <0.0025 38.03 >50 1 hour
 Z23 13521 188 616 369 340 175 438 4793 <100 (1/dil'n) 18 hour Z23 66074 262 1292 1396 614
 336 1054 9472 <100 (1/dil'n) *plateau **flat inhibition curve - probably <0.0025 with plateau
 very long, shallow slope *plateau with very long, shallow slope to curve

(550) TABLE-US-00024 TABLE 17A Neutralization Profile on a Diverse Panel of Viruses:
 IC.sub.50 Values PG9 PG16 PGC14 PGG14 PG20 b12 2G12 2F5 4E10 Clade A 94UG103 0.1731
 0.0080 >50 >50 >50 3.54 >50 3.79 9.7 92RW020 0.0637 0.0040**** 28.5960 >50 >50 >50 0.56
 3.37 3.38 93UG077 >50 >50 >50 >50 >50 41.12 >50 4.45 11.15 Clade B 92BR020 >50 >50
 0.6366 >50 >50 27.5 2.26 >50 41.44 APV-13 >50 >50 >50 >50 >50 >25 23.9 2.8 3.8 APV-17
 26.4465 >50 >50 >50 >50 >25 >50 2 5.1 APV-6 0.0869 0.08**** 7.4062 >50 25.7798 >25 5.3 0.1
 0.4 JRCSF <0.0025 <0.0025 >50 >50 >50 0.16 0.66 3.36 6 Clade C 93IN905 0.1400 0.1016***
 >50 >50 >50 34.15 >50 >50 1.55 IAVI-C18 0.0535 0.0067 >50 >50 >50 >50 >50 >50 IAVI-
 C22 24.8600 0.0687* 9.4999 >50 >50 3.6042 >50 >50 1.0229 IAVI-C3 12.9103 14.8372 >50 >50
 >50 5.0000 >50 >50 5.0000 Clade D 92UG024 10.9552 >50 >50 >50 >50 49.06 0.59 1.27 1.32

92UG005 >50 >50 >50 >50 >50 11.75 8.86 CRF01_AE 92TH021 0.1105 0.1273*** >50
 >50 >50 9.99 >50 1.51 1.9 CMU02 >50 >50 >50 >50 >50 4.25 >50 0.38 0.59 Pos C NL43 N/A
 <0.0025** 0.3727 0.1717 0.1880 0.06 0.75 2.41 4.95 Neg C aMLV >50 >50 >50 >50 >50 >50
 >50 >50 NA Not Applicable IC.sub.50: Inhibitory concentration to inhibit 50% of the virus
 (551) TABLE-US-00025 TABLE 17B Neutralization Profile on a Diverse Panel of Viruses:
 IC.sub.90 Values for mAbs PG9 and PG16 PG9 PG16 b12 2G12 2F5 4E10 Clade A 94UG103
 3.3736 1.5915 47.29 >50 46.63 >50 92RW020 6.5462 >50 >50 6.23 27.74 36.11 93UG077 >50
 >50 >50 >50 33.44 >50 Clade B 92BR020 >50 >50 >50 24.09 >50 >50 APV-13 >50 >50 >50 N/A
 N/A N/A APV-17 >50 >50 >50 N/A N/A N/A APV-6 1.9591 44.2600 >50 N/A N/A N/A JRCSF
 <0.0025 0.0130 1.17 5.38 25.31 44.07 Clade C 93IN905 1.8945 >50 >50 >50 >50 12.82 IAVI-C18
 0.8659 0.2074 >50 >50 N/A >50 IAVI-C22 >50 >50 29.6187 >50 >50 16.405 IAVI-C3 >50 >50
 >50 N/A N/A Clade D 92UG024 >50 >50 >50 7.57 34.44 23.71 92UG005 >50 >50 >50 >50 >50
 >50 CRF01_AE 92TH021 1.9871 23.4110 >50 >50 18.78 23.52 CMU02 >50 >50 34.2 >50 12.25
 13.4 Pos C NL43 N/A >50 0.28 15.75 19.32 29.56 Neg C aMLV >50 >50 >50 >50 >50 >50 NA—
 Not Applicable IC.sub.90: Inhibitory concentration to inhibit 90% of the virus ***Plateau effect

Example 5: Binding Specificity of Monoclonal Antibodies for HIV Gp120 by ELISA Assay
 (552) The purified anti-gp120 monoclonal antibodies, 1456_P20 (PG20), 1460_G14 (PGG14), and
 1495_C14 (PGC14), were confirmed for binding reactivity to gp120 in ELISA assays. When
 titrated in serial dilutions, all three antibodies exhibited similar binding profiles that suggest
 significantly higher relative avidity than control anti-gp120 (b12). MAb b12 is directed against an
 epitope overlapping the CD4 binding site. (Burton D R et al. 1994. Efficient neutralization of
 primary isolates of HIV-1 by a recombinant human monoclonal antibody. *Science* 266:1024-1027).
 (553) FIG. 5 shows dose response curves of 1456_P20 (PG20), 1460_G14 (PGG14), and
 1495_C14 (PGC14) binding to recombinant gp120 in ELISA as compared to control anti-gp120
 (b12). Data shown represented average OD values of triplicate ELISA wells obtained on the same
 plate.

(554) The monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) were tested for binding
 to soluble recombinant envelope proteins derived from several HIV strains in ELISA assay. ELISA
 assays were performed as described in Pantophlet, R., et al. *J Virol* 77, 642-658 (2003). For antigen
 binding ELISAs, serial dilutions of PG9 were added to antigen coated wells and binding was
 probed with alkaline phosphatase-conjugated goat anti-human immunoglobulin G (IgG) F(ab')₂ Ab
 (Pierce). For competition ELISAs, competitor mAbs were added to ELISA wells and incubated for
 15 min prior to adding 15 µg/mL biotinylated PG9 to each well. Biotinylated PG9 was detected
 using alkaline phosphatase conjugated streptavidin (Pierce) and visualized using p-nitrophenol
 phosphate substrate (Sigma). HIV-HXB2 gp120 was used for competition ELISA assays.

(555) FIG. 6 shows results from ELISA binding assays of monoclonal antibodies 1443_C16
 (PG16) and 1496_C09 (PG9) to HIV-1 YU2 gp140, JR-CSFgp120, membrane-proximal external
 regions (MPER) peptide of gp41 and V3 polypeptide. Specificity of the monoclonal antibodies
 1443_C16 (PG16) and 1496_C09 (PG9) for gp120 was then confirmed, but it was noted that the
 binding to soluble envelope glycoprotein was weak.

Example 6: Binding Reactivity of Monoclonal Antibodies 1443_C16 (PG16) and 1496_C09 (PG9)
 to Envelope Proteins Expressed on Transfected Cell Surface and Competition by Soluble CD4
 (sCD4)

(556) MAb cell binding assays were performed as described in Pancera, M. & Wyatt, R. *Virology*
 332, 145-156 (2005). Titrating amounts of PG9 and PG16 were added to HIV-1 Env transfected
 293T cells, incubated for 1 hr at 4° C., washed with FACS buffer, and stained with goat anti-human
 IgG F(ab')₂ conjugated to phycoerythrin. For competition assays, competitor antibodies were added
 to the cells 15 min prior to adding 0.1 µg/mL biotinylated PG9 or PG16. For sCD4 inhibition
 assays, 40 µg/mL sCD4 was added to the cells and incubated for 1 h at 4° C. prior to adding
 titrating amounts of antibodies. Binding was analyzed using flow cytometry, and binding curves

were generated by plotting the mean fluorescence intensity of antigen binding as a function of antibody concentration.

(557) Ninety-six-well ELISA plates were coated overnight at 4° C. with 50 µL PBS containing 100 ng gp120 or gp140 per well. The wells were washed four times with PBS containing 0.025% TWEEN® 20 (polysorbate 20) and blocked with 3% BSA at room temperature for 1 h. Serial dilutions of PG9 were added to antigen coated wells, incubated for 1 h at room temperature, and washed 4× with PBS supplemented with 0.025% TWEEN® 20 (polysorbate 20). Binding was probed with alkaline phosphatase-conjugated goat anti-human immunoglobulin G (IgG) F(ab')₂ Ab (Pierce) diluted 1:1000 in PBS containing 1% BSA and 0.025% TWEEN® 20 (polysorbate 20). The plate was incubated at room temperature for 1 h, washed four times, and the plate was developed by adding 50 µL of alkaline phosphatase substrate (Sigma) to 5 mL alkaline phosphatase staining buffer (pH 9.8), according to the manufacturer's instructions. The optical density at 405 nm was read on a microplate reader (Molecular Devices). For competition ELISAs, competitor mAbs were added to gp120Hx132 or gp140Yu2 coated ELISA wells and incubated for 15 min prior to adding 15 µg/mL biotinylated PG9 to each well. Biotinylated PG9 was detected using alkaline phosphatase conjugated streptavidin (Pierce) and visualized using p-nitrophenol phosphate substrate (Sigma). For sCD4 inhibition ELISAs, 5 µg/mL sCD4 was added to antigen-coated wells and incubated for 15 min at room temperature prior to adding titrating amounts of PG9. A FACSArray™ plate reader (BD Biosciences, San Jose, CA) was used for flow cytometric analysis and FlowJo™ software was used for data interpretation.

(558) HIV gp160 derived from YU2 was transfected in 293 cells. Binding of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) were detected in transfected cells (FIG. 7). The preincubation of transfected cells with soluble CD4 (sCD4) partially inhibited binding of monoclonal antibody for 1496_C09 (PG9), and for 1443_C16 (PG16) suggesting that antibody binding is effected by the presence of sCD4. Binding is inhibited by at least 15%, at least 20%, at least 25%, or at least 30%. Binding of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to 293 cells transfected with gp160 derived from JR-CSF and ADA strains was also detected (FIG. 8). The binding of both monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to JR-CSF transfected cells was blocked by sCD4. Results further confirm that binding activities of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) are affected by the presence of sCD4.

Example 7: Binding Reactivity of Monoclonal Antibodies 1443_C16 (PG16) and 1496_C09 (PG9) to Pseudoviruses

(559) In vitro virus capture assay was used to test if monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) bind to intact entry competent pseudoviruses. The monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) were coated at the bottom of 96-well plate via anti-human Fc. JR-CSF pseudovirus was added and captured by the monoclonal antibody 1443_C16 (PG16) or 1496_C09 (PG9) in a dose dependent manner. Target cells were added to initiate infection. Infection measured in RLU then represented the binding and capture activity of monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9). FIG. 9 shows the binding and capture of JR-CSF pseudovirus by both monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) in a dose dependent manner, which is similar or better than another known broad and potent neutralizing antibody 2G12.

Example 8: Monoclonal Antibodies 1443_C16 (PG16) and 1496_C09 (PG9) Cross-Compete with Each Other and with sCD4 in Binding to JR-CSF Pseudovirus

(560) In a competition version of virus capture assay where JR-CSF pseudovirus was captured by monoclonal antibodies 1443_C16 (PG16), competition of the capture by either monoclonal antibodies 1443_C16 (PG16), 1496_C09 (PG9) and sCD4 was measured. FIG. 10B shows that binding of monoclonal antibody 1443_C16 (PG16) to JR-CSF pseudovirus was blocked by itself, monoclonal antibody 1496_C09 (PG9) and sCD4 in a dose dependent manner. In a corresponding

manner, FIG. 10B shows that binding of monoclonal antibody 1496_C09 (PG9) to JR-CSF pseudovirus was blocked by itself, monoclonal antibody 1443_C16 (PG16) and sCD4 in a dose dependent manner. Results indicated that the monoclonal antibodies 1443_C16 (PG16) and 1496_C09 (PG9) bind to closely related epitopes on gp120 and their binding is affected by the presence of sCD4 presumably due to conformational changes induced on HIV-1 envelope by sCD4.

Example 9: Antigen Binding Properties of PG9 and PG16

(561) Antigen binding properties of PG9 and PG16 were determined by ELISA assays as shown in FIG. 11A-B. Binding of PG9 and PG16 to monomeric gp120 and artificially trimerized gp140 constructs were determined (FIG. 11A). Binding of PG9 and PG16 to Env expressed on the surface of 293T cells as determined by flow cytometry. (FIG. 11B). b12 was used as a control for ELISA assays. The bNAb b12 and the non-neutralizing antibody b6 were included in the cell surface binding assays to show the expected percentages of cleaved and uncleaved Env expressed on the cell surface.

Example 10: Binding of PG9 and PG16 to Cleavage-Defective HIV-1.SUB.YU2 .Trimers

(562) Binding of PG9 and PG16 to cleavage-defective HIV-1.sub.YU2 trimers was determined by flow cytometry. PG9 and PG16 bind with high affinity to cleavage-defective HIV-1.sub.YU2 trimers as shown in FIG. 12. Binding curves were generated by plotting the mean fluorescence intensity (MFI) of antigen binding as a function of antibody concentration.

Example 11: Mapping the PG9 and PG16 Epitopes

(563) Mapping the epitopes of PG9 and PG16 epitopes was performed by a competitive binding assay as shown in FIG. 13. PG9 and PG16 competed with each other for cell surface Env binding and neither antibody competed with the CD4bs antibody b12 for Env binding. Competitor antibody is indicated at the top of each graph. (FIG. 13A). Ligation of cell surface Env with sCD4 diminished binding of PG9 and PG16. 2G12 was included to control for CD4-induced shedding of gp120. (FIG. 13B). sCD4 inhibited binding of PG9 to artificially trimerized gp140.sub.JR-CSF as determined by ELISA. (FIG. 13C). PG9 competed with 10/76b (anti-V2), F425/b4e8 (anti-V3) and X5 (CD4i) for gp120 binding in competition ELISA assays. (FIG. 13D). PG9 and PG16 failed to bind variable loop deleted HIV-1.sub.JR-CSF variants expressed on the surface of 293T cells. 2G12 was included to control for cell surface Env expression. (FIG. 13E).

Example 12: Competition ELISA Assays Using PG9

(564) When competition ELISA assays using PG9 were performed, PG9 competed with c108g (anti-V2) and partially competed with 17b (CD4i). No competition was observed with A32 (anti-C1/C2/C4/CD4i), C11 (C1), 2G12 (glycan shield), b6 (CD4bs), b3 (CD4bs) or 23b (C1/C5) for gp120Dxs2 binding as shown in FIG. 14.

Example 13: Binding of PG9 and PG16 to HIV-1.SUB.JR_FL .E168K

(565) Antibody binding to HIV-1JR-FLACT E168K Env expressed on the surface of 293T cells as determined by flow cytometry is shown in FIG. 15. A cytoplasmic tail deleted construct was used to increase cell surface expression. The bNAb b12 and the non-neutralizing antibody b6 were included in the cell surface binding assays to show the expected percentages of cleaved and uncleaved Env expressed on the cell surface. (Pancera M., et al. Virology 332:145 (2005). HIV-1JR-FL E168K was generated by site-directed mutagenesis. Binding curves were generated by plotting the MFI of antigen binding as a function of antibody concentration.

Example 14: PG9 binding to deglycosylated gp120

(566) gp120.sub.DU422 was treated with 40 mU/μg Endoglycosidase H (Endo H, New England Biolabs) in sodium acetate buffer for 24 hr at 37° C. Mock treated gp120 was treated under same conditions, but the enzyme was omitted from the reaction. Binding of PG9 and b6 to EndoH treated and mock treated gp120 was determined by ELISA as shown in FIG. 16.

Example 15: Neutralization Activity Against HIV-1.SUB.SF162.K160N

(567) Neutralization activity of PG9 and PG16 against HIV-1 SF162 and HIV-1 SF162K160N was determined using a single-round replication luciferase reporter assay of pseudotyped virus. HIV-1

SF162 K160N was generated by site-directed mutagenesis as shown in FIG. 17.

Example 16: Binding of PG9 and PG16 to Mixed Trimers

(568) Alanine substitutions at positions 160 and 299 were introduced into HIV-1.sub.YU2 Env to abolish binding of PG9 and PG16. An alanine substitution at position 295 was also introduced into the same construct to abrogate binding of 2G12. Co-transfection of 293T cells with WT and mutant plasmids in a 1:2 ratio resulted in the expression of 29% mutant homotrimers, 44% heterotrimers with two mutant subunits, 23% heterotrimers with one mutant subunit, and 4% wild-type homotrimers. These proportions were calculated using the formula described in Yang, X., Kurteva, S., Lee, S., and J. Sodroski, J Virol 79(6):3500-3508 (March 2005), and assumes that mutant and wild-type gp120s mix randomly to form trimers. Binding of mAbs to Env trimers was determined by flow cytometry as shown in FIG. 18. b12 was included as control for Env cell surface expression.

Example 17: PG9 or PG16 Neutralization Activity on HIV with Alanine Mutations within Gp120

(569) Alanine mutations within gp120 of HIV decrease PG9 or PG16 neutralization activity as shown in Table 21. In the table, amino acid numbering is based on the sequence of HIV-1HxB2. Boxes are color coded as follows: white, the amino acid is identical among 0 to 49% of all HIV-1 isolates; light grey, the amino acid is identical among 50 to 90% of isolates; dark grey, the amino acid is identical among 90 to 100% of isolates. Amino acid identity was determined based on a sequence alignment of HIV-1 isolates listed in the HIV sequence database at hiv-web.lanl.gov/content/hiv-db/mainpage.html. C refers to constant domains and V refers to variable loops. Neutralization activity is reported as fold increase in IC.sub.50 value relative to WT JR-CSF and was calculated using the equation (IC.sub.50 mutant/IC.sub.50 WT). Boxes are color coded as follows: white, substitutions which had a negative effect on neutralization activity; light grey, 4-9 fold IC.sub.50 increase; medium grey, 10-100 fold IC.sub.50 increase; dark grey, >100 fold IC.sub.50 increase. Experiments were performed in triplicate and values represent an average of at least three independent experiments.

(570) TABLE-US-00026 TABLE 18A IC50 (1/Dil'n).sup.b IC50 (µg/ml).sup.a Donor Clade Virus
b12 2G12 2F5 4E10 PG9 PG16 PGC14 Serum A MGRM-A-001 >50 >50 >50 15.08 >50 >50 >50
<100 MGRM-A-002 >50 >50 >50 6.45 0.02 0.004 >50 804 MGRM-A-003 >50 >50 7.37 5.94 0.65
2.65 >50 <100 MGRM-A-004 >50 >50 7.49 3.14 0.02 0.04 >50 523 MGRM-A-005 3.64 >50 5.70
4.09 0.28 0.09 >50 175 MGRM-A-006 13.62 13.75 15.73 9.87 >50 >50 >50 131 MGRM-A-007
>50 >50 16.33 1.82 0.37 5.91 >50 142 MGRM-A-008 >50 >50 >50 7.59 >50 >50 >50 142
MGRM-A-009 4.34 7.47 9.40 12.01 0.03 0.01 >50 941 MGRM-A-010 >50 17.01 20.75 14.44 0.02
0.004 >50 1430 MGRM-A-011 4.01 >50 >50 2.88 0.02 0.24 >50 404 MGRM-A-012 >50 >50 2.36
4.27 11.18 20.72 >50 <100 MGRM-A-013 7.04 >50 0.66 1.46 0.16 0.09 >50 350 MGRM-A-014
>50 >50 1.43 1.74 0.62 20.33 >50 158 94UG103 6.92 48.12 1.92 4.97 0.24 0.04 >50 350
92RW020 >50 0.48 3.36 4.54 0.08 0.28 46 282 93UG077 46.95 >50 3.30 10.60 >50 >50 >50 206
94KE105 >50 7.22 >50 7.63 29.56 6.13 >50 <100 93RW029 >50 >50 >50 15.52 1.19 3.83 42 256
92RW009 >50 26.14 39.08 >50 0.03 0.11 >50 254 92UG031 >50 >50 3.81 4.94 3.08 0.43 >50 259
92RW026 >50 17.20 8.63 12.88 0.27 0.03 >50 361 92UG037 >50 45.24 3.24 8.84 0.02 0.01 >50
1252 92RW008 9.46 22.47 10.41 14.53 0.01 0.002 37 4067 92RW021* >50 >50 4.16 4.87 0.05
0.11 >50 316 VLGCA1 >50 >50 3.90 4.58 0.07 0.18 >50 197 92RW024 >50 >50 8.22 8.88 0.18
0.08 >50 241

(571) TABLE-US-00027 TABLE 18B IC50 (1/Dil'n).sup.b IC50 (µg/ml).sup.a Donor Clade Virus
b12 2G12 2F5 4E10 PG9 PG16 PGC14 Serum B 6535.3 1.93 3.85 2.76 1.23 0.22 36.88 35 387
(Acute) QH0692.42 0.73 4.39 5.42 12.67 >50 >50 >50 <100 (Acute) SC422661.8 6.11 0.84 >50
6.35 0.79 1.13 >50 182 (Acute) PVO.4 >50 0.80 >50 18.32 4.01 5.43 >50 171 (Acute) TRO.11 >50
0.29 >50 1.39 5.43 0.22 >50 222 (Acute) CAAN.A2 >50 >50 23.05 17.89 5.67 8.83 >50 <100
(Acute) TRJ0.58 >50 >50 >50 11.94 0.43 1.16 >50 171 (Acute) THR0.18 3.62 >50 >50 4.68 12.39
1.34 >50 <100 (Acute) 92BR020 >50 4.84 >50 >50 >50 >50 4 <100 APV_13 >50 9.24 3.81 7.33

>50 >50 <100 APV_17 >50 >50 10.53 14.59 24.78 >50 <100 APV_6 >50 1.90 0.25 1.10
0.12 0.29 23 394 93TH305 4.17 0.55 7.61 12.33 2.08 19.34 6 133 VLGCB3 0.15 7.90 >50 5.76
0.02 0.40 21 244 JRCSF 0.21 0.37 1.85 3.30 0.003 0.001 15 8425 NL43 0.17 0.49 2.02 4.67 0.32
0.02 40 1488 MGRM- 0.75 0.08 0.55 1.46 >50 >50 >50 <100 Chronic- B-001 MGRM- 0.86 >50
1.25 2.19 1.41 3.06 >50 220 Chronic- B-002 MGRM- >50 0.06 1.00 3.50 50.00 0.19 >50 280
Chronic- B-003 MGRM- 0.26 8.65 2.41 3.70 0.11 0.01 >50 1316 Chronic- B-004 MGRM- 2.82
0.55 >50 16.70 6.66 0.73 >50 140 Chronic- B-008 MGRM- >50 1.50 0.96 1.69 0.004 0.01 27 1640
Chronic- B-010 MGRM- 2.11 >50 0.81 1.07 >50 >50 >50 249 Chronic- B-011 MGRM- >50 0.22
17.65 48.05 0.91 3.74 >50 304 Chronic- B-012 MGRM- 2.59 >50 >50 2.77 0.32 0.02 >50 644
Chronic- B-017 MGRM- 0.66 >50 10.80 23.19 0.16 0.70 >50 180 Chronic- B-018 MGRM- 6.16
0.20 0.78 2.45 >50 >50 >50 <100 Chronic- B-020 MGRM- >50 0.16 0.10 27.92 0.04 0.13 >50 286
Chronic- B-023 MGRM- >50 >50 >50 9.19 0.18 0.01 >50 884 Chronic- B-024 JRFL 0.02 1.45
3.54 18.91 >50 >50 >50 <100 SF162 0.02 1.67 2.52 4.28 >50 >50 <0.0025 9777

(572) TABLE-US-00028 TABLE 18C IC50 (1/Dil'n).sup.b IC50 (µg/ml).sup.a Donor Clade Virus
b12 2G12 2F5 4E10 PG9 PG16 PGC14 Serum C MGRM-C-001 >50 2.93 >50 5.66 >50 16.79 >50
175 MGRM-C-002 >50 >50 44.68 18.19 >50 28.30 >50 <100 MGRM-C-004 5.46 >50 >50 24.24
1.18 5.09 >50 183 MGRM-C-005 2.66 >50 >50 16.41 2.98 2.55 >50 306 MGRM-C-006 >50 >50
>50 4.94 0.23 2.62 >50 224 MGRM-C-007 >50 >50 >50 5.84 0.09 0.05 >50 598 MGRM-C-008
1.51 >50 >50 2.97 >50 >50 >50 160 MGRM-C-009 >50 >50 >50 0.56 >50 >50 >50 <100 MGRM-
C-010 >50 >50 >50 10.96 12.45 >50 >50 <100 MGRM-C-012 >50 >50 >50 0.44 0.24 0.48 >50
432 MGRM-C-013 >50 >50 18.35 2.10 >50 >50 >50 105 MGRM-C-014 >50 >50 >50 2.48 0.64
>50 >50 124 MGRM-C-015 13.30 1.75 >50 2.52 0.50 0.26 >50 365 MGRM-C-017 >50 >50 >50
1.47 1.52 1.80 >50 190 MGRM-C-019 >50 >50 >50 3.49 0.01 0.002 12 6894 MGRM-C-020 >50
18.58 >50 2.80 >50 >50 >50 <100 MGRM-C-022 >50 >50 >50 5.71 0.19 0.25 >50 126 MGRM-C-
023 13.88 >50 >50 1.95 0.51 0.09 >50 220 MGRM-C-024 >50 >50 >50 22.61 0.22 0.04 >50 494
MGRM-C-025 >50 >50 >50 5.58 0.17 0.04 >50 434 93IN905 21.38 >50 >50 1.26 0.03 0.25 19
647 LAVIC_18 >50 >50 >50 >50 0.10 0.02 >50 577 LAVI_C22 7.64 >50 >50 2.02 0.14 0.02 25
1002 LAVI_C3 0.94 >50 >50 2.85 1.45 9.55 12 443 98IN022 0.42 >50 >50 0.53 0.006 0.003 9
2708 93MW959 >50 >50 >50 4.55 0.04 0.007 >50 976 97ZA012 >50 >50 >50 4.70 1.27 2.55 >50
188 CRF08 98CN006 >50 >50 >50 1.91 >50 >50 >50 397 BC CRF07 98CN009 1.52 >50 >50 2.46
1.07 5.76 43 289 BC

(573) TABLE-US-00029 TABLE 18D IC50 (1/Dil'n).sup.b IC50 (µg/ml).sup.a Donor Clade Virus
b12 2G12 2F5 4E10 PG9 PG16 PGC14 Serum D MGRM-D-001 >50 >50 0.63 1.84 >50 >50 >50
<100 MGRM-D-002 >50 >50 24.64 9.44 0.027 0.01 29 515 MGRM-D-003 >50 >50 >50 2.49 0.02
0.01 >50 363 MGRM-D-004 >50 >50 2.30 1.58 0.03 0.01 >50 616 MGRM-D-005 >50 25.66 >50
35.16 0.59 19.66 >50 <100 MGRM-D-008 >50 >50 >50 42.90 6.86 >50 >50 <100 MGRM-D-011
7.75 1.50 >50 0.91 0.06 0.01 >50 298 MGRM-D-012 0.13 >50 1.70 1.13 9.31 0.35 >50 <100
MGRM-D-013 >50 >50 2.12 5.38 0.06 0.11 >50 <100 MGRM-D-014 >50 >50 2.22 3.24 0.02
0.003 48 5127 MGRM-D-016 1.12 >50 9.85 15.45 0.10 0.02 >50 364 MGRM-D-018 1.39 0.12
4.05 3.90 0.02 0.004 >50 883 MGRM-D-019 >50 >50 0.14 0.04 0.03 0.01 >50 497 MGRM-D-020
>50 >50 >50 >50 2.03 16.27 >50 <100 MGRM-D-021 5.23 22.98 >50 13.26 >50 >50 >50 <100
MGRM-D-022 17.63 >50 8.45 16.92 >50 >50 >50 <100 MGRM-D-024 5.92 >50 >50 3.60 0.03
0.02 >50 239 MGRM-D-026 1.55 >50 4.37 2.95 17.51 >50 >50 <100 MGRM-D-028 0.78 >50 >50
1.28 4.39 >50 >50 <100 MGRM-D-029 >50 >50 >50 5.30 >50 >50 >50 <100 92UG024 45.64 0.42
0.95 2.17 1.91 23.98 >50 112 92UG005 >50 >50 8.61 7.46 >50 >50 >50 <100 92UG046 0.07 >50
>50 12.15 0.64 1.42 >50 114 92UG001 1.01 >50 12.98 13.58 41.79 >50 >50 <100 94UG114 >50
13.92 >50 9.72 >50 >50 >50 <100

(574) TABLE-US-00030 TABLE 18E IC50 (1/Dil'n).sup.b IC50 (µg/ml).sup.a Donor Clade Virus
b12 2G12 2F5 4E10 PG9 PG16 PGC14 Serum CRF01_AE MGRM-AE-001 25.95 >50 0.29 0.85
2.97 4.33 >50 <100 MGRM-AE-002 17.10 >50 0.31 0.55 0.04 0.01 >50 653 MGRM-AE-003 >50

>50 0.24 0.34 0.02 0.03 >50 211 MGRM-AE-004 >50 >50 0.98 1.27 0.01 0.002 >50 1773
MGRM-AE-005 0.63 >50 0.14 0.47 0.16 0.02 >50 233 MGRM-AE-006 >50 >50 0.18 0.23 0.05
0.03 >50 151 MGRM-AE-007 >50 >50 0.07 0.45 0.11 0.04 >50 176 MGRM-AE-008 >50 >50 >50
0.94 10.58 3.25 >50 141 92TH021 N/A >50 N/A 1.17 0.09 0.10 >50 192 CMU02 29.32 >50 0.60
0.72 7.69 43.63 >50 142 CRF_AG MGRM-AG-001 11.87 0.69 0.75 1.12 8.83 0.03 >50 388
MGRM-AG-002 0.89 0.54 0.54 0.80 0.04 0.03 >50 147 MGRM-AG-003 >50 >50 0.14 0.64 9.71
>50 >50 <100 MGRM-AG-005 >50 >50 >50 2.13 29.67 >50 >50 150 MGRM-AG-006 >50 3.92
0.85 1.76 >50 >50 >50 <100 MGRM-AG-008 >50 >50 0.54 1.48 0.02 0.002 45 1518 MGRM-AG-
009 >50 >50 24.80 31.39 >50 >50 >50 <100 MGRM-AG-011 >50 >50 >50 1.36 0.01 0.002 >50
1427 MGRM-AG-012 10.40 1.94 0.33 0.86 1.37 25.13 >50 <100 MGRM-AG-013 >50 0.95 1.79
2.61 0.23 0.31 >50 <100 G MGRM-G-001 >50 >50 4.1 2.04 0.16 0.15 >50 <100 MGRM-G-004
>50 >50 >50 1.47 >50 >50 >50 <100 MGRM-G-006 >50 >50 1.33 1.23 0.51 2.42 >50 116
MGRM-G-009 >50 >50 7.21 1.34 4.90 >50 >50 <100 MGRM-G-011 >50 >50 1.16 1.44 0.19 0.04
>50 150 MGRM-G-013 >50 >50 0.59 1.15 >50 >50 >50 <100 MGRM-G-014 >50 >50 9.65 13.67
6.32 6.98 >50 <100 MGRM-G-015 >50 >50 0.43 1.07 1.51 5.33 >50 <100 MGRM-G-016 >50 >50
16.82 1.02 0.40 11.35 >50 <100 MGRM-G-017 >50 >50 0.60 1.14 0.03 0.02 >50 453 MGRM-G-
019 3.77 31.03 >50 6.53 0.67 1.21 >50 <100 MGRM-G-024 2.38 >50 1.07 1.57 0.07 0.01 >50 236
MGRM-G-025 >50 31.94 >50 1.70 >50 >50 >50 <100 MGRM-G-027 >50 >50 0.28 1.19 0.01 0.01
>50 351 MGRM-G-028 >50 28.25 2.24 6.32 0.13 3.09 >50 <100

(575) TABLE-US-00031 TABLE 18F IC50 (1/Dil'n).sup.b IC50 (µg/ml).sup.a Donor Clade Virus
b12 2G12 2F5 4E10 PG9 PG16 PGC14 Serum F MGRM- >50 >50 4.31 2.74 0.11 0.43 >50 104
F1-004 MGRM- >50 >50 1.10 1.01 1.45 0.27 >50 <100 F1-006 MGRM- >50 >50 1.61 2.75 >50
>50 >50 <100 F1-008 MGRM- >50 N/A 14.56 3.69 0.03 0.01 >50 634 F1-010 MGRM- >50 1.81
>50 0.37 0.01 0.003 >50 866 F1-012 MGRM- >50 >50 4.57 N/A 0.56 N/A 6 142 F1-013 MGRM-
>50 >50 15.13 7.36 0.01 0.01 >50 437 F1-014 MGRM- >50 >50 0.10 0.53 >50 >50 >50 <100 F1-
015 MGRM- >50 >50 21.47 7.61 0.58 1.12 >50 <100 F1-016 MGRM- >50 >50 >50 4.92 >50 >50
>50 <100 F1-017 MGRM- >50 >50 3.91 3.60 0.03 0.01 >50 432 F1-018 MGRM- >50 >50 0.59
0.66 4.55 4.35 >50 <100 F1-020 MGRM- >50 14.09 1.37 1.87 >50 >50 46 <100 F1-021 MGRM-
>50 >50 1.26 1.01 0.06 0.08 >50 246 F1-022 MGRM- >50 9.23 1.78 0.44 >50 >50 >50 101 F1-
023 neg. aMLV >50 >50 >50 >50 >50 >50 >50 <100 control

(576) TABLE-US-00032 TABLE 19A Neutralization Potency. Median IC.sub.90 (µg/mL) against
viruses neutralized with an IC.sub.90 <50 µg/ml Clade .sup.a # viruses b12 2G12 2F5 4E10 PG9
PG16 PGC14 A 27 48.45 17.77 28.82 40.62 0.99 0.81 >50 B 31 2.30 4.65 25.85 32.38 0.11 0.01
9.45 C 27 28.41 28.67 >50 23.37 2.94 5.10 >50 D 25 12.68 8.76 9.02 23.45 0.34 0.44 >50
CRF01_AE 10 12.68 >50 8.14 12.95 0.36 1.51 >50 CRF_AG 10 16.97 7.04 13.49 15.78 0.28 1.86
>50 G 15 23.62 >50 17.54 16.67 1.91 1.96 >50 F 15 >50 21.49 17.77 7.64 0.25 0.55 >50 total 162
20.30 13.27 17.54 23.37 0.36 1.16 9.45 * White boxes indicate a medium potency of >50 µg/mL,

darkest grey between 20 and 50 µg/mL, lightest grey between 2 and 20 µg/mL, medium grey
between 0.2 and 2 µg/mL, and darker grey <0.2 µg/mL. CRF_07BC and CRF_08BC viruses not
included in the clade analysis because there was only one virus tested from each of these clades.

(577) TABLE-US-00033 TABLE 19B Neutralization Breadth. # viru- % viruses neutralized with an
IC.sub.90 <50 µg/ml Clade .sup.a ses b12 2G12 2F5 4E10 PG9 PG16 PGC14 A 27 4 4 33 22 74 41
0 B 31 45 52 45 23 42 26 6 C 27 15 4 0 41 52 41 0 D 25 28 12 12 20 44 36 0 CRF01_AE 10 11 0
67 70 60 60 0 CRF_AG 10 10 30 70 60 40 40 0 G 15 13 0 53 53 47 27 0 F 15 0 7 47 43 47 29 0
total 162 19 15 33 36 51 35 4 # vir- % viruses neutralized with an IC.sub.90 <1.0 µg/ml Clade
.sup.a uses b12 2G12 2F5 4E10 PG9 PG16 PGC14 A 27 0 0 0 0 36 27 0 B 31 10 6 0 0 13 19 3 C 27
0 0 0 0 15 15 0 D 25 0 4 0 0 32 20 0 CRF_01AE 10 0 0 0 0 40 30 0 CRF_AG 10 0 0 0 0 30 10 0 G
15 0 0 0 0 13 7 0 F 15 0 0 0 0 33 21 0 total 162 2 2 0 0 25 18 <1 * White boxes indicate that no
viruses were neutralized, darkest grey indicate 1 to 30% of viruses were neutralized, lightest grey
indicate 30 to 60% of viruses were neutralized, medium grey indicate 60 to 90% of viruses were

neutralized, and darker grey indicate 90 to 100% of viruses were neutralized. CRF_07BC and CRF_08BC viruses not included in the clade analysis because there was only one virus tested from each of these clades.

(578) TABLE-US-00034 TABLE 20 Neutralization activity of PG9 and PG16 against JR-CSF pseudovirus containing alanine point mutations. Fold IC₅₀ increase Fold IC₅₀ increase gp120 relative to wild-type.sup.d gp120 relative to wild-type.sup.d Mutation.sup.ab domain.sup.c PG9 PG16 Mutation.sup.a, b domain.sup.c PG9 PG16 W112A C1 1 1 P299A V3 (base) 200 1400 V120A C1 2 1 N301A V3 (base) 9 3 K121A C1 1 1 N302A V3 (stem) 1 1 (V1/V2 stem) L122A C1 2 1 R304A V3 (stem) 2 3 (V1/V2 stem) L125A C1 1 1 K305A V3 (stem) 50 2800 (V1/V2 stem) V127A C1 30 57 S306A V3 (tip) 1 1 (V1/V2 stem) N134A V1 5 23 I307A V3 (tip) 10 3000 N156A C1 280 1500 H308A V3 (tip) 3 1 (V1/V2 stem) S158A C1 >2000 >2000 I309A V3 (tip) 9 150 (V1/V2 stem) F159A C1 >2000 >2500 P313A V3 (tip) 1 1 (V1/V2 stem) N160k V2 >2000 >2500 R315A V3 (tip) 1 1 T162A V2 >2000 >2500 F317A V3 (tip) 3 1400 I165A V2 1 1 Y318A V3 (tip) 2 1000 R166A V2 2 1 T319A V3 (tip) 1 1 D167A V2 5 30 T320A V3 (tip) 2 1 K168A V2 1 3 E322A V3 (stem) 2 3 K171A V2 1 1 D325A V3 (stem) 1 1 E172A V2 1 1 H330A V3 (base) 1 1 Y173A V2 1400 1000 N332A V3 (base) 1 1 F176A V2 ≥5000 ≥7000 Q337A C3 1 1 Y177A V2 1 5 N339A C3 1 1 L179A V2 1 3 K343A C3 1 1 D180A V2 1 4 R350A C3 1 1 V181A V2 200 250 N355A C3 9 3 V182A V2 1 3 S365A C3 2 3 I184A V2 1 1 N386A C3 1 1 D185A V2 1 1 T388A C3 1 1 N188A V2 3 3 N392A V4 7 23 T190A V2 2 4 W395A V4 1 1 N197K C2 1 1 R419A C4 3 3 (V1/V2 stem) T198A C2 2 1 I420A C4 9 11 (V1/V2 stem) S199A C2 2 1 K421A C4 1 1 (V1/V2 stem) T202A C2 1 1 Q422A C4 9 5 (V1/V2 stem) F210A C2 3 1 I423A C4 40 14 I213A C2 1 1 I424A C4 10 9 N241A C2 4 3 I439A C4 2 3 N262A C2 1 1 T450A C4 1 1 N276A C2 1 1 L452A C4 1 1 N295A C2 2 1 P470A V5 1 1 T297A V3 (base) 1 1 .sup.aAmino acid number is based on the sequence of HIV-1.sub.HxB2. .sup.bWhite boxes indicate that the amino acid is identical among 0 to 49% of all HIV isolates, light gray boxes indicate that the amino acid is identical among 50-90% of all HIV isolates, and dark gray boxes indicate that the amino acid is identical among 90-100% of all HIV isolates. Amino acid identity was determined based upon a sequence alignment of HIV-1 isolates listed in the HIV sequence database <http://hiv-gov/content/hiv-db/mainpage.html>. .sup.cC refers to constant domains and V refers to variable loops. .sup.dNeutralization activity is reported as fold increase in IC₅₀ value relative to WT JR-CSF and was calculated using the equation (IC₅₀ mutant/IC₅₀ WT). White: substitutions which had a negligible effect on neutralization activity, lightest grey: 4-9 fold IC₅₀ increase, dark grey: 10-100 fold IC₅₀ increase, darkest grey: >100 fold IC₅₀ increase. Experiments were performed in triplicate and values represent an average of at least three independent experiments.

(579) TABLE-US-00035 TABLE 21 Alanine mutations that decrease PG9 and PG16 neutralization activity. Fold IC₅₀ increase relative to wild-type.sup.d Mutation.sup.a, b gp120 domain.sup.c PG9 PG16 V127A C1 (V1/V2 stem) 30 57 N134A V1 5 23 N156A C1 (V1/V2 stem) 280 1500 S158A C1 (V1/V2 stem) >2000 >2000 F159A C1 (V1/V2 stem) >2000 >2500 N160K V2 >2000 >2500 T162A V2 >2000 >2500 D167A V2 5 30 Y173A V2 1400 1000 F176A V2 >5000 >7000 V181A V2 200 250 P299A V3 (base) 200 1400 K305A V3 (stem) 50 2800 I307A V3 (tip) 10 3000 I309A V3 (tip) 9 150 F317A V3 (tip) 3 1400 Y318A V3 (tip) 2 1000 N392A V4 7 23 I420A C4 9 11 I423A C4 40 14 I424A C4 10 9 .sup.aAmino acid numbering is based on the sequence of HIV-1.sub.HxB2. .sup.bBoxes are color coded as follows: white, the amino acid is identical among 0 to 49% of all HIV-1 isolates; light grey, the amino acid is identical among 50 to 90% of isolates; dark grey, the amino acid is identical among 90 to 100% of isolates. Amino acid identity was determined based on a sequence alignment of HIV-1 isolates listed in the HIV sequence database at <http://hiv-web.lanl.gov/content/hiv-db/mainpage.html>. .sup.cC refers to constant domains and V refers to variable loops. .sup.dNeutralization activity is reported as fold increase in IC₅₀ value relative to WT JR-CSF and was calculated using the equation (IC₅₀.sub.50 mutant/IC₅₀.sub.50 WT). Boxes are color coded as follows: white, substitutions which had a negative effect on neutralization activity;

light grey, 4-9 fold IC.sub.50 increase; medium grey, 10-100 fold IC.sub.50 increase; dark grey, >100 fold IC.sub.50 increase. Experiments were performed in triplicate and values represent an average of at least three independent experiments.

Example 18: Identification of 14443_C16 (PG16) Sister Clones

(580) 1443_C16 sister clones were identified by screening clonal transfection of rescued variable region genes for JR-CSF neutralization. Thus, antibodies that were identified as sister clones of 1443_C16 (PG16) have the similar HIV neutralization profiles as the human monoclonal 1443_C16 (PG16). Moreover, the nucleic acid or amino acid sequences of the sister clone antibodies are at least 50%, 55%, 60%, 70%, 75%, 80%, 85%, 90%, 95%, 99%, 100% or any percentage point in between, identical to those of 1443_C16 (PG16).

(581) TABLE-US-00036 TABLE 22 Antibody JRCSF 1443 C16 concentration Neutralization Sister mAbs Gamma Chain Clone Light Chain Clone (µg/ml) Index 1456 A12

1456_A12_G3_01_002	1456_A12_L2_01_023	0.006	0.90	1456_A12_L2_01_036	0.012	0.82
1456_A12_L2_01_040	0.016	2.79	1456_A12_G3_01_004	1456_A12_L2_01_023	<0.005	1.00
1456_A12_L2_01_036	<0.005	1.02	1456_A12_L2_01_040	0.005	6.95	1469 M23
1469_M23_G3_01_005	1469_M23_L2_01_001	2.624	215.74	1469_M23_G3_01_006	0.000	10.05
1480 I08	1480_I08_G3_01_012	1480_I08_L2_01_005	<0.005	10.34	1480_I08_G3_01_016	10
223.14	1480_I08_G3_01_021	<0.005	2.98	1480_I08_G3_01_032	<0.005	3.83
1480_I08_G3_01_037	34	1.36	1480_I08_G3_01_055	<0.005	1.16	1489 I13
1489_I13_L2_01_007	0.0000	2.02	1489_I13_G3_01_004	0.0009	22.86	1489_I13_G3_01_007
1.455	139.35	1503 H05	1503_H05_G1_01_001	1503_H05_L2_01_021	0.013	0.96
1503_H05_G1_01_006	0.000	3.75	1503_H05_G3_01_005	1.108	91.41	1503_H05_G3_01_007
0.567	155.54	Note that the constant region of the 1456_A12 heavy chain clones used in transfection contains an error generated during the cloning process that lead to no full-length IgG production.				

Example 19: 1443_C16 (PG16) Antibody Sister Clones and the 1443_C16 (PG16) Antibody Exhibit Similar Neutralization Specificity

(582) Antibodies 1456 A12, 1503_H05, 1489_I13 and 1469 M23 were tested for neutralization activity against several pseudoviruses containing distinct mutations that map the reactivity epitope of 1443_C16 (PG16) on gp120 in a standard TZM-bl assay (Table 23). Like 1443_C16 (PG16), which does not bind or neutralize wild-type JR-FL, but instead, neutralizes JR-FL with the E168K mutation, all 1443_C16 (PG16) sister clones neutralize JR-FL(E168K) with low IC50 values. Similarly, all 1443_C16 (PG16) sister clones do not neutralize the Y318A mutants and 1309A mutants of JR-CSF, where the part of the putative binding epitope is mapped on the V3 tip.

(583) TABLE-US-00037 TABLE 23 Neutralization specificity of 1443 C16 (PG16) sister clones as shown with specific mutations on gp120. IC50 (ug/ml) JR- JR-CSF JR-CSF JR-FL mAb CSF (Y318A) (I309A) (E168K) ADA 92RW020 1503 H05 0.001 >1.0 >1.0 0.002 0.003 0.020 1456 A12 0.001 >1.0 >1.0 0.003 0.005 0.050 1469 M23 0.002 >1.0 >1.0 0.005 0.005 0.050 1489 I13 0.002 >1.0 >1.0 0.005 0.008 0.030 1443 C16 0.001 >1.0 >1.0 0.006 0.004 0.090 1496 C09 0.006 0.001 0.001 0.020 0.200 0.100

Example 20: 1443_C16 (PG16) Sister Clones Exhibit Similar Neutralization Breadth and Potency as 1443_C16 (PG16) for Clade B and Clade C Viruses

(584) The antibodies 1456 A12, 1503_H05, 1489_I13 and 1469 M23 exhibit neutralization activity against a panel of Glade B and Glade C pseudoviruses with similar breadth as does 1443 C16 (PG16) in a standard TZM-bl assay (Table 24). The neutralization potency of each sister clone for each pseudovirus is comparable to that for 1443_C16 (PG16). When the IC50 value is determined, the value for the sister clone is within a 0.5 log range from that for 1443_C16 (PG16).

(585) TABLE-US-00038 TABLE 24 Neutralization breadth and potency of 1443 C16 (PG16) sister clones. IC50 (ug/ml) 1443 1456 1469 1503 1489 Virus C16 A12 M23 H05 I13 Clade CAAN 6.37 10.61 17.72 13.46 24.87 B REJ04541 <0.01 <0.01 0.39 0.22 0.34 THRO.18 2.19 2.08 7.01

4.12	7.41	PVO.4	12.3	10.42	21.25	11.01	20.57	TR0.11	3.61	3.05	7.52	4.30	10.94	AC10
<0.01	<0.01	<0.01	<0.01	<0.01	Clade	DU156	<0.01	<0.01	<0.01	<0.01	<0.01	C	DU422	0.59
0.36	0.97	0.71	1.87	Du172	<0.01	<0.01	<0.01	<0.01	<0.01	ZM214	>25	>25	>25	
>25	>25		ZM233	<0.01	<0.01	<0.01	<0.01	<0.01	CAP45	<0.01	<0.01	<0.01	<0.01	<0.01
ZM249	<0.01	<0.01	<0.01	<0.01	<0.01	Control	MuLV	>25	>25	>25	>25	>25	>25	>25

Example 21: Primary and Confirmatory Screening Results for Selected Antibodies Isolated from B-Cell Cultures Established from Human Donors

(586) The screening strategy used in the isolation of the monoclonal antibodies PGT-121 (corresponding to clones 4838_L06 and 4873_E03), PGT-122 (corresponding to clone 4877_D15), PGT-123 (corresponding to clone 4858_P08), PGT-125 (corresponding to clone 5123_A06), PGT-126 (corresponding to clone 5141_B17), PGT-130 (corresponding to clone 5147_N06), PGT-135 (corresponding to clones 5343_B08 and 5344_E16), and PGT-136 (corresponding to clones 5329_C19 and 5366_P21) is the same as the PG9 and PG16 mAbs, except that functional neutralization was the only primary screening assay used (i.e. no ELISA was used to screen these antibodies).

(587) Moreover, the strategy use to identify these mAbs following reverse transcription polymerase chain reaction (RT-PCR) rescue differs from previous protocols. Specifically, in addition to performing a primary neutralization screening step, a confirmatory screening step was performed for some of the positive hits identified from the primary screening step (Tables 25-27). The confirmatory screening step was performed using the same assay as the primary screening step. Following functional screening, the B cell culture lysates were subjected to variable gene family-specific RT-PCR, as performed previously to identify the PG9 and PG16 mAbs. However, instead of directly cloning into IgG.sub.1 expression vector, the PCR products representing the rescued heavy and light chains were subjected to deep sequencing, which is also known as “next-generation sequencing”, “454 sequencing” or “pyrosequencing.”

(588) In the process of deep sequencing, a B cell well location-specific sequence tag was built into the second round of PCR to enable the identification of B cell well origin of each sequence determined in the subsequent pooled sequencing reaction. One or more consensus variable gene sequences were generated from each B cell culture well by an informatics algorithm. The consensus sequences from an individual B cell well were then compared among all consensus sequences generated from other B cell culture wells. Similar heavy chains or light chain sequences were “clustered” because similar mAbs may be derived from the same precursor B cell. Selected variable genes were then cloned into an IgG.sub.1 expression vector to produce and purify monoclonal antibodies. Unlike the previous rescue strategy, polyclonal transfection was not performed to screen for neutralization activity to identify potential variable genes from the PCR product pool prior to proceeding to monoclonal transfection.

(589) The similarity among variable genes that were “clustered” is apparent in the alignment of nucleotide and amino acid sequence alignments (Tables 28-31). For instance, all three mAbs from donor 517, i.e. PGT-121, PGT-122 and PGT-123 are in the same cluster. Donor 196 provided two distantly related clusters of mAbs, with one cluster including PGT-125 and PGT-126, and another including PGT-130. Donor 039 provided two distantly related clusters of mAbs, each including PGT-135 or PGT-136.

(590) TABLE-US-00039 TABLE 25 Donor 517 Confirmatory Neutralization B Cell Primary Neutralization Index Index Range Culture MGRM- MGRM- mAb ID Well ID 92BR020 94UG103 JRCSF C-26 C-026 PGT-121 4838_L06 4.9 1.4 3.2 996.3 high >50 4873_E03 3.6 0.8 2.2 371.4 high >50 N/A 4869_K15 3.9 1.5 2.1 103.9 high >50 PGT-122 4877_D15 5.4 1.3 2.3 37.5 moderate 10-50 PGT-123 4858_P8 2.8 1.1 1.6 33.8 moderate 10-50 N/A 4834_C11 2.5 1.1 2.0 28.3 moderate 10-50

(591) TABLE-US-00040 TABLE 26 Donor 196 Primary Confirmatory Polyclonal Transfectant B Cell Neutralization Index Neutralization Neutralization Index Culture MGRM- Index MGRM-

mAb ID	Well ID	92TH021 JRCFSF C-26 Virus Range	92TH021 JRCFSF C-026 PGT-125 5123_A6
133.0	1727.6	511.5 JRCFSF high >50 50.65 118.60 88.31 PGT-126 5141_B17 2.3	
1410.1	653.9	JRCFSF high >50 2.74 101.51 102.41 PGT-127 5145_B14 1.0 31.1 86.9	
MGRM- high >50	0.94 1.61 2.62 C-026 PGT-128 5114_A19 6.6 77.5 17.1 JRCFSF high >50		
10.02 136.49 32.19 PGT-130 5147_N6 538.2 19.3 3.0 92TH021 high >50 4.20 1.24 1.05			
PGT-131 5136_H1 354.0 6.2 1.2 92TH021 high >50 PGT-132 5113_D22 51.0 3.0 6.0			
92TH021 high >50 PGT-133 5117_E22 42.5 3.6 3.4 92TH021 high >50			

(592) TABLE-US-00041 TABLE 27 Donor 039 Primary Confirmatory B Cell Neutralization Index Neutralization Index Culture MGRM- Index mAb ID Well ID 93IN905 JRCFSF C-26 Virus Range PGT-135 5343_B8 43.0 1.8 6.5 93IN905 moderate 10-50 PGT-137 5345_I1 3.3 1.1 11.3 MGRM-C-026 moderate 10-50 PGT-136 5366_P21 5.5 1.1 6.2 MGRM-C-026 mod low 5-10 5329_C19 5.6 0.9 6.0 MGRM-C-026 mod low 5-10 PGT-135 5344_E16 2.3 1.5 3.6 MGRM-C-026 low 1.5-5

Example 22: Neutralization Values (IC50, IC80, IC90, and IC95) Against 23 HIV Viruses for Selected Antibodies Isolated from B-Cell Cultures Established from Human Donors

(593) Table 32 shows neutralization profiles (IC50 values) of monoclonal antibodies PGT 121, PGT 122, PGT 123, PGT 125, PGT 126, PGT 130, PGT 135, PGT 136, and PG9 on a diverse panel of 23 HIV viruses from different clades (A, B, C, D, AE, and AG). PGT 121, PGT 122, PGT 123, PGT 125, PGT 126, PGT 130, PGT 135, and PGT 136 all neutralize virus in clades A, B, C, and D. Moreover, PGT 121, PGT 122, and PGT 123 also neutralize virus in Glade AG. PGT 125 neutralizes clades A, B, C, D, AE, whereas PGT 126 and PGT 130 neutralize all clades, i.e. A, B, C, D, AE, and AG.

(594) Table 33 shows IC.sub.50, IC.sub.50, IC.sub.90, and IC.sub.95 values of the PGT 121 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PGT 121 monoclonal antibody. This table further demonstrates that the PGT 121 monoclonal antibody neutralizes HIV virus from clades A, B, C, and D strongly, as evidenced by low IC.sub.95 values shown for these clades.

(595) Table 34 shows IC.sub.50, IC.sub.50, IC.sub.90, and IC.sub.95 values of the PGT 122 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PGT 122 monoclonal antibody. This table further demonstrates that the PGT 122 monoclonal antibody neutralizes HIV virus from clades A, B, C, and D strongly, as evidenced by low IC.sub.95 values shown for these clades.

(596) Table 35 shows IC.sub.50, IC.sub.50, IC.sub.90, and IC.sub.95 values of the PGT 123 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PGT 123 monoclonal antibody. This table further demonstrates that the PGT 123 monoclonal antibody neutralizes HIV virus from clades A, B, C, and D strongly, as evidenced by low IC.sub.95 values shown for these clades.

(597) Table 36 shows IC.sub.50, IC.sub.50, IC.sub.90, and IC.sub.95 values of the PGT 125 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PGT 125 monoclonal antibody. This table further demonstrates that the PGT 125 monoclonal antibody neutralizes HIV virus from clades A, B, C, D, and AE strongly, as evidenced by low IC.sub.95 values shown for these clades.

(598) Table 37 shows IC.sub.50, IC.sub.50, IC.sub.90, and IC.sub.95 values of the PGT 126 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PGT 126 monoclonal antibody. This table further demonstrates that the PGT 126 monoclonal antibody neutralizes HIV virus from clades A, B, C, and D strongly, as evidenced by low IC.sub.95 values shown for these clades.

(599) Table 38 shows IC.sub.50, IC.sub.80, IC.sub.90, and IC.sub.95 values of the PGT 130 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PGT 130 monoclonal antibody. This table further

demonstrates that the PGT 130 monoclonal antibody neutralizes HIV virus from clades A, B, C, and AE strongly, as evidenced by low IC.sub.95 values shown for these clades.

(600) Table 39 shows IC.sub.50, IC.sub.80, IC.sub.90, and IC.sub.95 values of the PGT 135 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PGT 135 monoclonal antibody. This table further demonstrates that the PGT 135 monoclonal antibody neutralizes HIV virus from clades B, C, and D strongly, as evidenced by low IC.sub.95 values shown for these clades.

(601) Table 40 shows IC.sub.50, IC.sub.80, IC.sub.90, and IC.sub.95 values of the PGT 136 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PGT 136 monoclonal antibody. This table further demonstrates that the PGT 136 monoclonal antibody neutralizes HIV virus from clades C and D strongly, as evidenced by low IC.sub.95 values shown for these clades.

(602) Table 41 shows IC.sub.50, IC.sub.80, IC.sub.90, and IC.sub.95 values of the PG9 monoclonal antibody on the same panel of viruses shown in Table 32. The results shown on this table recapitulate those shown on Table 32 for the PG9 monoclonal antibody. This table further demonstrates that the PG9 monoclonal antibody neutralizes HIV virus from clades A, B, C, D, and AE strongly, as evidenced by low IC.sub.95 values shown for these clades.

(603) For Tables 32-41, the following color-coding scheme applies regarding the concentration of the appropriate antibody:

(604) TABLE-US-00042 10 ng/ml 100 ng/ml 1000 ng/ml 1000-10,000 ng/ml Negative
= >10

(605) TABLE-US-00043 TABLE 32 IC.sub.50 Neutralization Values for PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, and PGT-136. IC50 (ug/mL) IC50 (Reciprocal Donor of dilution) Donor 517 Donor 196 Donor 039 064 Reference Serum Clade Virus PGT-121 PGT-122 PGT-123 PGT-125 PGT-126 PGT-130 PGT-135 PGT-136 PG9 Z23-6X Z23 Z23 A 92RW020
0.0063 0.0072 0.0038 0.0028 0.0046 0.0140 0.1573 0.7383 0.1614 219 327 322 93UG077
0.0459 0.1913 0.0303 0.0262 0.0256 >10* >10 >10 >10 336 392 365 94UG103 1.3778 1.4643
0.8461 0.0124 0.0091 0.6355 >10 >10 0.3098 179 238 235 MGRM- 2.6078 1.6836 0.4921
0.0055 0.0033 0.0054 >10 >10 0.0375 236 258 308 A-010 B 92BR020 0.0177 0.0337 0.0130
0.0214 0.0163 1.1242 0.1043 >10 >10 308 364 336 APV13 0.3157 0.9923 0.2215 0.0128
0.0120 0.0345 0.5452 >10 >10 217 297 232 APV17 0.0950 0.5220 0.1798 7.3065 0.5013
>10 >10 >10 >10 297 283 273 APV6 0.0423 0.0851 0.0328 0.0359 0.0249 >10 >10 >10
0.2139 396 458 682 JRFL 0.0276 0.0471 0.0283 0.0156 0.0140 0.0346 >10 >10 >10 533 608
612 JRCSF 0.0343 0.0727 0.0423 0.0060 0.0061 0.0089 0.1131 >10 0.0048 425 401 472 NL43
>10 >10 >10 >10 >10 >10 6.7910 >10 0.6871 3905 3823 3577 C 93IN905 0.0082 0.0138
0.0071 0.0137 0.0194 0.0182 0.0183 0.0135 0.0480 366 405 395 MGRM- 0.0034 0.0089
0.0031 0.0106 0.0076 0.0173 0.0123 0.0065 0.1130 426 299 298 C-026 MGRM- 0.0094
0.0388 0.0169 >10 1.1404 0.0052 >10 >10 2.4538 355 340 472 C-027 MGRM- 1.1929 2.0600
0.4433 5.7772 0.2827 >10 2.1608 >10 0.1211 168 145 195 C-028 D 92UG005 >10 >10 2.4924
>10 0.0181 0.4741 >10 >10 >10 350 361 379 92UG024 >10 >10 >10 >10 >10 >10 0.0139
0.0915 1.9142 343 338 433 MGRM- 0.7442 0.8673 0.2154 2.3116 0.0639 >10 >10 >10 >10
<100 114 150 D-001 MGRM- 0.0086 0.0115 0.0047 0.0382 0.0093 0.0378 0.0433 >10 0.0444
414 367 329 D-018 AE 92TH021 >10 >10 >10 0.0066 0.1147 0.0082 >10 >10 0.1026 287 290
304 CMU02 >10 >10 >10 >10 >10 >10 >10 >10 >10 309 308 277 AG MGRM- 1.1719 >10
0.1036 >10 1.3488 0.9466 >10 >10 >10 134 197 209 AG-005 aMLV >10 >10 >10 >10 >10
>10 >10 >10 >10 <100 <100 <100

(606) TABLE-US-00044 TABLE 33 Neutralization Values for PGT-121. Neutralization by PGT-121 (ug/ml) Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.0063 0.0188 0.0361 0.0671 P 93UG077 0.0459 0.1716 0.3691 0.7402 P 94UG103 1.3778 8.8543 >10.0000 >10.0000 P MGRM- 2.6078 >10.0000 >10.0000 >10.0000 P A-010 B 92BR020 0.0177 0.0532 0.1011 0.1828 P APV13

0.3157 1.3067 2.7989 5.1522 P APV17 0.0950 0.3707 0.7991 1.5541 P APV6 0.0423 0.1447
0.2960 0.5688 P JRFL 0.0276 0.0806 0.1506 0.2674 P JRCSF 0.0343 0.1119 0.2226 0.4169 P
NL43 >10.0000 >10.0000 >10.0000 >10.0000 N C 93IN905 0.0082 0.0302 0.0645 0.1298 P
MGRM- 0.0034 0.0140 0.0320 0.0687 P C-026 MGRM- 0.0094 0.0829 0.3739 3.1434 P C-027
MGRM- 1.1929 4.9098 >10.0000 >10.0000 P C-028 D 92UG005 >10.0000 >10.0000 >10.0000
>10.0000 N 92UG024 >10.0000 >10.0000 >10.0000 >10.0000 N MGRM- 0.7442 3.2881 6.2560
9.3894 P D-001 MGRM- 0.0086 0.0231 0.0414 0.0715 P D-018 AE 92TH021 >10.0000 >10.0000
>10.0000 >10.0000 N CMU02 >10.0000 >10.0000 >10.0000 >10.0000 N AG MGRM- 1.1719
>10.0000 >10.0000 >10.0000 P AG-005 aMLV >10.0000 >10.0000 >10.0000 >10.0000 N
(607) TABLE-US-00045 TABLE 34 Neutralization Values for PGT-122. Neutralization by PGT-
122 (ug/ml) Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.0072 0.0407 0.1127 0.2891 P
93UG077 0.1913 0.9312 2.0994 3.8823 P 94UG103 1.4643 6.5793 >10.0000 >10.0000 P MGRM-
1.6836 >10.0000 >10.0000 >10.0000 P A-010 B 92BR020 0.0337 0.0929 0.1680 0.2895 P APV13
0.9923 4.1434 8.6010 >10.0000 P APV17 0.5220 1.7580 3.5831 6.9228 P APV6 0.0851 0.3508
0.7787 1.5492 P JRFL 0.0471 0.1655 0.3422 0.6590 P JRCSF 0.0727 0.2455 0.4945 0.9261 P
NL43 >10.0000 >10.0000 >10.0000 >10.0000 N C 93IN905 0.0138 0.0505 0.1080 0.2172 P
MGRM- 0.0089 0.0370 0.0849 0.1811 P C-026 MGRM- 0.0388 0.4995 >10.0000 >10.0000 P C-
027 MGRM- 2.0600 >10.0000 >10.0000 >10.0000 P C-028 D 92UG005 >10.0000 >10.0000
>10.0000 >10.0000 N 92UG024 >10.0000 >10.0000 >10.0000 >10.0000 N MGRM- 0.8673
4.4756 >10.0000 >10.0000 P D-001 MGRM- 0.0115 0.0350 0.0674 0.1247 P D-018 AE 92TH021
>10.0000 >10.0000 >10.0000 >10.0000 N CMU02 >10.0000 >10.0000 >10.0000 >10.0000 N AG
MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N AG-005 aMLV >10.0000 >10.0000 >10.0000
>10.0000 N
(608) TABLE-US-00046 TABLE 35 Neutralization Values for PGT-123. Neutralization by PGT-
123 (ug/ml) Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.0038 0.0106 0.0194 0.0343
P 93UG077 0.0303 0.1433 0.3504 0.7798 P 94UG103 0.8461 3.2922 8.5552 >10.0000 P MGRM-
0.4921 2.7917 6.4406 >10.0000 P A-010 B 92BR020 0.0130 0.0390 0.0741 0.1339 P APV13
0.2215 0.8787 1.8718 3.5039 P APV17 0.1798 0.5389 1.0082 1.7530 P APV6 0.0328 0.1447
0.3394 0.7264 P JRFL 0.0283 0.0904 0.1782 0.3318 P JRCSF 0.0423 0.1428 0.2890 0.5467 P
NL43 >10.0000 >10.0000 >10.0000 >10.0000 N C 93IN905 0.0071 0.0281 0.0625 0.1306 P
MGRM- 0.0031 0.0094 0.0183 0.0338 P C-026 MGRM- 0.0169 0.1495 >10.0000 >10.0000 P C-
027 MGRM- 0.4433 2.8001 7.1057 >10.0000 P C-028 D 92UG005 2.4924 >10.0000 >10.0000
>10.0000 P 92UG024 >10.0000 >10.0000 >10.0000 >10.0000 N MGRM- 0.2154 0.6779 1.3056
2.3279 P D-001 MGRM- 0.0047 0.0175 0.0381 0.0791 P D-018 AE 92TH021 >10.0000 >10.0000
>10.0000 >10.0000 N CMU02 >10.0000 >10.0000 >10.0000 >10.0000 N AG MGRM- 0.1036
1.2780 6.3844 >10.0000 P AG-005 aMLV >10.0000 >10.0000 >10.0000 >10.0000 N
(609) TABLE-US-00047 TABLE 36 Neutralization Values for PGT-125. Neutralization by PGT-
125 (ug/ml) Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.0028 0.0110 0.0248 0.0526 P
93UG077 0.0262 0.0926 0.2057 0.4866 P 94UG103 0.0124 0.0373 0.0710 0.1287 P MGRM-
0.0055 0.0240 0.0575 0.1313 P A-010 B 92BR020 0.0214 0.0738 0.1557 0.3228 P APV13 0.0128
0.0414 0.0821 0.1543 P APV17 7.3065 >10.0000 >10.0000 >10.0000 P APV6 0.0359 0.1330
0.2885 0.5974 P JRFL 0.0156 0.0462 0.0873 0.1571 P JRCSF 0.0060 0.0196 0.0392 0.0741 P
NL43 >10.0000 >10.0000 >10.0000 >10.0000 N C 93IN905 0.0137 0.0494 0.1056 0.2165 P
MGRM- 0.0106 0.0350 0.0705 0.1342 P C-026 MGRM- >10.0000 >10.0000 >10.0000 >10.0000
N C-027 MGRM- 5.7772 >10.0000 >10.0000 >10.0000 P C-028 D 92UG005 >10.0000 >10.0000
>10.0000 >10.0000 N 92UG024 >10.0000 >10.0000 >10.0000 >10.0000 N MGRM- 2.3116
>10.0000 >10.0000 >10.0000 P D-001 MGRM- 0.0382 0.1440 0.3389 0.8924 P D-018 AE
92TH021 0.0066 0.0292 0.0701 0.1572 P CMU02 >10.0000 >10.0000 >10.0000 >10.0000 N AG
MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N AG-005 aMLV >10.0000 >10.0000 >10.0000
>10.0000 N

(610) TABLE-US-00048 TABLE 37 Neutralization Values for PGT-126. Neutralization by PGT-126 (ug/ml) Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.0046 0.0168 0.0359 0.0728 P 93UG077 0.0256 0.1015 0.2385 0.5771 P 94UG103 0.0091 0.0316 0.0655 0.1280 P MGRM- 0.0033 0.0137 0.0316 0.0687 P A-010 B 92BR020 0.0163 0.0457 0.0846 0.1521 P APV13 0.0120 0.0411 0.0840 0.1621 P APV17 0.5013 4.4290 >10.0000 >10.0000 P APV6 0.0249 0.0681 0.1230 0.2134 P JRFL 0.0140 0.0454 0.0900 0.1688 P JRCSF 0.0061 0.0180 0.0339 0.0608 P NL43 >10.0000 >10.0000 >10.0000 >10.0000 N C 93IN905 0.0194 0.0676 0.1404 0.2751 P MGRM- 0.0076 0.0345 0.0831 0.1849 P C-026 MGRM- 1.1404 >10.0000 >10.0000 >10.0000 P C-027 MGRM- 0.2827 2.6247 >10.0000 >10.0000 P C-028 92UG005 0.0181 0.1455 0.5529 2.4508 P D 92UG024 >10.0000 >10.0000 >10.0000 >10.0000 N MGRM- 0.0639 0.3334 0.8321 1.7951 P D-001 MGRM- 0.0093 0.0391 0.0918 0.2053 P D-018 AE 92TH021 0.1147 1.6162 6.9070 >10.0000 P CMU02 >10.0000 >10.0000 >10.0000 >10.0000 N AG MGRM- 1.3488 >10.0000 >10.0000 >10.0000 P AG-005 aMLV >10.0000 >10.0000 >10.0000 >10.0000 N

(611) TABLE-US-00049 TABLE 38 Neutralization Values for PGT-130. Neutralization by PGT-130 (ug/ml) Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.0140 0.5079 3.6823 >10.0000 P 93UG077 >10.0000 >10.0000 >10.0000 >10.0000 N 94UG103 0.6355 >10.0000 >10.0000 >10.0000 P MGRM- 0.0054 0.0251 0.0643 0.1642 P A-010 B 92BR020 1.1242 >10.0000 >10.0000 >10.0000 P APV13 0.0345 0.5731 2.8823 >10.0000 P APV17 >10.0000 >10.0000 >10.0000 N APV6 >10.0000 >10.0000 >10.0000 >10.0000 N JRFL 0.0346 0.7510 5.7157 >10.0000 P JRCSF 0.0089 0.0317 0.0669 0.1341 P NL43 >10.0000 >10.0000 >10.0000 >10.0000 N C 93IN905 0.0182 0.2716 1.3069 5.4770 P MGRM- 0.0173 0.2629 1.3244 6.1260 P C-026 MGRM- 0.0052 0.0318 0.1061 0.4638 P C-027 MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N C-028 D 92UG005 0.4741 >10.0000 >10.0000 >10.0000 P 92UG024 >10.0000 >10.0000 >10.0000 >10.0000 N MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N D-001 MGRM- 0.0378 0.2892 2.3096 >10.0000 P D-018 AE 92TH021 0.0082 0.0261 0.0513 0.0963 P CMU02 >10.0000 >10.0000 >10.0000 >10.0000 N AG MGRM- 0.9466 >10.0000 >10.0000 >10.0000 P AG-005 aMLV >10.0000 >10.0000 >10.0000 >10.0000 N

(612) TABLE-US-00050 TABLE 39 Neutralization Values for PGT-135. Neutralization by PGT-135 (ug/ml) Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.1573 3.4582 >10.0000 >10.0000 P 93UG077 >10.0000 >10.0000 >10.0000 >10.0000 N 94UG103 >10.0000 >10.0000 >10.0000 >10.0000 N MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N A-010 B 92BR020 0.1043 0.2526 0.4298 0.7203 P APV13 0.5452 >10.0000 >10.0000 >10.0000 P APV17 >10.0000 >10.0000 >10.0000 >10.0000 N APV6 >10.0000 >10.0000 >10.0000 >10.0000 N JRFL >10.0000 >10.0000 >10.0000 >10.0000 N JRCSF 0.1131 0.5033 1.4664 8.2996 P NL43 6.7910 >10.0000 >10.0000 >10.0000 P C 93IN905 0.0183 0.0648 0.1356 0.2677 P MGRM- 0.0123 0.0562 0.1390 0.3312 P C-026 MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N C-027 MGRM- 2.1608 >10.0000 >10.0000 >10.0000 P C-028 D 92UG005 >10.0000 >10.0000 >10.0000 >10.0000 N 92UG024 0.0139 0.0540 0.1220 0.2673 P MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N D-001 MGRM- 0.0433 0.1999 >10.0000 >10.0000 P D-018 AE 92TH021 >10.0000 >10.0000 >10.0000 >10.0000 N CMU02 >10.0000 >10.0000 >10.0000 >10.0000 N AG MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N AG-005 aMLV >10.0000 >10.0000 >10.0000 >10.0000 N

(613) TABLE-US-00051 TABLE 40 Neutralization Values for PGT-136. Neutralization by PGT-136 (ug/ml) Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.7383 >10.0000 >10.0000 >10.0000 P 93UG077 >10.0000 >10.0000 >10.0000 >10.0000 N 94UG103 >10.0000 >10.0000 >10.0000 >10.0000 N MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N A-010 B 92BR020 >10.0000 >10.0000 >10.0000 >10.0000 N APV13 >10.0000 >10.0000 >10.0000 >10.0000 N APV17 >10.0000 >10.0000 >10.0000 >10.0000 N APV6 >10.0000 >10.0000 >10.0000 >10.0000 N JRFL >10.0000 >10.0000 >10.0000 >10.0000 N JRCSF >10.0000 >10.0000 >10.0000 >10.0000 N NL43 >10.0000 >10.0000 >10.0000 >10.0000 N C 93IN905 0.0135 0.0623 0.1562 0.3791 P MGRM- 0.0065 0.0285 0.0687 0.1590 P C-026 MGRM- >10.0000 >10.0000 >10.0000 >10.0000

N C-027 MGRM- >10.0000 >10.0000 >10.0000 N C-028 D 92UG005 >10.0000
 >10.0000 >10.0000 >10.0000 N 92UG024 0.0915 0.6193 1.9558 5.9700 P MGRM- >10.0000
 >10.0000 >10.0000 >10.0000 N D-001 MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N D-018
 AE 92TH021 >10.0000 >10.0000 >10.0000 >10.0000 >10.0000 N CMU02 >10.0000 >10.0000 >10.0000
 >10.0000 N AG MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N AG-005 aMLV >10.0000
 >10.0000 >10.0000 >10.0000 N

(614) TABLE-US-00052 TABLE 41 Neutralization Values for PG9. Neutralization by PG9 (ug/ml)
 Clade Virus IC50 IC80 IC90 IC95 Overall A 92RW020 0.1614 1.0383 3.4024 >10.0000 P
 93UG077 >10.0000 >10.0000 >10.0000 >10.0000 N 94UG103 0.3098 1.9524 5.5354 >10.0000 P
 MGRM- 0.0375 0.1215 0.2418 0.4561 P A-010 B 92BR020 >10.0000 >10.0000 >10.0000
 >10.0000 N APV13 >10.0000 >10.0000 >10.0000 >10.0000 N APV17 >10.0000 >10.0000
 >10.0000 >10.0000 N APV6 0.2139 1.1316 3.2130 9.6532 P JRFL >10.0000 >10.0000 >10.0000
 >10.0000 N JRCSF 0.0048 0.0181 0.0391 0.0798 P NL43 0.6871 >10.0000 >10.0000 >10.0000 P
 C 93IN905 0.0480 0.3077 0.9807 3.3063 P MGRM- 0.1130 >10.0000 >10.0000 >10.0000 P C-026
 MGRM- 2.4538 >10.0000 >10.0000 >10.0000 P C-027 MGRM- 0.1211 0.6455 1.6522 3.7075 P
 C-028 D 92UG005 >10.0000 >10.0000 >10.0000 >10.0000 N 92UG024 1.9142 >10.0000
 >10.0000 >10.0000 P MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N D-001 MGRM- 0.0444
 0.1805 0.4300 1.0466 P D-018 AE 92TH021 0.1026 0.4475 1.0694 2.4256 P CMU02 >10.0000
 >10.0000 >10.0000 >10.0000 N AG MGRM- >10.0000 >10.0000 >10.0000 >10.0000 N AG-005
 aMLV >10.0000 >10.0000 >10.0000 >10.0000 N

Example 23: Heavy and Light Chain Usage for Selected Antibodies Isolated from B-Cell Cultures
 Established from Human Donors

(615) Monoclonal antibodies PGT-121 (corresponding to clones 4838_L06 and 4873 E03), PGT-
 122 (corresponding to clone 4877_D15), PGT-123 (corresponding to clone 4858_P08), PGT-125
 (corresponding to clone 5123 A06), PGT-126 (corresponding to clone 5141_B17), PGT-130
 (corresponding to clone 5147_N06), PGT-135 (corresponding to clones 5343_B08 and 5344_E16),
 and PGT-136 (corresponding to clones 5329_C19 and 5366_P21) are derived from related germline
 genes.

(616) The similarity of the variable genes is apparent based on the gene usage (Tables 42 and 43).
 Although the exact gene alleles used may not be definitive, the alleles that are most likely used are
 provided with the percentage identity to the germline gene noted.

(617) TABLE-US-00053 TABLE 42 Heavy Chain Germline Gene Usage for PGT-
 121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, and PGT-136. mAb
 Germline V- Identity Germline J- Identity Clone gene allele to V-gene gene allele to J-
 gene Heavy chain CDR3 PGT-121 IGHV4-59*01 80.35% IGHJ6*03 82.26%
 TLHGRRIGIVAFNEW or IGHV4- (229/285 nt) (51/62 nt) FTYFYMDV (SEQ ID
 59*07 NO: 143) IGHV4-59*02 80.00% or IGHV4- (228/285 nt) 59*03 or IGHV4-61*08
 PGT-122 IGHV4-61*08 80.35% IGHJ6*03 83.87% TKHGRRIGVVAFKE (229/285 nt)
 (52/62 nt) WFTYFYMDV (SEQ ID IGHV4-59*02 80.00% NO: 262) (228/285 nt)
 IGHV4-59*01 79.65% (227/285 nt) PGT-123 IGHV4-59*03 77.54% IGHJ6*03 83.87%
 ALHGKRIYGIVALGEL (221/285 nt) (52/62 nt) FTYFYMDV (SEQ ID IGHV4-59*01
 77.19% NO: 171) (220/285 nt) IGHV4-59*02 76.84% or IGHV4- (219/285 nt) 59*07 or
 IGHV4-61*08 PGT-125 IGHV4-b*02 80.21% IGHJ5*02 66.67% FDGEVLVYNHWPKPA
 (231/288 nt) (34/51 nt) WVDL (SEQ ID NO: IGHV4-b*01 79.86% IGHJ5*01 64.71%
 187) (230/288 nt) (33/51 nt) IGHV4-39*07 79.38% IGHJ4*03 62.50% (231/291 nt) (30/48
 nt) PGT-126 IGHV4-b*02 82.29% IGHJ5*02 72.55% FDGEVLVYHDWPKPA (237/288 nt)
 (37/51 nt) WVDL (SEQ ID NO: IGHV4-39*07 81.79% IGHJ5*01 68.63% 203)
 (238/291 nt) (35/51 nt) IGHV4-b*01 81.94% IGHJ4*03 64.58% (236/288 nt) (31/48 nt)
 PGT-130 IGHV4-39*07 79.38% IGHJ5*02 72.55% SGGDILYYEWQKPH (231/291 nt)
 (37/51 nt) WFSP (SEQ ID NO: 219) IGHV4-59*04 80.00% IGHJ5*01 68.63%

(228/285 nt) (35/51 nt) PGT-135 IGHV4-39*07 82.46% IGHJ5*02 72.55%
 HRHHDVFMVLPIAGW (235/285 nt) (37/51 nt) FDV (SEQ ID NO: 235) IGHV4-39*03 82.04% IGHJ5*01 70.59% (233/284 nt) (36/51 nt) PGT-136 IGHV4-39*07 83.86% IGHJ5*02 78.43% HKYHDIFRVVPVAGW (239/285 nt) (40/51 nt) FDP (SEQ ID NO: 252) IGHV4-39*03 83.45% IGHJ5*01 74.51% (237/284 nt) (38/51 nt)
 (618) TABLE-US-00054 TABLE 43 Light Chain Germline Gene Usage for PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, and PGT-136. mAb Germline V- Identity Germline J- Identity Clone gene allele to V-gene gene allele to J-gene Light chain CDR3 PGT-121 IGLV3-21*01 81.01% IGLJ3*02 86.49% HIWDSRVPTKWV or IGLV3- (209/258 nt) (32/37 nt) (SEQ ID NO: 152) 21*02 or IGLV3-21*03 PGT-122 IGLV3-21*02 82.56% IGLJ3*02 81.08% HIWDSRRPTNWV IGLV3-21*01 (213/258 nt) (30/37 nt) (SEQ ID NO: 164) or IGLV3- 82.17% 21*03 (212/258 nt) PGT-123 IGLV3-21*01 76.74% IGLJ3*02 83.78% HIYDARGGTNWV or IGLV3- (198/258 nt) (31/37 nt) (SEQ ID NO: 180) 21*02 or IGLV3-21*03 PGT-125 IGLV2-8*01 84.62% IGLJ3*01 86.49% GSLVGNWDVI IGLJ2*01 or (231/273 nt) (32/37 nt) (SEQ ID NO: 196) IGLV2-8*02 84.25% (230/273 nt) PGT-126 IGLV2-8*01 91.21% IGLJ2*01 or 89.19% SSLVGNWDVI IGLV2-8*02 (249/273 nt) IGLJ3*01 (33/37 nt) (SEQ ID NO: 212) 90.84% (248/273 nt) PGT-130 IGLV2-8*01 88.19% IGLJ2*01 or 89.19% SSLFGRWDVV IGLV2-8*02 IGLJ3*01 (33/37 nt) (SEQ ID NO: 228) 87.85% (253/288 nt) PGT-135 IGKV3-15*01 82.44% IGKJ1*01 94.44% QQYEEWPRT IGKV3D- (230/279 nt) (34/36 nt) (SEQ ID NO: 245) 15*01 82.08% (229/279 nt) PGT-136 IGKV3-15*01 86.38% IGKJ1*01 97.22% QQYEEWPRT IGKV3D- (241/279 nt) (35/36 nt) (SEQ ID NO: 245) 15*01 86.02% (240/279 nt)

Example 24: Heavy and Light Chain Usage for Selected Antibodies Isolated from B-Cell Cultures Established from Human Donors

(619) Monoclonal antibodies PGT-141 (corresponding to clones 4964_G22 and 4993_K13), PGT-142 (corresponding to clone 4995_E20), PGT-143 (corresponding to clone 4980_N08), and PGT-144 (corresponding to clone 4970_K22) are derived from related germline genes.

(620) The similarity of the variable genes is apparent based on the gene usage (Tables 44 and 45). Although the exact gene alleles used may not be definitive, the alleles that are most likely used are provided with the percentage identity to the germline gene noted.

(621) TABLE-US-00055 TABLE 44 Heavy Chain Germline Gene Usage for PGT-141, PGT-142, PGT-143, and PGT-144. mAb Germline V- Identity Germline J- Identity Clone gene allele to V-gene gene allele to J-gene Heavy chain CDR3 PGT-141 IGHV1-84.03% IGHJ6*02 74.19% GSKHRLRDYVLYDDYGLINYQEWND (4964_G22; 8*01 (242/288 nt) IGHJ6*01 (46/62 nt) YLEFLDV (SEQ ID NO: 279) 4993_K13) IGHV1-81.60% or 72.58% 2*02 or (235/288 nt) IGHJ6*03 (45/62 nt) IGHV1- or 2*04 IGHJ6*04 PGT-142 IGHV1- 83.68% IGHJ6*02 74.19% GSKHRLRDYVLYDDYGLINYQEWND (4995_E20) 8*01 (241/288 nt) IGHJ6*01 (46/62 nt) YLEFLDV (SEQ ID NO: 279) IGHV1- 81.60% or 72.58% 2*02 or (235/288 nt) IGHJ6*03 (45/62 nt) IGHV1- or 2*04 IGHJ6*04 PGT-143 IGHV1- 84.03% IGHJ6*02 74.19% GSKHRLRDYVLYDDYGLINYQEWND (4980_N08) 8*01 (242/288 nt) IGHJ6*01 (46/62 nt) YLEFLDV (SEQ ID NO: 279) IGHV1- 81.60% or 72.58% 2*02 or (235/288 nt) IGHJ6*03 (45/62 nt) IGHV1- or 2*04 IGHJ6*04 PGT-144 IGHV1- 83.33% IGHJ6*02 74.19% GSKHRLRDYVLYDDYGLINQQEWND (4970_K22) 8*01 (240/288 nt) or (46/62 nt) YLEFLDV (SEQ ID NO: 308) IGHV1- 80.90% IGHJ6*01 72.58% 2*02 or (233/288 nt) IGHJ6*03 (45/62 nt) IGHV1- or 2*04 IGHJ6*04

(622) TABLE-US-00056 TABLE 45 Light Chain Germline Gene Usage for PGT-141, PGT-142, PGT-143, and PGT-144. mAb Germline V- Identity Germline J- Identity Clone gene allele to V-gene gene allele to J-gene Light chain CDR3 PGT-141 IGKV2-

28*01 86.05% IGKJ1*01 89.19% MQGLNRPWT (4964_G22; or IGKV2D- (253/294 nt) (33/37 nt) (SEQ ID NO: 288) 4993_K13) 28*01 PGT-142 IGKV2-28*01 86.05% IGKJ1*01 89.19% MQGLNRPWT (4995_E20) or IGKV2D- (253/294 nt) (33/37 nt) (SEQ ID NO: 288) 28*01 PGT-143 IGKV2-28*01 86.05% IGKJ1*01 89.19% MQGLNRPWT (4980_N08) or IGKV2D- (253/294 nt) (33/37 nt) (SEQ ID NO: 288) 28*01 PGT-144 IGKV2-28*01 86.73% IGKJ1*01 86.49% MQGLNRPWT (4970_K22) or IGKV2D- (255/294 nt) (32/37 nt) (SEQ ID NO: 288) 28*01

Example 25: Heavy and Light Chain Alignments for Selected Antibodies (PGT-141, PGT-142, PGT-143, and PGT-144)

(623) Alignments of the genes (nucleic acid sequences) and proteins (amino acid sequence) for variable regions of both the heavy and light chains of the PGT-141, PGT-142, PGT-143, and PGT-144 antibodies are provided in Tables 46-49.

(624) Moreover, gene relationship trees that depict the relatedness of either the heavy or light chains of these antibodies to one another are provided in FIGS. 25 and 26, respectively.

Example 26: High Through put Functional Screening of Activated B Cells From 4 African Elite Neutralizers Yields a Panel of Novel Broadly Neutralizing Antibodies

(625) Antibodies PGT-121, PGT-122, PGT-123, PGT-125, PGT-126, PGT-130, PGT-135, PGT-136, PGT-141, PGT-142, PGT-143, and PGT-144 were generated according to the I-STAR™ Human bNAb (broadly Neutralizing Antibody) Discovery Platform depicted in FIG. 28. The isolation process involves identifying multiple neutralizing hits from IgG Memory B cells (as shown in FIG. 29). Once the recombinant monoclonal antibody is generated, then the neutralizing ability of the monoclonal antibody is confirmed. As a consequence of these methods, the recombinant antibodies of the invention are highly related (as shown in Examples 21-24). Moreover, these methods identify clusters of related sequences with increased neutralization activity. Furthermore, the antibodies of the invention bind to highly conserved regions of the HIV viral spike (FIGS. 30 and 31).

(626) Thirteen new monoclonal antibodies were isolated from 4 Protocol G elite neutralizers. Table 50 provides information regarding characteristics of each antibody.

(627) Preliminary mapping indicates that the antibodies from donors 17, 36 and 39 provided in Table 50 define a collection of overlapping and highly conserved epitopes at the viral spike. Evidence of the overlapping nature of these epitopes is provided by, for instance, the results of competition studies (Table 54). As an example, PGT-121 and PGT-125 demonstrate strong competition for binding to the spatially overlapping epitopes.

(628) TABLE-US-00057 TABLE 50 Donor Cluster Mabs HCDR3 Length Identity to V-gene 17 #1 3 (PGT-121-3) 24 77-80% 36 #1 2 (PGT-125-6) 19 79-82% #2 2 (PGT-130-1) 19 80% 39 #1 1 (PGT-135) 18 82% #2 1 (PGT-136) 18 83% 84 #1 4 (PGT-141-4) 32 83-84%

(629) TABLE-US-00058 TABLE 51 IAVI Protocol G: elite neutralizers score ≥ 2.5 Clade A Clade B Clade C CRF01_AE Rank Score Country 94UG103 92BR020 JRCSF IAVI C22 93IN905 92TI1021 1 3.67 Ivory Coast 900 900 2700 2700 2700 2700 2 3 Zambia 300 300 2700 300 2700 2700 5 2.83 Ivory Coast 300 300 900 300 2700 2700 5 2.83 Ivory Coast 300 900 2700 900 2700 100 5 2.83 Kenya 300 900 900 900 2700 300 5 2.83 South Africa 300 900 900 2700 2700 100 5 2.83 Rwanda 300 2700 900 2700 2700 <100 8 2.69 Zambia 345 345 1190 1190 1190 345 10 2.67 UK 300 900 900 2700 900 100 10 2.67 Zambia 900 900 900 300 2700 100 10 2.67 Uganda 900 900 900 2700 900 <100 15 2.5 Ivory Coast 300 900 300 900 900 300 15 2.5 South Africa 100 300 300 2700 900 900 15 2.5 South Africa 300 300 300 2700 2700 100 15 2.5 UK 300 900 300 900 900 300 15 2.5 South Africa 2700 100 300 2700 2700 <100 15 2.5 Uganda 900 900 900 900 900 <100 15 2.5 Zambia 300 <100 900 300 2700 2700

(630) Table 52 provides quantitative values for the neutralizing activity of each monoclonal antibody isolated from the 4 protocol G elite neutralizers.

(631) TABLE-US-00059 TABLE 52 1050 (ug/ml) Donor 17 Donor 36 PG9 PGT121 PGT122

PGT123 PGT125 PGT126 PGT130 92BR020 >10 0.0177 0.0337 0.0130 0.0214 0.0163 1.1242
92RW020 0.1614 0.0063 0.0072 0.0038 0.0028 0.0046 0.0140 92TH021 0.1026 >10 >10 >10
0.0066 0.1147 0.0082 92UG005 >10 >10 >10 2.4924 >10 0.0181 0.4741 92UG024 1.9142 >10
>10 >10 >10 >10 >10 93IN905 0.0480 0.0082 0.0138 0.0071 0.0137 0.0194 0.0182 93UG077 >10
0.0459 0.1913 0.0303 0.0262 0.0256 >10* 94UG103 0.3098 1.3778 1.4643 0.8461 0.0124 0.0091
0.6355 CMU02 >10 >10 >10 >10 >10 >10 >10 MGRM-C-026 0.1130 0.0034 0.0089 0.0031
0.0106 0.0076 0.0173 APV13 >10 0.3157 0.9923 0.2215 0.0128 0.0120 0.0345 APV17 >10 0.0950
0.5220 0.1798 7.3065 0.5013 >10 APV6 0.2139 0.0423 0.0851 0.0327 0.0359 0.0249 >10 JRFL
>10 0.0276 0.0471 0.0283 0.0156 0.0140 0.0346 MGRM-A-010 0.0375 2.6078 1.6836 0.4921
0.0025 0.0033 0.0054 MGRM-AG-005 >10 1.1719 >10 0.1036 >10 1.3488 0.9466 MGRM-C-027
2.4538 0.0094 0.0388 0.0169 >10 1.1404 0.0052 MGRM-C-028 0.1211 1.1929 2.0600 0.4433
5.7772 0.2827 >10 MGRM-D-001 >10 0.7442 0.8673 0.2154 2.3116 0.0639 >10 MGRM-D-018
2.0444 0.0086 0.0115 0.0047 0.0382 0.0093 0.0378 JRCSF 0.0048 0.0343 0.0727 0.0423 0.0060
0.0061 0.0089 NL43 0.6871 >10 >10 >10 >10 >10 >10 aMLV >10 >10 >10 >10 >10 >10
median 0.12 0.04 0.08 0.04 0.01 0.02 0.03 IC50 % 59 77 73 82 73 86 64 neutralized 1050 (ug/ml)
Donor 39 Donor 84 PGT135 PGT136 PGT141 PGT142 PGT143 PGT144 92BR020 0.1043 >10
>10 >10 >10 >10 92RW020 0.1573 0.7383 >10 >10 >10 >10 92TH021 >10 >10 0.0019 0.0022
0.0028 0.0934 92UG005 >10 >10 >10 >10 >10 >10 92UG024 0.0139 0.0915 >10 >10 >10 >10
93IN905 0.0183 0.0135 0.0016 0.0017 0.0021 0.1581 93UG077 >10 >10 >10 >10 >10 >10
94UG103 >10 >10 >10 >10 >10 >10 CMU02 >10 >10 >10 >10 >10 >10 MGRM-C-026 0.0123
0.0065 >10 >10 >10 >10 APV13 0.5452 >10 >10 >10 >10 >10 APV17 >10 >10 0.6159 0.3929
0.6179 5.1207 APV6 >10 >10 5.0838 5.4170 4.4341 >10 JRFL >10 >10 >10 >10 >10 >10
MGRM-A-010 >10 >10 0.0563 0.0578 0.0657 4.0160 MGRM-AG-005 >10 >10 0.3915 0.2196
0.2441 >10 MGRM-C-027 >10 >10 0.8219 0.2722 0.5049 >10 MGRM-C-028 2.1608 >10 2.3504
1.2264 1.8230 >10 MGRM-D-001 >10 >10 >10 >10 >10 >10 MGRM-D-018 0.0433 >10 >10 >10
>10 >10 JRCSF 0.1131 >10 0.0047 0.0085 0.0064 0.1279 NL43 6.7910 >10 0.8562 0.7123 0.5849
>10 aMLV >10 >10 >10 >10 >10 >10 median 0.11 0.05 0.50 0.25 0.37 0.16 IC50 % 54 18 45 45
45 23 neutralized

Example 27: PGT Monoclonal Anti HIV Antibody Potency

Example 28: Isolation of Anti-HIV Antibodies PGT-127, PGT-128, PGT-131, PGT-137, and PGT-145

in Walker L. M. et al, 2009, Science 326: 285-9. Specifically, surface IgG B cells seeded at near clonal density in 384-well microplates were activated in short-term culture. Supernatants were screened for neutralization activity against 2-4 pseudotyped viruses for which neutralization activity was detected at higher titers in the donor serum. Heavy and light chain variable regions were isolated from B cell lysates of selected neutralizing hits by reverse transcription from RNA followed by multiplex PCR amplification using family-specific V-gene primer sets. Amplicons from each lysate were uniquely tagged with multiplex identifier (MID) sequences and 454 sequencing regions (Roche). A normalized pooling of gamma, kappa and lambda chains was performed based on agarose gel image quantitation and the pool was analyzed by 454 Titanium® sequencing. Consensus sequences of the V.sub.H and V.sub.L chains were generated using the Amplicon Variant Analyzer (Roche) and assigned to specific B cell culture wells by decoding the MID tags. Clonally related sequences were identified by Clustal analysis. Selected V.sub.H and V.sub.L chains were synthesized and cloned in expression vectors with the appropriate IgG.sub.1, IgK or IgL constant domain. Monoclonal antibodies were reconstituted by transient transfection in HEK293 cells followed by purification from serum-free culture supernatants.

(636) Table 55 provides the gene usage data for the heavy chains of monoclonal anti-HIV antibodies PGT-127, PGT-128, PGT-131, PGT-137, and PGT-145.

(637) TABLE-US-00062

TABLE	55 mAb	Germline	V- Identity	Germline	J- Identity	to
Heavy	chain	Donor	Clone	gene	allele	to
V-gene	gene	allele	J-gene	CDR3	196	PGT-127
IGHV4-39*07	84.19%	IGHJ5*02	68.63%	FGGEVLVYRDW (5145_B14)	(245/291 nt)	(35/51 nt)
PKPAWVDL IGHV4-39*03	83.45%	(SEQ ID NO: 322)	(242/290 nt)	IGHV4-39*06	83.51%	IGHJ5*01 64.71% (243/291 nt) (33/51 nt)
IGHV4-b*02	83.68%	(241/288 nt)	PGT-128 IGHV4-39*07	79.73%	IGHJ5*02 74.51%	FGGEVLRVYTDW (5114_A19) (232/291 nt) (38/51 nt)
PKPAWVDL IGHV4-b*01	79.86%	(SEQ ID NO: 336)	(230/288 nt)	IGHV4-39*03	79.31%	IGHJ5*01 70.59% (230/290 nt) (36/51 nt)
IGHV4-b*02	79.51%	(229/288 nt)	PGT-131 IGHV4-39*07	78.69%	IGHJ5*02 76.47%	SGGDILYYIEWQ (5136_H01) (229/291 nt) (39/51 nt)
KPHWFYP (SEQ IGHV4-28*05	78.69%	ID NO: 350)	(229/291 nt)	IGHV4-39*03	78.62%	GHJ5*01 72.55% (228/290 nt) (37/51 nt)
IGHV4-39*06	78.35%	(228/291 nt)	IGHV4-59*04	78.95%	(225/285 nt)	039 PGT-137 IGHV4-39*03
IGHJ5*02 78.00%		HKYHDIVMVVPI (5345_I01) (220/284 nt) (39/50 nt)	AGWFDP (SEQ IGHV4-39*01	77.19%	GHJ5*01 74.00%	ID NO: 366) or IGHV4- (220/285 nt) (37/50 nt)
39*02 or IGHV4-39*07	584	PGT-145 IGHV1-8*01	83.33%	IGHJ3*01 73.47%	GSKHRLRDYFL (4995_P16) (240/288 nt) (36/49 nt)	YNEYGPNYEEW IGHJ6*02
66.13%		GDYLATLDV (41/62 nt) (SEQ ID NO: 380)	IGHV1-46*01	80.21%	IGHJ3*02 71.43%	or IGHV1- (231/288 nt) (35/49 nt)
46*02 or IGHJ6*01 or	64.52%	IGHV1-46*03 IGHJ6*04	(40/62 nt)			

Example 29: Broad Neutralization Coverage of HIV by Multiple High Potent Antibodies

(638) Broadly cross-reactive neutralizing antibodies (bnMAbs) against highly variable viral pathogens are much sought-after to treat or protect against global circulating viruses. The neutralizing antibody repertoires of four HIV-infected donors with remarkably broad and potent neutralizing responses were probed and 17 new monoclonal antibodies (mAbs) were rescued that neutralize broadly across clades. Many of these new monoclonal anti-HIV antibodies are almost 10-fold more potent than the PG9, PG16, and VRC01 bnMAbs and 100-fold more potent than the original prototype bnMAbs (Wu, X., et al. Science 329, 856-861 (2010); Walker, L. M., et al. Science 326, 285-289 (2009); Binley, J. M., et al. J Virol 78, 13232-13252 (2004)). The MAbs largely recapitulate the neutralization breadth and potency found in the corresponding donor serum and many recognize previously undescribed epitopes on envelope (Env) glycoprotein gp120, illuminating new targets for vaccine design. Analysis of neutralization by the full complement of anti-HIV bnMAbs now available reveals that certain combinations of antibodies provide significantly more favorable coverage of the enormous diversity of global circulating viruses than

others and these combinations might be sought in active or passive immunization regimes. Overall, the isolation of multiple HIV bnMAbs, from several donors, that, in aggregate, provide broad coverage at low concentrations is a highly positive indicator for the eventual design of an effective antibody-based HIV vaccine.

(639) Most successful anti-viral vaccines elicit neutralizing antibodies as a correlate of protection (Amanna, I. J., et al. *Hum Vaccin* 4, 316-319 (2008); Plotkin, S. A. *Pediatr Infect Dis J* 20, 63-75 (2001)). For highly variable viruses, such as HIV, HCV and, to a lesser extent influenza, vaccine design efforts have been hampered by the difficulties associated with eliciting neutralizing antibodies that are effective against the enormous diversity of global circulating isolates (i.e. broadly neutralizing antibodies, also referred to as bnAbs) (Barouch, D. H. *Nature* 455, 613-619 (2008); Karlsson Hedestam, G. B., et al. *Nat Rev Microbiol* 6, 143-155 (2008)). However, for HIV for example, 10-30% of infected individuals do, in fact, develop broadly neutralizing sera, and protective bnMAbs have been isolated from infected donors (Wu, X., et al. *Science* 329, 856-861 (2010); Walker, L. M., et al. *Science* 326, 285-289 (2009); Stamatatos, L., et al. *Nat Med* 15, 866-870 (2009); Trkola, A., et al. *J Virol* 69, 6609-6617 (1995); Stiegler, G., et al. *AIDS Res Hum Retroviruses* 17, 1757-1765 (2001); Burton, D. R., et al. *Science* 266, 1024-1027 (1994); Kwong, P. D. & Wilson, I. A. *Nat Immunol* 10, 573-578 (2009)). It has been suggested that, given the appropriate immunogen, it should be possible to elicit these types of responses by vaccination (Schief, W. R., et al. *Curr Opin HIV AIDS* 4, 431-440 (2009)) and understanding the properties of bnMAbs has become a major thrust in research on highly variable viruses.

(640) Sera from approximately 1,800 HIV-1 infected donors was previously screened for neutralization breadth and potency, designating the top 1% as “elite neutralizers”, based on a score incorporating both breadth and potency (Simek, M. D., et al. *J Virol* 83, 7337-7348 (2009)). In this study, bnMAbs were isolated from the top four elite neutralizers (Tables 56) by screening antibody-containing memory B cell supernatants for broad neutralizing activity using a recently described high-throughput functional approach (Walker, L. M., et al. *Science* 326, 285-289 (2009)). Antibody variable genes were rescued from B cell cultures that displayed cross-clade neutralizing activity and expressed as full-length IgGs. Analysis of the sequences revealed that all of the mAbs isolated from each individual donor belong to a distant, but clonally related cluster of antibodies (FIG. 49). Since it has been proposed that antibodies from HIV-1 infected patients are often polyreactive (Haynes, B. F., et al. *Science* 308, 1906-1908 (2005); Mouquet, H., et al. *Nature* 467, 591-595 (2010)), the new mAbs were tested for binding to a panel of antigens and showed that they were not polyreactive (FIG. 36).

(641) TABLE-US-00063 TABLE 56 Serum neutralizing activity of selected donors. presumed Clade A Clade B Clade C CRF01_AE Donor clade Score 94UG103 92BR020 JRCSF MGRM-C26 93IN905 92TH021 #36 CRF02_AG 3.67 900 900 ≥ 2700 ≥ 2700 ≥ 2700 ≥ 2700 #84 A or D 3.00 300 300 ≥ 2700 300 ≥ 2700 ≥ 2700 #17 A 2.83 300 ≥ 2700 900 ≥ 2700 ≥ 2700 <100 #39 C 2.83 300 900 900 ≥ 2700 ≥ 2700 100

(642) TABLE-US-00064 TABLE 61 Neutralization activity of the newly identified PGT antibodies. Median Percent viruses neutralized IC.sub.50 IC.sub.50 < 50 IC.sub.50 < 1 IC.sub.50 < 0.1 (µg/ml) µg/ml µg/ml µg/ml PGT121 0.03 70 57 44 PGT122 0.05 65 48 36 PGT123 0.03 67 54 40 PGT125 0.04 52 40 32 PGT126 0.04 60 50 40 PGT127 0.08 50 37 27 PGT128 0.02 72 60 50 PGT130 0.16 52 35 23 PGT131 0.52 40 23 13 PGT135 0.17 33 23 13 PGT136 7.81 16 6 3 PGT137 3.46 22 8 4 PGT141 0.35 56 36 15 PGT142 0.21 57 40 23 PGT143 0.31 56 37 17 PGT144 2.06 38 16 3 PGT145 0.29 78 52 27 PG9 0.23 77 54 29 VRC01 0.32 93 74 20 PGV04 0.20 88 65 25 b12 2.82 34 10 2 2G12 2.38 32 11 1 4E10 3.41 96 13 1 Median neutralization potency against viruses neutralized with an IC.sub.50 < 50 µg/ml is color-coded as follows: green, 20-50 µg/ml; yellow, 2-20 µg/ml; orange, 0.2-2 µg/ml; red, <0.2 µg/ml. Neutralization breadth is color-coded as follows: green, 1% to 30%; yellow, 30% to 60%; orange, 60% to 90%; red, >90%.

(643) The potency and breadth of the mAbs were next assessed on a 162-pseudovirus panel

0.82 0.98 29.56 0.82 0.58 0.18 50.00 50.00 2.26 G 50.00 4.69 5.95 4.77 50.00 0.01 0.51 0.12
0.13 50.00 50.00 1.44 AE 50.00 50.00 50.00 50.00 50.00 1.17 0.13 0.56 1.26 50.00 50.00 0.63 AG
50.00 31.02 16.43 41.24 50.00 1.27 3.47 0.16 0.10 50.00 14.00 1.42 All 50.00 18.01 9.46 13.76
50.00 0.88 0.82 0.34 0.30 50.00 50.00 3.50 D) Median IC.sub.50 ($\mu\text{g/ml}$) against viruses
neutralized with an IC.sub.50 <50 $\mu\text{g/ml}$.sup.3 Clade n PGT121 PGT122 PGT123 PGT125
PGT126 PGT127 PGT128 PGT130 PGT131 PGT135 PGT136 A 26 0.20 0.33 0.18 0.07 0.06 0.16
0.04 4.99 0.73 1.01 na B 31 0.11 0.15 0.14 0.13 0.17 0.36 0.07 0.77 0.50 2.87 2.09 C 27 0.12 0.17
0.08 0.18 0.96 7.99 0.17 2.67 15.80 0.28 0.28 D 25 0.12 0.11 0.13 0.87 0.33 0.18 0.12 3.18 2.67
1.13 2.10 F 15 0.17 5.53 0.74 0.37 1.00 0.86 1.17 4.84 1.92 1.26 na G 15 0.22 0.65 0.19 0.05 0.17
0.67 0.12 28.06 na 1.01 7.54 AE 10 na na na 0.22 19.58 na 1.47 0.08 0.21 na na AG 10 0.56 0.97
2.27 0.05 0.06 0.24 0.13 0.07 0.16 na na All 162 0.13 0.26 0.18 0.78 0.17 0.46 0.32 1.88 0.54 1.13
2.00 Clade PGT137 PGT141 PGT142 PGT143 PGT144 PGT145 PG9 VRC01 PGV04 b12 2G12
4E10 A 2.54 2.88 2.37 1.57 4.31 4.30 2.24 2.20 1.45 48.45 17.77 40.61 B na 4.62 9.52 2.84 7.06
0.54 2.00 2.05 1.06 5.96 4.52 30.31 C 1.32 3.83 1.69 2.02 12.43 1.76 7.00 6.18 3.83 27.42 28.67
21.21 D 16.10 25.10 9.75 8.36 19.06 5.27 0.80 5.72 3.23 12.08 3.77 23.46 F 21.09 4.55 3.45 6.08
7.79 9.74 0.52 2.72 2.34 na 21.50 7.63 G 1.45 1.58 1.99 1.85 7.57 2.18 6.35 1.44 1.74 22.81 na
16.57 AE na 0.03 0.03 0.05 3.41 11.09 1.02 4.08 8.18 12.08 na 12.96 AG na 3.66 5.15 4.97 na 0.87
0.36 1.17 1.97 16.96 7.04 15.36 All 2.10 3.80 3.75 2.60 8.37 2.10 1.22 2.70 2.00 18.01 8.06 22.37
E) Percent viruses neutralized with an IC.sub.50 <50 $\mu\text{g/ml}$.sup.3 Clade n PGT121 PGT122
PGT123 PGT125 PGT126 PGT127 PGT128 PGT130 PGT131 PGT135 PGT136 A 26 50 50 50 54
50 46 50 31 13 4 0 B 31 81 77 77 58 74 05 77 42 29 23 3 C 27 67 59 63 26 48 38 67 26 21 33 13 D
25 36 32 32 28 36 28 48 16 8 8 4 F 15 00 73 53 20 27 13 53 20 7 33 0 G 15 00 00 47 7 20 20 47 7
0 20 7 AE 10 0 0 0 40 20 0 50 50 40 0 0 AG 10 30 30 40 30 30 30 50 30 20 0 0 All 162 56 54 52 37
46 36 57 28 10 17 5 Clade PGT137 PGT141 PGT142 PGT143 PGT144 PGT145 PG9 VRC01
PGV04 b12 2G12 4E10 A 12 38 46 35 4 46 69 88 88 4 4 23 B 0 32 39 29 13 61 29 100 84 45 52 16
C 21 48 58 50 29 58 48 81 48 17 4 42 D 4 20 20 16 8 36 40 72 56 28 12 20 F 7 47 53 47 27 67 40
100 93 0 7 40 G 13 33 47 40 7 67 40 93 87 13 0 53 AE 00 30 30 30 20 60 70 100 80 10 0 70 AG 0
20 30 20 0 50 40 90 100 10 30 60 All 7 36 40 33 14 53 48 80 77 10 15 34 F) Median IC.sub.50
($\mu\text{g/ml}$) against all viruses.sup.3 Clade n PGT121 PGT122 PGT123 PGT125 PGT126 PGT127
PGT128 PGT130 PGT131 PGT135 PGT136 A 26 47.89 48.07 23.26 26.63 6.27 50.00 9.38 50.00
50.00 50.00 50.00 B 31 0.23 0.41 0.42 0.49 0.29 1.17 0.12 50.00 50.00 50.00 50.00 C 27 0.47 7.99
7.29 50.00 50.00 50.00 2.54 50.00 50.00 50.00 50.00 D 25 50.00 50.00 50.00 50.00 50.00
50.00 50.00 50.00 50.00 50.00 F 15 1.86 23.47 5.05 50.00 50.00 50.00 38.14 50.00 50.00 50.00
50.00 G 15 2.10 26.35 50.00 50.00 50.00 50.00 50.00 50.00 50.00 AE 10 50.00 50.00
50.00 50.00 50.00 50.00 29.21 50.00 50.00 50.00 50.00 AG 10 50.00 50.00 50.00 50.00 50.00
50.00 42.87 50.00 50.00 50.00 50.00 All 162 10.40 38.11 18.25 50.00 50.00 50.00 3.73 50.00 5000
50.00 50.00 Clade PGT137 PGT141 PGT142 PGT143 PGT144 PGT145 PG9 VRC01 PGV04 b12
2G12 4E10 A 50.00 50.00 50.00 50.00 50.00 50.00 2.71 2.38 1.62 50.00 50.00 50.00 B 50.00 50.00
50.00 50.00 50.00 6.51 50.00 2.05 2.05 50.00 44.23 50.00 C 50.00 50.00 43.11 50.00 50.00 30.57
22.11 6.71 50.00 50.00 50.00 50.00 D 50.00 50.00 50.00 50.00 50.00 50.00 50.00 8.77 7.58 50.00
50.00 50.00 F 50.00 50.00 34.51 50.00 50.00 16.34 50.00 2.72 2.52 50.00 50.00 50.00 G 50.00
50.00 50.00 50.00 50.00 9.93 50.00 1.69 2.07 50.00 50.00 29.65 AE 50.00 50.00 50.00 50.00 50.00
39.37 0.78 4.08 11.00 50.00 50.00 19.49 AG 50.00 50.00 50.00 50.00 50.00 35.44 50.00 1.40 1.97
50.00 50.00 26.15 All 50.00 50.00 50.00 50.00 50.00. 31.20 46.50 3.31 4.11 50.00 50.00 50.00 a)
Neutralization potency is color-coded as follows: white, median potency >50 $\mu\text{g/ml}$; green, median
potency between 20 and 50 $\mu\text{g/ml}$; yellow, median potency between 2 and 20 $\mu\text{g/ml}$; orange,
median potency between 0.2 and 2 $\mu\text{g/ml}$; red, median potency <0.2 $\mu\text{g/ml}$. b) Neutralization
breadth is color-coded as follows: white, no virus neutralized; green, 1% to 30% of viruses
neutralized; yellow, 30% to 60% of viruses neutralized; orange, 60% to 90% of viruses neutralized;
red, >90% of viruses neutralized.

(645) TABLE-TABLE 58B IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-121 PGT-122
PGT-123 PGT-125 PGT-126 PGT-127 92RW008 A 0.003 0.003 0.003 0.004 0.008 0.012 92WR009
1.739 42.406 15.680 0.144 >50 >50 92WR020 0.004 0.009 0.002 0.004 0.006 0.010 92WR021
0.009 0.021 0.005 0.005 0.006 0.011 92WR024 >50 >50 >50 35.530 41.995 >50 92WR026 0.014
0.036 0.012 0.007 0.008 0.024 92UG031 >50 >50 >50 >50 >50 >50 92UG037 0.031 0.068 0.023
0.005 0.011 0.014 93WR029 >50 >50 >50 >50 >50 >50 93UG077 0.019 0.084 0.028 0.012 0.020
0.055 94UG103 2.518 2.041 0.678 0.008 0.008 0.017 MG RM-A-001 >50 >50 >50 12.392 >50
>50 MG RM-A-002 0.013 0.017 0.011 0.013 0.015 9.422 MG RM-A-003 0.435 3.794 0.495 >50
>50 >50 MG RM-A-004 >50 >50 >50 >50 >50 >50 MG RM-A-005 >50 >50 >50 >50 >50
MG RM-A-006 0.396 >50 >50 >50 >50 >50 MG RM-A-007 >50 >50 >50 >50 >50 MG RM-
A-008 >50 >50 >50 >50 >50 >50 MG RM-A-009 0.006 0.012 0.004 0.009 0.004 0.013 MG RM-
A-010 2.444 2.509 0.705 0.004 0.002 0.017 MG RM-A-011 >50 >50 >50 >50 >50 >50 MG RM-
A-012 0.092 0.257 0.192 0.177 0.097 0.105 MG RM-A-013 0.387 2.474 0.565 0.014 0.006 0.029
MG RM-A-014 0.003 0.007 0.003 13.913 0.010 0.040 VLGCA1 >50 >50 >50 0.078 >50 >50 94
KE 105 AC 0.029 0.056 0.024 0.004 0.006 0.023 92TH021 AE >50 >50 >50 0.006 0.197 >50
CMU02 >50 >50 >50 >50 >50 >50 MGRM-AE-001 >50 >50 >50 2.948 9.684 >50 MGRM-AE-
002 >50 >50 >50 >50 >50 >50 MGRM-AE-003 >50 >50 >50 0.092 >50 >50 MGRM-AE-004 >50
>50 >50 >50 >50 >50 MGRM-AE-005 >50 >50 >50 0.010 0.324 >50 MGRM-AE-006 >50 >50
>50 0.009 17.253 >50 MGRM-AE-007 >50 >50 >50 0.876 >50 >50 MGRM-AE-008 >50 >50 >50
>50 >50 >50 MG RM-AG-001 AG 2.096 14.472 2.509 >50 >50 >50 MG RM-AG-002 >50 >50
>50 >50 >50 MG RM-AG-003 0.465 43.663 >50 >50 >50 >50 MG RM-AG-005 0.649 33.861
0.149 >50 1.207 >50 MG RM-AG-006 0.014 0.028 0.014 0.008 0.010 0.021 MG RM-AG-008
0.494 >50 19.162 >50 >50 >50 MG RM-AG-009 0.080 0.128 0.073 0.026 0.030 0.059 MG RM-
AG-011 >50 >50 >50 >50 24.51 >50 MG RM-AG-012 >50 >50 >50 >50 >50 >50 MG RM-AG-
013 0.557 0.521 3.371 0.004 0.008 0.018 IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-128 PGT-
130 PGT-131 PGT-135 PGT-136 PGT-137 92RW008 A 0.003 0.039 1.011 1.417 >50 0.148
92WR009 0.812 1.274 3.115 >50 >50 >50 92WR020 0.005 0.039 0.168 0.067 2.335 0.005
92WR021 0.005 0.004 0.012 >50 >50 3.516 92WR024 >50 >50 >50 >50 >50 >50 92WR026 0.010
0.037 0.041 0.068 >50 0.092 92UG031 >50 >50 >50 >50 >50 >50 92UG037 0.006 0.061 0.048
3.672 >50 >50 93WR029 >50 34.264 >50 >50 >50 >50 93UG077 0.014 21.392 >50 >50 >50 >50
94UG103 0.011 1.402 1.097 >50 >50 >50 MG RM-A-001 >50 >50 >50 >50 >50 >50 MG RM-A-
002 0.149 0.010 >50 >50 >50 >50 MG RM-A-003 >50 >50 >50 >50 >50 >50 MG RM-A-004 >50
>50 >50 >50 >50 >50 MG RM-A-005 >50 >50 >50 >50 >50 >50 MG RM-A-006 >50 >50 >50
>50 >50 >50 MG RM-A-007 >50 >50 >50 >50 >50 >50 MG RM-A-008 >50 >50 >50 >50 >50
>50 MG RM-A-009 0.005 0.308 0.035 >50 >50 >50 MG RM-A-010 0.005 0.005 0.010 >50 >50
>50 MG RM-A-011 >50 >50 >50 >50 >50 >50 MG RM-A-012 0.025 >50 >50 >50 >50 >50 MG
RM-A-013 0.008 >50 >50 >50 >50 49.188 MG RM-A-014 0.007 0.161 15.444 0.636 >50 18.576
VLGCA1 0.838 >50 >50 >50 >50 >50 94 KE 105 AC 0.007 0.004 0.013 0.063 >50 18.548
92TH021 AE 0.010 0.009 0.014 >50 >50 >50 CMU02 >50 >50 >50 >50 >50 >50 MGRM-AE-001
0.069 1.005 4.805 >50 >50 >50 MGRM-AE-002 0.522 1.534 >50 >50 >50 >50 MGRM-AE-003
0.180 0.337 1.407 >50 >50 >50 MGRM-AE-004 >50 2.429 >50 >50 >50 >50 MGRM-AE-005
0.032 0.006 0.017 >50 >50 >50 MGRM-AE-006 0.044 0.009 0.026 >50 >50 >50 MGRM-AE-007
>50 0.083 0.026 >50 >50 >50 MGRM-AE-008 >50 >50 >50 >50 >50 >50 MG RM-AG-001 AG
4.730 >50 >50 >50 >50 >50 MG RM-AG-002 4.987 >50 >50 >50 >50 >50 MG RM-AG-003
34.787 0.041 2.917 >50 >50 >50 MG RM-AG-005 0.037 0.540 1.041 >50 >50 >50 MG RM-AG-
006 0.009 0.009 0.019 17.890 29.108 28.298 MG RM-AG-008 0.668 >50 >50 >50 >50 >50 MG
RM-AG-009 0.021 9.461 >50 >50 >50 >50 MG RM-AG-011 3.931 >50 >50 >50 >50 >50 MG
RM-AG-012 >50 >50 >50 41.885 >50 >50 MG RM-AG-013 0.004 0.010 0.030 >50 >50 >50
IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-141 PGT-142 PGT-143 PGT-144 PGT-145 VRC01
92RW008 A 0.579 0.447 0.385 >50 0.400 0.037 92WR009 0.022 0.006 0.010 0.893 0.026 0.187

92WR020 >50 31.523 >50 >50 0.997 0.130 92WR021 >50 >50 >50 >50 >50 1.620 0.066 92WR022
0.175 0.069 0.122 16.346 0.069 0.327 92WR026 1.781 1.780 5.364 32.691 1.453 N/A 92UG031
0.093 0.050 0.095 2.175 3.220 0.469 92UG037 1.276 0.557 0.854 8.241 3.148 0.245 93WR029
0.019 0.014 0.053 4.391 >50 1.570 93UG077 >50 >50 >50 >50 >50 0.097 94UG103 >50 >50 >50
>50 0.331 0.285 MG RM-A-001 0.209 0.169 0.274 1.111 0.062 >50 MG RM-A-002 0.217 0.007
0.016 0.085 0.019 N/A MG RM-A-003 7.961 9.239 11.154 >50 3.315 0.422 MG RM-A-004 0.007
0.002 0.007 0.855 0.068 0.064 MG RM-A-005 11.112 9.695 7.483 18.590 0.128 0.095 MG RM-A-
006 >50 >50 >50 >50 >50 0.521 MG RM-A-007 >50 >50 >50 >50 >50 12.584 MG RM-A-008
>50 >50 >50 >50 >50 0.303 MG RM-A-009 0.029 0.058 0.050 2.834 0.235 0.118 MG RM-A-010
0.022 0.015 0.019 2.791 0.028 0.325 MG RM-A-011 >50 >50 >50 >50 0.484 0.273 MG RM-A-
012 >50 >50 >50 >50 0.230 >50 MG RM-A-013 9.856 3.775 9.241 >50 1.761 0.152 MG RM-A-
014 0.628 0.367 1.071 >50 1.357 0.135 VLGCA1 2.067 2.754 17.389 >50 8.009 0.163 94 KE 105
AC 0.204 0.190 0.368 0.290 1.317 0.499 92TH021 AE 0.001 0.002 0.003 0.052 0.013 0.616
CMU02 >50 >50 >50 >50 1.425 0.515 MGRM-AE-001 >50 >50 >50 >50 20.916 0.167 MGRM-
AE-002 >50 >50 >50 >50 1.868 0.217 MGRM-AE-003 >50 >50 >50 >50 0.552 0.685 MGRM-
AE-004 >50 0.001 0.002 0.104 0.006 0.187 MGRM-AE-005 >50 >50 >50 >50 0.967 0.029
MGRM-AE-006 >50 >50 >50 >50 >50 4.648 MGRM-AE-007 33.133 6.697 13.558 >50 0.181
0.835 MGRM-AE-008 0.101 0.080 0.135 3.553 1.464 1.715 MG RM-AG-001 AG 18.030 12.423
18.933 >50 0.164 0.228 MG RM-AG-002 0.049 0.044 0.076 7.780 0.056 0.950 MG RM-AG-003
>50 >50 >50 >50 >50 0.043 MG RM-AG-005 0.497 0.383 1.114 >50 0.045 0.091 MG RM-AG-
006 >50 >50 >50 >50 6.603 0.344 MG RM-AG-008 19.242 12.125 34.009 >50 0.244 0.124 MG
RM-AG-009 >50 >50 >50 >50 >50 >50 MG RM-AG-011 0.118 0.081 0.156 44.563 0.003 0.084
MG RM-AG-012 >50 >50 >50 >50 >50 0.031 MG RM-AG-013 >50 21.719 >50 >40 14.387 0.197
IC.sub.50 (µg/mL).sup.a IC.sub.50 (l/dil'n).sup.b Isolate Subtype PGV04 PG9 #17 #84 #36 #39
92RW008 A 0.014 0.006 6500.7 9301.5 2084.1 644.0 92WR009 0.108 0.050 463.6 1247.6 241.0
<100 92WR020 0.055 0.141 5671.1 428.3 2617.1 2730.0 92WR021 0.038 0.027 2343.6 1092.1
4204.8 428.0 92WR024 0.310 0.258 <100 507.7 149.7 178.0 92WR026 0.140 0.164 1561.0 717.3
2183.4 1046.0 92UG031 4.463 3.605 <100 282.5 422.2 459.0 92UG037 0.145 0.014 1120.4 501.8
1431.8 410.0 93WR029 0.540 0.690 <100 589.2 <100 270.0 93UG077 0.083 >50 1082.8 326.4
707.2 436.0 94UG103 0.202 0.340 193.4 208.3 1138.4 568.0 MG RM-A-001 6.781 >50 <100
231.7 <100 149.0 MG RM-A-002 0.093 0.020 839.3 533.0 2782.2 264.0 MG RM-A-003 0.128
1.942 <100 612.2 <100 375.0 MG RM-A-004 0.175 0.020 <100 4567.2 <100 319.0 MG RM-A-
005 0.134 0.750 <100 151.2 <100 282.0 MG RM-A-006 0.244 <50 260.6 646.5 120.0 564.0 MG
RM-A-007 7.213 1.061 <100 <100 <100 556.0 MG RM-A-008 0.555 <50 <100 <100 <100 196.0
MG RM-A-009 0.033 0.023 4535.8 2228.7 1611.4 286.0 MG RM-A-010 0.205 0.022 271.3 1704.2
2779.2 279.0 MG RM-A-011 0.090 0.033 <100 242.3 <100 169.0 MG RM-A-012 0.225 19.294
229.9 <100 770.6 215.0 MG RM-A-013 0.086 0.122 892.0 162.8 835.9 776.0 MG RM-A-014
0.172 1.248 7222.6 412.6 705.2 681.0 VLGCA1 0.024 0.069 <100 311.2 203.2 <100 94 KE 105
AC 2.981 37.875 955.0 1035.2 4540.1 1633.0 92TH021 AE 1.289 0.059 <100 5298.1 3301.2 212.0
CMU02 1.238 >50 <100 101.3 <100 312.0 MGRM-AE-001 0.063 20.586 <100 <100 222.6 223.0
MGRM-AE-002 0.569 0.040 <100 169.1 254.0 249.0 MGRM-AE-003 6.401 0.044 <100 150.0
<100 181.0 MGRM-AE-004 0.183 0.009 <100 6926.5 131.7 123.0 MGRM-AE-005 0.051 0.282
<100 208.2 2772.2 265.0 MGRM-AE-006 >50 0.063 <100 117.5 1352.7 194.0 MGRM-AE-007
5.753 0.180 <100 270.9 444.5 184.0 MGRM-AE-008 1.513 31.482 <100 611.7 <100 341.0 MG
RM-AG-001 AG 0.218 17.125 174.8 292.1 302.2 499.0 MG RM-AG-002 0.879 0.076 125.5 748.0
139.4 206.0 MG RM-AG-003 0.058 >50 <100 505.3 <100 121.0 MG RM-AG-005 0.027 >50
517.3 775.6 865.9 708.0 MG RM-AG-006 1.230 >50 1105.3 321.1 4106.0 536.0 MG RM-AG-008
2.484 0.013 125.1 827.8 <100 692.0 MG RM-AG-009 0.148 >50 525.2 101.4 220.2 128.0 MG
RM-AG-011 0.071 0.007 <100 2785.8 228.1 274.0 MG RM-AG-012 0.040 22.845 104.9 <100
136.6 235.0 MG RM-AG-013 0.030 0.259 321.2 319.1 2055.6 303.0

(646) TABLE-00067 TABLE IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-121 PGT-122 PGT-123 PGT-125 PGT-126 PGT-127 6535.3 B 0.005 0.006 0.004 0.010 0.022 0.020 92 BR 020 0.014 0.021 0.008 0.016 0.015 0.059 93TH305 0.007 0.013 0.008 0.008 0.013 0.017 APV 13 0.251 0.778 0.138 0.007 0.012 0.026 APV_17 0.066 0.250 0.114 8.063 0.353 10.378 APV 6 0.018 0.019 0.023 0.021 0.007 0.040 CAAN.A2 0.011 0.011 0.015 5.467 0.273 0.371 JRFL 0.021 0.026 0.014 0.009 0.014 0.029 MGRM-Chronic-B-001 0.102 0.138 0.150 >50 >50 >50 MGRM-Chronic-B-002 0.386 1.801 0.234 0.064 0.078 0.279 MGRM-Chronic-B-003 0.011 0.008 0.008 0.157 0.041 0.120 MGRM-Chronic-B-004 0.009 0.008 0.008 0.007 0.010 0.018 MGRM-Chronic-B-008 0.007 0.017 0.010 >50 5.890 21.515 MGRM-Chronic-B-010 0.014 0.018 0.009 0.006 0.005 0.019 MGRM-Chronic-B-011 >50 >50 >50 3.228 0.108 0.297 MGRM-Chronic-B-012 0.036 0.066 0.227 0.036 0.027 0.075 MGRM-Chronic-B-017 >50 5.664 >50 >50 >50 >50 MGRM-Chronic-B-018 >50 >50 >50 >50 0.052 1.896 MGRM-Chronic-B-020 0.004 0.005 0.005 0.087 0.021 0.050 MGRM-Chronic-B-023 0.005 0.010 0.055 >50 >50 >50 MGRM-Chronic-B-024 0.195 2.308 >50 >50 >50 >50 PVO.4 0.137 0.689 0.105 0.042 0.017 0.164 QH0692.42 0.823 0.493 0.158 0.048 0.048 0.129 SC422661.8 0.098 0.103 0.039 30.138 0.119 36.688 SF162 0.005 0.009 0.005 0.004 0.003 0.019 THR0.18 >50 >50 >50 >50 >50 >50 TRJ04551.58 7.095 >50 16.913 0.019 0.025 0.063 TRO.11 0.008 0.011 0.008 0.111 0.041 0.079 VLGCB3 0.005 0.008 0.004 0.015 0.007 0.022 NL43 >50 >50 >50 >50 >50 JRCSF 0.027 0.057 0.046 0.004 0.008 0.018 93IN905 C 0.005 0.013 0.004 0.009 0.015 0.024 93MW959 0.013 0.016 0.011 37.481 9.441 6.951 97ZA012 0.002 0.004 0.002 2.465 0.042 >50 98IN022 0.007 0.032 0.011 22.057 0.279 19.189 MG RM-C-001 17.482 19.103 42.406 >50 >50 >50 MG RM-C-002 0.019 0.022 0.010 0.008 0.011 0.057 MG RM-C-004 0.011 0.020 0.009 1.825 0.033 0.532 MG RM-C-005 0.015 0.027 0.012 2.087 0.298 0.570 MG RM-C-006 0.017 0.018 0.021 0.015 0.014 0.319 MG RM-C-007 0.196 3.348 0.547 >50 25.298 >50 MG RM-C-008 0.012 0.028 0.011 >50 2.118 >50 MG RM-C-009 >50 >50 >50 >50 >50 >50 MG RM-C-010 >50 >50 >50 >50 >50 >50 MG RM-C-012 0.123 2.002 0.605 >50 0.111 1.073 MG RM-C-013 7.228 32.575 14.879 >50 >50 >50 MG RM-C-014 26.941 >50 1.003 >50 >50 >50 MG RM-C-015 25.071 >50 13.141 >50 3.140 >50 MG RM-C-017 0.009 0.015 0.010 2.358 0.170 3.397 MG RM-C-019 >50 0 >50 >50 0.130 0.005 0.017 MG RM-C-020 >50 13.955 >50 >50 >50 >50 MG RM-C-022 0.015 0.012 0.010 1.233 0.960 0.410 MG RM-C-023 0.017 0.027 0.011 1.314 0.065 1.082 MG RM-C-024 0.052 0.102 0.195 >50 >50 >50 MG RM-C-025 46.753 8.691 13.304 42.541 0.786 >50 MG RM-C-026 0.002 0.006 0.001 0.005 0.007 0.014 MG RM-C-027 0.011 0.039 0.020 >50 15.557 >50 MG RM-C-028 0.472 2.216 0.402 14.023 0.196 7.567 IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-128 PGT130 PGT-131 PG1-135 PGT-136 PGT-137 6535.3 B 0.011 0.031 0.056 >50 >50 >50 92 BR 020 0.009 1.395 6.157 0.073 8.608 7.811 93TH305 0.006 0.021 0.016 >50 20.214 >50 APV 13 0.008 0.052 0.189 0.716 >50 >50 APV_17 0.016 10.661 >50 >50 >50 >50 APV 6 0.007 >50 >50 >50 >50 >50 CAAN.A2 1.482 >50 >50 5.154 >50 >50 JRFL 0.007 0.046 0.454 >50 >50 >50 MGRM-Chronic-B-001 >50 >50 >50 >50 >50 >50 MGRM-Chronic-B-002 0.036 >50 >50 >50 >50 >50 MGRM-Chronic-B-003 0.007 >50 >50 >50 >50 >50 MGRM-Chronic-B-004 0.009 0.010 0.040 >50 >50 >50 MGRM-Chronic-B-008 0.143 5.183 >50 >50 >50 >50 MGRM-Chronic-B-010 0.004 0.006 0.016 >50 >50 >50 MGRM-Chronic-B-011 0.020 0.153 6.468 >50 >50 >50 MGRM-Chronic-B-012 0.026 0.373 15.708 0.166 43.142 >50 MGRM-Chronic-B-017 >50 >50 >50 >50 >50 >50 MGRM-Chronic-B-018 0.020 31.873 >50 >50 >50 >50 MGRM-Chronic-B-020 0.007 0.014 0.053 0.057 >50 32.851 MGRM-Chronic-B-023 >50 >50 >50 0.329 >50 >50 MGRM-Chronic-B-024 >50 5.776 >50 >50 >50 >50 PVO.4 0.005 5.171 >50 >50 >50 QH0692.42 0.029 >50 >50 >50 >50 >50 SC422661.8 1.578 >50 >50 21.751 >50 >50 SF162 0.007 0.007 0.025 0.023 0.482 3.407 THR0.18 >50 >50 >50 >50 >50 >50 TRJ04551.58 0.019 0.023 0.088 >50 >50 >50 TRO.11 0.018 0.257 1.913 0.030 0.073 9.349 VLGCB3 0.012 4.596 >50 0.142 >50 3.361 NL43 >50 >50 >50 8.034 27.695 17.444 JRCSF 0.003 0.010 0.029 0.131 >50 >50 93IN905 C 0.009 0.020 0.177 0.011 0.011 0.042 93MW959 0.045 8.548 >50 >50 >50 >50 97ZA012 0.019 1.318 1.814 >50 >50 >50 98IN022 0.014 >50 >50 0.014 13.622 0.408 MG RM-C-

001 >50 >50 >50 10.121 7.518 21.503 MG RM-C-002 0.019 >50 >50 >50 8.034 MG RM-C-004 0.025 2.756 3.714 0.046 0.034 0.026 MG RM-C-005 0.030 0.935 1.356 >50 >50 >50 MG RM-C-006 0.014 0.095 0.611 >50 >50 >50 MG RM-C-007 0.111 >50 >50 >50 >50 >50 MG RM-C-008 0.373 7.277 >50 >50 >50 >50 MG RM-C-009 >50 >50 >50 >50 >50 MG RM-C-010 >50 >50 >50 >50 >50 >50 MG RM-C-012 0.058 >50 >50 0.017 >50 0.102 MG RM-C-013 >50 >50 >50 >50 >50 MG RM-C-014 >50 >50 >50 >50 >50 MG RM-C-015 0.196 0.344 1.015 0.104 11.066 1.983 MG RM-C-017 1.065 >50 >50 >50 >50 >50 MG RM-C-019 0.010 0.010 0.021 0.264 20.100 >50 MG RM-C-020 5.846 >50 >50 >50 >50 >50 MG RM-C-022 0.042 0.127 0.780 >50 >50 >50 MG RM-C-023 0.026 0.151 0.143 0.051 3.198 0.230 MG RM-C-024 >50 >50 >50 >50 >50 MG RM-C-025 10.489 >50 >50 >50 >50 >50 MG RM-C-026 0.007 0.036 1.674 0.004 0.006 0.011 MG RM-C-027 0.081 0.008 0.065 >50 >50 >50 MG RM-C-028 14.423 >50 >50 1.474 33.676 >50 IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-141 PGT-142 PGT-143 PGT-144 PGT-145 VRC01 6535.3 B >50 >50 >50 >50 >50 1.755 92 BR 020 20.299 15.146 25.892 44.036 1.051 0.290 93TH305 >50 5.856 17.270 >50 0.032 0.243 APV 13 >50 >50 >50 >50 0.337 0.657 APV_17 0.590 0.628 1.108 10.298 0.488 0.477 APV 6 14.216 4.725 13.968 >50 0.104 1.812 CAAN.A2 >50 >50 >50 >50 7.085 0.722 JRFL >50 >50 >50 >50 30.401 0.034 MGRM-Chronic-B-001 >50 >50 >50 >50 3.223 0.066 MGRM-Chronic-B-002 >50 >50 >50 >50 3.672 0.117 MGRM-Chronic-B-003 0.400 0.484 0.854 7.408 0.045 0.100 MGRM-Chronic-B-004 0.586 0.172 0.395 9.415 0.023 0.055 MGRM-Chronic-B-008 >50 >50 >50 >50 1.903 0.155 MGRM-Chronic-B-010 >50 >50 >50 >50 >50 0.156 MGRM-Chronic-B-011 2.241 0.994 1.557 5.637 0.206 0.587 MGRM-Chronic-B-012 >50 >50 >50 >50 9.662 1.033 MGRM-Chronic-B-017 0.147 0.076 0.145 0.712 0.054 0.141 MGRM-Chronic-B-018 0.572 0.323 0.377 2.397 0.013 0.328 MGRM-Chronic-B-020 >50 >50 >50 >50 >50 0.424 MGRM-Chronic-B-023 0.318 0.486 0.236 >50 0.009 0.275 MGRM-Chronic-B-024 0.011 0.005 0.010 0.040 0.018 0.377 PVO.4 0.113 0.066 0.162 0.535 0.299 0.218 QH0692.42 >50 >50 >50 >50 >50 1.194 SC422661.8 1.656 3.153 2.017 >50 0.079 0.179 SF162 >50 >50 >50 >50 >50 0.421 THR0.18 0.007 0.012 0.016 0.029 0.013 2.461 TRJ04551.58 >50 >50 >50 >50 >50 0.060 TRO.11 0.383 0.270 0.219 4.728 0.044 0.186 VLGC3 >50 >50 >50 >50 >50 0.108 NL43 >50 >50 >50 >50 0.006 0.100 JRCSF 0.009 0.010 0.007 0.149 0.002 0.164 93IN905 C 0.001 0.001 0.002 0.132 0.002 0.138 93MW959 0.003 0.001 0.002 0.332 1.203 0.053 97ZA012 >50 >50 >50 >50 0.915 0.088 98IN022 0.001 0.001 0.001 0.171 0.005 0.342 MG RM-C-001 >50 >50 >50 >50 >50 1.369 MG RM-C-002 7.834 10.535 6.766 >50 0.008 0.577 MG RM-C-004 >50 >50 >50 >50 >50 1.077 MG RM-C-005 8.332 1.854 9.192 >50 >50 5.417 MG RM-C-006 0.330 0.478 0.369 48.894 0.010 2.202 MG RM-C-007 >50 >50 >50 >50 >50 MG RM-C-008 >50 >50 >50 >50 >50 1.445 MG RM-C-009 0.143 0.061 0.121 0.561 0.193 0.261 MG RM-C-010 0.411 0.070 0.171 7.616 0.141 0.085 MG RM-C-012 >50 >50 >50 >50 6.984 0.073 MG RM-C-013 >50 >50 >50 >50 >50 >50 MG RM-C-014 11.807 6.024 8.811 >50 5.816 >50 MG RM-C-015 0.033 0.038 0.085 0.236 0.120 1.569 MG RM-C-017 4.018 1.111 5.170 23.035 >50 0.366 MG RM-C-019 0.008 0.003 0.004 0.513 0.005 1.126 MG RM-C-020 1.336 0.446 0.797 1.288 0.712 0.396 MG RM-C-022 0.381 0.086 0.103 2.578 0.492 12.989 MG RM-C-023 0.153 0.087 0.148 0.380 0.201 0.077 MG RM-C-024 0.101 0.030 0.050 2.064 0.174 3.395 MG RM-C-025 0.247 0.183 0.336 17.550 0.044 2.364 MG RM-C-026 >50 >50 >50 >50 >50 0.707 MG RM-C-027 3.275 0.792 0.598 >50 0.095 0.125 MG RM-C-028 13.771 2.117 12.367 >50 0.099 0.319 IC.sub.50 (µg/mL).sup.a IC.sub.50 (l/dil'n).sup.b Isolate Subtype PGV04 PG9 #17 #84 #36 #39 6535.3 B 0.757 1.053 3712.0 477.8 848.0 387.0 92 BR 020 0.250 >50 1623.7 262.2 741.3 421.0 93TH305 0.255 8.124 3675.2 828.0 1520.7 306.0 APV 13 1.227 >50 177.8 211.6 1460.1 173.0 APV_17 0.309 34.778 470.0 161.5 279.4 482.0 APV 6 24.187 0.181 914.6 253.2 512.9 706.0 CAAN.A2 2.262 19.267 1743.1 173.1 148.8 222.0 JRFL 0.032 >50 1291.5 348.5 858.2 471.0 MGRM-Chronic-B-001 0.048 >50 330.1 160.6 246.0 390.0 MGRM-Chronic-B-002 0.142 3.439 200.1 <100 344.4 320.0 MGRM-Chronic-B-003 0.258 >50 3664.9 307.2 370.4 103.0 MGRM-Chronic-B-004 0.047 0.533 1874.9 796.5 1463.1 304.0 MGRM-Chronic-B-008 0.166 8.784 2805.1

251.8 260.2 685.0 MGRM-Chronic-B-010 0.744 0.004 1145.5 443.0 3234.8 383.0 MGRM-Chronic-B-011 0.497 >50 130.8 458.4 1030.2 759.0 MGRM-Chronic-B-012 >50 0.200 903.5 480.6 634.2 664.0 MGRM-Chronic-B-017 0.129 0.688 132.7 317.6 <100 383.0 MGRM-Chronic-B-018 0.164 0.216 <100 612.7 289.3 375.0 MGRM-Chronic-B-020 1.262 >50 4132.2 390.1 858.8 305.0 MGRM-Chronic-B-023 0.139 0.027 2216.5 950.0 196.7 523.0 MGRM-Chronic-B-024 2.517 0.222 157.1 1060.2 104.1 110.0 PVO.4 0.454 24.752 212.9 239.4 420.8 123.0 QH0692.42 1.904 >50 175.1 <100 227.6 359.0 SC422661.8 0.110 1.477 367.3 228.7 <100 220.0 SF162 0.0028 >50 5329.3 2049.0 5103.0 3043.0 THR0.18 >50 26.379 <100 1036.0 107.7 126.0 TRJ04551.58 0.028 0.858 152.5 400.2 1416.1 617.0 TRO.11 0.121 16.865 2706.9 861.3 454.5 2740.0 VLGC3 0.045 0.022 2961.7 238.5 1333.0 754.0 NL43 0.028 >50 418.2 3308.8 3033.8 1184.0 JRCSF 0.078 0.003 863.0 6255.4 2371.1 433.0 93IN905 C 0.332 0.035 4396.9 14817.0 1313.8 3667.0 93MW959 >50 0.054 1602.2 6171.8 176.1 506.0 97ZA012 0.041 3.400 8150.0 1083.1 433.0 594.0 98IN022 5.693 0.003 3711.8 15197.1 729.6 1918.0 MG RM-C-001 >50 >50 <100 117.0 <100 174.0 MG RM-C-002 >50 >50 1799.2 733.6 704.1 556.0 MG RM-C-004 0.899 2.011 2177.9 234.5 327.9 2064.0 MG RM-C-005 23.898 8.742 1323.5 405.1 383.0 600.0 MG RM-C-006 >50 0.427 1681.6 1231.3 532.3 564.0 MG RM-C-007 >50 0.064 330.7 244.2 <100 442.0 MG RM-C-008 0.791 >50 1742.3 223.3 <100 903.0 MG RM-C-009 1.042 >50 <100 460.9 <100 224.0 MG RM-C-010 0.299 >50 <100 836.6 <100 316.0 MG RM-C-012 0.025 0.610 716.3 521.6 169.8 2150.0 MG RM-C-013 >50 >50 <100 <100 <100 453.0 MG RM-C-014 21.190 1.099 <100 702.6 <100 <100 MG RM-C-015 0.983 0.427 141.6 544.2 253.4 1095.0 MG RM-C-017 0.383 2.688 3154.6 225.7 266.4 1587.0 MG RM-C-019 9.632 0.005 <100 4576.4 1383.6 424.0 MG RM-C-020 0.112 >50 110.9 245.6 115.3 <100 MG RM-C-022 18.283 0.346 1972.8 921.3 725.8 658.0 MG RM-C-023 0.128 0.614 1046.6 354.6 285.3 1358.0 MG RM-C-024 >50 0.227 718.0 607.2 <100 170.0 MG RM-C-025 >50 0.148 228.2 886.0 129.6 651.0 MG RM-C-026 0.303 0.060 12635.8 2201.2 2438.1 9554.0 MG RM-C-027 2.762 5.358 2368.0 859.3 1636.4 869.0 MG RM-C-028 0.149 0.067 115.0 363.0 <100 502.0 (647) TABLE-US-00068 TABLE 58D IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-121 PGT-122 PGT-123 PGT-125 PGT-126 PGT-127 98CN009 CRF07_BC 0.009 0.013 0.007 0.030 0.019 0.090 98CN006 CRF08 BC 0.010 0.021 0.008 0.201 0.046 0.067 92UG001 D >50 >50 >50 >50 >50 >50 92UG005 18.292 >50 9.794 >50 0.037 1.107 92UG024 >50 >50 >50 >50 >50 >50 92UG046 >50 >50 >50 >50 >50 92UG114 0.004 0.011 0.005 0.349 0.038 7.091 MG RM-D-001 0.615 1.130 0.147 2.837 0.084 2.234 MG RM-D-002 0.006 0.005 0.012 0.118 0.039 0.600 MG RM-D-003 10.685 >50 3.736 0.065 0.021 0.029 MG RM-D-004 >50 >50 >50 0.371 >50 >50 MG RM-D-005 >50 >50 >50 >50 0.792 6.024 MG RM-D-008 >50 >50 >50 >50 43.139 >50 MG RM-D-011 0.018 0.019 0.012 0.008 0.009 0.015 MG RM-D-012 >50 >50 >50 >50 >50 >50 MG RM-D-013 >50 >50 >50 >50 >50 >50 MG RM-D-014 0.009 0.012 0.009 >50 0.030 0.393 MG RM-D-016 >50 >50 >50 0.120 >50 >50 MG RM-D-018 0.005 0.007 0.007 >50 0.009 0.021 MG RM-D-019 >50 >50 >50 0.049 >50 >50 MG RM-D-020 0.010 0.007 0.018 >50 0.023 0.015 MG RM-D-021 5.334 11.735 11.120 0.015 0.047 0.056 MG RM-D-022 >50 >50 >50 0.040 >50 >50 MG RM-D-024 0.038 0.185 0.023 >50 0.376 1.945 MG RM-D-026 >50 >50 >50 >50 >50 >50 MG RM-D-028 0.009 3.338 0.132 >50 >50 >50 MG RM-D-029 >50 >50 >50 >50 >50 >50 MGRM-F1-004 F 1.665 1.291 1.783 >50 0.063 0.068 MGRM-F1-006 5.328 3.266 6.894 0.035 0.078 7.172 MGRM-F1-008 >50 >50 >50 0.361 >50 >50 MGRM-F1-010 0.016 0.024 0.016 >50 >50 >50 MGRM-F1-012 1.907 >50 22.222 >50 0.300 >50 MGRM-F1-013 0.083 0.152 0.080 >50 >50 >50 MGRM-F1-014 0.082 0.195 0.111 >50 >50 >50 MGRM-F1-015 28.359 31.437 >50 >50 >50 >50 MGRM-F1-016 0.931 4.808 4.948 >50 >50 >50 MGRM-F1-017 16.891 >50 >50 >50 >50 >50 MGRM-F1-018 0.040 0.563 0.054 >50 14.941 3.586 MGRM-F1-020 0.167 0.321 0.051 >50 >50 >50 MGRM-F1-021 0.099 2.301 0.307 >50 >50 >50 MGRM-F1-022 0.104 0.277 0.075 >50 >50 >50 MGRM-F1-023 0.097 0.154 0.053 0.12 0.009 0.119 IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-128 PGT-130 PGT-131 PGT-135 PGT-136 PGT-137 98CN009 CRF07_BC 0.025 0.136 4.051 0.071 0.428 5.059 98CN006 CRF08 BC 0.015 0.258 35.286 0.429 0.406 1.217 92UG001 D >50 >50 >50 >50 >50

>50 92UG001 0.018 >50 0.959 0.854 >50 >50 92UG024 >50 >50 >50 0.010 0.052 0.106
92UG046 >50 >50 >50 >50 >50 >50 92UG114 0.083 >50 >50 1.289 >50 >50 >50 MG RM-D-001
0.138 >50 >50 >50 >50 >50 MG RM-D-002 0.011 0.869 11.150 >50 >50 >50 MG RM-D-003
0.019 0.651 19.624 >50 >50 >50 MG RM-D-004 24.887 1.208 2.007 >50 >50 >50 MG RM-D-005
0.051 >50 >50 >50 >50 >50 MG RM-D-008 0.033 >50 >50 >50 >50 >50 MG RM-D-011 0.007
0.019 0.756 23.478 >50 6.297 MG RM-D-012 >50 >50 >50 >50 >50 >50 MG RM-D-013 8.847
0.429 4.141 >50 >50 >50 MG RM-D-014 0.013 0.061 0.516 2.209 >50 >50 MG RM-D-016 >50
>50 >50 >50 >50 >50 MG RM-D-018 0.008 0.049 0.148 0.039 >50 15.100 MG RM-D-019 >50
>50 >50 >50 >50 >50 MG RM-D-020 0.011 24.677 >50 >50 >50 >50 MG RM-D-021 0.014 0.156
0.106 >50 >50 >50 MG RM-D-022 >50 >50 >50 >50 >50 >50 MG RM-D-024 0.067 >50 >50 >50
>50 >50 MG RM-D-026 >50 >50 >50 >50 >50 >50 MG RM-D-028 >50 0.825 >50 >50 >50 >50
MG RM-D-029 >50 >50 >50 30.849 >50 >50 MGRM-F1-004 F 0.027 0.023 0.114 0.211 >50
40.343 MGRM-F1-006 0.763 >50 >50 >50 >50 >50 MGRM-F1-008 >50 0.065 2.742 >50 >50 >50
MGRM-F1-010 16.663 >50 >50 >50 >50 >50 MGRM-F1-012 0.007 >50 >50 0.010 >50 >50
MGRM-F1-013 12.723 >50 >50 21.080 >50 >50 MGRM-F1-014 0.277 11.795 >50 >50 >50 >50
MGRM-F1-015 >50 >50 >50 1.114 >50 >50 MGRM-F1-016 1.209 >50 >50 >50 >50 >50 MGRM-
F1-017 11.659 >50 >50 7.788 >50 >50 MGRM-F1-018 0.023 >50 >50 >50 >50 >50 MGRM-F1-
020 0.355 9.728 >50 0.168 9.311 1.931 MGRM-F1-021 >50 >50 >50 0.065 >50 >50 MGRM-F1-
022 4.545 >50 >50 0.964 2.418 6.231 MGRM-F1-023 0.007 0.050 >50 0.088 >50 >50 IC.sub.50
(µg/mL).sup.a Isolate Subtype PGT-141 PGT-142 PGT-143 PGT-144 PGT-145 VRC01 98CN009
CRF07_BC 1.277 3.999 10.206 >50 0.136 0.346 98CN006 CRF08 BC >50 >50 >50 >50 >50
0.644 92UG001 D >50 >50 >50 >50 >50 1.159 92UG005 >5.0 >50 >50 >50 >50 0.576 92UG024
>50 >50 >50 >50 1.600 0.576 92UG046 >50 >50 >50 >50 >50 12.687 92UG114 >50 >50 >50 >50
20.113 0.330 MG RM-D-001 >50 >50 >50 >50 0.560 >50 MG RM-D-002 >50 >50 >50 >50 >50
0.252 MG RM-D-003 >50 >50 >50 >50 0.006 2.064 MG RM-D-004 >50 >50 >50 >50 10.881
1.250 MG RM-D-005 5.154 1.153 1.414 14.807 10.398 >50 MG RM-D-008 0.371 0.174 0.153
9.849 0.025 0.854 MG RM-D-011 >50 >50 >50 >50 14.655 0.075 MG RM-D-012 >50 >50 >50
>50 >50 0.216 MG RM-D-013 0.179 0.221 0.386 >50 3.480 0.087 MG RM-D-014 0.010 0.005
0.009 0.113 2.511 0.332 MG RM-D-016 >50 >50 >50 >50 >50 0.452 MG RM-D-018 >50 >50 >50
>50 0.065 >50 MG RM-D-019 1.655 0.155 0.277 >50 0.111 0.119 MG RM-D-020 >50 >50 >50
>50 10.003 0.192 MG RM-D-021 >50 >50 >50 >50 >50 33.445 MG RM-D-022 >50 >50 >50 >50
34.305 >50 MG RM-D-024 >50 >50 >50 >50 0.224 0.366 MG RM-D-026 32.480 12.512 44.566
>50 0.410 0.435 MG RM-D-028 <50 >50 >50 >50 0.721 0.377 MG RM-D-029 0.057 0.192 0.200
0.459 0.030 3.224 MGRM-F1-004 F 0.021 0.013 0.036 0.105 0.361 0.305 MGRM-F1-006 8.993
12.874 12.341 20.563 13.642 2.856 MGRM-F1-008 0.193 0.124 0.189 0.507 0.228 1.114 MGRM-
F1-010 >50 >50 >50 >50 3.321 0.097 MGRM-F1-012 >50 >50 >50 >50 0.113 0.488 MGRM-F1-
013 0.043 0.023 0.013 0.509 0.402 0.170 MGRM-F1-014 0.001 0.001 0.001 0.014 0.009 0.056
MGRM-F1-015 >50 >50 >50 >50 1.764 0.387 MGRM-F1-016 >50 >50 >50 >50 17.531 0.517
MGRM-F1-017 30.829 23.465 26.279 >50 0.286 1.077 MGRM-F1-018 0.785 0.404 0.701 >50
1.020 0.032 MGRM-F1-020 1.152 0.824 0.980 8.929 0.816 1.890 MGRM-F1-021 >50 >50 >50
>50 8.038 0.039 MGRM-F1-022 0.114 0.040 0.082 0.382 0.009 0.055 MGRM-F1-023 0.307 0.602
0.972 1.163 5.879 0.392 IC.sub.50 (µg/mL).sup.a IC.sub.50 (l/dil'n).sup.b Isolate Subtype PGV04
PG9 #17 #84 #36 #39 98CN009 CRF07_BC 0.0125 0.371 2551.0 1469.5 756.3 1430.0 98CN006
CRF08 BC 1.320 >50 1908.7 365.2 367.0 980.0 92UG001 D 0.553 >50 <100 <100 <100 260.0
92UG005 0.440 >50 265.5 <100 590.3 480.0 92UG024 0.201 1.861 <100 202.0 138.1 1930.0
92UG046 22.362 1.861 <100 200.5 <100 137.0 92UG114 0.461 40.608 5390.2 <100 292.1 548.0
MG RM-D-001 >50 >50 <100 <100 112.8 <100 MG RM-D-002 0.322 0.021 5960.5 <100 726.4
442.0 MG RM-D-003 >50 0.033 183.1 695.8 414.7 176.0 MG RM-D-004 43.200 0.060 <100
125.9 108.5 397.0 MG RM-D-005 >50 1.802 <100 235.9 109.3 249.0 MG RM-D-008 3.564
11.454 <100 348.8 <100 356.0 MG RM-D-011 0.025 0.071 1037.5 123.2 769.0 498.0 MG RM-D-

012 0.125 16.182 <100 109.0 109.1 261.0 MG RM-D-001 0.088 0.021 <100 745.4 422.8 377.0
 MG RM-D-014 0.194 0.023 2249.9 1526.5 428.0 322.0 MG RM-D-016 0.201 0.098 <100 282.7
 203.4 253.0 MG RM-D-018 17.528 0.025 6632.6 259.2 1730.4 642.0 MG RM-D-019 0.350 0.038
 <100 296.7 260.0 467.0 MG RM-D-020 0.097 1.665 2488.0 <100 393.5 190.0 MG RM-D-021
 >50 >50 <100 <100 457.5 148.0 MG RM-D-022 >50 >50 <100 <100 <100 <100 MG RM-D-024
 >50 0.038 355.3 144.4 124.7 399.0 MG RM-D-026 1.761 23.664 <100 401.4 103.2 <100 MG RM-
 D-028 1.008 20.234 2028.2 259.0 <000 563.0 MG RM-D-029 1.557 >50 <100 291.4 <100 387.0
 MGRM-F1-004 F 0.580 0.175 260.7 874.0 687.6 366.0 MGRM-F1-006 >50 7.249 <100 147.3
 110.8 487.0 MGRM-F1-008 1.031 >50 <100 242.1 <100 308.0 MGRM-F1-010 0.184 0.012 825.3
 420.4 <100 564.0 MGRM-F1-012 0.250 0.023 573.3 332.3 299.8 1366.0 MGRM-F1-013 0.084
 1.253 283.9 2403.4 377.4 625.0 MGRM-F1-014 0.020 0.020 314.2 6934.5 <100 538.0 MGRM-F1-
 015 0.179 >50 <100 145.7 <100 234.0 MGRM-F1-016 0.268 0.518 110.3 103.0 <100 407.0
 MGRM-F1-017 2.383 >50 <100 279.4 <100 832.0 MGRM-F1-018 0.027 0.019 770.0 518.6 151.2
 391.0 MGRM-F1-020 5.987 6.396 305.4 220.8 <100 306.0 MGRM-F1-021 0.031 >50 227.6 505.7
 218.7 919.0 MGRM-F1-022 0.072 0.028 350.0 1296.4 221.3 638.0 MGRM-F1-023 0.171 >50
 362.3 337.3 674.5 2177.0
 (648) TABLE-US-00069 TABLE 58E IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-121 PGT-122
 PGT-123 PGT-125 PGT-126 PGT-127 MGRM-G-001 G 0.004 0.005 0.009 0.033 0.038 0.113
 MGRM-G-004 0.115 2.146 0.181 >50 >50 >50 MGRM-G-006 >50 >50 >50 >50 >50 >50 MGRM-
 G-009 0.020 0.514 33.546 >50 >50 >50 MGRM-G-011 3.167 31.299 14.900 >50 7.318 >50
 MGRM-G-013 3.991 3.448 3.969 >50 48.845 7.699 MGRM-G-014 0.009 0.008 0.005 0.007 0.009
 0.009 MGRM-G-015 >50 >50 >50 >50 >50 >50 MGRM-G-016 >50 >50 >50 >50 >50 >50
 MGRM-G-017 0.004 0.032 0.026 >50 >50 >50 MGRM-G-019 0.011 0.016 0.008 >50 0.018 0.429
 MGRM-G-024 0.014 0.033 0.032 >50 >50 >50 MGRM-G-025 0.063 0.079 0.046 >50 17.054
 19.279 MGRM-G-027 0.124 0.606 0.182 34.414 0.875 >50 MGRM-G-028 0.014 0.010 0.008
 0.012 0.012 0.032 aMLV negative >50 >50 >50 >50 >50 >50 IC.sub.50 (µg/mL).sup.a Isolate
 Subtype PGT-128 PGT-130 PGT-131 PGT-135 PGT-136 PGT-137 MGRM-G-001 G 0.019 0.0198
 0.404 0.393 8.113 41.703 MGRM-G-004 >50 >50 >50 >50 >50 >50 MGRM-G-006 >50 >50 >50
 >50 >50 >50 MGRM-G-009 >50 >50 >50 >50 >50 >50 MGRM-G-011 8.212 >50 >50 >50 >50
 >50 MGRM-G-013 0.995 >50 >50 >50 >50 >50 MGRM-G-014 0.004 2.226 4.313 >50 >50 >50
 MGRM-G-015 >50 >50 >50 >50 >50 >50 MGRM-G-016 >50 >50 >50 >50 >50 >50 MGRM-G-
 017 0.319 >50 >50 0.137 >50 34.905 MGRM-G-019 0.010 27.938 >50 0.145 44.285 0.126
 MGRM-G-024 0.095 5.838 10.730 0.062 0.152 0.051 MGRM-G-025 0.017 8.452 7.750 0.592 >50
 >50 MGRM-G-027 0.313 4.137 2.904 0.062 45.222 0.228 MGRM-G-028 0.008 0.074 0.466 >50
 >50 >50 aMLV negative >50 >50 >50 >50 >50 >50 IC.sub.50 (µg/mL).sup.a Isolate Subtype PGT-
 141 PGT-142 PGT-143 PGT-144 PGT-145 VRC01 MGRM-G-001 G 0.392 0.362 0.423 >50 0.028
 0.0371 MGRM-G-004 >50 >50 >50 >50 >50 >50 MGRM-G-006 0.043 0.019 0.031 3.843 0.286
 0.055 MGRM-G-009 1.890 2.811 2.968 >50 0.982 0.073 MGRM-G-011 0.086 0.046 0.074 1.577
 0.028 0.065 MGRM-G-013 >50 >50 >50 >50 >50 0.553 MGRM-G-014 0.222 0.076 0.107 0.562
 0.144 0.056 MGRM-G-015 >50 >50 >50 >50 10.465 0.548 MGRM-G-016 4.690 8.170 7.600 >50
 0.081 0.063 MGRM-G-017 0.164 0.125 0.107 9.972 0.205 0.124 MGRM-G-019 20.310 5.949
 4.773 >50 0.041 0.060 MGRM-G-024 0.051 0.054 0.048 7.923 0.060 0.025 MGRM-G-025 >50
 >50 >50 >50 >50 0.292 MGRM-G-027 >50 >50 >50 >50 0.433 0.360 MGRM-G-028 24.888
 30.309 42.988 >50 0.046 0.164 aMLV negative >50 >50 >50 >50 >50 >50 IC.sub.50
 (µg/mL).sup.a IC.sub.50 (l/dil'n).sup.b Isolate Subtype PGV04 PG9 #17 #84 #36 #39 MGRM-G-
 001 G >50 0.166 5296.0 557.6 635.9 169.0 MGRM-G-004 >50 >50 600.5 157.8 107.3 574.0
 MGRM-G-006 0.083 1.983 <100 548.1 <100 <100 MGRM-G-009 0.043 10.704 160.3 336.3 <100
 345.0 MGRM-G-011 0.780 0.335 <100 641.1 <100 891.0 MGRM-G-013 0.168 >50 100.5 561.4
 480.3 629.0 MGRM-G-014 0.056 16.478 3593.6 389.5 1058.0 295.0 MGRM-G-015 0.333 6.599
 <100 <100 <100 266.0 MGRM-G-016 0.019 1.363 <100 681.8 106.0 440.0 MGRM-G-017 0.065

0.104 1582.3 701.7 282.9 420.0 MGRM-G-019 0.027 14.559 1416.7 253.3 354.8 524.0 MGRM-G-024 0.033 0.163 1077.4 388.2 131.5 516.0 MGRM-G-025 1.608 >50 396.1 129.5 164.2 213.0 MGRM-G-027 0.125 0.007 289.0 140.8 143.4 676.0 MGRM-G-028 0.352 0.891 2769.4 522.4 1201.4 621.0 aMLV negative >50 >50 <100 <100 <100 <100 .sup.a White squares indicate an IC.sub.50 of >50 Pg/mL, green squares indicate 50 Pg/mL >IC.sub.50 >10 Pg/mL, yellow squares indicate 10 Pg/mL >IC.sub.50 >1 Pg/mL, orange squares indicate 1 Pg/mL >IC.sub.50 >0.1 Pg/mL, and red squares indicate IC.sub.50 <0.1 Pg/mL. .sup.b White squares indicate an IC.sub.50 of <1:100 dilution, green squares indicate 1:100 >IC.sub.50 >1:150, yellow squares indicate 1:150 >IC.sub.50 >1:500, orange squares indicate 1:500 >IC.sub.50 >1:1000, and red squares indicate IC.sub.50 >1:1000 dilution.

(649) TABLE-US-00070 TABLE 59 Binding activity PGT mAbs. EC.sub.50 (µg/ml).sup.a WT JR-FL JR-FL gp120 JR-FL GP120 Endo H treated Donor mAb gp120 Δ V1/Δ V2 Δ V3 JR-FL gp120 #17 121 0.2 0.2 >10 >10 122 0.3 0.3 >10 >10 123 0.2 0.3 >10 >10 #36 125 0.2 0.6 >10 >10 126 0.1 0.3 >10 >10 127 0.1 0.7 >10 >10 128 0.1 0.3 7.8 >10 130 0.2 >10 >10 >10 131 0.4 >10 >10 >10 #39 135 0.1 0.1 0.2 >10 136 0.2 0.3 0.1 >10 137 0.1 0.2 0.1 >10 .sup.a Binding was evaluated by ELISA. EC.sub.50 values were derived by nonlinear regression analysis. Boxes are color coded as follows: red, EC.sub.50 < 1.0 µg/ml; yellow, 1.0 µg/ml < EC.sub.50 < 10 µg/ml; gray, EC.sub.50 > 10 µg/ml. Experiments were performed in duplicate, and data represent an average of at least two independent experiments.

(650) Many of the clonally related mAbs exhibited differing degrees of overall neutralization potency. For example, the median IC.sub.50s of PGTs 131, 136, 137, and 144 were approximately 10- to 50-fold higher than those of their somatically-related sister clones (Table 61). Also, in some cases, the somatically-related mAbs exhibited similar neutralization potency, but differing degrees of neutralization breadth, against the panel of viruses tested (Tables 58A-E and Table 61). For example, PGT-128 neutralized with comparable overall potency but significantly greater neutralization breadth than the clonally related PGT-125, -126, and -127 mAbs (Tables 58A-E and Table 61). Overall, these observations suggest that serum neutralization breadth may develop from the successive selection of somatic mutants that bind to a modified epitope or a slightly different envelope (Env) conformation expressed on virus escape variants. Additionally, these results indicate that the full serum neutralization breadth and potency may be mediated by a small number of sequentially selected mAbs that bind to distinct, but overlapping, epitopes differentially expressed on various isolates. In this respect, antibody somatic variants could in effect “slide” around the Env spike surface. Comparison of the neutralization profiles of the mAbs isolated from a given donor with the profiles from the sera revealed that the mAbs isolated could largely recapitulate the corresponding serum neutralization breadth and potency (FIGS. 33a and 37).

(651) The epitopes recognized by the newly isolated bnMAbs were determined. ELISA binding assays indicated that PGTs 121-123, 125-128, 130, 131, and 135-137 bound to monomeric gp120 (Table 59). In contrast, the PGT 141-145 bnMAbs exhibited a strong preference for membrane-bound, trimeric HIV-1 Env (FIG. 38). Based on this result, it was postulated that these bnMAbs bound to quaternary epitopes similar to those of the recently described PG9 and PG16 bnMAbs (Walker, L. M., et al. Science 326, 285-289 (2009)). Indeed, this hypothesis was confirmed by competition studies, N160K sensitivity, and, for PGTs 141-144, an inability to neutralize JR-CSF pseudoviruses expressing homogenous Man.sub.9GlcNAc.sub.2 glycans (Walker, L. M., et al. PLoS Pathog 6(2010)) (FIG. 39).

(652) To define the epitopes recognized by the remaining PGT antibodies, competition ELISA assays were carried out with a panel of well-characterized neutralizing and non-neutralizing antibodies (FIG. 34a). Unexpectedly, all of the remaining antibodies (PGTs 121-123, 125-128, 130, 131, 135-137) competed with the glycan-specific bnMAb 2G12. This result was surprising given that 2G12 had previously formed its own unique competition group. All of the mAbs except for PGTs 135, 136 and 137 also competed with a V3 loop-specific mAb and failed to bind to gp120

ΔV3, suggesting their epitopes were in proximity to or contiguous with the V3 loop (FIG. 34a and Table 59). Deglycosylation of gp120 with Endo H abolished binding by all the mAbs, indicating that certain oligomannose glycans were important for epitope recognition (Table 59). Competition of these mAbs with 2G12 and lack of binding to deglycosylated gp120 prompted us to investigate whether these antibodies contacted glycans directly. Glycan array analysis revealed that PGTs 125-128, and 130 bound specifically to both Man.sub.8GlcNAc.sub.2 and Man.sub.9GlcNAc.sub.2, whereas the remaining antibodies showed no detectable binding to high-mannose glycans (FIG. 34b). Interestingly, the binding of PGTs 125-128, 130 to gp120 was competed by Man.sub.9 but, unlike 2G12, was not competed by monomeric mannose or Man.sub.4 (D1 arm of Man.sub.9GlcNAc.sub.2) (FIGS. 34c and 34d), suggesting a different mode of glycan recognition. Furthermore, in contrast to 2G12, no evidence was found for domain exchange and monomeric Fab fragments exhibited potent neutralizing activity (FIG. 41).

(653) To further define the epitopes recognized by the mAbs, neutralizing activity against a large panel of HIV-1.sub.JR-CSF variants incorporating single alanine substitutions was assessed using a single round of replication pseudovirus assay (Table 60). In the panel of mutants, the N-linked glycans at positions 332 and/or 301 were important for neutralization by PGTs 125-128, 130, and 131 suggesting their direct involvement in epitope formation. The apparent dependency on so few glycans suggests that, although PGTs 125-128, 130, and 131 contact Man.sub.8-9GlcNAc2 glycans directly, their arrangement in the context of gp120 is critical for high affinity glycan recognition and neutralization potency. This is further highlighted by the inability of PGT Mabs to neutralize SIVmac239, HIV-2 or HCV, which display a high level of glycosylation. Although PGTs 121-123 failed to exhibit detectable binding to high-mannose glycans and be competed by mannose sugars (FIG. 40), the only substitutions that completely abolished neutralization by these mAbs were those that resulted in removal of the glycan at position 332. Although structural studies will be required to fully define the epitopes recognized by these antibodies, the above result suggests either that the PGT 121-123 mabs bind to a protein epitope along the gp120 polypeptide backbone that is conformationally dependent on the N332 glycan or that the glycan contributes more strongly to binding in the context of the intact protein.

(654) TABLE-US-00071 TABLE 60A Neutralizing activity of PGT mAbs against a panel of JR-CSF alanine mutants. Fold IC.sub.50 increase relative to wild-type.sup.c Mutation.sup.a gp120 domain.sup.b PGT-121 PGT-122 PGT-123 PGT-125 PGT-126 PGT-127 PGT-128 PGT-130 PGT-131 PGT-135 D113A C1 1.2 1.0 1.1 1.1 1.2 0.3 0.5 0.8 ND 0.8 V120A 0.8 1.2 1.1 0.5 0.7 1.5 1.4 0.8 0.7 1.4 L125A 0.9 2.4 2.5 1.6 1.2 1.4 2.9 2.5 2.5 0.9 V127A 0.7 1.0 1.2 0.9 1.3 2.5 1.1 0.8 0.6 1.4 N134A V1 0.2 0.3 0.4 0.5 0.6 1.0 1.2 0.6 0.5 0.2 N156A V2 0.9 1.1 6.2 3.6 1.8 1.6 0.4 1.5 1.0 0.3 N160K 1.1 1.1 1.0 0.5 0.5 0.8 1.0 3.4 0.4 7.7 T162A 0.3 0.6 0.7 0.5 0.6 1.1 1.1 0.6 0.6 0.9 I165A 3.3 3.0 2.2 2.3 2.5 1.7 2.1 3.2 2.4 4.0 R166A 1.4 0.9 0.9 1.5 1.0 1.6 2.7 1.2 1.6 0.9 D167A 1.0 1.1 1.2 1.7 1.6 1.6 0.3 1.2 1.2 0.7 K168A 1.5 0.8 0.8 0.9 0.7 1.9 1.1 1.2 1.7 0.4 E172A 1.4 1.0 1.1 1.8 0.7 2.1 2.5 1.4 ND 1.3 Y177A 1.4 2.4 2.4 1.1 0.9 2.9 1.4 3.6 5.3 0.3 L179A 1.9 1.1 1.2 1.3 2.6 2.5 2.8 2.4 4.8 1.2 V182A 2.1 2.2 2.8 1.3 1.4 1.6 1.2 1.2 1.1 1.5 D185A 1.0 1.8 1.9 0.6 1.0 2.0 1.5 0.9 0.7 1.3 N188A 0.8 1.0 1.1 0.7 0.8 0.8 1.3 1.0 0.9 0.4 N197A C2 1.1 1.4 1.4 1.0 1.1 0.6 1.0 1.2 1.1 0.3 S199A (V1/V2 stem) 0.8 1.0 1.1 0.7 0.8 3.0 1.0 0.9 1.6 0.3 T202A 1.2 1.5 1.2 0.8 0.8 1.8 1.1 1.2 2.7 1.8 F210A C2 1.5 2.6 2.8 1.4 1.5 0.8 0.8 1.7 1.0 1.3 N241A 0.5 1.4 0.8 0.5 0.6 0.6 0.7 0.6 0.7 0.1 N262A 1.1 1.1 0.5 0.9 1.0 1.4 1.6 0.7 0.4 2.1 N276A 0.8 1.4 1.1 0.9 0.9 1.6 0.6 1.0 0.8 2.5

(655) TABLE-US-00072 TABLE 60B Fold IC.sub.50 Increase relative to wild-type.sup.c Mutation.sup.a gp120 domain.sup.b PGT-121 PGT-122 PGT-123 PGT-125 PGT-126 PGT-127 PGT-128 PGT-130 PGT-131 PGT-135 V292A 2.7 1.3 1.7 3.3 5.0 0.9 2.1 0.9 1.4 2.0 N295A V3 0.6 0.7 1.2 0.8 0.9 2.3 1.0 0.7 2.1 >18 T297A 0.7 0.6 0.3 0.9 0.8 1.7 1.1 0.9 1.2 >30 P299A 0.2 0.5 0.7 0.2 0.2 0.2 1.2 0.8 2.0 0.36 N301A 0.4 0.4 1.0 213 314 >250 10.7 >1000 >170 0.5 N302A 0.7 1.1 1.3 0.7 0.9 3.0 1.5 1.1 1.0 0.8 T303A 0.7 2.5 1.1 >250 >250 >500 >500 >250 >170 0.3 R304A 1.7

0.7 0.7 1.9 1.8 0.4 0.7 2.5 1.5 0.7 K305A 1.9 3.0 1.9 1.9 1.0 2.6 1.8 6.9 ND 0.2 S306A 0.7 0.5 0.5
1.2 0.7 0.6 1.1 1.5 0.7 0.4 I307A 3.7 3.8 5.9 5.0 1.8 10.2 3.7 1000.0 >170 0.3 I309A 1.1 2.1 1.6 1.0
1.2 6.0 2.8 >77.9 8.8 0.3 P313A 0.5 0.4 0.2 0.3 0.4 1.3 1.0 0.3 0.52 0.5 R315A 0.9 0.5 0.7 0.9 1.0
0.3 0.4 0.6 0.7 1.8 F317A 1.7 3.4 3.1 1.8 1.0 0.2 0.8 17.0 96 0.5 T319A 1.3 0.5 0.4 1.1 0.8 1.0 0.5
1.6 0.9 0.4 T320A 1.2 0.7 0.7 0.8 0.6 1.8 1.0 0.9 4.5 0.5 E321aA 0.3 0.4 0.6 1.0 0.6 1.9 1.3 1.7 1.8
0.8 I323A 1.9 2.2 1.0 0.9 0.9 4.0 3.0 0.8 38.0 1.5 G324A 16.6 12.5 14.5 1.7 3.3 ND ND >50 ND
0.4 D325A 2.0 44.0 64.0 0.4 0.5 0.93 0.5 >1000 >170 0.5 I326A 0.7 0.6 0.4 0.9 0.8 3.5 1.6 1.3 1.7
1.2 R327A 2.5 1.7 1.7 2.1 2.2 1.8 2.5 2.4 19.0 0.67 H330A 0.8 0.5 0.3 0.7 0.5 0.5 0.3 0.9 0.5 >250
N332A >200 >200 >200 2.0 181 >100 1.4 0.6 0.5 >250 S334A >50 >50 >50 0.8 >30 >500 1.4 0.5
0.4 >30 Q337A C3 1.0 0.5 0.5 0.8 0.4 0.5 0.7 1.0 1.3 2.1 N339A 0.5 0.4 0.8 0.9 0.6 1.1 0.8 0.2 1.0
1.2 T341A 0.5 0.3 0.3 1.1 0.5 1.7 1.3 0.8 1.3 0.6 K343A 0.6 0.3 0.3 0.7 1.2 1.0 1.0 1.1 1.1 1.1
R350A 0.8 0.6 0.5 0.6 0.7 1.1 1.0 0.5 1.3 0.9 N355A 1.4 0.8 0.5 1.2 0.2 1.4 0.7 1.3 2.3 0.8 N386A
1.3 1.2 1.4 0.8 0.9 0.6 1.1 0.1 1.1 0.3 S387A 0.5 0.9 0.9 0.6 0.6 1.0 1.2 0.7 1.1 0.6 T388A 0.9 0.3
0.2 0.6 0.6 0.5 1.0 1.0 0.6 0.1

(656) TABLE-US-00073 TABLE 60C Fold IC.sub.50 increase relative to wild-type.sup.c

Mutation.sup.a gp120 domain.sup.b PGT-121 PGT-122 PGT-123 PGT-125 PGT-126 PGT-127
PGT-128 PGT-130 PGT-131 PGT-135 N392Q V4 0.9 1.0 1.0 1.2 1.6 1.1 0.9 1.2 0.4 >250 S393A
0.6 0.5 0.6 0.5 0.5 1.2 1.4 0.7 1.1 0.9 T394A 0.2 0.6 0.4 0.6 0.6 1.6 3.0 1.2 0.9 >75 W395A 0.5 0.4
0.4 0.9 0.6 1.6 0.8 0.6 0.5 0.6 N396A 1.3 1.1 1.2 1.6 1.7 2.4 0.9 1.0 0.9 1 N411A 0.4 0.7 0.5 0.5 0.7
0.8 0.7 0.6 0.6 0.2 T413A 0.5 0.8 0.4 0.5 0.7 0.8 1.0 0.6 0.6 0.3 I414A 0.5 0.5 0.3 0.3 0.4 2.2 1.4
0.7 1.0 1.2 I415A 0.6 0.4 0.4 0.3 0.3 2.1 0.7 0.6 0.6 5 L416A 0.6 1.0 0.9 0.8 0.8 1.0 1.1 1.0 1.0 1.2
D417A 0.7 0.5 0.7 0.5 1.0 0.4 0.5 1.0 1.0 5.0 R419A C4 1.8 0.8 0.7 1.0 1.2 2.5 0.5 2.6 1.7 3.9
I420A 3.3 3.5 3.2 1.1 0.6 1.8 0.8 9.8 153 1.4 K421A 1.3 1.2 0.9 1.2 0.7 1.5 2.0 2.9 11.0 0.3 Q422A
1.2 1.1 0.8 1.2 1.1 0.1 0.2 1.0 1.9 0.8 I4234 3.3 2.8 1.5 5.0 1.1 0.5 2.6 >1000 >80 0.1 I424A 0.9 0.8
0.5 0.7 0.4 2.0 1.3 3.9 4.8 0.1 E466A V5 1.2 0.9 0.60 1.6 1.4 0.9 0.6 1.7 1.7 0.5 F468A 0.8 1.0 0.50
1.0 0.7 3.0 0.5 0.4 1.6 1.1 P470A 0.9 2.1 0.8 1.6 2.7 1.0 0.6 2.4 1.0 0.8 G471A 0.8 1.8 1.3 1.0 1.2
1.2 0.9 0.8 1.2 3 D474A 1.3 1.8 1.1 1.4 3.3 1.5 0.8 2.0 1.6 1.2 R476A 1.1 0.9 2.0 1.9 1.5 1.2 0.7 1.3
1.0 0.8 D477A 1.9 0.9 2.3 3.4 1.1 1.7 0.7 1.1 1.0 1.5 N478A 0.8 0.9 0.6 1.5 1.0 1.2 0.8 0.8 1.0 0.7
R480A 0.8 1.4 1.7 2.1 1.9 0.2 0.1 1.0 1.5 1.4 .sup.aAmino acid numbering is based on the sequence
of HIV-1.sub.HxB2 .sup.bC refers to constant domains and V refers to variable loops.

cNeutralization activity is reported as fold increase in IC.sub.50 value relative to WT JR-CSF and
was calculated using the equation (IC.sub.50mutant/IC.sub.50WT). Gray: substitutions which had a
negligible effect on neutralization activity, yellow: 10-40 fold IC.sub.50 increase, red: >40 fold
IC.sub.50 increase. Experiments were performed in duplicate and values represent an average of at
least two independent experiments.

(657) Vaccines against pathogens with low antigenic diversity, such as hepatitis B virus (HBV) or
measles, commonly achieve 90-95% efficacy (Plotkin. Vaccines (Elsevier Health Sciences,
Philadelphia, 2008)). Similarly, the influenza vaccine achieves 85-90% efficacy in years when the
vaccine and circulating strain are well-matched (Bridges, C. B., et al. JAMA 284, 1655-1663
(2000); Herrera, G. A., et al. Vaccine 25, 154-160 (2007)). However, efficacy drops severely in
years when there is a mismatch between the vaccine and circulating strain. In the case of HIV, the
global diversity of circulating viruses is such that the match between the prophylactic antibodies
and the circulating viruses, i.e. the antibody viral coverage, will be crucial for the degree of
efficacy of active or passive prophylaxis approaches. To date, although the recent RV144 trial has
led to speculation that some degree of protection against HIV may be achieved through extra-
neutralizing activities of antibodies, such as antibody-dependent cell-mediated cytotoxicity or
phagocytosis, the strongest evidence for protection is for neutralizing antibodies in non-human
primate models using simian-human immunodeficiency virus (SHIV) challenge (Parren, P. W., et
al. J Virol 75, 8340-8347 (2001); Nishimura, Y., et al. J Virol 76, 2123-2130 (2002); Hessel, A. J.,
et al. Nat Med 15, 951-954 (2009); Hessel, A. J., et al. PLoS Pathog 5, e1000433 (2009); Willey,

R., et al. *AIDS Res Hum Retroviruses* 26, 89-98 (2010)).

(658) Passive administration of neutralizing antibodies in animal models suggest that a serum titer of approximately or greater than 100 times the IC_{sub}.50 is often required to achieve a meaningful level of protection (Parren, P. W., et al. *J Virol* 75, 8340-8347 (2001); Nishimura, Y., et al. *J Virol* 76, 2123-2130 (2002); Hessel, A. J., et al. *Nat Med* 15, 951-954 (2009); Hessel, A. J., et al. *PLoS Pathog* 5, e1000433 (2009); Willey, R., et al. *AIDS Res Hum Retroviruses* 26, 89-98 (2010)).

Therefore, if a vaccine elicits a serum bNAbs concentration on the order of 10 µg/ml, and if an IC_{sub}.50: protective serum ratio of 1:100 is assumed, then protection would be then be achieved by bNAbs IC_{sub}.50 is lower than 0.1 µg/ml. As a second conservative scenario, for an IC_{sub}.50: protective serum ratio of 1:500, protection would be achieved against viruses for which the bNAbs IC_{sub}.50 is lower than 0.02 µg/ml. As shown in FIG. 33b-d, although various bnMAbs display breadth at high concentrations, viral coverage often drops sharply at lower concentrations.

Therefore, if elicited or delivered singly, only the most potent Abs, such as 121 and 128, would be able to achieve a meaningful level of viral coverage, in particular at concentrations corresponding to the more conservative scenario given above. As bnMAbs display different and in some cases complementary breadth, we further looked at the coverage achieved by antibody combinations. For the two IC_{sub}.50: protective serum concentration ratios above, a combination of PGV04 and VRC01, the two most potent CD4bs bnMAbs, would provide protection against 50% and 3% of viruses, respectively (FIG. 33c). In contrast, for a vaccine eliciting antibodies with high potency and favorable non-overlapping breadth, such as 128 and 145, coverage would be achieved against 70% and 40% of viruses for the two scenarios (FIG. 33d). Several combinations of two bnMAbs, including those directed to overlapping epitopes, can yield this degree of coverage (FIG. 44). In addition, a combination of all of the bnMAbs would cover 89% and 62% of viruses, correspondingly. Coverage against such a large proportion of viruses would likely have an important impact on the pandemic.

(659) In summary, an effective vaccine against HIV-1 may require the elicitation of a combination of complementary potent neutralizing antibodies. The demonstration that large numbers of potent and diverse bNAbs can be isolated from several different individuals provides grounds for renewed optimism that an antibody-based vaccine is achievable. Critically, the instant invention provides the required large number of potent and diverse bNAbs that comprise an antibody-based vaccine.

(660) Methods Summary

(661) Activated memory B cell supernatants were screened in a high throughput format for neutralization activity using a micro-neutralization assay, as described (Walker, L. M., et al. *Science* 326, 285-289 (2009)). Heavy and light chain variable regions were isolated from B cell lysates of selected neutralizing hits by reverse transcription from RNA followed by multiplex PCR amplification using family-specific V-gene primer sets. For some antibodies, traditional cloning methods were used for antibody isolation, as described (Walker, L. M., et al. *Science* 326, 285-289 (2009)). For other antibodies, amplicons from each lysate were uniquely tagged with multiplex identifier (MID) sequences and 454 sequencing regions (Roche). Single round of replication pseudovirus neutralization assays and cell surface binding assays were performed as described previously (Walker, L. M., et al. *Science* 326, 285-289 (2009); Pantophlet, R., et al. *J Virol* 77, 642-658 (2003); Li, M., et al. *J Virol* 79, 10108-10125 (2005)). Glycan reactivities were profiled on a printed glycan microarray (version 5.0 from the Consortium for Functional Glycomics (CFG)) as described previously (Blixt, O., et al. *Proc Natl Acad Sci USA* 101, 17033-17038 (2004)).

(662) Antibodies and Antigens

(663) The following antibodies and reagents were procured by the IAVI Neutralizing Antibody Consortium: antibody 2G12 (Polymun Scientific, Vienna, Austria), antibody F425/b4E8 (provided by Lisa Cavacini, Beth Israel Deaconess Medical Center, Boston, MA), soluble CD4 (Progenics, Tarrytown, NY), HxB2 gp120, SF162 gp120, BaL gp120, JR-FL gp120, JR-CSF gp120 and YU2 gp120 (provided by Guillaume Stewart-Jones, Oxford University). Purified ADA gp120 was

produced in the laboratory of Robert Doms, University of Pennsylvania. Fab X5 was expressed in *E. coli* and purified using an anti-human Fab specific affinity column. Deglycosylated gp120 JRFL was expressed in HEK 293S GnT.sup.-/- cells and treated with Endo H (Roche).

(664) Donors

(665) The donors identified for this study were selected from the IAVI sponsored study, Protocol G (Simek, M. D., et al. J Virol 83, 7337-7348 (2009)). Eligibility for enrolment into Protocol G was defined as: male or female at least 18 years of age with documented HIV infection for at least three years, clinically asymptomatic at the time of enrolment, and not currently receiving antiretroviral therapy. Selection of individuals for monoclonal antibody generation was based on a rank-order high throughput screening and analytical algorithm (Simek, M. D., et al. J Virol 83, 7337-7348 (2009)). Volunteers were identified as elite neutralizers based on broad and potent neutralizing activity against a cross-clade pseudovirus panel (Simek, M. D., et al. J Virol 83, 7337-7348 (2009)).

(666) Isolation of MAbs

(667) The method for isolating human MAbs from memory B cells in circulation has previously been described (Walker, L. M., et al. Science 326, 285-289 (2009)). Surface IgG B cells seeded at near clonal density in 384-well microplates were activated in short-term culture. Supernatants were screened for neutralization activity against 2-4 pseudotyped viruses for which neutralization activity was detected at high titers in the donor serum. Heavy and light chain variable regions were isolated from B cell lysates of selected neutralizing hits by reverse transcription from RNA followed by multiplex PCR amplification using family-specific V-gene primer sets. Amplicons from each lysate were uniquely tagged with multiplex identifier (MID) sequences and 454 sequencing regions (Roche, Indianapolis, IN). A normalized pooling of gamma, kappa and lambda chains was performed based on agarose gel image quantitation and the pool was analysed by 454 Titanium® sequencing. Consensus sequences of the V.sub.H and V.sub.L chains were generated using the Amplicon Variant Analyzer (Roche) and assigned to specific B cell culture wells by decoding the MID tags. Selected V.sub.H and V.sub.L chains were synthesized and cloned in expression vectors with the appropriate IgG.sub.1, IgG.sub.3 or IgG.sub.4 constant domain. Monoclonal antibodies were reconstituted by transient transfection in HEK293 cells followed by purification from serum-free culture supernatants.

(668) TABLE-US-00074 TABLE 62 Sequence (Direction is 5'-3' for forward/sense or reverse/ SEQ Donor Antibody Chain Primer antisense primers) ID NO: 584 PGT-141 Heavy Forward-VH1 ACTATGGACTGGATTTGGAGGATC 585 584 PGT-142 Heavy Forward-VH1 ACTATGGACTGGATTTGGAGGATC 585 584 PGT-143 Heavy Forward-VH1 ACTATGGACTGGATTTGGAGGATC 585 584 PGT-144 Heavy Forward-VH1 ACTATGGACTGGATTTGGAGGATC 585 584 PGT-145 Heavy Forward-VH1 ACTATGGACTGGATTTGGAGGATC 585 517 PGT-121 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 517 PGT-122 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 517 PGT-123 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 517 PGT-124 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 517 PGT-133 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 517 PGT-134 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-125 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-126 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-127 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-128 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-130 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-131 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-132 Heavy Forward-VH4 AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-135 Heavy Forward-VH4

AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-138 Heavy Forward-VH4
AACATGAAACACCTGTGGTTCTTCCT 586 196 PGT-139 Heavy Forward-VH4
AACATGAAACACCTGTGGTTCTTCCT 586 039 PGT-135 Heavy Forward-VH4
AACATGAAACACCTGTGGTTCTTCCT 586 039 PGT-136 Heavy Forward-VH4
AACATGAAACACCTGTGGTTCTTCCT 586 039 PGT-137 Heavy Forward-VH4
AACATGAAACACCTGTGGTTCTTCCT 586 584 PGT-141 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 584 PGT-142 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 584 PGT-143 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 584 PGT-144 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 584 PGT-145 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 517 PGT-121 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 517 PGT-122 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 517 PGT-123 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 517 PGT-124 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 517 PGT-133 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 517 PGT-134 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-125 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-126 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-127 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-128 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-130 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-131 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-132 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-135 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-138 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 196 PGT-139 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 039 PGT-135 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 039 PGT-136 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 039 PGT-137 Heavy Reverse
GGAAGTAGTCCTTGACCAGGCAGC 587 584 PGT-141 Light Forward-VK2
ATGAGGCTCCCTGCTCAGCT 588 584 PGT-142 Light Forward-VK2
ATGAGGCTCCCTGCTCAGCT 588 584 PGT-143 Light Forward-VK2
ATGAGGCTCCCTGCTCAGCT 588 584 PGT-144 Light Forward-VK2
ATGAGGCTCCCTGCTCAGCT 588 584 PGT-145 Light Forward-VK2
ATGAGGCTCCCTGCTCAGCT 588 039 PGT-135 Light Forward-VK3
CCCCAGCTCAGCTTCTCTTCC 589 039 PGT-136 Light Forward-VK3
CCCCAGCTCAGCTTCTCTTCC 589 039 PGT-137 Light Forward-VK3
CCCCAGCTCAGCTTCTCTTCC 589 584 PGT-141 Light Reverse
CCTTGGATAGAAGTTATTCAGC 590 584 PGT-142 Light Reverse
CCTTGGATAGAAGTTATTCAGC 590 584 PGT-143 Light Reverse
CCTTGGATAGAAGTTATTCAGC 590 584 PGT-144 Light Reverse
CCTTGGATAGAAGTTATTCAGC 590 584 PGT-145 Light Reverse
CCTTGGATAGAAGTTATTCAGC 590 039 PGT-135 Light Reverse
CCTTGGATAGAAGTTATTCAGC 590 039 PGT-136 Light Reverse
CCTTGGATAGAAGTTATTCAGC 590 039 PGT-137 Light Reverse
CCTTGGATAGAAGTTATTCAGC 590 196 PGT-125 Light Forward-VL2
CATGGCCTGGGCTCTGCT 591 196 PGT-126 Light Forward-VL2 CATGGCCTGGGCTCTGCT
591 196 PGT-127 Light Forward-VL2 CATGGCCTGGGCTCTGCT 591 196 PGT-128 Light
Forward-VL2 CATGGCCTGGGCTCTGCT 591 196 PGT-130 Light Forward-VL2
CATGGCCTGGGCTCTGCT 591 196 PGT-131 Light Forward-VL2 CATGGCCTGGGCTCTGCT

591 196 PGT-132 Light Forward-VL2 CATGGCCTGGGCTCTGCT 591 196 PGT-135 Light
 Forward-VL2 CATGGCCTGGGCTCTGCT 591 196 PGT-138 Light Forward-VL2
 CATGGCCTGGGCTCTGCT 591 196 PGT-139 Light Forward-VL2 CATGGCCTGGGCTCTGCT
 591 517 PGT-121 Light Forward-VL3 CCATGGCCTGGATCCCTCT 592 517 PGT-122 Light
 Forward-VL3 CCATGGCCTGGATCCCTCT 592 517 PGT-123 Light Forward-VL3
 CCATGGCCTGGATCCCTCT 592 517 PGT-124 Light Forward-VL3
 CCATGGCCTGGATCCCTCT 592 517 PGT-133 Light Forward-VL3
 CCATGGCCTGGATCCCTCT 592 517 PGT-134 Light Forward-VL3
 CCATGGCCTGGATCCCTCT 592 196 PGT-125 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-126 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-127 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-128 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-130 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-131 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-132 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-135 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-138 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 196 PGT-139 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 517 PGT-121 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 517 PGT-122 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 517 PGT-123 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 517 PGT-124 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 517 PGT-133 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593 517 PGT-134 Light Reverse
 CCTTCATGCGTGACCTGGCAGC 593

PGT Antibody Expression and Purification

(669) Antibody genes were cloned into an expression vector and transiently expressed with the FreeStyle 293 Expression System (Invitrogen, Carlsbad, CA). Antibodies were purified using affinity chromatography (Protein A Sepharose Fast Flow, GE Healthcare, UK). Purity and integrity were checked with SDS-PAGE.

(670) Neutralization Assays

(671) Neutralization by monoclonal antibodies and donor sera was performed by Monogram Biosciences using a single round of replication pseudovirus assay as previously described (Richman, D. D., et al. Proc Natl Acad Sci USA 100, 4144-4149 (2003)). Briefly, pseudoviruses capable of a single round of infection were produced by co-transfection of HEK293 cells with a subgenomic plasmid, pHIV-1luc3, that incorporates a firefly luciferase indicator gene and a second plasmid, pCXAS that expressed HIV-1 Env libraries or clones. Following transfection, pseudoviruses were harvested and used to infect U87 cell lines expressing co-receptors CCR5 or CXCR4. Pseudovirus neutralization assays using HIV-1JR-csf alanine mutants are fully described elsewhere (Walker, L. M., et al. Science 326, 285-289 (2009)). Neutralization activity of MAbs against HIV-1JR-csf alanine mutants was measured using a TZM-BL assay, as described (Walker, L. M., et al. Science 326, 285-289 (2009)). Kifunensine-treated pseudoviruses were produced by treating 293T cells with 25 μ M kifunensine on the day of transfection. Memory B cell supernatants were screened in a micro-neutralization assay against a cross-clade panel of HIV-1 isolates and SIVmac239 (negative control). This assay was based on the 96-well pseudotyped HIV-1 neutralization assay (Monogram Biosciences) and was modified for screening 15 μ l of B cell culture supernatants in a 384-well format.

(672) Cell Surface Binding Assays

(673) Titrating amounts of antibodies were added to HIV-1 Env transfected 293T cells, incubated for 1 hr at 37° C., washed with FACS buffer, and stained with goat anti-human IgG F(ab')₂

conjugated to phycoerythrin (Jackson ImmunoResearch, West Grove, PA). Binding was analyzed using flow cytometry, and binding curves were generated by plotting the mean fluorescence intensity of antigen binding as a function of antibody concentration. For competition assays, titrating amounts of competitor antibodies were added to the cells 30 min prior to adding biotinylated PGT MAbs at a concentration required to give EC₅₀.

(674) ELISA Assays

(675) For antigen-binding ELISAs, serial dilutions of MAbs were added to antigen-coated wells and binding was probed with alkaline phosphatase-conjugated goat anti-human immunoglobulin G (IgG) F(ab')₂ Ab (Pierce, Rockford, IL). For competition ELISAs, titrating amounts of competitor MAbs were added to gp120-coated ELISA wells and incubated for 30 min prior to adding biotinylated PGT MAbs at a concentration required to give IC₅₀. Biotinylated PGT MAbs were detected using alkaline phosphatase conjugated streptavidin (Pierce) and visualized using p-nitrophenol phosphate substrate (Sigma, St. Louis, MO).

(676) Glycan Microarray Analysis

(677) Monoclonal antibodies were screened on a printed glycan microarray version 5.0 from the Consortium for Functional Glycomics (CFG) as described previously (Blixt, O., et al. Proc Natl Acad Sci USA 101, 17033-17038 (2004)). Antibodies were used at a concentration of 30 µg/ml and were precomplexed with 15 µg/ml secondary antibody (goat-anti-human-Fc-rPE, Jackson ImmunoResearch) before addition to the slide. Complete glycan array data sets for all antibodies may be found at www.functionalglycomics.org in the CFG data archive under “cfg_rRequest_2250”.

(678) Oligomannose Dendron Synthesis

(679) The oligomannose dendrons (Man.sub.4D and Man.sub.9D) were synthesized by Cu(I) catalyzed alkyne-azide cycloaddition between azido oligomannose and the second generation of AB3 type alkynyl dendron. Detailed procedures and characterization were previously reported (Wang, S. K., et al. Proc Natl Acad Sci USA 105, 3690-3695 (2008)).

(680) Fabrication of Gp120 Microarray

(681) NETS-activated glass slides (Nexterion slide H, Schott North American) were printed with robotic pin (Arrayit 946) to deposit gp120 JRFL at concentrations of 750 or 250 µg/ml in printing buffer (120 mM phosphate, pH 8.5; containing 5% glycerol and 0.01% TWEEN® 20 (polysorbate 20)). 12 replicates were used for each concentration. The printed slides were incubated in relative humidity 75% chamber overnight and treated with blocking solution (superblock blocking buffer in PBS, Thermo) at room temperature for 1 h. The slides were then rinsed with PBS-T (0.05% Tween 20) and PBS buffer, and centrifuged at 200 g to remove residual solution from slide surface.

(682) Oligomannose Dendron-Gp120 Competition Assay with MAbs

(683) Serial diluted oligomannose dendrons were mixed with MAb (40 µg/ml) in PBS-BT buffer (1% BSA and 0.05% TWEEN® 20 (polysorbate 20) in PBS). The mixtures were applied directly to each sub-array on slide. After incubation in a humidified chamber for 1 h at RT, the slides were rinsed sequentially with PBS-T and PBS buffer, and then centrifuged at 200 g. Each sub-array was then stained with Cy3 labeled goat anti-human Fc IgG (7.5 m/ml in PBSBT) for 1 h in a humidified chamber. The slides were then rinsed sequentially with PBS-T and demonized water and centrifuged at 200 g. The fluorescence of the final arrays was imaged at 10 µm resolution (Ex: 540 nm; Em: 595 nm) with an ArrayWorx microarray reader (Applied Precision).

(684) Sequence Analysis

(685) Germ line genes were predicted using the immunoglobulin sequence alignment tools IMGT/V-QUEST (Brochet, X., et al. Nucleic Acids Res 36, W503-508 (2008)) and SoDA2 (Munshaw, S. & Kepler, T. B. Bioinformatics 26, 867-872 (2010)). Clonally-related sequences were identified by common germ line V-genes and long stretches of identical N-nucleotides.

(686) Statistical Analysis

(687) Statistical analyses were done with Prism 5.0 for Mac (GraphPad, La Jolla, CA). Viruses that

are not neutralized at an IC_{sub}.50 or IC_{sub}.90 < 50 µg/ml were given a value of 50 µg/ml for median calculations. For combinations of antibodies, a virus was counted as covered if at least one of the MAbs was neutralized depending on individual concentrations (IC_{sub}.50). This approach does not take additivity into account and therefore underestimates the neutralization potency of antibody combinations.

(688) Having thus described in detail preferred embodiments of the present invention, it is to be understood that the invention defined by the above paragraphs is not to be limited to particular details set forth in the above description as many apparent variations thereof are possible without departing from the spirit or scope of the present invention.

Claims

1. A non-naturally occurring PGT-124 monoclonal antibody comprising (a) a light chain variable region comprising three complementarity determining regions having the amino acid sequences of SEQ ID NOs: 415, 151, and 416 and (b) a heavy chain variable region comprising three complementarity determining regions having the amino acid sequences of SEQ ID NOs: 406, 407, and 408, wherein the CDRs are per Kabat definition, wherein the antibody comprises an engineered Fc region incorporating a salvage receptor binding epitope, thereby exhibiting increased serum half-life through enhanced interaction with the neonatal Fc receptor (FcRn).
2. The antibody of claim 1, comprising a light chain variable region having the amino acid sequence of SEQ ID NO: 414 and a heavy chain variable region having the amino acid sequence of SEQ ID NO: 405.
3. The antibody of claim 1, comprising a light chain having the amino acid sequence of SEQ ID NO: 413 and a heavy chain having the amino acid sequence of SEQ ID NO: 404.
4. An antigen-binding fragment of the antibody according to claim 1.
5. The antigen-binding fragment of the antibody according to claim 4, which is selected from the group consisting of the Fab, Fab', F(ab')₂, Fv, single chain Fv, diabody, and domain antibody (dAb) fragments.
6. A nucleic acid molecule encoding the antibody according to claim 1, wherein the nucleic acid molecule encodes a light chain variable region comprising the nucleic acid sequence of SEQ ID NO: 412 and a heavy chain variable region comprising the nucleic acid sequence of SEQ ID NO: 403.
7. A nucleic acid molecule encoding the antibody according to claim 1, wherein the nucleic acid molecule encodes a light chain sequence comprising the nucleic acid sequence of SEQ ID NO: 411 and a heavy chain sequence comprising the nucleic acid sequence of SEQ ID NO: 402.
8. The nucleic acid molecule according to claim 6, which is selected from the group consisting of cDNA, hnRNA, and mRNA.
9. A vector comprising the nucleic acid molecule according to claim 6.
10. A cell comprising the vector according to claim 8.
11. An immortalized B cell clone expressing the antibody according to claim 1.
12. A pharmaceutical composition comprising the antibody of claim 1 and a pharmaceutically acceptable carrier.
13. A pharmaceutical composition comprising the nucleic acid molecule of claim 6 and a pharmaceutically acceptable carrier.
14. A method of treating an HIV-1 infection, or for post-exposure prophylaxis comprising administering a pharmaceutical composition, in an effective amount, comprising the antibody of claim 1 to a patient in need thereof.
15. A method of treating an HIV-1 infection, or for post-exposure prophylaxis comprising administering a pharmaceutical composition, in an effective amount comprising the nucleic acid molecule of claim 6 to a patient in need thereof; wherein the nucleic acid molecule comprises a

DNA expression system comprising a promoter and leader sequence operably linked to the variable region sequence, and/or operably linked to an origin of replication, and/or operably linked to one or more selectable markers, and/or operably linked to one or more control elements/regulatory sequences.

16. A method for producing the antibody of claim 1, comprising the steps of (i) culturing a host cell that expresses the antibody of claim 1 and (ii) isolating the antibody.
